

NEW EDITION

New General Mathematics 4

O' Level Course

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With Answers

New General Mathematics 4

An 'O' Level Course

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General arithmetic (4)

Approximations, estimates, limits of accuracy

Approximation and estimation (revision)

Approximations are very useful when calculating. Numbers can be approximated to obtain rough estimates of calculations. For example, if someone pays \$93.86 tax each month, then the amount paid in a year is $\$93.86 \times 12$. A rough estimate of the total amount paid in a year is

$$\begin{aligned} \$90 \times 10 &= \$900 \\ &\text{(rounding to 1 significant figure)} \end{aligned}$$

$$\begin{aligned} \$95 \times 10 &= \$950 \\ &\text{(rounding to the nearest 5)} \end{aligned}$$

$$\begin{aligned} \$90 \times 12 &= \$1080 \\ &\text{(rounding one number, but not the other)} \end{aligned}$$

Rough estimates are *not* accurate. (For example, the true amount in the above calculation is \$1126.32.) However, they give some idea of the size or **order of magnitude** of the correct results of calculations. They are often used to check the correctness of answers to calculations. They are especially valuable when using calculators.

Example 1

Use a calculator to find the value of $\frac{4.386 \times 0.0894}{18.17}$.

Check your result by making a rough estimate.

$$\begin{aligned} \frac{4.386 \times 0.0894}{18.17} &= \frac{0.3921084}{18.17} \\ &= 0.0215799 \end{aligned}$$

$$\text{Rough estimate: } \frac{4 \times 0.09}{20} = \frac{0.36}{20} = 0.018$$

The answer and the rough estimate are of the same order of magnitude (both are about 0.02). If the answer and the rough estimate are *not* of the same order of magnitude then the data should be re-entered into the calculator.

Rounding off numbers (revision)

Numbers can be rounded off to the nearest hundred, ten, whole number, etc., or to a given number of decimal places.

The digits 1, 2, 3 and 4 are rounded down and the digits 5, 6, 7, 8 and 9 are rounded up.

Example 2

Round off the number 163,864 (a) to 2 decimal places; (b) to 1 decimal place; (c) to the nearest whole number; (d) to the nearest hundred.

- | | |
|------------------------|-----------------------------|
| (a) $163,864 = 163.86$ | to 2 d.p. |
| (b) $163,864 = 163.9$ | to 1 d.p. |
| (c) $163,864 = 164$ | to the nearest whole number |
| (d) $163,864 = 200$ | to the nearest hundred |

Significant figures (revision)

Numbers are sometimes rounded off to a given number of significant figures. The significance of a digit depends on its position in the number. Thus in the number 146,83, the 4 is more significant than the 6, the 6 is more significant than the 8 and the 1 is the most significant digit.

The first significant figure in a decimal fraction is the first non-zero digit in the fraction. For example, the first significant figure in the decimal 0.002487 is 2.

Example 3

Round off 146,83 (a) to 1 significant figure, (b) to 2 significant figures, (c) to 3 significant figures, (d) to 4 significant figures.

- (a) $146,83 = 100$ to 1 s.f.
- (b) $146,83 = 150$ to 2 s.f.
- (c) $146,83 = 147$ to 3 s.f.
- (d) $146,83 = 146,8$ to 4 s.f.

Example 4

Round off 0,002 487 to (a) 1 s.f., (b) 2 s.f., (c) 3 s.f.

- (a) 0,002 to 1 s.f.
- (b) 0,002 5 to 2 s.f.
- (c) 0,002 49 to 3 s.f.

Example 5

Round off (a) 8,026 to 3 s.f., (b) 50,95 to 2 s.f., (c) 18,057 to 1 decimal place, (d) \$450 170 to 3 s.f.

- (a) $8,026 = 8,03$ to 3 s.f.
- (b) $50,95 = 51$ to 2 s.f.
- (c) $18,057 = 18,1$ to 1 d.p.
- (d) $\$450\ 170 = \$450\ 000$ to 3 s.f.

Example 6

In 1987 Zimbabwe's exports to Britain were \$240 million to 2 s.f. What is the range of values of these exports?

The range of values is between \$235 million and \$245 million.

Exercise 1a

1 Round off the following to the nearest whole number.

- (a) 3,24 m (b) 0,97 m
- (c) 86,02 ha (d) 341,77 cm
- (e) 496 km (f) 164,90 mm
- (g) 52,84 (h) \$18,07
- (i) 129,7 litres (j) \$99,50

2 Round off the following to 1 decimal place.

- (a) 0,078 6 (b) 18,04 (c) 18,954
- (d) 0,786 (e) 7,926 (f) 20,08

3 Round off the following to 2 significant figures.

- (a) 36,9 (b) 109,4
- (c) 0,009 281 (d) 4,98

- (e) 0,024 39 (f) 86 125
- (g) 144 (h) 9,04
- (i) 2,99 (j) 0,030 46

4 Round off the following to 3 significant figures.

- (a) 7 579 (b) 52 069
- (c) 352 289 (d) 1 789
- (e) 0,089 21 (f) 170,08
- (g) 83,352 (h) 0,906 4
- (i) 827 502 (j) 8,007

5 In 1987 there was an estimated \$389 400 000 in notes and coin in circulation in Zimbabwe. Give at least three ways in which a newspaper might have reported this amount.

6 In the 1982 census, the populations of Zimbabwe's three largest towns were given as shown in Table 1.1.

Table 1.1

town	population
Bulawayo	495 300
Chitungwiza	172 000
Harare	658 400

Round off these numbers to (a) 1 s.f., (b) 2 s.f.

7 A newspaper headline reads 'Government rejects a loan of \$1,5 billion from IMF.' Between what two amounts does this figure lie?

8 The numbers of foreign visitors to Zimbabwe between 1984 and 1988 are given in Table 1.2.

Table 1.2

1984	1985	1986	1987	1988
339 598	389 465	433 372	487 716	491 721

(a) Round off the number to 1 s.f.

(b) Use your rounded numbers to estimate the total number of foreign visitors to Zimbabwe for the five years.

9 Find the order of magnitude of the outcomes of the following calculations. (Round the given numbers to 1 s.f.)

(a) $67,09 \times 4,38 \times 0,1178$

(b) $8\,956 \div 27,31$

(c)
$$\frac{55,73 \times 8\,607}{645,8}$$

(d)
$$\frac{3,705}{22,73 \times 76,35}$$

- 10 Use a calculator to find the true outcomes of the calculations in question 9. Give all your answers correct to 4 s.f.

Limits of accuracy

No measurement, however carefully made, is exact. The best measuring instruments are usually accurate to three figures only. Most measurements, therefore, are approximate. For example, if the length of a line is given as 23,8 cm, its true length will lie between a minimum, or **lower bound**, of 23,75 cm and a maximum, or **upper bound**, of 23,85 cm. This gives a possible **error** of $\pm 0,05$ cm in the given length. Figure 1.1 shows the end of the line (magnified) and the lower and upper bounds on its length. The difference between the upper and lower bound gives the range of measurement.



Fig. 1.1

Percentage error

$$\text{Percentage error} = \frac{\text{error}}{\text{measurement}} \times 100\%$$

In Fig. 1.1,

$$\begin{aligned} \text{percentage error} &= \pm \frac{0,05}{23,8} \times 100\% \\ &= \pm 0,210\,084\% \\ &= \pm 0,21\% \quad (\text{to 2 s.f.}) \end{aligned}$$

Example 7

The length of a pole is measured as 5 metres to the nearest metre. State (a) the upper and lower bounds of the length of the pole, (b) the greatest possible error, and calculate (c) the percentage error in the measurement.

(a) Since the measurement is given to the nearest metre, the true length of the pole will be between 4,5 m and 5,5 m.

$$\text{upper bound} = 5,5 \text{ m}$$

$$\text{lower bound} = 4,5 \text{ m}$$

(b) Greatest possible error = $\pm 0,5$ m

$$\begin{aligned} \text{(c) Percentage error} &= \pm \frac{0,5 \text{ m}}{5 \text{ m}} \times 100\% \\ &= \pm 10\% \end{aligned}$$

Example 8

A girl and a boy estimate the length of a line to be 9 cm and 12 cm respectively. If the true length of the line is 9,6 cm, find (a) the absolute error, (b) the percentage error, for each student.

(a) Absolute error

$$= \text{approximate value} - \text{true value}$$

For the girl,

$$\begin{aligned} \text{absolute error} &= 9 \text{ cm} - 9,6 \text{ cm} \\ &= -0,6 \text{ cm} \end{aligned}$$

The girl has underestimated by 0,6 cm.

For the boy,

$$\begin{aligned} \text{absolute error} &= 12 \text{ cm} - 9,6 \text{ cm} \\ &= +2,4 \text{ cm} \end{aligned}$$

The boy has overestimated by 2,4 cm.

$$\text{(b) Percentage error} = \frac{\text{absolute error}}{\text{true value}} \times 100\%$$

For the girl,

$$\begin{aligned} \text{percentage error} &= \frac{-0,6}{9,6} \times 100\% \\ &= -6,25\% \end{aligned}$$

The girl's estimate was 6,25% too low.

For the boy,

$$\begin{aligned} \text{percentage error} &= \frac{+2,4}{9,6} \times 100\% \\ &= 25\% \end{aligned}$$

The boy's estimate was 25% too high.

Example 9

The length of a running track is measured and given as 400 m. Find the percentage error if the length is measured (a) to the nearest metre, (b) to the nearest 10 m, (c) to 1 significant figure.

(a) The range of actual measurement is between 399,5 m and 400,5 m.

$$\text{Error} = \pm 0,5 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\pm 0,5 \text{ m}}{400 \text{ m}} \times 100\% \\ &= \pm 0,125\%\end{aligned}$$

(b) The range of actual measurement is between 395 m and 405 m.

$$\text{Error} = \pm 5 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\pm 5 \text{ m}}{400 \text{ m}} \times 100\% \\ &= \pm 1\frac{1}{4}\%\end{aligned}$$

(c) The range of actual measurement is between 350 m and 450 m.

$$\text{Error} = \pm 50 \text{ m}$$

$$\begin{aligned}\text{Percentage error} &= \frac{\pm 50 \text{ m}}{400 \text{ m}} \times 100\% \\ &= \pm 12\frac{1}{2}\%\end{aligned}$$

Notice in Example 9 that the percentage error becomes greater as the method of estimation becomes rougher.

Example 10

Given the calculation $\$8,70 \times 2,4$, a student gets a rough answer by rounding the given values to the nearest whole number. Calculate the resulting percentage error.

$$\begin{aligned}\text{Rough answer} &= \$9 \times 2 \\ &= \$18\end{aligned}$$

$$\begin{aligned}\text{True answer} &= \$8,70 \times 2,4 \\ &= \$20,88\end{aligned}$$



$$\begin{aligned}\text{Absolute error} &= \$18 - \$20,88 \\ &= -\$2,88\end{aligned}$$

$$\begin{aligned}\text{Percentage error} &= \frac{-\$2,88}{\$20,88} \times 100\% \\ &= -13,8\% \text{ to 3 s.f.}\end{aligned}$$



Example 11

The length and breadth of a rectangular plot are measured to the nearest metre as being 16 m and 11 m respectively. Calculate the upper and lower bounds of the area of the plot.

The upper bound of the area of the rectangle occurs when its length and breadth have their greatest possible values:

$$\begin{aligned}\text{greatest possible length} &= 16,5 \text{ m} \\ \text{greatest possible breadth} &= 11,5 \text{ m} \\ \text{corresponding area (upper bound)} &= 16,5 \times 11,5 \text{ m}^2 \\ &= 189,75 \text{ m}^2\end{aligned}$$

The lower bound of the area of the rectangle occurs when its length and breadth have their least possible values:

$$\begin{aligned}\text{least possible length} &= 15,5 \text{ m} \\ \text{least possible breadth} &= 10,5 \text{ m} \\ \text{corresponding area (lower bound)} &= 15,5 \times 10,5 \text{ m}^2 \\ &= 162,75 \text{ m}^2\end{aligned}$$

In Example 11, notice how errors become compounded when approximate values are used in computations. The upper and lower bounds of the area of the rectangle in the example differ by 27 m^2 , quite a considerable amount.

Exercise 1b

1 What is the range of values of each of the following measurements?

(a) The length of a line segment is 9 cm to the nearest cm.

(b) A man is 1,75 m tall to the nearest cm.

(c) The radius of the earth is 6 400 km to 2 s.f.

(d) The distance from the school to the market is 9,8 km to 1 d.p.

(e) The population of Zimbabwe (1989) was 9 million to 1 s.f.

(f) A container holds 7,5 litres of petrol to the nearest 0,1 litre.

(g) A woman spent \$20 000 to the nearest thousand dollars.

(h) The mass of a wrestler is 75,6 kg to the nearest 0,1 kg.

- 3 The maximum temperature for a particular day in Kadoma was $30,2^{\circ}\text{C}$ to the nearest $0,1^{\circ}\text{C}$.
- 4 The volume of a sphere is 860 cm^3 to the nearest cm^3 .
- 2 Calculate the percentage error in each of the following measurements.
- The capacity of a bucket is 7,5 litres to 1 d.p.
 - The distance between two towns is 60 km to the nearest km.
 - A pole is 125 cm high to the nearest cm.
 - The volume of a box is 25 cm^3 to the nearest cm^3 .
 - The thickness of a book is 20 mm to the nearest mm.
 - The mass of a girl is 62 kg to 2 s.f.
 - The speed of an aircraft is 800 km/h to 1 s.f.
 - The radius of a ball is 21 cm to the nearest cm.
 - The area of a classroom is 400 m^2 to 2 s.f.
 - The University of Zimbabwe has 9 000 students to 1 s.f.
- 3 There are 36 candles in a box. Each candle has a mass of 115 g.
- Round the given values to 1 s.f. and hence estimate the mass of the candles in the box.
 - Calculate the true mass of the candles.
 - Hence find, to 1 d.p., the percentage error of your estimation.
- 4 A French tourist has 980 Francs which she wishes to exchange for Dollars. The rate of exchange is \$0,347 to 1 Franc. She roughly estimates the value of her Francs by calculating $0,35 \times 1000$.
- Write down her rough estimate.
 - Calculate the true value in Dollars and cents.
 - Hence find, to 1 d.p., the percentage error in the estimation.
- 5 Given the calculation $783 \div 1,8$, a student gets a rough answer by rounding each number to 1 s.f. Calculate, to 1 d.p., the percentage error in his result.
- 6 13 g of salt is dissolved in 88 g of water to make a salt solution. If each amount is given to the nearest gram, calculate the upper and

lower bounds of the masses of the solution.

- 7 A box contains 960 packets of coffee. Each packet is marked '250 g to the nearest 5 g'. Calculate, in kg, the upper and lower bounds of the masses of coffee in the box.
- 8 The length and breadth of a rectangular field are given as 100 m and 70 m to the nearest 5 m. Calculate the least and greatest possible areas of the field.
- 9 The radius of a circle is given as 6 cm to the nearest whole cm. Calculate, in terms of π , the lower and upper bounds of (a) the circumference, (b) the area, of the circle.
- 10 The edge of a cube is given as 8 cm to the nearest whole cm. Calculate the lower and upper bounds of its (a) surface area in cm^2 , (b) volume in cm^3 .

Degree of accuracy

Many calculations involve measurements. The degrees of accuracy of such measurements affect the degree of accuracy of the results of the calculations. Therefore, the degree of accuracy of measurements in a calculation must be taken into consideration when determining the answer to the calculation. The final result of a calculation should not be given to a number of significant figures more than the number of significant figures in any of the given data.

Rounded-off values are sometimes used in calculations. For example, π is often taken as 3,14 or 3,142.

Sometimes it seems that an answer found at an intermediate stage of a calculation may need to be rounded off before it is used in a subsequent stage. However, since such rounded-off values affect the degree of accuracy of the results of calculations, it is generally advisable *not* to round off intermediate values.

Example 12

Calculate the area of a circle of radius 3,5 cm.

$$\begin{aligned}
 \text{Area of circle} &= \pi (3,5)^2 \text{ cm}^2 \\
 &= 3,14 \times (3,5)^2 \text{ cm}^2 \\
 &= 38,465 \text{ cm}^2 \\
 &= 38 \text{ cm}^2 \text{ to 2 s.f.}
 \end{aligned}$$



Note that the radius, 3.5 cm, is given to 2 s.f. and 3.14 is a rounded-off value of π . In this case the answer should not be given to more than 2 significant figures.

Rough estimate: $3 \times 4 \times 4 \text{ cm}^2 = 48 \text{ cm}^2$

The rough estimate is of the same degree as the calculated answer. (To make rough estimates, first express each number to 1 s.f. before calculating.)

Exercise 1c

Give answers to reasonable degrees of accuracy unless otherwise stated.

- 1 A car travels a distance of 100 km for 1 h 38 min 45 s.
 - (a) Round off the time to the nearest $\frac{1}{4}$ hour.
 - (b) Use the rounded time to find the average speed for the journey in km/h. Give your answer to 1 significant figure.
- 2 In an experiment, the radius, r , of a spherical balloon is measured as 21 cm. Calculate the volume, V , of the balloon. Take π to be 3.14 and use $V = \frac{4}{3}\pi r^3$.
- 3 In the right-angled triangle, ABC, in Fig. 1.2, $\hat{B} = 90^\circ$ and the lengths of AB and BC are given to the nearest centimetre. Calculate AC.

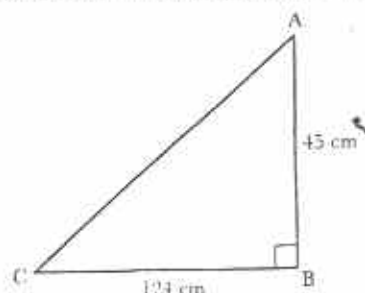


Fig. 1.2

- 4 Use four-figure tables or a calculator to calculate the following. Take π to be 3.142.

(a) $\pi\sqrt{3}$

(b) $2\pi \times 5,289$

(c) $(2,41)^3$

(d) $\frac{\pi}{28}$

- 5 Using square root and reciprocal tables, or a calculator, calculate the values of the following.

(a) $\frac{1}{\sqrt{110}}$

(b) $\frac{\sqrt{56}}{9}$

(c) $\frac{80}{\sqrt{12,5}}$

- 6 If a sheet of cardboard is 0.8 mm thick, calculate the range of values of the height of 328 sheets of the same cardboard. Also calculate the maximum and the minimum number of sheets which are in a pile 50 cm high.
- 7 A room is 4.6 m long, 3.7 m wide and 3.2 m high. Calculate
 - (a) the diagonal of the longer wall,
 - (b) the diagonal of the floor.
- 8 An aircraft travels 5 000 km (to 1 s.f.) in 6 hours (to the nearest hour). Calculate its speed in km/h.
- 9 Calculate the volume of a cylinder of radius 4 cm and height 20 cm. (Take π to be 3.14.)
- 10 Taking the value of π to be 3.14, calculate the area of a circle of radius 30 cm.

Geometrical constructions (3) Locus

Basic constructions

Fig. 2.1 shows the basic constructions using set square, ruler and compasses.

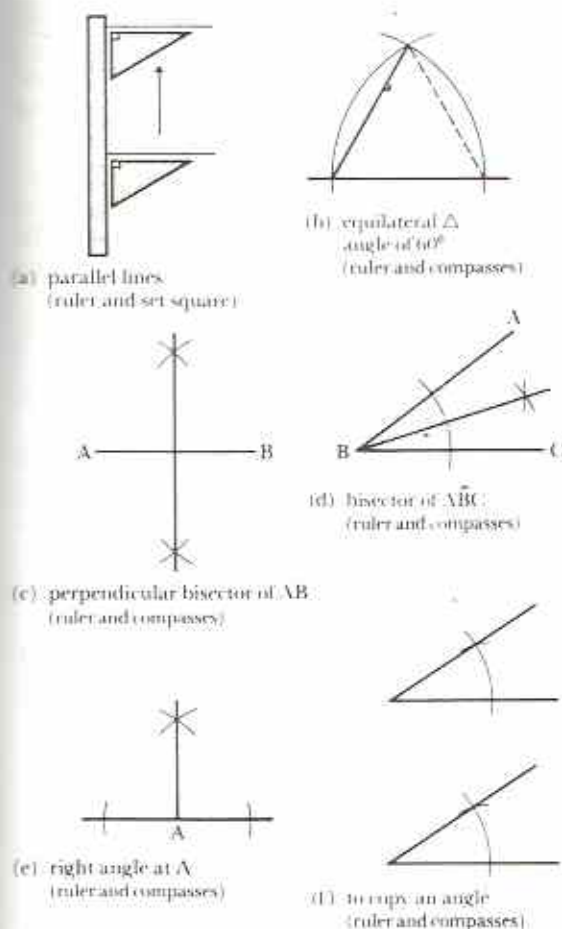


Fig. 2.1

To construct angles of 45° and 30° , bisect angles of 90° and 60° respectively (Fig. 2.2).

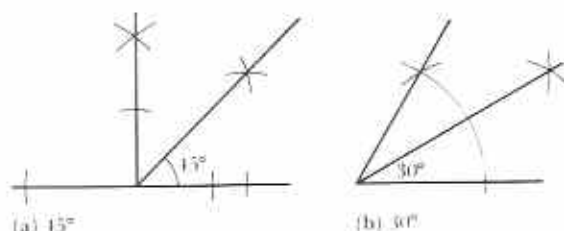


Fig. 2.2

Exercise 2a (Revision)

Make a sketch before constructing the figure.

- (a) Construct $\triangle ABC$ such that $AB = 8$ cm, $BC = 11$ cm and $AC = 10$ cm.

(b) Mark a point X on AB such that $BX = 5$ cm. Use ruler and set square to construct a line through X parallel to BC to meet AC at Y .

(c) Measure XY and CY .
- (a) Construct an equilateral triangle XYZ such that $XY = 5$ cm.

(b) With YZ as base, construct isosceles $\triangle AYZ$ such that $AY = AZ = 8$ cm.

(c) Similarly construct isosceles $\triangle BYZ$ such that B is on the opposite side of YZ to A with $BY = BZ = 8$ cm.

(d) What kind of quadrilateral is $BYAZ$?

(e) Measure AX and BX .
- (a) Construct $\triangle ABC$ such that $AB = 7$ cm, $BC = 6$ cm and $\angle ABC = 60^\circ$.

(b) The bisector of $\angle C$ meets the perpendicular bisector of AB at X . Find the point X by construction.

(c) Measure BX .

- 4 (a) Use ruler and compasses to construct $\triangle PQR$ in which $\hat{Q} = 90^\circ$, $QR = 5$ cm and $PR = 10$ cm.
 (b) Measure PQ .
 (c) By using Pythagoras' theorem now check the result.
- 5 (a) Use ruler and compasses to construct the parallelogram $PQRS$ in which $QR = 5$ cm, $RS = 11$ cm and $\hat{QRS} = 135^\circ$.
 (b) Measure the length of the shorter diagonal of $PQRS$.
- 6 (a) Construct quadrilateral $ABCD$ such that $AB = 5$ cm, $BD = DC = 8$ cm, $\hat{ABD} = 30^\circ$ and $\hat{BCD} = 45^\circ$.
 (b) Measure the diagonal AC .

To construct a perpendicular to a given straight line from a point outside the line.
 Given a line AB and a point M outside the line,



It is required to construct a line through M perpendicular to AB .

(a) Place the sharp end of a pair of compasses on M . Open the compasses sufficiently to draw an arc which cuts AB at P and Q as in Fig. 2.4.

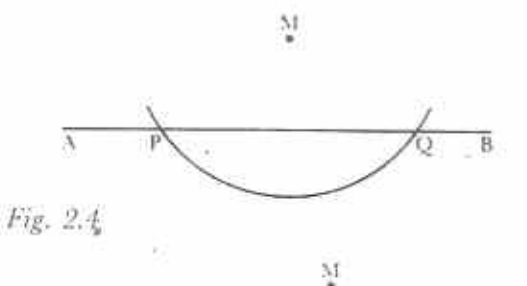


Fig. 2.4

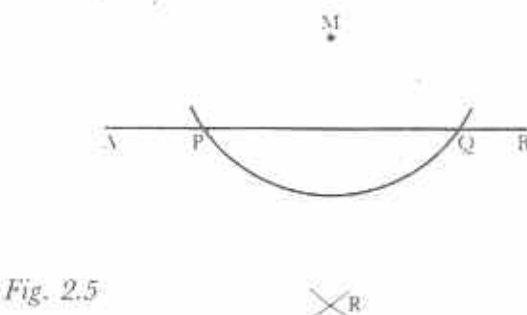


Fig. 2.5

- (b) With centres A and B and equal radii, draw arcs to cut each other at R as in Fig. 2.5.
 (c) Join MR , cutting AB at S (Fig. 2.6).
 In Fig. 2.6, $\hat{MSA} = \hat{MSB} = 90^\circ$. (Check by measurement.)

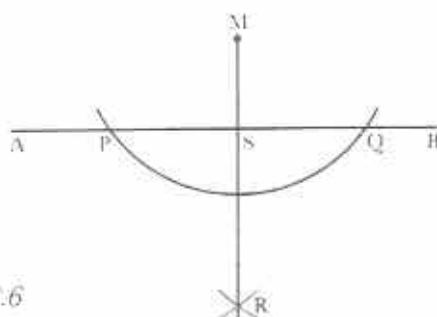


Fig. 2.6

Exercise 2b

Make a sketch before constructing the accurate figure.

- 1 (a) Construct a triangle with sides 6, 7, 9 cm.
 (b) Construct the three altitudes of the triangle. (c) What do you notice?
- 2 (a) Construct a triangle with sides 5, 6, 9 cm.
 (b) Construct the three altitudes as in question 1. (Two of the sides will have to be produced.) (c) Do the altitudes meet at one point?
- 3 (a) Construct a rectangle $ABCD$ in which $AB = 86$ mm, $BC = 58$ mm.
 (b) Construct the bisectors of A and B . Let the bisectors meet at point E .
 (c) Construct the perpendicular from E to AB and measure its length.
- 4 (a) Construct $\triangle LMN$ in which $LM = 7.6$ cm, $MN = LN = 9.7$ cm.
 (b) Through N , construct a line parallel to LM .
 (c) Construct MD perpendicular to the line constructed in (b), meeting it at D . Measure MD .
- 5 (a) Construct $\triangle ABC$ with sides 6, 8, 9 cm.
 (b) Bisect all three angles of $\triangle ABC$. The bisectors should meet in one point. Call this I .
 (c) Construct the perpendicular IX from I to BC .
 (d) Draw a circle with radius IX and centre I . Does this circle do anything special?
 (e) Measure the radius of the circle.

- 6 (a) Construct $\triangle XYZ$ with sides 5, 7, 9 cm.
 (b) Use the method of question 5 to construct a circle which touches all three sides of the triangle.
 (c) Measure the radius of the circle.
- 7 (a) Construct quadrilateral ABCD with $\angle D = 90^\circ$, $AD = 6$ cm, $DC = 9$ cm, $BC = 8.4$ cm and $AB = 5.4$ cm.
 (b) Bisect angles D and A and let the bisectors meet at X.
 (c) Construct the perpendicular XP to DA. Draw the circle with centre X and radius XP.
 (d) (i) Measure the radius of the circle.
 (ii) Does the circle do anything special?
- 8 (a) Draw a circle of radius 5 cm. From any point P on the circumference draw chords $PQ = 4$ cm and $PR = 6$ cm and complete the $\triangle PQR$.
 (b) Mark any point A on the circumference of the circle. Construct perpendiculars AX, AY, AZ from A to QR, RP, PQ respectively.
 (c) Is there anything special about the points X, Y, Z?
 (d) Is this always true? (Test by drawing another circle with chords of any convenient length.)



Fig. 2.7

Locus

Fig. 2.7 shows what happens to a raindrop which hits an umbrella on its way to the ground. The dotted line shows various positions of the raindrop.

The dotted line in Fig. 2.7 is called the **locus** of the raindrop. The locus can be thought of as the path traced out by the raindrop as it moves.

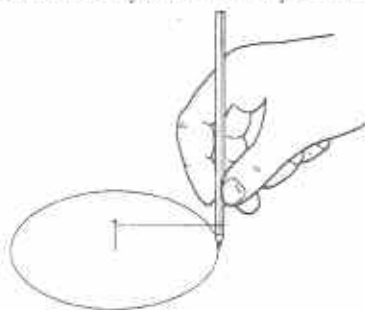


Fig. 2.8

In Fig. 2.8, a pencil point moves on a plane surface so that it is always a constant distance from a fixed point in the plane. The path that it traces out is a circle with the fixed point as centre. The locus of the pencil point is a circle.

The examples in Figs 2.7 and 2.8 demonstrate the simple definition of a locus: the path traced out by a moving point. However, this definition is not strictly correct. A point describes a *position* only, it has no size and cannot really be said to move. Hence, instead of describing a locus as the path traced out by a moving point, use the following definition:

A locus is the set of all possible positions occupied by an object which varies its position according to some given law.

In Fig. 2.7, the shape of the umbrella and the law of gravity determine the locus of the raindrop. In Fig. 2.8 the law which controls the pencil point is that it must be a constant distance from a fixed point in the same plane.

Common loci

Loci is the plural of locus.

Locus of points at a given distance from a fixed point.

In Fig. 2.9 overleaf, O is a fixed point, P_1, P_2, P_3, P_4 are points which are a constant distance

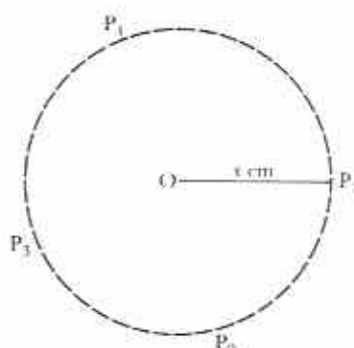


Fig. 2.9

x cm from O. The locus of the points is a **circle** of radius x cm (shown by the dotted line).

However, the locus need not necessarily lie in a plane. In 3 dimensions, the locus of P is a **spherical surface** of radius x cm and centre O (Fig. 2.10),

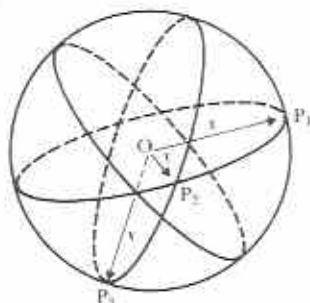


Fig. 2.10

Locus of points at a given distance from a straight line.

In Fig. 2.11, AB is a straight line which continues indefinitely in both directions. Points P_1, P_2, P_3, P_4 are each a distance x cm from AB. In 2 dimensions, the locus of the points consists of two straight lines parallel to AB, each a distance x cm from AB (shown by the dotted lines). Notice that this locus consists of **two separate lines**.

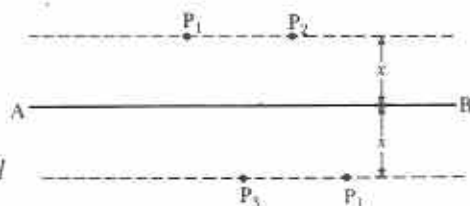


Fig. 2.11

In 3 dimensions, the locus is a **cylindrical surface** of radius x cm with AB as the central axis of the surface (Fig. 2.12).

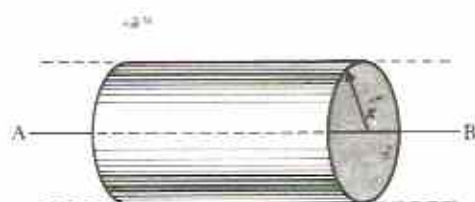


Fig. 2.12

Locus of points equidistant from two given points.

In Fig. 2.13, X and Y are two fixed points. Points P_1, P_2, P_3 are such that $P_1X = P_1Y$, $P_2X = P_2Y$ and $P_3X = P_3Y$. P_1, P_2, P_3 lie on the perpendicular bisector of XY. The locus of the points is the **perpendicular bisector** of XY (shown by the dotted line).

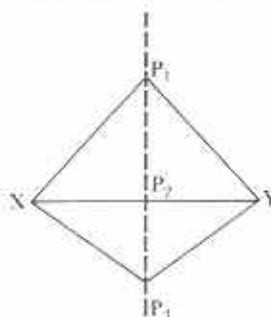


Fig. 2.13

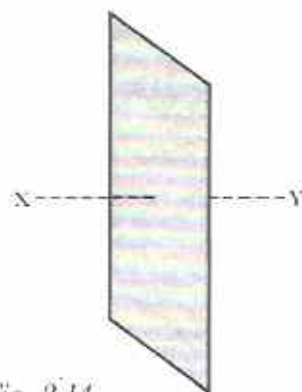


Fig. 2.14

In 3 dimensions, the locus is a **plane surface** which bisects XY perpendicularly (Fig. 2.14).

Exercise 2c

- 1 Place a cotton reel on a large sheet of paper. Unwind about 15 cm of thread and tie a loop at the free end. Put a pencil through the loop as in Fig. 2.15. Keeping the thread tight, move the pencil so that the thread winds back onto the reel. What is the shape of the locus traced by the pencil point?



Fig. 2.15

- 2 Push two drawing pins through a sheet of paper so that they are about 5 cm apart. Make a loop of thread about 15 cm in length. Place the loop over the drawing pins and use a pencil to keep it tight as in Fig. 2.16.

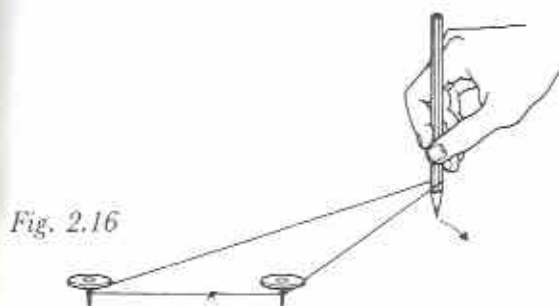


Fig. 2.16

Keeping the thread tight, move the pencil so that the point draws its locus on the paper. Describe the shape of the locus.

- 3 Draw two points A and B 4 cm apart. Use a 30° set square to plot a point P so that $\angle APB = 30^\circ$. See Fig. 2.17.

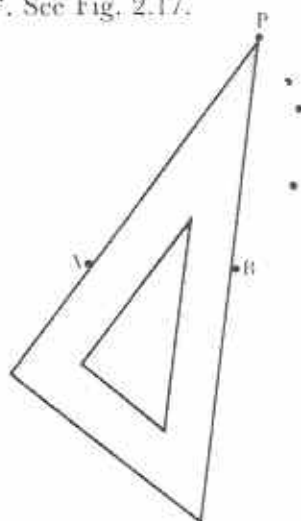


Fig. 2.17

Use this method to plot about 12 different positions of P. Hence draw the locus of P.

- 4 Repeat the method of question 3 so that $\angle APB =$ (a) 45° , (b) 60° , (c) 90° .
- 5 Draw a circle of radius 4 cm and mark a fixed point A on its circumference. P can take any position on the circumference and Q is the mid-point of AP. For various positions of P, plot the corresponding positions of Q. Hence find the locus of Q.
- 6 Describe the following loci as accurately as possible.
 - (a) The locus of a door handle when the door opens through 180° .
 - (b) The locus of the tip of the minute hand of a clock during 1 hour.
 - (c) The locus of the mid-point of the hour hand of a clock during 1 hour.
 - (d) The locus of the centre of a coin which rolls in a straight line across a floor.
 - (e) The locus of a stone swinging on the end of a string.
 - (f) The locus of a table-tennis ball when served.
- 7 Describe the locus of a ball which rolls across the floor of a car at a constant speed, if the car is also travelling along a straight road at a constant speed.
- 8 A pencil slides inside a hemispherical bowl so that both ends are in contact with the bowl. Describe the locus of the mid-point of the pencil.
- 9 P is any position on a straight line AB. Q is a fixed point 3 cm from AB. Describe the locus of the mid-point of PQ.
- 10 A wire stretches from the top of a vertical pole to a point on the horizontal ground some distance from the foot of the pole. If the wire is tight, describe the locus of the lower end of the wire.
- 11 A set square lies flat on a table. It is rotated about its hypotenuse until it is flat on the table again. What is the path traced out by its right-angled corner?
- 12 A number of circles are drawn so that their circumferences pass through two fixed points. What is the locus of the centres of the circles?
- 13 A and B are fixed points. P is a variable point such that the area of $\triangle APB$ is constant. Find the locus of P.

- 14 In Fig. 2.18, the goat is tied by a rope $3\frac{1}{2}$ m long to a fixed wire 10 m long. A ring at the end of the rope can slide along the wire.

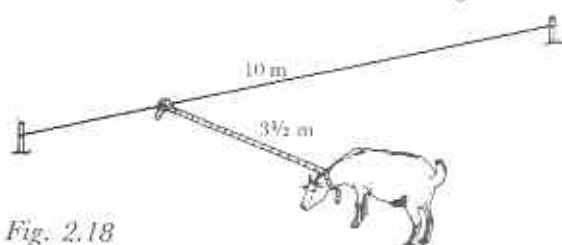


Fig. 2.18

Sketch the locus of the goat's head if the goat moves so as to keep as far as possible from the wire. Use the value $3\frac{1}{2}$ for π to calculate the approximate length of the locus.

- 15 In Fig. 2.19, a wheel of radius 14 cm starts with its centre in position A and rolls up two steps until its centre reaches position B. Sketch the locus of the centre of the wheel. Calculate the length of the locus using the value $\frac{22}{7}$ for π .

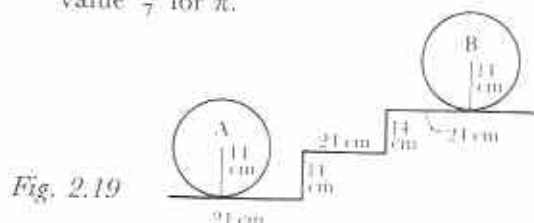


Fig. 2.19

Construction of loci in 2 dimensions

Example 1

$\triangle ABC$ is such that $\hat{B} = 70^\circ$, $AB = 5$ cm and $BC = 7.5$ cm. Find by construction the positions of a point P in the plane of $\triangle ABC$ which is equidistant from B and C and 3.5 cm from A .

Fig. 2.20 is a scale drawing showing $\triangle ABC$ and the main details of the required construction. Since P is equidistant from B and C , it must lie on the perpendicular bisector of BC . P lies on locus l_1 in Fig. 2.20.

Since P is 3.5 cm from A , it must lie on a circle of radius 3.5 cm, centre A . P lies on locus l_2 in Fig. 2.20.

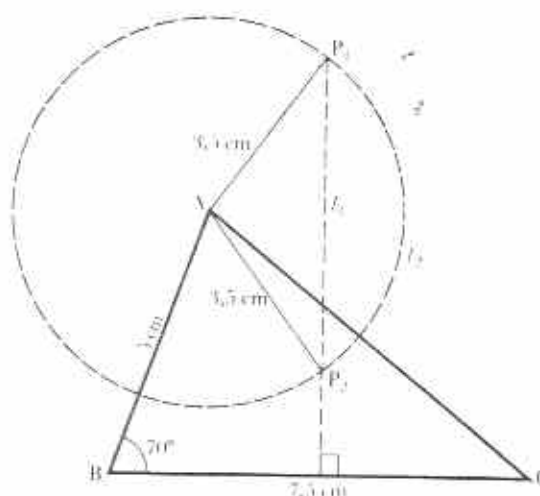


Fig. 2.20

To satisfy both conditions, P lies at the point(s) of intersection of these two loci, i.e. at P_1 and P_2 in Fig. 2.20.

Note: To make Fig. 2.20 clearer, some essential construction lines have been omitted. In practice these should be included. Also the two loci have been shown by dotted lines. Normally these would be solid lines.

Example 2

AB is a straight line. A circle is drawn with centre A and radius 2 cm. Construct the points in the plane of the circle which are 2.5 cm away from AB and from the circumference of the circle.

Fig. 2.21 is a scale drawing of the required construction.

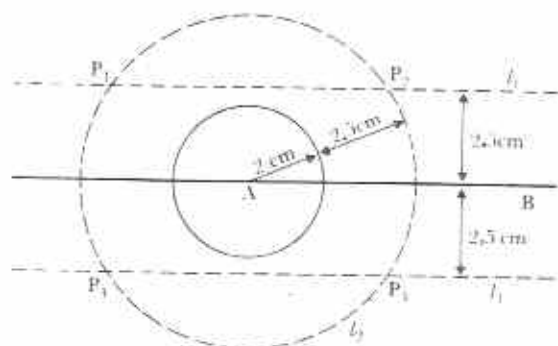


Fig. 2.21

Since the points are 2.5 cm away from AB, they must lie on lines which are parallel to AB and 2.5 cm from it, i.e. on locus l_1 in Fig. 2.21.

Since the points are 2.5 cm from the circumference of the circle, they must lie on the circumference of a circle of radius 4.5 cm, centre A, i.e. on locus l_2 in Fig. 2.21.

To satisfy both conditions, the required points are the points of intersection of these two loci, i.e. P_1, P_2, P_3, P_4 in Fig. 2.21.

Exercise 2d

In this exercise all loci are in 2 dimensions. Make a rough sketch first.

- 1 A and B are two points which are 7 cm apart. Construct the positions of a point P which is 4.2 cm from A and 5.6 cm from B. How many possible positions for P are there? Measure the distance between them.
- 2 AB and CD are two intersecting straight lines. Show how to construct the position of a point P which is 2 cm from AB and 3 cm from CD. How many possible positions for P are there?
- 3 $\triangle ABC$ is isosceles with $AB = AC = 6$ cm and $BC = 4$ cm. Construct points which are equidistant from B and C, and 3 cm from A. Measure their distances from BC.
- 4 $\triangle ABC$ has a right angle at C and $AC = BC = 4$ cm. Construct points which are equidistant from B and C, and 3 cm from A. Measure their distance from BC.
- 5 An aircraft flies at a height of 1 000 m on a straight, level course. The course takes the aircraft directly over two points A and B which are 1 500 m apart on the horizontal ground. On a scale drawing, construct two positions of the aircraft when its angle of elevation from A is 50° . In each case measure the angle of elevation from B.
- 6 A is a fixed point 3 cm from a straight line BC. Construct points which are 1 cm from BC and 3.5 cm from A. Measure the distance between them.
- 7 O is a fixed point on a straight line AB. P is a point which is 4 cm from AB and 5 cm from O. Construct 2 positions of P on the same side of AB. Measure the distance between them.
- 8 A boat is 20 m from a straight river bank and 29 m from a tree on the edge of the bank. By scale drawing, construct two possible positions of the boat. Measure the distance between them.
- 9 Show how to construct a quadrilateral PQRS in which $QR = 4$ cm, $\hat{R} = 110^\circ$, $RS = 5$ cm, $\hat{Q} = 90^\circ$ and $PQ = PS$.
- 10 A and B are two fixed points 6 cm apart. A circle is drawn with centre B and radius 2 cm. Construct the positions of points which are equidistant from A and B, and 3 cm from the circumference of the circle. Measure the distance between them.
- 11 (a) Using ruler and compasses only, construct $\triangle ABC$ such that $AC = 10$ cm, $BC = 8.5$ cm and $\hat{ACB} = 135^\circ$.
(b) Using *any* geometrical instruments, find a point P within $\triangle ABC$ which is at a distance 2.8 cm from AC and 6 cm from B. Measure the length of AP.
- 12 Construct $\triangle ABC$ so that $AB = 6$ cm, $BC = 9$ cm, $CA = 5$ cm. Construct a point P, equidistant from A and C such that the area of $\triangle APB$ is 12 cm^2 .

Further loci in 2 dimensions

Locus of points equidistant from two straight lines.

In Fig. 2.22, AB and CD are straight lines which intersect at O. P_1 is equidistant from AB and CD. Similarly P_2 is equidistant from the two lines. P_1 and P_2 lie on the bisector of the acute angle between the two lines.

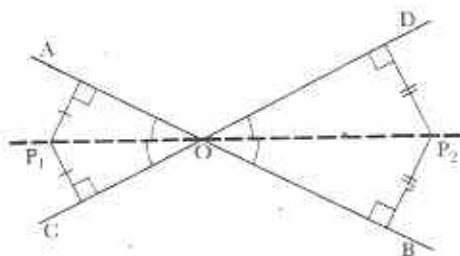


Fig. 2.22

In Fig. 2.23, P_3 is equidistant from AB and CD. P_3 lies on the bisector of the obtuse angle between the two lines.

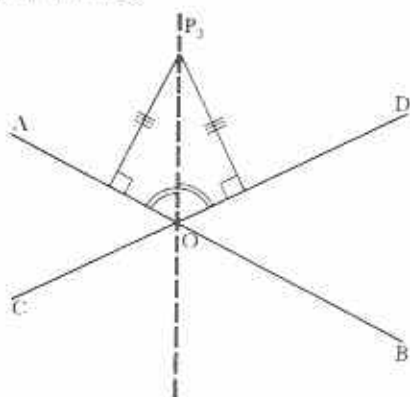


Fig. 2.23

The complete locus of points which are equidistant from two straight lines is the **pair of bisectors of the angles between the lines** (Fig. 2.24).

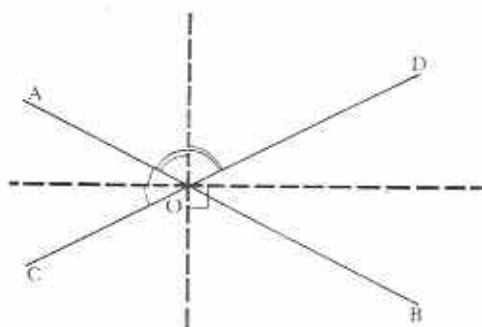


Fig. 2.24

Notice that the two parts of the locus intersect at right angles.

Locus of points which subtend a given angle on a given line segment.

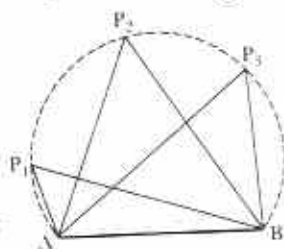


Fig. 2.25

In Fig. 2.25, AB is a given line segment. The arc $AP_1P_2P_3B$ is a major arc of a circle. It follows that $\angle AP_1B = \angle AP_2B = \angle AP_3B$ (angles in the same segment are equal). The arc is the locus of points which subtend a certain angle on a given line segment.

P can also be on the other side of AB. The complete locus of points which subtend a certain angle on AB is two circular arcs (Fig. 2.26).

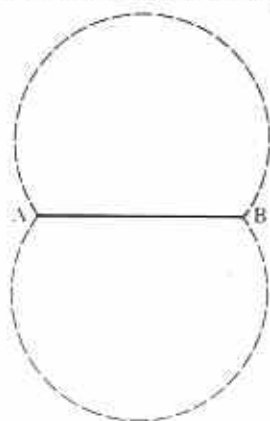


Fig. 2.26

Example 3

Using ruler and compasses only (a) construct $\triangle ABC$ such that $AB = 6$ cm, $AC = 8.5$ cm and $\angle BAC = 120^\circ$. (b) Construct the locus l_1 of points equidistant from A and B. (c) Construct the locus l_2 of points equidistant from AB and AC. (d) Find the points of intersection P_1 and P_2 , of l_1 and l_2 and measure P_1P_2 .

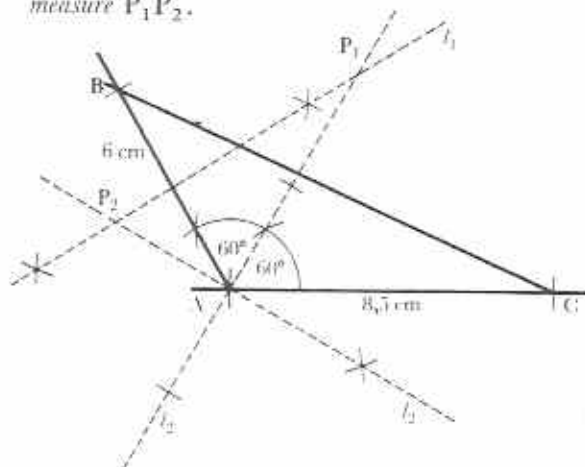


Fig. 2.27

Fig. 2.27 is a half-size scale drawing showing all construction lines and loci.

- (a) Note the construction of $\hat{BAC}$ ($= 120^\circ$).
 (b) l_1 is the perpendicular bisector of AB.
 (c) l_2 is in two parts. AP_1 is the bisector of $\hat{BAC}$. AP_2 is perpendicular to AP_1 . Notice that points on AP_2 are equidistant from AB and CA produced.
 (d) By measurement $P_1P_2 = 6.9$ cm.
 (The loci are shown by dotted lines in Fig. 2.27. In your work use solid lines.)

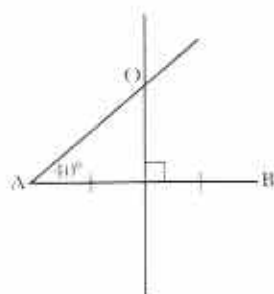


Fig. 2.29

Example 4

On a line AB, 3 cm long, construct the locus of points which subtend an angle of 50° on AB.

Method:

- 1 It is necessary to find the centre of the circular arc. Since $\hat{APB} = 50^\circ$, $\hat{AOB} = 100^\circ$ (angle at centre $= 2 \times$ angle at circumference). Hence the angles of $\triangle OAB$ are $100^\circ, 40^\circ, 40^\circ$.
- 2 Construct $\triangle OAB$.
- 3 With centre O and radius OA draw the arc APB.

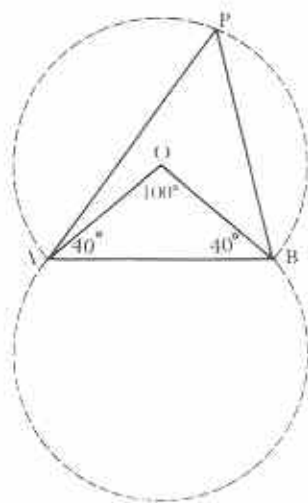


Fig. 2.28

- 4 Complete the second part of the locus on the other side of AB, using the same method.
- An alternative method of finding O is to use the fact that the centre of the arc lies on the perpendicular bisector of AB. Since $\hat{OAB} = 40^\circ$, O can be found as in Fig. 2.29.

Circumcircle of a triangle

A circle which passes through the three vertices of a triangle is the **circumcircle** of the triangle. A circumcircle can be drawn for any triangle (Fig. 2.30).

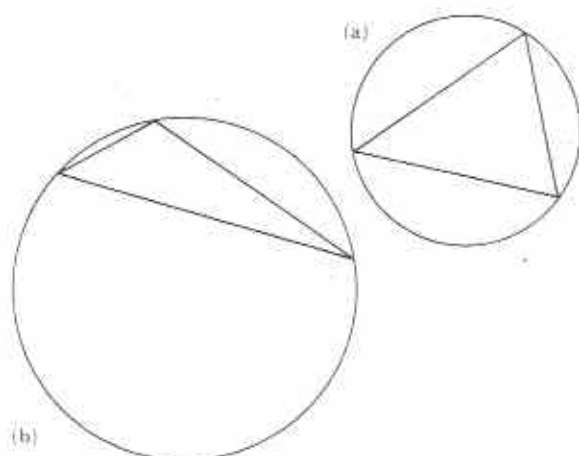


Fig. 2.30

The construction of the circumcircle of a triangle uses the fact that the perpendicular bisector of a chord of a circle passes through its centre. The three sides of the given triangle form three chords of its circumcircle.

If the perpendicular bisectors of the three sides of a triangle are constructed they will meet at a single point, the **circumcentre**, i.e. the centre of the circumcircle. The radius of the circumcircle is the distance of the circumcentre from any one of the vertices of the given triangle. The construction is shown in Fig. 2.31.

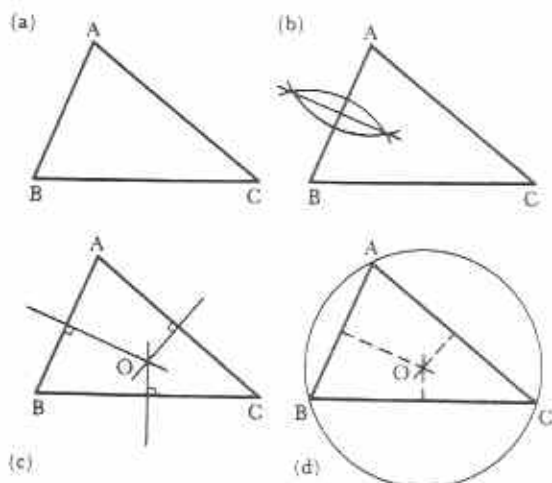


Fig. 2.31

Exercise 2e

In this exercise all loci are in 2 dimensions. Draw a rough sketch first.

- 1 Given any triangle XYZ, show how to construct a point P on YZ such that P is equidistant from XY and XZ.
- 2 Draw any triangle ABC. Construct the position of a point which is equidistant from AC and BC, and also equidistant from A and B. How many possible positions of the point are there?
- 3 On a line AB, 5 cm long, construct the locus of points which subtend an angle of 45° on AB.
- 4 On a line AB, 6 cm long, construct an arc of a circle which subtends an angle of 40° on AB. Find a point C on this arc such that AC = 8 cm. Measure BC.
- 5 Draw $\triangle ABC$ in which AB = 5.2 cm, BC = 9.2 cm and CA = 8 cm. Construct the possible positions of a point which is equidistant from AB and AC, and 3.2 cm from BC.
- 6 Draw two straight lines AOB and COD intersecting at O. On OA mark a point Q such that OQ = 2.5 cm. Construct all the possible positions of a point equidistant from AB and CD, and 4 cm from Q.
- 7 On a line XY, 8 cm long, construct the locus of all points which subtend an angle of 100° on XY.

- 8 On a base BC, 6 cm long, construct $\triangle ABC$ of area 12 cm^2 such that $\hat{A} = 65^\circ$. Measure AB and AC.
- 9 Construct a trapezium ABCD in which $AB \parallel DC$, AB = 4 cm, BC = 8 cm, CD = 11 cm, DA = 6 cm. (Hint: In a rough figure, divide the trapezium into parallelogram ABXD and $\triangle BCX$. First construct $\triangle BCX$.)
- 10 Using ruler, set square and compasses only, construct a quadrilateral ABCD, with AB = DA = 4 cm, BC = 6 cm, DC = 3 cm and $BC \parallel AD$. (Hint: Draw a rough sketch. Divide the quadrilateral into $\triangle DCX$ and parallelogram ADXB. Draw $\triangle DCX$ first and produce CX to B.)
 - (a) Draw the locus, l_1 , of points equidistant from the points C and D.
 - (b) Draw also the locus, l_2 , of points equidistant from AD and AB.
 - (c) If P is the intersection of l_1 and l_2 , measure AP.
- 11 Draw $\triangle ABC$ such that $\hat{A} = 75^\circ$, AB = 8 cm, AC = 7 cm. Construct two positions of a point P, equidistant from AC and BC, such that the area of $\triangle APB$ is 15.2 cm^2 .
- 12 Construct the circumcircle of any
 - (a) acute-angled scalene triangle,
 - (b) obtuse-angled scalene triangle,
 - (c) right-angled scalene triangle.
 Describe the positions of the circumcentres of the circumcircles that you have drawn.
- 13 Construct on a single diagram,
 - (a) triangle XYZ with base XY = 12 cm, XZ = 10 cm and YZ = 8.5 cm,
 - (b) the point P on the circumcircle of $\triangle XYZ$ such that P is equidistant from XY and XZ.
 Measure PZ to the nearest mm.
- 14 (a) Construct the parallelogram ABCD in which AB = 7 cm, BC = 4 cm and $\hat{ABC} = 120^\circ$.
 - (b) Measure and write down AC.
 - (c) On the same diagram, construct (i) the circumcircle of $\triangle ABC$, (ii) the locus of points equidistant from A and D.
 - (d) Mark the point P on the circumference of the circumcircle such that PA = PD and $\hat{APC} = 60^\circ$.

Circle geometry (2) Tangents

Tangent to a circle

In Fig. 3.1, a circle, centre O , is cut by a straight line MN at the two points X and Y . $\triangle OXY$ is isosceles and $\hat{OXY} = \hat{OYX}$. Hence $\hat{OXM} = \hat{OYN}$.

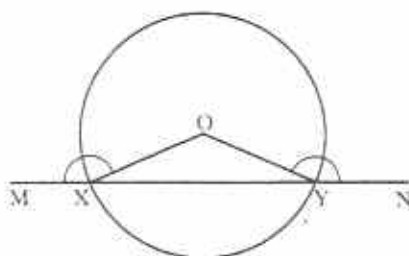


Fig. 3.1

Fig. 3.2 shows what happens to the positions of X and Y if MN moves downwards. X and Y occupy new positions such as X_1, Y_1 and X_2, Y_2 , etc., becoming closer to each other.

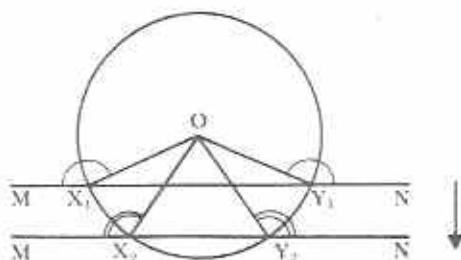


Fig. 3.2

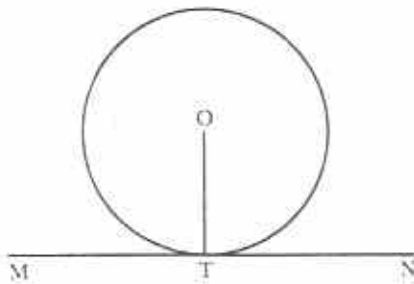


Fig. 3.3

Eventually the points X and Y will coincide at a single point T as in Fig. 3.3. Similarly the radii OX and OY will coincide to become one radius, OT .

Since $\hat{OXM} = \hat{OYN}$ (Fig. 3.1), it follows that in Fig. 3.3,

$$\hat{OTM} = \hat{OTN}$$

and since MTN is a straight line,

$$\hat{OTM} = \hat{OTN} = 90^\circ.$$

Hence $OT \perp MN$.

In Fig. 3.3, the line MN is said to be a **tangent** to the circle. The tangent *does not cut* the circle; it *touches* the circle.

Remember the following:

1 A tangent to a circle is perpendicular to the radius drawn to its point of contact.

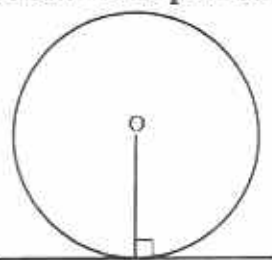


Fig. 3.4

2 The perpendicular to a tangent at its point of contact passes through the centre of the circle.

Example 1

TA is a tangent at A to a circle, centre O , AB is a chord. If $\hat{BAT} = x^\circ$, show that $\hat{BOA} = 2x^\circ$.

Fig. 3.5, overleaf, is a sketch of the circle.

If $\hat{BAT} = x^\circ$
 then $\hat{BAO} = (90 - x)^\circ$ (radius $\perp$ tangent)
 and $\hat{ABO} = (90 - x)^\circ$ (isos. $\triangle AOB$)

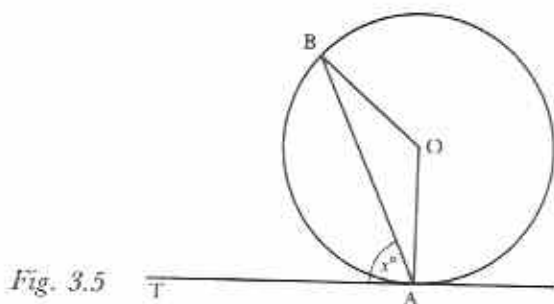


Fig. 3.5

$$\begin{aligned}\therefore \hat{BOA} &= 180^\circ - 2(90^\circ - x^\circ) \\ &\quad (\text{sum of angles of } \triangle ABO) \\ &= 180^\circ - 180^\circ + 2x^\circ \\ &= 2x^\circ\end{aligned}$$

Exercise 3a

- 1 Calculate the size of angle α in each part of Fig. 3.6. In each figure, O is the centre of the circle.

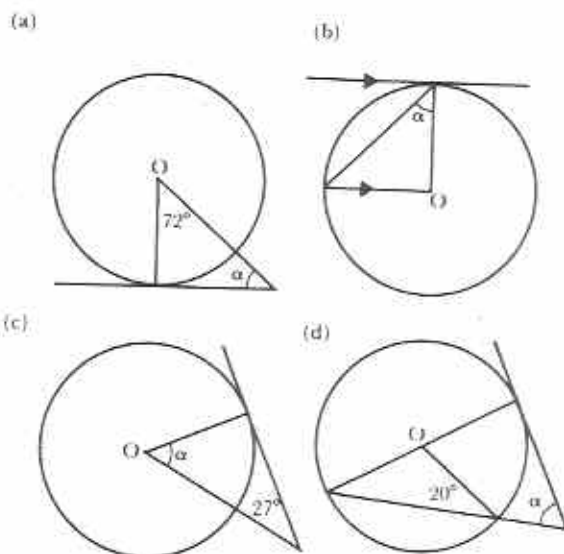


Fig. 3.6

- 2 Calculate the size of θ in each part of Fig. 3.7. In each figure, O is the centre of the circle.

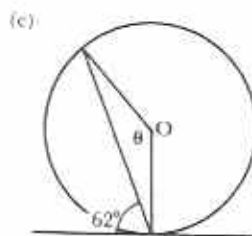
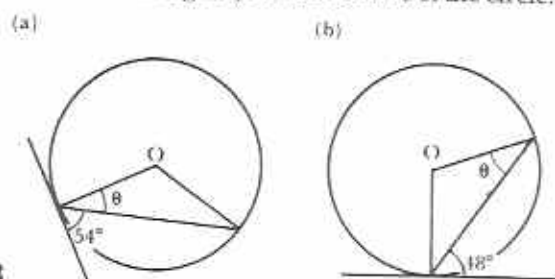
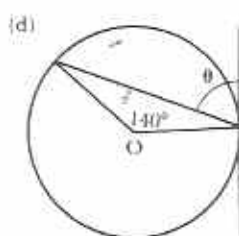


Fig. 3.7



- 3 Calculate OA in each part of Fig. 3.8. In each figure, O is the centre of the circle and the dimensions are in cm.

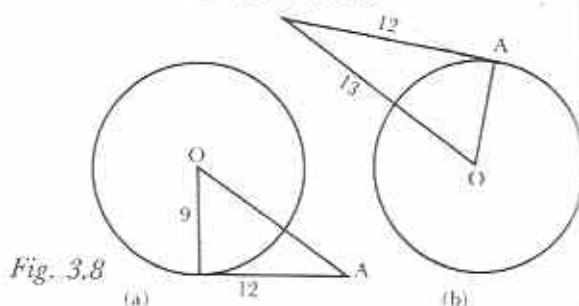


Fig. 3.8

- 4 In Fig. 3.9, O is the centre of the circle and TA is a tangent at A.

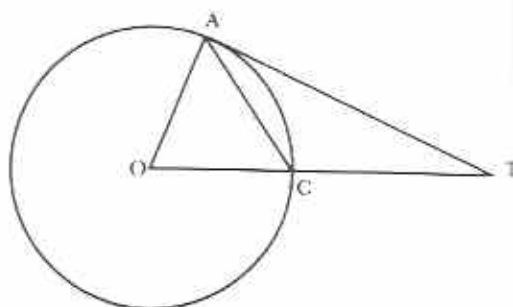


Fig. 3.9

Use Fig. 3.9 to answer the following.

- If $\hat{AOC} = 86^\circ$, calculate $\hat{CAT}$.
 - If $\hat{ATO} = 36^\circ$, calculate $\hat{ACO}$.
 - If $\hat{OAC} = 69^\circ$, calculate $\hat{ATC}$.
 - If $\hat{ACT} = 122^\circ$, calculate $\hat{CAT}$.
- 5 The tangent from a point T touches a circle at R. If the radius of the circle is 2.8 cm and T is 5.3 cm from the centre, calculate TR.
- 6 A point P is 6.5 cm from the centre of a circle, and the length of the tangent from P is 5.6 cm. Calculate the radius of the circle.

- 7 AB is a chord and O is the centre of a circle. If $\angle AOB = 78^\circ$, calculate the obtuse angle between AB and the tangent at B.
- 8 Two circles have the same centre and their radii are 15 cm and 17 cm. A tangent to the inner circle at P cuts the outer circle at Q. Calculate PQ.
- 9 In Fig. 3.10 AC is a tangent to the circle, centre O, and $\angle BCA = 90^\circ$.

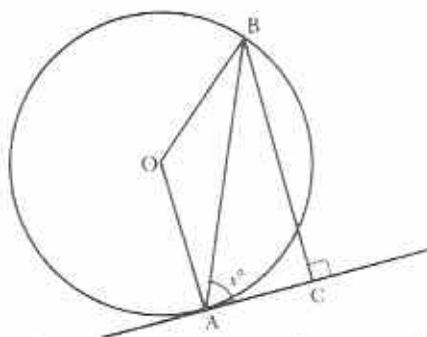


Fig. 3.10

- (a) If $\angle BAC = x^\circ$, find (i) $\angle OBA$, (ii) $\angle ABC$ in terms of x .
- (b) Hence show that AB bisects $\angle OBC$.
- 10 AD is a diameter of a circle, AB is a chord and AT is a tangent. (a) State the size of $\angle ABD$. (b) If $\angle BAT$ is an acute angle of x° , find the size of $\angle DAB$ in terms of x . (c) Hence prove that $\angle BAT = \angle ADB$.

Tangents from an external point

Theorem

The tangents to a circle from an external point are equal.

Given: A point T outside a circle, centre O. TA and TB are tangents to the circle at A and B.

To prove: $TA = TB$.

Construction: Join OA, OB and OT.

Proof:

In $\triangle$ s OAT and OBT (Fig. 3.11),

$$\angle A = \angle B = 90^\circ \quad (\text{radius} \perp \text{tangent})$$

$$OA = OB \quad (\text{radii})$$

$$OT = OT \quad (\text{common side})$$

$$\therefore \triangle OAT \equiv \triangle OBT \quad (\text{RHS})$$

$$\therefore TA = TB$$

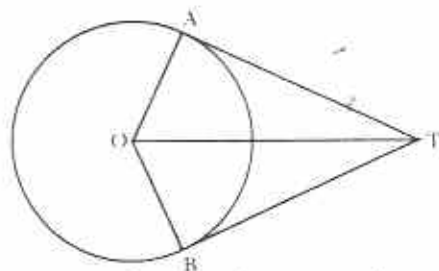


Fig. 3.11

Notice also that $\angle AOT = \angle BOT$ and $\angle ATO = \angle BTO$. Hence the line joining the external point to the centre of the circle bisects the angle between the tangents and the angle between the radii drawn to the points of contact of the tangents, i.e. OT, is on the line of symmetry of Fig. 3.11.

Example 2

In Fig. 3.12 O is the centre of the circle and TA and TB are tangents. If $\angle ATO = 39^\circ$, calculate $\angle TBX$.

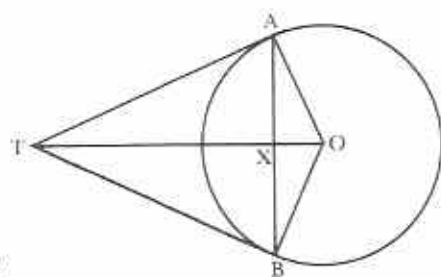


Fig. 3.12

$$\begin{aligned} \text{In } \triangle TAX, \\ \angle AXT &= 90^\circ && (\text{symmetry of Fig. 3.12}) \\ \therefore \angle TAX &= 180^\circ - (90^\circ + 39^\circ) && (\text{sum of angles of } \triangle) \\ &= 180^\circ - 129^\circ \\ &= 51^\circ \\ \therefore \angle TBX &= 51^\circ && (\text{symmetry}) \end{aligned}$$

In Example 2 there are many ways of showing that $\angle TBX = 51^\circ$, e.g. by noticing that $\angle ATX$ is the semi-vertical angle of isosceles triangle ATB.

Example 3

X, Y, Z are three points on a circle, centre O. The tangents at X and Y meet at T. If $\angle XTY = 58^\circ$, calculate $\angle XZY$.

See Fig. 3.13.

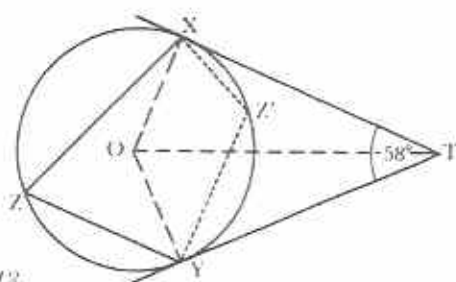


Fig. 3.13

Join OX and OY.

In quadrilateral TXOY,

$$\angle OXT = \angle OYT = 90^\circ \quad (\text{radius} \perp \text{tangent})$$

$$\therefore \angle OXT + \angle OYT = 180^\circ$$

$$\therefore \angle XTY + \angle XOY = 180^\circ \quad (\text{angles of quad.})$$

$$\therefore \angle XOY = 180^\circ - 58^\circ$$

$$= 122^\circ$$

$$\therefore \angle XYZ = \frac{1}{2} \text{ of } 122^\circ \quad (\text{angle at centre} \\ = 2 \times \text{angle} \\ \text{at circumference})$$

$$= 61^\circ$$

If Z is on the minor arc XY (at Z') then

$$\angle XZY = 119^\circ \quad (\text{opp. angles of cyclic quad.})$$

Hence $\angle XYZ$ is 61° or 119° .

Exercise 3b

- Use Fig. 3.12 to answer the following.
 - If $\angle ATO = 36^\circ$, calculate $\angle ABO$.
 - If $\angle ABT = 57^\circ$, calculate $\angle AOT$.
 - If $\angle BTO = 44^\circ$, calculate $\angle TAX$.
 - If $AB = 18\text{ cm}$, and $TB = 15\text{ cm}$, calculate TX .
- In Fig. 3.12, prove that $TAOB$ is a cyclic quadrilateral.
- In Fig. 3.12, if $\angle AOT = 47^\circ$, calculate $\angle ABO$.
- A, B, C are three points on a circle, centre O, such that $\angle BAC = 37^\circ$. The tangents at B and C meet at T. Calculate $\angle BTC$.
(Hint: Make a sketch and join BO and CO.)
- P, Q, R are three points on a circle, centre O. The tangents at P and Q meet at T. If $\angle PTQ = 62^\circ$, calculate $\angle PRQ$.

- In Fig. 3.14 AB is a diameter of circle ABC centre O. TA and TC are tangents.

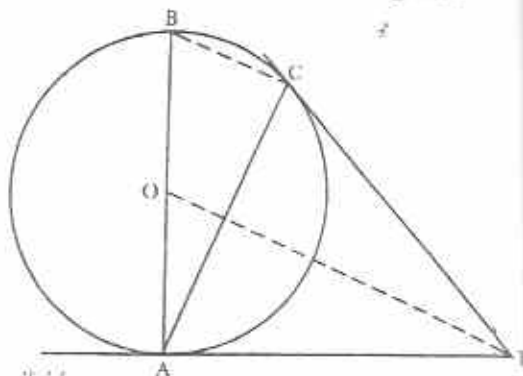


Fig. 3.14

If $\angle ATC = 2x$, show that $\angle BAC = x$. (Hint: Join OT.)

- In Fig. 3.14 prove that BC is parallel to OT.
- A quadrilateral PQRS is such that a circle can be drawn inside it to touch all four sides. Prove that $PQ + RS = PS + QR$.
- A, B, C are three points on a circle. The tangents at A and B meet at T, and $BC \parallel TA$. Prove that AB bisects $\angle TBC$.
- O is the centre of a circle, and two tangents from a point T touch the circle at A and B. BT is produced to C. If $\angle AOT = 67^\circ$, calculate $\angle ATC$.

Contact of circles

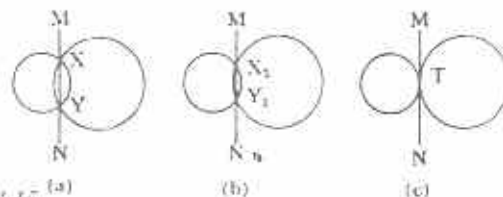


Fig. 3.15

Fig. 3.15(a) shows two circles with a common chord XY. If line MN is fixed and the circles move away from each other, the points X and Y become closer together, such as X_1, Y_1 in Fig. 3.15(b). Eventually the points will coincide at a single point as in Fig. 3.15(c). In Fig. 3.15(c), T is the point of contact of the circles and MN is a tangent to both circles at T.

Two circles touch each other if they both touch the same straight line at the same point.

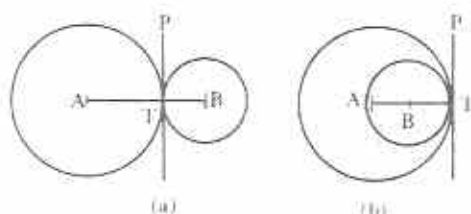


Fig. 3.16

In Fig. 3.16(a) the two circles touch each other **externally**, and in Fig. 3.16(b) they touch **internally**.

Let their centres be A and B and let the common tangent at T be PT.

In Fig. 3.16(a):

$$\hat{ATP} = \hat{BTP} = 90^\circ \quad (\text{tangent} \perp \text{radius})$$

$$\therefore \hat{ATB} = 180^\circ$$

$\therefore$ ATB is a straight line.

In Fig. 3.16(b):

$$\hat{ATP} = \hat{BTP} = 90^\circ$$

$\therefore$ AT and BT are both at right angles to TP.

$\therefore$ A, B and T lie in a straight line.

The straight line joining A and B in Fig. 3.16 is called the **line of centres**. In both parts of Fig. 3.16, **if two circles touch each other, the point of contact lies on the line of centres.**

The distance between their centres is the **sum** of their radii if the circles touch externally, and the **difference** of their radii if they touch internally.

Example 4

A and B are the centres of two circles which touch each other externally. Both circles touch a third circle, centre C, internally. If $AB = 13$ cm, $BC = 14$ cm, $CA = 11$ cm, calculate the radii of the circles.

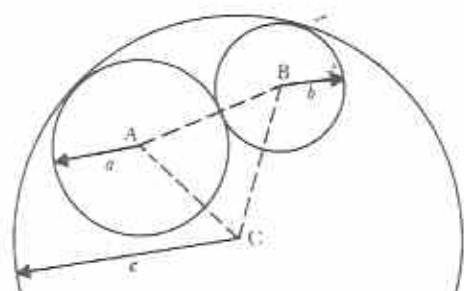


Fig. 3.17

Fig. 3.17 is a sketch of the three circles.

Let the radii of the circles centres A, B, C be a , b , c cm respectively. Then:

$$\text{from AB, } a + b = 13 \quad (1)$$

$$\text{from BC, } c - b = 14 \quad (2)$$

$$\text{from CA, } c - a = 11 \quad (3)$$

Subtract (3) from (2)

$$-b + a = 3 \quad (4)$$

Add (1) to (4)

$$2a = 16$$

$$\therefore a = 8$$

$$\therefore b = 5$$

$$\therefore c = 19$$

The circles are of radii, 8 cm, 5 cm and 19 cm respectively.

Exercise 3c

1 Fig. 3.18 shows three circles which touch each other externally.

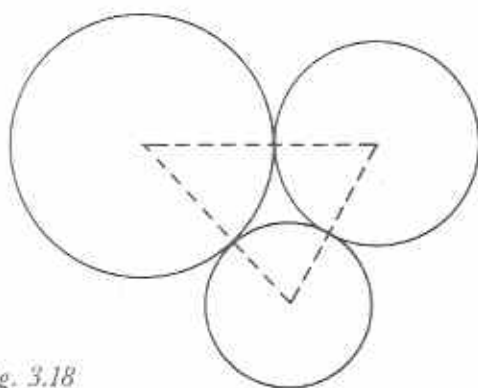


Fig. 3.18

If the centres of the circles form a triangle with sides of length 9 cm, 7 cm, 6 cm, calculate the radii of the circles.

- In Fig. 3.18, find the radii of the circles if their centres form a triangle of sides 5 cm, 6 cm, 7 cm.
- In Fig. 3.17, calculate the radii of the circles if $AB = 11$ cm, $BC = 8$ cm and $CA = 9$ cm.
- Two circles, centres X and Y , touch externally at T . A is a point on their common tangent such that $AT = 12$ cm, $AX = 13$ cm and $AY = 15$ cm. (a) Calculate XY . (b) If the circles touch internally instead of externally, what is XY ?
- In Fig. 3.19, two circles touch at T and a line through T cuts the circles at A and B . OA and QB are radii.

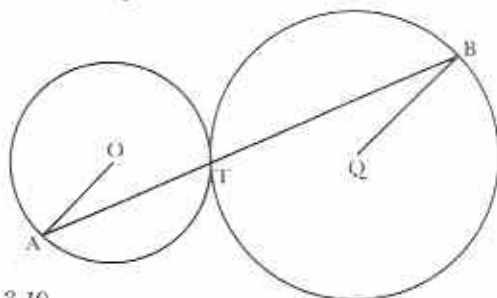


Fig. 3.19

Prove that $OA \parallel QB$. (Hint: Join OQ .)

- Two circles touch at a point A . T is any point of their common tangent. Tangents from T touch one of the circles at P and the other at Q . Prove that $TP = TQ$.
- Two circles touch each other at T . A straight line touches one circle at A and the other at B . Prove that $\angle ATB = 90^\circ$.
- Fig. 3.20 is a plan view of three equal cylindrical tins held together by an elastic band.

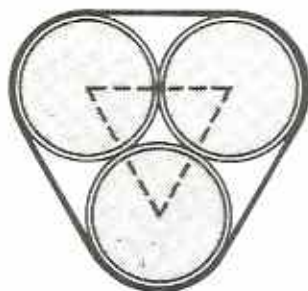


Fig. 3.20

If the centres of the tins form an equilateral triangle of side 14 cm, use the value $3\frac{1}{2}$ for π to calculate the length of the elastic band in this position.

Alternate segment

In both parts of Fig. 3.21 SAT is a tangent to the circle at A . The chord AB divides the circle into two segments APB and AQB .

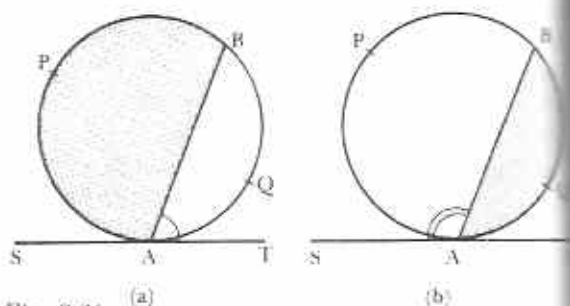


Fig. 3.21

In Fig. 3.21(a) the segment APB is the **alternate segment** to $\angle SAT$, i.e. it is on the other side of AB from $\angle SAT$. Similarly, in Fig. 3.21(b), segment AQB is the alternate segment to $\angle SAT$.

Sometimes the word **opposite** is used instead of **alternate**.

Theorem

If a straight line touches a circle, and from the point of contact a chord is drawn, the angles which the chord makes with the tangent are equal to the angles in the alternate segments.

Given: A circle, with SAT a tangent at A and chord AB dividing the circle into two segments APB and AQB . Segment APB is alternate to $\angle SAT$.

To prove: $\angle SAT = \angle APB$ and $\angle SAT = \angle AQB$.

Construction: Draw diameter AD . Join BD .

Proof:

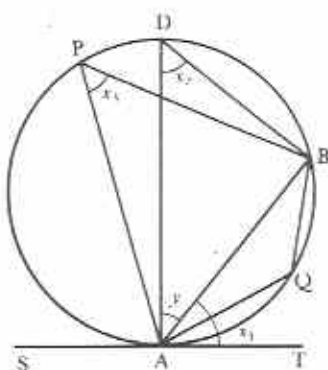


Fig. 3.22

With the lettering of Fig. 3.22,

$$x_1 + y = 90^\circ \quad (\text{tangent} \perp \text{radius})$$

$$\text{also } \angle ABD = 90^\circ \quad (\text{angle in semicircle})$$

$$\therefore x_2 + y = 90^\circ \quad (\text{sum of angles of } \triangle)$$

$$\therefore x_1 = x_2 \quad (\text{angles in same segment})$$

$$\therefore \angle TAB = \angle APB$$

$$\text{Also } \angle SAB = 180^\circ - x_1 \quad (\text{angles on str. line})$$

$$= 180^\circ - x_3 \quad (x_1 = x_3 \text{ proved})$$

$$= \angle AQB \quad (\text{opp. angles of cyclic quad.})$$

Example 5

In Fig. 3.23 PQX is a tangent to the circle QRS. Calculate $\angle SQX$.

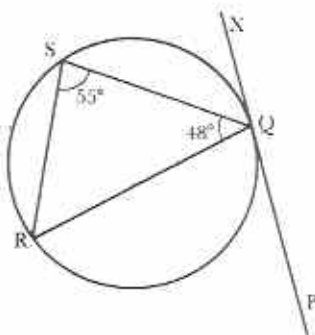


Fig. 3.23

In $\triangle QRS$,

$$\angle SQX = 180^\circ - (55^\circ + 48^\circ) \quad (\text{sum of angles of } \triangle)$$

$$= 180^\circ - 103^\circ$$

$$= 77^\circ$$

$$\therefore \angle SQX = 77^\circ \quad (\text{alternate segment})$$

Example 6

In Fig. 3.24 PT is a tangent to circle ABCT, $BA = BT$ and $\angle ATP = 82^\circ$. Calculate $\angle BCT$.

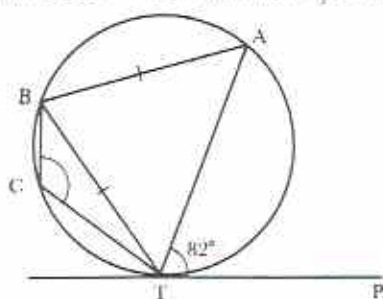


Fig. 3.24

$$\angle ABT = 82^\circ \quad (\text{alternate segment})$$

In $\triangle ABT$,

$$\begin{aligned} \angle BAT &= \frac{1}{2}(180^\circ - 82^\circ) \quad (\text{sum of angles of} \\ &= \frac{1}{2} \times 98^\circ \quad \text{isos. } \triangle) \\ &= 49^\circ \end{aligned}$$

$$\therefore \angle BCT = 180^\circ - 49^\circ \quad (\text{opp. angles of cyclic quad.})$$

$$= 131^\circ$$

Exercise 3d

1 In Fig. 3.25 XYZ is a tangent to circle ABCY.

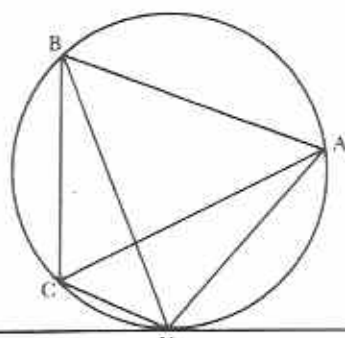


Fig. 3.25

- Name two angles equal to $\angle AYZ$.
- Name two angles equal to $\angle CYX$.
- Name an angle equal to $\angle BYZ$.
- Name an angle equal to $\angle BYX$.
- If $\angle AYZ = 58^\circ$ what is $\angle ACY$?
- If $\angle BCY = 112^\circ$ what is $\angle BYZ$?
- If $\angle BCY = 125^\circ$ what is $\angle BYX$?
- If $\angle BYZ = 100^\circ$ what is $\angle BAY$?

2 In Fig. 3.26 TAX and TBY are tangents to the circle and C is a point on the major arc AB.

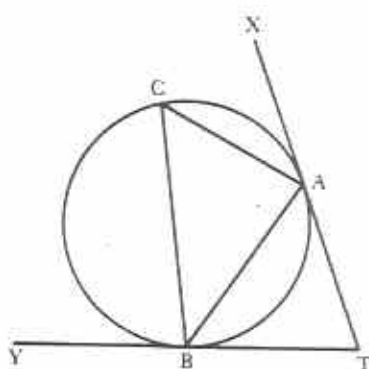


Fig. 3.26

- (a) If $\angle ATB = 68^\circ$, calculate $\angle ACB$.
 - (b) If $\angle ABC = 73^\circ$, $\angle BAC = 83^\circ$, calculate $\angle ATB$.
 - (c) If $\angle ACB = 59^\circ$, $\angle CBY = 78^\circ$, calculate $\angle CAX$.
 - (d) If $\angle CAX = 65^\circ$, $\angle CBY = 76^\circ$, calculate $\angle ATB$.
 - (e) If $\angle ABC = 48^\circ$, $\angle ATB = 72^\circ$, calculate $\angle BAC$.
- 3 In Fig. 3.27, XYZ is a tangent to the circle at Y. Name an angle equal to $\angle YQP$.

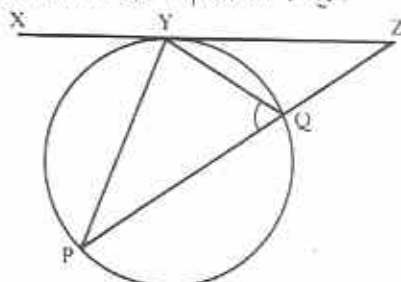


Fig. 3.27

- 4 In Fig. 3.28, TY is a tangent to the circle TVS. If $\angle SVT = 48^\circ$ and $VS = ST$, what is $\angle VTY$?

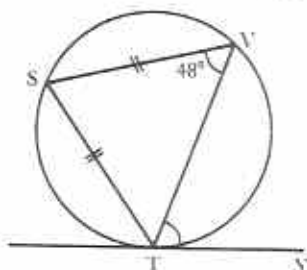


Fig. 3.28

- 5 In Fig. 3.29 the tangents from T touch a circle at A and B and BC is a chord parallel to TA. If $\angle BAT = 54^\circ$, calculate $\angle BAC$.
- 6 In Fig. 3.29, if $\angle ATB = 82^\circ$, calculate the angles of $\triangle ABC$.
- 7 In Fig. 3.30, TS is tangent to circle PQRS. If $PR = PS$ and $\angle PQR = 117^\circ$, calculate $\angle RST$.

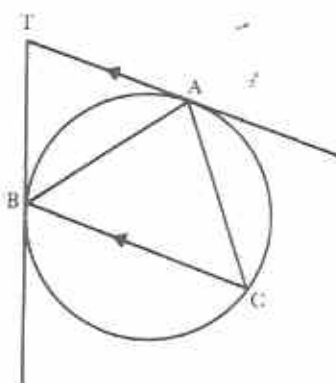


Fig. 3.29

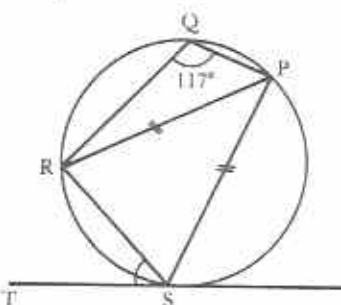


Fig. 3.30

- 8 AB is a chord of a circle and the tangents at A and B meet at T. C is a point on the minor arc AB. If $\angle ATB = 54^\circ$ and $\angle CBT = 23^\circ$, calculate $\angle CAT$.
- 9 In Fig. 3.31, if $\angle ACB = 37^\circ$ and $\angle ATB = 42^\circ$, calculate $\angle ART$ and $\angle AEB$.

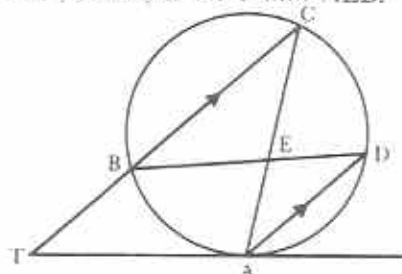


Fig. 3.31

- 10 A, B, C are three points on a circle. The tangent at C meets AB produced at T. If $\angle ACT = 103^\circ$, $\angle ATC = 43^\circ$, calculate the angles of $\triangle ABC$.
- 11 The angles of a triangle are 40° , 60° , 80° , and a circle touches its sides at P, Q, R. Calculate the angles of $\triangle PQR$.
- 12 AT is a tangent to the circle. ABCD. $\angle BAC = 64^\circ$ and $\angle CAT = 72^\circ$. Calculate $\angle BCA$ and $\angle CDA$.

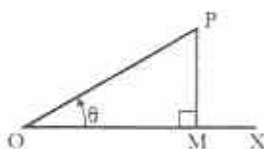
Chapter 4

The sine rule

Trigonometrical ratios of obtuse angles

The trigonometrical ratios of an acute angle in a right-angled triangle have already been defined.

Fig. 4.1



In Fig. 4.1,

$$\sin \theta = \frac{MP}{OP} \quad \left(\frac{\text{opp}}{\text{hyp}} \right)$$

$$\cos \theta = \frac{OM}{OP} \quad \left(\frac{\text{adj}}{\text{hyp}} \right)$$

$$\tan \theta = \frac{MP}{OM} \quad \left(\frac{\text{opp}}{\text{adj}} \right)$$

If OX in Fig. 4.1 is kept fixed and OP allowed to rotate anticlockwise, there will come a stage when θ becomes obtuse (Fig. 4.2).

Fig. 4.2

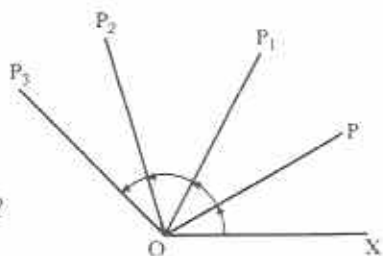
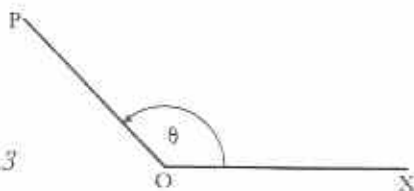


Fig. 4.3



When θ is obtuse, it is no longer in a right-angled triangle. It is therefore impossible to define $\sin \theta$, $\cos \theta$ and $\tan \theta$ in terms of the

ratios of the hypotenuse, adjacent and opposite sides of a right-angled triangle (Fig. 4.3).

It is necessary, therefore, to define the trigonometrical ratios in such a way as to be suitable for obtuse angles as well as acute angles. Fig. 4.4 shows acute and obtuse angles θ within cartesian axes OX, OY.

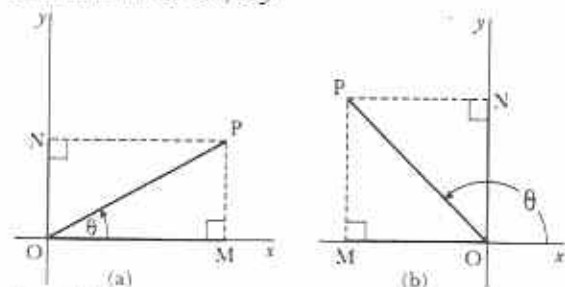


Fig. 4.4

Notice that the lettering of Fig. 4.4(a) is very much the same as that of Fig. 4.1.

In Fig. 4.4(a),

$$\sin \theta = \frac{MP}{OP} = \frac{ON}{OP}$$

$$\cos \theta = \frac{OM}{OP}$$

$$\tan \theta = \frac{MP}{OM} = \frac{ON}{OM}$$

In Fig. 4.4, OM is called the **projection of OP on OX**. ON is the projection of OP on OY. This makes it possible to define the trigonometrical ratios in a new way:

$$\sin \theta = \frac{\text{projection of OP on OY}}{OP}$$

$$\cos \theta = \frac{\text{projection of OP on OX}}{OP}$$

$$\tan \theta = \frac{\text{projection of OP on OY}}{\text{projection of OP on OX}}$$

With these definitions in Fig. 4.4(b),

$$\sin \theta = \frac{ON}{OP} \quad \cos \theta = \frac{OM}{OP} \quad \tan \theta = \frac{ON}{OM}$$

The ratios are now the same for both parts of Fig. 4.4. In Fig. 4.5, O is the centre of the circle and the origin of axes Ox, Oy. $\angle POM = \angle QOL = \theta$ (acute).

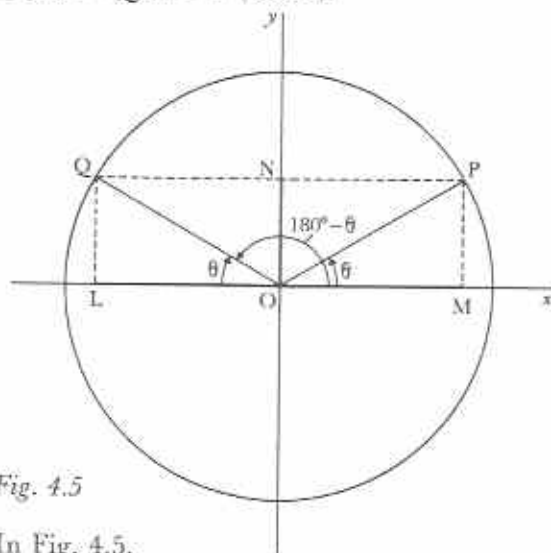


Fig. 4.5

In Fig. 4.5,

$$\angle MOQ = 180^\circ - \theta \text{ (obtuse)}$$

$$OP = OQ \text{ (radii, both taken to be positive lengths)}$$

ON is a positive length since it is on the positive part of Oy. OM is a positive length since it is on the positive part of Ox. OL is a negative length since it is on the negative part of Ox.

From the symmetry of the figure,

$$OL = -OM.$$

Hence

$$\sin(180^\circ - \theta) = \frac{ON}{OQ} = \frac{ON}{OP} = \sin \theta$$

$$\cos(180^\circ - \theta) = \frac{OL}{OQ} = \frac{-OM}{OP} = -\cos \theta$$

$$\tan(180^\circ - \theta) = \frac{ON}{OL} = \frac{ON}{-OM} = -\tan \theta$$

The following examples show how to use the above statements.

$$\sin 160^\circ = \sin(180 - 20)^\circ = \sin 20^\circ = 0.3420$$

$$\begin{aligned} \cos 160^\circ &= \cos(180 - 20)^\circ = -\cos 20^\circ \\ &= -0.9397 \end{aligned}$$

$$\begin{aligned} \tan 160^\circ &= \tan(180 - 20)^\circ = -\tan 20^\circ \\ &= -0.3640 \end{aligned}$$

Exercise 4a

Use tables to find the values of the following

- | | |
|-------------------------|-------------------------|
| 1 $\sin 110^\circ$ | 2 $\cos 110^\circ$ |
| 3 $\tan 110^\circ$ | 4 $\sin 153^\circ$ |
| 5 $\sin 98^\circ$ | 6 $\cos 106^\circ$ |
| 7 $\cos 142^\circ$ | 8 $\tan 167^\circ$ |
| 9 $\tan 93^\circ$ | 10 $\cos 128^\circ$ |
| 11 $\sin 156^\circ 30'$ | 12 $\tan 173.5^\circ$ |
| 13 $\cos 161.4^\circ$ | 14 $\tan 131^\circ 42'$ |
| 15 $\sin 93^\circ 12'$ | 16 $\cos 135.6^\circ$ |
| 17 $\cos 103^\circ 6'$ | 18 $\sin 118^\circ 42'$ |
| 19 $\sin 178.35^\circ$ | 20 $\tan 92^\circ 40'$ |
| 21 $\sin 164^\circ 13'$ | 22 $\cos 118.83^\circ$ |
| 23 $\cos 121^\circ 31'$ | 24 $\sin 95.17^\circ$ |

Example 1

Find the values of θ lying between 0° and 180° each of the following.

(a) $\cos \theta = 0.2874$ (b) $\sin \theta = 0.9361$

(c) $\cos \theta = -0.8224$ (d) $\tan \theta = -2.164$

(a) $\cos \theta = 0.2874$

From tables, $\theta = 73.3^\circ$

Since $\cos \theta$ is positive, θ is acute.

(b) $\sin \theta = 0.9361$

From tables, $\theta = 69.4^\circ$

But $\sin 69.4^\circ = \sin(180 - 69.4)^\circ$

$$= \sin 110.6^\circ$$

$$\theta = 69.4^\circ \text{ or } \theta = 110.6^\circ$$

(c) $\theta = -0.8224$

Since $\cos \theta$ is negative, θ is obtuse.

First find the acute angle whose cosine is 0.8224

From tables, $0.8224 = \cos 34.67^\circ$

$$\Rightarrow -0.8224 = \cos(180 - 34.67)^\circ$$

$$= \cos 145.33^\circ$$

$$\Rightarrow \theta = 145.33^\circ$$

Or by scientific calculator (set in degree mode)

Key

Display

AC

0

$\cdot$ 8 2 2 4 $\div$

-0.8224

SHIFT

-0.8224

$\cos^{-1}$

145.3257

$$\theta = 145.33^\circ$$

$$(d) \tan \theta = -2.164$$

Since $\tan \theta$ is negative, θ is obtuse.

From tables, $2.164 = \tan 65.2^\circ$

$$\Rightarrow \theta = 180^\circ - 65.2^\circ = 114.8^\circ$$

Exercise 4b

Find the values of θ lying between 0° and 180° in each of the following. Give the answers in degrees to 1 or 2 d.p. where appropriate.

- | | |
|----------------------------|----------------------------|
| 1 $\cos \theta = 0.8090$ | 2 $\cos \theta = -0.8090$ |
| 3 $\tan \theta = 3.732$ | 4 $\tan \theta = -3.732$ |
| 5 $\sin \theta = 0.9205$ | 6 $\tan \theta = -1.963$ |
| 7 $\cos \theta = -0.9397$ | 8 $\sin \theta = 0.4226$ |
| 9 $\cos \theta = -0.1392$ | 10 $\tan \theta = -0.6249$ |
| 11 $\cos \theta = -0.7278$ | 12 $\sin \theta = 0.6088$ |
| 13 $\sin \theta = 0.9646$ | 14 $\tan \theta = -2.106$ |
| 15 $\tan \theta = -0.3166$ | 16 $\cos \theta = -0.4253$ |
| 17 $\sin \theta = 0.8329$ | 18 $\tan \theta = -0.7230$ |
| 19 $\sin \theta = 0.9594$ | 20 $\cos \theta = -0.7846$ |
| 21 $\tan \theta = -1.678$ | 22 $\sin \theta = 0.2628$ |
| 23 $\sin \theta = 0.7449$ | 24 $\tan \theta = -12.43$ |

The sine rule

In any $\triangle ABC$, the angles are usually denoted by the capital letters A, B, C and the sides opposite these angles by a , b , c respectively.

Given: Any $\triangle ABC$ (acute and obtuse-angled $\triangle$ s are given in Fig. 4.6)

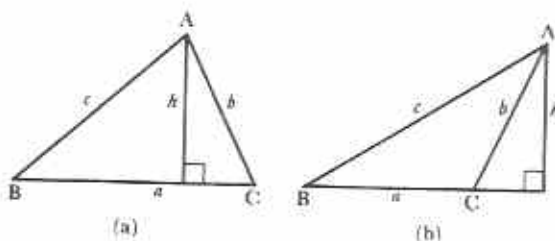


Fig. 4.6

To prove:
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Construction: Draw the perpendicular from A to BC (produced, if necessary)

Proof:

In Fig. 4.6(a) and (b),

$$\sin B = \frac{h}{c} \quad (1)$$

In Fig. 4.6(a),

$$\sin C = \frac{h}{b} \quad (2)$$

In Fig. 4.6(b),

$$\sin(180^\circ - C) = \frac{h}{b}$$

$$\Rightarrow \sin C = \frac{h}{b} \quad [\sin(180^\circ - \theta) = \sin \theta] \quad (2)$$

From (1) $h = c \sin B$

From (2) $h = b \sin C$

$$\Rightarrow c \sin B = b \sin C$$

$$\Leftrightarrow \frac{b}{\sin B} = \frac{c}{\sin C}$$

Similarly, by drawing a perpendicular from C to AB,

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

$$\Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

This formula is used for solving triangles which are not right-angled and in which either **two angles and any side** are given or **two sides and the angle opposite one of them** are given.

Example 2

In $\triangle ABC$, $B = 39^\circ$, $C = 82^\circ$, $a = 6.73$ cm. Find c .

First draw a sketch of the information (Fig. 4.7).

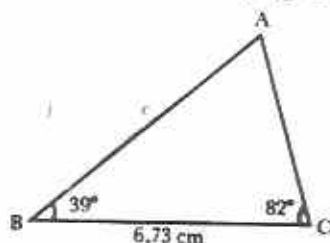


Fig. 4.7

In Fig. 4.7

$$A = 180^\circ - (39^\circ + 82^\circ) \\ = 59^\circ$$

$$\frac{c}{\sin C} = \frac{a}{\sin A}$$

$$\frac{c}{\sin 82^\circ} = \frac{6,73}{\sin 59^\circ}$$

$$\Leftrightarrow c = \frac{6,73 \times \sin 82^\circ}{\sin 59^\circ} \text{ cm}$$

$$= 7,775 \text{ cm} \\ = 7,78 \text{ cm to 2 d.p.}$$

Tables of logarithms of sines were used in the above working.

This calculation may also be done on a scientific calculator as follows:

Key	Display
AC	0
5 9 sin	0.8571673
M+ (or Min)	0.8571673
8 2 sin	0.990268
× 6 . 7 3	5.73
÷ MR	0.8571673
=	7.7750331

$$c = 7,78 \text{ cm}$$

In the above calculator sequence, note the following:

- 1 the use of the memory keys,
- 2 the data are entered in virtually the reverse order to that given in the worked example.

Example 3

Find the remaining angles of $\triangle ABC$ in which $a = 12,5 \text{ cm}$, $c = 17,7 \text{ cm}$ and $C = 116^\circ$.

First make a sketch of the information.

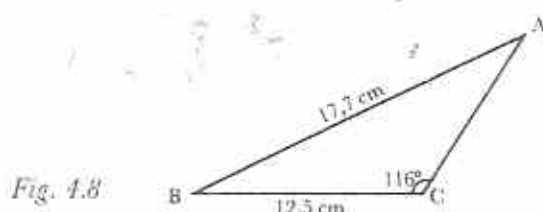


Fig. 4.8

In Fig. 4.8, c and C are known. Since a is also given, A can be found using the sine rule. The formula of the sine rule can be arranged so that the unknown comes first:

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{12,5} = \frac{\sin 116^\circ}{17,7}$$

$$\Leftrightarrow \sin A = \frac{12,5 \times \sin 116^\circ}{17,7} \\ = \frac{12,5 \times \sin 64^\circ}{17,7}$$

$$\Rightarrow A = 39,4^\circ \text{ or } (180 - 39,4)^\circ \\ = 39,4^\circ \text{ or } 140,6^\circ$$

But C is obtuse, therefore A cannot be obtuse.
 $\Rightarrow A = 39,4^\circ$ (or $39^\circ 24'$)

* Sequence for scientific calculator:

Key	Display
AC	0
6 4 sin	0.898794
× 1 2 . 5	12.5
÷ 1 7 . 7 =	0.6347415
SHIFT	0.6347415
sin 1	39.400819

$$A = 39,4^\circ$$

$$B = 180^\circ - (39,4 + 116)^\circ \\ = 180^\circ - 155,4^\circ \\ = 24,6^\circ \text{ (or } 24^\circ 36')$$

The calculator sequences given in Examples 2

and 3 may be applied to the remaining examples and exercises in this chapter. See also Chapter 20 for further advice regarding the use of the scientific calculator to solve triangles.

Example 4

In $\triangle ABC$, $a = 7,1$ cm, $b = 9,5$ cm and $B = 63^\circ 18'$. Solve the triangle completely.

To solve a triangle *completely* means to calculate all the unknown sides and angles. Make a sketch of the given information.

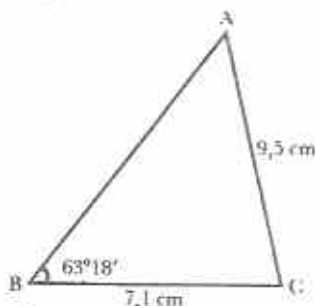


Fig. 4.9

$$63^\circ 18' = 63\frac{18}{60}^\circ = 63\frac{3}{10}^\circ = 63,3^\circ$$

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

$$\frac{\sin A}{7,1} = \frac{\sin 63,3^\circ}{9,5}$$

$$\Leftrightarrow \sin A = \frac{7,1 \times \sin 63,3^\circ}{9,5}$$

$$A = 41,89^\circ \text{ or } (180 - 41,89)^\circ = 138,11^\circ$$

But $a < b$

$$\Leftrightarrow A < B$$

$$\Rightarrow A = 41,89^\circ$$

$$\begin{aligned} \Rightarrow C &= 180^\circ - (63,3 + 41,89)^\circ \\ &= 180^\circ - 105,19^\circ \\ &= 74,81^\circ \end{aligned}$$

$$\frac{c}{\sin C} = \frac{b}{\sin B}$$

$$\frac{c}{\sin 74,81^\circ} = \frac{9,5}{\sin 63,3^\circ}$$

$$\begin{aligned} \Leftrightarrow c &= \frac{9,5 \times \sin 74,81^\circ}{\sin 63,3^\circ} \text{ cm} \\ &= 10,26 \text{ cm} \end{aligned}$$

working:

No.	Log
7,1	0,8513
$\sin 63,3^\circ$	1,9510
9,5	0,8023
$\sin 41,89^\circ$	0,9777
	1,8246

working:

No.	Log
9,5	0,9777
$\sin 74,81^\circ$	1,9845
	0,9622
$\sin 63,3^\circ$	1,9510
10,26	1,0112

Answers, correct to 1 d.p.

$$A = 41,9^\circ, C = 74,8^\circ, c = 10,3 \text{ cm}$$

Answers, with angles correct to the nearest minute:

$$A = 41^\circ 53', C = 74^\circ 49', c = 10,3 \text{ cm}$$

In general, four-figure tables give answers which are correct to 3 s.f. Answers should be rounded to 3 s.f. If a question gives angles in degrees and minutes, then answers should be given correct to the nearest minute. Note that $41,89^\circ = 41^\circ + (0,89 \times 60)' = 41^\circ + 53,4' = 41^\circ 53'$ to the nearest minute.

Exercise 4c

1

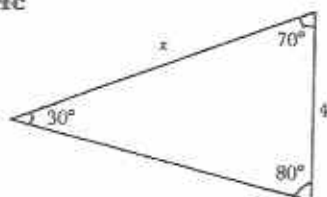


Fig. 4.10

Using the information given in Fig. 4.10, write down, but do not solve, an equation which can be used to find x .

- In $\triangle ABC$, $A = 29^\circ$, $B = 36^\circ$, $b = 15,8$ cm. Find a .
- In $\triangle ABC$, $A = 54^\circ 12'$, $B = 71^\circ 30'$, $a = 12,4$ cm. Find b .
- In $\triangle ABC$, $B = 104,3^\circ$, $C = 31,3^\circ$, $a = 29,0$ cm. Calculate c .
- In $\triangle PQR$, $P = 83^\circ$, $p = 285$ m, $r = 216$ m. Calculate R .
- In $\triangle ABC$, $C = 53^\circ$, $b = 3,56$ m, $c = 4,28$ m. Calculate B .
- In $\triangle ABC$, $A = 115^\circ$, $a = 65$ m, $b = 32$ m. Solve the triangle completely.
- In $\triangle ABC$, $B = 25^\circ 36'$, $C = 124^\circ 24'$, $c = 39,2$ m. Solve the triangle completely.
- In $\triangle XYZ$, $Y = 29,8^\circ$, $Z = 51,4^\circ$, $x = 19,6$ cm. Solve the triangle completely.
- In $\triangle ABC$, $A = 38^\circ 18'$, $a = 252$ m, $b = 198$ m. Solve the triangle completely.
- In $\triangle ABC$, $C = 96,2^\circ$, $b = 11,2$ cm, $c = 39,4$ cm. Solve the triangle completely.
- Calculate the values of angles A and C of $\triangle ABC$, where $b = 14,35$ cm, $a = 7,82$ cm and $B = 115^\circ 36'$.

Example 5

Two ships A and B leave a port P at the same time. A travels on a bearing of 159° and B travels on a bearing of 215° . After some time, A is 9 km from P, and the bearing of B from A is 256° . Calculate the distance of B from P.

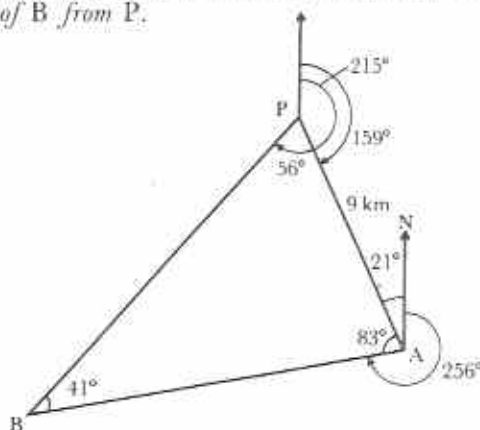


Fig. 4.11

Fig. 4.11 shows the positions of P, A and B. The arrows at P and A point northwards.

In $\triangle PAB$,

$$\hat{BPA} = 215^\circ - 159^\circ = 56^\circ$$

$$\hat{NAP} = 180^\circ - 159^\circ = 21^\circ$$

$$\Rightarrow \hat{PAB} = 360^\circ - (256^\circ + 21^\circ)$$

$$= 360^\circ - 277^\circ = 83^\circ$$

$$\Rightarrow \hat{B} = 180^\circ - (56^\circ + 83^\circ)$$

$$= 180^\circ - 139^\circ = 41^\circ$$

$$\frac{PB}{\sin 83^\circ} = \frac{9}{\sin 41^\circ}$$

$$\Leftrightarrow PB = \frac{9 \times \sin 83^\circ}{\sin 41^\circ} \text{ km}$$

$$= 13.61 \text{ km}$$

$$= 13.6 \text{ km to 3 s.f.}$$

Ship B is 13.6 km from P.

In Example 5, the bearings 159° , 215° , 256° are examples of **three-figure true bearings**. Each bearing is measured clockwise from north and is given in the range 000° to 360° .

Example 6

A tree is on a bearing $S 36^\circ W$ from a point X and $S 73^\circ E$ from a point Y. If X is 200 m due east of

Y, calculate the distance of the tree from Y to the nearest metre.

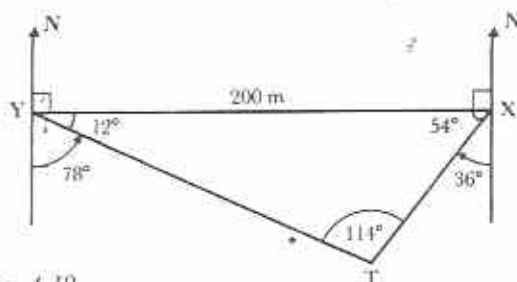


Fig. 4.12

In Fig. 4.12, T represents the position of the tree.

In $\triangle XYT$,

$$\hat{XYT} = 90^\circ - 78^\circ = 12^\circ$$

$$\hat{YXT} = 90^\circ - 36^\circ = 54^\circ$$

$$\hat{YTX} = 180^\circ - (12^\circ + 54^\circ)$$

$$= 180^\circ - 66^\circ = 114^\circ$$

$$\frac{YT}{\sin 54^\circ} = \frac{200}{\sin 114^\circ}$$

$$\Leftrightarrow YT = \frac{200 \times \sin 54^\circ}{\sin 114^\circ} \text{ m}$$

$$= \frac{200 \sin 54^\circ}{\sin 66^\circ} \text{ m}$$

$$= 177.1 \text{ m}$$

$$= 177 \text{ m to the nearest m}$$

The tree is 177 m from Y.

working:

No.	Log
200	2.3010
$\sin 54^\circ$	1.9080
$\sin 66^\circ$	2.2090
177.1	2.2483

The bearing $S 36^\circ W$ is an example of a **compass bearing**. To find the direction $S 36^\circ W$, first face south then turn through 36° in a westerly direction. In Fig. 4.12 it can be seen that $S 36^\circ W$ is the same as the three-figure bearing 216° (i.e. $180^\circ + 36^\circ$). Similarly, the bearing $S 78^\circ E$ is equivalent to the three-figure bearing 102° (i.e. $180^\circ - 78^\circ$).

Examples 5 and 6 show the importance of drawing a fully labelled diagram.

Exercise 4d

- 1 A point X is 34 m due east of a point Y. The bearings of a flagpole from X and Y are $N 18^\circ W$ and $N 40^\circ E$ respectively. Calculate the distance of the flagpole from Y.

- 2 A man walks due west for 4 km. He then changes direction and walks on a bearing of 197° until he is south-west of his starting point. How far is he then from his starting point?
- 3 A girl starts from a point A and walks 285 m to B on a bearing of 078° . She then walks due south to a point C which is 307 m from A. What is the bearing of A from C, and what is BC?
- 4 The bearing of a house from a point A is 319° . From a point B, 317 m due east of A, the bearing of the house is 288° . How far is the house from A?
- 5 A mass is hung from a horizontal beam by two strings as in Fig. 4.13. Calculate the length of the longer string.

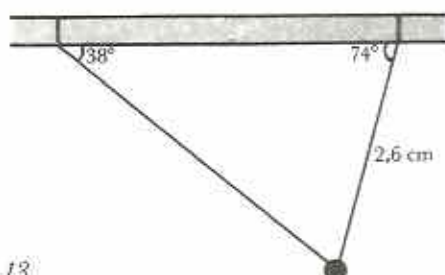


Fig. 4.13

- 6 An aircraft is timetabled to travel from A to B. Due to bad weather it flies from A to C then from C to B, where AC and CB make angles of 27° and 66° respectively with AB. If AC = 220 km, calculate AB.

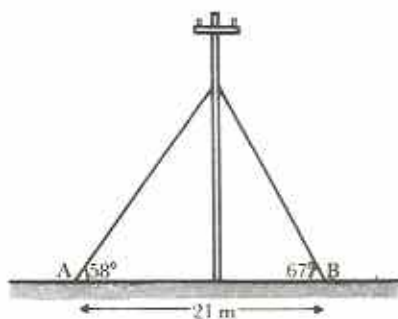


Fig. 4.14

- 7 Two wires support an electricity pole as shown in Fig. 14.14. If the wires make angles of 58° and 67° with the ground and AB = 21 m, calculate the lengths of the wires.
- 8 From points A and B on level ground, the angles of elevation of the top of a building are 25° and 37° respectively. If AB = 57 m, calculate, to the nearest metre, the distances of the top of the building from A and B.
- 9 The bearings of ships A and B from a port P are 225° and 116° respectively. Ship A is 3.9 km from ship B on a bearing of 258° . Calculate the distance of ship A from P.
- 10 An aeroplane flies from a town X on a bearing of 045° to another town Y, a distance of 200 km. It then changes course and flies to another town Z on a bearing of 120° . If Z is due east of X, calculate (a) the distance from X to Z, (b) the distance from Y to XZ.
- 11 PQR is an acute-angled triangle in which $\hat{PQR} = 42^\circ 14'$. S is the point on QR such that $\hat{PSQ} = 90^\circ$. Given that PR = 5.23 cm and SR = 2.37 cm, calculate (a) PRQ, (b) QR.
- 12 A ship sails directly from a port P to a point Q, which is 10 nautical miles due east of a lighthouse L.

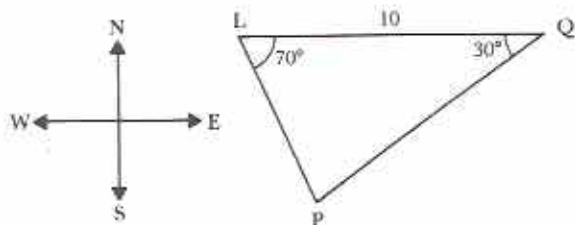


Fig. 4.15

Given that $\hat{PLQ} = 70^\circ$ and $\hat{PQL} = 30^\circ$, calculate

- (a) the bearing of Q from P,
 (b) the bearing of L from P,
 (c) the distance PQ,
 (d) the shortest distance between the lighthouse and the ship during its journey.

Graphs (3) Gradient

Gradient of a straight line

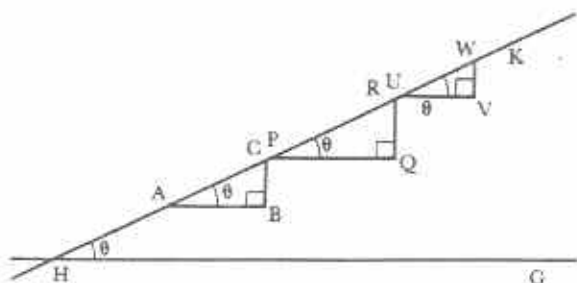


Fig. 5.1

In Fig. 5.1, HG is a horizontal line and HK is a line which makes an angle θ with HG. The $\triangle$ s ABC, PQR, UVW are similar.

$$\text{Hence } \frac{BC}{AB} = \frac{QR}{PQ} = \frac{VW}{UV}.$$

Each of these fractions measures the **gradient** of the line HK. Hence the gradient of a straight line is the same at any point on it.

Also $\tan \theta = \frac{BC}{AB} = \frac{QR}{PQ} = \frac{VW}{UV}$, so $\tan \theta$ is also a measure of the gradient.

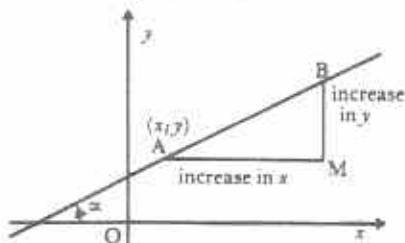


Fig. 5.2

In Fig. 5.2 the point A has coordinates $(x; y)$. In going from A to B the *increase* in x is AM. The corresponding *increase* in y is MB.

Gradient of AB

$$= \frac{\text{increase in } y \text{ from A to B}}{\text{increase in } x \text{ from A to B}} = \frac{MB}{AM}$$

Since y *increases* as x *increases*, the gradient is *positive*. AB makes an acute angle α with the positive direction of the x -axis and $\tan \alpha$ is positive.

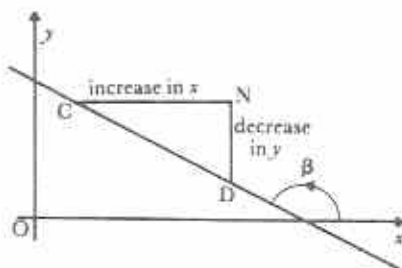


Fig. 5.3

In Fig. 5.3 the point C has coordinates $(x; y)$. In going from C to D the *increase* in x is CN. The corresponding *decrease* in y is ND. Consider a decrease to be a negative increase:

Gradient of CD

$$= \frac{\text{increase in } y \text{ from C to D}}{\text{increase in } x \text{ from C to D}} = \frac{-ND}{CN}$$

Since y *decreases* as x *increases*, the gradient is *negative*. CD makes an obtuse angle β with the positive direction of the x -axis and $\tan \beta$ is negative.

In algebraic graphs, the gradient of a straight line is the **rate of change of y compared with x** . For example, if the gradient is 3, then for any increase in x , y increases 3 times as much. Compare this with rates of change in distance–time and velocity–time graphs.

Example 1

Find the gradients of the lines joining (a) A(-1; 2) and B(3; -2), (b) C(0; -1) and D(4; 1).

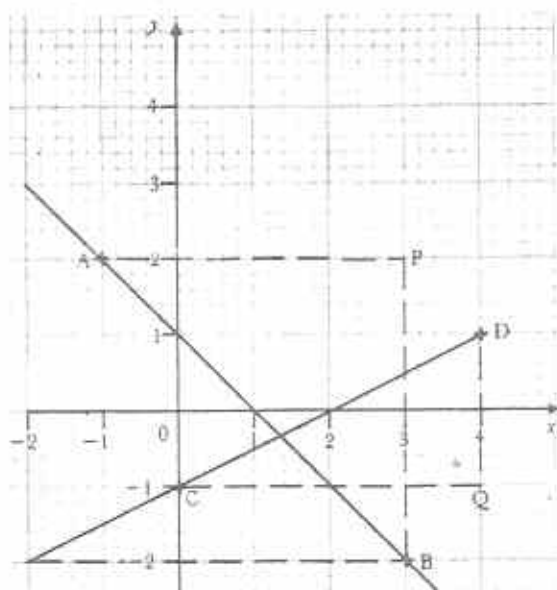


Fig. 5.4

Fig. 5.4 shows the points A, B, C, D and the lines AB and CD.

$$\begin{aligned} \text{(a) Gradient of AB} &= \frac{\text{increase in } y}{\text{increase in } x} = \frac{-PB}{AP} \\ &= \frac{-4}{4} = -1 \end{aligned}$$

Notice that the increase in y is negative (i.e. a decrease).

$$\begin{aligned} \text{(b) Gradient of CD} &= \frac{\text{increase in } y}{\text{increase in } x} = \frac{QD}{CQ} \\ &= \frac{2}{4} = \frac{1}{2} \end{aligned}$$

The gradient can also be calculated without drawing the graph: consider a line which passes through the points $(x_1; y_1)$ and $(x_2; y_2)$.
Gradient of the line

$$\begin{aligned} &= \frac{\text{increase in } y}{\text{increase in } x} \\ &= \frac{\text{difference in the } y \text{ coordinates}}{\text{difference in the } x \text{ coordinates}} = \frac{y_2 - y_1}{x_2 - x_1} \end{aligned}$$

For example, in Example 1,

$$\text{Gradient of AB} = \frac{(-2) - (2)}{(3) - (-1)} = \frac{-4}{4} = -1$$

$$\text{Gradient of CD} = \frac{(1) - (-1)}{(4) - (0)} = \frac{2}{4} = \frac{1}{2}$$

Exercise 5a

Find the gradients of the lines joining the following pairs of points.

- | | |
|---------------------|---------------------|
| 1 (9; 7), (2; 5) | 2 (2; 5), (4; 8) |
| 3 (5; 3), (0; 0) | 4 (6; 1), (1; 5) |
| 5 (0; 4), (3; 0) | 6 (-3; 2), (4; 4) |
| 7 (2; 3), (6; -5) | 8 (-4; 3), (8; -6) |
| 9 (-4; -4), (-1; 5) | 10 (7; -2), (-1; 2) |

Example 2

(a) Draw a graph of the line represented by the equation $4x + 2y = 5$. (b) Find the gradient by taking measurements.

(a) First make a table of values.

When $x = 0$, $0 + 2y = 5 \Leftrightarrow y = 2\frac{1}{2}$;

when $x = 1$, $4 + 2y = 5 \Leftrightarrow y = \frac{1}{2}$;

when $x = 2$, $8 + 2y = 5 \Leftrightarrow y = -1\frac{1}{2}$.

x	0	1	2
y	$2\frac{1}{2}$	$\frac{1}{2}$	$-1\frac{1}{2}$

Fig. 5.5 shows the required graph.

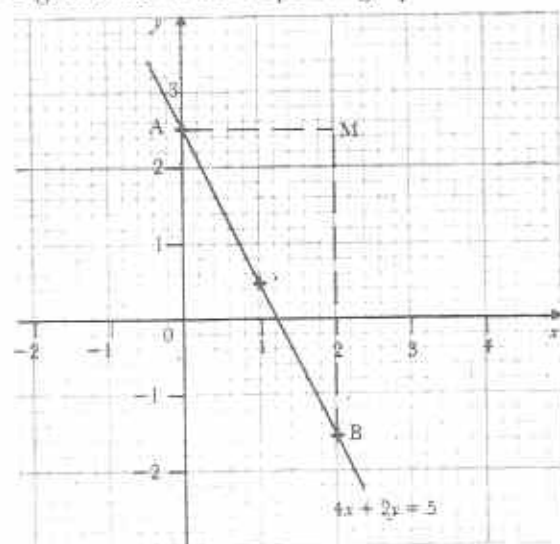


Fig. 5.5

(b) Choose two convenient points, such as A and B in Fig. 5.5.

$$\begin{aligned}\text{Gradient of AB} &= \frac{\text{increase in } y}{\text{increase in } x} = \frac{-MB}{AM} \\ &= \frac{-4}{2} = -2\end{aligned}$$

Exercise 5b

Draw the graphs of the lines represented by the following equations. In each case find the gradient by taking measurements.

- 1 $y = 3x + 1$
- 2 $y = 3x - 2$
- 3 $y = -2x + 3$
- 4 $4x - 2y + 1 = 0$
- 5 $2x + 3y = 0$
- 6 $2x + 3y = 6$
- 7 $4x - 3y = 5$
- 8 $2x - 5y = 6$
- 9 $5x - 2y = 5$
- 10 $7x + 4y - 8 = 0$

Sketching graphs of straight lines

From Exercise 5b it can be seen that the gradient of a line depends only on the coefficients of x and y in its equation. For example, in questions 1 and 2 the following results were obtained:

$$y = 3x + 1, \text{ gradient } 3$$

$$y = 3x - 2, \text{ gradient } 3$$

The results of questions 5 and 6 were as follows:

$$2x + 3y = 0, \text{ gradient } -\frac{2}{3}$$

$$2x + 3y = 6, \text{ gradient } -\frac{2}{3}$$

Notice that the last two equations can be rearranged:

$$2x + 3y = 0$$

$$\Leftrightarrow 3y = -2x$$

$$\Leftrightarrow y = -\frac{2}{3}x$$

$$2x + 3y = 6$$

$$\Leftrightarrow 3y = -2x + 6$$

$$\Leftrightarrow y = -\frac{2}{3}x + 2$$

Hence, when y is the subject of the equation, the coefficient of x gives the gradient. An equation in the form $y = mx + c$ is that of a straight line with gradient m .

If the gradient and one point on the line are known, it is possible to make a rough sketch of the graph.

Example 3

Make a rough sketch of the line whose equation is $2x + 4y = 9$.

First: Rearrange the equation to make y the subject.

$$2x + 4y = 9$$

$$4y = -2x + 9$$

$$y = -\frac{1}{2}x + \frac{9}{4}$$

Second: Find a point on the line. The simplest point is usually that where $x = 0$. When $x = 0$, $y = \frac{9}{4}$. $(0; \frac{9}{4})$ is a point on the line. Fig. 5.6 is a rough sketch of the line.

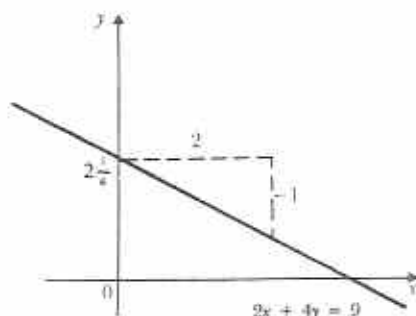


Fig. 5.6

Notice that the axes x and y should always be shown on a rough sketch.

An alternative method of sketching a straight line is to find the two points where the line crosses the axes.

Example 4

Sketch the graph of the line whose equation is $4x - 3y = 12$.

$$\text{When } x = 0, -3y = 12$$

$$\Leftrightarrow y = -4$$

The line crosses the y -axis at $(0; -4)$.
 When $y = 0$, $4x = 12$
 $\Rightarrow x = 3$

The line crosses the x -axis at $(3, 0)$.
 Fig. 5.7 is a rough sketch of the line.

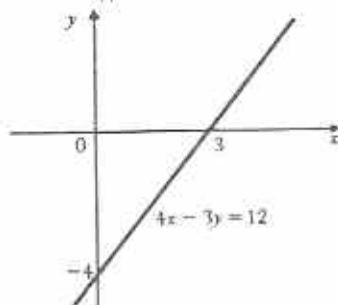


Fig. 5.7

Any line which is parallel to the x -axis has a **zero gradient**. The equations of such lines are always in the form $y = c$, where c may be any number. Fig. 5.8 shows the graphs of $y = 5$ and $y = -3$.

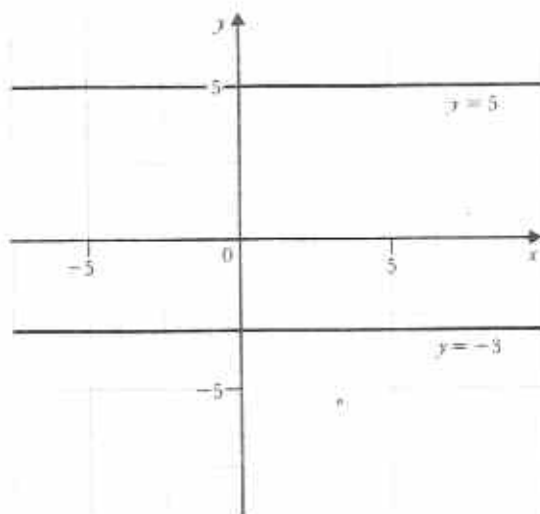


Fig. 5.8

Notice that the equation of the x -axis is $y = 0$.
 The gradient of a line which is parallel to the y -axis is undefined (i.e. cannot be found). The equations of such lines are always in the form $x = a$, where a may be any number. Fig. 5.9 shows the graphs of the lines $x = 2$ and $x = -4$.

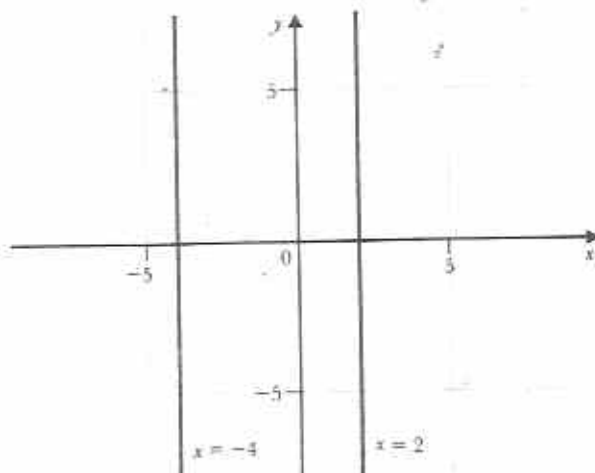


Fig. 5.9

Notice that the equation of the y -axis is $x = 0$.

Exercise 5c

1 Sketch the lines which pass through the following points with the given gradients:

- point $(2; 1)$, gradient 3
- point $(5; 0)$, gradient -2
- point $(1; -3)$, gradient -3
- point $(-4; -2)$, gradient $\frac{2}{3}$
- point $(5; -2)$, gradient $-\frac{4}{3}$

2 Write down the gradients of the lines represented by the following equations. Hence sketch the graphs of the lines.

- $y = 2x + 3$
- $y = \frac{1}{3}x$
- $y = \frac{5}{4}x - 2$
- $3x + 7y = 5$
- $4x - 7y = 7$

3 Find the coordinates of the points where the lines represented by the following equations cross the axes. Hence sketch the graphs of the lines.

- $y = 2x - 2$
- $y = \frac{1}{3}x + 1$
- $3x - 5y = 30$
- $4x + 3y = 2$
- $8x + 5y = 4$

- 4 Write down the gradients of the lines represented by the sketches in Fig. 5.10.

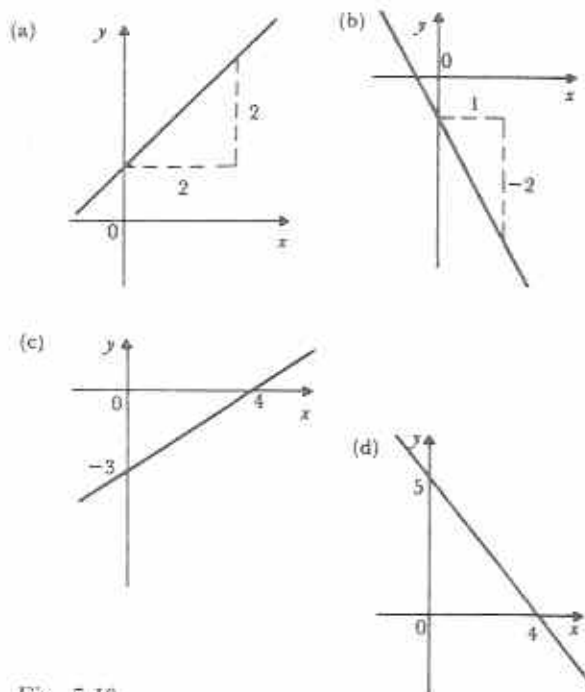


Fig. 5.10

- 5 Write down the gradients of the lines (a), (b), (c), (d), (e) in Fig. 5.11.

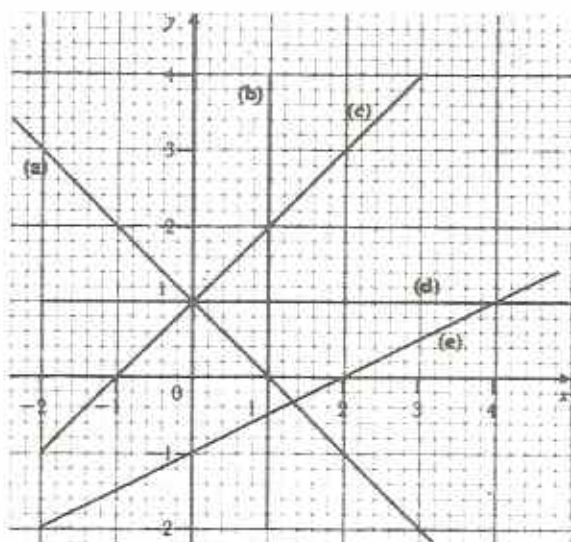


Fig. 5.11

Equation of a straight line

- (a) Given its gradient and a point on the line

Example 5

A straight line of gradient 5 passes through the point $B(3; -8)$. Find the equation of the line.

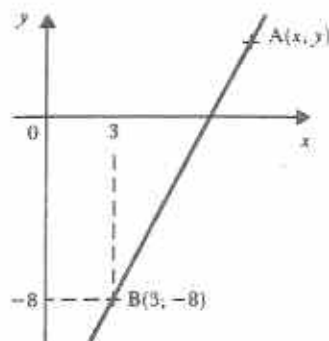


Fig. 5.12

Fig. 5.12 is a sketch of the line.

In Fig. 5.12 the point $A(x; y)$ is any general point on the line.

$$\begin{aligned}\text{Gradient of AB} &= \frac{y - (-8)}{x - 3} \\ &= \frac{y + 8}{x - 3}\end{aligned}$$

Hence $\frac{y + 8}{x - 3} = 5$ since the gradient of AB is 5.

$$\begin{aligned}y + 8 &= 5(x - 3) \\ &= 5x - 15 \\ y &= 5x - 23\end{aligned}$$

The equation of the line is $y = 5x - 23$.

In general, the equation of a straight line of gradient m which passes through the point $(a; b)$ is given by

$$\frac{y - b}{x - a} = m$$

- (b) Given two points on the line

Example 6

Find the equation of the straight line which passes through the points $Q(-1; 7)$ and $R(3; -2)$.

Figure 5.13 is a sketch of the line through Q and R.

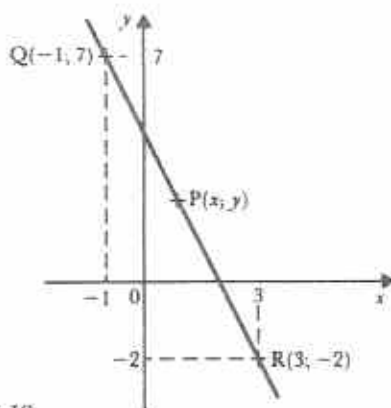


Fig. 5.13

In Fig. 5.13 the point $P(x, y)$ is any general point on the line.

$$\text{Gradient of } QR = \frac{7 - (-2)}{(-1) - 3} = \frac{9}{-4} = -2\frac{1}{4}$$

$$\text{Gradient of } PR = \frac{y - (-2)}{x - 3} = \frac{y + 2}{x - 3}$$

But PQR is a straight line, hence

$$\text{gradient of } PR = \text{gradient of } QR$$

$$\frac{y + 2}{x - 3} = -2\frac{1}{4}$$

$$y + 2 = -2\frac{1}{4}(x - 3)$$

$$y + 2 = -2\frac{1}{4}x + 6\frac{3}{4}$$

$$y = -2\frac{1}{4}x + 4\frac{3}{4}$$

The equation of the line is $y = -2\frac{1}{4}x + 4\frac{3}{4}$.

In general, the equation of a straight line which passes through the points $(a; b)$ and $(c; d)$ is given by:

$$\frac{y - b}{x - a} = \frac{d - b}{c - a}$$

Exercise 5d

- 1 Find the equation of the line which passes through the point

- (a) $(4; 9)$ and has a gradient of 3,
(b) $(0; 0)$ and has a gradient of 3,

- (c) $(-2; 8)$ and has a gradient of -1 ,
(d) $(6; 0)$ and has a gradient of $-\frac{3}{4}$,
(e) $(0; -5)$ and has a gradient of -4 ,
(f) $(-1; 2)$ and has a gradient of $2\frac{1}{2}$.

- 2 Find the equation of the line which passes through the points

- (a) $(0; 0)$ and $(3; 7)$,
(b) $(0; 0)$ and $(3; -7)$,
(c) $(-1; 4)$ and $(5; -2)$,
(d) $(-6; -6)$ and $(4; -3)$,
(e) $(7; 2)$ and $(-9; 7)$,
(f) $(2; -11)$ and $(-4; 4)$.

- 3 A straight line is drawn through the points $(7; 0)$ and $(-2; 3)$. Find (a) its gradient, (b) its equation.

- 4 A straight line of gradient $4\frac{1}{2}$ passes through the point $(4; -3)$. Write down (a) the equation of the line, (b) the equation of a parallel line which passes through the point $(0; \frac{1}{2})$.

- 5 Line l passes through the point $(10; -1)$. Line m passes through the point $(-1\frac{1}{2}; -4\frac{1}{2})$. Find the equations of l and m if both lines pass through the point $(1; 2)$.

- 6 (a) Find the gradient of the straight line through the points $(2; -2)$ and $(7; 8)$.
(b) Find the equation of the straight line which passes through the point $(3; 5)$ and has gradient -4 . [Camb]

- 7 (a) Find the equation of the straight line which passes through the points $(0; 5)$ and $(5; 0)$.
(b) Show that the equation of the straight line which passes through $(0; a)$ and $(a; 0)$ is $x + y = a$.

- 8 (a) Write down an expression for the gradient of the line joining the points $(6; k)$ and $(4; 1)$. Find the value of k if this gradient is $\frac{3}{5}$.
(b) Find the equation of the line through the point $(-4; 5)$ with gradient -2 . [Camb]

- 9 Find the equations of the sides of a triangle which has vertices $A(0; 0)$, $B(7; 0)$, $C(5; 6)$.

- 10 Lines r and s both pass through the point $(k; 9)$. Line r has a gradient of $-\frac{4}{3}$ and passes through the point $(5; -3)$.

- (a) Find the value of k .
(b) Find the equation of line s given that it crosses the x -axis at $(-14; 0)$.

Gradient of a curve

The gradient at any particular point on a curve is defined as being the gradient of the tangent to the curve at that point.

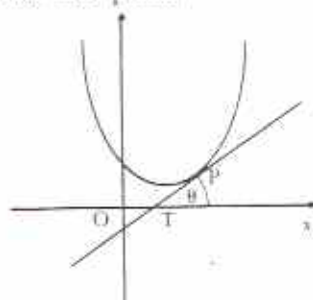


Fig. 5.14

In Fig. 5.14 the gradient of the curve at point P is the gradient of the tangent TP, i.e. $\tan \theta$. The tangent is drawn by placing a ruler against the curve at P and drawing a line, taking care that the 'angles' between the line and the curve appear equal.

Notice that the gradient of a straight line is the same at any point on the line, but that the gradient of a curve changes from point to point.

Example 7

Fig. 5.15 is the graph of the curve $y = 2 + x - x^2$ for values of x from -2 to 3 . Use the given tangents to find the gradient of the curve at (a) P, (b) Q.

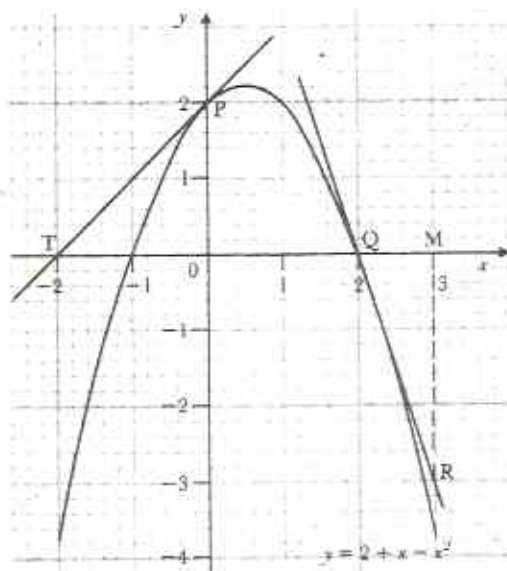


Fig. 5.15

$$\begin{aligned} \text{(a) Gradient of the curve at P} &= \text{gradient of tangent TP} \\ &= \frac{OP}{TO} = \frac{2}{2} = 1 \end{aligned}$$

$$\begin{aligned} \text{(b) Gradient of the curve at Q} &= \text{gradient of tangent QR} \\ &= \frac{-MR}{QM} = \frac{-3}{1} = -3 \end{aligned}$$

Notice that $\triangle$ s TOP and QMR were used to find the gradients of the tangents. Any suitable right-angled triangles could have been used. In this case the intercepts TO and QM were convenient lengths.

Example 8

Draw the graph of $y = \frac{1}{2}x^2$ for values of x from -2 to 3 . Find the gradient of the curve at the point where x has the value (a) 3, (b) -2 .

Table 5.1 is the table of values.

Table 5.1

x	-2	1	0	1	2	3
y	1	$\frac{1}{4}$	0	$\frac{1}{4}$	1	$2\frac{1}{2}$

Fig. 5.16 shows the required graph.

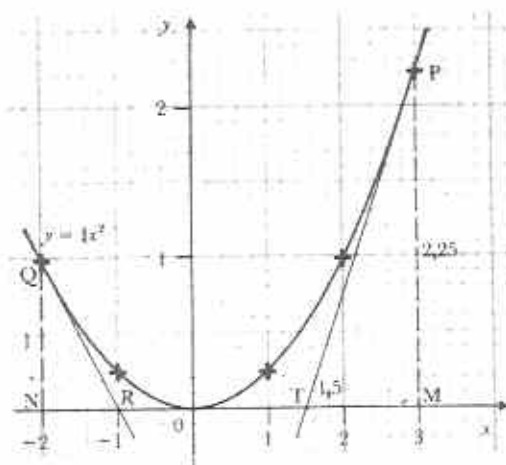


Fig. 5.16

Using the tangents drawn to the curve where $x = 3$ and $x = -2$:

$$\begin{aligned} \text{(a) Gradient of the curve where } x = 3 \\ &= \text{gradient of tangent PT} \\ &= \frac{MP}{TM} = \frac{2,25}{1,5} = \frac{2\frac{1}{4}}{1\frac{1}{2}} = \frac{9}{6} = 1\frac{1}{2} \end{aligned}$$

When $x = 3$, the gradient of the curve is $1\frac{1}{2}$ (i.e. at P, y is increasing $1\frac{1}{2}$ times as fast as x).

$$\begin{aligned} \text{(b) Gradient of curve where } x = -2 \\ &= \text{gradient of tangent QR} \\ &= \frac{-QN}{NR} = \frac{-1}{1} = -1 \end{aligned}$$

When $x = -2$, the gradient of the curve is -1 (i.e. at Q, y is decreasing at the same rate as x is increasing).

Notice the following points:

- 1 In Example 8, the lengths MP, TM, QN, NR are measured according to the scales of the axes.
- 2 In Examples 7 and 8, the method of drawing tangents using a ruler can give inaccurate results. Gradients found by this method must only be taken as approximate.

Fig. 5.17 shows the tangents drawn at the **turning points** of two quadratic functions.

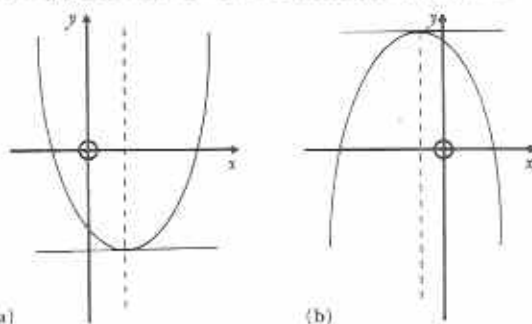


Fig. 5.17

In each case the tangent is parallel to the x -axis. Hence the gradient at a turning point is zero. In Fig. 5.17(a) the turning point corresponds to the **minimum** value of the function. In Fig. 5.17(b) the turning point corresponds to the **maximum** value of the function. In each figure the line of symmetry of the curve is shown by a broken line.

Exercise 5e

- 1 Write down the equation of the line of symmetry of the curve in (a) Fig. 5.15, (b) Fig. 5.16.
- 2 Fig. 5.18 is the graph of the function $x^2 - 6x + 4$.

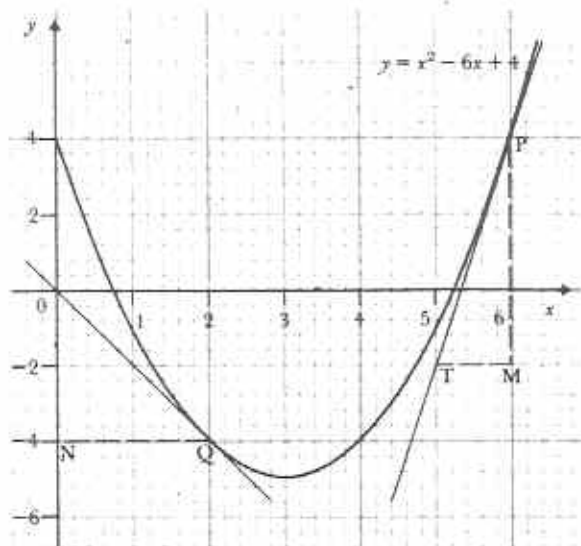


Fig. 5.18

- (a) Use the given tangents to find the gradient of the curve (i) at P, (ii) at Q.
 - (b) Find the minimum value of the function.
 - (c) Write down the equation of the line of symmetry of the curve.
- 3 Fig. 5.19 is the graph of $y = 3 - 2x - x^2$.

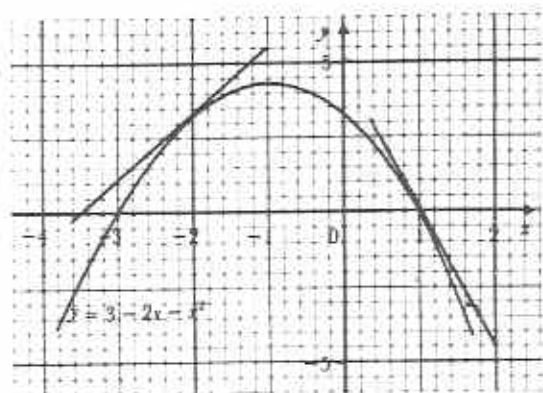


Fig. 5.19

(a) Use the given tangents to find the gradient of the curve (i) when $x = -2$, (ii) when $x = 1$.

(b) What is the maximum value of $3 - 2x - x^2$?

(c) Write down the equation of the line of symmetry of the curve.

- 4 (a) Copy and complete Table 5.2 for the relation $y = 3x - x^2$.

Table 5.2

x	-1	0	1	2	3	4
y	-4		2			

(b) Draw the graph of $y = 3x - x^2$ from $x = -2$ to $x = 4$, using a scale of 2 cm to 1 unit on both axes.

(c) Find the gradient of the curve at (i) $x = 0$, (ii) $x = 2$.

(d) Write down the equation of the line of symmetry of the curve

(e) Find the maximum value of $3x - x^2$.

- 5 Draw the graph of $y = x^2$ for values of x from -4 to 4 . Use a scale of 1 cm to 1 unit on the x -axis and 2 cm to 1 unit on the y -axis. Find the gradient at the point where (a) $x = 3$, (b) $x = 1.5$, (c) $x = -2$.

- 6 Copy and complete Table 5.3 giving values for the function $y = 2x^2 - 4x + 3$ from $x = -2$ to $x = 4$.

Table 5.3

x	-2	-1	0	1	2	3	4
y	19		3	1			

Draw the graph of $y = 2x^2 - 4x + 3$, using a scale of 2 cm to 1 unit on the x -axis and 1 cm to 2 units on the y -axis.

From your graph, find

(a) the equation of the line of symmetry of the curve,

(b) the gradient of the curve at $x = 3$,

(c) the minimum value of y .

- 7 Draw the graph of $y = x^2 - 4x$ from $x = -1$ to $x = 5$. Use a scale of 2 cm to 1 unit on both axes. Find the gradient at the point where (a) $x = 4$, (b) $x = 2$, (c) $x = 0$.

- 8 Draw the graph of $y = 5x - 2x^2$ from $x = -1$ to $x = 4$. Use a scale of 2 cm to 1 unit on the x -axis and 1 cm to 1 unit on the y -axis. (a) Find the gradient of the curve at the point where $x =$ (i) 0, (ii) 1, (iii) 3. (b) Write down the equation of the line of symmetry of the curve.

- 9 Draw the graph of $y = x^2 - 3x + 2$ for values of x from -1 to 4 . Find the gradient at the point where x has the value (a) $2\frac{1}{2}$, (b) $1\frac{1}{2}$, (c) 0, (d) $-\frac{1}{2}$.

- 10 Draw the graph of $y = 1 + x - x^2$ from $x = -2$ to $x = 3$. Find the gradient at the point where x has the value (a) $2\frac{1}{2}$, (b) $1\frac{1}{2}$, (c) $\frac{1}{2}$, (d) -1 .

Chapter 6

Lengths and angles in solids

Angles between lines and planes

Fig. 6.1 shows a flagpole standing on horizontal ground. It is kept vertical by three straight wires attached to the pole at A and to the ground at K, L, M.

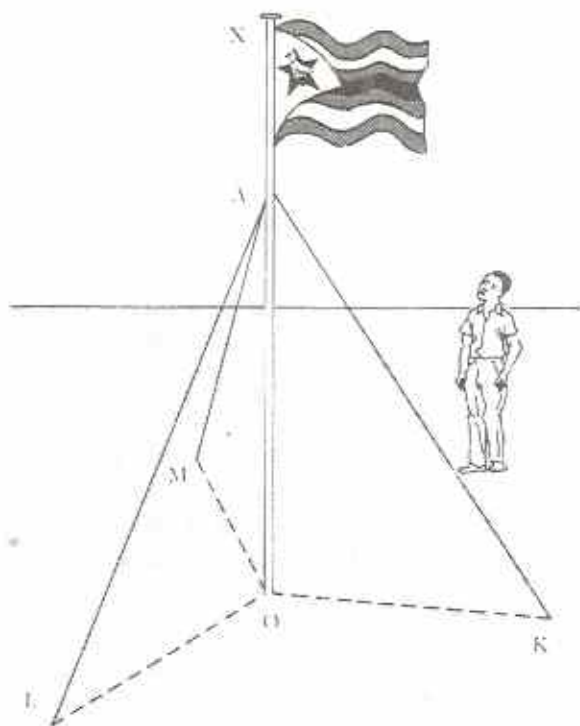


Fig. 6.1

The pole XO in Fig. 6.1 is perpendicular to the ground. The pole is said to meet the ground **normally** (i.e. perpendicularly).

The wires AK, AL, AM in Fig. 6.1 meet the ground **obliquely** (i.e. not perpendicularly). The angles between the wires and the ground are $\angle AKO$, $\angle ALO$, $\angle AMO$.

In Fig. 6.2 the line XY cuts the plane surface normally at O. Since XY is perpendicular to the plane, it is at right angles to every line drawn on the plane through O.

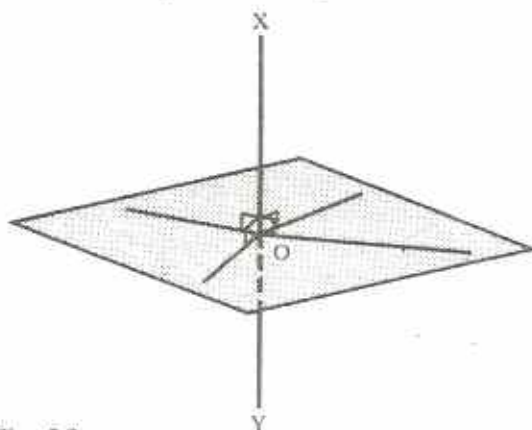


Fig. 6.2

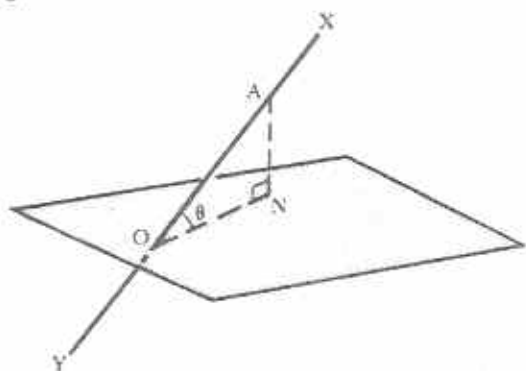


Fig. 6.3

In Fig. 6.3 the line XY cuts the plane obliquely at O. The angle between XY and the plane is found by drawing any perpendicular AN from XY to the plane. The angle $\angle AON$ (θ) is the angle between the line and the plane. Figs. 6.4 and 6.5, overleaf, show another way of finding the angle between a line and a plane.

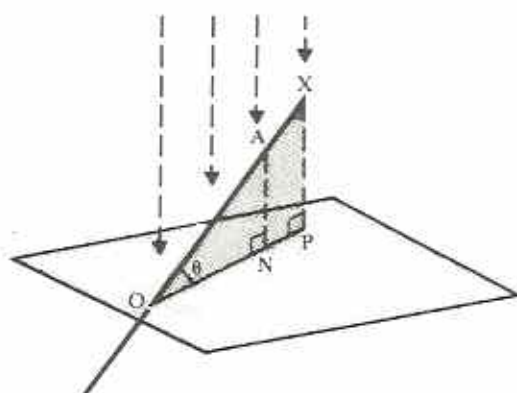


Fig. 6.4

In Fig. 6.4 OP is the **projection** of OX on the horizontal plane. Think of OP as the shadow of OX when the sun is vertically above the plane. $\angle XOP = \angle AON = \theta$ is the angle between OX and the horizontal plane.

In many cases the plane is not horizontal. In Fig. 6.5 OQ is the projection of OX on the vertical plane (i.e. the shadow of OX when the plane is lit from the side). $\angle XOQ = \angle AOM = \alpha$ is the angle between OX and the vertical plane.

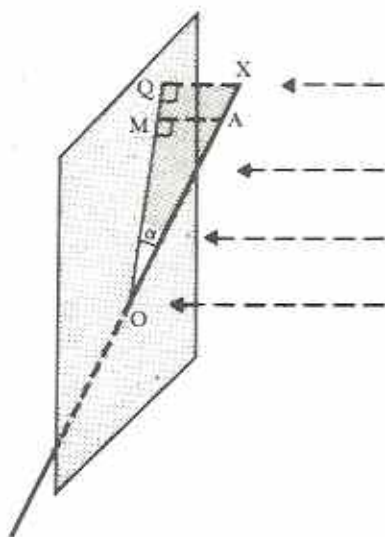


Fig. 6.5

Figs. 6.4 and 6.5 show that **the angle between a line and a plane is the angle between the line and its projection on the plane.**

Example 1

Fig. 6.6 shows a cuboid $ABCDEFGH$.

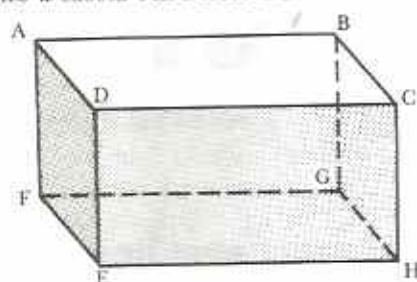


Fig. 6.6

- Name four edges which are perpendicular to plane $BCHG$.
- Which one of the following is not a right angle?
 $\angle AFG$, $\angle AFH$, $\angle AFE$, $\angle CGF$, $\angle CBE$, $\angle BDE$.
- Name the projection of DG on (i) plane $EFGH$, (ii) plane $ADEF$, (iii) plane $ABGF$.
- Name the angle between AH and (i) plane $EFGH$, (ii) plane $BCHG$, (iii) plane $CDEH$.

(a) AB , DC , EH , FG

(b) $\angle CBE$ (since $BC \perp$ plane $CDEH$, $\angle BCE = 90^\circ$. Hence $\triangle CBE$ is right-angled at C , not B . [See Fig. 6.7.]

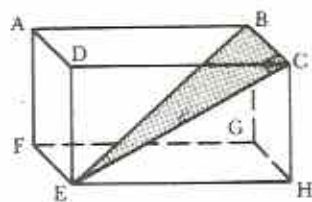


Fig. 6.7

- (i) EG , (ii) DF , (iii) AG .
- (i) $\angle AHF$, (ii) $\angle AHB$, (iii) $\angle AHD$.

Exercise 6a (discussion)

- Name three right angles in Fig. 6.1.
- In Fig. 6.8, O is the centre of the base of the right cone, vertex V .

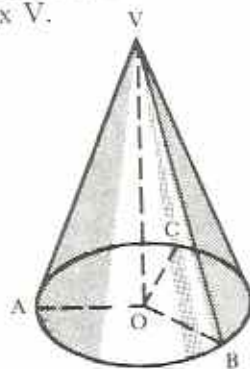


Fig. 6.8

- (a) Name three right angles in Fig. 6.8.
 (b) Name the angle between VB and the base of the cone.
 3 In Fig. 6.9, VABCD is a square-based right pyramid. K, L, M, N are the mid-points of the edges shown in the figure.

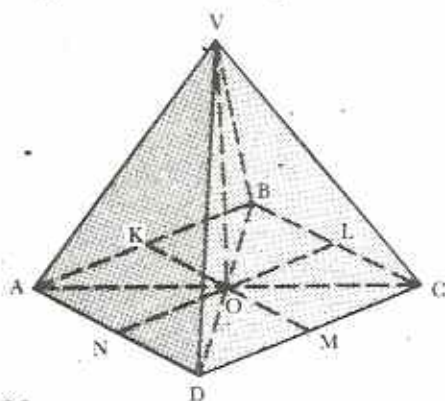


Fig. 6.9

- (a) Name eight line segments which are perpendicular to VO.
 (b) Which plane does AC meet perpendicularly?
 (c) LN is a normal to which plane?
 (d) Name the angle between VB and plane ABCD.
 (e) Name three other angles which are equal in size to that in part (d).
 (f) Name the angle between VK and plane ABCD.
 (g) Name three other angles which are equal in size to that in part (f).
 (h) Name the angle between VO and plane VCD.
 4 In Fig. 6.10, PQRSTUVW is a cuboid.

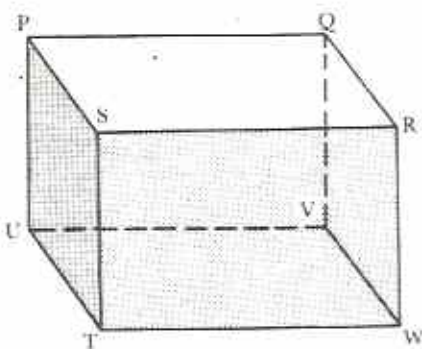


Fig. 6.10

- (a) Name two planes which are perpendicular to edge RW.
 (b) Name four normals to the plane PQVU.
 (c) Which of the following are right angles: $\hat{P}UT$, $\hat{Q}RT$, $\hat{Q}VS$, $\hat{R}WU$, $\hat{V}TP$, $\hat{P}UW$.
 (d) Name the projection of QT on (i) plane TUVW, (ii) plane QRWY, (iii) plane RSTW.
 (e) Name the angle between RU and (i) plane PQRS, (ii) plane PSTU, (iii) plane PQVU.

- 5 In Fig. 6.11, ABCDEF is a prism whose cross-section is a right-angled triangle.

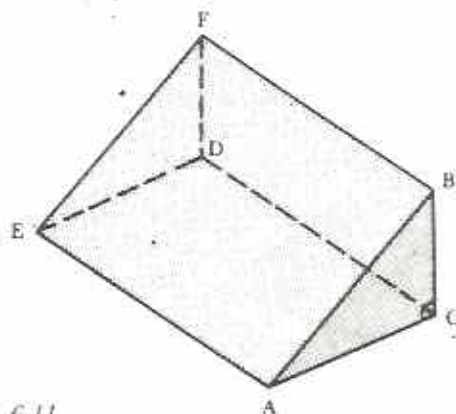


Fig. 6.11

- (a) Name the plane to which ED is a normal.
 (b) Which one of the following is *not* a right angle? $\hat{F}DC$, $\hat{F}DA$, $\hat{F}DE$, $\hat{E}DB$, $\hat{A}FD$.
 (c) Name the projection of AF on (i) plane ACDE, (ii) plane BCDF.
 (d) Name the angle that EB makes with (i) plane ACDE, (ii) plane BCDF.
 6 In Fig. 6.12, PQRSTWXYZ is a cuboid.

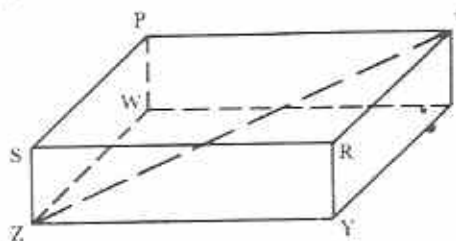


Fig. 6.12

Decide which of the following statements about ZQ are true and which are false.

- (a) The projection of ZQ on plane QXY is YQ.

- (b) ZQ is the longest line segment that can be drawn in the cuboid.
 (c) ZQ makes an angle of 45° with plane ZWPS.
 (d) The angle between ZQ and plane QRYX is $\hat{ZQY}$.
 (e) The projection of ZQ on the plane ZWPS is ZP.

Angles between planes

When one plane cuts another, they intersect along a straight line. Fig. 6.13 shows two examples of planes which meet. The dotted lines are the lines of intersection.

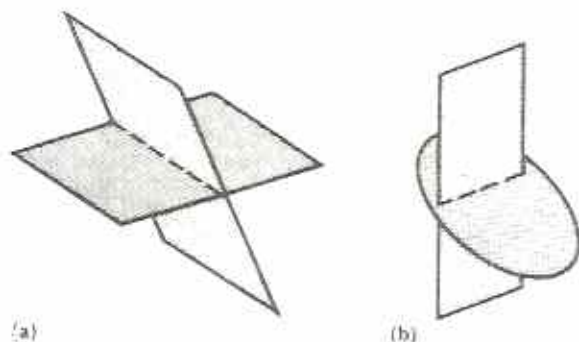


Fig. 6.13

The angle between two planes is found as follows. Choose a point on their line of intersection. From this point, draw a line on each plane at right angles to the line of intersection. The angle between these lines is the angle between the planes. This is shown in Fig. 6.14 where α and β are the angles between the planes.

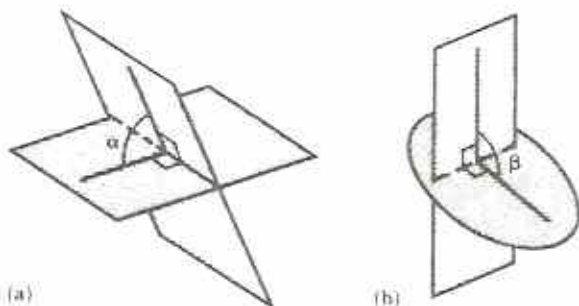


Fig. 6.14

Example 2

Name the angle between the shaded planes in each part of Fig. 6.15.

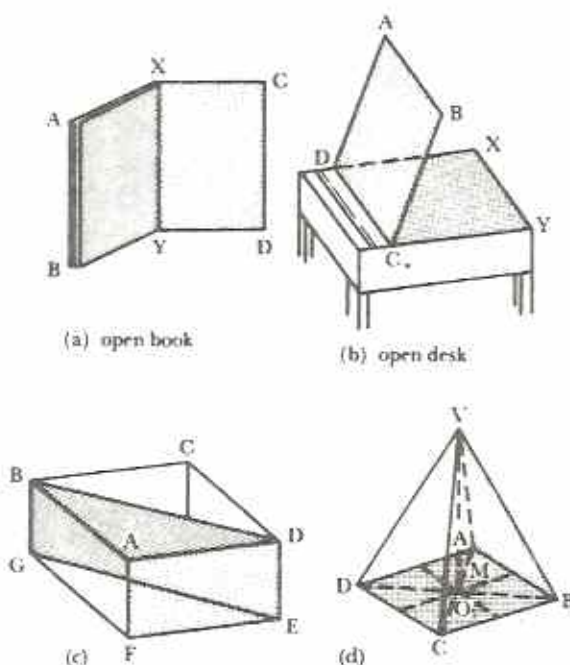


Fig. 6.15

- (a) $\hat{AXC}$ or $\hat{BYD}$
 (b) $\hat{BCY}$ or $\hat{ADX}$
 (c) $\hat{BDA}$ or $\hat{GEF}$
 (d) $\hat{VMO}$ (Note that MV is at right angles to AB ; AV and BV are *not* at right angles to AB .)

Exercise 6b (discussion)

- In Fig. 6.6 name the angle between the two shaded planes. What is the size of this angle?
- In Fig. 6.7 name the angle between plane BCE (shaded) and plane $BCHG$.
- In Fig. 6.8 name the angle between plane VOA and plane VOB .
- In Fig. 6.9 name the angle between the following planes.
 - $ABCD$ and VAC
 - $ABCD$ and VNL
 - $ABCD$ and VDC
 - $ABCD$ and VAD
 - VAC and VNL
 - VAD and VBC

5 In Fig. 6.11 name the angle between the following planes.

- ACDE and ABC
- ACDE and ABFE
- ACDE and BCDF
- BCDF and ABFE

6 (a) Trace Fig. 6.12 into your exercise book.
(b) Draw and shade the triangular plane ZQW.

(c) Name the angle that plane ZQW makes with plane WXYZ.

(d) Name the angle that plane ZQW makes with plane PWZS.

(e) What is the size of the angle that plane ZQW makes with plane PQXW?

7 Fig. 6.16 shows a cuboid with a dividing plane.

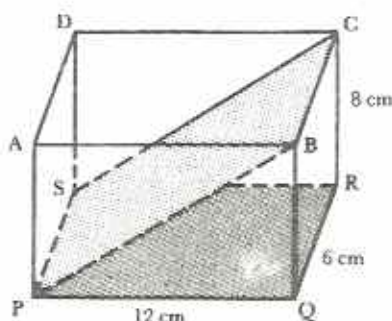


Fig. 6.16

(a) Name the angle between the shaded planes.

(b) Use the given dimensions to state the tangent of this angle in its simplest terms.

8 Fig. 6.17 is a view of a market stall.

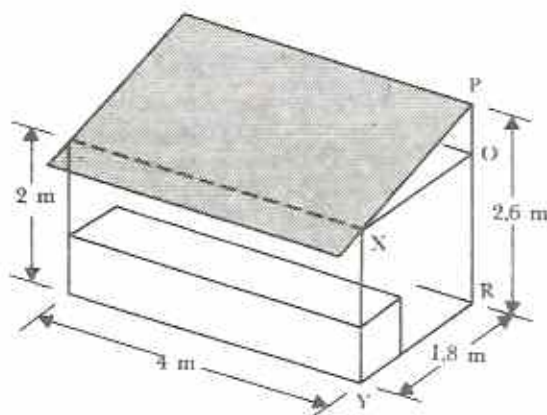


Fig. 6.17

(a) Name the angle that the roof makes with the horizontal.

(b) Use the given dimensions to state the tangent of this angle as a decimal fraction.

9 Fig. 6.18 represents a roof of a building. XOY and MON are lines of symmetry of the horizontal base of the roof.

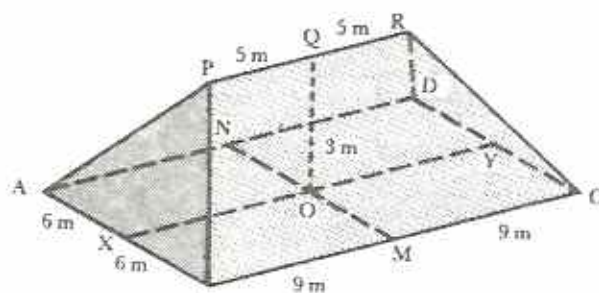


Fig. 6.18

(a) Name the angle between plane PBCR and the horizontal.

(b) Use the given dimensions to find the tangent of that angle.

(c) Name the angle between plane PAB and the horizontal.

(d) Find the tangent of that angle.

10 Fig. 6.19 is the net of a square-based pyramid.

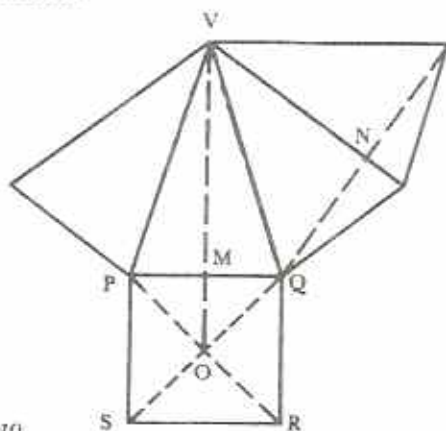


Fig. 6.19

(a) Sketch a view of the pyramid. Include all the given letters and lines on your sketch.

(b) Name the angle between plane VPQ and plane PQRS.

(c) Name the angle between planes VQR and VRS.

Calculating lengths and angles in solids

In most solids, unknown lengths and angles can be found by solving right-angled triangles.

Fig. 6.20 shows some of the right-angled triangles contained in a cuboid.

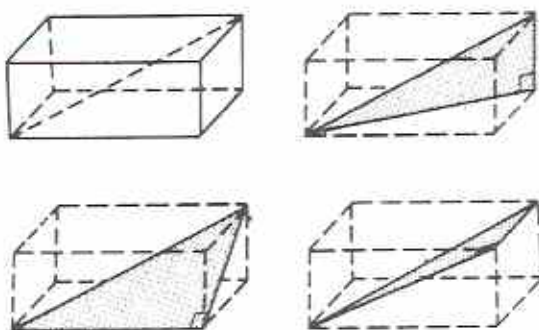


Fig. 6.20

Fig. 6.21 shows some of the right-angled triangles contained in a right pyramid.

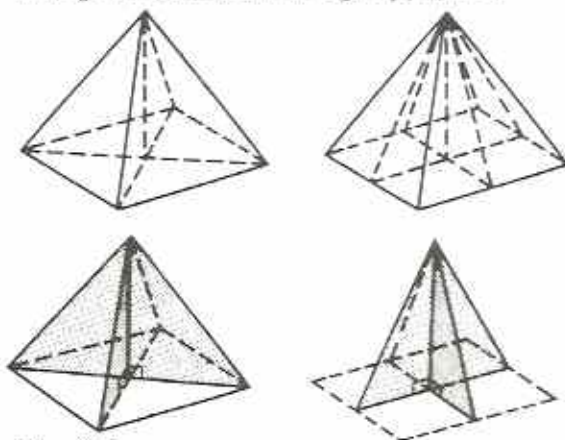


Fig. 6.21

The following examples show that when a right-angled triangle is used, it is advisable to sketch it separately from the solid.

Example 3

One end of a rectangular tank of length 6 m is a square ABCD of side 2 m. If AP is a diagonal of the tank, calculate (a) AP correct to 1 decimal place, (b) the angle between AP and plane ABCD, (c) the shortest distance between plane APD and BC.

Fig. 6.22 is a view of the tank.

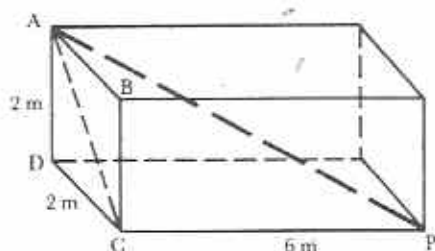


Fig. 6.22

(a) Fig. 6.23 shows the triangles used to calculate AP.

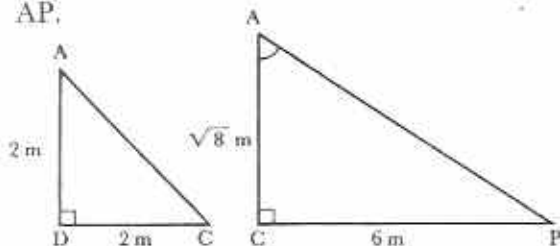


Fig. 6.23

In $\triangle ADC$,
 $AC^2 = AD^2 + DC^2$ (Pythagoras)
 $= 2^2 + 2^2 = 4 + 4 = 8$

$AC = \sqrt{8} \text{ m}$
 In $\triangle ACP$,
 $AP^2 = AC^2 + CP^2$ (Pythagoras)

$$= (\sqrt{8})^2 + 6^2 = 8 + 36 = 44$$

$$AP = \sqrt{44} \text{ m} = 6.6 \text{ m to 1 d.p.}$$

(b) AC is the projection of AP on plane ABCD. Hence $\angle CAP$ is the required angle. In $\triangle ACP$

$$\tan \hat{CAP} = \frac{CP}{AC}$$

$$= \frac{6}{\sqrt{8}} = \frac{6}{2\sqrt{2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$= \frac{3 \times 1.414}{2} = 2.121$$

$$\hat{CAP} = 64.75^\circ$$

AP meets plane ABCD at an angle of 65° (the nearest degree).

(c) In Fig. 6.24(a), CQ is the shortest distance between BC and plane APD.

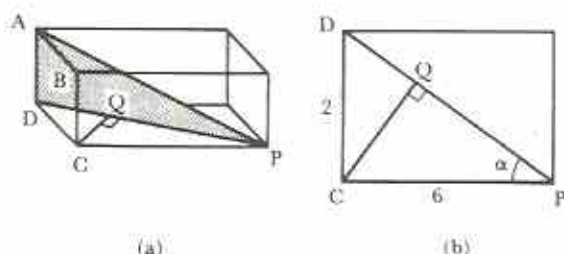


Fig. 6.24

In Fig. 6.24(b),

$$DP^2 = DC^2 + CP^2$$

$$= 2^2 + 6^2 = 4 + 36 = 40$$

$$DP = \sqrt{40} \text{ m}$$

$$\text{In } \triangle PQC, \sin \alpha = \frac{CQ}{CP} = \frac{CQ}{6}$$

$$\text{In } \triangle PCD, \sin \alpha = \frac{DC}{DP} = \frac{2}{\sqrt{40}}$$

$$\text{Hence } \frac{CQ}{6} = \frac{2}{\sqrt{40}}$$

$$CQ = \frac{2 \times 6}{\sqrt{40}} = \frac{12\sqrt{40}}{40} \text{ m}$$

$$= \frac{3\sqrt{40}}{10} = \frac{3 \times 6,325}{10} \text{ m}$$

$$= 1,8975 \text{ m}$$

$$= 1,9 \text{ m to 1 d.p.}$$

BC is 1,9 m from plane APD.

Notice the value of sketching the various triangles (Figs. 6.23, 6.24). This makes it easy to place the right angle correctly and to solve the triangle using Pythagoras' theorem and trigonometry. Also notice in parts (b) and (c) how rationalising the denominators simplifies the arithmetic.

Example 4

A pyramid with vertex V and edges VA, VB, VC, VD each 13 cm long has a rectangular base ABCD where AB = CD = 8 cm and AD = BC = 6 cm. Calculate (a) the height VO of the pyramid, (b) the angle between the base and an edge, (c) the angle between the base and $\triangle VBC$, (d) the angle between the base and $\triangle VCD$.

Since the pyramid is symmetrical, the point O vertically below V is at the centre of the rectangle ABCD. This is shown in Fig. 6.25.

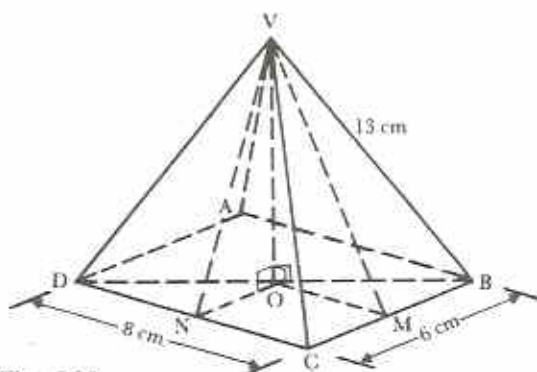


Fig. 6.25

The triangles in Fig. 6.26 are used to calculate VO.

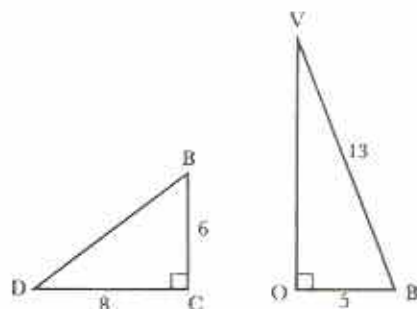


Fig. 6.26

(a) In $\triangle BCD$,

$$BD^2 = 6^2 + 8^2$$

$$= 36 + 64 = 100$$

$$BD = \sqrt{100} \text{ cm} = 10 \text{ cm}$$

$$BO = \frac{1}{2}BD = 5 \text{ cm}$$

In $\triangle VOB$,

$$VO^2 = 13^2 - 5^2$$

$$= 169 - 25 = 144$$

$$VO = 12 \text{ cm}$$

(b) Since VO is perpendicular to the base, $\angle VBO$ is one of the angles between the base and an edge.

In $\triangle VBO$,

$$\tan \angle VBO = \frac{VO}{BO} = \frac{12}{5} = 2,4$$

$$\therefore \angle VBO = 67,38^\circ$$

(c) Let M be the mid-point of BC. Since MV and MO are both perpendicular to edge BC, $\angle VMO$ is the angle between the base and face VBC. $\angle VMO$ can be found from $\triangle VMO$ (Fig. 6.27).

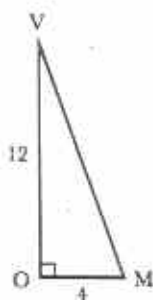


Fig. 6.27

$$\text{In } \triangle VMO, \tan \angle VMO = \frac{12}{4} = 3$$

$$\angle VMO = 71.57^\circ$$

(d) Similarly, $\angle VNO$ is the angle between the base and face VCD.

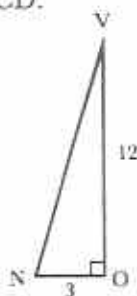


Fig. 6.28

$$\text{In } \triangle VNO, \tan \angle VNO = \frac{12}{3} = 4$$

$$\angle VNO = 75.97^\circ$$

Example 5

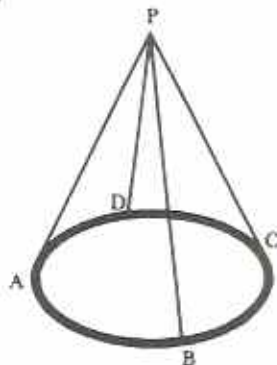


Fig. 6.29

In Fig. 6.29 a hoop of radius 80 cm is suspended horizontally by four strings each 160 cm long and each attached to a nail vertically above the hoop at P. The strings are attached to points A, B, C, D which are equally spaced on the hoop. Calculate (a) the angle which PA makes with the horizontal, (b) $\angle BPD$, (c) the angle between PA and PB.

Since the strings are of equal length, the centre of the hoop, O, is vertically below P. Fig. 6.30 shows, (a) the position of O and, (b) $\triangle PAC$.

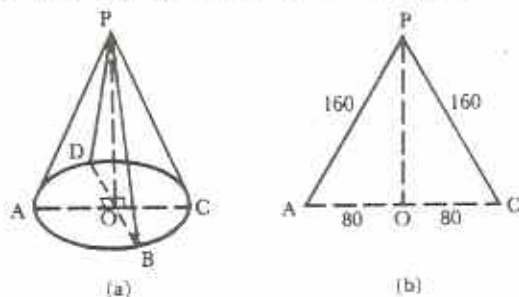
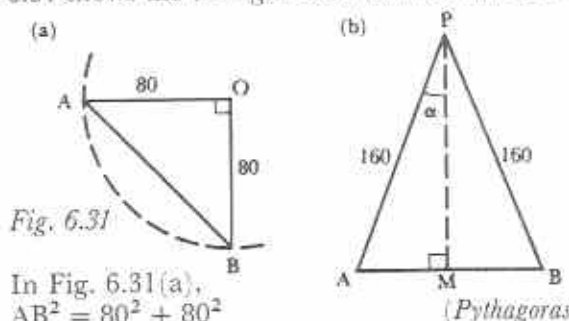


Fig. 6.30

(a) In $\triangle PAC$,
 $AC = 80 \text{ cm} + 80 \text{ cm} = 160 \text{ cm}$
 Therefore $\triangle PAC$ is equilateral.
 $\therefore \angle PAC = 60^\circ$ (angles of equilateral $\triangle$)
 PA makes an angle of 60° with the horizontal.

(b) Since the strings are equally spaced
 $\angle BPD = \angle APC$.
 $\angle ABC = 60^\circ$ (angles of equilateral $\triangle$)
 $\Rightarrow \angle BPD = 60^\circ$

(c) $\angle APB$ is the angle between PA and PB. Fig. 6.31 shows the triangles used to calculate $\angle APB$.



In Fig. 6.31 (a),
 $AB^2 = 80^2 + 80^2$
 $= 2 \times 80^2$
 $AB = \sqrt{2 \times 80^2} = 80\sqrt{2} \text{ cm}$

In Fig. 6.31(b), M is the mid-point of AB.

$$\sin \alpha = \frac{AM}{AP} = \frac{\frac{1}{2} \times 80\sqrt{2}}{160} = \frac{\sqrt{2}}{4}$$

$$= \frac{1,414}{4} = 0,3535$$

$$\alpha = 20,7^\circ$$

$$\angle APB = 2\alpha = 2 \times 20,7^\circ = 41,4^\circ$$

PA and PB meet at $41,4^\circ$.

Exercise 6c

Draw as many sketches as are necessary.

- The length, breadth and height of various cuboids are given below. Calculate the length of the long diagonal of each cuboid.
 - 6 cm, 10 cm, 15 cm
 - 4 cm, 4 cm, 2 cm
 - 2 m, 5 m, 14 m
 - 4 cm, 5 cm, 20 cm
- Use the data of Fig. 6.16 to calculate
 - CQ,
 - PC,
 - the angle between the shaded faces.
- Use the data of Fig. 6.17 to calculate the angle that the roof makes with the horizontal.
- Fig. 6.32 shows a desk lid which is kept open with a 30 cm ruler.

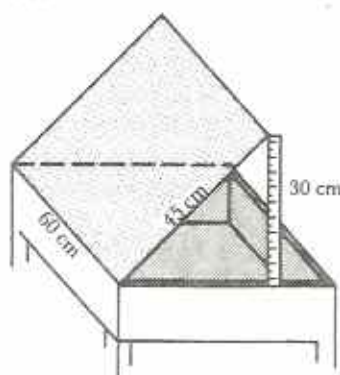


Fig. 6.32

Calculate the angle that the lid makes with the horizontal.

- Fig. 6.33 shows another way of keeping a desk lid open with a ruler.
 - Calculate the angle that the lid makes with the horizontal.
 - Hence find the perpendicular distance between the top edge of the lid and the desk.

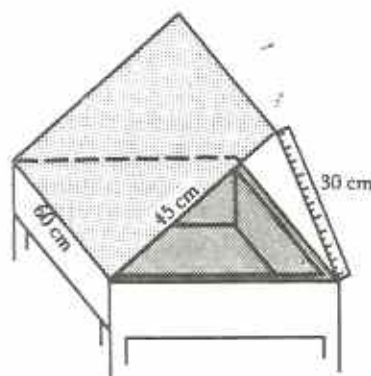


Fig. 6.33

- In Fig. 6.34, PQRS is a horizontal rectangular assembly ground 80 m by 60 m and PT is a tower 47 m high.

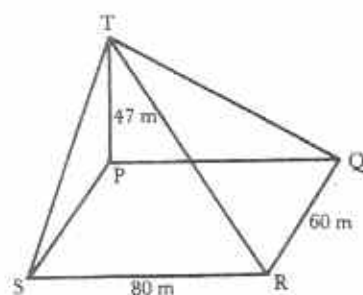


Fig. 6.34

Find, to the nearest degree, the angle of elevation of T from R.

- Fig. 6.35 shows a stick, AB, leaning against the corner of a room. The foot of the stick, A, is 30 cm from one wall and 40 cm from the other. The top of the stick, B, is 120 cm above the floor. Calculate
 - AO,
 - the angle that AB makes with the floor,
 - the length of the stick.

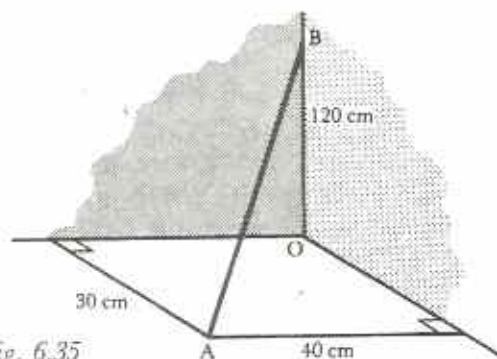


Fig. 6.35

- 8 A 50 cm stick leans, as in question 7, so that its foot is 10 cm from one wall and 20 cm from the other. Calculate (a) the height of the top of the stick above the floor, (b) the angle that the stick makes with the floor.
- 9 A pyramid has a square base of side 8 cm and sloping edges 9 cm long. Calculate (a) the height of the pyramid, (b) the angle, to the nearest $\frac{1}{10}$ th of a degree, (i) between a sloping edge and the base, (ii) between a sloping face and the base.
- 10 A pyramid is 4 cm high and stands on a base which is a regular hexagon of side 3 cm. Calculate (a) the length of an edge of the pyramid, (b) the angle that the edge makes with the base, (c) the angle that a triangular face makes with the base.
- 11 The cuboid in Fig. 6.36 is 28 m long, 21 m wide and 12 m high. Calculate the angle between (a) PC and the plane CDSR, (b) the planes PBCS and QBCR.

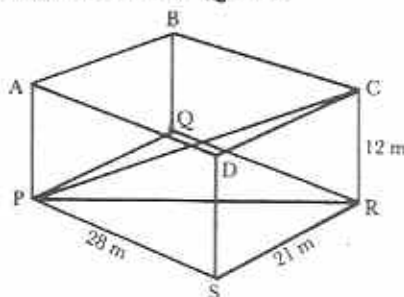


Fig. 6.36

- 12 Fig. 6.37 shows an 8 cm by 9 cm by 12 cm cuboid.

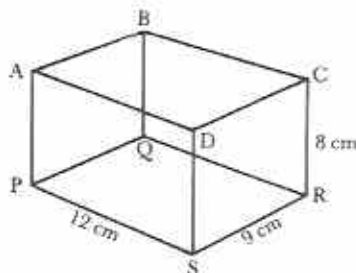


Fig. 6.37

Calculate (a) PR, (b) PC, (c) the angle between PC and (i) PQRS, (ii) DCRS, (iii) BCRQ, (d) the angle between ABCD and PBCS, (e) the angle between PQCD and ABQP.

- 13 A door 2 m high by 1.5 m wide is opened to an angle of 60° as shown in Fig. 6.38.

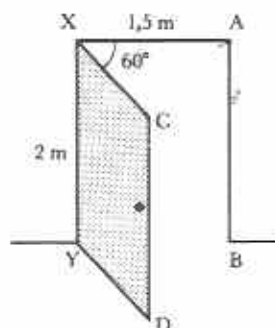


Fig. 6.38

- Calculate (a) BD, (b) DX, (c) the angle that plane DXB makes with the horizontal.
- 14 O is the vertex of a right pyramid on a square base ABCD, the sides of the base being 10 cm long. The edges OA, OB, OC, OD are each 12.5 cm. Calculate (a) the height of the pyramid, (b) the angle between OAB and the base ABCD, (c) the volume of the pyramid in cm^3 , correct to 3 s.f.
- 15 A right pyramid on a square base PQRS has a vertex T. Each of the sloping faces is an equilateral triangle of side 12 cm. Calculate (a) the base area of the pyramid, (b) the height of the pyramid, correct to 1 decimal place, (c) the volume of the pyramid, correct to 3 significant figures, (d) the angle which TS makes with the base PQRS.
- 16 The net of a pyramid consists of a square of side 16 cm and four isosceles triangles whose equal sides are each 17 cm. (a) Sketch the pyramid. (b) Calculate the height of the pyramid. (c) Calculate the angle between one of the 17 cm edges and the square base. (d) Calculate the angle between a triangular face and the square face.
- 17 Use the data of Fig. 6.18 to calculate the angle between the plane ABCD and (a) plane PRCB, (b) plane APB.

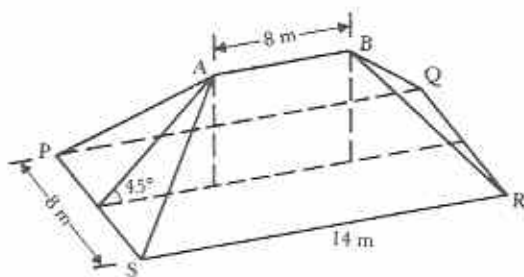


Fig. 6.39

- 18 Fig. 6.39 is a sketch of a symmetrical roof which has both triangular faces inclined at 45° to the horizontal.

Use the given dimensions to calculate
(a) the height of AB above the plane PQRS,
(b) the angle between the planes ABRS and PQRS.

- 19 In Fig. 6.40, which is not drawn to scale, TPQRSU is a solid which has a horizontal rectangular base PQRS in which PQ = 8 cm and PS = 10 cm. The triangular faces TPQ and URS are equilateral triangles and each makes an angle of 60° with the plane PQRS.

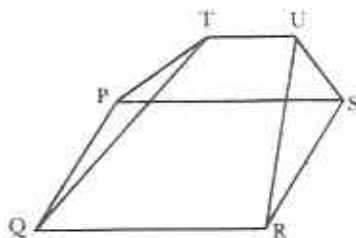


Fig. 6.40

Calculate (a) the height of TU above the plane PQRS, (b) TU, (c) the angle which TQ makes with the plane PQRS.

- 20 ABC is an equilateral triangle inscribed in a circle centre O, radius 80 cm, on a horizontal plane. A rod OE of length 60 cm is fixed vertically at O and stayed by wires from the top E to A, B and C. Calculate (a) the length of one of the wires, (b) the angle AEB correct to the nearest degree, (c) the angle between the planes BEC and BAC, correct to the nearest degree. (Hint: The property that the altitudes of an equilateral triangle trisect each other may be found useful.)

Inclined planes

Example 6

In Fig. 6.41, ABCD and XYCD are rectangular planes such that XYCD is horizontal and B is 10 cm above CY. EF || CB and EC = 16 cm, CP = 12 cm. Calculate the angle that (a) EF, (b) EB makes with the horizontal.

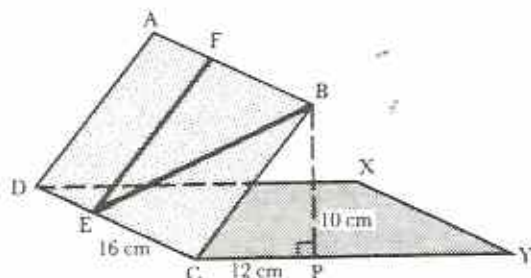


Fig. 6.41

- (a) EF || CB so $\angle BCP$ equals the required angle.

$$\text{In } \triangle BCP, \tan \angle BCP = \frac{10}{12} = \frac{5}{6} = 0.8333$$

$$\Rightarrow \angle BCP = 39.8^\circ$$

EF makes an angle of 39.8° with the horizontal.

- (b) PE is the projection of BE on plane XYCD. Hence BEP is the required angle. Fig. 6.42 shows the triangles used to find BEP.

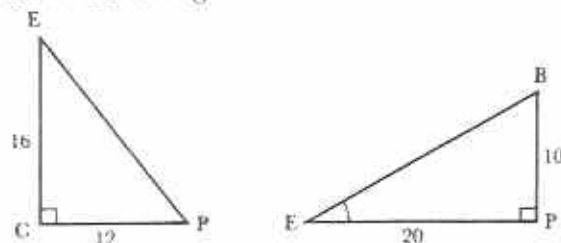


Fig. 6.42

$$\text{In } \triangle ECP,$$

$$EP^2 = 16^2 + 12^2$$

$$= 256 + 144 = 400$$

(Pythagoras)

$$EP = \sqrt{400} = 20 \text{ cm}$$

(Note: $\triangle ECP$ is a 3; 4; 5 $\triangle$.)

$$\text{In } \triangle BEP,$$

$$\tan \angle BEP = \frac{BP}{EP} = \frac{10}{20} = 0.5$$

$$\Rightarrow \angle BEP = 26.57^\circ$$

EB makes an angle of 26.6° with the horizontal. The **slope** of a line or a plane is the angle that it makes with the horizontal. In Fig. 6.41 it has been shown that the slope of EF is 39.8° and the slope of EB is 26.6° . The slope of EF is the same as the slope of plane ABCD. In this case EF is said to be a **line of greatest slope**. DA and CB are also lines of greatest slope. EB is not a line of greatest slope; its slope (26.6°) is less than that of EF (39.8°).

Example 7

A rectangular lid 25 cm by 20 cm is kept open at an angle of 65° to the horizontal, the hinges being on one of the long edges. Calculate the slope of a diagonal of the lid.

Fig. 6.43 shows the lid ABCD. θ is the required angle.

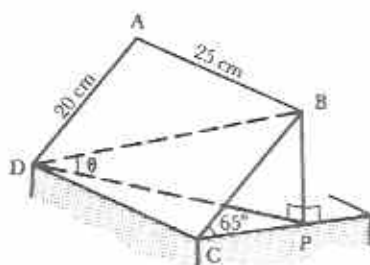


Fig. 6.43

Since θ is in the right-angled triangle BPD, it is necessary to find two sides of this triangle. The triangles in Fig. 6.44 each contain a side of $\triangle BPD$.

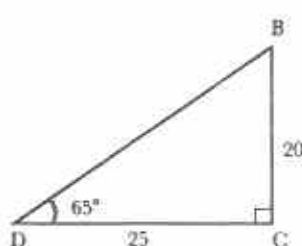
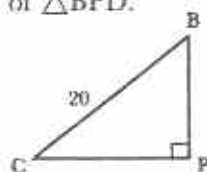


Fig. 6.44

$$\begin{aligned} \text{In } \triangle BPC, \sin 65^\circ &= \frac{BP}{20} \\ \Rightarrow BP &= 20 \sin 65^\circ \text{ cm} \end{aligned}$$

$$\text{In } \triangle BCD, \quad BD^2 = 20^2 + 25^2 = 400 + 625 = 1025$$

$$BD = \sqrt{1025} = 32.02 \text{ cm}$$

In $\triangle BPD$,

$$\sin \theta = \frac{BP}{BD} = \frac{20 \sin 65^\circ}{32.02}$$

$$\theta = 34.49^\circ$$

working:	
No.	Log
20	1.3010
$\sin 65^\circ$	1.9573
	1.2583
32.02	1.5054
$\sin 34.49^\circ$	1.7529

The slope of the diagonal is 34.5° .

Notice in Example 7 that tables of log-sines were used. This is quicker than using natural sines first, then using logarithms.

Alternatively, the calculation can be done on a scientific calculator as follows:

Key

AC

6 5 sin

$\times$ 2 0

$\times$ 3 2 0 2 =

SHIFT

sin

$\theta = 34.48^\circ$

Display

0

0.9063077

20

0.5660885

0.5660885

34.477917

Exercise 6d

Draw as many sketches as are necessary.

1 Name the lines of greatest slope in the following.

- Fig. 6.11, plane ABFE
- Fig. 6.16, plane CSPB
- Fig. 6.18, (i) plane PAB, (ii) plane RCBP
- Fig. 6.25, (i) plane VBC, (ii) plane VCD
- Fig. 6.34, plane TSR
- Fig. 6.43, plane ABCD

2 Name any lines which are at an angle to the line of greatest slope in the following.

- Fig. 6.9, plane VAD
- Fig. 6.18, plane PAB
- Fig. 6.22, plane APC
- Fig. 6.39, plane ABRS

3 In Fig. 6.41, calculate the slopes of EF and EB when $EC = 20$ cm and $CP = BP = 15$ cm.

4 Calculate the slope of a diagonal of the lid of the desk, (a) in Fig. 6.32, (b) in Fig. 6.33.

5 In Fig. 6.43, calculate the slope of DB when $DC = 24$ cm, $CB = 7$ cm and $\angle BCP = 30^\circ$.

6 Fig. 6.45 shows a prism such that M is the mid-point of AD. Calculate the slopes of BM and BD.

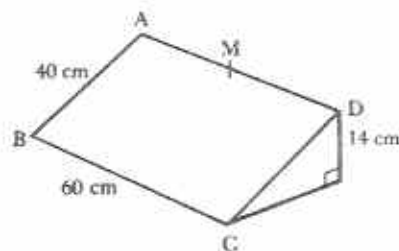


Fig. 6.45

Fractions in algebra

Simplification of fractions

Lowest terms

When simplifying algebraic fractions, always fully factorise the numerators and denominators. It may then be possible to divide the numerator and denominator by any factors which they have in common.

Example 1

Reduce $\frac{6m^2u^2 - 4mu^3}{9m^3u - 4mu^3}$ to its lowest terms.

$$\begin{aligned} & \frac{6m^2u^2 - 4mu^3}{9m^3u - 4mu^3} \\ &= \frac{2mu^2(3m - 2u)}{mu(9m^2 - 4u^2)} && \text{(taking out common factors)} \\ &= \frac{2mu^2(3m - 2u)}{mu(3m + 2u)(3m - 2u)} && \text{(difference of two squares)} \\ &= \frac{2u}{3m + 2u} && \text{[dividing numerator and denominator by } mu(3m - 2u)\text{]} \end{aligned}$$

Example 2

Simplify $\frac{a^2 + ax - 6x^2}{2x^2 + ax - a^2}$.

$$\begin{aligned} \frac{a^2 + ax - 6x^2}{2x^2 + ax - a^2} &= \frac{(a - 2x)(a + 3x)}{(2x - a)(x + a)} \\ &= -\frac{a + 3x}{x + a} \end{aligned}$$

In Example 2, notice that

$$a - 2x = -(2x - a)$$

so that $\frac{a - 2x}{2x - a} = \frac{-(2x - a)}{(2x - a)} = -1$.

In general,

$$\frac{c}{-d} = -\frac{c}{d} \text{ and } \frac{-c}{d} = -\frac{c}{d}$$

so that $\frac{c}{-d} = \frac{-c}{d} = -\frac{c}{d}$.

Remember also that $\frac{-c}{-d} = \frac{c}{d}$ since two negative quantities, divided one by the other, give a positive result.

In the same way,

$$\begin{aligned} \frac{a - m}{2m - a} &= \frac{a - m}{-(a - 2m)} = -\frac{a - m}{a - 2m} \\ \frac{a - m}{2m - a} &= \frac{-(m - a)}{2m - a} = -\frac{m - a}{2m - a} \\ \frac{a - m}{2m - a} &= \frac{-(m - a)}{-(a - 2m)} = \frac{m - a}{a - 2m} \end{aligned}$$

Hence if the sign of the numerator *or* the denominator is changed, the sign of the fraction is changed. However, if the signs of *both* the numerator *and* the denominator are changed, the sign of the fraction is unchanged. Because of this, there will sometimes be alternative answers to those given in the examples and exercises in this chapter. (Notice that to change the signs of a term is equivalent to multiplying it by -1 .)

Example 3

Simplify $\frac{a^2 - 5a + 6}{2 - 3a + a^2}$.

$$\begin{aligned}\frac{a^2 - 5a + 6}{2 - 3a + a^2} &= \frac{(a-2)(a-3)}{(2-a)(1-a)} \\ &= -\frac{a-3}{1-a} \text{ or } \frac{a-3}{a-1} \\ &\text{or } \frac{3-a}{1-a} \text{ or } -\frac{3-a}{a-1}\end{aligned}$$

Exercise 7a

Simplify the following fractions. If there is no simpler form, say so.

1 $\frac{mnu}{nuv}$ 2 $\frac{8x^2z}{10xyz}$ 3 $\frac{u+m}{u+n}$ 4 $\frac{ab+ac}{ad+ae}$

5 $\frac{a^2+ab}{a^2+ac}$ 6 $\frac{a^2+b^2}{a+b}$ 7 $\frac{u^2+uv}{uv+v^2}$ 8 $\frac{h^2-hk}{hk}$

9 $\frac{5d^2nv^3}{15d^3n^2v^4}$ 10 $\frac{c^2-cd}{d^2-cd}$ 11 $\frac{a^2-b^2}{b^2-ab}$

12 $\frac{x^2+xy}{x^2-y^2}$ 13 $\frac{28c^2d^2e^2}{35ce^4}$ 14 $\frac{x^2-4x}{x^2-4}$

15 $\frac{c^2-2cd+d^2}{c^2-cd}$ 16 $\frac{m^2+2mn+n^2}{m^2-n^2}$

17 $\frac{c^2-2c-15}{c^2-3c-10}$ 18 $\frac{d^2-9}{d^2-7d+12}$

19 $\frac{m^3n-2m^2n^2}{2mn^3-m^2n^2}$ 20 $\frac{xy-y^2}{(x-y)^2}$ 21 $\frac{xy-y^2}{(x+y)^2}$

22 $\frac{h^2+k^2}{(h+k)^2}$ 23 $\frac{h^2-k^2}{(h-k)^2}$

24 $\frac{u^2-5uv+6v^2}{u^2+uv-12v^2}$ 25 $\frac{x^2+xy-6y^2}{x^2-3xy+2y^2}$

26 $\frac{15+2x-x^2}{x^2-25}$ 27 $\frac{9a^2-m^2}{m^2-2am-3a^2}$

28 $\frac{8-2a-a^2}{2a^2-3a-2}$ 29 $\frac{a^2-am-an+mn}{a^2-am+an-mn}$

30 $\frac{a^2+am-an-mn}{a^2+am+an+mn}$ 31 $\frac{(u+w)^2-v^2}{(v+w)^2-u^2}$

32 $\frac{a^2-(b+c)^2}{c^2-(a-b)^2}$ 33 $\frac{2a^2m-3am^2+m^3}{am^2-a^2m-2a^3}$

34 $\frac{(2m-u)^2-(m-2u)^2}{5m^2-5u^2}$

35 $\frac{a(b+c)+(b+c)^2}{b^2-c^2+ab-ac}$ 36 $\frac{b^2+ac-ab-bc}{c^2-ac+ab-bc}$

Multiplication and division of fractions

Factorise fully first, then divide the numerator and denominator by any common factors.

Example 4

Simplify $\frac{a^2+2a-3}{a^2-16} \times \frac{a+4}{a^2+8a+15}$

Given expression

$$\begin{aligned}&= \frac{(a+3)(a-1)}{(a-4)(a+4)} \times \frac{a+4}{(a+5)(a+3)} \\ &= \frac{a-1}{(a-4)(a+5)}\end{aligned}$$

The answer should be left in the form given. Do not multiply out the brackets.

Example 5

Simplify $\frac{m^2-a^2}{m^2+bm+am+ab} \div \frac{m^2-2am+a^2}{cm+bc}$

To divide by a fraction, multiply by its reciprocal. Given expression

$$\begin{aligned}&= \frac{m^2-a^2}{m^2+bm+am+ab} \times \frac{cm+bc}{m^2-2am+a^2} \\ &= \frac{(m-a)(m+a)}{(m+b)(m+a)} \times \frac{c(m+b)}{(m-a)(m-a)} \\ &= \frac{c}{m-a} \quad [\text{dividing above and below by } (m-a), (m+a), (m+b)]\end{aligned}$$

Example 6

Simplify

$$\frac{a^2 + ab}{a^2 - 2ab + b^2} \div \frac{a + 3b}{a + 2b} \times \frac{ab - a^2}{a^2 + 3ab + 2b^2}$$

Given expression

$$\begin{aligned} &= \frac{a^2 + ab}{a^2 - 2ab + b^2} \times \frac{a + 2b}{a + 3b} \times \frac{ab - a^2}{a^2 + 3ab + 2b^2} \\ &= \frac{a(a+b)}{(a-b)(a-b)} \times \frac{a+2b}{a+3b} \times \frac{a(b-a)}{(a+b)(a+2b)} \\ &= -\frac{a^2}{(a-b)(a+3b)} \quad \begin{array}{l} \text{[dividing above and below} \\ \text{by } (a+b), (a+2b), \\ (a-b)] \end{array} \end{aligned}$$

Notice that $(a-b)$ divides into $(b-a)$ to give -1 . This is because $-1 \times (a-b) = (b-a)$.

Exercise 7b

Simplify the following.

$$1 \quad \frac{18ab}{15bc} \times \frac{20cd}{24de}$$

$$2 \quad \frac{12dn^3}{15cd^3} \div \frac{9c^3n}{10c^2d^2}$$

$$3 \quad \frac{m+n}{m} \times \frac{mn}{3m+3n}$$

$$4 \quad \frac{uv}{3u-6v} \times \frac{4u-8v}{u^2v}$$

$$5 \quad \frac{a^2 - b^2}{a^2 + ab} \div \frac{2a - 2b}{ab}$$

$$6 \quad \frac{a^2 - 4}{a^2 - 3a + 2} \div \frac{a}{a-1}$$

$$7 \quad \frac{m^2 - 9}{m^2 - m - 6} \times \frac{m^2 + 2m}{m^2}$$

$$8 \quad \frac{18m^2u}{16n^3v^2} \div \frac{24m}{15nu^3} \times \frac{8n^2v^3}{30m^3u}$$

$$9 \quad \frac{a^2 - b^2}{ab + a^2} \times \frac{2a^3}{ab - a^2}$$

$$10 \quad \frac{3d^2 - 12}{9d^2} \times \frac{6d^3}{4d + 8}$$

$$11 \quad \frac{2a - 2b + 2c}{8bc} \times \frac{10abc}{5a - 5b + 5c}$$

$$12 \quad \frac{n^2 - 9}{n^2 - n} \times \frac{n^2 - 3n + 2}{n^2 + n - 6}$$

$$13 \quad \frac{m^2 - n^2}{m^2 - 2mn + n^2} \div \frac{m^2 + mn}{n^2 - mn}$$

$$14 \quad \frac{a^2 - ab - 6b^2}{a^2 + ab - 6b^2} \times \frac{a^2 - ab - 2b^2}{a^2 - 2ab - 3b^2}$$

$$15 \quad \frac{5abc^2 - 10abcd}{3b^2c^2 - 6b^2cd} \times \frac{12bc^2d}{10acd}$$

$$16 \quad \frac{c^2 - cd}{d^2 - de} \div \frac{d^2 - cd}{cd - ce}$$

$$17 \quad \frac{e^2 - 5e + 6}{e^2 + 2e - 3} \div \frac{3e - 9}{2e^2 + 6e}$$

$$18 \quad \frac{u^2 + 3u - 10}{3u^2 + 12u} \div \frac{u^2 - 25}{u^2 - u - 20}$$

$$19 \quad \frac{x^2 - 3x - 4}{x^2 - 4x} \div \frac{x^2 - 4x + 4}{x^2 - 4}$$

$$20 \quad \frac{a^2 + ab - 2b^2}{a^2 - 2ab - 3b^2} \times \frac{a^2 - b^2}{ab + 2b^2} \div \frac{a^2 - 2ab + b^2}{a^2 - 3ab}$$

Addition and subtraction of fractions**Example 7**

$$\text{Simplify } 2 + \frac{6a^2 + 2b^2}{3ab} - \frac{4a - b}{2b}$$

The denominators are $3ab$ and $2b$. The LCM of $3ab$ and $2b$ is $6ab$. Express each fraction in the expression with the denominator of $6ab$.

$$\begin{aligned} &2 + \frac{6a^2 + 2b^2}{3ab} - \frac{4a - b}{2b} \\ &= \frac{2 \times 6ab}{6ab} + \frac{2(6a^2 + 2b^2)}{6ab} - \frac{3a(4a - b)}{6ab} \\ &= \frac{12ab + 2(6a^2 + 2b^2) - 3a(4a - b)}{6ab} \\ &= \frac{12ab + 12a^2 + 4b^2 - 12a^2 + 3ab}{6ab} \\ &= \frac{15ab + 4b^2}{6ab} \end{aligned}$$

$$= \frac{b(15a + 4b)}{6ab}$$

$$= \frac{15a + 4b}{6a}$$

Example 8

Simplify $\frac{3}{m^2 + mn - 2n^2} - \frac{2}{m^2 - 4mn + 3n^2}$.

Factorise the denominators so that their LCM can be used as the common denominator. Given expression

$$= \frac{3}{(m-n)(m+2n)} - \frac{2}{(m-n)(m-3n)}$$

The LCM of the denominators is $(m-n)(m+2n)(m-3n)$.

Given expression

$$= \frac{3(m-3n) - 2(m+2n)}{(m-n)(m+2n)(m-3n)}$$

$$= \frac{3m - 9n - 2m - 4n}{(m-n)(m+2n)(m-3n)}$$

$$= \frac{m - 13n}{(m-n)(m+2n)(m-3n)}$$

Example 9

Simplify $\frac{x+4}{x^2-3x} - \frac{x-1}{9-x^2}$.

$$\frac{x+4}{x^2-3x} - \frac{x-1}{9-x^2}$$

$$= \frac{x+4}{x(x-3)} - \frac{x-1}{(3-x)(3+x)}$$

$$= \frac{x+4}{x(x-3)} + \frac{x-1}{(x-3)(3+x)}^*$$

$$= \frac{(x+4)(x+3) + x(x-1)}{x(x-3)(x+3)}$$

$$= \frac{x^2 + 7x + 12 + x^2 - x}{x(x-3)(x+3)}$$

$$= \frac{2x^2 + 6x + 12}{x(x-3)(x+3)}$$

$$= \frac{2(x^2 + 3x + 6)}{x(x-3)(x+3)}$$

* Notice that the sign in front of the fraction changed since $(3-x) = -(x-3)$. This gives an LCM of $x(x-3)(x+3)$.

Example 10

Simplify

$$\frac{3a-5m}{a^2-5am+6m^2} + \frac{1}{a-2m} - \frac{2}{a-3m}$$

Given expression

$$= \frac{3a-5m}{(a-2m)(a-3m)} + \frac{1}{a-2m} - \frac{2}{a-3m}$$

$$= \frac{3a-5m + (a-3m) - 2(a-2m)}{(a-2m)(a-3m)}$$

$$= \frac{3a-5m+a-3m-2a+4m}{(a-2m)(a-3m)}$$

$$= \frac{2a-4m}{(a-2m)(a-3m)}$$

$$= \frac{2(a-2m)}{(a-2m)(a-3m)}$$

$$= \frac{2}{a-3m}$$

Exercise 7c

Simplify the following.

$$1 \quad \frac{3}{2ab} + \frac{4}{3bc} \qquad 2 \quad 5 - \frac{a-b}{c}$$

$$3 \quad \frac{3a-b}{5ab} - \frac{2b+3c}{6bc} + \frac{3c-2a}{15ac}$$

$$4 \quad \frac{3}{2(x+y)} - \frac{1}{3(x+y)}$$

$$5 \quad \frac{6}{a-2b} + \frac{4}{2b-a}$$

$$6 \quad 3 + \frac{2b}{a-b} \qquad 7 \quad 2 - \frac{x}{x+2y}$$

$$8 \frac{1}{4(u-v)} - \frac{1}{5(v-u)}$$

$$9 \frac{7u}{2u+3v} - 3$$

$$10 \frac{3a}{2a+b} - \frac{b}{4a+2b}$$

$$11 \frac{3mn}{2m^2+2n^2} + \frac{5mn}{3m^2+3n^2}$$

$$12 \frac{1}{4x-2y} - \frac{1}{y-2x}$$

$$13 \frac{a+b}{ab} - \frac{b+c}{bc}$$

$$14 \frac{u^2-v^2}{uv} + \frac{v}{u} - \frac{3uv-u^2}{v^2}$$

$$15 \frac{d+1}{2d-8} - \frac{d+2}{12-3d}$$

$$16 \frac{2}{a+1} + \frac{3}{a+2}$$

$$17 \frac{3x}{x-1} - \frac{4}{x+2}$$

$$18 \frac{e-2}{e+2} - \frac{e-1}{e+3}$$

$$19 \frac{3}{m-n} + \frac{m+3n}{(m-n)^2}$$

$$20 \frac{7c+2d}{(2c+d)^2} - \frac{3}{2c+d}$$

$$21 \frac{a+b}{(a-2b)^2} - \frac{1}{2b-a}$$

$$22 \frac{3c}{c^2-d^2} - \frac{3d}{d^2-c^2}$$

$$23 \frac{4m-9n}{16m^2-9n^2} + \frac{1}{4m-3n}$$

$$24 \frac{(m+n)^2}{m^2-n^2} + \frac{m^2+mn}{n^2-mn}$$

$$25 \frac{c(3-c)}{c^2+3c-10} + \frac{c-1}{c+5}$$

$$26 \frac{5}{d^2-2d-8} + \frac{2}{d^2-6d+8}$$

$$27 \frac{4}{3a+3b} - \frac{3}{2a-2b} + \frac{b}{a^2-b^2}$$

$$28 \frac{5}{d^2-16} + \frac{2}{(d+4)^2}$$

$$29 \frac{4}{x-3} - \frac{1}{x+2} - \frac{x+7}{x^2-x-6}$$

$$30 \frac{2a-1}{a^2-5a+6} - \frac{a+3}{a^2+2a-8}$$

Example 11

Given that $x:y = 9:4$, evaluate $\frac{8x-3y}{x-\frac{3}{4}y}$.

If $x:y = 9:4$, then $\frac{x}{y} = \frac{9}{4}$.

Divide numerator and denominator of

$\frac{8x-3y}{x-\frac{3}{4}y}$ by y .

$$\frac{8x-3y}{x-\frac{3}{4}y} = \frac{8\left(\frac{x}{y}\right) - 3}{\frac{x}{y} - \frac{3}{4}}$$

Substitute $\frac{9}{4}$ for $\frac{x}{y}$ in the expression.

Value of expression

$$\begin{aligned} &= \frac{8 \times \frac{9}{4} - 3}{\frac{9}{4} - \frac{3}{4}} = \frac{18 - 3}{1\frac{1}{2}} = \frac{15}{1\frac{1}{2}} \\ &= 15 \div \frac{1}{2} = 15 \times \frac{2}{1} \\ &= 10 \end{aligned}$$

Example 12

If $x = \frac{2a+3}{3a-2}$, express $\frac{x-1}{2x+1}$ in terms of a .

Substitute $\frac{2a+3}{3a-2}$ for x in the given expression.

$$\frac{x-1}{2x+1} = \frac{\frac{2a+3}{3a-2} - 1}{2 \times \frac{2a+3}{3a-2} + 1}$$

Multiply the numerator and denominator by $(3a-2)$.

$$\begin{aligned}\frac{x-1}{2x+1} &= \frac{(2a+3) - (3a-2)}{2(2a+3) + (3a-2)} \\ &= \frac{2a+3-3a+2}{4a+6+3a-2} \\ &= \frac{-a+5}{7a+4}\end{aligned}$$

Exercise 7d

- 1 If $\frac{x}{y} = \frac{3}{4}$, evaluate $\frac{2x-y}{2x+y}$.
- 2 Given $p:q = 9:5$, evaluate $\frac{15p-2q}{5p+16q}$.
- 3 If $a:b = 5:3$, evaluate $\frac{6a+b}{a-\frac{1}{3}b}$.
- 4 Given $\frac{x}{y} = \frac{2}{7}$, evaluate $\frac{7x+y}{x-\frac{1}{7}y}$.
- 5 If $a = \frac{d+1}{d-1}$, express $\frac{a+1}{a-1}$ in terms of d .
- 6 If $x = \frac{a+3}{2a-1}$, express $\frac{2x+1}{3x+1}$ in terms of a .
- 7 If $x = \frac{3m-5}{3m+5}$, express $\frac{x-1}{x+1}$ in terms of m .
- 8 If $x = \frac{3w-1}{w+2}$, express $\frac{2x-3}{3x-1}$ in terms of w .
- 9 If $X = \frac{2a+3}{3a-2}$, express $\frac{X-1}{2X+1}$ in terms of a .
- 10 If $h = \frac{m+1}{m-1}$, express $\frac{2h-1}{2h+1}$ in terms of m .

Equations with fractions

Example 13

Solve the equation $\frac{1}{3a-1} = \frac{2}{a+1} - \frac{3}{8}$.

The LCM of the denominators is $8(3a-1)(a+1)$. To clear fractions, multiply

the terms on both sides of the equation by $8(3a-1)(a+1)$.

$$\text{If } \frac{1}{3a-1} = \frac{2}{a+1} - \frac{3}{8}$$

$$\text{then } \frac{1}{\cancel{3a-1}} \times 8(\cancel{3a-1})(a+1)$$

$$= \frac{2}{\cancel{a+1}} \times 8(3a-1)(\cancel{a+1})$$

$$- \frac{3}{8} \times 8(3a-1)(a+1)$$

$$\text{i.e. } 8(a+1) = 16(3a-1) - 3(3a-1)(a+1)$$

$$8a+8 = 48a-16-3(3a^2+2a-1)$$

$$8a+8 = 48a-16-9a^2-6a+3$$

$$8a+8-48a+16+9a^2+6a-3=0$$

$$9a^2-34a+21=0$$

$$(a-3)(9a-7)=0$$

$$\therefore a=3 \quad \text{or} \quad 9a=7$$

$$\therefore a=3 \quad \text{or} \quad \frac{7}{9}$$

$$\text{Check: If } a=3, \frac{1}{3a-1} = \frac{1}{9-1} = \frac{1}{8}$$

$$\text{and } \frac{2}{a+1} - \frac{3}{8} = \frac{2}{4} - \frac{3}{8} = \frac{1}{2} - \frac{3}{8} = \frac{1}{8}$$

$$\text{If } a = \frac{7}{9},$$

$$\frac{1}{3a-1} = \frac{1}{\frac{7}{3}-1} = \frac{1}{\frac{4}{3}} = \frac{3}{4}$$

$$\text{and } \frac{2}{a+1} - \frac{3}{8} = \frac{2}{1\frac{2}{9}} - \frac{3}{8}$$

$$= \frac{18}{16} - \frac{3}{8} = \frac{9}{8} - \frac{3}{8} = \frac{3}{4}$$

Example 14

Solve the equation $\frac{3}{x^2-5x+6} = \frac{2}{x^2-x-6}$.

Factorise the denominators of the fractions.

$$\frac{3}{(x-2)(x-3)} = \frac{2}{(x-3)(x+2)}$$

Multiply both sides by $(x-2)(x-3)(x+2)$

Then $3(x+2) = 2(x-2)$

$$\begin{aligned}\Leftrightarrow 3x + 6 &= 2x - 4 \\ \Leftrightarrow 3x - 2x &= -4 - 6 \\ \Leftrightarrow x &= -10\end{aligned}$$

Check: If $x = -10$,

$$\frac{3}{x^2 - 5x + 6} = \frac{3}{100 + 50 + 6} = \frac{3}{156} = \frac{1}{52}$$

$$\text{and } \frac{2}{x^2 - x - 6} = \frac{2}{100 + 10 - 6} = \frac{2}{104} = \frac{1}{52}$$

Compare Examples 13 and 14 with Examples 7, 8, 9 and 10. In Examples 13 and 14, both sides of the equations are multiplied by the LCM of the denominators. Hence every denominator becomes 1 and the fractions are cleared. However, in Examples 7, 8, 9 and 10, the common denominators must stay in the given expressions, so that the expressions remain the same size. This is an important difference between **solving equations** with fractions and **simplifying expressions** with fractions.

Exercise 7c

Solve the following equations.

$$1 \quad \frac{3}{a} = a - 2 \qquad 2 \quad 5 - 2d = \frac{2}{d}$$

$$3 \quad \frac{7}{3} + \frac{2}{e} = e \qquad 4 \quad e = \frac{3}{e + 2}$$

$$5 \quad m = \frac{8}{3m + 2} \qquad 6 \quad a + 3 = \frac{6}{a + 4}$$

$$7 \quad 3x - 2 = \frac{4}{x - 1} \qquad 8 \quad n - 3 = \frac{4}{n - 3}$$

$$9 \quad \frac{x - 2}{x + 4} = x \qquad 10 \quad \frac{3n}{2n - 1} = n$$

$$11 \quad \frac{a - 4}{7} = \frac{2}{3a - 1}$$

$$12 \quad \frac{4}{w + 3} - \frac{3}{w + 2} = 0$$

$$13 \quad \frac{3}{2b - 5} - \frac{4}{b - 3} = 0$$

$$14 \quad \frac{c}{c - 2} = \frac{3}{2c - 5}$$

$$15 \quad \frac{2}{d - 2} = \frac{3d}{4d + 12}$$

$$16 \quad \frac{4n - 3}{6n + 1} = \frac{2n - 1}{3n + 4}$$

$$17 \quad \frac{2m + 3}{2m + 5} - \frac{m - 1}{m - 2} = 0$$

$$18 \quad \frac{3}{c + 2} - \frac{2}{2c - 3} = \frac{1}{7}$$

$$19 \quad \frac{3}{x - 4} = \frac{2}{x - 1} - 4$$

$$20 \quad \frac{1}{2a - 5} + \frac{7}{9} = \frac{2}{a + 5}$$

$$21 \quad \frac{2}{d + 3} = \frac{3}{2d - 1} - \frac{4}{15}$$

$$22 \quad \frac{11}{m + 3} = \frac{5}{2m} - \frac{1}{m - 4}$$

$$23 \quad \frac{1}{2n - 3} + \frac{1}{2n + 1} - \frac{1}{n - 1} = 0$$

$$24 \quad \frac{2}{u + 2} = \frac{2}{u + 1} - \frac{1}{u + 4}$$

$$25 \quad \frac{3}{2d + 3} - \frac{1}{2d + 1} = \frac{1}{d + 1}$$

$$26 \quad \frac{4a - 1}{a + 4} - 2 = \frac{2a - 1}{a + 2}$$

$$27 \quad \frac{2}{x + 3} - \frac{x - 6}{x^2 - 9} = 0$$

$$28 \quad \frac{4}{m^2 + 3m + 2} - \frac{3}{m^2 + 5m + 6} = 0$$

$$29 \quad \frac{2}{u - 2} = \frac{2u - 1}{u^2 + u - 6} - \frac{3}{u + 3}$$

$$30 \quad \frac{3}{v - 4} - \frac{v + 2}{v^2 - 3v - 4} = \frac{1}{2v + 2}$$

Undefined fractions

Table 7.1 and Fig. 7.1 give a table of values and the corresponding graph of the function

$$\frac{1}{x-1}$$

Table 7.1

x	-1	0	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{9}{10}$	$1\frac{1}{10}$	$1\frac{1}{4}$	$1\frac{1}{2}$	2
y	$-\frac{1}{2}$	-1	-2	-4	-10	10	4	2	1

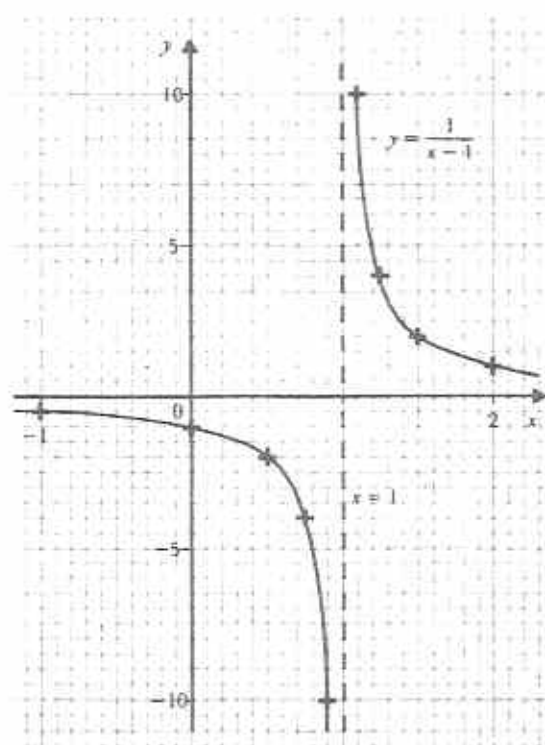


Fig. 7.1

Notice the following:

- 1 As the value of x approaches 1 from below, the value of $\frac{1}{x-1}$ decreases rapidly. For example, when $x = 0.999$,

$$\frac{1}{x-1} = \frac{1}{0.999-1} = \frac{1}{-0.001} = -1\,000$$

- 2 As the value of x approaches 1 from above, the value of $\frac{1}{x-1}$ increases rapidly. For example, when $x = 1.001$,

$$\frac{1}{x-1} = \frac{1}{1.001-1} = \frac{1}{0.001} = 1\,000$$

When $x = 1$, it is impossible to say what the value of $\frac{1}{x-1}$ is.

$$\text{Let } y = \frac{1}{x-1}$$

$$\text{When } x = 1, y = \frac{1}{1-1} = \frac{1}{0}$$

Division by zero is impossible. The fraction $\frac{1}{x-1}$ is said to be **undefined** when $x = 1$.

Fig. 7.1 shows that $\frac{1}{x-1}$ is undefined when $x = 1$. If the denominator of a fraction has the value zero, the fraction will be undefined. If an expression contains an undefined fraction, the whole expression is undefined.

Example 15

Find the values of x for which the following fractions are not defined.

(a) $\frac{3}{x+2}$ (b) $\frac{2x+13}{3x-12}$ (c) $\frac{5x}{x(5-x)}$

(d) $\frac{x^2-2x+3}{(x+3)(x-8)}$

(a) $\frac{3}{x+2}$ is undefined when $x+2 = 0$.

$$\text{If } x+2 = 0$$

$$\text{then } x = -2$$

The fraction is not defined when $x = -2$.

(b) $\frac{2x+13}{3x-12}$ is undefined when $3x-12 = 0$

$$\text{If } 3x-12 = 0$$

$$\text{then } 3x = 12$$

$$x = 4$$

The fraction is undefined when $x = 4$.

(c) $\frac{5x}{x(5-x)}$ is undefined when $x(5-x) = 0$.

If $x(5-x) = 0$

then either $x = 0$ or $5-x = 0$, i.e. $x = 5$

The fraction is undefined when $x = 0$ or when $x = 5$.

(d) $\frac{x^2 - 2x + 3}{(x+3)(x-8)}$ is undefined when

$$(x+3)(x-8) = 0$$

$$\text{If } (x+3)(x-8) = 0$$

then either $x+3 = 0$ or $x-8 = 0$

i.e. either $x = -3$ or $x = 8$

The fraction is undefined when $x = -3$ or when $x = 8$.

If part of an expression is undefined, then the whole expression is not defined.

Example 16

Find the values of x for which the expression

$$\frac{a}{x} - \frac{b}{x^2 + 6x - 7} \text{ is not defined.}$$

$$\frac{a}{x} - \frac{b}{x^2 + 6x - 7} = \frac{a}{x} - \frac{b}{(x-1)(x+7)}$$

The expression is not defined if any of its fractions has a denominator of 0.

$\frac{a}{x}$ is undefined when $x = 0$.

$\frac{b}{(x-1)(x+7)}$ is undefined when

$$(x-1)(x+7) = 0.$$

$$\text{If } (x-1)(x+7) = 0$$

then either $(x-1) = 0$ or $(x+7) = 0$

i.e. either $x = 1$ or $x = -7$

The expression is not defined when $x = 0, 1$ or -7 .

Exercise 7f

Find the values of x for which the following expressions are not defined.

1 $\frac{7}{x-3}$

2 $\frac{2x}{4-x}$

3 $\frac{3x+2}{x+7}$

4 $\frac{3+x}{x}$

5 $\frac{6x}{2x-5}$

6 $\frac{8}{15+3x}$

7 $\frac{y}{20-3x}$

8 $\frac{2a}{x(x+2)}$

9 $\frac{5b}{(1-2x)x}$

10 $\frac{3x+1}{(x+4)(x+3)}$

11 $\frac{7x^2}{(x+1)(x-1)}$

12 $\frac{4}{(x-6)(x-6)}$

13 $\frac{4x-3}{x(x+4)(x-9)}$

14 $\frac{1}{x^2-3x+2}$

15 $\frac{6x-1}{x^2-8x-20}$

16 $\frac{x^2+12x+36}{x^2-3x-10}$

17 $\frac{x^2-3x-10}{x^2+12x+36}$

18 $\frac{18}{x} + \frac{x^2+1}{x^2-9}$

19 $\frac{a}{x-2} + \frac{b}{x^2-2x} - \frac{c}{x+2}$

20 $\frac{x^2+10}{x^2+4x-5} - \frac{2x-1}{x^2+8x+15}$

Example 17

(a) For what value(s) of x is the expression $\frac{x^2+15x+50}{x-5}$ not defined? (b) Find the value(s) of x for which the expression is zero.

(a) The expression is not defined when its denominator is zero.

i.e. when $x-5 = 0$

$$x = 5$$

(b) Let $\frac{x^2+15x+50}{x-5} = 0$

Multiply both sides by $x-5$.

$$x^2+15x+50 = 0$$

$$(x+5)(x+10) = 0$$

either $x+5 = 0$ or $x+10 = 0$

i.e. either $x = -5$ or $x = -10$

The expression is zero when $x = -5$ or $x = -10$.

Exercise 7g (miscellaneous practice)

- Simplify $\frac{y^2 + 2yz + z^2}{y^2 - z^2}$.
- Simplify the expression $\frac{2m^2 + m - 15}{m^2 - 9}$ and state the value of m for which the simplified expression is not defined.
- Simplify $\frac{3}{a^2 - 3a + 2} \div \frac{2}{2a^2 - 5a + 2}$ and state the value of a for which the simplified expression is not defined.
- Reduce $\frac{3}{x} - \frac{x}{2} + 5$ to a single fraction.
- Simplify $\frac{3}{2x-4} + \frac{2}{6-3x}$.
- Simplify
 - $\frac{2b}{a^2 - b^2} + \frac{a}{b^2 - ab}$,
 - $\frac{2a}{a^2 - b^2} + \frac{b}{b^2 - ab}$.
- Simplify $\frac{3x+2}{3} - \frac{x-1}{4} - \frac{5}{12}$.
- Simplify $\frac{3y}{x^2 - xy - 2y^2} - \frac{2y}{x^2 - 2xy} + \frac{2x+y}{x^2 + xy}$.
- Simplify $\left(\frac{2}{x} - \frac{5}{y}\right) \div \frac{4}{xy}$.
- If $a = \frac{2m+1}{2m-1}$, express $\frac{2a+1}{2a-1}$ in terms of m .
- Solve $\frac{1}{2-m} + \frac{3m}{m^2-4} = 0$.
- Solve the equation $\frac{2}{2x-3} + \frac{3}{2x+3} = \frac{2}{4x^2-9}$.
- Solve $\frac{1}{3-a} - \frac{1}{3} + \frac{4}{2a-5} = 0$.
- If $x:y = 9:1$, evaluate $\frac{11x+y}{x+y}$.
- If $A = \frac{3x+2}{x+3}$, (a) for what value of x is A undefined; (b) for what range of values of x is $A < 2$?
- (a) For what value(s) of x is the expression $\frac{2x+11}{x^2+x-20}$ not defined?
(b) For what value(s) of x is the expression zero?
- (a) Solve $\frac{5}{a+4} - \frac{2}{a-2} = 0$.
(b) Simplify $\frac{5}{a+4} - \frac{2}{a-2}$.
- (a) Simplify $\frac{5}{2b+2} - \frac{2b+1}{b^2-2b-3} - \frac{1}{3-b}$.
(b) Solve $\frac{5}{2b+2} - \frac{2b+1}{b^2-2b-3} = \frac{1}{3-b}$.
- Simplify $\frac{2}{a+2} + \frac{1}{a+1} - \frac{2a+3}{a^2+3a+2}$ and find the value of a for which the simplified expression is not defined.
- (a) Simplify $\frac{(x+1)(x-2) - (x-1)(x+4)}{2x-1}$.
(b) (i) If k is a constant not equal to zero find the value(s) of x for which the expression $\frac{k}{x} + \frac{b}{x-3} + \frac{c}{x(x-3)}$ is not defined.
(ii) If x is not equal to any of the values obtained in (i), find the value of k such that $\frac{k}{x} + \frac{2}{x-3} - \frac{6}{x(x-3)} = 0$.

Graphs (4) Velocity-time curves

Area under a curve

It is often necessary to estimate the area contained between a curve and the axes, or other lines. There are many ways of doing this. Example 1 gives two of the simplest methods.

Example 1

Estimate the area enclosed between the curve $y = 20 + x - x^2$ and the axes in the positive quadrant.

By counting squares:

Draw the curve on suitable graph paper (Fig. 8.1).

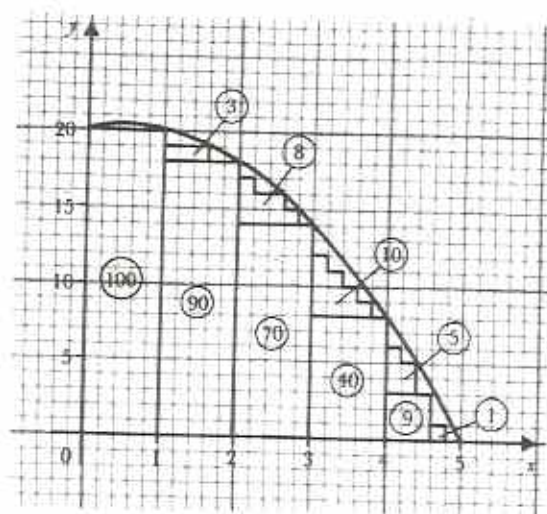


Fig. 8.1

Divide the area under the curve into convenient shapes, based on the squares of the graph paper. In each shape write the number of 'small squares' that it contains (in Fig. 8.1 the small squares have sides of 2 mm).

Estimate the remaining area by counting anything more than half a small square as one

square, ignoring those bits which are less than half a small square.

Number of whole squares = 336

Estimated part squares = 19

Total area = 355 small squares

On the x-axis, 2 mm represents $\frac{1}{5}$ of a unit of length.

On the y-axis, 2 mm represents 1 unit of length.

Hence 1 small square represents $\frac{1}{5} \times 1$ unit of area = $\frac{1}{5}$ unit²

$\Rightarrow 355$ small squares represent $\frac{355}{5}$ unit² = 71 unit²

Area under the graph ≈ 71 unit².

Notice the importance of the relationship between the small squares and the units of area.

By measuring trapeziums:

Draw the lines $x = 1$, $x = 2$, $x = 3$, $x = 4$ to divide the area under the curve into a rectangle (A), trapeziums (B), (C), (D) and triangle (E), Fig. 8.2.

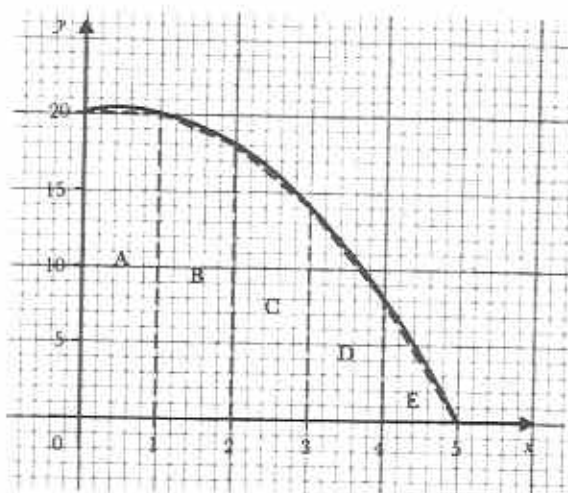


Fig. 8.2

Fig. 8.2 shows that the combined area of shapes (A), (B), (C), (D), (E) is approximately that contained between the curve and the axes.

$$\begin{aligned}\text{Total area} &\approx 20 \times 1 + \frac{1}{2}(20 + 18)1 \\ &\quad + \frac{1}{2}(18 + 14)1 + \frac{1}{2}(14 + 8)1 \\ &\quad + \frac{1}{2} \times 8 \times 1 \text{ unit}^2 \\ &= 20 + 19 + 16 + 11 + 4 \text{ unit}^2 \\ &= 70 \text{ unit}^2\end{aligned}$$

Area under the graph $\approx 70 \text{ unit}^2$.

The true area between the curve $y = 20 + x - x^2$ and the axes is 70.8 unit^2 to 1 d.p. Hence the method of counting squares is extremely accurate. Measuring trapeziums is slightly less accurate but is usually quicker.

Exercise 8a

In this exercise, give all answers to 2 s.f.

- 1 Use the method of counting squares to estimate the area bounded by the curve and the x -axis in Fig. 5.15 on page 38.
- 2 Look at Fig. 5.16 on page 38. Estimate the area bounded by the lines OM, MP and the curve OP.
- 3 Measure trapeziums to find the area below the curve but above the x -axis in Fig. 5.19 on page 39.
- 4 Use a suitable method to estimate the area above the curve but below the x -axis in Fig. 5.18 on page 39.
- 5 Draw the curve $y = 6x - x^2$ for values of x from 0 to 6, taking 2 cm for each x -unit and 1 cm for each y -unit. Find the area between the curve and the x -axis, both by counting squares and by measuring trapeziums.
- 6 Draw the curve $y = 2x(3 - x)$ for values of y corresponding to $x = 0, \frac{1}{2}, 1, 1\frac{1}{2}, 2, 2\frac{1}{2}, 3$. Use scales of 4 cm to 1 unit on the x -axis and 2 cm to 1 unit on the y -axis. Estimate the area included between the x -axis and the curve.

Velocity-time curves

Curved graphs can be drawn to represent many physical situations. They are very commonly used to show the relationship between time and the velocity of moving objects.

Gradient/acceleration

In Chapter 5 it was shown that if a graph connects x and y , its gradient is the rate of change of y compared with x .

Hence if a graph connects velocity and time, its gradient is the rate of change of velocity compared with time. This rate of change is called **acceleration**.

Area under curve/distance

Fig. 8.3 is a simple graph of the motion of a car which travels at 15 m/s for 3 s .

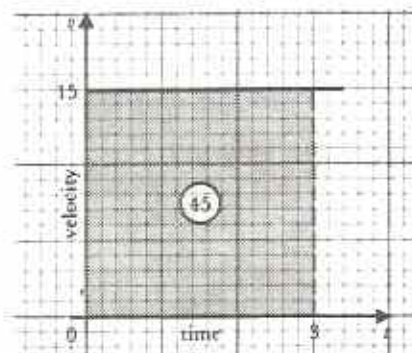


Fig. 8.3

In Fig. 8.3 the area under the graph is the area of the shaded rectangle
 $= \text{length} \times \text{breadth} = 15 \text{ m/s} \times 3 \text{ s} = 45 \text{ m}$
 $= \text{distance travelled by the car}$

In general, the area under a velocity-time graph measures the distance travelled in a given time interval (Fig. 8.4).

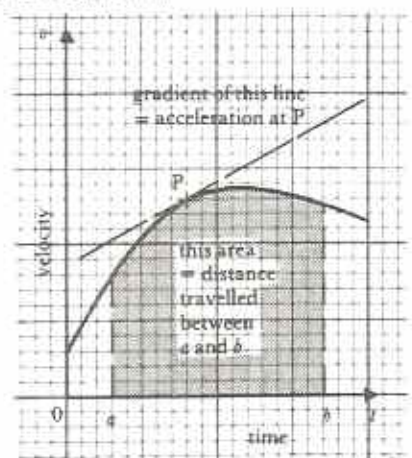


Fig. 8.4

Example 2, which follows, illustrates the use of gradient to represent acceleration and area to represent distance travelled.

Example 2

The speed of a car is noted at $\frac{1}{2}$ -minute intervals and the results are as given in Table 8.1.

Table 8.1

time (minutes)	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3
velocity (km/h)	20	$36\frac{1}{2}$	42	37	20	20	$26\frac{1}{2}$

(a) Estimate the distance travelled during the three minutes. (b) Estimate the acceleration in m/s^2 at $\frac{1}{2}$ min and 2 min from the start.

Fig. 8.5 is a graph of the journey,

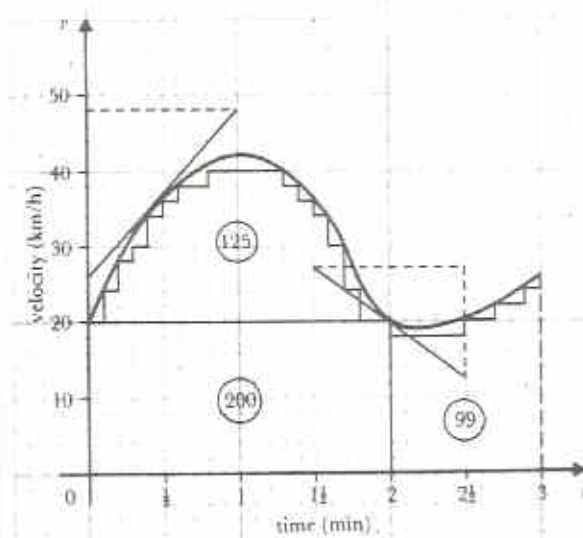


Fig. 8.5

(a) Estimated number of small squares = 444
1 small square represents $2 \text{ km/h} \times 0.1 \text{ min}$

$$= \frac{2 \times 1000}{60 \times 60} \text{ m/s} \times 6 \text{ s} = 3\frac{1}{3} \text{ m}$$

$$\text{Distance travelled} = 444 \times 3\frac{1}{3} \text{ m} = 1480 \text{ m} \\ \approx 1500 \text{ m}$$

The car travels about 1500 m.

(b) Draw the tangent to the curve at the point $(\frac{1}{2}; 36\frac{1}{2})$. Its gradient shows an increase of velocity of 22 km/h in 1 min. (Compare the

following sketch, Fig. 8.6, with the construction in Fig. 8.5.)

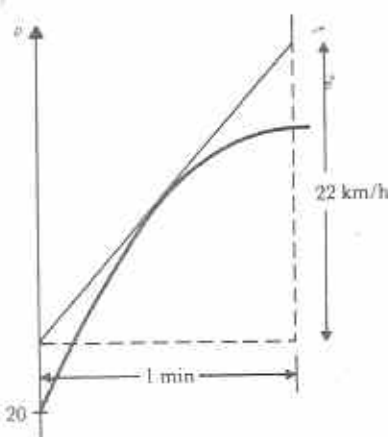


Fig. 8.6

$$\begin{aligned} \text{Acceleration} &= \frac{22000}{60 \times 60} \text{ m/s in } 60 \text{ s} \\ &= \frac{22000}{60 \times 60 \times 60} \text{ m/s per second} \\ &\approx 0.102 \text{ m/s}^2 \end{aligned}$$

Similarly, at the point $(2; 20)$, the rate of change of velocity is $-15 \text{ km/h per minute}$.

$$\begin{aligned} \text{Deceleration}^* &= \frac{15000}{60 \times 60 \times 60} \text{ m/s}^2 \\ &\approx 0.069 \text{ m/s}^2 \end{aligned}$$

* **Deceleration** is the rate at which velocity decreases; it may be thought of as negative acceleration. Fig. 8.7 shows that the acceleration of a moving body can be positive, negative or zero (at those points where the tangents to the velocity-time curve are parallel to the time-axis).

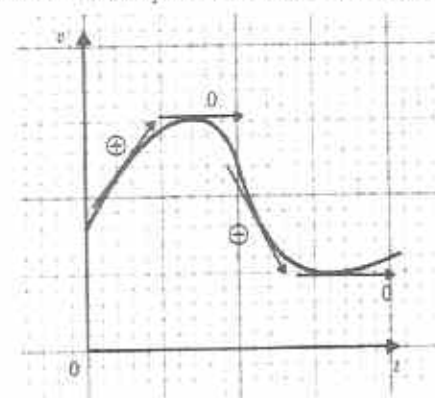


Fig. 8.7

Notice that just as the gradient on a velocity–time graph gives acceleration, so the gradient on a distance–time graph gives velocity, since velocity is the rate of change of distance with time. This definition is needed in questions 3 and 4 of the following exercise.

Exercise 8b

- The velocity of a car accelerating from rest is taken at 10-second intervals. The results are given in Table 8.2.

Table 8.2

t (seconds)	0	10	20	30	40	50	60	70
v (m/s)	0	14.3	20.6	24.4	26.9	28.4	29.4	30

- Draw a v – t graph. (b) Estimate the accelerations 10 s and 45 s from the start. (c) Also estimate the total distance covered in the 70 seconds.
- The depth of a river 60 m wide is found at 5-m intervals from one side to the other, straight across the river. The depths are 0; 11; 16.5; 18; 19.5; 22.5; 24.5; 25.5; 26; 24.5; 20; 11; 0 metres. (a) Draw the cross-section of the river. (b) Measure trapeziums to estimate the area of the cross-section in m^2 . (c) If the river is flowing at 4 km/h, calculate its flow in litres/second giving the answer in standard form, correct to 2 s.f.
- An object moves along a straight line so that its distance, s m, from a fixed point after t seconds is given by the formula $s = 5t - t^2$.
 - Plot s against t for values of t from 0 to 5, taking a 2-cm scale on both axes.
 - Estimate the gradients when $t = 1$ and $t = 3$.
 - What do the gradients represent? What happens when the gradient is zero?
- Water is poured at a steady rate into a pot of irregular shape. The depths of the water in the pot at 5-second intervals are 0; 2.2; 3.8; 5.0; 6.0; 6.8; 7.4; 7.9; 8.4; 8.9; 9.5; 10.6; 15.2 cm.
 - Draw a graph showing depth against time.

(b) Estimate the rate, in cm/s, at which the level is rising in the pot 20 s and 35 s after the start.

(c) Use your graph to guess the probable shape of the pot.

- A particle moves along a straight line AB so that, after t seconds, the velocity v m/s in the direction AB is given by $v = 2t^2 - 9t + 5$. Corresponding values of t and v are given in Table 8.3 below.

Table 8.3

t	0	1	2	3	4	5	6	7
v	5		-5	-4	1	10	23	

Calculate the value of v when $t = 1$ and the value of v when $t = 7$.

Taking 2 cm to represent 1 second on the horizontal axis, and 2 cm to represent 5 m/s on the vertical axis, draw the graph $v = 2t^2 - 9t + 5$ for the range $0 \leq t \leq 7$.

Use your graph to estimate

- the values of t when the velocity is zero
- the time at which the acceleration is zero
- the acceleration after 6 seconds.

[Cam]

- A particle moves along a straight line so that its velocity, v m/s, after t seconds is given by

$$v = 6 + 5t - t^2$$

(a) Copy and complete Table 8.4, giving corresponding values of t and v .

Table 8.4

t	0	1	2	3	4	5	6
v	6			12	10		

(b) Choose a suitable scale and draw a graph to show the relationship between v and t .

Use your graph to estimate

- the speed and time when the acceleration is zero,
- the acceleration after 5 seconds,
- the distance travelled in the first 6 seconds.

- 7 A particle moves on a straight line so that its velocity v m/s at time t seconds from the start is given by $v = 0,1t^3$.
- Draw the graph of v against t for values of t from 0 to 4.
 - Estimate the acceleration when $t = 2$ and when $t = 3,5$.
 - Find the distance covered in the 4 seconds.
- 8 A car travels along a straight road and its speed v m/s when it is t seconds past a certain point is given in Table 8.5.

Table 8.5

t	0	5	10	15	20	25	30	35	40	45	50
v	19,7	23	25,7	27,7	29	29,7	29,6	29	28,2	27,2	25,8

- Find the maximum speed of the car in m/s and the time when it occurs.
- Calculate the acceleration when $t = 13$.
- Find the total distance travelled to the nearest 10 m.

- 9 A car changes its speed smoothly over 6 min of continuous running. Its speeds, in km/h, at 1-min intervals are successively 80; 65,4; 54,8; 47; 44,8; 52; 70.

By drawing a suitable graph, estimate, in m/s^2 , the acceleration 5 min after the first observation and the deceleration after 2 min. Find also the total distance covered to the nearest 100 m.

- 10 The speed of a heavy lorry accelerating smoothly from rest is shown in Table 8.6.

Table 8.6

time (s)	0	10	20	30	40	50	60
speed (km/h)	0	24,0	36,0	42,5	45,0	44,2	39,0

By drawing a suitable graph, estimate the distance covered during this time to the nearest 10 m. Find the acceleration 5 s after the start and the deceleration 55 s after the start (in m/s^2). What is the highest speed and when is it reached?

Variation

Direct variation

Fig. 9.1 shows a new pencil cut into a number of pieces.



Fig. 9.1

The mass of each piece is proportional to its length. The ratio of mass to length is the same for all the pieces.

If a person walks at a steady speed, the distance travelled is proportional to the time taken.

These are both examples of **direct proportion**, or **direct variation**. In the first example the mass, M , **varies directly** with the length, L . In the second, the distance, D , varies directly with the time, T .

The symbol $\propto$ means 'varies with' or 'is proportional to'. The statements in the previous paragraph are written:

$$M \propto L$$

$$D \propto T$$

$D \propto T$ really means that the ratio $\frac{D}{T}$ is **constant** (i.e. stays the same).

Example 1

If $D \propto T$ and $D = 80$ when $T = 5$, find (a) the relationship between D and T , (b) the value of T when $D = 56$.

(a) If $D \propto T$
then $D = kT$, where k is a constant.
 $D = 80$ when $T = 5$,
hence $80 = k \times 5$

$$\Leftrightarrow k = \frac{80}{5} = 16$$

The relationship between D and T is $D = 16T$.

$$(b) \quad D = 16T$$

When $D = 56$,

$$56 = 16T$$

$$\Leftrightarrow T = \frac{56}{16} = \frac{7}{2} = 3\frac{1}{2}$$

Fig. 9.2 is a **sketch graph** of the relation $D \propto T$.

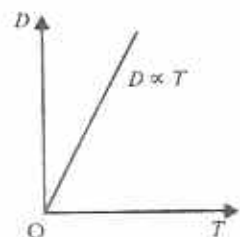


Fig. 9.2

Since $D = 16T$, the graph is a straight line of gradient 16 passing through the origin.

The graph of the relation between any two linear quantities which vary directly is always a straight line through the origin.

Example 2

Table 9.1 shows the extension E cm in an elastic string when it is pulled by a force of T newtons.

Table 9.1

T	5	8	11
E	7.5		16.5

- (a) Show that E is directly proportional to T .
(b) Find the value of E when $T = 8$.

(a) By calculation:
if $E \propto T$
then $E = kT$, where k is a constant,

$$\text{or } \frac{E}{T} = k$$

i.e. If $E \propto T$, then $\frac{E}{T}$ should have a constant value for the results given in Table 9.1.

When $T = 5$, $E = 7,5$

$$\frac{E}{T} = \frac{7,5}{5} = 1,5$$

When $T = 11$, $E = 16,5$

$$\frac{E}{T} = \frac{16,5}{11} = 1,5$$

Since $\frac{E}{T} = 1,5$ in both cases, E is directly proportional to T .

Graphically: plot the given values (Fig. 9.3).

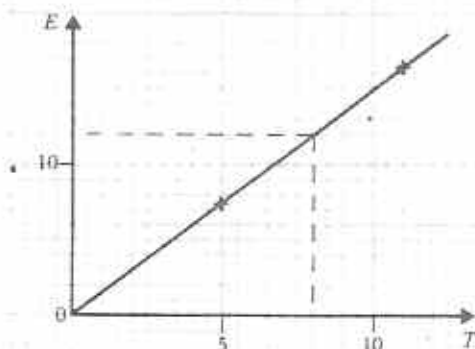


Fig. 9.3

Since the graph of E against T is a straight line passing through the origin, $E \propto T$.

(b) By calculation:

$$E = 1,5T$$

When $T = 8$,

$$E = 1,5 \times 8 = 12$$

or, from the graph;

when $T = 8$, $E = 12$

Exercise 9a

Questions 1–5 may be done orally.

- 1 If 1 m of wire has a mass of x g, what will be the mass of 25 m of the same wire?
- 2 If 1 jar of coffee costs \$ y , what will be the cost of 4 jars of coffee?

- 3 If a man cycles 15 km in 1 hour, how far will he cycle in t hours if he keeps up the same rate?
- 4 If eggs cost n cents each, how much will 16 eggs cost?
- 5 If a cup holds d ml of water, how much water will 8 of these cups hold?
- 6 If $C \propto n$ and $C = 28$ when $n = 4$, find the formula connecting C and n .
- 7 If $D \propto t$ and $D = 32$ when $t = 2$, find the relationship between D and t .
- 8 If $x \propto y$ and $x = 3$ when $y = 12$, find the relationship between x and y .
- 9 If $d \propto s$ and $d = 120$ when $s = 30$, find the formula connecting d and s .
- 10 $a \propto b$ and $a = 2,4$ when $b = 3$. Find the relationship between a and b .
- 11 If $D \propto S$ and $D = 140$ when $S = 35$, find (a) the relationship between D and S , (b) the value of S when $D = 176$.
- 12 $x \propto y$ and $x = 30$ when $y = 12$. Find (a) the formula connecting x and y , (b) x when $y = 10$, (c) y when $x = 14$.
- 13 $P \propto Q$ and $P = 4,5$ when $Q = 12$. Find (a) the relationship between P and Q , (b) P when $Q = 16$, (c) Q when $P = 2,4$.
- 14 $A \propto B$ and $A = 1\frac{7}{8}$ when $B = \frac{5}{6}$. Find (a) A when $B = 0,4$, (b) B when $A = 7,5$.
- 15 $d \propto P$ and $d = 0,2$ when $P = 10$. Find (a) d when $P = 18$, (b) P when $d = 1,1$.
- 16 D is directly proportional to S .
(a) If $D = 120$ when $S = 30$, find the relationship between D and S .
(b) Find S when $D = 192$.
- 17 y varies directly with x .
(a) If $y = 9$ when $x = 45$ find the equation which connects x and y .
(b) Find y when $x = 40$.
(c) Find x when $y = 10$.
- 18 If $p \propto q$ and $p = 0,7$ when $q = 0,028$, find the relationship between p and q .
- 19 $x \propto y$ and $x = 17\frac{1}{2}$ when $y = 10\frac{1}{2}$.
(a) Find the equation which connects x and y .
(b) Find x when $y = 12$.
- 20 The mass of a plastic disc is proportional to its area. A disc of area 180 cm^2 has a mass of 200 g. If a similar disc has a mass of 250 g, what is its area?

- 21 What is the relation illustrated by the sketch graph in Fig. 9.4?

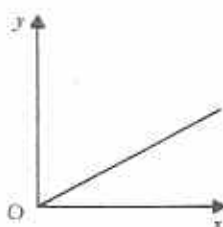


Fig. 9.4

- 22 Sketch the relation $N \propto s$.
23 Two variables A and B have corresponding values as shown in Table 9.2.

Table 9.2

A	6	10	12
B	34		68

- (a) Either graphically, or by calculation, show that $B \propto A$.
(b) Find the value of B when $A = 10$.
24 The number of US Dollars (US\$) exchanged for a number of Pounds sterling (£) is given in Table 9.3.

Table 9.3

£	2	4	6	8	10
US\$	3.20	6.40	9.60	12.80	16.00

- (a) Show that $\text{US\$} \propto \text{£}$.
(b) Find the law connecting US\$ and £.
(c) Find the value of £7 in US Dollars.
(d) Find the value of US\$8 in £.
25 The height (H cm) of liquid in a tube and the volume (V cm³) of the liquid are as given in Table 9.4.

Table 9.4

V	3	6	12
H	2.6	5.2	10.4

- (a) Show that $H \propto V$.
(b) Find the law of variation in the form $H = kV$.
(c) Find V when $H = 6.5$.
(d) Find H when $V = 4.5$.

Direct variation between non-linear quantities

Quantities which vary directly are not always in linear form. For example, the mass, m , of a cardboard square is directly proportional to its area, A .

$$m \propto A$$

However, in Fig. 9.5, $A = x^2$ so it follows that $m \propto x^2$.



Fig. 9.5

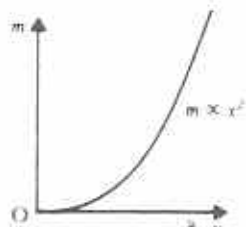


Fig. 9.6

$m \propto x^2$ is an example of direct proportion which one of the variables is in quadratic form.

Fig. 9.6 is a sketch graph of $m \propto x^2$. Notice that the curve is similar to the curve obtained in the graph of $y = x^2$.

Similarly, the volumes of spheres are directly proportional to the cubes of their radii:

$$V = \frac{4}{3}\pi r^3$$

or $V \propto r^3$, since $\frac{4}{3}\pi$ is a constant.

Fig. 9.7 is a sketch of the graph of $V \propto r^3$.

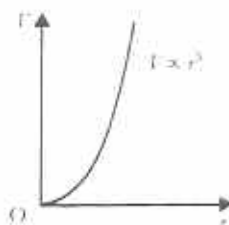


Fig. 9.7

Notice that the curve in Fig. 9.7 rises much more steeply than that of Fig. 9.6 (also see Chapter 8).

Example 3

$y \propto \sqrt{x}$ and $y = 4\frac{1}{2}$ when $x = 9$. (a) Find the relationship between x and y . (b) Find y when $x = 16$. (c) Find x when $y = 6$. (d) Sketch the graph of $y \propto \sqrt{x}$.

- (a) If $y \propto \sqrt{x}$
 then $y = k\sqrt{x}$, where k is a constant.
 When $x = 9$, $y = 4\frac{1}{2}$

$$\Rightarrow 4\frac{1}{2} = k\sqrt{9}$$

$$= 3k$$

$$k = \frac{4\frac{1}{2}}{3} = \frac{3}{2}$$

$$\Rightarrow y = \frac{3}{2}\sqrt{x}$$

- (b) When $x = 25$

$$y = \frac{3}{2}\sqrt{25} = 7\frac{1}{2}$$

- (c) When $y = 6$

$$6 = \frac{3}{2}\sqrt{x}$$

$$\Rightarrow \sqrt{x} = \frac{12}{3} = 4$$

$$\Rightarrow x = 16$$

- (d) Fig. 9.8 is a sketch graph of $y \propto \sqrt{x}$.

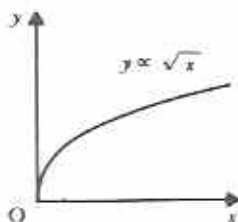


Fig. 9.8

The shape of the curve in part (d) is explained as follows:

$$\begin{aligned} \text{If } y &\propto \sqrt{x} \\ \text{then } y^2 &\propto x \\ \text{or } x &\propto y^2 \end{aligned}$$

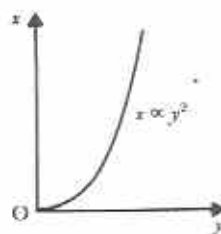
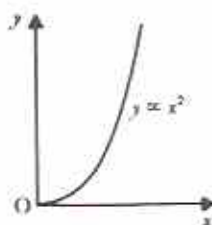


Fig. 9.9 (a)

(b)

Fig. 9.9(a) is a sketch of $y \propto x^2$. By interchanging the axes, Fig. 9.9(b) is a sketch of $x \propto y^2$. Fig. 9.8, $y \propto \sqrt{x}$, is equivalent to Fig. 9.9(b), $x \propto y^2$, with the x and y -axes in the standard positions.

The next example shows how a variation problem may be solved when the value of the constant k is not required.

Example 4

x is directly proportional to the square of y . What is the percentage change in x if y increases by 20%?

From the first sentence,

$$x \propto y^2$$

$$\text{Let } x = ky^2 \quad (1)$$

Let x become X if y increases to $\frac{120}{100}y$.

$$\text{Then } X = k\left(\frac{120}{100}y\right)^2 \quad (2)$$

Dividing (2) by (1),

$$\frac{X}{x} = \frac{k\left(\frac{120}{100}y\right)^2}{ky^2} = \left(\frac{120}{100}\right)^2 = \frac{144}{100}$$

$$\Leftrightarrow X = \frac{144}{100}x$$

Hence x increases by 44%.

Exercise 9b

- A particle moves in such a way that its displacement, s metres, at time t seconds is given by the relation $s = at^2$, where a is a constant. Calculate a if $s = 32$ when $t = 4$.
- y varies directly as the square of x . If $y = 98$ when $x = 7$, calculate y when $x = 5$.
- $x \propto y^2$ and $x = 45$ when $y = 3$.
 - Find the relationship between x and y .
 - Find x when $y = 4$.
 - Find y when $x = 125$.
- $A \propto B^3$ and $A = 32$ when $B = 4$.
 - Find the formula connecting A and B .
 - Find A when $B = 6$.
 - Find B when $A = 13.5$.
- P varies directly as the square root of Q and $P = 10$ when $Q = 16$.
 - Find the equation in P and Q .
 - Find P when $Q = 9$.
 - Find Q when $P = 1\frac{7}{8}$.
- $z^2 \propto Y$ and $z = 9$ when $Y = 27$.
 - Find the relation between z and Y .
 - Find z when $Y = 48$.
 - Find Y when $z = 6$.

- 7 V varies directly as the cube of D and $V = 108$ when $D = 6$.
- Find the formula connecting V and D .
 - Find V when $D = 3$.
 - Find D when $V = 2,048$.
- 8 $D \propto \sqrt{H}$ and $D = 6$ when $H = 24$.
- Find the relation between D and H .
 - Find D when $H = 150$.
 - Find H when $D = 10\frac{1}{2}$.
- 9 State the relation which is illustrated by the sketch graph in Fig. 9.10.

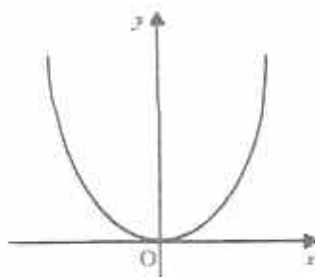


Fig. 9.10

- 10 Sketch the following curves, showing both positive and negative values of x .
- $y \propto x^2$
 - $y \propto x^3$
 - $y \propto \sqrt{x}$
- 11 Draw a sketch graph of the relation y^2 is proportional to x .
- 12 y varies directly with x^2 . Table 9.5 shows some corresponding values of x and y .

Table 9.5

x	-2	-1	$\frac{1}{2}$
y	16	4	

Find the relation between x and y and complete the table.

- 13 The power, P watts, used in an electric circuit is proportional to the square of the current, C amps. When the current is 4 amps the circuit uses 500 watts. Find the current when the circuit uses 2 420 watts.
- 14 The distance of the horizon from an observer varies directly with the square root of the height of the observer above ground level. At a height of 8 metres the horizon is 10 km away. Find the distance of the horizon from an observer at a height of 98 metres.

- 15 x varies directly with the square of y . Find the percentage change in x if y is
- increased by 10%.
 - decreased by 10%.
- 16 x is directly proportional to the square root of y . What is the percentage change in x if y is increased by 44%?
- 17 If $V \propto R^3$, what is the percentage increase in V if R increases by 20%?
- 18 If $W \propto D^2$, what is the percentage decrease in W if D decreases by 15%?

Inverse variation

Fig. 9.11 shows a circle cut into (a) 5 equal sectors, (b) 12 equal sectors.

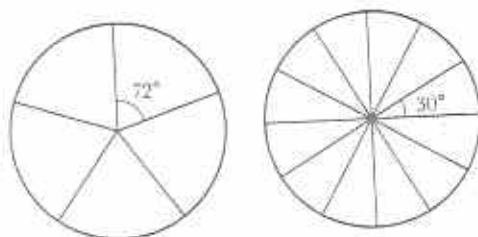


Fig. 9.11

The greater the number of sectors, the smaller the angle of each sector.

If a pot of tea is shared between some people the greater the number of people, the less tea each will receive.

These are examples of **inverse proportion** or **inverse variation**. In the first example, the size of the angle, θ , **varies inversely** with the number of sectors, n . In the second, the volume of tea received, V , is inversely proportional to the number of people, n . These statements are written:

$$\theta \propto \frac{1}{n} \quad V \propto \frac{1}{n}$$

Example 5

If V varies inversely with n and $V = 220$ when $n = 8$, find V when $n = 6$.

If $V \propto \frac{1}{n}$ then $V = \frac{k}{n}$, where k is a constant.
 $V = 220$ when $n = 6$.

$$220 = \frac{k}{6} \Leftrightarrow k = 6 \times 220 \quad \text{and} \quad V = \frac{6 \times 220}{n}$$

When $n = 8$,

$$\begin{aligned} V &= \frac{6 \times 220}{8} \\ &= 165 \end{aligned}$$

$$\begin{aligned} N &= \frac{2700 \times 8}{3^3} \\ &= \frac{2700 \times 8}{27} \\ &= 800 \end{aligned}$$

800 beads can be made.

Fig. 9.12 is a sketch graph of the relation $V \propto \frac{1}{n}$.

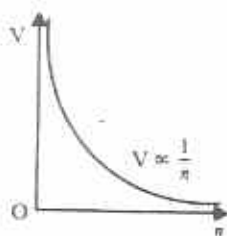


Fig. 9.12

Notice that the curve approaches both axes but does not reach them. This is because division by 0 is impossible. (See Fig. 7.1, Chapter 7; also see Chapter 15.)

Example 6

The number of spherical glass beads which can be made from a given volume of glass varies inversely with the cube of the diameter of the beads. When the diameter is 2 mm, the number of beads is 2700. How many beads of diameter 3 mm can be made from the glass?

Let N = number of beads, d = diameter of the beads.

From the first sentence,

$$N \propto \frac{1}{d^3} \quad \text{or} \quad N = \frac{k}{d^3}, \quad \text{where } k \text{ is a constant.}$$

From the second sentence,

$$\begin{aligned} 2700 &= \frac{k}{2^3} \\ \Leftrightarrow k &= 8 \times 2700 \\ \text{So } N &= \frac{2700 \times 8}{d^3} \end{aligned}$$

When $d = 3$,

Exercise 9c

- If d varies inversely as l , use the symbol $\propto$ to show a connection between d and l .
- A piece of string is cut into n pieces of equal length l .
 - Does n vary directly or inversely with l ?
 - Use the symbol $\propto$ to show a connection between n and l .
- A rectangle has a constant area, A . Its length is l and its breadth is b .
 - Write a formula for l in terms of A and b .
 - Write a formula for b in terms of A and l .
 - Does l vary inversely or directly with b ?
- If $x \propto \frac{1}{y}$ and $x = 22$ when $y = 3$, find the relationship between x and y .
- If $R \propto \frac{1}{T}$ and $T = 8$ when $R = 4$, find the relationship between R and T .
- If y varies inversely as x , and $y = 2$ when $x = 3$, find y when $x = 6$.
- P is inversely proportional to Q and $P = 5$ when $Q = 4$. What is the value of Q when $P = 25$?
- If x varies inversely as the square of y , and $x = 4$ when $y = \frac{1}{2}$, what is y when x is 5?
- Make x the subject of the relation $y \propto \frac{1}{\sqrt{x}}$.
- A quantity $(y - k)$ varies inversely as the square of x . Make y the subject of an equation in x , k and h , where k and h are constants.
- The electrical resistance R of a wire varies inversely as the square of the radius r . Use a constant k to show the relation between R and r .
- P varies inversely as the square root of v and $P = 4.5$ when $v = 25$. Find v when $P = 15$.

- 13 Sketch the following curves, showing both positive and negative values of x .

(a) $y \propto \frac{1}{x}$ (b) $y \propto \frac{1}{x^2}$ (c) $y \propto \frac{1}{\sqrt{x}}$

(Hint: For sketching purposes, it may help to substitute $=$ for $\propto$ and then plot some sample points.)

- 14 The variables X and Y are connected by the relation ' Y varies inversely as X '. Table 9.6 shows the values of Y for some selected values of X .

Table 9.6

X	10	20	30	40
Y	12	6	?	3

What is the missing value of Y ?

- 15 The length of wire that can be made from a mass of copper is inversely proportional to the square of the diameter of the wire. When the diameter is 3 mm the length of the wire is 1.8 km. Find the length of the wire when its diameter is 1.2 mm.

Joint variation

The mass, M , of a coin of radius r and thickness h depends on the volume, V , of metal in the coin, i.e. $M \propto V$

or $M \propto \pi r^2 h$ (since $V = \pi r^2 h$)

or $M \propto r^2 h$ (since π is a constant)

$M \propto r^2 h$ means that the mass of the coin **varies jointly** with the square of the radius and the thickness. This is an example of **joint variation**.

Example 7

The mass of a wire varies jointly with its length and the square of its diameter. 500 m of wire of diameter 3 mm has a mass of 31.5 kg. What is the mass of 1 km of wire of diameter 2 mm?

Let M = mass in kg, d = diameter in mm and L = length in m.

Then, from the first sentence,

$$M \propto Ld^2$$

or $M = kLd^2$, where k is a constant.

From the second sentence,

$$31.5 = k \times 500 \times 3^2 \quad (1)$$

From the third sentence,

$$M = k \times 1000 \times 2^2 \quad (2)$$

Dividing (2) by (1),

$$\frac{M}{31.5} = \frac{1000 \times 2^2}{500 \times 3^2} = \frac{2 \times 4}{9} = \frac{8}{9}$$

$$M = \frac{8}{9} \times 31.5 = 28$$

Hence the mass is 28 kg.

Notice that it was not necessary to find k . Notice also that it is possible to mix dimensions, such as mm and m, so long as this is done consistently.

Example 8

If $X \propto YZ^{\frac{1}{2}}$ and $Y \propto Z^{-2}$ show that $X \propto Y^{\frac{5}{6}}$.

If $Y \propto Z^{-2}$

then $Y = \frac{k}{Z^2}$ where k is a constant.

$$\Rightarrow Z^2 = \frac{k}{Y}$$

$$\Rightarrow Z = \sqrt{\frac{k}{Y}}$$

Given that $X \propto YZ^{\frac{1}{2}}$ substitute $\sqrt{\frac{k}{Y}}$ for Z .

$$X \propto Y \left(\sqrt{\frac{k}{Y}} \right)^{\frac{1}{2}}$$

$$\Rightarrow X \propto Y \left(\frac{k}{Y} \right)^{\frac{1}{4}}$$

$$\Rightarrow X \propto Y \times \frac{k^{\frac{1}{4}}}{Y^{\frac{1}{4}}}$$

$$\Rightarrow X \propto k^{\frac{1}{4}} Y^{\frac{3}{4}}$$

$\Rightarrow X \propto Y^{\frac{3}{4}}$ since $k^{\frac{1}{4}}$ is a constant.

The method in Example 8 is to eliminate the variable which is not required.

Exercise 9d

- $x \propto yz$. When $y = 2$ and $z = 3$, $x = 30$.
(a) Find the relation between x , y and z .
(b) Find x when $y = 4$ and $z = 6$.
- $x \propto \frac{y}{z}$. $x = 27$ when $y = 9$ and $z = 2$.
(a) Find the relation between x , y and z .
(b) Find x when $y = 14$ and $z = 12$.
- $p \propto \frac{q}{r^2}$ and $p = 3\frac{1}{3}$ when $q = 5$ and $r = 3$.
(a) Find the equation connecting p , q and r .
(b) Find p when $q = 9$ and $r = 1.2$.
- The height (h) of a cone varies directly as its volume (V) and inversely as the square of its radius (r). Use the constant k to show the relationship between h , V and r .
- $A \propto BC$ and $A = 6$ when $B = 4$ and $C = 9$.
(a) Find A when $B = 3$ and $C = 10$.
(b) Find C if $A = 20$ and $B = 15$.
(c) By what percentage does A change if B is increased by 10% and C is decreased by 10%?
- x varies directly with the square of y and with z . When $y = 2$ and $z = 3$, $x = 4\frac{1}{2}$.
(a) Find x when $y = 5$ and $z = 4$.
(b) Find y when $x = 21$ and $z = 3\frac{1}{2}$.
(c) What happens to x if y is doubled and z halved?
- x , y and z are related quantities such that x varies directly as y and inversely as the square root of z . When $x = 300$ and $y = 65$, $z = 25$. Calculate the value of x when $y = 468$ and $z = 144$.
- If $P \propto \frac{1}{V}$ and $V \propto R^2$, how does P vary with R ?
- $x \propto y$ and $y \propto z^3$. How does x vary with z ?
- $x \propto y^2$ and $y \propto z^2$. How does x vary with z ?
- $A \propto BC$ and $B \propto \frac{1}{C^2}$. How does A vary with C ?
- y varies directly as x and inversely as z . x varies inversely as y^2 . Prove that z^2 varies directly as x^3 .
- z varies directly as $\frac{x}{y^2}$ and y varies inversely as x . If $z = \frac{1}{3}$ when $x = 2$ and $y = \frac{1}{4}$, express

(a) y in terms of x ,

(b) z in terms of x .

If the value of x is increased by 10%, find the corresponding increase in the value of z .

- The mass, m , of a roller varies jointly with its length, L , and the square of its diameter, d . A roller of diameter 20 mm is 5 cm long and has a mass of 0.29 kg. Calculate the mass, in kg, of a roller 30 mm in diameter and 2 m long.
- Given that the energy E varies directly as the resistance R and inversely as the square of the distance d , obtain an equation connecting E , R and d .

If $E = \frac{32}{25}$ when $R = 8$ and $d = 5$, calculate

- the value of R when $E = 16$ and $d = 3$;
- the value of d when $R = 5$ and $E = \frac{5}{6}$;
- the percentage increase in the value of R when each of E and d increases by 3%.

Partial variation

When feeding a large number of people, as in a hotel, the total cost depends on two separate factors: first the cost of the overheads (such as fuel and wages), secondly the cost of the food used.

The cost of the overheads generally remains constant, but the cost of the food is proportional to the number of people being fed. Hence the total cost is **partly constant and partly varies** as the number of people.

In algebraic terms, $C = a + kN$, where C is the total cost, N the number of people and a and k are constants. This is an example of **partial variation**.

Example 9

C is partly constant and partly varies as N . $C = 45$ when $N = 10$ and $C = 87$ when $N = 24$. (a) Find the formula connecting C and N . (b) Find C when $N = 18$.

(a) From the first sentence,

$C = a + kN$ where a and k are constants.

From the second sentence,

$$45 = a + 10k \quad (1)$$

$$\text{and } 87 = a + 24k \quad (2)$$

Subtract (1) from (2),

$$42 = 14k$$

$$k = \frac{42}{14} = 3$$

Substitute 3 for k in (1),

$$45 = a + 30$$

$$a = 15$$

Hence $C = 15 + 3N$ is the required formula.

(b) When $N = 18$,

$$C = 15 + 3 \times 18$$

$$= 15 + 54$$

$$= 69$$

Example 10

The resistance to motion of a car is partly constant and partly varies as the square of the speed. At 40 km/h the resistance is 530 N, and at 60 km/h it is 730 N. What will be the resistance at 70 km/h?

Let R = resistance in newtons, V = speed in km/h, then, from the first sentence,

$$R = a + kV^2,$$

where a and k are constants.

From the second sentence,

$$530 = a + 1600k \quad (1)$$

$$\text{and } 730 = a + 3600k \quad (2)$$

Subtract (1) from (2),

$$200 = 2000k$$

$$k = \frac{1}{10}$$

Substituting in (1),

$$530 = a + 1600 \times \frac{1}{10}$$

$$= a + 160$$

$$a = 370$$

$$\text{Hence } R = 370 + \frac{1}{10}V^2$$

When $V = 70$

$$R = 370 + \frac{1}{10} \times 4900$$

$$= 370 + 490$$

$$= 860$$

Hence the resistance is 860 N.

Notice in Examples 9 and 10, since there are two unknowns, two equations must be formed. These are then solved simultaneously.

Exercise 9e

- x is partly constant and partly varies as y . When $y = 2$, $x = 30$, and when $y = 6$, $x = 50$. (a) Find the relationship between x and y . (b) Find x when $y = 3$.
- x is partly constant and partly varies with y . When $y = 3$, $x = 11$, and when $y = 4$, $x = 14$. (a) Find the relationship between x and y . (b) Find x when $y = 10$.
- x is partly constant and partly varies with y . When $y = 3$, $x = 7$, and when $y = 6$, $x = 9$. (a) Find the relationship between x and y . (b) Find x when $y = 4$.
- D is partly constant and partly varies with V . When $V = 40$, $D = 150$, and when $V = 54$, $D = 192$. (a) Find the formula connecting D and V . (b) Find D when $V = 73$.
- The cost of making a dress is partly constant and partly varies with the amount of time it takes to make. If it takes 3 hours to make, it costs \$40. If it takes 5 hours to make, it costs \$46. Find the cost if it takes $1\frac{1}{2}$ hours.
- A varies partly as B and partly as the square root of B . When $B = 4$, $A = 22$ and when $B = 9$, $A = 42$. Find A when $B = 25$.
- Table 9.7 is an incomplete table for the relation $y = \frac{1}{2}x^2 + k$, where k is a constant.

Table 9.7

x	-3	-2	1	2	3	4
y	6.5	4	2.5	4	6.5	

- (a) What is the value of k ?
(b) Find the value of y at $x = 4$.
- Two quantities, P and Q , are connected by a linear relation of the form $P = kQ + c$ where k and c are constants.
(a) If $Q = 60$ when $P = 10$, and $Q = 240$ when $P = 100$, find the equation connecting P and Q .
(b) Sketch the graph of the relation, indicating where it cuts the two axes.
- The fixed costs of a manufacturing business are \$30 000. The variable costs, which are proportional to the sales, are \$40 000 when the sales are \$80 000. Calculate the total costs and the profit when the sales are \$90 000.

- 10 The cost of feeding a number of students is partly constant and partly varies directly as the number of students. Feeding 75 students during a certain period costs \$875 and feeding 100 students during the same period of time costs \$1000. Find the cost of feeding 220 students over the same period of time.
- 11 The cost of making computers is partly constant and partly varies as the number of computers produced. The total cost of making 4 computers is \$18 700 and of making 10 computers is \$35 500. Find the total cost of making 20 computers.
- 12 The resistance R to the motion of a car is partly constant and partly proportional to the square of the speed v . When the speed is 30 km/h the resistance is 190 newtons and when the speed is 50 km/h the resistance is 350 newtons. Find for what speed the resistance is 302,5 newtons.
- 13 The diameter of a reel of sticky tape is partly constant and partly varies as the square root of the length of tape on the reel. When new, the reel contains 250 m of tape and is 8 cm in diameter. When all the tape has been used, the diameter of the empty reel is 2 cm. What length of tape is on the reel when its diameter is 5,6 cm?
- 14 The cost of running a hotel is partly constant and partly varies as the square of the number of people staying in the hotel. Accommodating 5 people costs \$200 and the cost for 8 people is \$395. Calculate the cost of accommodating 12 people in that hotel.
- 15 n is the algebraic sum of two terms, one of which varies directly as u and the other inversely as u^2 . If $n = 11$ when $u = 2$ and $n = 25,16$ when $u = 5$, calculate n when $u = 3$.

Mensuration of solid shapes

Surface area and volume of solids

Formulae for the areas and volumes of common solids already found in earlier books of this course are given below.

Prisms

In general,
 volume = area of constant cross-section
 $\times$ perpendicular height
 = area of base $\times$ height (Fig. 10.1)

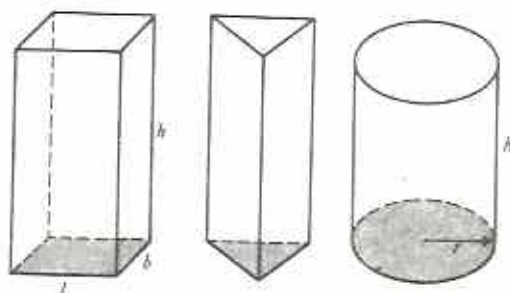


Fig. 10.1

cuboid

$$\begin{aligned}\text{volume} &= lbh \\ \text{surface area} &= 2(lb + lh + bh)\end{aligned}$$

cylinder

$$\begin{aligned}\text{volume} &= \pi r^2 h \\ \text{curved surface area} &= 2\pi rh \\ \text{total surface area} &= 2\pi rh + 2\pi r^2 \\ &= 2\pi r(h + r)\end{aligned}$$

Pyramid and cone

In general
 volume = $\frac{1}{3} \times$ base area $\times$ height (Fig. 10.2)

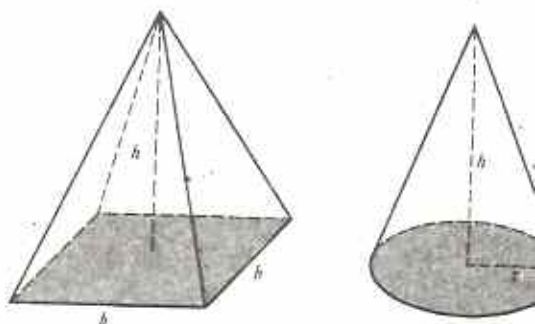


Fig. 10.2

square-based pyramid

$$\text{volume} = \frac{1}{3}b^2h$$

cone

$$\begin{aligned}\text{volume} &= \frac{1}{3}\pi r^2 h \\ \text{curved surface area} &= \pi rl \\ \text{total surface area} &= \pi rl + \pi r^2 \\ &= \pi r(l + r)\end{aligned}$$

Example 1

A car petrol tank is 0.8 m long, 25 cm wide and 20 cm deep. How many litres of petrol can it hold?

Working in cm,

$$\text{volume of tank} = 80 \times 25 \times 20 \text{ cm}^3$$

$$1 \text{ litre} = 1000 \text{ cm}^3$$

$$\begin{aligned}\text{capacity of tank} &= \frac{80 \times 25 \times 20}{1000} \text{ litres} \\ &= 40 \text{ litres}\end{aligned}$$

The tank can hold 40 litres of petrol.

Example 2

A circular metal sheet 48 cm in diameter and 2 mm thick is melted and recast into a cylindrical bar 6 cm in diameter. How long is the bar?

$$\text{Radius of sheet} = \frac{48}{2} \text{ cm} = 24 \text{ cm}$$

Radius of bar = $\frac{6}{2}$ cm = 3 cm

Let the bar be x cm long.

Then its volume = $\pi \times 3^2 \times x$ cm³

Volume of circular sheet = $\pi \times 24^2 \times \frac{1}{5}$ cm³

Hence $\pi \times 3^2 \times x = \pi \times 24^2 \times \frac{1}{5}$

$$\begin{aligned} \Rightarrow x &= \frac{\pi \times 24^2 \times \frac{1}{5}}{\pi \times 3^2} \\ &= \frac{576}{9 \times 5} \\ &= \frac{64}{5} = 12,8 \end{aligned}$$

The bar is 12,8 cm long.

Notice in Example 2 that no numerical value of π was needed. Never substitute a value for π unless it is necessary.

Example 3

A 216° sector of a circle of radius 5 cm is bent to form a cone. Find the radius of the base of the cone and its vertical angle.

In Fig. 10.3, the radius of the base of the cone is r cm and the vertical angle is 2α .

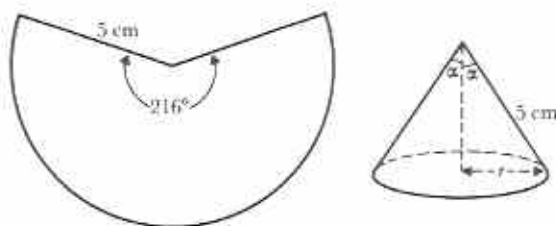


Fig. 10.3

circumference of base of cone
= length of arc of sector

$$\Rightarrow 2\pi r = \frac{216}{360} \times 2\pi \times 5$$

$$\Leftrightarrow r = \frac{216}{360} \times 5 = 3$$

$$\sin \alpha = \frac{3}{5} = 0,6000$$

$$\Rightarrow \alpha = 36,87^\circ$$

$$\Rightarrow 2\alpha = 73,74^\circ$$

Radius of base = 3 cm

Vertical angle = 73,7° (to 0,1°)

Example 4

Fig. 10.4 shows a wooden block in the form of a prism. PQRS is a trapezium with $PQ \parallel SR$, $PQ = 7$ cm, $PS = 5$ cm and $SR = 4$ cm. If the block is 12 cm long, calculate its volume.

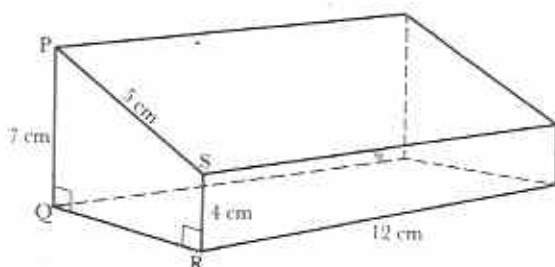


Fig. 10.4

Volume of block = area of PQRS $\times$ 12 cm³

$$\text{area of PQRS} = \frac{1}{2}(7 + 4) \times QR \text{ cm}^2$$

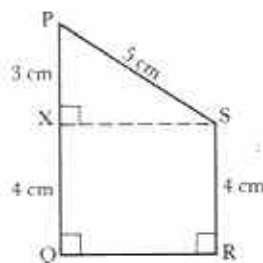


Fig. 10.5

With the construction of Fig. 10.5,

$$SX = QR$$

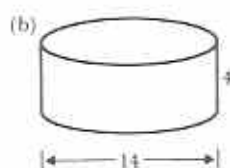
But $SX = 4$ cm (the sides of $\triangle PXS$ form a 3; 4; 5 Pythagorean triple)

$$\begin{aligned} \text{Volume of block} &= \frac{1}{2}(7 + 4) \times 4 \times 12 \text{ cm}^3 \\ &= 11 \times 2 \times 12 \text{ cm}^3 \\ &= 264 \text{ cm}^3 \end{aligned}$$

Exercise 10a

Use the value $3\frac{1}{7}$ for π where necessary.

1 Calculate the volumes of the solids in Fig. 10.6(a) – (f). All lengths are in cm.



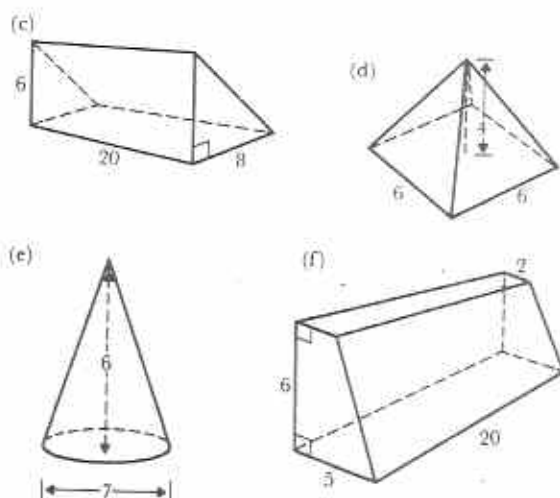


Fig. 10.6

- 2 Calculate the total surface areas of the solids in parts (a), (b), (c) of Fig. 10.6.
- 3 A rectangular tank is 76 cm long, 50 cm wide and 40 cm high. How many litres of water can it hold?
- 4 A water tank is 1.2 m square and 1.35 m deep. It is half full of water. How many times can a 9-litre bucket be filled from the tank?
- 5 $2\frac{1}{2}$ litres of oil are poured into a container whose cross-section is a square of side $12\frac{1}{2}$ cm. How deep is the oil in the container?
- 6 The diagrams in Fig. 10.7 show the cross-sections of steel beams. All dimensions are in cm. Calculate the volumes, in cm^3 , of 5-metre lengths of the beams.

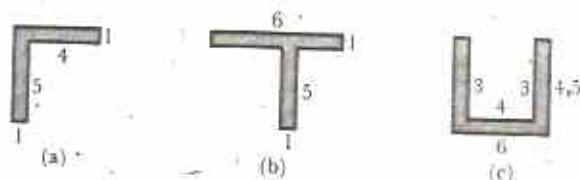


Fig. 10.7

- 7 Fig. 10.8 shows the cross-section of a steel rail, dimensions being given in cm. Calculate the mass, in tonnes, of a 20-metre length of the rail if the mass of 1 cm^3 of the steel is 7.5 g.

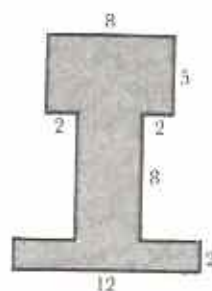


Fig. 10.8

- 8 Fig. 10.9 shows the cross-section of a ruler. (a) Calculate the volume of the ruler in cm^3 if it is 30 cm long. (b) If the ruler is made of plastic and has a mass of 45 g, what is the density of the plastic in g/cm^3 ? (c) Find the mass, to the nearest g, of the ruler if it is made of wood of density 0.7 g/cm^3 .

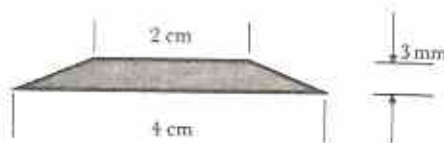


Fig. 10.9

- 9 Calculate, in terms of π , the total surface area of a solid cylinder of radius 3 cm and height 4 cm.
- 10 A paper label just covers the curved surface of a cylindrical tin of diameter 12 cm and height $10\frac{1}{2}$ cm. Calculate the area of the paper label.
- 11 A cylindrical tin full of engine oil has a diameter of 12 cm and a height of 14 cm. The oil is poured into a rectangular tin 16 cm long and 11 cm wide. What is the depth of the oil in the tin?
- 12 A cylindrical shoe polish tin is 10 cm in diameter and 3.5 cm deep. (a) Calculate the capacity of the tin in cm^3 . (b) When full, the tin contains 300 g of polish. Calculate the density of the polish in g/cm^3 correct to 2 d.p.
- 13 A wire of circular cross-section has a diameter of 2 mm and a length of 350 m. If the mass of the wire is 6.82 kg, calculate its density in g/cm^3 .

- 14 A measuring cylinder of radius 3 cm contains water to a height of 49 cm. If this water is poured into a similar cylinder of radius 7 cm, what will be the height of the water column?
- 15 A metal disc 12 cm in diameter and 5 cm thick is melted down and cast into a cylindrical bar of diameter 5 cm. How long is the bar?
- 16 A solid metal cylinder, 8 cm in diameter and 8 cm long, is to be made into discs 4 cm in diameter and 5 mm thick. Assuming no wastage, how many discs can be made?
- 17 Water flows through a 7-cm diameter pipe at the rate of 4 metres/second.
- (a) How many cm^3 of water flow through the pipe in one second?
- (b) Express the flow of water as a rate in litres/minute.
- 18 How many cylindrical glasses 6 cm in diameter and 10 cm deep can be filled from a cylindrical jug 10 cm in diameter and 18 cm deep?
- 19 A cylindrical container 30 cm in diameter holds approximately 30 litres of oil. How far does the oil level fall after 1 litre of oil has been used?
- 20 A pyramid 8 cm high stands on a rectangular base 6 cm by 4 cm. Calculate the volume of the pyramid.
- 21 A paper cone has a base diameter of 8 cm and a height of 3 cm.
- (a) Calculate the volume of the cone in terms of π .
- (b) Make a sketch of the cone and hence use Pythagoras' theorem to calculate its slant height.
- (c) Calculate the curved surface area of the cone in terms of π .
- (d) If the cone is cut and opened out into a sector of a circle, what is the angle of the sector?
- 22 Calculate the volume and curved surface area of a cone 14 cm in base diameter and 24 cm high.
- 23 If the cone in question 22 is made of paper, and the paper is flattened out into a sector of a circle, what is the angle of the sector?

- 24 Fig. 10.10 shows a cross-section of a dam wall. How many m^3 of concrete will it take to build a 100-m length of this wall?

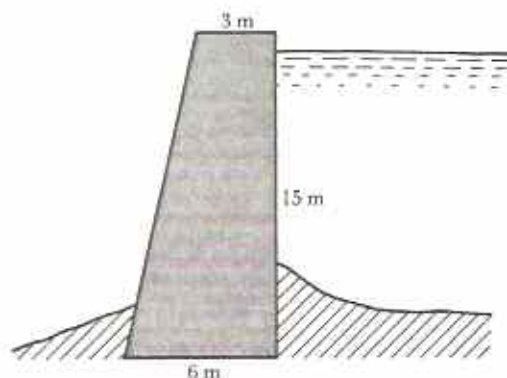


Fig. 10.10

- 25 A cone of height 9 cm has a volume of $n \text{ cm}^3$ and a curved surface area of $n \text{ cm}^2$. Find the vertical angle of the cone.

Sphere

- Fig. 10.11 represents a solid sphere of radius r .
- volume = $\frac{4}{3}\pi r^3$
- curved surface area = $4\pi r^2$

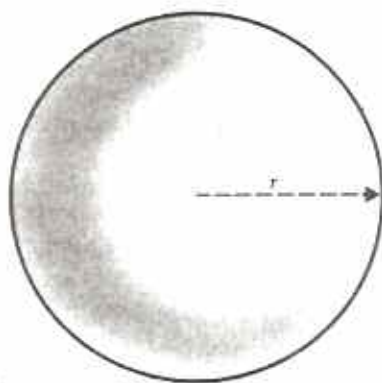


Fig. 10.11

(The proof of these formulae is beyond the scope of this course.)

Example 5

A solid sphere has a radius of 5 cm and is made of metal of density 7.2 g/cm^3 . Calculate the mass of the sphere in kg.

$$\begin{aligned}\text{Volume of sphere} &= \frac{4}{3}\pi \times 5^3 \text{ cm}^3 \\ &= \frac{4 \times \pi \times 125}{3} \text{ cm}^3 \\ &= \frac{500\pi}{3} \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Mass of sphere} &= \frac{500\pi}{3} \times 7,2 \text{ g} \\ &= \frac{500\pi \times 7,2}{3 \times 1000} \text{ kg} = \frac{7,2\pi}{3 \times 2} \text{ kg} \\ &= 1,2\pi = 1,2 \times 3,142 \\ &= 3,7704 \text{ kg} \\ &= 3,77 \text{ kg to 3 s.f.}\end{aligned}$$

Example 6

Calculate the total surface area of a solid hemisphere of radius 6,8 cm. Use the value 0,4971 for $\log \pi$.



Fig. 10.12

In Fig. 10.12, total surface area
= curved surface area + plane surface area
= $2\pi r^2 + \pi r^2$
= $3\pi r^2$

When $r = 6,8$
total surface area
= $3 \times \pi \times 6,8^2 \text{ cm}^2$
= $435,7 \text{ cm}^2$
= 436 cm^2 to 3 s.f.

working:

No	Log
6,8 ²	0,8325 $\times 2$
	= 1,6650
π	0,4971
3	0,4771
435,7	2,6392

Exercise 10b

Use the value 3,142 for π or 0,4971 for $\log \pi$, whichever is more convenient.

- Calculate the volume and surface area to 3 s.f. of each of the following.
 - A sphere, radius 10 cm
 - A sphere, diameter 16 cm
 - A hemisphere, radius 2 cm
 - A hemisphere, diameter 9 cm

- The diameter of an iron ball used in 'putting the shot' is 12 cm. If the density of iron is $7,8 \text{ g/cm}^3$, calculate the mass of the ball in kg to 3 s.f.
- A cylinder and sphere both have the same diameter and the same volume. If the height of the cylinder is 36 cm, find their common radius.
- A metal sphere 6 cm in diameter is melted and cast into balls of diameter $\frac{1}{2}$ cm. How many of the smaller balls will there be?
- A sphere has a volume of 1000 cm^3 .
 - Use tables to calculate its radius correct to 3 s.f.
 - Hence calculate the surface area of the sphere.

Addition and subtraction of volumes

Many **composite solids** can be made by joining basic solids together.

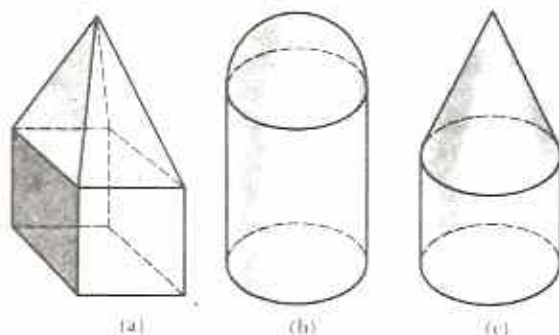


Fig. 10.13

In Fig. 10.13, the composite solids are made as follows:

- a cube and a square-based pyramid.
- a cylinder and a hemisphere,
- a cylinder and a cone.

Example 7

Fig. 10.14 represents a gas tank in the shape of a cylinder with a hemispherical top. The internal height and diameter are 1 m and 30 cm respectively. Calculate the capacity of the tank to the nearest litre.



Fig. 10.14

With the lettering of Fig. 10.14,
volume of tank

$$= \text{volume of cylinder} \\ + \text{volume of hemisphere}$$

$$= \pi r^2 h + \frac{2}{3} \pi r^3 = \pi r^2 \left(h + \frac{2}{3} r \right)$$

In Fig. 10.14

$$r = 15$$

$$h = 100 - 15 = 85$$

volume of tank

$$= \pi 15^2 \left(85 + \frac{2}{3} \times 15 \right) \text{ cm}^3$$

$$= 225\pi (85 + 10) \text{ cm}^3$$

$$= 225\pi \times 95 \text{ cm}^3$$

capacity in litres

$$= \frac{225 \times \pi \times 95}{1000}$$

$$= 67.14 = 67 \text{ to the nearest litre}$$

working:

No.	Log
225	2.3522
π	0.4971
95	1.9777
	4.8270
1000	3.0000
67.14	1.8270

Hollow shapes, such as boxes and pipes, have space inside.

The volume of material in a hollow object is found by subtracting the volume of the space inside from the volume of the shape as if it were solid.

Example 8

Fig. 10.15 represents an open rectangular box made of wood 1 cm thick. If the external dimensions of the box are 42 cm long, 32 cm wide and 15 cm deep, calculate the volume of wood in the box.



Fig. 10.15

The internal measurements of the box are 40 cm long, 30 cm wide and 14 cm deep.

$$\text{External volume} = 42 \times 32 \times 15 \text{ cm}^3 \\ = 20160 \text{ cm}^3$$

$$\text{Internal volume} = 40 \times 30 \times 14 \text{ cm}^3 \\ = 16800 \text{ cm}^3$$

$$\text{Volume of wood} = 20160 \text{ cm}^3 - 16800 \text{ cm}^3 \\ = 3360 \text{ cm}^3$$

Example 9

Find the mass of a cylindrical iron pipe 2.1 m long and 12 cm in external diameter, if the metal is $\frac{1}{4}$ cm thick and of density 7.8 g/cm³. Take π to be $\frac{22}{7}$.

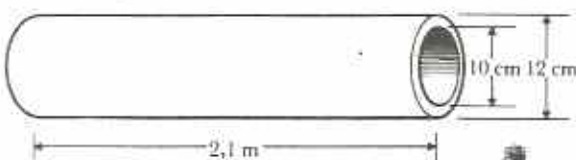


Fig. 10.16

$$\text{Volume of outside cylinder} = \pi \times 6^2 \times 210 \text{ cm}^3$$

$$\text{Volume of inside cylinder} = \pi \times 5^2 \times 210 \text{ cm}^3$$

Volume of iron

$$= \pi \times 6^2 \times 210 - \pi \times 5^2 \times 210 \text{ cm}^3$$

$$= 210\pi (6^2 - 5^2) \text{ cm}^3$$

$$= 210\pi (6 + 5)(6 - 5) \text{ cm}^3$$

$$= 210\pi \times 11 \text{ cm}^3$$

Mass of iron

$$= 210\pi \times 11 \times 7.8 \text{ g}$$

$$= \frac{210 \times 22 \times 11 \times 7.8}{7 \times 1000} \text{ kg}$$

$$= 56.6 \text{ kg to 3 s.f.}$$

(The working of the final line should be checked using tables.)

Example 10

Calculate, to one place of decimals, the volume in cm³ of the metal in a hollow sphere 10 cm in external diameter, the metal being 1 mm thick.

$$\text{Outer radius} = 5 \text{ cm}$$

$$\text{Inner radius} = 4.9 \text{ cm}$$

Volume of metal

$$= \frac{4}{3} \pi \times 5^3 - \frac{4}{3} \pi \times 4.9^3 \text{ cm}^3$$

$$= \frac{4}{3} \pi (125 - 4.9^3) \text{ cm}^3$$

$$= \frac{4}{3} \pi (125 - 117.7) \text{ cm}^3$$

working:

No.	Log
4.9 ³	0.6902
117.7	2.0706

$$= \frac{4 \times \pi \times 7.3}{3} \text{ cm}^3$$

$$= \frac{29.2\pi}{3} \text{ cm}^3$$

$$= 30.6 \text{ cm}^3 \text{ to 1 d.p.}$$

No.	Log
29.2	1.4654
π	0.4971
	1.9625
3	0.4771
30.58	1.4854

* This can be done on a calculator by treating it as $1.333 \times 3.142 \times (125 \div 4.9^3)$:

Key

AC

Display

0

4 $\div$ 9 $\times$ $\times$ $\times$ $=$

117.649

Min

117.649

1 2 5 $\div$ MR $=$

7.351

$\times$ 3 $\div$ 1 4 2 $=$

3.142

$\times$ 1 $\div$ 3 3 3 $=$

30.78809

$$\text{volume} = 30.8 \text{ cm}^3 \quad (3)$$

(1) Notice how 4.9^3 is done on the calculator.

(2) Many calculators have a π button which automatically enters the value of π .

(3) The calculator value (30.8) is more accurate than the tables value (30.6). This is because the intermediate value in the tables calculation (7.3) was slightly inaccurate.

In Examples 7, 8, 9, 10, notice how factorisation simplifies the calculation.

Example 11

A right circular cylinder of height 12 cm and radius 4 cm is filled with water. A heavy circular cone of height 9 cm and base-radius 6 cm is lowered, with vertex downwards and axis vertical, into the cylinder until the cone rests on the rim of the cylinder. Find (a) the volume of water which spills over from the cylinder, and (b) the height of the water in the cylinder after the cone has been removed.

Fig. 10.17 shows the position of the cone and cylinder.

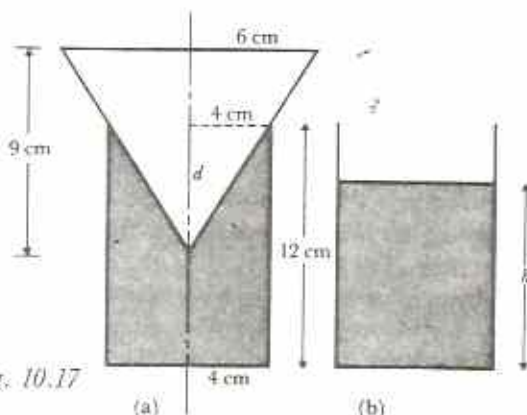


Fig. 10.17

(a) Let the cone be immersed to a depth d cm. By similar triangles,

$$\frac{d}{4} = \frac{9}{6}$$

$$d = \frac{9 \times 4}{6} = 6$$

$$\begin{aligned} \text{Volume of water which spills over} \\ &= \text{volume displaced by end of cone} \\ &= \frac{1}{3} \pi \times 4^2 \times d \text{ cm}^3 = \frac{1}{3} \times \frac{22}{7} \times 16 \times 6 \text{ cm}^3 \\ &= \frac{704}{7} \text{ cm}^3 = 100\frac{4}{7} \text{ cm}^3 \end{aligned}$$

(b) Let the height of the water after the cone has been removed be h cm.

$$\begin{aligned} \text{Volume of water in Fig. 10.17(a)} \\ &= \text{volume of water in Fig. 10.17(b)} \\ \pi \times 4^2 \times 12 - \frac{1}{3} \times \pi \times 4^2 \times 6 &= \pi \times 4^2 \times h \\ \Leftrightarrow 12 - \frac{1}{3} \times 6 &= h \\ \Leftrightarrow h &= 12 - 2 = 10 \end{aligned}$$

The height of the water will be 10 cm.

Exercise 10c

Use the value $\frac{22}{7}$ for π or 0.4971 for $\log \pi$, whichever is more convenient.

1 An open rectangular box has internal dimensions 2 m long, 20 cm wide and 22.5 cm deep. If the box is made of wood 2.5 cm thick, find the volume of the wood in cm^3 .

2 An open concrete tank is internally 1 m wide, 2 m long and 1.5 m deep, the concrete being 10 cm thick. Calculate

- (a) the capacity of the tank in litres, and
(b) the volume of concrete used in m^3 .

- 3 Fig. 10.18 shows the plan of a foundation which is of uniform width 1 m.

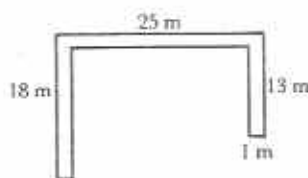


Fig. 10.18

If 1 m^3 of earth has a mass of $1\frac{1}{2}$ tonnes, what mass of earth will be removed when digging the foundation to a depth of $1\frac{1}{2}$ m?

- 4 A cast iron pipe has a cross-section as shown in Fig. 10.19, the iron being 1 cm thick. The mass of 1 cm^3 of cast iron is 7.2 g. Calculate the mass of a 2-metre length of the pipe.

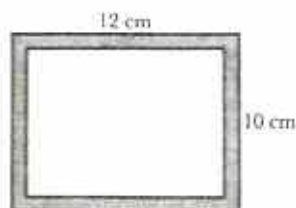


Fig. 10.19

- 5 A fish tank is in the shape of an open glass cuboid 30 cm deep with a base of 16 cm by 17 cm, these measurements being external. If the glass is 0.5 cm thick and its mass is 3 g/cm^3 , find (a) the capacity of the tank in litres, and (b) the mass of the tank in kg.
- 6 Fig. 10.20 shows a storage tank made from a cylinder with a hemispherical end. Use the dimensions in Fig. 10.20 to calculate the volume of the tank in m^3 .

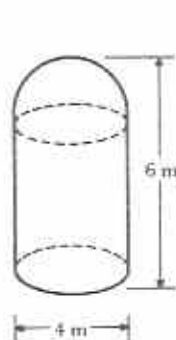


Fig. 10.20

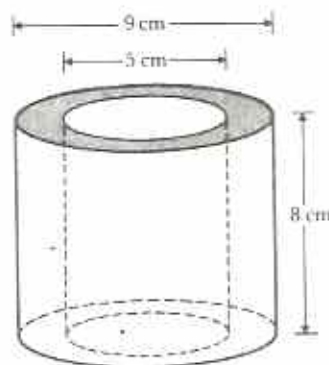


Fig. 10.21

- 7 Fig. 10.21 shows a cylindrical casting of height 8 cm and external and internal diameters 9 cm and 5 cm respectively. Calculate the volume of metal in the casting.
- 8 The outer radius of a cylindrical metal tube is R and t is the thickness of the metal.
- (a) Show that the volume, V , of metal in a length, l units, of the tube is given by $V = \pi l(2R - t)$.
- (b) Hence calculate V when $R = 7.5$, $t = 1$ and $l = 20$.
- 9 A rectangular box, 18 cm by 12 cm by 6 cm, contains six tennis balls, each of diameter 6 cm. Calculate the percentage of the volume of the box occupied by the tennis balls.
- 10 Calculate the volume in cm^3 of the material in a cylindrical pipe 1.8 m long, the internal and external diameters being 16 cm and 18 cm respectively.
- 11 Calculate the approximate mass in kg of a 2-m length of cylindrical clay pipe of external and internal diameters 15 cm and 12 cm. The density of clay is 1.3 g/cm^3 .
- 12 How far does the water level drop in a cylindrical tank of internal diameter 35 cm if 11 litres are drawn off?
- 13 A conical funnel 12 cm deep and 15 cm in diameter is full of liquid. It is emptied into a cylindrical tin 10 cm in diameter. Calculate the height of the liquid in the tin.
- 14 A spherical container 15 cm in diameter is half full of acid. The acid is poured into a tall cylindrical beaker of diameter 6 cm. How deep is the acid in the beaker?
- 15 A cylindrical tin of internal diameter 8 cm contains water to a depth of 6 cm. How far does the water level rise when a heavy ball of diameter 6 cm is placed in the tin?
- 16 A solid cube of side 8 cm is dropped into a cylindrical tank of radius 7 cm. Calculate the rise in the water level if the original depth of water was 9 cm.
- 17 An iron ball, 6 cm in diameter, is placed in a cylindrical tin 12 cm in diameter. Water is poured into the tin until its depth is 8 cm. If the ball is now removed, how far does the water level drop?
- 18 If, instead of the ball in question 17, an iron rod 8 cm long and 6 cm in diameter had

been placed flat in the tin, how far would the level have dropped when the rod was removed? (Assume that the other conditions remained the same.)

- 19 A wooden bowl is in the shape of a hollow hemisphere of external diameter 20 cm. The wood is 1 cm thick. Find the mass of the bowl if the wood has a density of 0.74 g/cm^3 .
- 20 A solid aluminium casting for a pulley consists of 3 discs, each $1\frac{1}{2} \text{ cm}$ thick, of diameters 4 cm, 6 cm and 8 cm. A central hole 2 cm in diameter is drilled out as in Fig. 10.22. If the density of aluminium is 2.8 g/cm^3 , calculate the mass of the casting.

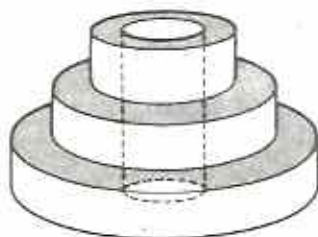


Fig. 10.22

Frustum of a cone or pyramid

If a cone or pyramid standing on a horizontal table is cut through parallel to the table, the top part is a smaller cone or pyramid. The other part is called a **frustum** (Fig. 10.23).

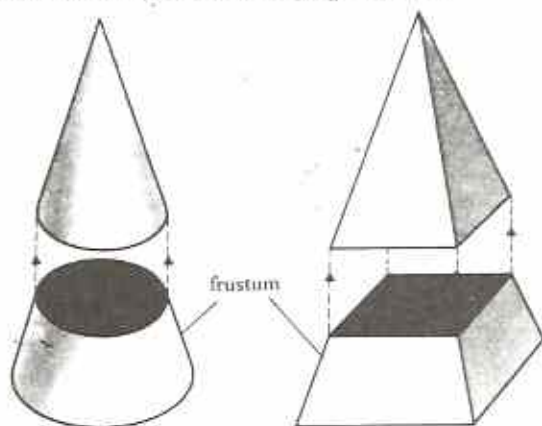


Fig. 10.23

To find the volume or surface area of a frustum, it is necessary to consider the frustum as a complete cone (or pyramid) with the smaller cone removed. Read Examples 12 and 13 carefully.

Example 12

Find the capacity in litres of a bucket 24 cm in diameter at the top, 16 cm in diameter at the bottom and 20 cm deep.

Complete the cone of which the bucket is a frustum, i.e. add a cone of height x cm and base diameter 16 cm as in Fig. 10.24.

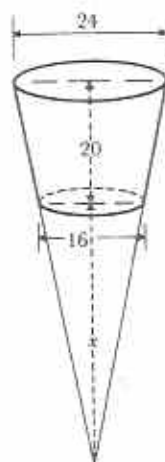


Fig. 10.24

By similar triangles,

$$\frac{x}{8} = \frac{x + 20}{12}$$

$$12x = 8x + 160$$

$$4x = 160$$

$$x = 40$$

Volume of bucket

$$= \frac{1}{3}\pi 12^2 \times 60 - \frac{1}{3}\pi 8^2 \times 40 \text{ cm}^3$$

$$= \frac{1}{3}\pi (8640 - 2560) \text{ cm}^3$$

$$= \frac{1}{3}\pi \times 6080 \text{ cm}^3 = 6366 \text{ cm}^3$$

Capacity of bucket = 6.37 litres to 3 s.f.

Example 13

Find, in cm^2 , the area of material required for a lampshade in the form of a frustum of a cone of which the top and bottom diameters are 20 cm and 30 cm respectively, and the vertical height is 12 cm.

Complete the cone of which the lampshade is a frustum as in Fig. 10.25.

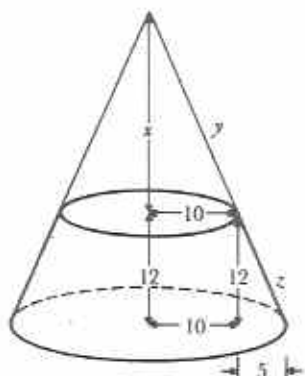


Fig. 10.25

With the lettering of Fig. 10.25, by similar triangles,

$$\frac{x}{10} = \frac{12}{5}$$

$$x = 24$$

By Pythagoras' theorem, $y = 26$ and $z = 13$

Surface area of frustum

$$= \pi \times 15 \times 26 - \pi \times 10 \times 13 \text{ cm}^2$$

$$= 13\pi(45 - 20) \text{ cm}^2$$

$$= 13\pi \times 25 \text{ cm}^2 = 1021 \text{ cm}^2$$

Area of material required = 1020 cm^2 to 3 s.f

Exercise 10d

- 1 A frustum of a cone has top and bottom diameters of 14 cm and 10 cm respectively and a depth of 6 cm. Find the volume of the frustum in terms of π .
- 2 A right pyramid on a base 10 m square is 15 m high.
 - (a) Find the volume of the pyramid.
 - (b) If the top 6 m of the pyramid are removed, what is the volume of the remaining frustum?
- 3 A frustum of a pyramid is 16 cm square at the bottom, 6 cm square at the top, and 12 cm high. Find the volume of the frustum.
- 4 A lampshade like that of Fig. 10.25 has a height of 12 cm and upper and lower diameters of 10 cm and 20 cm.
 - (a) What area of material is required to cover the curved surface of the frustum?
 - (b) What is the volume of the frustum? (Give both answers in terms of π .)
- 5 The volume of a right circular cone is 5 litres. Calculate the volumes of the two parts into which the cone is divided by a plane parallel

to the base, one-third of the way down from the vertex to the base. Give your answers to the nearest cm^3 .

- 6 A storage container is in the form of a frustum of a right pyramid 4 m square at the top and 2.5 m square at the bottom. If the container is 3 m deep, what is its capacity in m^3 ?
- 7 The cone in Fig. 10.26 is exactly half full of water by volume. How deep is the water in the cone?

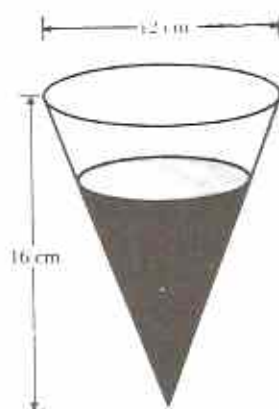


Fig. 10.26

- 8 A bucket is 20 cm in diameter at the open end, 12 cm in diameter at the bottom, and 16 cm deep. To what depth would the bucket fill a cylindrical tin 28 cm in diameter?



Fig. 10.27 This is one of the pyramids at Meroë in the Sudan. In what form is it?

The cosine rule

The cosine rule

Given: Any $\triangle ABC$ (acute-angled and obtuse-angled triangles are given in Fig. 11.1).

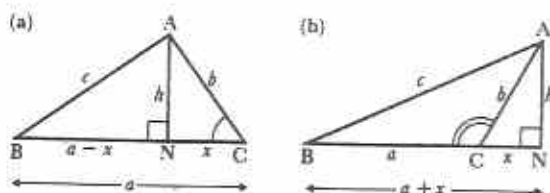


Fig. 11.1

To prove: $c^2 = a^2 + b^2 - 2ab \cos C$

Construction: Draw the perpendicular from A to BC (produced if necessary).

Proof:

In Fig. 11.1(a), with $\hat{C}$ acute,

$$c^2 = (a-x)^2 + h^2 \quad (\text{Pythagoras})$$

$$= a^2 - 2ax + x^2 + h^2$$

$$= a^2 - 2ax + b^2 \quad (\text{In } \triangle ACN, x^2 + h^2 = b^2)$$

$$= a^2 + b^2 - 2ab \cos C \quad (\text{In } \triangle ACN, \frac{x}{b} = \cos C,$$

$$x = b \cos C)$$

In Fig. 11.1(b), with $\hat{C}$ obtuse,

$$c^2 = (a+x)^2 + h^2 \quad (\text{Pythagoras})$$

$$= a^2 + 2ax + x^2 + h^2$$

$$= a^2 + 2ax + b^2 \quad (\text{In } \triangle ACN, x^2 + h^2 = b^2)$$

$$= a^2 + b^2 + 2a(-b \cos C)$$

$$(\text{In } \triangle ACN, \frac{x}{b} = \cos \hat{ACN}$$

$$= \cos(180^\circ - C)$$

$$= -\cos C$$

$$x = -b \cos C)$$

$$= a^2 + b^2 - 2ab \cos C$$

In either case, $c^2 = a^2 + b^2 - 2ab \cos C$

similarly, $b^2 = a^2 + c^2 - 2ac \cos B$

and, $a^2 = b^2 + c^2 - 2bc \cos A$

This formula is for solving triangles which are not right-angled in which **two sides and the included angle** are given.

Example 1

Find AB in Fig. 11.2.

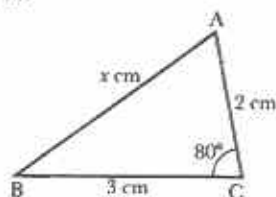


Fig. 11.2

By the cosine rule,

$$x^2 = 2^2 + 3^2 - 2 \times 2 \times 3 \times \cos 80^\circ$$

$$= 4 + 9 - 12 \times 0,1736$$

$$= 13 - 2,0832$$

$$= 10,9168 = 10,92 \text{ to 4 s.f.}$$

$$x = \sqrt{10,92}$$

$$x = 3,305 = 3,3 \text{ to 2 s.f.}$$

$$AB \approx 3,3 \text{ cm}$$

Example 2

Find y in Fig. 11.3.

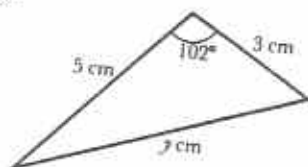


Fig. 11.3

$$y^2 = 5^2 + 3^2 - 2 \times 5 \times 3 \times \cos 102^\circ$$

$$= 25 + 9 - 30(-\cos 78^\circ)$$

$$= 34 + 30 \cos 78^\circ$$

$$= 34 + 30 \times 0,2079$$

$$= 34 + 6,237$$

$$= 40,237$$

$$= 40,24 \text{ to 4 s.f.}$$

$$y = \sqrt{40,24} = 6,343$$

$$= 6,34 \text{ to 3 s.f.}$$

Notice that in Example 1, $x^2 < 2^2 + 3^2$ and in Example 2, $y^2 > 5^2 + 3^2$.

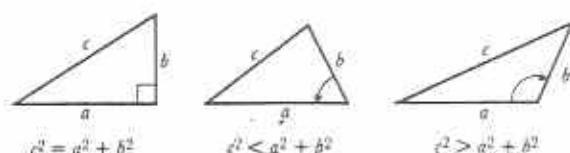


Fig. 11.4

Fig. 11.4 shows that

$c^2 = a^2 + b^2$ when c is opposite a *right* angle
 $c^2 < a^2 + b^2$ when c is opposite an *acute* angle
 $c^2 > a^2 + b^2$ when c is opposite an *obtuse* angle

Example 3

In $\triangle ABC$, $c = 8,44$ m, $a = 7,92$ m and $B = 151,3^\circ$. Calculate AC.

First, make a sketch of the data (Fig. 11.5).

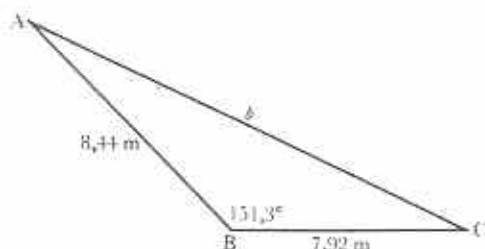


Fig. 11.5

By the cosine rule,

$$\begin{aligned} b^2 &= 8,44^2 + 7,92^2 - 2 \times 8,44 \times 7,92 \times \cos 151,3^\circ \\ &= 71,23 + 62,73 + 2 \times 8,44 \times 7,92 \times \cos 28,7^\circ \\ &= 133,96 + 16,88 \times 7,92 \times \cos 28,7^\circ \\ &= 133,96 + 117,3 \\ &= 251,26 \\ &= 251,3 \text{ to 4 s.f.} \\ b &= \sqrt{251,3} \\ &= 15,85 = 15,9 \text{ to 3 s.f.} \\ AC &= 15,9 \text{ m} \end{aligned}$$

working:

No	Log
16,88	1,2274
7,92	0,8987
$\cos 28,7^\circ$	0,8743
117,3	2,0692

Notice the use of tables of squares, logarithms and square roots in Example 3.

* A scientific calculator may be used to evaluate this expression. Work from right to left, starting by evaluating the trigonometrical function:

Key

1 5 1 - 3 cos

Display

- 0.8771461

x 7 - 9 2

7.92

x 8 - 4 4

8.44

x 2 +/- = M+

117.26532

7 - 9 2 x = M+

62.7264

8 - 4 4 x - M+

71.2336

MR sqrt

15.850089

$b = 15,9$

Note the use of the calculator's memory to store the three products. The **M+** key automatically adds the products.

Exercise 11a

Calculate the length of the side opposite the given angle in each of the $\triangle$ s ABC. Give answers correct to 3 s.f.

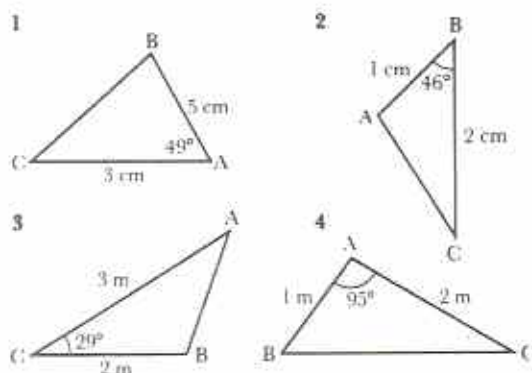


Fig. 11.6

- 1 $A = 120^\circ$, $b = 7$ cm, $c = 12$ cm
- 2 $B = 54^\circ$, $c = 4$ cm, $a = 5$ cm
- 3 $C = 13^\circ$, $a = 10$ m, $b = 15$ m
- 4 $B = 135,5^\circ$, $c = 8$ cm, $a = 5$ cm
- 5 $A = 125,4^\circ$, $b = 2,4$ cm, $c = 5$ cm
- 6 $C = 47,8^\circ$, $a = 13,1$ m, $b = 24,2$ m

Solving triangles using the sine and cosine rules

Example 4

In $\triangle ABC$, $a = 6,7$ cm, $c = 2,3$ cm and $B = 46,6^\circ$. Find b , A and C .

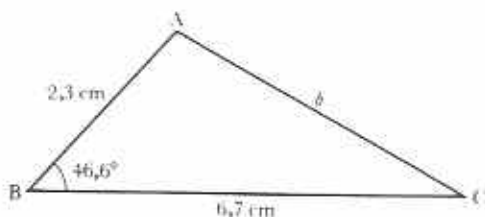


Fig. 11.7

In Fig. 11.7, using the cosine rule,

$$\begin{aligned} b^2 &= 2,3^2 + 6,7^2 - 2 \times 2,3 \times 6,7 \cos 46,6^\circ \\ &= 5,29 + 44,89 - 4,6 \times 6,7 \times \cos 46,6^\circ \\ &= 50,18 - 21,17 \\ &= 29,01 \end{aligned}$$

$$b = \sqrt{29,01} = 5,386 \text{ cm}$$

Using the sine rule,

$$\frac{\sin C}{2,3} = \frac{\sin 46,6^\circ}{5,386}$$

$$\sin C = \frac{2,3 \sin 46,6^\circ}{5,386}$$

$$C = 18,08^\circ$$

$$= 180^\circ - (46,6 + 18,08)^\circ$$

$$= 180^\circ - 64,68^\circ$$

$$= 115,32^\circ$$

To 1 d.p.

$$b = 5,4 \text{ cm}, C = 18,1^\circ, A = 115,3^\circ$$

Notice the following points in Example 4:

- 1 The cosine rule is used to find b .
- 2 The sine rule is used to find one of the remaining angles.
- 3 The *smaller* of the two unknown angles is found first, since this must be an acute angle. If the sine rule had been used to find A , $\log \sin A$ would have been $\bar{1},9561$, which gives $A = 64,7^\circ$ or $115,3^\circ$. To avoid this ambiguity,

always find the smaller angle first, since the angle must be acute.

- 4 Rounding off numbers only takes place at the *final* answer stage. Do *not* use rounded numbers at intermediate stages of the calculation.

Exercise 11b

Calculate the unknown side and angles in each of the $\triangle$ s ABC given. Give final answers correct to 3 s.f.

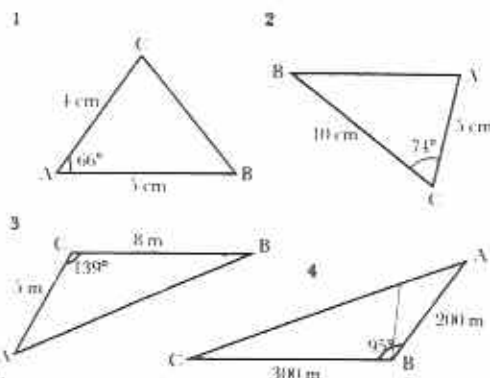


Fig. 11.8

- 5 $A = 58,1^\circ$, $b = 10$ m, $c = 8,5$ m
- 6 $B = 126^\circ$, $c = 5,6$ cm, $a = 5$ cm
- 7 $C = 25,7^\circ$, $b = 3,5$ cm, $a = 6$ cm
- 8 $A = 140,15^\circ$, $b = 45$ m, $c = 24$ m
- 9 $C = 143,3^\circ$, $b = 3,8$ cm, $a = 2,3$ cm
- 10 $B = 34,5^\circ$, $c = 2,8$ cm, $a = 5,1$ cm

Using the cosine rule to calculate angles

The formula $a^2 = b^2 + c^2 - 2bc \cos A$ can be rearranged with $\cos A$ as the subject of the formula:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\text{similarly, } \cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\text{and } \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

This formula can be used to calculate the angles of a triangle in which **all three sides** are given.

Example 5

Calculate the angles of a triangle with sides of length 4 m, 5 m and 7 m.

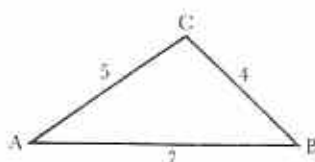


Fig. 11.9

With the lettering of Fig. 11.9,

$$\cos A = \frac{5^2 + 7^2 - 4^2}{2 \times 5 \times 7} = \frac{58}{70} = 0,8286$$

$$\Rightarrow A = 34,04^\circ$$

$$\cos B = \frac{4^2 + 7^2 - 5^2}{2 \times 4 \times 7} = \frac{40}{56} = \frac{5}{7} = 0,7143$$

$$\Rightarrow B = 44,42^\circ$$

$$\cos C = \frac{4^2 + 5^2 - 7^2}{2 \times 4 \times 5} = \frac{-8}{40} = -\frac{1}{5} = -0,2000$$

$$\Rightarrow C = 180^\circ - 78,46^\circ = 101,54^\circ$$

Check:

$$A + B + C = 34,04^\circ + 44,42^\circ + 101,54^\circ = 180^\circ$$

In questions such as Example 5, it is advisable to use the cosine formula to find every angle, then to check the results by addition.

Example 6

Calculate the angles of triangles which have sides (a) 400 m, 500 m, 700 m, (b) 2,8 cm, 4,2 cm, 5,6 cm.

In both cases, the calculation is simplified by considering similar triangles (i.e. equiangular) which have less complex numbers as sides.

$$(a) 400:500:700 = 4:5:7$$

Solve the triangle with sides of 4, 5 and 7 units, as in Example 5.

$$(b) 2,8:4,2:5,6 = 28:42:56 \\ = 2:3:4$$

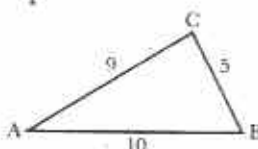
Solve the triangle with sides of 2, 3 and 4 units.

This is left as an exercise.

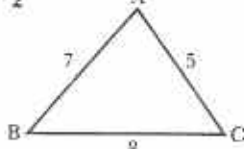
Exercise 11c

Calculate the angles of the $\triangle$ s ABC whose sides are given in cm. Give the final answers to the nearest $0,1^\circ$.

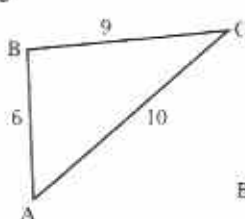
1



2



3



4

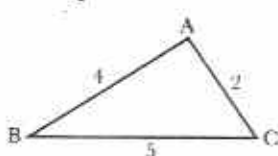


Fig. 11.10

$$5 \ a = 5, \ b = 7, \ c = 9$$

$$6 \ a = 45, \ b = 33, \ c = 21$$

$$7 \ a = 5,2, \ b = 6,5, \ c = 7,8$$

$$8 \ a = 7,2, \ b = 6,3, \ c = 9,9$$

$$9 \ a = 14,4, \ b = 11,2, \ c = 7,6$$

$$10 \ a = 2,7, \ b = 3,7, \ c = 3,1$$

Example 7

The sides of a parallelogram are 7 cm and 10 cm and one of its diagonals is 15 cm. Use the cosine formula to find the length of the other diagonal.

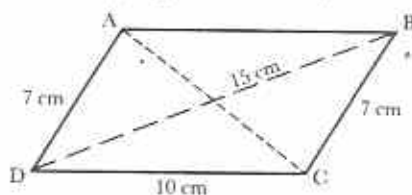


Fig. 11.11

In $\triangle BDC$,

$$\begin{aligned} \cos C &= \frac{10^2 + 7^2 - 15^2}{2 \times 10 \times 7} \\ &= \frac{149 - 225}{140} \\ &= -\frac{76}{140} \end{aligned}$$

In parallelogram ABCD,

$$\hat{ADC} + \hat{DCB} = 180^\circ \quad (\text{adjacent angles of } \parallel^{\text{m}})$$

$$\Rightarrow \hat{ADC} = 180^\circ - \hat{DCB}$$

$$\Rightarrow \cos \hat{ADC} = \cos (180^\circ - \hat{DCB})$$

$$\Rightarrow \cos \hat{ADC} = -\cos \hat{DCB} = -\frac{76}{140}$$

In $\triangle ADC$,

$$AC^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \times \cos \hat{ADC}$$

$$= 49 + 100 - 140 \times \frac{76}{140}$$

$$= 149 - 76$$

$$= 73$$

$$AC = \sqrt{73} \text{ cm}$$

$$= 8.544 \text{ cm}$$

The other diagonal is approximately 8.54 cm long.

Notice in Example 7 that it was not necessary to find the value of C in degrees.

Exercise 11d

- 1 In $\triangle ABC$ in Fig. 11.12, M is the mid-point of BC.

(a) Calculate $\cos B$ in $\triangle ABC$.

(b) Hence calculate AM.

(Note: Do not find B in degrees.)

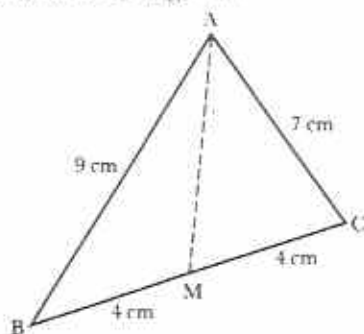


Fig. 11.12

- 2 With the data of Fig. 11.13, calculate

(a) $\hat{ABC}$, (b) AC, all lengths being given in cm.

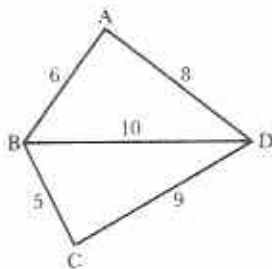


Fig. 11.13

- 3 In Fig. 11.14, find x and θ . All lengths are given in cm.

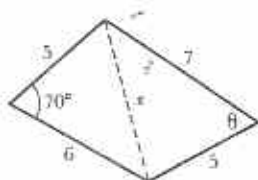


Fig. 11.14

- 4 In Fig. 11.15, PQRS is a cyclic quadrilateral. $PQ = 7$ cm, $QR = 8$ cm and $PR = 7.5$ cm.

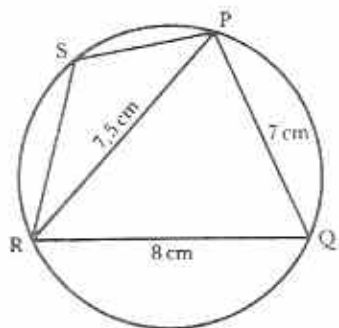


Fig. 11.15

(a) Calculate $\hat{PSR}$.

(b) Hence if $SR = SP$, calculate $\hat{SPR}$. Give your answers correct to the nearest tenth of a degree.

- 5 In a $\triangle ABC$, $AB = 8$ cm, $BC = 4$ cm, $CA = 5$ cm and BC is produced to P so that $CP = 4$ cm. Use the cosine rule to find $\cos \hat{ACB}$. Hence find AP.
- 6 Given Fig. 11.16, find $\cos \hat{P}$ in $\triangle PQX$. Hence find QX.

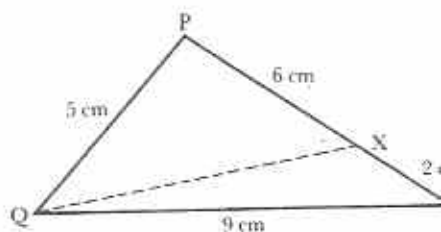


Fig. 11.16

- 7 The sides of a parallelogram are 3 cm and 5 cm and include an angle of 144° . Find the lengths of the diagonals of the parallelogram.
- 8 Calculate y in Fig. 11.17. (First find $\cos \theta$ but do not work out θ .)

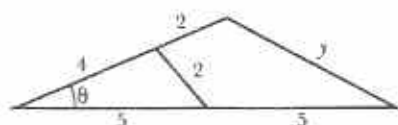


Fig. 11.17

- 9 In $\triangle PQR$ $p:q:r = \sqrt{3}:1:1$. Calculate the ratio $\hat{P}:\hat{Q}:\hat{R}$ in its simplest form.
- 10 In Fig. 11.18, ABCD is a trapezium with sides of lengths as shown.

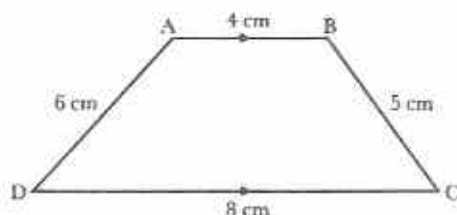


Fig. 11.18

Copy Fig. 11.18 and draw a line BX parallel to AD to cut CD in X. Hence calculate (a) $\hat{C}$, (b) BD.

Bearings and distances

Example 8

Three towns, A, B and C, are situated so that $AB = 60$ km and $AC = 100$ km. The bearing of B from A is 060° and the bearing of C from A is 290° . Calculate (a) the distance BC, (b) the bearing of B from C.

Fig. 11.19 is a sketch of the positions of A, B and C.

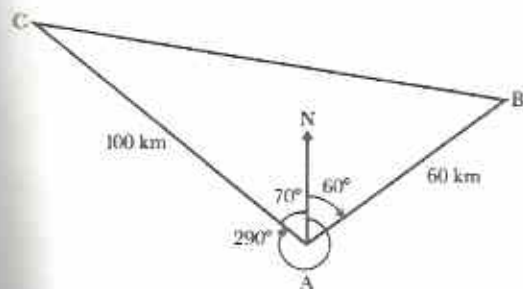


Fig. 11.19

In $\triangle ABC$,

$$\begin{aligned}\hat{CAB} &= 60^\circ + (360 - 290)^\circ \\ &= 60^\circ + 70^\circ \\ &= 130^\circ\end{aligned}$$

By the cosine rule,

$$\begin{aligned}BC^2 &= 100^2 + 60^2 - 2 \times 100 \times 60 \times \cos 130^\circ \\ &= 10\,000 + 3\,600 + 12\,000 \cos 50^\circ \\ &= 13\,600 + 12\,000 \times 0.6428 \\ &= 13\,600 + 7\,713.6 \\ &= 21\,313.6 = 21\,310 \text{ to 4 s.f.}\end{aligned}$$

$$\begin{aligned}BC &= \sqrt{21\,310} \text{ km} \\ &= 146 \text{ km}\end{aligned}$$

By the sine rule,

$$\frac{\sin C}{60} = \frac{\sin 130^\circ}{146}$$

$$\sin C = \frac{60 \times \sin 130^\circ}{146}$$

$$= \frac{60 \times \sin 50^\circ}{146}$$

$$C = 18.35^\circ$$

working:

No.	Log
60	1.7782
$\sin 50^\circ$	1.8843
	1.6625
146	2.1644
$\sin 18.35^\circ$	1.4981

Fig. 11.20 shows the angles at C:

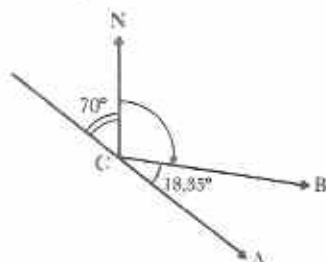


Fig. 11.20

$\hat{NCB}$ gives the bearing of B from C.

$$\begin{aligned}\hat{NCB} &= \hat{NCA} - 18.35^\circ \\ &= 110^\circ - 18.35^\circ = 91.65^\circ\end{aligned}$$

To 3 s.f.,

- (a) $BC = 146$ km,
(b) The bearing of B from C is 091.7° .

Exercise 11e

Give all distances correct to 3 s.f. Give all angles and bearings correct to 0.1° .

- 1 From a point on the edge of the sea, one ship is 5 km away on a bearing $S 50^\circ E$ and another is 2 km away on a bearing $S 60^\circ W$. Find the distance between the ships.

- 2 A girl walks 50 m on a bearing 025° and then 200 m due east. How far is she from her starting point?
- 3 Two goal posts are 8 m apart. A footballer is 34 m from one post and 38 m from the other. Within what angle must he kick the ball if he is to score a goal?
- 4 City A is 300 km due east of city B. City C is 200 km on a bearing of 123° from city B. How far is it from C to A?
- 5 A triangular field has two sides 50 m and 60 m long, and the angle between these sides is 96° . How long is the third side?
- 6 Two boats A and B left a port C at the same time along different routes. B travelled on a bearing of 150° (S 30° E) and A travelled on the north side of B. When A had travelled 8 km and B had travelled 10 km, the distance between the two boats was found to be 12 km. Calculate the bearing of A's route from C.
- 7 A man prospecting for oil in the desert leaves his base camp and drives 42 km on a bearing of 032° . He then drives 28 km on a bearing of 154° . How far is he then from his base camp and what is his bearing from it?
- 8 Two ships leave port at the same time. One travels at 5 km/h on a bearing of 046° . The other travels at 9 km/h on a bearing of 127° . How far apart are the ships after 2 hours?
- 9 A boat sails 4 km on a bearing of 038° and then 5 km on a bearing of 067° .
 - (a) How far is the boat from its starting point?
 - (b) Calculate the bearing of the boat from its starting point.
- 10 A photographer is 350 m away from a lion and wants to get closer before she takes a photograph. There is a water-hole in the direct line between the lion and herself, so she moves at an angle of 8° to this line to a better position 200 m further on. Calculate her distance from the lion.

- 11 Two planks, of lengths 1 m and 1.2 m, lean against each other as shown in Fig. 11.21. If the angle between the planks is 36°
 - (a) how far apart are the bottom edges of the planks, and
 - (b) what angle does the longer one make with the floor?

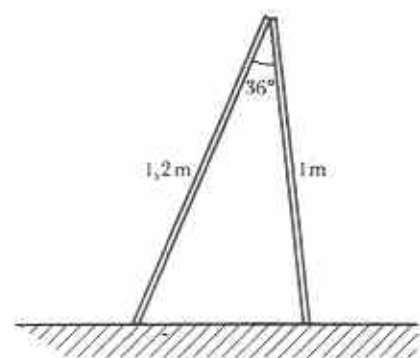


Fig. 11.21

- 12 An aeroplane flies due north from Harare Airport for 500 km. It then flies on a bearing of 060° for a further distance of 300 km before overflying a road junction. Calculate
 - (a) the distance of the aeroplane from Harare Airport when it was directly above the road junction,
 - (b) the bearing of the aeroplane from Harare Airport at this instant.
- 13 A ship leaves port and travels 21 km on a bearing of 032° and then 45 km on a bearing of 287° .
 - (a) Calculate its distance from the port.
 - (b) Calculate the bearing of the port from the ship.
- 14 An aircraft flies along a triangular course. The first leg is 200 km on a bearing of 115° and the second leg is 150 km on a bearing of 230° . How long is the third leg of the course and on what bearing must the aircraft fly?
- 15 Villages A, B, C, D are such that B is 4 km due east of A, C is 3 km due south of B and D is 4 km S 50° W from C. Calculate the distance and bearing of A from D.

Chapter 12

Consumer arithmetic (2)

Taxation

A **tax** is a financial contribution which people are legally obliged to make to the State. The Government, acting on behalf of the State, decides the various rates of tax. It uses taxes to pay for services such as education, health, public transport and national defence.

Sales tax

A proportion of money paid for goods is given to the Government. The part which is given to the Government is called **sales tax**. At the time of going to press, sales tax was 12,5% (or one-eighth) of the selling price of consumable goods (such as clothing and petrol) and 20% (or one-fifth) of the selling price of durable goods (such as cars and furniture).

Example 1

A gold and diamond ring is advertised as

\$1 495, excluding sales tax

What will be the total cost of the ring?

A ring is a durable item; sales tax is 20%.

True cost = 120% of \$1 495

$$= \frac{6}{5} \times \$1\,495$$

$$= \$1\,794$$

Example 2

A coat costs \$229,50 including sales tax. How much tax does the Government receive?

A coat is a consumable item; sales tax on it is 12,5%.

$$112,5\% \text{ of basic cost} = \$229,50$$

$$1\% \text{ of basic cost} = \$ \frac{229,50}{112,5}$$

$$\begin{aligned} 12,5\% \text{ of basic cost} &= \$ \frac{229,50 \times 12,5}{112,5} \\ &= \$ \frac{229,50}{9} \\ &= \$25,50 \end{aligned}$$

The Government receives \$25,50 sales tax.

Income tax

Most people pay a part of their earnings to the Government. The part they pay is called **income tax**. In Zimbabwe the way of calculating income tax is roughly as follows.

- 1 Tax is paid on **taxable income**. The rates of taxable income for married and single people are given in Table 12.1.

Table 12.1

taxable income per annum	rate of tax (%)	
	married	single
first \$1 000	10	14
second \$1 000	12	16
third \$1 000	14	18
fourth \$1 000	16	20
fifth \$1 000	18	22
and so on until ...	⋮	⋮
fifteenth \$1 000	38	42,5
sixteenth \$1 000	40	45
seventeenth \$1 000	42,5	45
over \$17 000	45	45

- 2 Abatements are given to help to meet the cost of personal and family commitments.

Here are some typical abatements:

single person	\$1 800
married person	\$3 000
children	\$600 per child
dependants	\$400 max

(Note: other abatements are available, but only the above will be used in this book.)

A taxpayer's total abatements are called the **abatable amount**. The maximum abatable amounts are \$6 600 and \$3 800 for married and single people respectively.

3 Annual tax is calculated as follows:

Chargeable income tax

= tax on gross income
- tax on abatable amount

Total tax payable

= chargeable income tax
+ 15% surcharge

(Note: in practice, the rate of the surcharge increases as total tax payable increases. However, for illustrative purposes, 15% will be used in this book.)

Example 3

A married teacher has an income of \$11 560. He has 4 children and a dependent relative. How much income tax should he pay?

First:

gross income = \$11 560

Second:

abatements:

married allowance	\$3 000
4 children	\$2 400
dependant	\$400
abatable amount =	\$5 800

Third:

tax on gross income of \$11 560
= \$100 + \$120 + \$140 + \$160
+ \$180 + \$200 + \$220 + \$240
+ \$260 + \$280 + \$300 + 32% of \$560
= \$2 379,20

tax on abatable amount of \$5 800
= \$100 + \$120 + \$140 + \$160
+ \$180 + 20% of \$800
= \$860

chargeable income tax
= \$2 379,20 - \$860
= \$1 519,20

Finally:

chargeable income tax = \$1 519,20
surcharge of 15% = \$227,88
Total tax payable = \$1 747,08

Exercise 12a

Unless otherwise stated, use the rates given above in this exercise.

1 Find out how much customers pay for each item in the advertisement in Fig. 12.1.

FANTASTIC SAVINGS ON SELECTED GOODS	
Three-piece suite	\$800
Kitchen table and 4 chairs	\$350
Black pots	\$14
Bicycle	\$349
All prices subject to Sales Tax of 20%	

Fig. 12.1

2 Each price in the advertisement in Fig. 12.1 includes a sales tax of 12,5%. How much money does the Government receive in each case?

JANUARY SALE	
back to school	
BOY'S SHIRTS	\$18.00
BOY'S SHORTS	\$9.00
BOY'S TROUSERS	\$36.00
GIRL'S BLOUSES	\$13.50
GIRL'S SKIRTS	\$27.00
GIRL'S DRESSES	\$39.60
TRACKSUITS	\$45.99
SOCKS	\$3.33

Fig. 12.2

- 3 The 'Young Fashion' department of a large store advertises stone-washed denim tops at \$63.99 with skirts to match at \$72.99. A young woman buys one of each.

How much sales tax does she pay if the advertised prices are inclusive of sales tax at 12.5%?

- 4 A 3-band radio can be bought by paying '12 easy instalments of \$22.50'. How much sales tax is included in the total price? (Note: a radio is a durable item.)
- 5 Calculate the gross income tax on each of the following salaries when earned by (i) married, (ii) single people.
(a) \$6 000 (b) \$9 000 (c) \$14 000
(d) \$7 500 (e) \$8 600 (f) \$10 780
- 6 Calculate the abatable amounts for the following:
(a) a single woman with a dependent parent;
(b) a married man with 3 children;
(c) a married woman with 1 child and a dependent mother;
(d) a married man with 5 children and dependent parents.
- 7 A single taxpayer has an annual income of \$7 000 and total abatements of \$2 000. Calculate the amount of tax paid.
- 8 A graduate trainee accountant has a salary of \$13 320. If the trainee is single and has no dependants, find: (a) her abatable amount, (b) the tax on the gross income, (c) the tax on the abatable amount, (d) the chargeable income tax (i.e. the tax before surcharge), (e) the surcharge, (f) the total tax that she pays annually, (g) her remaining income after tax is paid.
- 9 An experienced trained accountant has a salary of \$27 800. He is married, with 4 children and a dependent mother.
(a) Calculate his abatable amount.
(b) Calculate the total tax that he pays.
(c) He is paid monthly and tax is taken from his pay in equal instalments. Find his monthly 'take home pay'.
- 10 Companies pay tax of 45% on all taxable income. In addition, their chargeable income tax is subject to a surcharge of 17.5%. Find the tax paid by a company which has a taxable income of \$88 724.

Household bills

Discount

Sometimes a trader will reduce the price of an item in order to sell it. The reduction in price is called a **discount**. Discounts are often given to customers who can pay in cash.

Example 4

A refrigerator costs \$899. A 10% discount is given for cash. What is the discount price?

Either:

$$\begin{aligned}\text{Discount} &= 10\% \text{ of } \$899 = \frac{10}{100} \times \$899 \\ &= \$89.90\end{aligned}$$

$$\begin{aligned}\text{Cash price} &= \$899 - \$89.90 \\ &= \$809.10\end{aligned}$$

Or:

$$\begin{aligned}\text{Cash price} &= (100\% - 10\%) \text{ of } \$899 \\ &= 90\% \text{ of } \$899 \\ &= \frac{90}{100} \times \$899 \\ &= 0.9 \times \$899 \\ &= \$809.10\end{aligned}$$

Example 5

A stationary shop sells notebooks at \$1.40 each or 5 for \$6. It gives a 35% discount to schools on orders of 100 notebooks or more. Calculate the unit costs of the notebooks at the three rates.

A 'unit cost' is the cost of a single item. Unit costs are used for comparing prices.

Non-discount rate:

$$\text{Cost of 1 notebook} = \$1.40$$

5 notebooks for \$6 rate:

$$\text{Cost of 1 notebook} = \$6 \div 5 = \$1.20$$

35% discount rate:

$$\begin{aligned}\text{Cost of 1 notebook} &= 65\% \text{ of } \$1.40 \\ &= \frac{65}{100} \times \$1.40 \\ &= 91 \text{ cents}\end{aligned}$$

Example 5 shows the savings that can be made when buying 'in bulk' (i.e. buying a lot of a particular item).

Hire purchase

Expensive items such as cars and television sets are too costly for most people to buy outright. People find it easier to buy such items by paying **instalments**. An instalment is a part payment. Paying by instalments is called **hire purchase**. The buyer hires the use of the item before paying for it completely. It costs money to hire an item. This is why hire purchase costs more than paying in cash.

Example 6

A motorbike costs \$2 676 cash. Alternatively it can be bought for 25% deposit and 24 monthly instalments of \$115. How much more expensive is it to buy the motorbike by hire purchase?

Hire purchase price

$$\begin{aligned} &= \text{deposit} + \text{instalments} \\ &= 25\% \text{ of } \$2\,676 + 24 \times \$115 \\ &= \$669 + \$2\,760 \\ &= \$3\,429 \end{aligned}$$

Price difference

$$\begin{aligned} &= \$3\,429 - \$2\,676 \\ &= \$753 \end{aligned}$$

In Example 6:

- 1 The difference in price, \$753, represents the cost of hiring the motorbike during the time it was being paid for.
- 2 The deposit is given as a percentage of the cost price. This is commonly done.

Exercise 12b

- 1 Find the price if a discount of
 - (a) 10% is given on a cost price of \$59;
 - (b) $12\frac{1}{2}\%$ is given on a cost price of \$420;
 - (c) 15 cents in the dollar is given on a book marked at \$28;
 - (d) 20% is given on a \$45 shirt.
- 2 The selling price of an armchair is \$220. The shop gives a 25% discount for cash. What is the cash price?
- 3 During a sale a shop takes 35 cents in the \$ off all marked prices.
 - (a) What percentage discount does this represent?
 - (b) What is the sale price of a handbag marked at \$39?

- 4 A Co-operative sells eggs at the prices shown on the notice in Fig. 12.3.

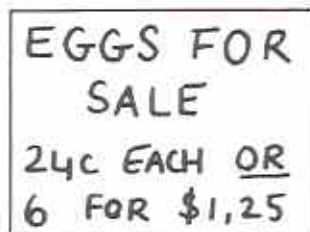


Fig. 12.3

How much is saved by buying four dozen eggs in sixes instead of separately?

- 5 A 500 g packet of rice costs 98c. A 50 kg sack of the rice costs \$65. Calculate the two unit costs of the rice if 1 kg is taken as the unit. What is the saving per kg when buying in bulk?
- 6 The hire purchase price of a hi-fi set is \$678 spread over 24 equal fortnightly payments. How much is each payment?
- 7 The hire purchase price of a vehicle is a deposit of \$4 500 down and 36 monthly payments of \$950. What is the total paid for the hire purchase?
- 8 A carpet either costs \$699 cash or 30 weekly payments of \$27.50.
 - (a) Find the cost of the carpet when paying by hire purchase,
 - (b) Find the cost of the hire of the carpet for the 30 weeks.
- 9 A television set costs either \$675 cash, or
 - 52 weekly payments of \$16.50, or
 - 104 weekly payments of \$9.50.Find the total cost of each of the hire purchase methods of payment. (Why should the two differ in cost?)
- 10 A new computer costs \$2 335. An 8% discount is given for cash. The hire purchase price of the computer is 15% down and 24 monthly payments of \$109.65. Calculate the difference between paying cash and paying by hire purchase.

Electricity and water charges

People who receive public services such as water and electricity also receive bills to pay for them. The following give some typical charges. Remember, however, that rates and methods of payment for services vary from time to time and from place to place.

Electricity and water charges are sometimes shown on the same bill. Fig. 12.4 shows a typical bill for these charges issued by the City of Harare and Zimbabwe Electricity Supply Authority (ZESA).



City of Harare/ZESA

P.O. Box 1000 TELEPHONE 707011

ELECTRICITY AND/OR WATER CHARGES

Previous Account	Last Payment	1 credit	1 debit	Balance
82,01	27 SEP	82,01	12 DEC	0,00
Code	This Reading	Last Reading	Units	
ZESA CHARGES				
T A FIXED MONTHLY CHARGE				7,30
E A 23705 22712 993				66,03
CITY OF HARARE CHARGES				
W 1 6128 6071 57				28,15
10% SURCHARGE ON ELEC ACC				7,33
Total				108,81

Property Reference: 49 19 123
 Address:

Last Reading Date: 12 NOV
 Due and Payable: 31 DEC 91

Please read this account with reference to the City Tariffsheet.
If this account remains unpaid after the due date, suppliers may be disconnected without further notice.

A RECEIPT ON THIS FORM IS NOT VALID UNLESS MACHINE PRINTED

Fig. 12.4

Although the electricity and water charges are shown on the same bill they are calculated differently as follows:

Electricity charges:

Fixed monthly charge: \$7,30
 Consumption rate: 6,65c per kWh
 Surcharge: 10% of bill

The 10% surcharge is a handling charge and goes to the City Council. The rest goes to ZESA.

kWh is short for **kilowatt-hour**. 1 kWh is the amount of electricity that is used when 1 000 watts of electricity are consumed in 1 hour. The kWh is the basic 'unit' of electrical consumption.

Water charges:

Fig. 12.5 gives some typical water charges.

WATER TARIFF

Please note that the Water Tariffs have been amended in respect of water consumed after the normal reading date in October 1990. Details of the amended Tariffs are given below.

MUNICIPAL AREA

Type	Scale	Rate per month per cubic metre
Single Family Dwelling Units	W1	first 13 m ³ at 36,5 c/m ³
		next 26 m ³ at 48,5 c/m ³
		next 31 m ³ at 60,0 c/m ³
		over 70 m ³ at 71,5 c/m ³
Commercial and Industrial	W2	48,5 c/m ³

OUTSIDE MUNICIPAL AREA

All consumers	W3	57,5 c/m ³
---------------	----	-----------------------

N.B. As water accounts are calculated in metric units all consumptions shown are cubic metres unless otherwise stated.

Fig. 12.5

The unit of water is the cubic metre (m³). 1 m³ is equivalent to 1 000 litres, or 1 kℓ, of water.

Example 7

Check the bill shown in Fig. 12.4.

Electricity charges:

$$\begin{aligned} \text{No. of units used} &= 23\,705 - 22\,712 \\ &= 993 \end{aligned}$$

$$\begin{aligned} \text{Charge for units} &= 993 \times 6,65\text{c} \\ &= \$66,0345 \end{aligned}$$

$$\text{Fixed monthly charge} = \$7,30$$

$$\text{Total of electricity} = \$73,33$$

$$\begin{aligned} 10\% \text{ surcharge} &= 10\% \text{ of } \$73,33 \\ &= \$7,33 \text{ to nearest c} \end{aligned}$$

Water charges:

$$\begin{aligned} \text{No. of units used} &= 6\,128 - 6\,071 \\ &= 57 \\ 57 &= 13 + 26 + 18 \end{aligned}$$

Charge for units

$$\begin{aligned} &= (13 \times 13,5\text{c}) + (26 \times 48,5\text{c}) + (18 \times 60\text{c}) \\ &= \$4,745 + \$12,61 + \$10,80 \\ &= \$28,155 = \$28,15^* \end{aligned}$$

Grand total:

$$\begin{aligned} &= \$73,33 + \$7,33 + \$28,15 \\ &= \$108,81 \end{aligned}$$

*Note that the 0,5c is rounded down.

Example 8

A household uses 716 units of electricity. What will be the cost of the electricity?

$$\text{Cost of units used} = 716 \times 6,65c$$

$$= 4\,761,4c$$

$$= \$47,61$$

$$\text{Fixed monthly charge} = \$7,30$$

$$\text{Subtotal} = \$54,91$$

$$10\% \text{ surcharge} = 10\% \text{ of } \$54,91$$

$$= \$5,49$$

$$\text{Total cost} = \$54,91 + \$5,49$$

$$= \$60,40$$

Example 9

Find the bill for 48 m^3 of water when it is calculated by (a) Scale W1, (b) Scale W2, (c) Scale W3.

(a) Scale W1

$$48 = 13 + 26 + 9$$

Water bill

$$= (13 \times 36,5c) + (26 \times 48,5c) + (9 \times 60c)$$

$$= 474,5c + 1261c + 540c = 2\,275,5c^*$$

$$= \$22,75$$

(b) Scale W2

$$\text{Water bill} = 48 \times 48,5 = 2\,328c$$

$$= \$23,28$$

(c) Scale W3

$$\text{Water bill} = 48 \times 57,5 = 2\,760c$$

$$= \$27,60$$

*In practice the 0,5c is rounded down. Examples 7, 8, 9 show that a calculator, if available, is a very useful aid for checking household bills.

Household rates (owners charges)

Property owners must pay for services which are supplied to their houses and families. Such services include road construction and maintenance, refuse removal and supply of public amenities such as civic centres, sports stadiums, street lighting and parks. Bills for this are called **owners charges**. They are also commonly called **household rates**, or simply just **rates**.

Fig. 12.6 shows a typical 6-monthly rates bill for a property in Harare.

Note that the bill contains three components

1 Land

The plot of land (known as a *stand*) is evaluated and charged at a rate of 1,166 cents per S.

2 Improvements

Any improvements to the stand are evaluated and charged at a rate of 0,818 cents per S. Note that such improvements normally include the main house and outbuildings.

3 Refuse removal

A fixed charge of \$25,25 for the regular disposal of rubbish.

City of Harare					Business Hours	
P.O. Box 1600 HARARE					Monday	8.00 - 4.45
					Tuesday	8.00 - 3.45
					Wednesday	8.00 - 3.45
					Thursday	8.00 - 4.15
					Friday	8.00 - 4.15
IN RESPECT OF STAND 1903 MOUNT PLEASANT						
Please quote Rates Clerk in communication						
Rates Code 044968						
Refuse payments to 19 MAR.						
ITEM	DATE FROM	DATE TO	VALUATION S	RATE cents per S		
LAND	1 JAN 91	30 JUN 91	6610	1,166	77,07	
IMPROVEMENTS	1 JAN 91	30 JUN 91	15490	0,818	126,71	
REFUSE REMOVAL	1 JAN 91	30 JUN 91			25,25	
Due date				Outpaid balance after	will attract interest	
30 APR 91				31 MAY 91	10,75 PER CENT FROM DUE DATE	
					229,03	

Fig. 12.6

Example 10

Check that the rates bill shown in Fig. 12.6 is correct.

Land charge:	$6,610 \times 1,166c =$	\$77,07
Improvements:	$15\,490 \times 0,818c =$	\$126,71
Refuse removal (fixed charge)	=	\$25,25
Total owner's charges	=	\$229,03

Exercise 12c

Unless told otherwise, use the water charges, electricity charges and household rates as given above when doing this exercise.

- 1 The electricity bill for a household which uses 580 units is \$50,46. This includes the fixed monthly charge and the 10% surcharge. Check that this bill is correct.
- 2 What would be the bill for a household which used half as many units as in question 1?
- 3 The reading on an electricity meter changes from 18 091 to 19 172 in one month. Calculate the bill.
- 4 The water bill for a house charged on the W1 Scale was \$24,55 when 51 m³ of water was used. Check that this bill is correct.
- 5 What would be the bill for the house in question 4 if, next month, it used only two-thirds as much water?
- 6 At the beginning and end of a month the readings on a water meter are 2 391 m³ and 2 448 m³ respectively. Calculate the bill if charges are made on the W1 Scale.
- 7 70 m³ of water are used in a month. Calculate the bill when charged on (a) Scale W1, (b) Scale W2, (c) Scale W3.
(Give possible reasons why the rates are different for different types of users.)
- 8 A property is valued as follows:
Land \$3 000
Improvements \$10 000
Calculate the 6-monthly owners charges for the property. Include charges for refuse removal.
- 9 The rates for the property in question 8 are increased to the following:
Land: 1,225c per \$
Improvements: 0,922c per \$
Refuse removal: \$27,85 (fixed)
Assuming that the valuations remain as

before, calculate the new 6-monthly owners charges for the property.

- 10 A house in Harare has a land valuation of \$6,500, an improvements valuation of \$18 250 and is subject to Scale W1 water charges. Readings of the water and electricity meter show that in the month of June the household used 64 m³ of water and 905 units of electricity.
Find the total amount that the bills will come to (i.e. the water and electricity bill for June and the owners charges for January to 30 June).

Insurance

Insurance is a financial arrangement for the payment of a sum of money to compensate for an unfortunate loss or injury. Those who wish to insure themselves make payments, or **premiums**, to insurance companies. For example, the Summit Insurance Company charges an annual premium of \$66 to householders for insuring the contents of their houses against loss or theft of amounts up to \$5 000. If a fire were to destroy or damage the house's contents, the householder would receive up to \$5 000 in compensation.

It is possible to insure yourself for nearly any kind of loss. Specialist statisticians, called **actuaries**, are employed by insurance companies to calculate the premiums for the various risks involved. Table 12.2 shows the premiums to be paid by travellers to ensure that they obtain compensation in the form of benefits which are shown overleaf.

Table 12.2

Period of cover	Premium per person (\$)
1-6 days	16,20
7-11 days	20,70
12-17 days	23,40
18-23 days	26,10
24-31 days	28,80
6 weeks	37,80
8 weeks	46,80
10 weeks	56,70

Benefits per person	
Benefit A	
Death by accident	\$5 000
Benefit B	
Permanent total disablement following accident	\$20 000
Benefit C	
Medical and other expenses. All necessary costs up to	\$1 000 000
Benefit D	
Loss of luggage	up to \$800
Benefit E	
Personal liability	\$500 000
Benefit F	
Hijack of aircraft	£500
Benefit G	
Loss of money	up to £500

Example 11

Referring to Table 12.2, what would be the premium if a traveller wishes to insure himself for 2 weeks?

2 weeks = 14 days

This falls within the 12–17 days' range.

Premium = \$23.40

Note: There is no reduction in premium if the time period is less than the upper limit of the range.

Example 12

The traveller in Example 11 has luggage valued at \$450. If this, together with \$210 worth of travellers cheques are stolen, what compensation could he expect?

Benefit D:

Compensation for lost luggage = \$450

Benefit G

Compensation for stolen money = \$210

Total compensation = \$660

Note: Compensation is paid only on the value of the losses, not on the maximum amount insurable.

Mortgages

A **mortgage** is a loan given for the purpose of buying land and any property (buildings) which is on the land. Companies which give mortgages are called **building societies**. A building society owns the property until the loan on it is repaid.

Interest is charged on the loan. Typical rates of interest are 13.25% on residential property (e.g. houses) and 14.75% on commercial properties (e.g. offices, shops).

In practice, building societies seldom give a mortgage for the full value of a property. The maximum mortgage obtainable is usually 95% of the price of the property.

A mortgage is usually repaid over a long period of time: anything up to 25 years or more is quite common. During this time, both the interest and the loan are repaid.

Example 13

A house buyer borrows \$50 000 from the B&H Building Society to buy a \$70 000 property. The mortgage is given over 25 years and the monthly repayments are \$576.39.

(a) How much was the house buyer's deposit on the property?

(b) What is the total amount that is repaid to the B&H Building Society over the 25 years?

(c) How much does the house buyer have to pay for the property altogether?

(a) Deposit = \$70 000 – \$50 000
= \$20 000

(b) Total repayments per year
= \$576.39 × 12

Total repayment over 25 years
= \$576.39 × 12 × 25
= \$172 917

(c) Total cost of property
= \$172 917 + \$20 000
= \$192 917



Note that the monthly payment includes both the interest and some repayment of the capital borrowed.

Exercise 12d

- 1 Use the data in Table 12.2 to calculate the premiums for a husband and wife who insure themselves against travel losses during a 6-week trip to Europe.
- 2 The wife in question 1 falls ill during her holiday. She incurs hospital bills of \$2 215 and is flown home to Zimbabwe at a further cost of \$1 207. Which of the benefits will she be compensated from and how much should she receive?
- 3 A household insurance company charges an annual premium of \$6 per \$100 sum assured. What will be the total premium for a householder who insures his house contents for \$14 300?
- 4 A car owner insures her car for 'third party, accident and theft', meaning that any losses *caused* by her or *incurred* by her will be paid under the insurance.
 - (a) Premiums are charged at the rate of \$65 per \$2 000 or part thereof, based on the value of the car. What will be the full premium if the car is worth \$26 800?
 - (b) A 'no claims bonus' of 60% is given because the driver has made no claims against the insurance company within the past 5 years. What will be her actual premium?
- 5 Table 12.3 shows the premiums to be paid for insurance of home possessions in rural,

urban and city areas for various amounts of cover.

- (a) What would be the premium to insure home possessions for \$4 000 in an urban area?
 - (b) What would be the premium to insure home possessions for \$5 000 in the city of Bulawayo?
 - (c) Additional cover is given at a premium rate of \$4 per \$100 or part thereof in all areas. What would be the premium for a rural farmer who insures his possessions for \$7 850?
- 6 A building society charges a monthly repayment of \$11.54 for each \$1 000 borrowed.
 - (a) Calculate the monthly repayment on a mortgage of \$15 000.
 - (b) What is the yearly repayment?
 - (c) What will be the total repayment over 25 years?
 - 7 If the Social Union Building Society's interest rate is 13.25% per annum, how much interest is paid on each \$1 000 borrowed
 - (a) per year, (b) per month?
 - 8 The City of Harare sells a stand (plot of land) to Betterbuilder Brick Company for \$56 750. The company obtains an 80% mortgage and makes monthly repayments of \$738.13 over 10 years.
 - (a) Calculate the amount of the loan.
 - (b) Calculate the total amount repaid to the building society over the 10 years.
 - 9 An 'interest only' mortgage is one on which only interest is paid. Payment of the capital takes place at the end of the period of the loan. A company obtained an 'interest only' mortgage of \$90 000 at a rate of 14.75% per annum. After 5 years it repaid the capital. How much interest did the company pay?
 - 10 A chicken farm is sold for \$230 000. By selling her previous farm, the buyer can afford a deposit of \$185 000.
 - (a) What size of mortgage does she need?
 - (b) The G&C Building Society offers her a 20-year mortgage at a monthly payment rate of \$13.83 per \$1 000 borrowed. Calculate the total monthly repayments.
 - (c) Calculate the total amount that is repaid over the 20 years.

Table 12.3

sum covered	Premiums		
	rural area	urban area	city area
\$2 500	\$34	\$49	\$54
\$3 000	\$37	\$57	\$68
\$4 000	\$43	\$67	\$82
\$5 000	\$48	\$79	\$99

Budgeting

In daily life it is important to keep accurate accounts of income and expenditure. This is true whether at a household, business, Co-operative or State level. Accurate accounts enable individuals, groups and countries to plan their spending. The planning of expenditure is called **budgeting**.

Budgeting is greatly assisted by keeping **cash accounts** of transactions. Fig. 12.7 is a simple household cash account which shows the income and expenditure for a household during a month.

Income		Expenditure	
Wages	1005.60	Food	226.11
Farm	247.17	Mortgage	354.25
		Insurance	26.44
		Elec/Water	94.67
		School costs	60.00
		Car running	56.80
		Entertainment	64.30
		Clothing	86.54
		Farm costs	58.00
		Savings	128.66
Total	1255.77	Total	1255.77

Fig. 12.7

Although Fig. 12.7 is a simplified account, it contains the main elements of all cash accounts:

Ace Public Transport Co-op Ltd. Accounts for June 1991

CASH RECEIVED			CASH SPENT		
Date	Details	\$	Date	Details	\$
01/06/91	Cash brought fwd	8176.38	07/06/91	Wages	3412.80
08/06/91	Weekly takings	5038.63	08/06/91	Fuel bill	926.55
15/06/91	Weekly takings	4499.47	12/06/91	Bus repairs	417.50
20/06/91	Bus hire	550.00	14/06/91	Wages	3412.80
22/06/91	Weekly takings	3883.61	21/06/91	Wages	3412.80
29/06/91	Weekly takings	6171.62	22/06/91	Fuel bill	1514.83
30/06/91	School bus hire	430.00	25/06/91	6 new tyres	749.40
			28/06/91	Wages	3412.80
			30/06/91	Garage rent	1750.00
			30/06/91	Petty cash	76.16
			30/06/91	Cash carried fwd	9664.07
		28749.71			28749.71

Fig. 12.8

- 1 A statement with details of income.
- 2 A statement with details of expenditure.
- 3 Total income and expenditure (which should balance).

All businesses and Co-operatives are required by law to keep cash accounts which show income and expenditure. Such accounts must show the cash which comes in and the cash which is paid out. Fig. 12.8 is a cash account showing one month's trading figures for a transport Co-operative.

Notice the following:

- 1 The cash account is in two parts:
cash received (income) on the left.
cash spent (expenditure) on the right.
- 2 The total on the left-hand side gives the total cash received.
- 3 The total on the right-hand side must balance the total on the left-hand side. To do this:
 (a) the actual expenditure is added,
 (b) the resulting sub-total is subtracted from the total cash received to give *cash carried forward*.
- 4 This cash is brought forward as cash received into the account for the next period.

Example 14

Use Fig. 12.7 to calculate the profit that the household made from farming.

Profit from farming

$$= \text{income from farming} - \text{farm costs}$$

$$= \$247,17 - \$58 = \$189,17$$

Example 14 shows how keeping accounts can highlight the efficiency of a small project such as selling farm produce.

Example 15

Use Fig. 12.8 to find the operating profit or loss of Ace Public transport Co-op during the month of June.

Operating profit

$$= \text{cash carried fwd (to July)}$$

$$- \text{cash brought fwd (from May)}$$

$$= \$9\,664,07 - \$8\,176,38$$

$$= \$1\,487,69$$

Note:

- 1 'fwd' is a common abbreviation of 'forward'.
- 2 If cash carried fwd < cash brought fwd, then a trading loss would have occurred.

Exercise 12e

- 1 Refer to Fig. 12.7.
 - (a) Which of the items under 'expenditure' is likely to be the same each month?
 - (b) Food, electricity and water are essential items. How much was spent on these essentials?

(c) Entertainment and clothing are non-essential items. How much was spent on non-essentials?

2 Refer to Fig. 12.8.

(a) The weekly takings come from bus fares. Find the total obtained from fares in June 1991.

(b) How much was obtained from hiring out buses?

(c) Find the total expenditure on fixed costs i.e. wages and garage rent.

(d) How much was spent on keeping the buses running during the month?

(e) How much did each new tyre cost?

3 Refer to Fig. 12.8. Next month the income was \$15 093,14 from bus fares and \$430,00 from school bus hire.

Expenditure amounted to 4-week's wages and garage rent as in June, \$2 674,37 for fuel and \$618,20 for bus maintenance. There was also \$112,74 in petty cash.

(a) Prepare a cash account for July 1991.

(b) What is the balance in hand at the end of July?

(c) Calculate the operating profit (or loss) for the month of July.

Fig. 12.9 is a summary account of the Central Government's budget for the financial years 1986/87 to 1988/89.

Government Budget (Z\$million) Expenditure and financing 1986/87 to 1988/89

	1986/87	1987/88	1988/89
REVENUE AND GRANTS	3 056,5	3 784,9	4 356,4
Income Tax	1 351,9	1 614,2	1 865,2
Sales Taxes	1 237,0	1 386,7	1 782,9
Other Revenues	467,6	784,0	708,3
EXPENDITURE AND NET LENDING	4 053,3	4 680,9	5 467,4
Current and Capital	3 822,1	4 295,7	4 960,2
Net Lending	231,2	385,2	507,2
DEFICIT	-996,8	-896,0	-1 111,0
FINANCING	+996,8	+896,0	+1 111,0
Foreign Loans	210,8	147,8	129,1
Domestic Loans	786,1	748,2	981,9

Fig. 12.9 (Source: Reserve Bank of Zimbabwe)

In Fig. 12.9:

The first row gives the total *revenue* or income. This is broken down to show revenue from income tax, sales tax and other sources.

The second row gives the main areas of Government *expenditure* on current and capital developments and on loans.

The third row gives the *budget deficit*, i.e. the excess of expenditure over revenue.

The fourth row shows how the deficit was *financed* or funded by foreign and domestic loans. Refer to Fig. 12.9 and the above information when answering questions 4–10.

- 4 (a) How much revenue was obtained from sales tax in 1987/88?
(b) In which year did net lending amount to \$507 200 000?
(c) Which item always produces the greatest source of revenue?
(d) Which year had the smallest budget deficit?
- 5 For the 3-year period, are the following statements true?
(a) The Government borrowed less and less foreign money to finance the budget deficit.
(b) Government expenditure rose steadily.
(c) The deficit each year was always between \$900 million and \$1 100 million.
(d) Each year income tax produced about twice as much revenue as sales tax.
- 6 In each of the main rows, the sub-totals usually add to give the main total. However, in the case of financing in 1986/87, there is a small discrepancy. Suggest how this arises.
- 7 What was the total revenue raised from income tax in the 3-year period?
- 8 What was the total spent on current and capital expenditure in the 3-year period?
- 9 (a) For 1986/87, express foreign loans as a percentage of the total amount needed to finance the deficit.
(b) Do likewise for 1988/89.
(c) What conclusion do you draw from your answers to parts (a) and (b)?
- 10 (a) Express the amount under revenue and grants in 1988/89 as a percentage of the 1986/87 figure.
(b) Do likewise for expenditure and net lending.
(c) What conclusion can you draw from your answers to parts (a) and (b)?

Matrices (2)

Matrix arithmetic

Addition and subtraction

Example 1

$$\text{If } \mathbf{A} = \begin{pmatrix} 3 & -2 \\ 1 & 0 \\ 0 & 4 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} -5 & 2 \\ 2 & 3 \\ -1 & 0 \end{pmatrix}$$

find (a) $\mathbf{A} + \mathbf{B}$, (b) $\mathbf{A} - \mathbf{B}$, (c) $3\mathbf{A}$.

$$\begin{aligned} \text{(a) } \mathbf{A} + \mathbf{B} &= \begin{pmatrix} 3 & -2 \\ 1 & 0 \\ 0 & 4 \end{pmatrix} + \begin{pmatrix} -5 & 2 \\ 2 & 3 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 3 + (-5) & (-2) + 2 \\ 1 + 2 & 0 + 3 \\ 0 + (-1) & 4 + 0 \end{pmatrix} \\ &= \begin{pmatrix} -2 & 0 \\ 3 & 3 \\ -1 & 4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{(b) } \mathbf{A} - \mathbf{B} &= \begin{pmatrix} 3 & -2 \\ 1 & 0 \\ 0 & 4 \end{pmatrix} - \begin{pmatrix} -5 & 2 \\ 2 & 3 \\ -1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 3 - (-5) & (-2) - 2 \\ 1 - 2 & 0 - 3 \\ 0 - (-1) & 4 - 0 \end{pmatrix} \\ &= \begin{pmatrix} 8 & -4 \\ -1 & -3 \\ 1 & 4 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{(c) } 3\mathbf{A} &= 3 \begin{pmatrix} 3 & -2 \\ 1 & 0 \\ 0 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 3 \times 3 & 3 \times (-2) \\ 3 \times 1 & 3 \times 0 \\ 3 \times 0 & 3 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 9 & -6 \\ 3 & 0 \\ 0 & 12 \end{pmatrix} \end{aligned}$$

Revision notes:

1. Matrices $\mathbf{A}$ and $\mathbf{B}$ are of the same *order*. They are both 3×2 matrices, i.e. 3 rows by 2 columns.
2. Matrices can be added or subtracted only if they are of the same order, in which case corresponding elements are added or subtracted as in parts (a) and (b) of Example 1.
3. When a matrix is multiplied by a *scalar*, as in part (c), the scalar multiplies every element of the matrix.

Multiplication

Example 2

If $\mathbf{M} = \begin{pmatrix} -2 & 4 \\ 3 & 5 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} 6 & 0 \\ -1 & 2 \end{pmatrix}$ find the matrix products (a) $\mathbf{MN}$, (b) $\mathbf{NM}$, (c) $\mathbf{M}^2$.

$$\text{(a) } \mathbf{MN} = \begin{pmatrix} -2 & 4 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} 6 & 0 \\ -1 & 2 \end{pmatrix}$$

$$\begin{aligned}
 &= \begin{pmatrix} (-2) \times 6 + 4 \times (-1) & (-2) \times 0 + 4 \times 2 \\ 3 \times 6 + 5 \times (-1) & 3 \times 0 + 5 \times 2 \end{pmatrix} \\
 &= \begin{pmatrix} -12 + (-4) & 0 + 8 \\ 18 + (-5) & 0 + 10 \end{pmatrix} \\
 &= \begin{pmatrix} -16 & 8 \\ 13 & 10 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) } \mathbf{NM} &= \begin{pmatrix} 6 & 0 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 3 & 5 \end{pmatrix} \\
 &= \begin{pmatrix} -12 + 0 & 24 + 0 \\ 2 + 6 & -4 + 10 \end{pmatrix} \\
 &= \begin{pmatrix} -12 & 24 \\ 8 & 6 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c) } \mathbf{M}^2 &= \begin{pmatrix} -2 & 4 \\ 3 & 5 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 3 & 5 \end{pmatrix} \\
 &= \begin{pmatrix} 4 + 12 & -8 + 20 \\ -6 + 15 & 12 + 25 \end{pmatrix} \\
 &= \begin{pmatrix} 16 & 12 \\ 9 & 37 \end{pmatrix}
 \end{aligned}$$

Revision notes:

- 4 In Example 2(a) $\mathbf{M}$ pre-multiplies $\mathbf{N}$. In Example 2(b) $\mathbf{M}$ post-multiplies $\mathbf{N}$.
- 5 Matrices can be multiplied only if there are as many columns in the first matrix as there are rows in the second matrix.
- 6 A $p \times q$ matrix will multiply a $q \times r$ matrix to give a $p \times r$ product:

$$(p \times \boxed{q \times q}) \times r \rightarrow (p \times r)$$

- 7 In general $\mathbf{AB} \neq \mathbf{BA}$ where $\mathbf{A}$ and $\mathbf{B}$ are matrices.

Exercises 13a (revision)

- 1 Simplify the following, giving each result as a single matrix.

$$\text{(a) } \begin{pmatrix} 3 & 9 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ -6 & 2 \end{pmatrix}$$

$$\text{(b) } \begin{pmatrix} 2 \\ 6 \end{pmatrix} + \begin{pmatrix} 10 \\ 3 \end{pmatrix}$$

$$\text{(c) } \begin{pmatrix} 5 & -1 \\ 1 & -1 \\ -2 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 6 \\ 1 & -7 \\ 3 & 0 \end{pmatrix}$$

$$\text{(d) } \begin{pmatrix} -4 & 2 \\ 1 & -2 \end{pmatrix} - \begin{pmatrix} 6 & 9 \\ -2 & -3 \end{pmatrix}$$

$$\text{(e) } \begin{pmatrix} 3 \\ 9 \end{pmatrix} - \begin{pmatrix} 0 \\ 5 \end{pmatrix} + \begin{pmatrix} 2 \\ 6 \end{pmatrix}$$

$$\text{(f) } \begin{pmatrix} 3 & -2 \\ 1 & 5 \end{pmatrix} + \begin{pmatrix} -1 & 3 \\ -2 & -1 \end{pmatrix} - \begin{pmatrix} 2 & 4 \\ 7 & -2 \end{pmatrix}$$

- 2 If $\mathbf{A} = \begin{pmatrix} -1 & 5 \\ 2 & 3 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 6 & 0 \\ 4 & -8 \end{pmatrix}$ find

$$\begin{array}{lll} \text{(a) } 3\mathbf{A} & \text{(b) } 2\mathbf{B} & \text{(c) } -2\mathbf{A} \\ \text{(d) } \frac{1}{2}\mathbf{B} & \text{(e) } 2\mathbf{A} + \mathbf{B} & \text{(f) } \mathbf{A} - 3\mathbf{B} \end{array}$$

- 3 Express each of the following products as a single element matrix.

$$\text{(a) } (4; 5) \begin{pmatrix} 6 \\ 1 \end{pmatrix}$$

$$\text{(b) } (2; 3; -1) \begin{pmatrix} \frac{1}{2} \\ -2\frac{1}{2} \\ 1\frac{1}{2} \end{pmatrix}$$

$$\text{(c) } (3; -1) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\text{(d) } (4; 5; 6) \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

- 4 Find x and y if

$$4 \begin{pmatrix} 5 & x \\ -1 & 1 \end{pmatrix} - 3 \begin{pmatrix} 6 & x \\ 2 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 4 \\ y & -1 \end{pmatrix}$$

- 5 Express each of the following products as a single matrix.

$$\text{(a) } \begin{pmatrix} 5 & 1 \\ 8 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

$$\text{(b) } (2; 5) \begin{pmatrix} 6 & 2 & 3 \\ 1 & 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 9 \\ -2 & 5 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 3 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 4 & 9 \\ -2 & 5 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 3 \\ -1 & 2 \end{pmatrix}^2$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 0 \\ 5 & -1 & 6 \end{pmatrix}$$

Algebra of 2×2 matrices

Zero matrix (null matrix)

Any 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is pre-multiplied or post-multiplied by the matrix $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, it is reduced to the matrix $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

Check that:

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

and

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ is called the **zero matrix** or **null matrix**.

Identity matrix

If any 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is pre-multiplied or post-multiplied by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ it remains unchanged.

Check that:

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

and

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is called the **identity matrix**. It is given the symbol **I**.

Inverse of a 2×2 matrix

If $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ is a matrix such that

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \mathbf{I} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix}$$

then $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ is the **inverse** of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\text{If } \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\text{then } pa + qc = 1 \quad (1)$$

$$pb + qd = 0 \quad (2)$$

$$ra + sc = 0 \quad (3)$$

$$rb + sd = 1 \quad (4)$$

$$(1) \times d: pad + qcd = d \quad (5)$$

$$(2) \times c: pbc + qcd = 0 \quad (6)$$

$$(5) - (6): pad - pbc = d$$

$$\Leftrightarrow p(ad - bc) = d$$

$$p = \frac{d}{ad - bc}$$

Similarly

$$q = \frac{-b}{ad - bc}$$

and

$$r = \frac{-c}{ad - bc}$$

and

$$s = \frac{a}{ad - bc}$$

Hence the inverse of any 2×2 matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is

$$\frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$ad - bc$ is called the **determinant** of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

To find the inverse of a 2×2 matrix:

- 1 interchange the top left-hand and bottom right-hand elements;
- 2 multiply the other two elements by -1 ;
- 3 divide the resulting matrix by the determinant of the original matrix.

Example 3

Find the inverse of

(a) $\begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$, (b) $\begin{pmatrix} 2 & 5 \\ 3 & 8 \end{pmatrix}$, (c) $\begin{pmatrix} 6 & 3 \\ 2 & 1 \end{pmatrix}$

(a) The determinant of $\begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix}$ is

$$3 \times 1 - 4 \times (-2) = 3 + 8 = 11$$

Its inverse is $\frac{1}{11} \begin{pmatrix} 1 & 2 \\ -4 & 3 \end{pmatrix}$

$$\begin{aligned} \text{Check: } \frac{1}{11} \begin{pmatrix} 1 & 2 \\ -4 & 3 \end{pmatrix} \begin{pmatrix} 3 & -2 \\ 4 & 1 \end{pmatrix} \\ = \frac{1}{11} \begin{pmatrix} 11 & 0 \\ 0 & 11 \end{pmatrix} \\ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

(b) The determinant of $\begin{pmatrix} 2 & 5 \\ 3 & 8 \end{pmatrix}$ is

$$2 \times 8 - 3 \times 5 = 16 - 15 = 1$$

Its inverse is $\frac{1}{1} \begin{pmatrix} 8 & -5 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 8 & -5 \\ -3 & 2 \end{pmatrix}$

$$\text{Check: } \begin{pmatrix} 8 & -5 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 5 \\ 3 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

(c) The determinant of $\begin{pmatrix} 6 & 3 \\ 2 & 1 \end{pmatrix}$ is

$$6 \times 1 - 2 \times 3 = 0$$

The inverse of the given matrix would contain the fraction $\frac{1}{0}$. Since division by 0 is impossible

it follows that $\begin{pmatrix} 6 & 3 \\ 2 & 1 \end{pmatrix}$ has *no* inverse.

Notice in Example 3 that a matrix whose determinant is zero has no inverse. Such matrices are called **singular** matrices.

Exercise 13b

Find the inverses of the following matrices where possible. Use multiplication to check each result.

1 $\begin{pmatrix} 6 & 3 \\ 1 & 2 \end{pmatrix}$

2 $\begin{pmatrix} 5 & 3 \\ 2 & 3 \end{pmatrix}$

3 $\begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix}$

4 $\begin{pmatrix} 4\frac{1}{2} & 3 \\ 5 & 3 \end{pmatrix}$

5 $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$

6 $\begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$

7 $\begin{pmatrix} 6 & 14 \\ 3 & 7 \end{pmatrix}$

8 $\begin{pmatrix} -5 & 3 \\ -1 & 4 \end{pmatrix}$

9 $\begin{pmatrix} 2 & -9 \\ 8 & -6 \end{pmatrix}$

10 $\begin{pmatrix} -5 & -4 \\ -2 & 0 \end{pmatrix}$

11 $\frac{1}{2} \begin{pmatrix} 4 & 1 \\ 16 & 3 \end{pmatrix}$

12 $\frac{1}{6} \begin{pmatrix} 0 & -3 \\ 2 & 0 \end{pmatrix}$

Matrices as operators

Simultaneous linear equations

If $\begin{pmatrix} 9 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 17 \\ 4 \end{pmatrix}$

then $\begin{pmatrix} 9x + 4y \\ 2x + y \end{pmatrix} = \begin{pmatrix} 17 \\ 4 \end{pmatrix}$

or $9x + 4y = 17$ (1)

and $2x + y = 4$ (2)

Hence the simultaneous equations (1) and (2) can be written as a single matrix equation:

$$\begin{pmatrix} 9 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 17 \\ 4 \end{pmatrix} \quad (3)$$

In (3) the matrix $\begin{pmatrix} 9 & 4 \\ 2 & 1 \end{pmatrix}$ multiplies $\begin{pmatrix} x \\ y \end{pmatrix}$.

Multiplication is an arithmetical operation; we say that the matrix acts as an **operator**.

The inverse of $\begin{pmatrix} 9 & 4 \\ 2 & 1 \end{pmatrix}$ is $\begin{pmatrix} 1 & -4 \\ -2 & 9 \end{pmatrix}$.

Pre-multiply both sides of (3) by this inverse matrix:

$$\begin{pmatrix} 1 & -4 \\ -2 & 9 \end{pmatrix} \begin{pmatrix} 9 & 4 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ -2 & 9 \end{pmatrix} \begin{pmatrix} 17 \\ 4 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\Leftrightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

or $x = 1$
and $y = 2$

Example 4

Solve the equations $3x - 4y = 1$ and $7x + y = 23$.

$$3x - 4y = 1$$

$$7x + y = 23$$

$$\Leftrightarrow \begin{pmatrix} 3 & -4 \\ 7 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 23 \end{pmatrix} \quad (1)$$

The determinant of $\begin{pmatrix} 3 & -4 \\ 7 & 1 \end{pmatrix}$ is:

$$3 \times 1 - 7 \times (-4) = 31$$

$$\text{Its inverse is } \frac{1}{31} \begin{pmatrix} 1 & 4 \\ -7 & 3 \end{pmatrix}.$$

Pre-multiply both sides of (1) by this inverse:

$$\frac{1}{31} \begin{pmatrix} 1 & 4 \\ -7 & 3 \end{pmatrix} \begin{pmatrix} 3 & -4 \\ 7 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{31} \begin{pmatrix} 1 & 4 \\ -7 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 23 \end{pmatrix}$$

$$\frac{1}{31} \begin{pmatrix} 31 & 0 \\ 0 & 31 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{31} \begin{pmatrix} 93 \\ 62 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

or $x = 3$
and $y = 2$

Check: $3 \times 3 - 4 \times 2 = 1$ and $7 \times 3 + 2 = 23$.

Exercise 13c

Use the matrix method to solve the following pairs of simultaneous equations.

1 $6x + 11y = 29$

$$x + 2y = 5$$

3 $2x + y = 3$

$$x + 2y = 9$$

5 $4x - 2y = 9$

$$x + y = 3$$

7 $2a - 3b = 3$

$$a + b = 4$$

9 $15s - 6t = 4$

$$6s + 18t = 5$$

2 $7x + 4y = 29$

$$4x + 2y = 16$$

4 $21x - 2y = 15$

$$13x - y = 10$$

6 $x - 3y = 2$

$$2x + 4y = -1$$

8 $\frac{1}{2}u - \frac{1}{3}v = 4$

$$\frac{1}{3}u + \frac{1}{2}v = 7$$

10 $6c - d = -2$

$$10c - 3d = -10$$

Further examples of the use of matrices as operators are given in the next chapter.

Exercise 13d (miscellaneous practice)

1 Evaluate as a single matrix

$$\begin{pmatrix} -3 & -8 \\ 6 & 2 \end{pmatrix} + 2 \begin{pmatrix} -2 & 1 \\ -3 & 5 \end{pmatrix}$$

2 If $\mathbf{M} = \begin{pmatrix} 2 & -6 \\ -1 & 4 \end{pmatrix}$, (a) find the value of the determinant of $\mathbf{M}$, (b) hence write down the inverse of $\mathbf{M}$.

3 Find the value of the determinant of the matrix $\begin{pmatrix} -2 & -4 \\ 5 & 3 \end{pmatrix}$. Hence write down the inverse of the matrix.

4 The value of the determinant of the matrix $\begin{pmatrix} 5 & -2 \\ -4 & x \end{pmatrix}$ is 7.

(a) Find the value of x .

(b) Hence write down the inverse of the matrix.

- 5 Find the value of k for which the matrix $\begin{pmatrix} 4 & k-2 \\ 8 & 6 \end{pmatrix}$ does not have an inverse.

- 6 If $\mathbf{A} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{B} = (-2 \ 5 \ 0)$,

evaluate

(a) $\mathbf{AB}$,

(b) $\mathbf{BA}$.

- 7 Express each of the following as a single matrix.

(a) $\begin{pmatrix} 1 & 5 \\ 4 & 0 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 3 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 3 & 3 \\ 0 & 3 & 3 & 0 \end{pmatrix}$

- 8 Given that

$$\begin{pmatrix} 3 & 2 & 4 \\ 6 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ m \\ -3 \end{pmatrix} = \begin{pmatrix} 10 \\ 7n \end{pmatrix}$$

find the values of m and n .

- 9 $\mathbf{P}$ is a 2×2 matrix such that

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \mathbf{P} - \mathbf{P} = \begin{pmatrix} -2 & 0 \\ 2 & 4 \end{pmatrix}$$

Find the matrix $\mathbf{P}$.

- 10 $\mathbf{M} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$ and $\mathbf{N} = \begin{pmatrix} -1 & 1 \\ 0 & 3 \end{pmatrix}$.

(a) Find $\mathbf{M} - 2\mathbf{N}$.

(b) Find the values of p and q if

$$\mathbf{M} \begin{pmatrix} 4 \\ p \end{pmatrix} = \mathbf{N} \begin{pmatrix} q \\ 6 \end{pmatrix}$$

- 11 Given that

$$\begin{pmatrix} 0 & 1 \\ -2 & 0 \end{pmatrix} \begin{pmatrix} q-7 \\ p & 0 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} + \begin{pmatrix} 0 & -2 \\ 6 & 3r \end{pmatrix}$$

find the value of p , of q and of r . [Camb]

- 12 Find a and b if

$$\begin{pmatrix} 3 & 7 \\ b & a \end{pmatrix} \begin{pmatrix} a & -7 \\ -2 & 3 \end{pmatrix} = \mathbf{I}$$

where $\mathbf{I}$ is the identity matrix.

- 13 $\mathbf{A} = \begin{pmatrix} 4 & 2 \\ 0 & 3 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} \frac{1}{4} & k \\ 0 & \frac{1}{3} \end{pmatrix}$ and

$$\mathbf{C} = \begin{pmatrix} 12 & 4 \\ -9 & m \end{pmatrix}.$$

(a) Evaluate $\mathbf{A}^2$.

(b) Find the value of k which makes $\mathbf{AB}$ the identity matrix.

(c) Find the value of m which makes the determinant of $\mathbf{A}$ equal to the determinant of $\mathbf{C}$. [Camb]

- 14 (a) Write down the inverse of the matrix

$$\begin{pmatrix} 3 & -4 \\ 5 & 7 \end{pmatrix}$$

(b) Hence find x and y if

$$\begin{pmatrix} 3 & -4 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 10 \\ 3 \end{pmatrix}$$

- 15 Express the simultaneous equations

$$3y = -5x + 3$$

$$2y = -3x - 1$$

as a single matrix equation in the form

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} p \\ q \end{pmatrix}$$

where a, b, c, d, p, q are integers.

Hence find the values of x and y .

Geometrical transformations (3)

Transformations and matrices

Translation

In Fig. 14.1 A is any point $(p; q)$ on the cartesian plane. OA or **a** is the **position vector** of A:

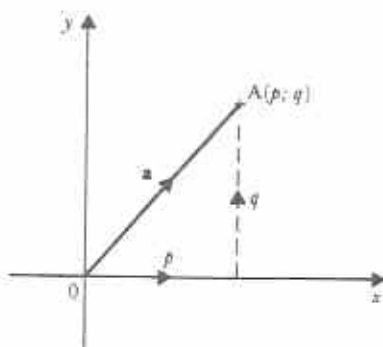


Fig. 14.1

$\vec{OA} = \mathbf{a} = \begin{pmatrix} p \\ q \end{pmatrix}$ = the displacement of A from the origin.

If A is the point $(1; 2)$ and it is translated by vector $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$, its final position B is shown in Fig. 14.2.

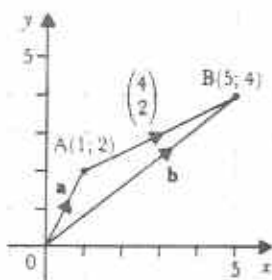


Fig. 14.2

The position of B is calculated by adding the translation vector to the position vector of A:

$$\vec{OB} = \mathbf{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

Translation is a geometrical transformation often represented by the letter T. Hence if T represents the translation $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$ then $T(\mathbf{a}) = \mathbf{b}$. Similarly,

$$\begin{aligned} T(\mathbf{b}) &= \mathbf{b} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} \\ &= \begin{pmatrix} 9 \\ 6 \end{pmatrix} = \mathbf{c} \end{aligned}$$

Hence $T(T(\mathbf{a})) = \mathbf{c}$.

This is usually written as $T^2(\mathbf{a}) = \mathbf{c}$.

Notice that if the cartesian plane is given a translation T then every point on the plane is translated through T.

Rotation

Fig. 14.3 shows successive rotations of a **unit square** through 90° clockwise about the origin. [Note: a unit square has vertices at $(0; 0)$, $(1; 0)$, $(1; 1)$, $(0; 1)$]

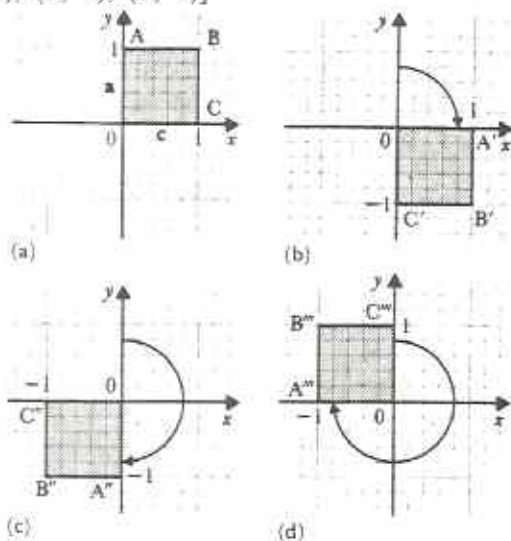


Fig. 14.3

In Fig. 14.3(a) the position vectors of the vertices A, B, C are $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ where

$$\mathbf{a} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

If R represents a rotation through 90° clockwise about the origin, then from Fig. 14.3, parts (a) and (b):

$$R(\mathbf{a}) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (1)$$

$$R(\mathbf{c}) = \begin{pmatrix} 0 \\ -1 \end{pmatrix} \quad (2)$$

It is possible to represent R by a matrix. Let this matrix be $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$.

From (1) and (2) above:

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{and } \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

By multiplying out the matrices,

$$0p + 1q = 1$$

$$\Leftrightarrow q = 1$$

Similarly $s = 0$, $p = 0$, $r = -1$.

$$\text{Hence } R = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

In Fig. 14.3(c), R^2 represents R followed by R , i.e. a rotation of 180° about 0.

$$R^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

In Fig. 14.3(d), R^3 represents R followed by R followed by R , i.e. a clockwise rotation of 270° about 0.

$$R^3 = R \times R^2$$

$$= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

R^4 represents four successive clockwise rotations of 90° , i.e. a rotation of 360° .

$$R^4 = R \times R^3 \text{ (or } R^2 \times R^2)$$

$$\begin{aligned} &= \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \text{ or } \\ &\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ in both cases.} \end{aligned}$$

Hence R^4 is equivalent to the identity matrix $\mathbf{I}$, as might be expected.

The matrices $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$,

$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ represent clockwise rotations of 90° , 180° , 270° , 360° about the origin.

Notice that if the cartesian plane is given a rotation R then every point, except the centre of rotation, is rotated through R . The centre of rotation is said to be an **invariant** point. *Invariant* means 'unchanging'.

Reflection

In Fig. 14.4, the unit square in part (a) is shown (b) reflected in the x -axis, and (c) reflected in the y -axis.

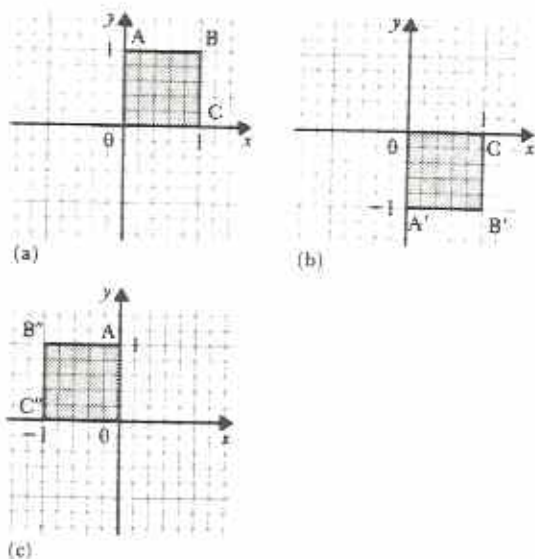


Fig. 14.4

If M represents the reflection in the x -axis, then by considering what happens to OA and OC ,

$$M(OA) = OA' \\ \text{and } M(OC) = OC$$

If M is the matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$, then

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1)$$

$$\text{and } \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad (2)$$

By multiplying out the matrices,
 $q = 0, s = -1, p = 1$ and $r = 0$.

$$\text{Hence } M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Similarly, by considering Fig. 10.4(c), if N represents reflection in the y -axis it can be shown that

$$N = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

Notice that

$$M^2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$N^2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Hence $M^2 = N^2 = I$ as might be expected.

If the cartesian plane is given a reflection M then every point on the plane, except those on the line of reflection is transformed. The line of reflection is an **invariant line**.

Example 1

The vertices of a triangle are $A(1; 2)$, $B(3; 1)$ and $C(-2; 1)$. If $\triangle ABC$ is reflected in the x -axis, calculate the coordinates of the vertices of its image.

The matrix M represents reflection in the x -axis where

$$M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

If the image of $\triangle ABC$ is $\triangle A'B'C'$, then

$$M(OA) = OA' \quad (1)$$

$$M(OB) = OB' \quad (2)$$

$$M(OC) = OC' \quad (3)$$

$$\text{From (1), } OA' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\text{From (2), } OB' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$\text{From (3), } OC' = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

The vertices of the image of $\triangle ABC$ are $A'(1; -2)$, $B'(3; -1)$, $C'(-2; -1)$.

Note: The working in Example 1 can be written more neatly by representing the vertices of $\triangle ABC$ by a single matrix,

$$\begin{pmatrix} 1 & 3 & -2 \\ 2 & 1 & 1 \end{pmatrix}.$$

Then, by matrix multiplication,

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} A & B & C \\ 1 & 3 & -2 \\ 2 & 1 & 1 \end{pmatrix} = \begin{pmatrix} A' & B' & C' \\ 1 & 3 & -2 \\ -2 & -1 & -1 \end{pmatrix}$$

Example 2

Triangle XYZ has vertices at $X(3; 4)$, $Y(5; -1)$, $Z(-2; 2)$. Find the coordinates of the image of X , Y , Z if the triangle is first translated by vector $\begin{pmatrix} -2 \\ 7 \end{pmatrix}$ and then rotated through 180° about the origin.

Let $\triangle X'Y'Z'$ be the image of $\triangle XYZ$ after translation $\begin{pmatrix} -2 \\ 7 \end{pmatrix}$.

$$OX' = \begin{pmatrix} 3 \\ 4 \end{pmatrix} + \begin{pmatrix} -2 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 11 \end{pmatrix}$$

$$OY' = \begin{pmatrix} 5 \\ -1 \end{pmatrix} + \begin{pmatrix} -2 \\ 7 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$

$$OZ' = \begin{pmatrix} -2 \\ 2 \end{pmatrix} + \begin{pmatrix} -2 \\ 7 \end{pmatrix} = \begin{pmatrix} -4 \\ 9 \end{pmatrix}$$

Let $\triangle X''Y''Z''$ be the image of $\triangle X'Y'Z'$ after rotation through 180° about the origin.

$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$ is the matrix of rotation:

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 3 & -4 \\ 11 & 6 & 9 \end{pmatrix} =$$

$$\begin{pmatrix} X'' & Y'' & Z'' \\ -1 & -3 & 4 \\ -11 & -6 & -9 \end{pmatrix}$$

The vertices of the final image of $\triangle XYZ$ are at $X''(-1; -11)$, $Y''(-3; -6)$ and $Z''(4; -9)$.

Exercise 14a

1 Copy and complete Table 14.1.

Table 14.1

transformation	matrix
identity	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
reflection in x -axis	
reflection in y -axis	
rotation of 180° about origin	

- Find the matrices which are equivalent to anticlockwise rotations of (a) 90° , (b) 270° about the origin.
- A triangle has vertices $(1; 1)$, $(2; 4)$ and $(3; 7)$. Find the coordinates of the images of its vertices if it is rotated through 90° clockwise about the origin.
- $P'Q'$ is the image of a line PQ after a translation of $\begin{pmatrix} 7 \\ -4 \end{pmatrix}$. If the coordinates of P' are $(6; 1)$, calculate the coordinates of P .
- A triangle has vertices $A(0; 0)$, $B(1; -1)$ and $C(1; 1)$. Calculate the position of its vertices after a rotation of 180° about the origin.
- Reflect the triangle of question 5 about the y -axis. Compare your answer with that of question 5.

7 A rhombus has vertices $(2; 2)$, $(1; -1)$, $(-2; -2)$ and $(-1; 1)$. Find the coordinates of its vertices when it is reflected in (a) the x -axis, (b) the y -axis.

8 In Fig. 14.5 there are three triangles, ABC , $A_1B_1C_1$ and $A_2B_2C_2$.

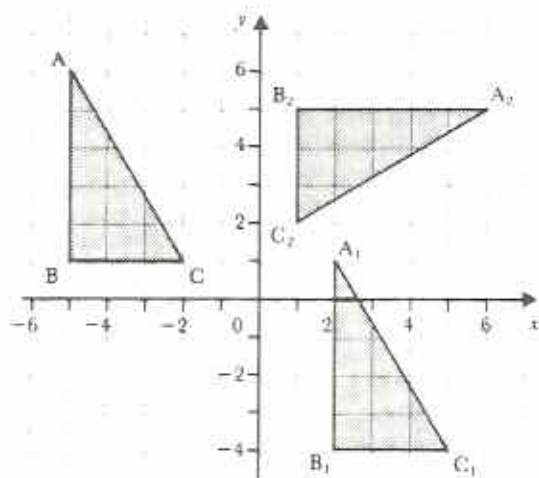


Fig. 14.5

- $A_1B_1C_1$ is the image of ABC under a single translation. Write down the column vector of this translation.
 - $A_2B_2C_2$ is the image of ABC under an anticlockwise rotation about the origin. Write down (i) the angle of the rotation, (ii) the matrix of the rotation.
 - $A_3B_3C_3$ (not shown in the diagram) is the image of ABC under a reflection represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. Write down the equation of the straight line through B_3 and C_3 .
- $\triangle XYZ$ has vertices at $X(1; 1)$, $Y(4; 2)$ and $Z(2; 3)$. Find the coordinates of the image of X , Y , Z if $\triangle XYZ$ is first rotated through 180° about the origin and then translated through $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$.
 - T is the translation $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ and R is a clockwise rotation of 90° about the origin.

A is the point $(-5; 2)$ and B is the point $(4; -3)$. Find the coordinates of

- (a) $T(A)$ (b) $R(B)$
 (c) the point C which when given translation T followed by rotation R has its image at $(6; -2)$.

Enlargement

In Fig. 14.6 $\triangle OAB$ is enlarged to $\triangle OPQ$ by scale factor 2 with the origin as the centre of enlargement.

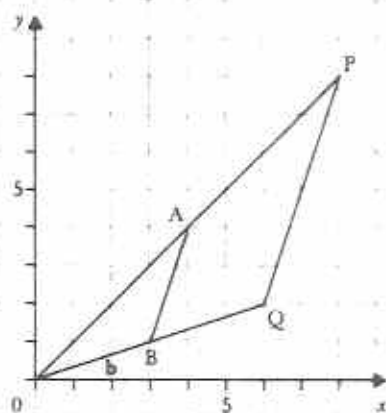


Fig. 14.6

From Fig. 14.6, $\mathbf{a} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$.

$$\mathbf{OP} = \begin{pmatrix} 8 \\ 4 \end{pmatrix} \text{ and } \mathbf{OQ} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$$

If operator E represents the enlargement, then $E(\mathbf{a}) = \mathbf{OP}$ and $E(\mathbf{b}) = \mathbf{OQ}$.

Let E be the matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$, then

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

$$\text{and } \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 3 \end{pmatrix}$$

from which $p = 2$, $q = 0$, $r = 0$, $s = 2$,

$$\text{and } E = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 2\mathbf{I}$$

In general, for any enlargement E with scale factor k and centre $(0; 0)$,

$$E = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = k \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = k\mathbf{I}$$

If the cartesian plane is given an enlargement, there is only one invariant point: the centre of enlargement.

Example 3

Quadrilateral OABC has vertices $O(0; 0)$, $A(6; -1)$, $B(-4; 2)$, $C(9; 9)$. OABC is enlarged with scale factor $-\frac{1}{3}$ with the origin as centre. Find the coordinates of its enlargement $O'A'B'C'$.

Enlargement matrix

$$= -\frac{1}{3}\mathbf{I} = -\frac{1}{3} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{3} & 0 \\ 0 & -\frac{1}{3} \end{pmatrix}$$

$$\begin{pmatrix} -\frac{1}{3} & 0 \\ 0 & -\frac{1}{3} \end{pmatrix} \begin{pmatrix} O & A & B & C \\ 0 & 6 & -4 & 9 \\ 0 & -1 & 2 & 9 \end{pmatrix}$$

$$\begin{pmatrix} O' & A' & B' & C' \\ 0 & -2 & \frac{4}{3} & -3 \\ 0 & \frac{1}{3} & -\frac{2}{3} & -3 \end{pmatrix}$$

The enlargement has coordinates $O'(0; 0)$, $A'(-2; \frac{1}{3})$, $B'(\frac{4}{3}; -\frac{2}{3})$ and $C'(-3; -3)$.

Shear

In Fig 14.7 the unit square OABC is mapped onto the parallelogram ODEC by a shear parallel to the x-axis.

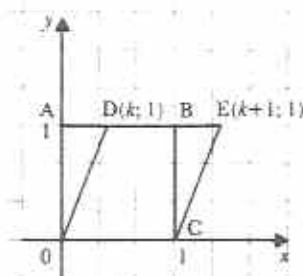


Fig. 14.7

Points on the x-axis remain where they are, while points above move by an amount proportional to their distance from the x-axis. If D

is the point $(k; 1)$, the shearing matrix, H , can be found as before:

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\text{and } \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} k \\ 1 \end{pmatrix}$$

from which $p = 1$, $r = 0$, $q = k$ and $s = 1$.

H is the matrix $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$.

This can be checked by considering the effect of the shear on point B :

$$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1+k \\ 1 \end{pmatrix}, \text{ as shown in Fig. 14.7.}$$

Just as $\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ is a shear parallel to the x -axis

with the x -axis invariant, so the matrix $\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$

is a shear parallel to the y -axis with the y -axis invariant.

Example 4

G is a transformation represented by the matrix

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}.$$

- Find the image of $(4; 3)$ under G .
- Find the image of $(4; -3)$ under G .
- Describe, completely, the transformation G .

$$(a) \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$$

The image of $(4; 3)$ is $(-2; 3)$.

$$(b) \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \end{pmatrix} = \begin{pmatrix} 10 \\ -3 \end{pmatrix}$$

The image of $(4; -3)$ is $(10; -3)$.

(c) Fig. 14.8 is a sketch showing how the transformation changes the line joining the given points.

From Fig. 14.8, G is a shear with the x -axis invariant. The shear factor is -2 which produces shearing to the left above the x -axis.

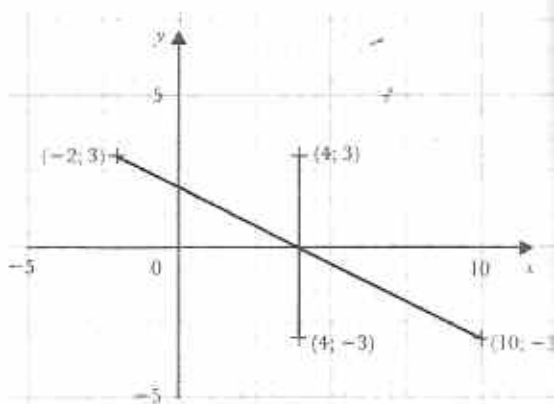


Fig. 14.8

Stretch

Consider the effect on the unit square of the operator $\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}$.

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The effect is to stretch the square in two directions to form a rectangle as shown in Fig. 14.9.

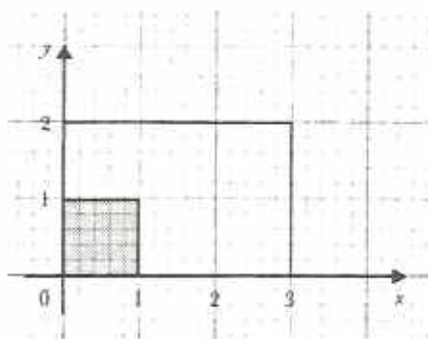


Fig. 14.9

Notice that the square is stretched by a factor of 3 in the direction of increase in x and a factor of 2 in the direction of increase in y . In general, a stretch, S , is given by

$$S = \begin{pmatrix} h & 0 \\ 0 & k \end{pmatrix}$$

which is a two-way stretch by factor h in the x -direction and k in the y -direction.

Notice that the origin is always mapped onto itself when multiplied by a 2×2 matrix.

Example 5

$A'(0;0)$, $B'(-5;6)$, $C'(-2;-9)$ are the images of $A(0;0)$, $B(5;2)$, $C(2;-3)$ under a transformation

represented by a matrix of the form $\begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$.

(a) Find the transformation matrix. (b) Find the matrix which will transform $\triangle A'B'C'$ back to $\triangle ABC$. (c) Complete the two matrices.

$$(a) \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} 5 \\ 2 \end{pmatrix} = \begin{pmatrix} -5 \\ 6 \end{pmatrix} \quad (1)$$

$$\text{and } \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \begin{pmatrix} 2 \\ -3 \end{pmatrix} = \begin{pmatrix} -2 \\ -9 \end{pmatrix} \quad (2)$$

From (1), $5a = -5$

$$\Leftrightarrow a = -1$$

$$\text{and } 2b = 6$$

$$\Leftrightarrow b = 3$$

The transformation matrix is $\begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$, a stretch. Check this result in equation (2).

(b) Let the matrix $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ transform $\triangle A'B'C'$ into $\triangle ABC$. Hence:

$$\begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} -5 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \end{pmatrix} \quad (3)$$

$$\text{and } \begin{pmatrix} p & q \\ r & s \end{pmatrix} \begin{pmatrix} -2 \\ -9 \end{pmatrix} = \begin{pmatrix} 2 \\ -3 \end{pmatrix} \quad (4)$$

$$\text{From (3): } -5p + 6q = 5$$

$$-5r + 6s = 2$$

$$\text{From (4): } -2p - 9q = 2$$

$$-2r - 9s = -3$$

The solution of these equations gives

$$p = -1, q = 0, r = 0, s = \frac{1}{3}$$

The matrix $\begin{pmatrix} -1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix}$ will transform $\triangle A'B'C'$ to $\triangle ABC$.

(c) Let the matrices in parts (a) and (b) be S and T respectively.

$$S = \begin{pmatrix} -1 & 0 \\ 0 & 3 \end{pmatrix}$$

$$T = \begin{pmatrix} -1 & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = -\frac{1}{3} \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}$$

= inverse of S .

Each matrix is the inverse of the other.

Notice in Example 5 that if a shape is mapped onto an image by a matrix A , then the image can be mapped back onto the original shape by the inverse of A , sometimes written as A^{-1} . Remember that $AA^{-1} = I$.

Exercise 14b

1 Use the matrix $\begin{pmatrix} -3 & 0 \\ 0 & -3 \end{pmatrix}$ to enlarge $\triangle ABC$ in Fig. 14.10.

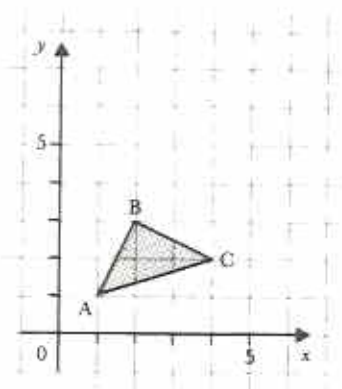


Fig. 14.10

2 Use the matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$ to shear $\triangle ABC$ in Fig. 14.10. Find the matrix which will map the image back onto $\triangle ABC$.

- 3 Use the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$ to transform $\triangle ABC$ in Fig. 14.10. Describe the transformation as fully as possible.
- 4 Find the matrix E which has the effect of enlarging plane shapes by a scale factor $1\frac{1}{2}$ with the origin as centre of enlargement.
- 5 Use E of question 4 to enlarge the rectangle whose vertices are $(0; 0)$, $(3; 0)$, $(0; 2)$, $(3; 2)$.
- 6 The matrix $\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$ represents a transformation H .
- (a) Find the image of $(2; 5)$ under H .
- (b) Find the image of $(-2; 5)$ under H .
- (c) Describe H as fully as possible.
- 7 In Fig. 14.11 $\triangle KLM$ is mapped onto $\triangle K'L'M'$ by a two-way stretch, S .

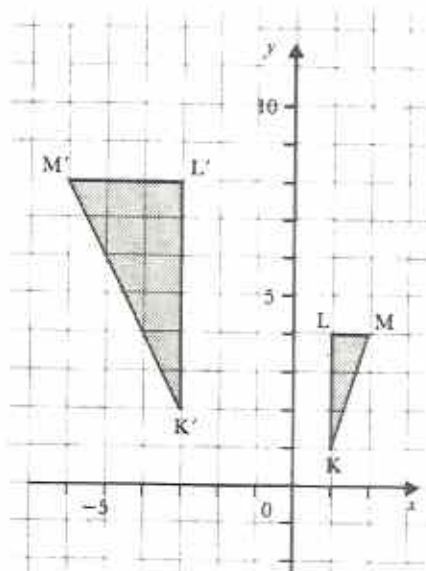


Fig. 14.11

- (a) What is the stretch factor in the x -direction?
- (b) What is the stretch factor in the y -direction?
- (c) Hence, or otherwise, find the matrix which represents S .
- 8 (a) A single transformation U maps $\triangle PQR$ of Fig. 14.12 onto $\triangle P_1Q_1R_1$ which has vertices $P_1(0; 16)$, $Q_1(12; 20)$, $R_1(8; 2)$. Describe U , writing down the matrix which represents the transformation.

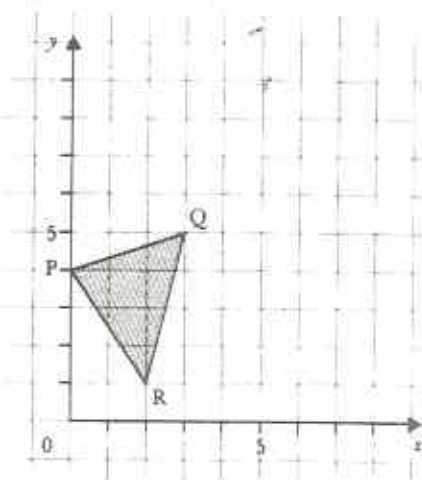


Fig. 14.12

- (b) $\triangle P_2Q_2R_2$ is the image of $\triangle PQR$ under a shear with the x -axis as the invariant line. If R_2 is the point $(6; 1)$ find the coordinates of P_2 and Q_2 .
- 9 A shear S is represented by the matrix $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$.
- (a) Calculate the coordinates of the image of the point $(2; -2)$ under S .
- (b) Calculate the coordinates of the point which will be mapped onto $(7; 4)$ by S .
- (c) Write down the equation of the invariant line. [Camb]
- 10 $\triangle PQR$ has vertices $P(0; 0)$, $Q(2; 1)$, $R(-1; 3)$. Find the coordinates of the images of P , Q , R if the triangle is enlarged by scale factor 2 with origin as centre. The enlargement is rotated through 90° clockwise about the origin. Calculate the coordinates of the final image of $\triangle PQR$.

Combined transformations

Example 6

Triangle OAB has vertices at $O(0; 0)$, $A(2; 1)$, $B(-1; 3)$. Find the vertices of the triangle if it is first enlarged by E , then translated through T where

$$E = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \text{ and } T = \begin{pmatrix} -3 \\ -5 \end{pmatrix}.$$

First, enlargement by E:

$$\begin{matrix} & O & A & B & & O' & A' & B' \\ \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} & \begin{pmatrix} 0 & 2 \\ 0 & 1 \end{pmatrix} & \begin{pmatrix} 2 & -1 \\ 1 & 3 \end{pmatrix} & = & \begin{pmatrix} 0 & 4 \\ 0 & 2 \end{pmatrix} & \begin{pmatrix} 4 & -2 \\ 2 & 6 \end{pmatrix} \end{matrix}$$

Second, translation of points O' , A' , B' through T:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} -3 \\ -5 \end{pmatrix} = \begin{pmatrix} -3 \\ -5 \end{pmatrix}$$

$$\begin{pmatrix} 4 \\ 2 \end{pmatrix} + \begin{pmatrix} -3 \\ -5 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$$

$$\begin{pmatrix} -2 \\ 6 \end{pmatrix} + \begin{pmatrix} -3 \\ -5 \end{pmatrix} = \begin{pmatrix} -5 \\ 1 \end{pmatrix}$$

The vertices of the transformed triangle are $O''(-3; -5)$, $A''(1; -3)$ and $B''(-5; 1)$. The combined transformation is shown in Fig. 14.13.

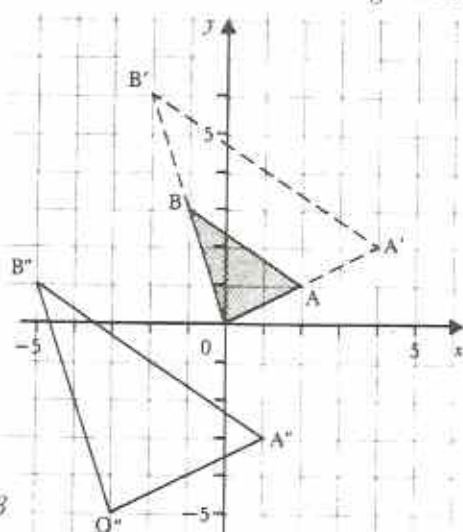


Fig. 14.13

The process in Example 6 can be written as

$$TE(\mathbf{a}) = \mathbf{b}$$

where TE means that E is done *before* T.

$$TE(\mathbf{a}) = \mathbf{b}$$

$$\Leftrightarrow T[E(\mathbf{a})] = \mathbf{b}$$

The order in which the operations are carried out is usually important. In general, $TE \neq ET$. For example, with the data of Example 6:

$$TE\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \end{pmatrix} \text{ as shown above}$$

$$\begin{aligned} \text{but } ET\begin{pmatrix} 2 \\ 1 \end{pmatrix} &= E\left[\begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 \\ -5 \end{pmatrix}\right] \\ &= E\begin{pmatrix} -1 \\ -4 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} -1 \\ -4 \end{pmatrix} = \begin{pmatrix} -2 \\ -8 \end{pmatrix} \end{aligned}$$

Example 7

The rhombus whose vertices are at $(0; 0)$, $(1; 2)$, $(3; 3)$ and $(2; 1)$ is first reflected in the x -axis and then sheared by the operator $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Find the vertices of the resulting figure.

The matrix of reflection is M where

$$M = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ for reflection in the } x\text{-axis.}$$

If H is the shearing matrix, then HM represents the combined transformation (i.e. M before H).

$$HM = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}$$

$\begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix}$ is the combined matrix of transformation:

$$\begin{aligned} \begin{pmatrix} 1 & -2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 3 & 2 \\ 0 & 2 & 3 & 1 \end{pmatrix} \\ = \begin{pmatrix} 0 & -3 & -3 & 0 \\ 0 & -2 & -3 & -1 \end{pmatrix} \end{aligned}$$

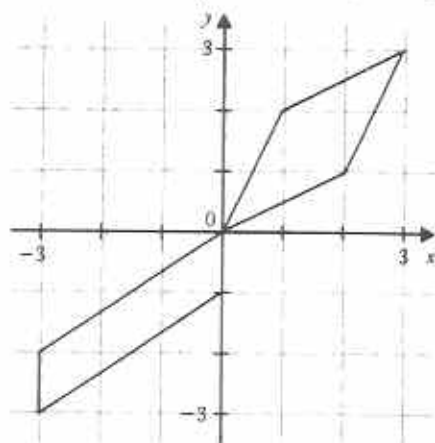


Fig. 14.14

The resulting figure has vertices at $(0;0)$, $(-3;-2)$, $(-3;-3)$ and $(0;-1)$. The effect of the combined transformation on the original rhombus is shown in Fig. 14.14.

Example 8

$(a'; b')$ is the image of a point $(a; b)$ after a transformation given by

$$\begin{pmatrix} a' \\ b' \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} -7 \\ -2 \end{pmatrix}$$

- Describe the transformation in words.
- A' is the image of point $A(-1; 2)$. Find the coordinates of A' .
- Find the coordinates of a point B which has an image at $B'(5; -5)$.

(a) First, the stretching matrix $\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}$ acts on $(a; b)$. Second, the result is translated by vector $\begin{pmatrix} -7 \\ -2 \end{pmatrix}$.

(b) Substituting -1 for a and 2 for b in the given equation:

$$\begin{aligned} \begin{pmatrix} a' \\ b' \end{pmatrix} &= \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} -7 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} -4 \\ 6 \end{pmatrix} + \begin{pmatrix} -7 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} -11 \\ 4 \end{pmatrix} \end{aligned}$$

A' is the point $(-11; 4)$.

(c) Let B have coordinates $(h; k)$, then:

$$\begin{aligned} \begin{pmatrix} 5 \\ -5 \end{pmatrix} &= \begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} h \\ k \end{pmatrix} + \begin{pmatrix} -7 \\ -2 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} 5 \\ -5 \end{pmatrix} &= \begin{pmatrix} 4h \\ 3k \end{pmatrix} + \begin{pmatrix} -7 \\ -2 \end{pmatrix} \\ \Rightarrow \begin{pmatrix} 5 \\ -5 \end{pmatrix} &= \begin{pmatrix} 4h - 7 \\ 3k - 2 \end{pmatrix} \\ \Rightarrow 5 &= 4h - 7 \\ \Leftrightarrow h &= 3 \\ \text{and } -5 &= 3k - 2 \\ \Leftrightarrow k &= -1 \end{aligned}$$

B is the point $(3; -1)$.

Exercise 14c

- Triangle XYZ in Fig. 14.15 is first rotated through 90° anticlockwise about the origin and then sheared by the operator $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

Calculate the coordinates of the vertices of its image.

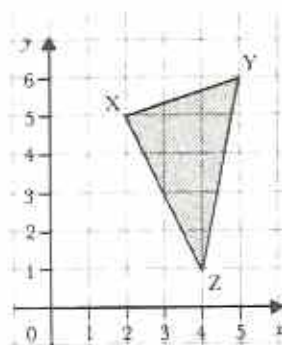


Fig. 14.15

- Use the matrix $\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$ to shear $\triangle XYZ$ in Fig. 14.15. Then reflect the result in the y -axis. Calculate the coordinates of its final image.
- Each of the following equations represents a transformation.

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -3 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

- Describe the transformations in words.
- For each transformation find the image of the point $(-1; 4)$.

- T is a translation $\begin{pmatrix} 2 \\ 8 \end{pmatrix}$ and S is a stretch represented by the matrix $\begin{pmatrix} -2 & 0 \\ 0 & 3 \end{pmatrix}$.

Calculate the image of $A(3; 2)$ under the following transformations:

- $ST(A)$
- $TS(A)$
- $S^{-1}(A)$

- 5 R is a clockwise rotation of 270° about the origin and H is a shear represented by the matrix $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$. Calculate the coordinates

of the point that P(-3;5) is mapped onto by the following combined transformations:

- (a) RH(P) (b) HR(P) (c) $H^2(P)$

- 6 In Fig. 14.16, semicircle A can be mapped onto semicircle B by an anticlockwise rotation about the origin followed by a translation.

- (a) State the angle of rotation.
(b) Find the matrix which represents the rotation.
(c) Find the column vector of the translation.
(d) Given that A can be mapped onto B by a single reflection in a line m , find the equation of m .

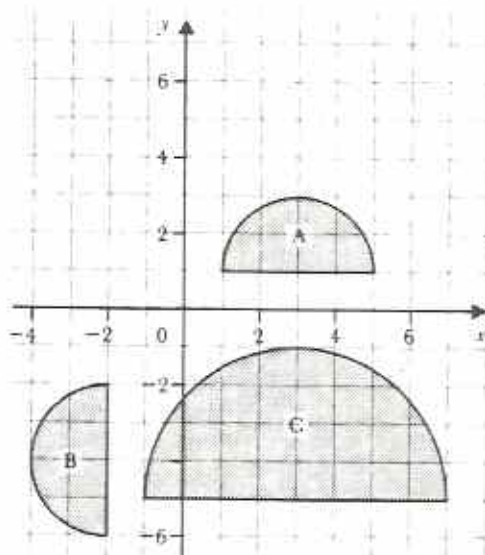


Fig. 14.16

- 7 In Fig. 14.16, semicircle C is the image of semicircle A under a transformation given by $TE(A) = C$, where E is an enlargement with the origin as centre and T is a translation.

- (a) State the scale factor of E.
(b) Write down the matrix representing E.
(c) Express T as a column vector.
(d) The transformation can also be given by $ET'(A) = C$ where T is a different translation and E is the same as before. Express T' as a column vector.

- (c) A can be mapped onto C by a single enlargement with centre $(h; k)$. State the values of h and k .

- 8 The cartesian plane is first translated by vector $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$ and then sheared by matrix

$$\begin{pmatrix} 1 & 0 \\ 3 & 0 \end{pmatrix}.$$

Show that the image of a point $(a; b)$ on the plane is $(a - 1; 3a + b)$. Hence write down the coordinates of the image of the origin under such a transformation.

- 9 $(x'; y')$ is the image of a point $(x; y)$ after a combination of transformations given by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -1 \\ 5 \end{pmatrix}$$

- (a) Find the coordinates of O' , the image of the point $O(0; 0)$.

- (b) Find the coordinates of A' , the image of the points $A(3; 2)$.

- (c) If $B'(7; 8)$ is the image of $B(m; n)$, form two equations which can be solved for m and n . Hence or otherwise find the values of m and n .

- 10 Answer this question on graph paper.

- (a) Using a scale of 1 cm to 1 unit on each axis, draw x and y axes, taking values of x from -8 to 12 and values of y from -6 to 14 . Draw and label the triangle X, with vertices $(2; 4)$, $(4; 4)$ and $(4; 1)$.

- (b) The single transformation U maps the triangle X onto the triangle $U(X)$ which has vertices $(6; 12)$, $(12; 12)$ and $(12; 3)$. Draw and label the triangle $U(X)$ and describe fully the transformation U.

- (c) The transformation R is a clockwise rotation of 90° about the origin. Draw and label the triangle $R(X)$.

- (d) The transformation T is the translation $\begin{pmatrix} -8 \\ 4 \end{pmatrix}$. Draw and label the triangle $T(X)$ and the triangle $RT(X)$.

- (e) The single transformation V is represented by the matrix $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. Draw

and label the triangle $V(X)$ and describe fully the transformation V. [Camb]

Graphs (5) Cubic and inverse functions, sketch graphs

Cubic functions

A **cubic function** of x is an expression in x in which 3 is the highest power of x . For example, $2x^3 + 5x^2 - x - 8$ is a cubic function of x .

Example 1

(a) Draw the graph of $y = x^3$ for values of x from -3 to $+3$. (b) Hence solve the equations $x^3 + 20 = 0$.

(a) Table 15.1 gives the necessary table of values. Notice that additional values of y for $x = \pm \frac{1}{2}$ have been calculated. These will be helpful when drawing the graph.

Table 15.1

x	-3	-2	-1	0	1	2	3	$\pm \frac{1}{2}$
y	-27	-8	-1	0	1	8	27	$\pm \frac{1}{8}$

Fig. 15.1 is the graph of $y = x^3$.

(b) If $x^3 + 20 = 0$, then $x^3 = -20$.
 $x^3 = -20$ when, in Fig. 15.1, $y = -20$.
 From Fig. 15.1, $y = -20$ at $x \approx -2.7$.
 $x = -2.7$ is the approximate solution of $x^3 + 20 = 0$.

Example 2

(a) Draw the graph of $y = x(x-2)(x+2)$ for values of x from -3 to $+3$. (b) Find the values of x at the points where the graph cuts the line $y = x + 2$.

Note: $x(x-2)(x+2) = x(x^2 - 4)$
 $= x^3 - 4x$

Hence $x(x-2)(x+2)$ is a cubic function in x .

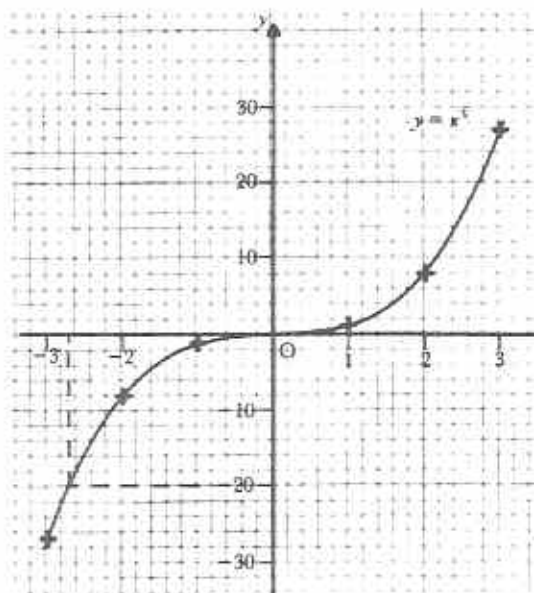


Fig. 15.1

(a) In Table 15.2, the values of y for integral values of x are first calculated. It then be seen that the values of y for $x = \pm \frac{1}{2}$ and $x = \pm \frac{1}{2}$ will be helpful when drawing graph. These values are also calculated.

Table 15.2

x	-3	-2	-1	0	1	2	3	$\pm \frac{1}{2}$
$x-2$	-5	-4	-3	-2	-1	0	1	$\pm \frac{1}{2}$
$x+2$	-1	0	1	2	3	4	5	$\pm \frac{3}{2}$
$y = x(x-2)(x+2)$	-15	0	3	0	-3	0	15	$\pm \frac{3}{4}$

The curve in Fig. 15.2 is the graph of $y = x(x - 2)(x + 2)$.

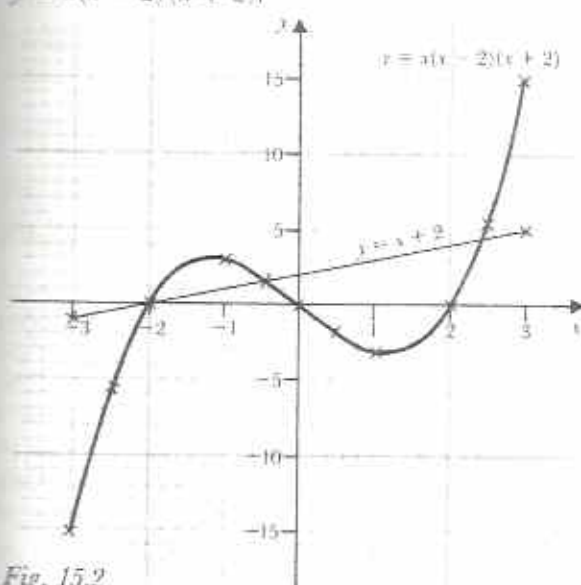


Fig. 15.2

(b) Draw the line $y = x + 2$ by plotting the values in Table 15.3.

Table 15.3

x	-3	0	3
y	-1	2	5

From Fig. 15.2, the values of x at the intersections of the curve and line are $x = -2$; -0.6 and 2.4 .

In part (b), since $y = x(x - 2)(x + 2)$ and $y = x + 2$, the values -2 ; -0.6 and 2.4 are the roots of the equation $x(x - 2)(x + 2) = x + 2$.

Remember that readings from graphs usually give approximate results only. A bigger scale in Fig. 15.2 would give more accurate results.

Example 3

(a) Draw the graph of $y = 2x^2(5 - x)$ in the range $x = 0$ to $x = 5$. Hence find (b) the maximum value of y in the given range, (c) the value of x when y is a maximum.

(a) The graph is drawn by plotting the values in Table 15.4.

Fig. 15.3 is the graph of $y = 2x^2(5 - x)$ within the given range.

Table 15.4

x	0	1	2	3	4	5
$2x^2$	0	2	8	18	32	50
$(5 - x)$	5	4	3	2	1	0
y	0	8	24	36	32	0

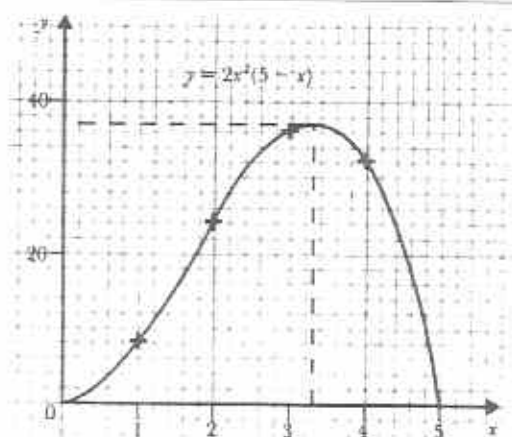


Fig. 15.3

(b) From Fig. 15.3, the maximum value of y is 37 (approximately).

(c) When $y = 37$, $x \approx 3.3$.

Notice the following:

- 1 $y = 2x^2(5 - x) = 10x^2 - 2x^3$. Hence the curve in Fig. 15.3 is that of a cubic function in x . It does *not* have a line of symmetry.
- 2 Fig. 15.4 is a sketch graph of $y = 2x^2(5 - x)$ extended to include values outside the given range.

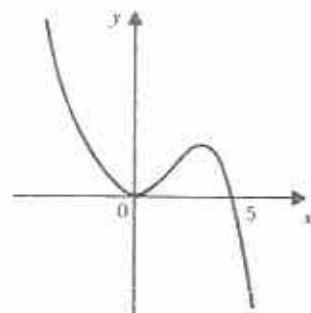


Fig. 15.4

It can be seen that y can have values greater than 37. The value of y found in part (b) is a maximum for the given range only.

Exercise 15a

- (a) Draw the graph of $y = x^3$ for values of x from -4 to $+4$. (b) On the same axes, draw the graph of $y = -x^3$. (c) Use either graph to solve the equation $x^3 + 50 = 0$.
- Solve the equation $x^3 = 5x + 2$ by drawing graphs of $y = x^3$ and $y = 5x + 2$ for values of x between -3 and $+3$.
- (a) Given that $y = x^3 + 2x - 1$, copy and complete Table 15.5.

Table 15.5

x	-3	-2	-1	0	1	2	3
x^3	-27	-8	-1	0	1		
$+2x$	-6	-4	-2	0			
-1	-1	-1	-1				
y	-34	-13	-4				

- (b) Use scales of 2 cm to 1 unit on the x -axis and 2 cm to 10 units on the y -axis and draw the graph of $y = x^3 + 2x - 1$. (c) Hence solve the equation $x^3 + 2x - 1 = 0$.
- Solve graphically the equation $x^3 = 5x - 3$. *Hint: Either use the method of question 2, drawing the graphs of $y = x^3$ and $y = 5x - 3$, or use the method of question 3, drawing the graph of $y = x^3 - 5x + 3$. In either case, use values of x in the range -3 to $+3$.*
 - Solve the equation $x^3 + 3x - 7 = 0$ graphically. Use values of x in the range 0 to 4.
 - Given that $y = x(x - 3)(x + 3)$, copy and complete Table 15.6.

Table 15.6

x	-4	-3	-2	-1	0	1	2	3	$\frac{3}{2}$
$x - 3$	-7	-6	-5	-4	-3				$-\frac{3}{2}$
$x + 3$	-1	0	1	2	3				$6\frac{1}{2}$
y	-28	0	10	8					$11\frac{1}{8}$

- (a) Using scales of 2 cm to 1 unit on the x -axis and 2 cm to 5 units on the y -axis, draw the graph of $y = x(x - 3)(x + 3)$.
- (b) Use the graph to find the roots of the equation $x(x - 3)(x + 3) = 2x + 4$.
- Given that $v = \frac{\pi x^2}{3}(30 - x)$, and taking the value of $\frac{\pi}{3}$ as approximately 1.05, complete

Table 15.7, giving values correct to the nearest whole number.

Table 15.7

x	1	2	3	4	5
v	30		255		656

Draw the graph showing the variation of v as x increases from 1 to 5. Hence find the value of x for which $v = 400$.

- Draw on the same axes the graphs of $y = x^3$ and $y = x(4x - 3)$ for values of x between 0 and 4. Use a scale of 2 cm to 1 unit on the x -axis and 2 cm to 10 units on the y -axis. From your graph, (a) find the values of x satisfying the equation $x^3 - 4x^2 + 3x = 0$; (b) find the range of values of x for which $x^3 < x(4x - 3)$; (c) find the gradient of the curve (i) $y = x^3$ at the point $x = 2$; (ii) $y = x(4x - 3)$ at $x = 2$.
- A particle moves such that its distance, d , from its starting point after t seconds is given by $d = 12t - \frac{1}{4}t^3$. Draw the graph of $d = 12t - \frac{1}{4}t^3$ for values of t from 0 to 5. Use the graph to find (a) the greatest distance that the particle is from its starting point during the first 5 seconds, (b) the time that it takes to return to the starting point.
- A skeleton box on a square base of side x cm is made from 36 cm of wire (Fig. 15.5).

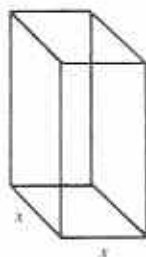


Fig. 15.5

- (a) Find the height of the box in terms of x and hence show that its volume, v , is given by $v = x^2(9 - 2x)$.
- (b) Draw the graph of $v = x^2(9 - 2x)$ for values of x from 0 to 4.
- (c) Hence find (i) the maximum volume of the box, (ii) the dimensions of the box when the volume is a maximum.

Inverse functions

An **inverse function** of x is an expression in which x appears in the denominator of a fraction. For example, $\frac{6}{x}$ and $\frac{2x^2}{1-3x}$ are inverse functions of x .

Example 4

(a) Draw the graph of $y = \frac{6}{x}$ for values of x from -4 to $+2$. (b) Find the values of x at the point where the line $y = 2x + 3$ cuts the graph. (c) Of what equation in x are these values the roots?

(a) In Table 15.8, the values of y for integral values of x are first calculated. The extra values of y for $x = \pm 1\frac{1}{2}$ are then added.

Table 15.8

x	-4	-3	-2	-1	0	1	2	$\pm 1\frac{1}{2}$
$y = \frac{6}{x}$	$-1\frac{1}{2}$	-2	-3	-6		6	3	± 4

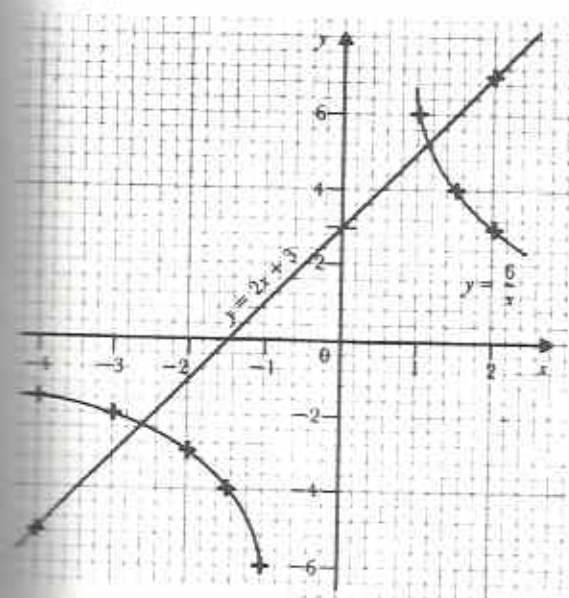


Fig. 15.6

Fig. 15.6 is the graph of $y = \frac{6}{x}$.

Notice that when $x = 0$, $\frac{6}{x}$ is undefined.

Notice that there is a break in continuity of the graph. It is in two branches which are separated by the axes. As x increases towards 0, y decreases in value; as x decreases towards 0, y increases in value.

This kind of curve is called a **hyperbola**.

(b) Draw the line $y = 2x + 3$ by plotting the points in Table 15.9.

Table 15.9

x	-4	0	2
y	-5	3	7

The line cuts the curve where $x \approx -2.6$ and $x \approx 1.1$.

(c) The curve and the line intersect where y simultaneously equals $\frac{6}{x}$ and $2x + 3$. Hence the values of x above are the roots of the equation: $\frac{6}{x} = 2x + 3$, i.e. $2x^2 + 3x - 6 = 0$.

Example 5

(a) Draw the graph of $y = 6x + \frac{20}{x}$ for values of x equal to $\frac{1}{2}$, 1, 2, 3, 4, 5. (b) Find the minimum value of y in the given range. (c) Find the corresponding value of x .

(a) In Table 15.10, values of y are calculated to the nearest whole number.

Table 15.10

x	$\frac{1}{2}$	1	2	3	4	5
$6x$	3	6	12	18	24	30
$\frac{20}{x}$	40	20	10	7	5	4
y	43	26	22	25	29	34

Fig. 15.7 is the required graph. (Note: to save space, the origin has not been included.)

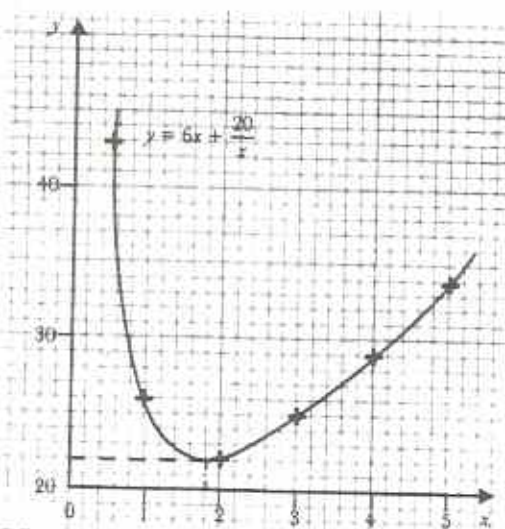


Fig. 15.7

- (b) The minimum value of y is 21.9.
(c) The corresponding value of x is 1.8.

Fig. 15.8 is a sketch graph of $y = 6x + \frac{20}{x}$, showing values outside the range of Example 5.

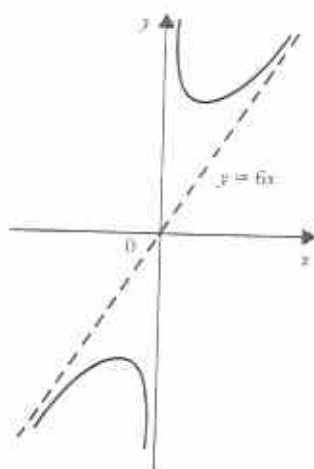


Fig. 15.8

It can be seen that y has values lower than 21.9 when $x < 0$.

Example 6

Solve the equation $2x^2 - x - 4 = 0$ by drawing appropriate inverse and linear graphs within the same axes.

Divide the given equation throughout by x .

$$2x - 1 - \frac{4}{x} = 0$$

Rearrange the resulting equation:

$$2x - 1 = \frac{4}{x}$$

Draw the graphs of $y = 2x - 1$ and $y = \frac{4}{x}$ on the same axes. See Fig. 15.9.

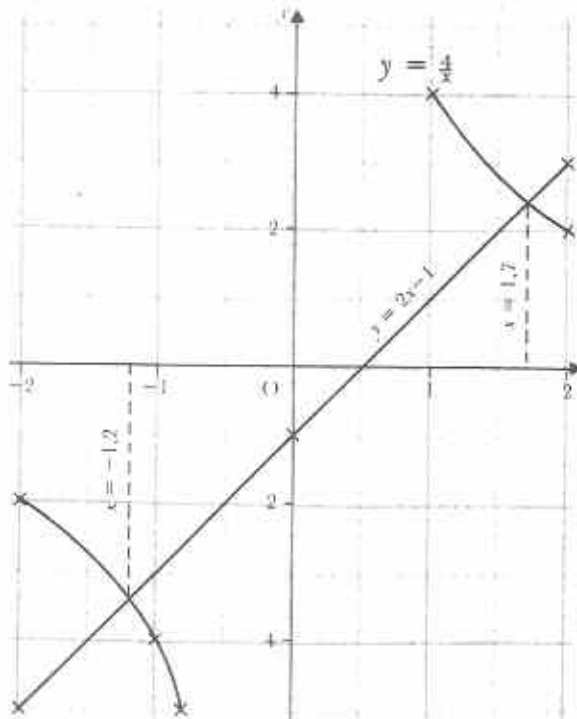


Fig. 15.9

The solution of the given quadratic equation is found by reading the values of x at the points of intersection of the graphs. From Fig. 15.9, the solutions of the equation are

$$x = -1.2 \text{ and } x = +1.7$$

Exercise 15b

- 1 Draw the graphs of (a) $y = \frac{1}{x}$, (b) $y = \frac{1}{x^2}$ for values of x equal to $\pm 4, \pm 2, \pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}$.

- 2 Draw the graphs of $y = \frac{5}{x}$ and $y = x(x-1)$ from $x = \frac{1}{2}$ to $x = 5$. (a) Read off the value of x at the intersection of the curves. (b) Of what equation in x is this value a root?

- 3 (a) Given that $y = x^2 + \frac{3}{x}$, copy and complete Table 15.11.

Table 15.11

x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	3	4
x^2	$\frac{1}{16}$	$\frac{1}{4}$	1	4	9	16
$\frac{3}{x}$	12	6	3	$1\frac{1}{2}$	1	
y	$12\frac{1}{16}$	$6\frac{1}{2}$	4	$4\frac{1}{2}$		

- (b) Draw the graph of $y = x^2 + \frac{3}{x}$.
 (c) Hence find the minimum value of y within the given range.
- 4 (a) Draw the graph of $y = x - \frac{5}{x}$ for values of x from -2 to $+4$.

- (b) Hence solve the equation $x - \frac{5}{x} = 2$.

- 5 On the same axes, draw the graphs of $y = \frac{1}{x}$ and $y = 2x + 5$. Use the points of intersection of the graphs to solve the equation $2x^2 + 5x - 1 = 0$.

- 6 Solve the equation $x^2 + 7x + 4 = 0$ by drawing appropriate inverse and linear graphs within the same axes.

- 7 (a) Draw the graph of $y = x^2 + \frac{7}{x}$ for values of x from -4 to $+2$.

- (b) Hence solve the equation $7 - x^2 = \frac{7}{x}$.

- 8 (a) Draw the graphs of $y = x + \frac{4}{x}$ and $y = x^2 - x$ for x equal to $1, 1\frac{1}{2}, 2, 2\frac{1}{2}, 3$.

(b) Show that the intersections of these graphs satisfy the equation $x^3 - 2x^2 - 4 = 0$, and use the graph to find a root of this equation.

- 9 (a) Draw the graph of $y = x + \frac{1}{x}$ for values of x from 0.1 to 10 .

(b) Use the graph to find approximately the roots of the equation $2x^2 - 9x + 2 = 0$. (Hint: Divide each term in the equation by $2x$.)

- 10 Copy and complete the following table of values for $y = x + \frac{25}{x}$ for $2 \leq x \leq 13$.

Table 15.12

(Values are rounded to one place of decimals.)

x	2	3	4	5	6	7	8	9	10	11	12	13
y	14.5			10.3		10.6	11.1		12.5		14.1	

Draw the graphs of $y = x + \frac{25}{x}$ and

$2y = x + 15$, using the same axes, and a scale of 2 cm to 2 units on each axis.

From the graphs, find:

- (a) the values of x for which

$$2x + \frac{50}{x} = x + 15$$

- (b) the gradient of the curve at $x = 3$.

- 11 Complete Table 15.13, which gives corresponding values of x and y for which

$$y = 30 - 3x - \frac{60}{x}$$

Table 15.13

x	2	$2\frac{1}{2}$	3	4	5	6	7	8	9	10
y	-6	$-1\frac{1}{2}$		3			0.4		-3.7	

Hence draw the graph of $y = 30 - 3x - \frac{60}{x}$.

- (a) Read off the greatest value of y .

- (b) To what value of x does this value of y correspond?

- 12 Copy and complete Table 15.14, overleaf, and use it to draw the graph of $y = \frac{3x}{x^2 + 1}$.

Use a scale of 2 cm to 1 unit along the x -axis and 2 cm to $\frac{1}{2}$ unit along the y -axis.

Table 15.14

x	-3	$-2\frac{1}{2}$	-2	$-1\frac{1}{2}$	-1	$-\frac{1}{2}$
y	-0.9		-1.2		-1.5	

x	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3
y	0	1.2		1.38		1.03	

Using the same axes, draw the graph of $y = \frac{2x}{3} + \frac{1}{3}$. From your graph find (a) the three roots of the equation $\frac{3x}{x^2 + 1} = \frac{2x}{3} + \frac{1}{3}$, (b) the maximum value of $\frac{3x}{x^2 + 1}$ within the given range, (c) the gradient of the curve at $x = -\frac{1}{2}$.

Sketch graphs

A **sketch graph** is a simple freehand drawing which shows the main features of a line or curve. Some examples of sketch graphs were given in Chapter 5.

Linear functions

Example 7

Sketch the graph of $2x - 3y = 24$.

Method: Find the **intercepts** on the axes; i.e. the positions where $x = 0$ and $y = 0$.

$$2x - 3y = 24$$

$$\text{When } x = 0, -3y = 24$$

$$y = -8$$

$$\text{When } y = 0, 2x = 24$$

$$x = 12$$

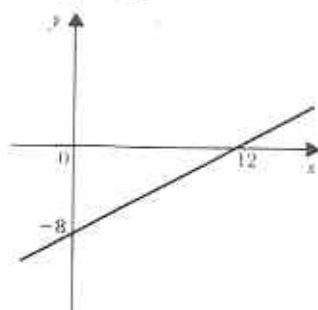


Fig. 15.10

The sketch graph can now be drawn (Fig. 15.11).

Always label the axes and origin. If possible show where the line crosses the axes.

Example 8

Find the equation of the line represented by the sketch graph in Fig. 15.11.

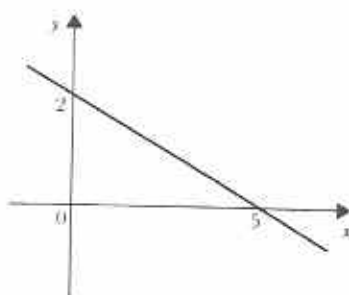


Fig. 15.11

Since the graph is a straight line, its equation is of the form $y = mx + c$, where m is the gradient of the line and c is the intercept on the y -axis. From Fig. 15.11, $m = -\frac{2}{5}$ *, $c = +2$

$$\text{The equation is } y = -\frac{2}{5}x + 2$$

$$\text{i.e. } 5y = -2x + 2$$

$$\text{or } 2x + 5y = 2$$

* As x increases by 5 units, y decreases by 2 units

Quadratic functions

A quadratic function has an equation in the form $y = ax^2 + bx + c$, where a , b and c are positive or negative constants. When a , the coefficient of x^2 , is positive, the graph is a cup-shaped parabola. When a is negative, the graph is a cap-shaped parabola (Fig. 15.12).

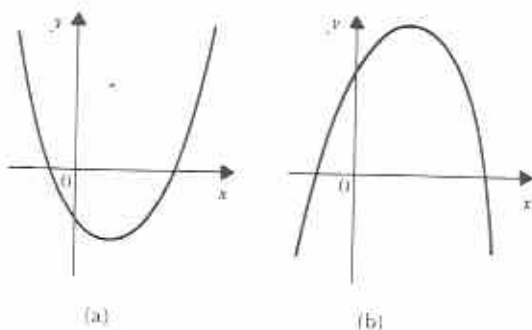


Fig. 15.12

Example 9

Sketch the graph of $y = x^2 - x - 12$, showing where the curve cuts the axes.

$$y = x^2 - x - 12$$

1 The curve cuts the y -axis when $x = 0$.

$$\text{When } x = 0, y = -12$$

2 The curve cuts the x -axis when $y = 0$.

$$\text{When } y = 0, x^2 - x - 12 = 0$$

$$\Leftrightarrow (x - 4)(x + 3) = 0$$

$$\Rightarrow x = 4 \text{ or } -3$$

3 The coefficient of x^2 is positive, hence the curve is a cup-shaped parabola.

The sketch in Fig. 15.13 is drawn using the data in 1, 2 and 3 above.

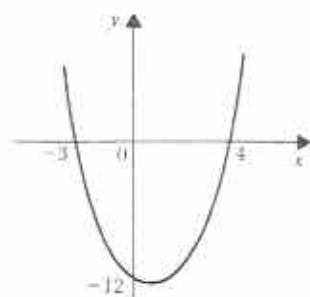


Fig. 15.13

Example 10

Find the equation of the curve represented by the sketch in Fig. 15.14.

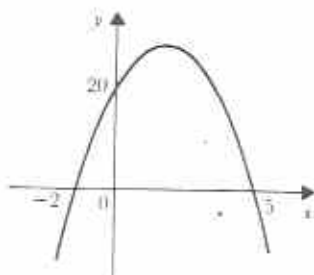


Fig. 15.14

Let the curve have the equation $y = ax^2 + bx + c$.

1 c is the intercept on the y -axis.

Hence, from Fig. 15.14, $c = +20$.

2 When $y = 0$, $ax^2 + bx + c = 0$. Hence from the intercepts on the x -axis in Fig. 15.14, the roots of the equation $ax^2 + bx + c = 0$ are -2 and $+5$.

$$(x + 2)(x - 5) = 0$$

$$x^2 - 3x - 10 = 0 \quad (1)$$

Since $c = +20$, multiply each term in (1) by -2 .

$$-2x^2 + 6x + 20 = 0$$

Hence $y = -2x^2 + 6x + 20$ is the required equation.

Notice that the coefficient of x^2 is negative and that Fig. 15.14 shows a cap-shaped parabola.

Inverse functions

Fig. 15.15 shows the graphs of (a) $y = \frac{1}{x+2}$

and (b) $y = \frac{1}{x-3}$.

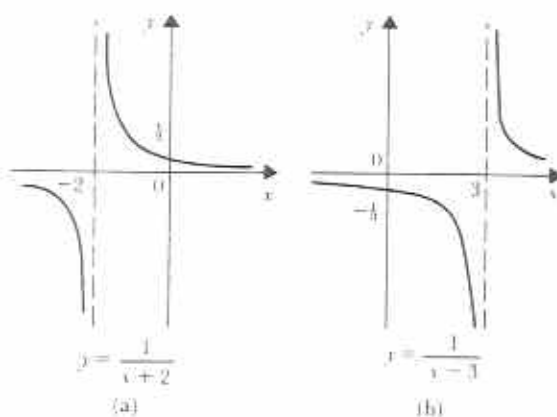


Fig. 15.15

In each graph, the continuity of the curve is broken at the line $x = k$, where k is the value of x for which the fraction is undefined. The intercept on the y -axis is found by substituting $x = 0$ in the given equation. Neither curve cuts the x -axis.

Exercise 15c

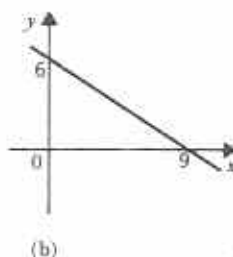
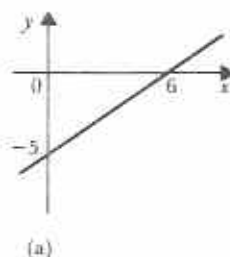
Draw freehand graphs throughout this exercise. Whenever possible, show where the graph cuts the axes.

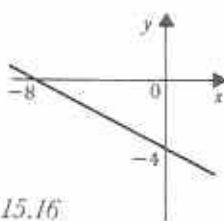
1 Sketch the graphs of the following.

(a) $3x - 2y = 12$ (b) $6x + 3y = 18$

(c) $y - 2x = 7$ (d) $y = 3x - 1$

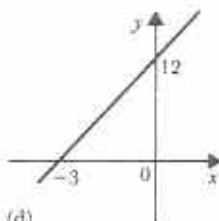
2 Find the equations of the lines shown by the sketch graphs in Fig. 15.16(a)–(d).





(c)

Fig. 15.16



(d)

- 3 What is the value of y at the point where the curve $y = x^2 + 3x - 11$ cuts the y -axis?
- 4 Sketch the graphs of the following, showing where the curve cuts the axes:
- $y = x^2 - 5x + 4$
 - $y = 15 - 2x - x^2$
 - $y = x^2 - 12x + 36$
 - $y = 16 - x^2$
 - $y = 3x^2 + 3x - 6$
 - $y = 10 - 8x - 2x^2$
- 5 Fig. 15.17 is a sketch graph of $y = 6 + x - x^2$.

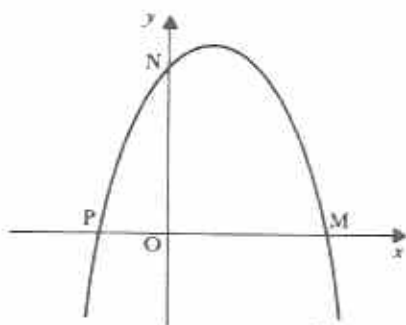


Fig. 15.17

- What is the value of x at M?
 - Find the tangent of the angle OMN.
- 6 What equation is represented by the sketch graph in Fig. 15.18?

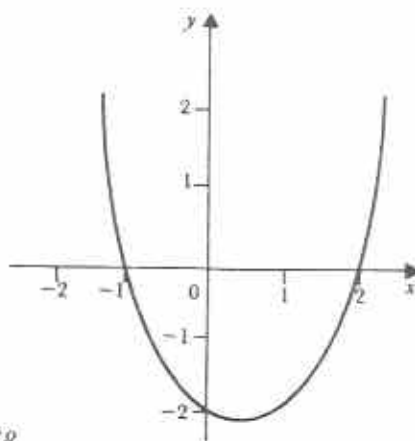
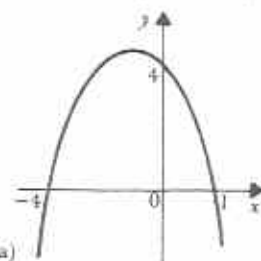
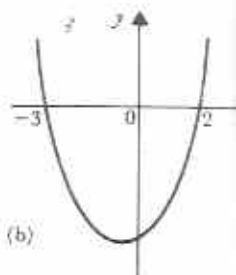


Fig. 15.18

- 7 What equations are represented by the sketch graphs in Fig. 15.19?



(a)



(b)

Fig. 15.19

- 8 Sketch the graphs of the following.

(a) $y = \frac{1}{x+1}$ (b) $y = \frac{1}{x-5}$

- 9 Find the equations of the curves shown by the sketch graphs in Fig. 15.20.

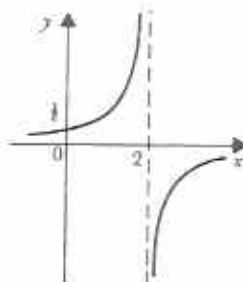
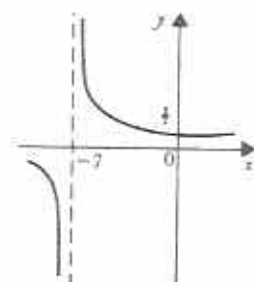


Fig. 15.20 (a)



(b)

- 10 Find the equations of the line, l , the parabola, p , and the hyperbola, h , shown in the sketch graph in Fig. 15.21.

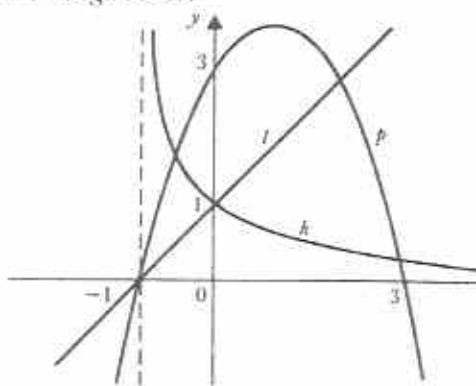


Fig. 15.21

Chapter 16

Statistics (5) Frequency distributions, histograms, cumulative frequency

Bar charts (revision)

A **bar chart** is a statistical graph in which bars are drawn such that their lengths or heights are proportional to the quantities they represent.

Example 1

Table 16.1 shows the number of schools in twelve towns.

Table 16.1

no. of schools	4	5	6	7	8
no. of towns	1	4	2	3	2

- Draw a bar chart to illustrate the information.
- State the mode and median of the distribution.
- Calculate the mean of the distribution to the nearest whole number.

- Fig. 16.1 is a bar chart showing the information in Table 16.1.

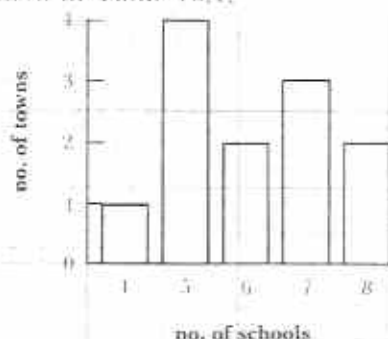


Fig. 16.1

- The mode is 5 schools (i.e. the most frequent number of schools; corresponding to the highest bar in Fig. 16.1). The median is 6 schools (i.e. the central value when the number of schools are arranged in order: 4 5 5 5 5 6 6 7 7 7 8 8).

- Mean number of schools per town

$$\begin{aligned}
 &= \frac{\text{total number of schools}}{\text{total number of towns}} \\
 &= \frac{1 \times 4 + 4 \times 5 + 2 \times 6 + 3 \times 7 + 2 \times 8}{12} \\
 &= \frac{4 + 20 + 12 + 21 + 16}{12} \\
 &= \frac{63}{12} = 5\frac{1}{4} \\
 &= 5 \text{ to the nearest whole number.}
 \end{aligned}$$

The number of times any particular number occurs is called its **frequency**. In Example 1, 3 towns have 7 schools; 3 is the frequency of 7 schools.

Exercise 16a (revision)

- Fig. 16.2 is a bar chart showing the numbers of hours of each kind of programme broadcast by a radio station on a certain day.
 - Which kind of programme was given most time?

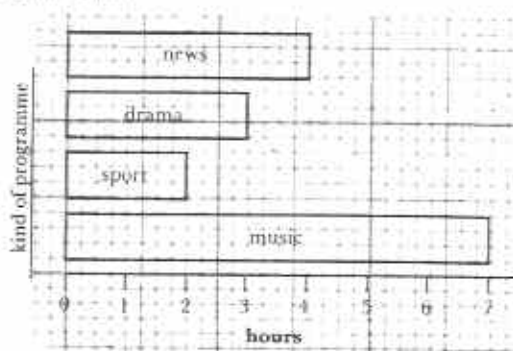


Fig. 16.2

- (b) How many hours were given to drama?
 (c) For how many hours did the radio station broadcast?
 (d) What fraction of the programme time was given to news?

- 2 Fig. 16.3 is a bar chart showing the numbers of kilometres a group of 75 students walked in a sponsored march.

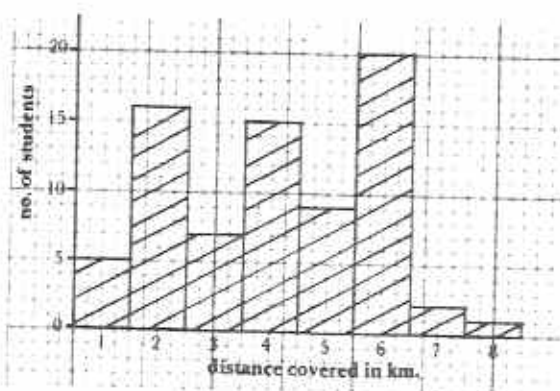


Fig. 16.3

- (a) What is the mode of the distances covered?
 (b) How many students covered less than 4 km?
 (c) What is the total distance covered by all the students?
 (d) If the students were listed in order according to the distances covered, how far would the student in the median position have walked?
- 3 Table 16.2 shows the numbers of seeds in 40 groundnuts.

Table 16.2

number of seeds	1	2	3	4
frequency	8	21	10	1

- (a) Draw a bar chart to illustrate Table 16.2.
 (b) State the mode and median number of seeds.
 (c) Calculate the mean number of seeds per groundnut.
- 4 Table 16.3 shows the distribution of marks in a test.

Table 16.3

marks	40	41	42	43	44	45	46
frequency	7	4	6	2	4	2	6

- (a) Draw a bar chart to show the distribution.
 (b) How many people took the test?
 (c) Find the median mark.
- 5 Table 16.4 shows the grades, in percentages, of 200 students in a test.

Table 16.4

grades (%)	10	20	30	40	50	60	70	80	90	100
no. of students	12	16	20	25	28	32	31	24	8	4

- (a) Draw a bar chart for this distribution of grades.
 (b) Find the mean of the distribution.
 (c) Find the mode and the median of these grades.

Grouped data

Frequency distributions

When statistical data contain a large number of values, it is impractical to draw a bar chart and often difficult to calculate averages. For example, Table 16.5 shows the weekly pay of 50 people.

Table 16.5

weekly pay of 50 people (\$)				
82	132	199	248	300
89	145	200	249	324
94	152	206	255	334
96	156	206	263	348
98	158	214	265	369
108	163	220	270	381
114	176	221	270	401
120	178	232	280	440
125	185	235	288	477
128	189	247	294	485

A bar chart of this data, if drawn, would contain 46 bars of height 1 unit and 2 bars of height 2 units (corresponding to the incomes \$206 and \$270). Such a graph would be difficult to draw and the result would show no pattern.

To overcome this problem, the data can be reduced to a **frequency distribution**. A frequency distribution is a table in which the given values are divided into **class intervals**. The number of values in each class interval is given as the **frequency** of the values in that interval. For example, the data in Table 16.5 can be grouped in equal class intervals of \$100 to give the frequency distribution shown in Table 16.6.

Table 16.6

class interval (weekly pay \$)	frequency (number of people)
0–99	5
100–199	16
200–299	19
300–399	6
400–499	4

It is necessary to define the limits of the class intervals very clearly. Otherwise it may be difficult to decide the class in which to include borderline values, such as \$199 and \$200.

Exercise 16b

- 1 Make a frequency distribution of the data in Table 16.5 taking equal class intervals \$1–\$100, \$101–\$200, \$201–\$300, etc. Compare this distribution with that of Table 16.6.
- 2 Make a frequency distribution of the data in Table 16.5 by taking ten equal class intervals \$0–\$49, \$50–\$99, ..., \$450–\$499.
- 3 Fifty students were asked to estimate the size of an angle to the nearest degree. Their results, arranged in order of size, are given in Table 16.7.

Make a frequency distribution table, taking six equal intervals, 55–59, 60–64, 65–69, ..., 80–84.

Table 16.7

estimations (degrees)									
56	58	58	60	60	61	62	63	64	64
65	65	65	66	66	66	66	67	67	68
68	68	68	69	69	69	70	70	70	70
70	72	72	72	72	73	73	74	74	74
75	75	75	76	78	79	80	80	81	83

Histogram

A frequency distribution can be represented by a block graph called a **histogram**. Figure 16.4 is the histogram of the frequency distribution in Table 16.6.

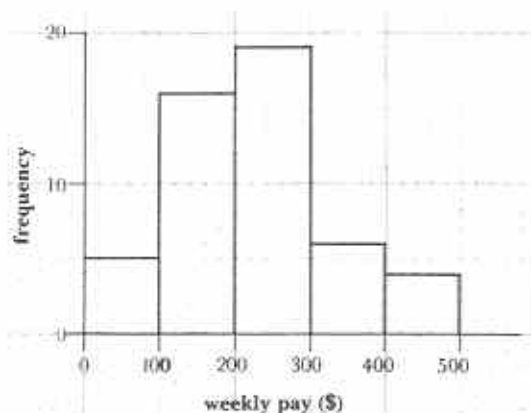


Fig. 16.4

A histogram consists of a number of rectangles. The horizontal width of each rectangle is given by the class interval.¹ The height is such that the area of the rectangle is proportional to the frequency in that interval.² Hence the areas of the rectangles show the frequency distribution.

Notes:

- 1 Since the true sizes of the class intervals are 0–99, 100–199, etc., there should be very small gaps between the rectangles. In practice these gaps are closed to give a continuous horizontal axis.
- 2 For a histogram with class intervals of equal widths, the vertical scale is proportional to the frequency of the distribution.

In Fig. 16.4, the **modal class** is 200–299, since this interval corresponds to the rectangle with the greatest area.

Frequency polygon

Grouped data can also be shown on a **frequency polygon**. Fig. 16.5 is a frequency polygon of the data shown in the histogram in Fig. 16.4.

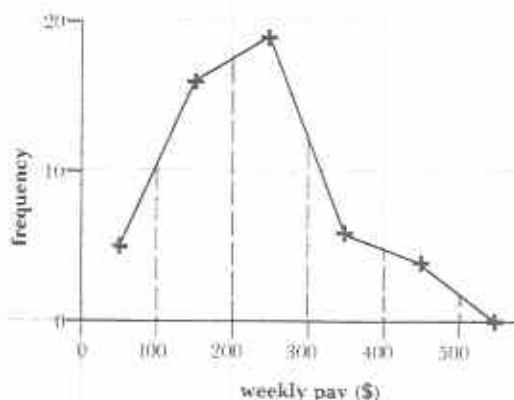


Fig. 16.5

In a frequency polygon, the frequencies are plotted at the *mid-points* of each class interval. The points are joined by straight lines.

Example 2

(a) Draw a histogram and frequency polygon for the frequency distribution given in Table 16.8.

Table 16.8

class interval	1–5	6–10	11–15	16–20	21–25
frequency	3	5	7	6	4

(b) Calculate the mean of the distribution.

(a) Fig. 16.6(a) and (b) shows the histogram and frequency polygon.

Note: The gaps between the intervals are 1 unit. These are too large to be ignored when drawing this histogram. To close the gaps, the widths of the rectangles are increased by $\frac{1}{2}$ unit on both

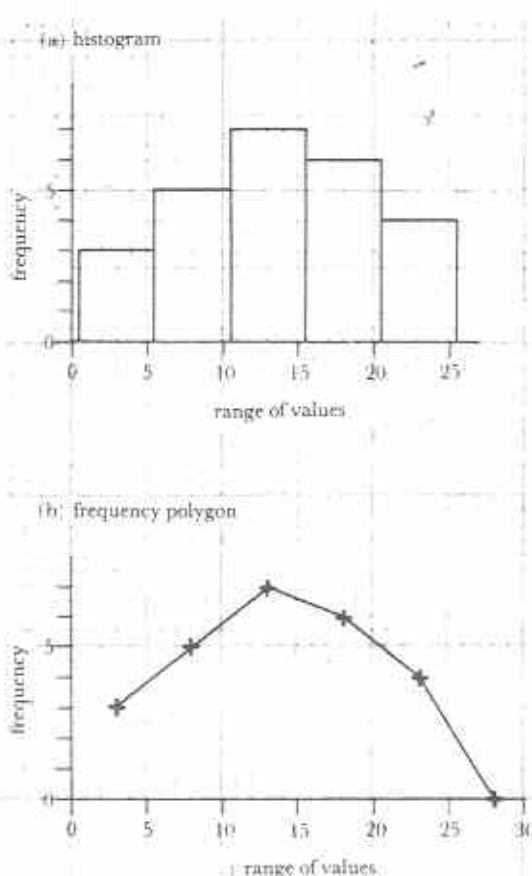


Fig. 16.6

sides. Compare this with Fig. 16.4 where the gaps were so small that they could be ignored. (b) Let the values in each class interval be represented by the mid-value of that class. Hence, all values in the class 1–5 are counted as 3, all values in the 6–10 class are counted as 8, and so on.

Mean value

$$\begin{aligned}
 &= \frac{3 \times 3 + 5 \times 8 + 7 \times 13 + 6 \times 18 + 4 \times 23}{3 + 5 + 7 + 6 + 4} \\
 &= \frac{9 + 40 + 91 + 108 + 92}{25}
 \end{aligned}$$

$$= \frac{340}{25}$$

$$= 13.6$$

The assumption in part (b) of Example 2 leads to some inaccuracy. However, when there is a large number of values, the error is likely to be very small and can be ignored.

Example 3

Table 16.9 gives the marks of 50 students in a test.

Table 16.9

35	51	83	60	61	73	44	90	70	93
56	34	52	61	43	57	40	58	88	64
52	71	25	86	79	35	73	44	71	95
63	53	48	78	65	98	28	72	67	82
46	54	62	35	70	41	63	73	50	68

(a) Construct the histogram, taking class intervals 21–30, 31–40, ..., 91–100. (b) What is the modal class? (c) Find the mean mark.

(a) When the data are not in numerical order, use a tally system to count the frequencies. Take each mark in turn and enter a tally stroke against the proper class interval. The frequency total, 50, gives a check on the accuracy of working. The resulting frequency distribution is given in Table 16.10.

Table 16.10

class	frequency	
21–30		2
31–40		5
41–50		7
51–60		9
61–70		11
71–80		8
81–90		5
91–100		3
	total	50

Fig. 16.7 is the histogram of the distribution. Notice that class intervals of $20\frac{1}{2}$ – $30\frac{1}{2}$, $30\frac{1}{2}$ – $40\frac{1}{2}$, ..., have been drawn in Fig. 16.7. This removes the gaps between the bars.

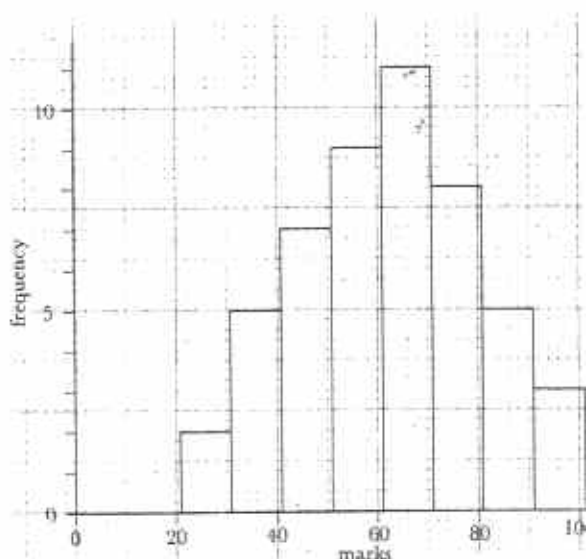


Fig. 16.7

(b) The modal class is 61–70. This can be seen in both the frequency distribution table and in the histogram.

The mode of the data is taken to be the central value of the modal class, 61–70, i.e. $65\frac{1}{2}$.

(c) To find the mean mark, choose a working mean and find the deviations from it.

In this example, $65\frac{1}{2}$ is taken as the working mean and the marks in each class interval are represented by the mid-mark of that class. For example, marks in the class interval 21–30 are counted as $25\frac{1}{2}$, and so on. The working is set out in columns as in Table 16.11.

Table 16.11

class interval	class centre	frequency (<i>f</i>)	deviation (<i>d</i>)	(<i>f</i> × <i>d</i>)
21–30	$25\frac{1}{2}$	2	–40	–80
31–40	$35\frac{1}{2}$	5	–30	–150
41–50	$45\frac{1}{2}$	7	–20	–140
51–60	$55\frac{1}{2}$	9	–10	–90
61–70	$65\frac{1}{2}$	11	0	0
71–80	$75\frac{1}{2}$	8	+10	+80
81–90	$85\frac{1}{2}$	5	+20	+100
91–100	$95\frac{1}{2}$	3	+30	+90
			total deviation	–190

From Table 16.11,
total deviation from working mean = –190

$$\begin{aligned}\text{mean deviation} &= \frac{-190}{50} \\ &= -3.8 \\ \text{mean mark} &= 65.5 - 3.8 \\ &= 61.7\end{aligned}$$

As in Example 2, the final result in part (c) is likely to be slightly inaccurate. (In fact, the true mean of the marks in Table 16.9 is 61.2.) Nevertheless, this method should be used when a large number of values is given.

Exercise 16c

- 1 Draw a histogram and a frequency polygon for the frequency distribution in Table 16.12.

Table 16.12

class	1-5	6-10	11-15	16-20	21-25
frequency	2	4	6	5	3

State the modal class of the distribution.

- 2 Draw a histogram of the frequency distribution in Table 16.13.

Table 16.13

class	1-5	6-10	11-15	16-20	21-25
frequency	4	6	11	6	1

Calculate the mean of the data.

- 3 Draw a histogram of the data in Table 16.14.

Table 16.14

class	1-10	11-20	21-30	31-40	41-50
frequency	5	12	17	10	6

Estimate the mode of the data.

- 4 Draw a histogram and frequency polygon of the frequency distribution in Table 16.15.

Table 16.15

class	0-14	15-21	22-28	29-35	36-42	43-49
frequency	3	5	8	18	9	7

Find the mode and the mean of the data.

- 5 Table 16.16 shows the heights of plants to the nearest 5 cm together with the corresponding numbers of plants.

Table 16.16

height in cm	20	25	30	35	40	45
no. of plants	5	3	1	1	3	2

Draw a histogram to illustrate this information. How many plants have heights greater than the mode by more than 10 cm?

- 6 Use the frequency distribution constructed for question 2, Exercise 16b, to estimate the mode of the data in Table 16.5.
- 7 Use the frequency distribution table constructed for question 3, Exercise 16b, to calculate the mean of the data in Table 16.7.
- 8 Students taking a teacher-training course are grouped by age as in Table 16.17.

Table 16.17

age group	19-20	20-21	21-22	22-23	23-24	24-25
number in group	4	5	10	16	12	3

Calculate the average age of the students.

- 9 Table 16.18 shows the numbers of absentees recorded each day of a school term.

Table 16.18

number absent	0-9	10-19	20-29	30-39	40-49	50-59
frequency	5	18	23	17	14	1

Calculate the average number of absentees per day.

- 10 The percentage marks of 100 students in a School Certificate examination are grouped as in Table 16.19.

Table 16.19

percentage	0-9	10-19	20-29	30-39	40-49
frequency	1	2	5	17	23

percentage	50-59	60-69	70-79	80-89	90-99
frequency	25	18	5	3	1

(a) Estimate the number of students who scored 15% less than the modal mark.

(b) Find the average percentage for the examination.

- 11 Small nails are sold in packets which have printed on them, 'Average contents 200 nails'. The contents of 100 packets, picked out at random, are counted and the results are given in Table 16.20.

Table 16.20

nails per packet	185-189	190-194	195-199
frequency	4	14	32

nails per packet	200-204	205-209	210-214
frequency	28	17	5

Is the statement on the packet true or not?

- 12 Table 16.21 gives the masses, in kg, of 30 students.

Table 16.21

43	45	50	47	51	58	52	47	42	54
61	50	45	55	57	41	46	49	51	50
59	44	53	57	49	40	48	52	51	48

- (a) Taking class intervals 40-44, 45-49, ..., construct the frequency distribution of the data.
 (b) Draw a histogram of the data.
 (c) Calculate the mean mass of the students.
- 13 Table 16.22 gives the heights, in cm, of the 30 students in question 12.

Table 16.22

145	163	149	152	166	156	159	139	145	141
150	158	150	149	143	159	154	167	146	147
152	162	144	169	162	150	173	160	167	171

- (a) Take class intervals 135-144, 145-154, ..., and construct the table of frequencies.
 (b) Calculate the mean height of the students.
- 14 Table 16.23 gives the masses, in kg, of 50 international athletes.

Table 16.23

67	75	79	56	59	60	64	76	58	80
54	65	78	66	65	65	70	62	70	62
70	61	83	51	74	69	59	73	71	74
73	81	69	82	71	53	67	72	66	70
85	63	58	69	75	61	62	68	52	68

Taking class intervals of 51-55, 56-60, ..., 81-85, construct (a) the frequency distribution, (b) a histogram, to show this information. (c) Choose a suitable working mean and hence find the average mass of the athletes.

- 15 Table 16.24 gives the diameters, to the nearest millimetre, of 90 tins.

Table 16.24

diameter (mm)	<70	<80	<90	<95	<100
no. of tins	0	4	13	25	41

diameter (mm)	<105	<110	<120	<130
no. of tins	63	78	86	90

- (a) State the limits between which the diameters of the 4 tins whose diameters are given as less than 80 mm must lie.
 (b) Construct a frequency table showing the number of tins in each group.
 (c) Draw a histogram to illustrate these frequencies.
 (d) State which is the modal group and use your histogram to obtain an estimate of the actual mode.

Example 4

Find the median of the marks given in Table 16.9 on page 137.

Since there are 50 marks, the median is the mean of the 25th and 26th marks when all the marks are arranged in order of size. Table 16.25 gives the marks arranged in order.

Table 16.25

1	25	11	44	21	57	31	67	41	78
2	28	12	46	22	58	32	68	42	79
3	34	13	48	23	60	33	70	43	82
4	35	14	50	24	61	34	70	44	83
5	35	15	51	25	61	35	71	45	86
6	35	16	52	26	62	36	71	46	88
7	40	17	52	27	63	37	72	47	90
8	41	18	53	28	63	38	73	48	93
9	43	19	54	29	64	39	73	49	95
10	44	20	56	30	65	40	73	50	98

$$\text{Median} = \frac{61 + 62}{2} = 61.5$$

Cumulative frequency

In Example 4 it was very time-consuming to arrange so many marks in order of size.

To save time and to avoid making errors it is more usual to make a **cumulative frequency table** and to draw a **cumulative frequency curve**.

Table 16.26 is a cumulative frequency table of the data in Table 16.9.

Table 16.26

class interval	frequency	cumulative frequency
21- 30	2	2
31- 40	5	$5 + 2 = 7$
41- 50	7	$7 + 7 = 14$
51- 60	9	$9 + 14 = 23$
61- 70	11	$11 + 23 = 34$
71- 80	8	$8 + 34 = 42$
81- 90	5	$5 + 42 = 47$
91-100	3	$3 + 47 = 50$

Table 16.26 shows that
2 students scored 30 marks or less,
7 students ($= 5 + 2$) scored 40 marks or less,
and so on.

Each number in the third column is found by adding the number in the second column to the previous total. This progressive increase in the total is what is meant by the word *cumulative*.

Cumulative frequency curve (ogive)

The data in Table 16.26 can be illustrated by plotting the cumulative frequencies against the corresponding upper limits of the class intervals. The points are joined by a smooth curve called an **ogive**.

Fig. 16.8 shows the ogive, or cumulative frequency curve, for the data which were given in Table 16.26.

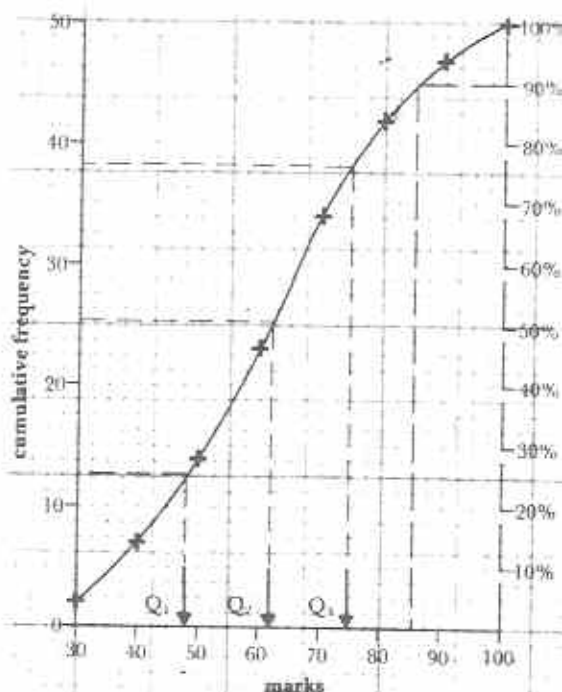


Fig. 16.8

Median and quartiles

The median is the mark that corresponds to the middle student. In Fig. 16.8 it can be seen that the $25\frac{1}{2}$ th student gets a mark of 62. This estimate is reasonably close to the result (61.5) obtained in Example 4.

Just as the median is half-way up the distribution, the **lower quartile** is one-quarter of the way up and the **upper quartile** is three-quarters of the way up. If the total frequency is n , then the lower quartile is the value of the $\frac{1}{4}(n+1)$ th item and the upper quartile is the value of the $\frac{3}{4}(n+1)$ th item. Hence, in Fig. 16.8, the quartiles are at $12\frac{3}{4}$ and $38\frac{1}{4}$ on the cumulative frequency axis. These correspond to marks of 48 and 75 respectively.

The lower quartile is usually called Q_1 .

The second quartile is the median, Q_2 .

The upper quartile is usually called Q_3 .

The **semi-interquartile range**, Q , is defined

as

$$Q = \frac{Q_3 - Q_1}{2}$$

In Fig. 16.8, $Q_3 = 75$ and $Q_1 = 48$, hence

$$Q = \frac{75 - 48}{2} = \frac{27}{2} = 13\frac{1}{2}$$

This shows that about half of the students scored within $13\frac{1}{2}$ marks of the median. Hence the semi-interquartile range gives a measure of the spread of the distribution.

Percentiles

Using the right-hand vertical axis in Fig. 16.8, it can be seen that:

the lower quartile is about $\frac{1}{4}$ of the way up;
the median is about $\frac{1}{2}$ of the way up;
the upper quartile is about $\frac{3}{4}$ of the way up.
These are sometimes called the 25th, 50th and 75th **percentiles**. For example, the 90th percentile corresponds to a mark of 85. This means that 90% of the students scored 85 marks or less, or that 10% of the students scored over 85 marks.

*Note: In this example the lower quartile is actually about $25\frac{1}{2}\%$ of the way up. However, when the total frequency is high, the first, second and third quartiles coincide with the 25th, 50th and 75th percentiles.

Example 5

In Example 3 it is given that students who score over 45 marks pass the test. Use Fig. 16.8 to estimate the percentage of students that passed.

In Fig. 16.8, 45 marks is at the 21st percentile. This means that 21% of the students scored 45 marks or less.

Percentage scoring over 45 marks

$$= 100\% - 21\% = 79\%$$

Percentage of students passing = 79%

Note: From Table 16.25 it can be seen that 39 students, or 78%, actually scored over 45 marks. Hence the estimate of 79% taken from the cumulative frequency curve is reasonably accurate.

Exercise 16d

- In the test in Example 3, the students are graded according to the marks scored as in Table 16.27.

Table 16.27

marks scored	0-50	51-70	71-90	91-100
grade	re-sit	pass	credit	distinction

Use Fig. 16.8 to estimate,

- the percentage of students required to re-sit,
 - the number of students that obtained a pass grade,
 - the percentage of students awarded a distinction.
- Draw a cumulative frequency curve of the data in Table 16.24 on page 139. Hence estimate
 - the median diameter of the tins,
 - the semi-interquartile range.
 - Table 16.28 shows the numbers of students who scored marks within 10-mark class intervals in a test.

Table 16.28

marks (class intervals)	1-10	11-20	21-30	31-40	41-50
number of students	2	7	9	11	13

marks (class intervals)	51-60	61-70	71-80	81-90	91-100
number of students	16	16	15	8	3

- Make a cumulative frequency table and hence draw an ogive showing the mark distribution.
- Estimate the median and upper and lower quartiles.
- Calculate the semi-interquartile range for the test.
- If any mark over 45 is a pass, estimate the percentage of students that passed.

- 4 Table 16.29 gives the mark distribution in a test.

Table 16.29

class interval	11-20	21-30	31-40	41-50	51-60
frequency	3	17	60	48	27

class interval	61-70	71-80	81-90	91-100
frequency	20	13	8	4

- (a) Draw a cumulative frequency curve for the test.
 (b) Find its median and semi-interquartile range.
 (c) Estimate the percentage of the candidates that obtained more than 56 marks.
 (d) Which mark is at the 70th percentile?
- 5 Table 16.30 shows the lives in hours, to the nearest hour, of 50 electric light bulbs.

Table 16.30

563	608	607	632	590	621	614	576	602	582
599	624	580	595	582	581	605	584	596	562
599	598	596	626	596	617	615	589	556	603
594	589	617	560	610	630	571	592	610	597
616	594	622	597	576	595	601	600	592	638

(a) Make a frequency distribution by grouping the values in Table 16.30 in 10-hour class intervals. Draw a histogram of the distribution and hence estimate the mode.

(b) By drawing a cumulative frequency curve, estimate the median and 80th percentile of the distribution.

- 6 36 girls were given a test in which the maximum mark available was 100. Table 16.31 shows the cumulative frequency of the results obtained.

Table 16.31

mark	10	20	30	40	50	60	70	80	90	100
number of girls scoring this mark or less	1	4	8	16	24	29	32	34	35	36

- (a) Calculate how many girls scored a mark between 61 and 70 inclusive.
 (b) Using a vertical scale of 2 cm to represent 5 girls and a horizontal scale of 1 cm to represent 10 marks, plot these values on graph paper and draw a smooth curve through your points.
 (c) Showing your method clearly, use your graph to estimate the median mark.

[Camb]

Inequalities (3) Linear programming

Solution of inequalities

Example 1

$y - x \leq 1$; $2x < 5$; $5y > -4x$ are simultaneous inequalities. (a) Show on a graph the region which contains the solution set of the inequalities. (b) If the solution set contains integral values of x and y only, list its members.

- (a) The boundary lines of the region are
 $p: y - x = 1$
 $q: 2x = 5$
 $r: 5y = -4x$

The unshaded region in Fig. 17.1 gives the solution set of all points (x, y) which satisfy the three inequalities.

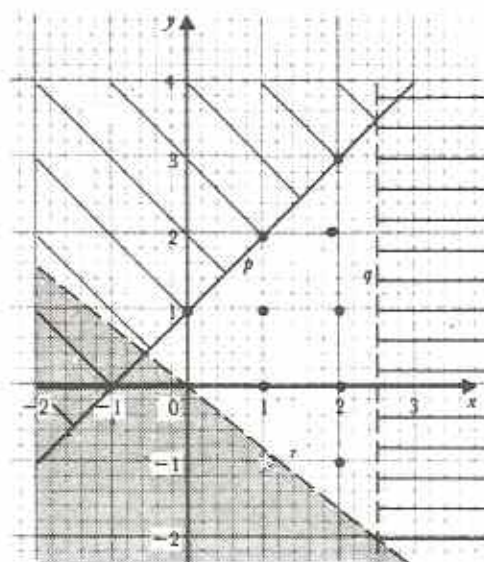


Fig. 17.1

(b) In Fig. 17.1 the solution set is shown by heavy points, indicating that the values of x and y are integral. The solution set is as follows: $\{(0; 1), (1; 0), (1; 1), (1; 2), (2; -1), (2; 0), (2; 1), (2; 2), (2; 3)\}$

Notes:

- Chapter 5 explains how to draw graphs of straight lines.
- The boundary line p is drawn solid to show that the set of points on the line is included in the required region.
- The boundary lines q and r are drawn using broken lines to show that the points on those lines are *not* included in the required region.
- Regions outside the boundaries are shaded to show that they are not required.

Example 2

Write down the three inequalities which define the unshaded area labelled A in Fig. 17.2.

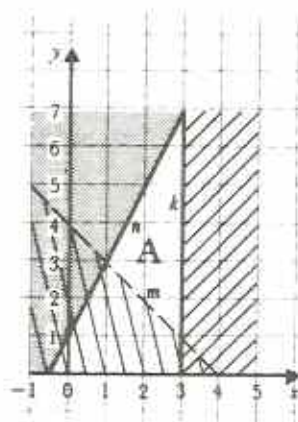


Fig. 17.2

The lines are labelled k, m, n for convenience. Line k :

k is the line $x = 3$, k is solid. Points to the right of k are not required. Hence the corresponding inequality is $x \leq 3$.

Line m :

m has a gradient of -1 and cuts the y -axis at $(0; 4)$. Its equation is $y = -x + 4$, m is a broken line. Points below m are not required. Hence $y > 4 - x$ is the corresponding inequality.

Line n :

n has a gradient of 2 and cuts the y -axis at $(0;1)$. Its equation is $y = 2x + 1$. n is solid. Points above n are not required. Hence $y \leq 2x + 1$ is the corresponding inequality. The inequalities which define the region A are $x \leq 3$; $y > 4 - x$ and $y \leq 2x + 1$.

Exercise 17a

- Using graph paper, show the regions defined by each of the following. (Use solid and broken lines where appropriate and leave each required region unshaded.)
 - $4y - x < 4$; $x - y < 3$; $x \geq -2$
 - $x - y > -2$; $x + y < 4$; $x \geq -1$; $y > 0$
 - $x < 4$; $y - 2x \leq 2$; $2 < y < 4$
 - $6 \leq 2x + 3y \leq 12$; $x - 2y < 8$; $y < 3$
- Solve each of the following graphically for integral values of x and y .
 - $y > x$; $y \leq 3x$; $y + 2x < 8$
 - $3x + 4y \leq 12$; $y - x \leq 2$; $y > 1$
- Write down the three inequalities which define the unshaded area labelled A in Fig. 17.3.

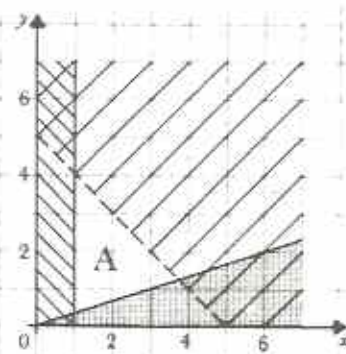


Fig. 17.3

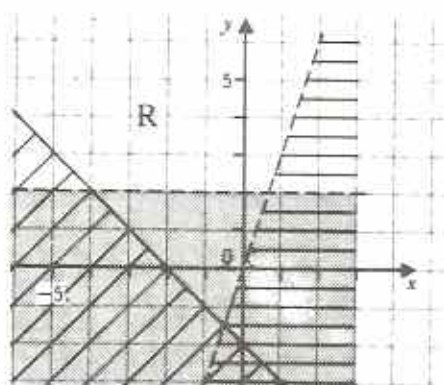


Fig. 17.4

- What are the three inequalities which define the unshaded region R in Fig. 17.4?

5

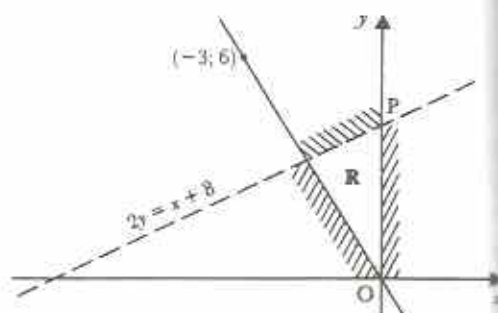


Fig. 17.5

In Fig. 17.5 find

- the coordinates of the point P where the line $2y = x + 8$ crosses the y -axis,
- the equation of the line which passes through the origin O and the point $(-3; 6)$
- the three inequalities which define the triangular region R in the diagram.

[Camb

Linear programming

Example 3

A student has \$2,50. She buys ballpens at 25c each and pencils at 10c each. She gets at least five of each and the money spent on ballpens is over 50c more than that spent on pencils.

- Find (a) how many ways the money can be spent
(b) the greatest number of ballpens that can be bought
(c) the greatest number of pencils that can be bought

Let the student buy x ballpens at 25c and y pencils at 10c.

Then, from the first two sentences,

$$25x + 10y \leq 250$$

Since she gets at least 5 of each,

$$x \geq 5$$

$$\text{and } y \geq 5$$

Also, from the third sentence,

$$25x - 10y > 50$$

These inequalities are shown in Fig. 17.6.

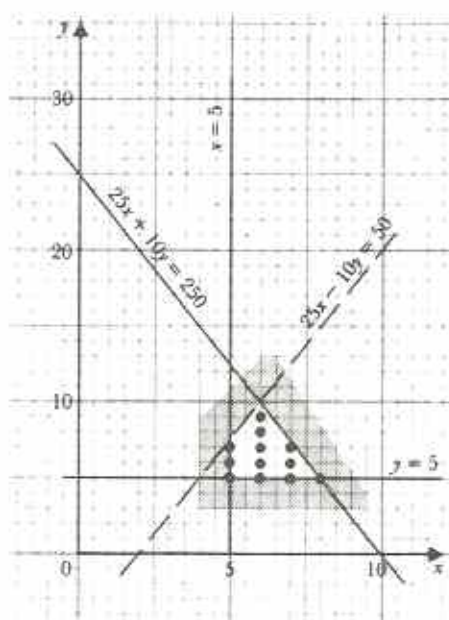


Fig. 17.6

(a) The solution set of the four inequalities is shown by the twelve points marked inside the unshaded region. For example the point (6; 8) shows that the student can buy 6 ballpens and 8 pencils. Hence there are 12 ways of spending the money.

(b) The greatest number of ballpens that can be bought is 8, corresponding to the point (8; 5).

(c) The greatest number of pencils is 9, corresponding to the point (6; 9).

Example 4

The student in Example 3 wants to buy as many items as possible. How many can she get and how much change will there be from the \$2,50?

The number of items bought is $x + y$.

If the total is n , then $x + y = n$.

The general equation $x + y = n$ can be represented graphically by a family of parallel lines.

Fig. 17.7 shows some members of the family when n has the values 5; 10; 12.

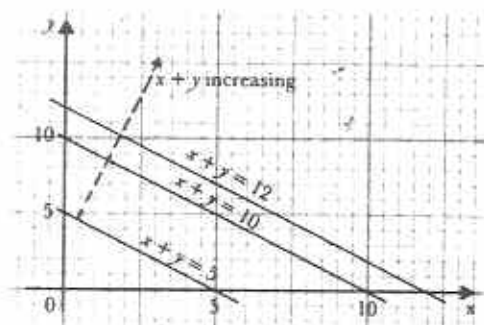


Fig. 17.7

Fig. 17.7 shows that as n increases, the lines appear to move to the right. Considering Fig. 17.6, the greatest possible value of n corresponds to the line which is parallel to $x + y = n$ and as far as possible to the right, but which also passes through the unshaded region.

Fig. 17.8 is a repeat of Fig. 17.6 with some of the family $x + y = n$ added.

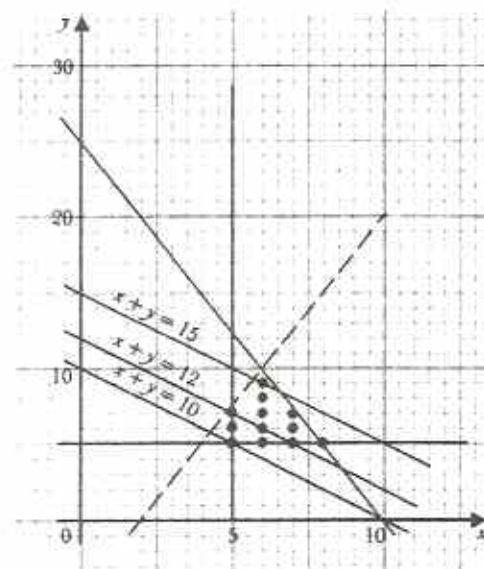


Fig. 17.8

From Fig. 17.8, the greatest value of n is 15, where the line $x + y = 15$ passes through the point (6; 9). The 15 items are made up as follows:

6 ballpens at 25c:	\$1.50
9 pencils at 10c:	\$0.90
total cost:	\$2.40

There will be 10c change from \$2.50.

The kind of problem solved in Examples 3 and 4 involves making decisions in a situation in which there are **restrictions**. Each restriction, such as the limit on the amount of money available, can be represented by a linear inequality. Hence the solution to the problem can be found graphically. This method is called **linear programming**. Linear programming can be used to solve a variety of realistic problems.

Example 5

To start a new bus company, a businessman needs at least 5 buses and 10 minibuses. He does not want to have more than 30 vehicles altogether. A bus takes up 3 units of garage space, a minibus takes up 1 unit of garage space and there are only 54 units of garage space available.

If x and y are the numbers of buses and minibuses respectively, (a) write down four inequalities which represent the restrictions on the businessman and (b) draw a graph which shows a region representing possible values of x and y .

Running costs are \$90 a day for a bus and \$48 a day for a minibus. (c) Write down an expression for the total cost per day, SC . (d) Find the maximum daily cost and the corresponding numbers of buses and minibuses.

(a) From the first sentence,

$$x \geq 5$$

$$y \geq 10$$

From the second sentence,

$$x + y \leq 30$$

From the third sentence,

$$3x + y \leq 54$$

(b) In Fig. 17.9, R is the region which contains the possible values of x and y .

(c) $C = 90x + 48y$

(d) In Fig. 17.9, m is the line $90x + 48y = 720$. As m moves to the right, the cost increases. When m reaches m' the line passes through R at the point (12; 18), the point of maximum cost.

$$\begin{aligned} \text{Maximum cost} &= 12 \times \$90 + 18 \times \$48 \\ &= \$1080 + \$864 \\ &= \$1944 \end{aligned}$$

It costs \$1944 to run 12 buses and 18 minibuses.

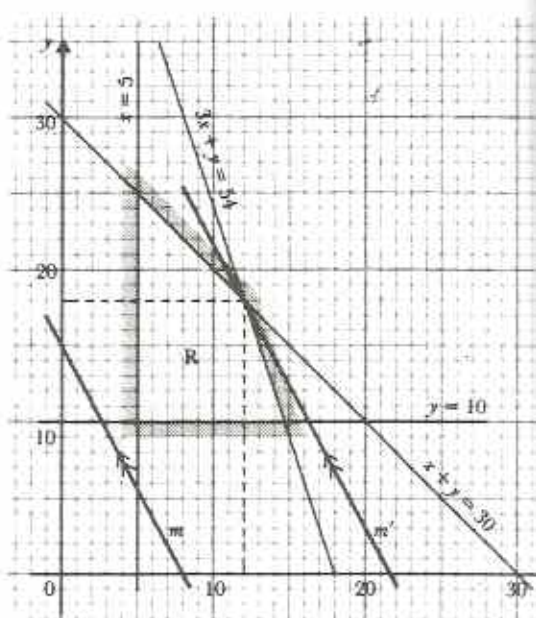


Fig. 17.9

Notes:

- In part (d), C has been chosen as 720, the LCM of 90 and 48. This value gives convenient line m through the points (8, 0) and (0, 15). m' is then drawn using a set square and ruler.
- The working would be clearer and more accurate if a larger scale had been used. Fig. 17.9.

Exercise 17b

- Redraw Fig. 17.9 using a scale of 2 cm to units on both axes. Use your graph to answer the following.

When the bus company is running at full efficiency, the daily profit on a bus is four times that on a minibus. Find the number of bus and minibuses the businessman should buy to maximise his profit.

- Notebooks cost 60c and pencils 36c. A girl has \$3.60 to spend and needs at least 2 notebooks and 3 pencils. She decides to spend as much as possible of her \$3.60. (a) How many ways can she spend her money? (b) Do any of the ways give her change? If so, how much?

- 3 A car repair workshop uses large numbers of two types of spare part, one costing \$3 and the other \$4. The workshop owner allows \$300 to buy spare parts and he needs twice as many cheap ones as dear ones. There must be at least 50 cheap and 20 expensive parts.

(a) What is the largest number of spare parts he can buy, and in what way?

(b) If he decides to get as many of the expensive parts as conditions allow, how many of each type can he get?

- 4 A storeman fills a new warehouse with two types of goods, A and B. They both come in tall boxes which cannot be stacked. A box of A takes up $\frac{1}{2} \text{ m}^2$ of floor space and costs \$50. A box of B takes up $1\frac{1}{2} \text{ m}^2$ of floor space and costs \$300. The storeman has up to 100 m^2 of floor space available and can spend up to \$15 000 altogether. He wants to buy at least 50 boxes of A and 20 boxes of B.

(a) How many boxes of each should he buy in order to (i) spend all the money available and also to use as much space as possible?

(ii) use all the space for the least cost?

(b) What is the cost in the second case?

- 5 Following an illness, a patient is required to take pills containing minerals and vitamins. The contents and costs of two types of pill, Feelgood and Getbetta, together with the patient's daily requirement, are shown in Table 17.1.

Table 17.1

	mineral	vitamin	cost
Feelgood	160 mg	4 mg	20c
Getbetta	40 mg	3 mg	10c
Daily requirement	800 mg	30 mg	

A daily prescription contains x Feelgood pills and y Getbetta pills.

(a) State the inequalities to be satisfied by x and y .

(b) Use a graphical method to show the solution set of x and y .

(c) Find the cheapest way of prescribing the pills and the cost.

- 6 While exploring for oil, it was necessary to carry at least 18 tonnes of supplies and 80 people into a desert region. There were two types of lorry available, Landmasters and Sandrovers. Each Landmaster could carry 900 kg of supplies and 6 people; each Sandrover could carry 1 350 kg of supplies and 5 people.

If there were only 12 of each type in good running order, find the smallest number of lorries necessary for the journey.

- 7 A shopkeeper orders packets of soap powder. The cost price of a large packet is \$2.70 and that of a small packet is \$1.20. She is prepared to spend up to \$60 altogether and needs twice as many small packets as large packets with a minimum of 10 large and 20 small packets.

(a) What is the greatest number of packets she can buy?

The profit is 30c on a large packet and 15c on a small packet.

(b) Which arrangement gives the greatest profit?

(c) What is that profit?

- 8 A dressmaker plans to buy new machines for her factory. Table 17.2 shows the cost, the necessary floor space and the output of each machine.

Table 17.2

machine	cost	floor space	output in components/hour
Machine A	\$300	3 m^2	10 per hour
Machine B	\$400	$2\frac{1}{2} \text{ m}^2$	15 per hour

She can spend \$3 600 altogether and she has 27 m^2 of floor space. Trade restrictions are such that she has to buy at least 3 of Machine A and 4 of Machine B.

(a) What is the maximum number of machines she can buy?

(b) What arrangement gives the biggest output?

- 9 A builder has \$960 000 and 8 ha of land available for building houses. Large houses cost \$24 000 each to build and need 0.25 ha;

small houses, cost \$15 000 each and occupy 0.1 ha. Permission to build is given so long as there are at least 16 large houses and 30 small houses.

(a) Find the greatest number of large houses that can be built.

(b) Find the distribution that (i) gives the greatest number of houses altogether, (ii) uses up all the land available.

- 10** A shopkeeper stocks two brands of drinks called Kula and Sundown, both of which are produced in cans of the same size. He wishes to order fresh supplies and finds that he has room for up to 1 000 cans. He knows that Sundown is more popular and so proposes to order at least twice as many cans of Sundown as Kula. He wishes,

however, to have at least 100 cans of Kula and not more than 800 cans of Sundown. Taking x to be the number of cans of Kula and y to be the number of cans of Sundown which he orders, write down the four inequalities involving x and/or y which satisfy these conditions.

The point $(x; y)$ represents x cans of Kula and y cans of Sundown. Using a scale of 1 cm to represent 100 cans on each axis, construct and indicate clearly, by shading the unwanted regions, the region in which $(x; y)$ must lie. The profit on a can of Kula is 3c and on a can of Sundown is 2c. Use your graph to estimate the number of cans of each that the shopkeeper should order to give the maximum profit. [Camb]

Chapter 18

Vectors (2)

Vectors (revision)

Naming vectors

A **vector** is any quantity which has direction as well as size. Displacement (or translation), velocity, force, acceleration are all examples of vectors. Fig. 18.1 shows a vector which moves a point from position A to position B.

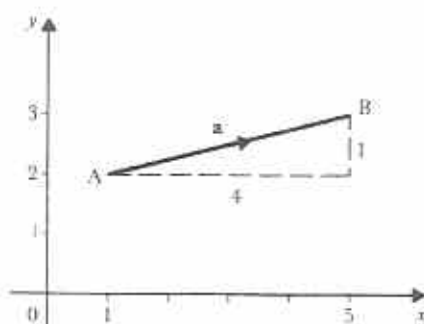


Fig. 18.1

The vector in Fig. 18.1 can be written in many ways:

either $\overrightarrow{AB}$, $\vec{AB}$, $\mathbf{AB}$, $\underline{AB}$
or $\vec{a}$, $\vec{a}$, $\mathbf{a}$, $\underline{a}$

Since the points are on a cartesian plane, $\mathbf{AB}$ can also be written as a column matrix, or **column vector**:

$$\mathbf{AB} = \mathbf{a} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

Direction is important. $\mathbf{BA}$ is in the opposite direction to $\mathbf{AB}$, although they are both parallel and have the same size:

$$\mathbf{BA} = -\mathbf{AB} = -\begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \end{pmatrix}$$

Magnitude

The **magnitude** or size of $\mathbf{AB}$ is represented by the length of the line segment AB. This is written as $|\mathbf{AB}|$ and is called the **modulus** of $\mathbf{AB}$. In Fig. 18.1,

$$\mathbf{AB} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$|\mathbf{AB}| = \sqrt{4^2 + 1^2} \quad (\text{Pythagoras})$$

$$= \sqrt{17}$$

The modulus of a vector is always positive.

Scalar multiplication

If a vector $\mathbf{AB}$ is multiplied by a **scalar** k , where k is any number, the result is a vector k times as big as $\mathbf{AB}$:

$$\text{If } \mathbf{AB} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\text{then } 3\mathbf{AB} = 3\begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 12 \\ 3 \end{pmatrix}$$

$$\text{and } -\frac{1}{2}\mathbf{AB} = -\frac{1}{2}\begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -\frac{1}{2} \end{pmatrix}$$

The effect of scalars can be summarised as follows:

1 If $\mathbf{a} = k\mathbf{b}$ then $\mathbf{a}$ is k times as big as $\mathbf{b}$ and parallel to it.

2 If $h\mathbf{a} = k\mathbf{b}$ then $\mathbf{a} \parallel \mathbf{b}$ or $h = 0$ and $k = 0$.

Addition and subtraction

Vectors may be added and subtracted. For example,

$$\text{If } \mathbf{p} = \begin{pmatrix} -9 \\ 0 \end{pmatrix} \text{ and } \mathbf{q} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}$$

$$\begin{aligned}\text{then } \mathbf{p} + \mathbf{q} &= \begin{pmatrix} -9 \\ 0 \end{pmatrix} + \begin{pmatrix} 6 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} -9 + 6 \\ 0 + (-2) \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\text{and } \mathbf{p} - \mathbf{q} &= \begin{pmatrix} -9 \\ 0 \end{pmatrix} - \begin{pmatrix} 6 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} -9 \\ 0 \end{pmatrix} + -\begin{pmatrix} 6 \\ -2 \end{pmatrix} \\ &= \begin{pmatrix} -9 + (-6) \\ 0 + (+2) \end{pmatrix} = \begin{pmatrix} -15 \\ 2 \end{pmatrix}\end{aligned}$$

Example 1

In Fig. 18.2, below, **AB**, **BC**, **CD**, **DE** are vectors as shown.

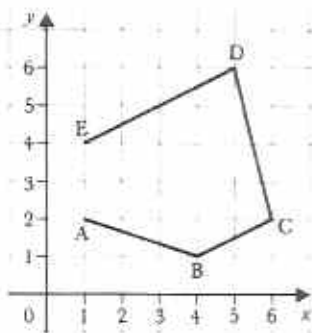


Fig. 18.2

- (a) Express each vector in the form $\begin{pmatrix} a \\ b \end{pmatrix}$.
 (b) Find $|\mathbf{DE}|$. (c) Show that $\mathbf{DE} = -2\mathbf{BC}$.
 (d) Express $\mathbf{BC} + \mathbf{CD}$ as a single column vector.
 (e) Express $\mathbf{BC} - \mathbf{CD}$ as a single column vector.

$$\begin{aligned}\text{(a) } \vec{AB} &= \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad \vec{BC} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \\ \vec{CD} &= \begin{pmatrix} -1 \\ 4 \end{pmatrix}, \quad \vec{DE} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\text{(b) } |\mathbf{DE}| &= \sqrt{(-4)^2 + (-2)^2} \\ &= \sqrt{16 + 4} = \sqrt{20}\end{aligned}$$

$$\text{(c) } \vec{DE} = \begin{pmatrix} -4 \\ -2 \end{pmatrix} = -2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = -2\vec{BC}$$

$$\text{(d) } \vec{BC} + \vec{CD} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}$$

$$\begin{aligned}\text{(e) } \vec{BC} - \vec{CD} &= \begin{pmatrix} 2 \\ 1 \end{pmatrix} - \begin{pmatrix} -1 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ -4 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \end{pmatrix}\end{aligned}$$

Exercise 18a (revision)

1 In Fig. 18.2 state the column vector which would displace E to A.

2 Express the following as positive vectors.

$$\text{(a) } -\begin{pmatrix} -3 \\ 1 \end{pmatrix} \quad \text{(b) } -\begin{pmatrix} 4 \\ -5 \end{pmatrix}$$

$$\text{(c) } -\begin{pmatrix} -8 \\ -6 \end{pmatrix} \quad \text{(d) } -\begin{pmatrix} 2 \\ 7 \end{pmatrix}$$

3 If $\mathbf{XY} = \begin{pmatrix} -8 \\ 5 \end{pmatrix}$ what is (a) $|\mathbf{XY}|$, (b) $\mathbf{YX}$?

4 A shape is translated through $\begin{pmatrix} 5 \\ -3 \end{pmatrix}$. It

then translated through $\begin{pmatrix} 2 \\ 8 \end{pmatrix}$.

(a) What single translation is this equivalent to? (b) How far is the shape from its starting position?

5 $\triangle PQR$ is such that $\mathbf{PQ} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}$ and

$\mathbf{QR} = \begin{pmatrix} -9 \\ 9 \end{pmatrix}$. Sketch $\triangle PQR$ and hence otherwise express $\mathbf{PR}$ and $\mathbf{RP}$ as column vectors.

6 If $\mathbf{p} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}$, $\mathbf{r} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}$, express each of the following as a single column vector.

- (a) $5\mathbf{p}$ (b) $-3\mathbf{q}$ (c) $\frac{1}{2}\mathbf{r}$
 (d) $\mathbf{p} + \mathbf{q}$ (e) $\mathbf{r} - \mathbf{p}$ (f) $\mathbf{p} - \mathbf{r}$
 (g) $3\mathbf{p} + \mathbf{r}$ (h) $\mathbf{p} - 2\mathbf{q}$
 (i) $5\mathbf{p} - 4\mathbf{q} + \mathbf{r}$ (j) $3\mathbf{p} + \mathbf{q} - 6\mathbf{r}$

7 Given $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ of question 6, evaluate the following, leaving the answers in surd form where necessary.

- (a) $|\mathbf{p}|$ (b) $|\mathbf{q}|$ (c) $|\mathbf{r}|$
 (d) $|\mathbf{p} + \mathbf{r}|$ (e) $|\mathbf{q} + \mathbf{r}|$ (f) $|\mathbf{p} - \mathbf{q}|$

3 Triangle ABC has coordinates $A(1;0)$, $B(0;2)$, $C(1;3)$. X is the point $(4;4)$. $\triangle ABC$ is displaced by vector $\mathbf{AX}$. Find

- (a) the coordinates of the image of $\triangle ABC$,
(b) the modulus of $\mathbf{AX}$.

4 (a) Find the vector $\mathbf{q}$ such that

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix} - \mathbf{q} = \begin{pmatrix} 9 \\ -3 \end{pmatrix}.$$

- (b) Hence find $|\mathbf{q}|$.

5 Copy Fig. 18.3 to show points A, B, C and the origin O.

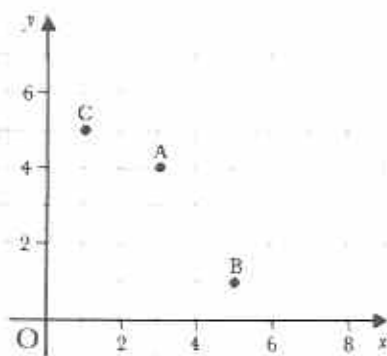


Fig. 18.3

Given that $\mathbf{AP} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, mark on the diagram and label clearly the point P. Given that $\mathbf{BQ} = 2\mathbf{AC}$, mark on the diagram and label clearly the point Q. Calculate $|\mathbf{OA}|$. [Camb]

Position vectors

In Fig. 18.4, P is a point $(x; y)$ on the cartesian plane, origin O.

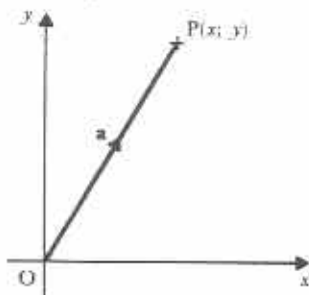


Fig. 18.4

Vector $\mathbf{a}$ is the displacement of P from O. Since this displacement gives the position of P relative to the origin, $\mathbf{a}$ is called the **position vector** of P.

In Fig. 18.4, $\mathbf{a} = \mathbf{OP} = \begin{pmatrix} x \\ y \end{pmatrix}$.

Hence if a point has coordinates $(x; y)$, its position vector is $\begin{pmatrix} x \\ y \end{pmatrix}$.

Position vectors can be used to find displacements between points.

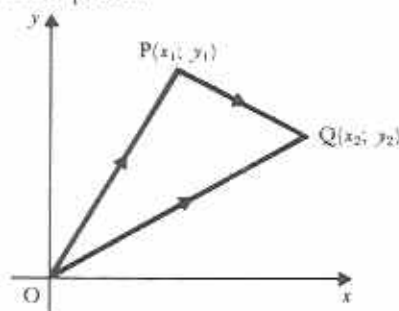


Fig. 18.5

In Fig. 18.5, by adding vectors,

$$\mathbf{OP} + \mathbf{PQ} = \mathbf{OQ}$$

$$\mathbf{PQ} = \mathbf{OQ} - \mathbf{OP}$$

$$\begin{aligned} \mathbf{PQ} &= \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} - \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \\ &= \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix} \end{aligned}$$

Also, by Pythagoras' theorem,

$$|\mathbf{PQ}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

The above results hold for any two general points $P(x_1; y_1)$ and $Q(x_2; y_2)$.

Example 2

If P and Q are the points $(3; 7)$ and $(11; 13)$ respectively, find $\mathbf{PQ}$ and $|\mathbf{PQ}|$.

In Fig. 18.6 overleaf,

$$\mathbf{PQ} = \mathbf{OQ} - \mathbf{OP}$$

$$= \begin{pmatrix} 11 \\ 13 \end{pmatrix} - \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$= \begin{pmatrix} 11 - 3 \\ 13 - 7 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$$

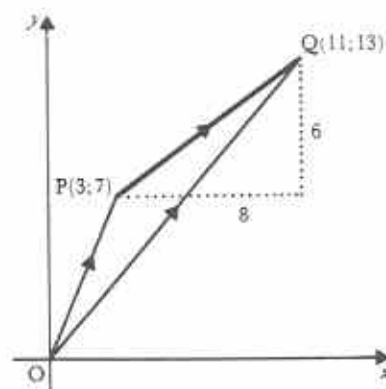


Fig. 18.6

$$\begin{aligned}
 |PQ| &= \sqrt{(11-3)^2 + (13-7)^2} \\
 &= \sqrt{8^2 + 6^2} = \sqrt{100} = 10
 \end{aligned}$$

Example 3

Quadrilateral OPQR is as shown in Fig. 18.7.

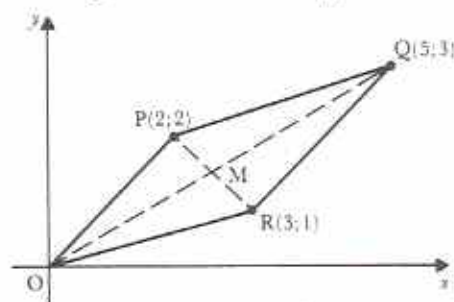


Fig. 18.7

(a) Show that OPQR is a parallelogram, and
(b) find the coordinates of the point of intersection of its diagonals.

(a) Using the position vectors of O, P, Q and R,

$$\vec{OP} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

$$\vec{PQ} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

$$\vec{RQ} = \begin{pmatrix} 5 \\ 3 \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix},$$

$$\vec{OR} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

Hence $\vec{OR} = \vec{RQ}$

and $\vec{PQ} = \vec{OR}$

Considering the sides OP and RQ:

$$\text{If } \vec{OP} = \vec{RQ}$$

then $|\vec{OP}| = |\vec{RQ}|$ and $\vec{OP} \parallel \vec{RQ}$ since equal vectors have the same magnitude and direction.

Hence OPQR is a parallelogram since it has a pair of opposite sides equal and parallel.

(b) Let the diagonals intersect at M.

$\vec{OM} = \frac{1}{2}\vec{OQ}$ (the diagonals of a parallelogram bisect each other)

$$\vec{OM} = \frac{1}{2} \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} 2\frac{1}{2} \\ 1\frac{1}{2} \end{pmatrix}$$

The diagonals intersect at the point $(2\frac{1}{2}; 1\frac{1}{2})$.

Example 3 makes use of the following important result:

$$\text{If } \mathbf{a} = \mathbf{b}$$

then $|\mathbf{a}| = |\mathbf{b}|$ and $\mathbf{a} \parallel \mathbf{b}$.

Example 4

In Fig. 18.8, A(5;6), B(1;8), C(p;4), D are the vertices of a rhombus in the positive quadrant of the cartesian plane.

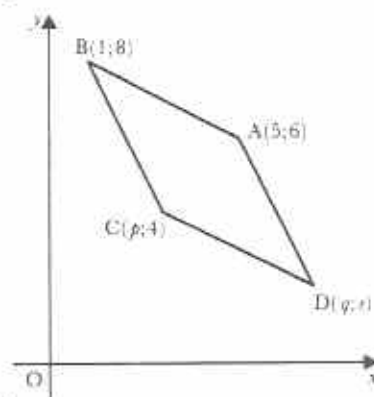


Fig. 18.8

Find p and hence find the coordinates of D.

If ABCD is a rhombus then adjacent sides are equal:

$$|\vec{AB}| = |\vec{BC}| \quad (1)$$

and opposite sides are equal and parallel:

$$\vec{AB} = \vec{DC} \quad (2)$$

From (1),

$$\begin{aligned} \sqrt{(1-5)^2 + (8-6)^2} &= \sqrt{(p-1)^2 + (4-8)^2} \\ \Rightarrow (-4)^2 + (2)^2 &= (p-1)^2 + (-4)^2 \\ \Rightarrow (p-1)^2 &= (2)^2 \\ \Rightarrow p-1 &= 2 \\ \Rightarrow p &= 3 \end{aligned}$$

Let D have coordinates $(q; r)$.

From (2),

$$\begin{pmatrix} 1-5 \\ 8-6 \end{pmatrix} = \begin{pmatrix} p-q \\ 4-r \end{pmatrix}$$

Hence $\begin{pmatrix} -4 \\ 2 \end{pmatrix} = \begin{pmatrix} 3-q \\ 4-r \end{pmatrix}$

$$\Rightarrow \begin{aligned} q &= 7 \\ r &= 2 \end{aligned}$$

$p=3$ and D is the point $(7; 2)$.

Exercise 18b

1 Given points A(7; 8) and B(2; -1), find
(a) **AB**, (b) **BA**.

2 The points O, P, Q, R, S have coordinates (0; 0), (1; 5), (3; 8), (7; 10), (10; 3) respectively. Express each of the following as a column vector.

- (a) **OQ** (b) **OS** (c) **PQ**
(d) **QR** (e) **QS** (f) **RP**

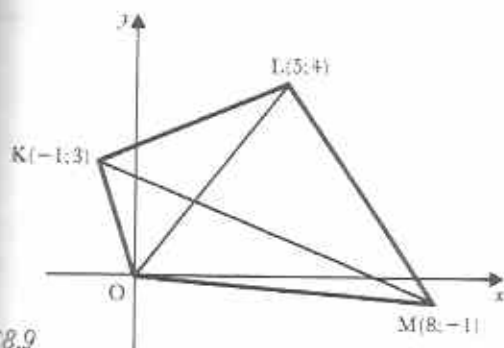


Fig. 18.9

3 Given Fig. 18.9, express each of the following as a single column vector.

- (a) **OK** (b) **OM** (c) **KL**
(d) **LM** (e) **OL** (f) **KM**
(g) **OK + KL** (h) **OL + LM**
(i) **MK + KL** (j) **ML + LO**

4 OABC is a parallelogram where O is the origin, $\mathbf{OA} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$, $\mathbf{OC} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}$.

(a) On graph paper mark and clearly label the points A, B and C.

(b) Express as a column vector (i) **OB**, (ii) **CA**. [Camb]

5 Use vectors to show that the quadrilateral P(-3; 0), Q(-1; 6), R(3; 5), S(5; -2) is a trapezium.

6 Use vectors to show that the quadrilateral A(3; -5), B(8; 5), C(6; 16), D(1; 6) is a rhombus.

7 Prove that the quadrilateral O(0; 0), A(4; 0), B(7; 5), C(3; 5) is a parallelogram.

8 Show that P(3; 2), Q(9; 4), R(11; 8), S(5; 6) is a parallelogram. Use a vector method to find the coordinates of the point of intersection of its diagonals.

9 O(0; 0), P(4; 6), Q, R(8; 2) are vertices of a quadrilateral. Find the coordinates of Q such that OPQR is a parallelogram. Find the coordinates of the point of intersection of its diagonals.

10 Points M and N have position vectors **m** and **n** respectively relative to the origin O.

If $\mathbf{m} = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$ and $\mathbf{MN} = \begin{pmatrix} 7 \\ 10 \end{pmatrix}$, find

- (a) **n** (b) **|n|**
(c) the coordinates of a point P such that OM is the short diagonal of parallelogram MNOP.

Properties of shapes

In the previous section, position vectors were restricted to the cartesian plane. However, vector methods can be used in any geometrical situation. They are often used to discover and prove properties of shapes.

In Fig. 18.10, PQRS is a parallelogram as shown.

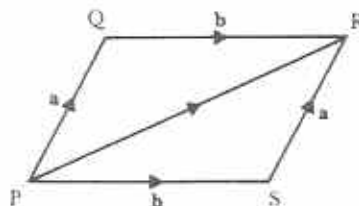


Fig. 18.10

$$\begin{aligned}\overline{PR} &= \overline{PQ} + \overline{QR} \text{ or } \overline{PS} + \overline{SR} \\ &= \mathbf{a} + \mathbf{b} \text{ or } \mathbf{b} + \mathbf{a}\end{aligned}$$

Hence $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$

This result shows that the addition of vectors is not affected by the order in which they are taken.

In Fig. 18.11 ABCD is any quadrilateral with vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ as shown.

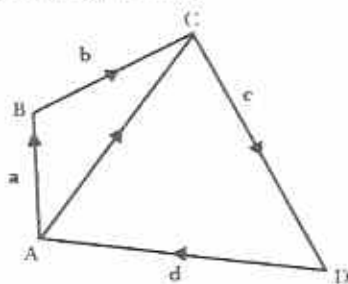


Fig. 18.11

$$\mathbf{a} + \mathbf{b} = \overline{AC}$$

$$\mathbf{c} + \mathbf{d} = \overline{CA}$$

adding,

$$\mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = \overline{AC} + \overline{CA}$$

$$\text{but } \overline{AC} + \overline{CA} = \mathbf{0}$$

$$\text{so } \mathbf{a} + \mathbf{b} + \mathbf{c} + \mathbf{d} = \mathbf{0}$$

If the vectors in Fig. 18.11 are taken to be displacements and the + sign is thought of as meaning 'followed by', the above result is hardly surprising. The total final displacement from the starting point, A, is zero when the vectors form the sides of a closed polygon.

Notice how the above results are used in the following examples.

Example 5

PQRS is any quadrilateral. A, B, C, D are the mid-points of PQ, QR, RS, SP respectively. Prove that ABCD is a parallelogram.

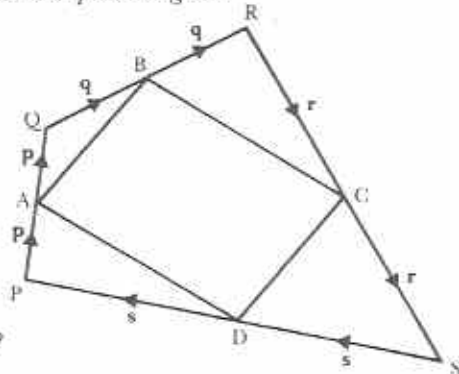


Fig. 18.12

Let $\overline{PQ} = 2\mathbf{p}$, $\overline{QR} = 2\mathbf{q}$, $\overline{RS} = 2\mathbf{r}$, $\overline{SP} = 2\mathbf{s}$, as shown in Fig. 18.12.

Considering the opposite sides AB and CD of quadrilateral ABCD:

$$\overline{AB} = \mathbf{p} + \mathbf{q}$$

$$\overline{CD} = \mathbf{r} + \mathbf{s}$$

$$\text{But } 2\mathbf{p} + 2\mathbf{q} + 2\mathbf{r} + 2\mathbf{s} = \mathbf{0}$$

$$\Leftrightarrow \mathbf{p} + \mathbf{q} + \mathbf{r} + \mathbf{s} = \mathbf{0}$$

$$\Leftrightarrow \mathbf{p} + \mathbf{q} = -\mathbf{r} - \mathbf{s}$$

$$\Leftrightarrow \mathbf{p} + \mathbf{q} = -(\mathbf{r} + \mathbf{s})$$

$$\text{Hence } \overline{AB} = \mathbf{p} + \mathbf{q} = -(\mathbf{r} + \mathbf{s}) = -\overline{CD}$$

$$\text{i.e. } \overline{AB} = \overline{DC}$$

If $\overline{AB} = \overline{DC}$, then $AB \parallel DC$ and $AB = DC$. ABCD is a parallelogram since it has a pair of opposite sides which are parallel and equal.

Example 6

In Fig. 18.13, P divides the line AB in the ratio $AP:PB = 7:3$. If $\overline{OA} = \mathbf{a}$ and $\overline{OB} = \mathbf{b}$, express $\overline{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

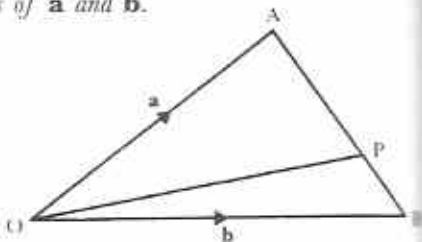


Fig. 18.13

In $\triangle OAB$,

$$\overline{OA} + \overline{AB} = \overline{OB}$$

$$\mathbf{a} + \overline{AB} = \mathbf{b}$$

$$\overline{AB} = \mathbf{b} - \mathbf{a}$$

Along $\overline{AB}$,

$$\overline{AP} = \frac{7}{10}\overline{AB}$$

$$= \frac{7}{10}(\mathbf{b} - \mathbf{a})$$

In $\triangle OAP$

$$\overline{OP} = \overline{OA} + \overline{AP}$$

$$= \mathbf{a} + \frac{7}{10}(\mathbf{b} - \mathbf{a})$$

$$= \mathbf{a} + \frac{7}{10}\mathbf{b} - \frac{7}{10}\mathbf{a}$$

$$= \frac{3}{10}\mathbf{a} + \frac{7}{10}\mathbf{b}$$

Example 7

In Fig. 18.14, $\mathbf{OA} = \mathbf{a}$ and $\mathbf{OB} = \mathbf{b}$.

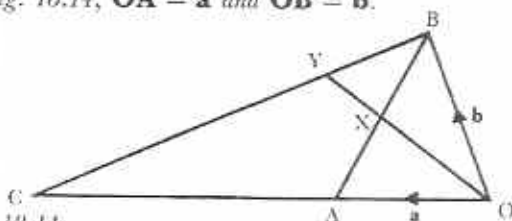


Fig. 18.14

- Express $\mathbf{BA}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
- If $\mathbf{X}$ is the midpoint of $\mathbf{BA}$, show that $\mathbf{OX} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$.
- Given that $\mathbf{OC} = 3\mathbf{a}$, express $\mathbf{BC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
- Given that $\mathbf{BY} = m\mathbf{BC}$, express $\mathbf{OY}$ in terms of $\mathbf{a}$, $\mathbf{b}$, m .
- If $\mathbf{OY} = n\mathbf{OX}$ use the results of (b) and (d) to evaluate m and n .

(a) In $\triangle OPA$,

$$\mathbf{OB} + \mathbf{BA} = \mathbf{OA}$$

$$\mathbf{b} + \mathbf{BA} = \mathbf{a}$$

$$\mathbf{BA} = \mathbf{a} - \mathbf{b}$$

(b) In $\triangle OBX$,

$$\mathbf{BX} = \frac{1}{2}\mathbf{BA} = \frac{1}{2}(\mathbf{a} - \mathbf{b})$$

$$\mathbf{OX} = \mathbf{OB} + \mathbf{BX}$$

$$= \mathbf{b} + \frac{1}{2}(\mathbf{a} - \mathbf{b}) = \frac{1}{2}(\mathbf{a} + \mathbf{b})$$

(c) In $\triangle OBC$,

$$\mathbf{BC} = \mathbf{BO} + \mathbf{OC}$$

$$= -\mathbf{b} + 3\mathbf{a} = 3\mathbf{a} - \mathbf{b}$$

(d) In $\triangle OBY$,

$$\mathbf{OY} = \mathbf{OB} + \mathbf{BY}$$

$$= \mathbf{b} + m\mathbf{BC} = \mathbf{b} + m(3\mathbf{a} - \mathbf{b})$$

$$\mathbf{OY} = 3m\mathbf{a} + (1 - m)\mathbf{b} \quad (1)$$

(e) $\mathbf{OY} = n\mathbf{OX}$

$$= n\left(\frac{1}{2}(\mathbf{a} + \mathbf{b})\right)$$

$$\mathbf{OY} = \frac{1}{2}n\mathbf{a} + \frac{1}{2}n\mathbf{b} \quad (2)$$

$$\text{also } \mathbf{OY} = 3m\mathbf{a} + (1 - m)\mathbf{b} \quad (1)$$

Since the vectors are identical, the scalars multiplying $\mathbf{a}$ and $\mathbf{b}$ can be equated:

$$\frac{1}{2}n = 3m \quad (\text{scalars of } \mathbf{a})$$

$$\frac{1}{2}n = 1 - m \quad (\text{scalars of } \mathbf{b})$$

$$3m = 1 - m \Leftrightarrow 4m = 1$$

$$\Leftrightarrow m = \frac{1}{4}$$

$$\text{If } m = \frac{1}{4}, \text{ then } \frac{1}{2}n = 3 \times \frac{1}{4}$$

$$\Leftrightarrow n = 1\frac{1}{2}$$

Notice the last stage of Example 7. In general if $h\mathbf{a} + k\mathbf{b} = n\mathbf{a} + m\mathbf{b}$ then $h = n$ and $k = m$ (or $\mathbf{a} = \mathbf{0}$ and $\mathbf{b} = \mathbf{0}$)

Exercise 18c

Make sketches where necessary.

1 Using Fig. 18.15, represent each of the following by a single vector.

- $\mathbf{PQ} + \mathbf{QR}$
- $\mathbf{PR} + \mathbf{RS}$
- $\mathbf{PS} + \mathbf{ST}$
- $\mathbf{PR} + \mathbf{RT}$
- $\mathbf{PQ} + \mathbf{QR} + \mathbf{RS}$
- $\mathbf{PQ} + \mathbf{QT} + \mathbf{TS}$
- $\mathbf{PQ} + \mathbf{QR} + \mathbf{RS} + \mathbf{ST}$
- $\mathbf{PQ} + \mathbf{QT} + \mathbf{TR} + \mathbf{RS}$

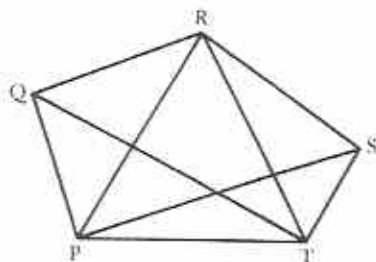


Fig. 18.15

2 In Fig. 18.16 $\mathbf{OA} = \mathbf{a}$, $\mathbf{OB} = \mathbf{b}$ and M is the mid-point of AB .

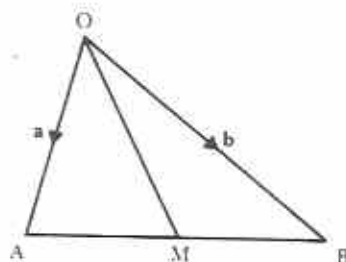


Fig. 18.16

Find $\mathbf{OM}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.

3 Given Fig. 18.17, overleaf, express $\mathbf{XY}$, $\mathbf{YZ}$ and $\mathbf{ZX}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

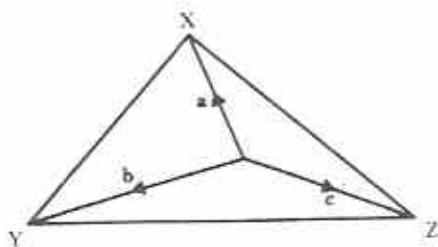


Fig. 18.17

- 4 In Fig. 18.18, $\vec{OP} = 2\mathbf{a}$, $\vec{PQ} = 2\mathbf{b} - 3\mathbf{a}$,
 $\vec{OR} = 3\mathbf{b}$.

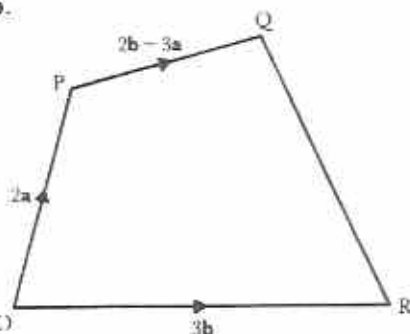


Fig. 18.18

Express (a) $\vec{OQ}$, (b) $\vec{QR}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ as simply as possible.

- 5 In Fig. 18.19, ORST is a parallelogram,
 $\vec{OR} = \mathbf{r}$ and $\vec{OT} = \mathbf{t}$.

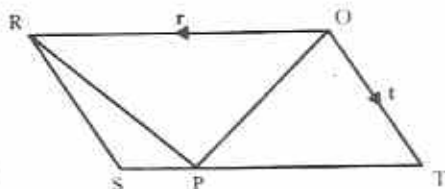


Fig. 18.19

If $\vec{ST} = 4\vec{SP}$, express the following in terms of $\mathbf{r}$ and/or $\mathbf{t}$.

- (a) $\vec{RS}$ (b) $\vec{ST}$ (c) $\vec{SP}$ (d) $\vec{RP}$ (e) $\vec{OP}$

- 6 In $\triangle PQR$, $\vec{PQ} = \mathbf{a}$, $\vec{PR} = \mathbf{b}$ and S is the mid-point of PR. Express the following in terms of $\mathbf{a}$ and/or $\mathbf{b}$.

- (a) $\vec{QR}$ (b) $\vec{PS}$ (c) $\vec{QS}$

- 7 PQRS is a trapezium in which $PQ \parallel SR$,
 $\vec{PQ} = \mathbf{a}$ and $\vec{QR} = \mathbf{b}$. E is the mid-point of PS and PQ is half as long as SR. Express the following in terms of $\mathbf{a}$ and/or $\mathbf{b}$.

- 8 ABCDEF is a regular hexagon. If $\vec{AB} = \mathbf{x}$ and $\vec{AF} = \mathbf{y}$, express the following in terms of $\mathbf{x}$ and $\mathbf{y}$.

- (a) $\vec{FC}$ (b) $\vec{BC}$ (c) $\vec{FE}$
 (d) $\vec{AE}$ (e) $\vec{AD}$ (f) $\vec{AC}$

- 9 In $\triangle PQR$, A is a point on PR such that $\vec{PA} = \frac{1}{3}\vec{PR}$ and B is the mid-point of QR. Point C lies on PQ produced so that $\vec{PC} = \frac{3}{2}\vec{PQ}$. If $\vec{PR} = \mathbf{x}$ and $\vec{PQ} = \mathbf{y}$, express the following in terms of $\mathbf{x}$ and $\mathbf{y}$.

- (a) $\vec{PA}$ (b) $\vec{PB}$ (c) $\vec{PC}$ (d) $\vec{AB}$ (e) $\vec{BC}$

- 10 In Fig. 18.20, M and N are the mid-points of AD and DC respectively.

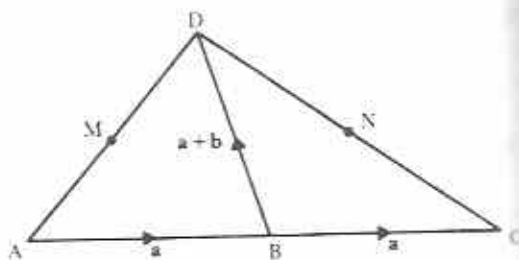


Fig. 18.20

It is given that $\vec{AB} = \vec{BC} = \mathbf{a}$ and that $\vec{BD} = \mathbf{a} + \mathbf{b}$. Write down as simply as possible in terms of $\mathbf{a}$ and/or $\mathbf{b}$ expressions for

- (a) $\vec{AD}$, (b) $\vec{DC}$, (c) $\vec{MN}$. [Camb]

- 11 Use vectors to show that if the diagonals of a quadrilateral bisect each other the quadrilateral is a parallelogram.
 12 Use vectors to show that the diagonals of a parallelogram bisect each other.
 13 In Fig. 18.21, OAB is any triangle, M and N are the mid-points of OA and OB respectively.

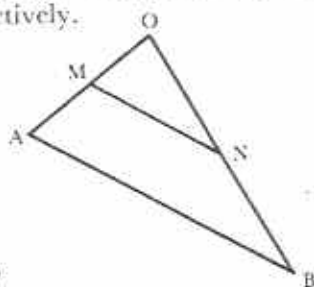


Fig. 18.21

If $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$,

- (a) express $\vec{AB}$, $\vec{OM}$, $\vec{ON}$, $\vec{MN}$ in terms of $\mathbf{a}$ and/or $\mathbf{b}$, (b) hence describe any relationship between line segments MN and AB.

- 14 ABCD is a quadrilateral whose diagonals are equal in length. The mid-points of AB

BC, CD, DA are joined in order to form a quadrilateral. Use a vector method to show that the quadrilateral so formed is a rhombus.

15 ABCD is a kite. The mid-points of AB, BC, CD, DA are joined to form a quadrilateral. Show that the quadrilateral so formed is a rectangle.

- 16 In trapezium PQRS, $\vec{QP} = \mathbf{a}$, $\vec{RQ} = \mathbf{b}$, $\vec{RS} = 3\mathbf{a}$ and the diagonals intersect at X.
- (a) Express $\vec{RP}$ and $\vec{QS}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
- (b) Show that $PX:PR = QX:QS = 1:4$.
- 17 In Fig. 18.22, P is a point on AB such that $\vec{BA} = 4\vec{BP}$ and Q is the mid-point of OA. OP and BQ intersect at X.

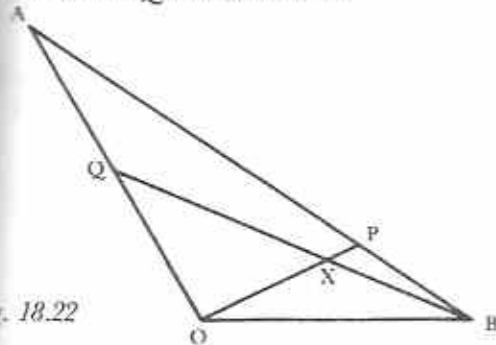


Fig. 18.22

Given $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$:

- (a) Express the following in terms of $\mathbf{a}$ and $\mathbf{b}$.
(i) $\vec{AB}$ (ii) $\vec{OP}$ (iii) $\vec{BQ}$
- (b) If $\vec{BX} = h\vec{BQ}$, express $\vec{OX}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and h .
- (c) If $\vec{OX} = k\vec{OP}$, use the previous result to find h and k .
- (d) Hence express $\vec{OX}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ only.
- 18 In Fig. 18.23, OABC is a parallelogram, M is the mid-point of $\vec{OA}$ and $\vec{AX} = \frac{2}{7}\vec{AC}$. $\vec{OA} = \mathbf{a}$ and $\vec{OC} = \mathbf{c}$.

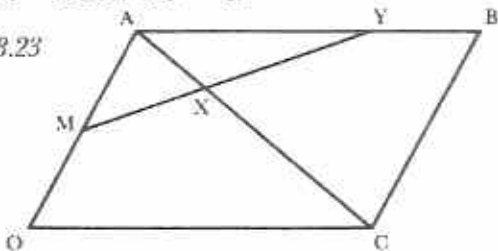


Fig. 18.23

- (a) Express the following in terms of $\mathbf{a}$ and $\mathbf{c}$.
(i) $\vec{MA}$ (ii) $\vec{AB}$ (iii) $\vec{AC}$ (iv) $\vec{AX}$

- (b) Using $\triangle MAX$, express $\vec{MX}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.
- (c) If $\vec{AY} = p\vec{AB}$, use $\triangle MAY$ to express $\vec{MY}$ in terms of $\mathbf{a}$, $\mathbf{c}$ and p .
- (d) Also if $\vec{MY} = q\vec{MX}$, use the result in (b) to express $\vec{MY}$ in terms of $\mathbf{a}$, $\mathbf{c}$ and q .
- (e) Hence find p and q and the ratio $AY:YB$.
- 19 In Fig. 18.24 $\vec{OP} = \mathbf{a}$ and $\vec{OS} = \mathbf{b}$.

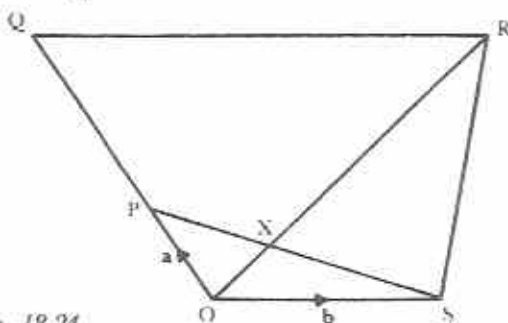


Fig. 18.24

- (a) Express $\vec{SP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
- (b) Given that $\vec{SX} = h\vec{SP}$, show that $\vec{OX} = ha + (1-h)\mathbf{b}$.
- (c) Given that $\vec{OQ} = 3\mathbf{a}$ and $\vec{QR} = 2\mathbf{b}$, write down an expression for $\vec{OR}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
- (d) Given that $\vec{OX} = k\vec{OR}$ use the results of parts (b) and (c) to find the values of h and k .
- (e) Find the numerical value of the ratio $\frac{PX}{XS}$. [Camb]

- 20 ABC is any triangle. M and N are the mid-points of BC and AC respectively and AM and BN intersect at G. $\vec{AB} = \mathbf{x}$ and $\vec{AC} = \mathbf{y}$.

- (a) Express $\vec{AM}$ in terms of $\mathbf{x}$ and $\mathbf{y}$.
- (b) If $\vec{AG} = h\vec{AM}$, express $\vec{AG}$ in terms of $\mathbf{x}$, $\mathbf{y}$ and h .
- (c) Express $\vec{BN}$ in terms of $\mathbf{x}$ and $\mathbf{y}$.
- (d) If $\vec{BG} = k\vec{BN}$, express $\vec{AG}$ in terms of $\mathbf{x}$, $\mathbf{y}$ and k .
- (e) Use the results of (b) and (d) to find h and k .
- (f) What can you deduce about the three lines joining the mid-points of the sides of a triangle to the opposite vertices?

Probability (2) Combined probabilities

Probability

Probability is a numerical measure of the likelihood of an event happening or not happening. For example, if it has rained in Gwanda in 9 out of the last 12 Septembers, then, statistically, the probability of rain falling in Gwanda next September is $\frac{9}{12}$ (or $\frac{3}{4}$ or 0,75). This is an example of **experimental probability**. Since experimental probability uses numerical records of past events to predict the future, its predictions cannot be taken to be absolutely accurate.

Alternatively, the probability of throwing a five on a fair six-sided die is $\frac{1}{6}$, since any one of the six faces is equally likely. This is an example of **theoretical probability**. Theoretical probabilities are exact values which can be calculated by considering the physical nature of the given situations.

Example 1

Tendai and Samuel have played each other at tennis 15 times this season. Tendai has won 12 of the matches. They play each other in a championship. What is the probability that (a) the match is drawn, (b) Tendai wins, (c) either Tendai or Samuel wins?

- (a) Tennis matches are either won or lost. They are never drawn.
Probability of a draw = 0
- (b) Tendai has won 12 of the last 15 matches.
Experimental probability of Tendai winning the match
 $= \frac{12}{15} = \frac{4}{5} = 0,8$
- (c) Since one or other of Tendai or Samuel must win, probability of either person winning = 1.

If p is the probability of an event happening then p lies in the range $0 \leq p \leq 1$. The probability

of an event *not* happening is p' where $p' = 1 - p$. For instance, in Example 1, the probability of Tendai *not* winning is $1 - 0,8$; i.e. 0,2.

Probability can also be described in a language. If $p(R)$ is the probability of a required outcome happening, then

$$p(R) = \frac{n(R)}{n(\mathcal{E})}$$

where R = (required outcomes)
 $\mathcal{E}$ = (all possible outcomes)

Example 2

A letter is chosen at random from the alphabet. Find the probability that it is (a) F, (b) F or T, (c) one of the letters of the word FREQUENCY, (d) one of the letters of the word TABLE.

In every case,

$$\mathcal{E} = \{A; B; C; \dots; Z\}$$

- (a) Let $A = \{F\}$

$$\text{then } p(A) = \frac{n(A)}{n(\mathcal{E})} = \frac{1}{26}$$

The probability that F is chosen is $\frac{1}{26}$.

- (b) Let $B = \{F; T\}$

$$\text{then } p(B) = \frac{n(B)}{n(\mathcal{E})} = \frac{2}{26} = \frac{1}{13}$$

There is a $\frac{1}{13}$ probability that F or T is chosen.

- (c) Let $C = \{F; R; E; Q; U; N; C; Y\}$

$$\text{then } p(C) = \frac{n(C)}{n(\mathcal{E})} = \frac{8}{26} = \frac{4}{13}$$

The probability of choosing one of the letters of frequency is $\frac{4}{13}$.

- (d) Let $D = \{T; A; B; L; E\}$

$$\text{then } p(D) = \frac{n(D)}{n(S)} = \frac{5}{26}$$

$$\text{and } p(D') = 1 - p(D) = 1 - \frac{5}{26} = \frac{21}{26}$$

There is a $\frac{21}{26}$ chance that none of the letters is in TABLE.

Notes:

- 1 'At random' means 'in a free irregular way'.
- 2 The letter E in part (c) of Example 2 was *not* counted twice.
- 3 Part (d) of Example 2 is most conveniently solved using the method of subtraction as shown. $p(D')$ means 'the probability of not being in D'.

Exercise 19a (revision)

- 1 A statistical survey shows that 28% of all men take size 9 shoes. What is the probability that your friend's father takes size 9 shoes?
- 2 A school contains 357 boys and 323 girls. If a student is chosen at random, what is the probability that a girl is chosen?
- 3 A State Lottery sells $1\frac{1}{2}$ million tickets of which 300 are prizewinners. What is the probability of getting a prize by buying just one ticket?
- 4 Statistics show that 92 out of every 100 adults are at least 150 cm tall. What is the probability that a person chosen at random from a large crowd is less than 150 cm tall?
- 5 Fig. 19.1 is a magic square.

16	2	3	13
5	11	10	8
9	7	6	12
4	14	15	1

Fig. 19.1

If a number is picked at random from Fig. 19.1 what is the probability that it is

- (a) odd,
- (b) prime,
- (c) less than 10,
- (d) exactly divisible by 3,
- (e) a perfect square,
- (f) a perfect cube?

- 6 A bag contains 2 black balls, 3 green balls, 4 red balls. A ball is picked from the bag at random. What is the probability that it is
 - (a) black,
 - (b) green,
 - (c) red,
 - (d) yellow,
 - (e) not black,
 - (f) either black or red?
- 7 A fair six-sided die is thrown. Find the probability of getting
 - (a) a 3
 - (b) a 4
 - (c) a 9
 - (d) either 1, 2 or 3
 - (e) a number divisible by 3
 - (f) a number less than 5
- 8 A letter is chosen at random from the alphabet. Find the probability that it is
 - (a) M
 - (b) not A or Z
 - (c) either P, Q, R or S
 - (d) one of the letters ZIMBABWE
- 9 Table 19.1 gives the numbers of students in age groups in a school.

Table 19.1

age	12	13	14	15	16	17	18
number	42	130	125	131	110	84	53

Find the probability that a student chosen at random is (a) 14, (b) 14 or less.

- 10 A card is picked at random from a pack of playing cards*. Find the probability of picking
 - (a) the 5 of
 - (b) the K of
 - (c) a 9
 - (d) a black Queen
 - (e) a diamond
 - (f) either a Jack, a 2 or an Ace
 - (g) a red card
 - (h) a red club

* A packet of playing cards contains 52 cards in 4 suits: clubs (), diamonds () hearts () spades (). There are 13 cards in each suit: Ace (A), 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack (J), Queen (Q), King (K). Clubs and spades are black, diamonds and hearts are red.

Mutually exclusive events

Example 3

Find the probability that a letter chosen at random from the alphabet is either a vowel or one of the letters X, Y, Z.

If $D = \{\text{desired outcomes}\}$

then $D = V \cup L$

where $V = \{\text{vowels}\} = \{A; E; I; O; U\}$

and $L = \{X; Y; Z\}$

Hence $D = \{A; E; I; O; U\} \cup \{X; Y; Z\}$
 $= \{A; E; I; O; U; X; Y; Z\}$

$$p(D) = \frac{n(D)}{n(\mathcal{E})} = \frac{8}{26} = \frac{4}{13}$$

In Example 3, if $p(V)$ and $p(L)$ are the probabilities of choosing a vowel and one of X, Y, Z respectively, then

$$p(V) = \frac{5}{26}$$

$$p(L) = \frac{3}{26}$$

and, by inspection,

$$\begin{aligned} p(D) &= p(V) + p(L) \\ &= \frac{5}{26} + \frac{3}{26} \\ &= \frac{8}{26} = \frac{4}{13} \text{ as before} \end{aligned}$$

The task of choosing a letter which is either a member of V or a member of L involves separate events which cannot happen together, i.e. one event excludes the other. They are said to be **mutually exclusive** events. In such cases the separate probabilities are added to give the combined probability.

The Venn diagram in Fig. 19.2 represents the situation in which sets V and L are mutually exclusive, or disjoint:

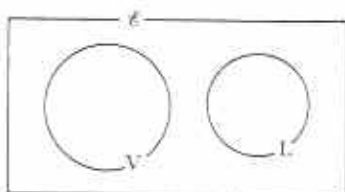


Fig. 19.2

In this case $p(V) + p(L) = p(V \cup L)$

Addition law

If events $A, B, C, \dots$ are mutually exclusive, the probability of A or B or C or $\dots$ happening is the sum of their individual probabilities $p(A) + p(B) + p(C) + \dots$

Example 4

A number is chosen at random from the set $\{2; 4; 6; \dots; 18; 20\}$. Find the probability that it is either a factor of 18 or a multiple of 5.

$$\mathcal{E} = \{2; 4; 6; \dots; 18; 20\}$$

$$\text{Let } F = \{\text{factors of 18}\} = \{2; 6; 18\}$$

$$\text{and } M = \{\text{multiples of 5}\} = \{10; 20\}$$

It follows that F and M are mutually exclusive.

$$p(F) = \frac{n(F)}{n(\mathcal{E})} = \frac{3}{10}$$

$$p(M) = \frac{n(M)}{n(\mathcal{E})} = \frac{2}{10}$$

The probability that either F or M happens

$$\begin{aligned} &= p(F) + p(M) \\ &= \frac{3}{10} + \frac{2}{10} = \frac{5}{10} \\ &= \frac{1}{2} \end{aligned}$$

Notice that the addition law is used to solve problems which contain the words *or* or *either/or*.

Exercise 19b

- 1 A card is chosen at random from a pack of playing cards. What is the probability that it is either a heart or the Queen of spades?
- 2 $F = \{2; 3; 7\}$ and $T = \{10; 20; 30; 40\}$.
 - (a) If one element is selected at random from F , write down the probability that it is odd.
 - (b) If one element is selected at random from T , write down the probability that it is a multiple of 5.
 - (c) If one element is selected at random from $F \cup T$ write down the probability that it is either a prime factor of 42 or a multiple of 4.
- 3 In a game of chance, an arrow spins at random. When it stops it points to one of eight sectors numbered as shown in Fig. 19.3.



Fig. 19.3

Find the probability that the arrow points at

- (a) a 3 or a 4, (b) a 1 or a 4, (c) a 1 or a 2.
 4 A bag contains 3 red balls, 4 blue balls, 5 white balls and 6 black balls. A ball is picked at random. What is the probability that it is either
 (a) red or blue,
 (b) red or white,
 (c) blue or white,
 (d) blue or black,
 (e) red, white or blue,
 (f) blue, white or black?
 5 A letter is chosen at random from the word COMPUTER. What is the probability that it is
 (a) either in the word CUT or in the word ROPE,
 (b) neither in the word MET nor in the word UP?

Independent events

Example 5

A die is thrown and a coin is tossed. What is the probability of getting both a six and a tail?

Table 19.2 contains all the possible outcomes of throwing the die and the coin at the same time.

Table 19.2

	die					
	1	2	3	4	5	6
head (h)	h1	h2	h3	h4	h5	h6
tail (t)	t1	t2	t3	t4	t5	t6

Table 19.2 shows that there are 12 possible outcomes of which one, ringed, gives a six and a tail.

Required probability = $\frac{1}{12}$

In Example 5 if $p(S)$ and $p(T)$ are the probabilities of getting a six and a tail then

$$p(S) = \frac{1}{6}$$

$$p(T) = \frac{1}{2}$$

Probability of getting both

$$= p(S) \times p(T)$$

$$= \frac{1}{6} \times \frac{1}{2} = \frac{1}{12} \text{ as before.}$$

The task of getting both a six and a tail involves two events which have no effect on each other. They are said to be **independent events**. In such cases, the separate probabilities are multiplied to give the combined probability.

Product law

If events A, B, C, ... are independent, the probability of A and B and C and ... happening is the product of their individual probabilities: $p(A) \times p(B) \times p(C) \times \dots$

The Venn diagram in Fig. 19.4 shows the intersecting sets S and T.

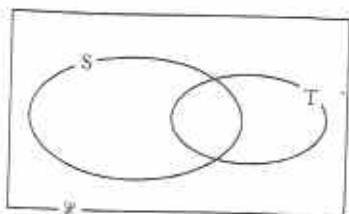


Fig. 19.4

Check that for any pair of intersecting sets,

$$n(S \cup T) = n(S) + n(T) - n(S \cap T)$$

Dividing each term by $n(\mathcal{E})$,

$$\frac{n(S \cup T)}{n(\mathcal{E})} = \frac{n(S)}{n(\mathcal{E})} + \frac{n(T)}{n(\mathcal{E})} - \frac{n(S \cap T)}{n(\mathcal{E})}$$

$$\Rightarrow p(S \cup T) = p(S) + p(T) - p(S \cap T)$$

The above probability equation may be used to simplify situations in which events are combined. If S and T had been mutually exclusive, i.e. disjoint, then $S \cap T = \emptyset$ and $p(S \cap T) = 0$, giving the addition law discussed earlier.

Example 6

Five girls and three boys put their names in a box. One name is picked out at random. Without replacing the first name, a second name is picked out at random. What is the probability that both are names of girls?

1st pick:

There are 5 girls and 8 names.

Probability that a girl's name is picked = $\frac{5}{8}$

2nd pick:

If a girl's name was picked, there now remain 4 girls and 7 names.

Probability that a girl's name is picked = $\frac{4}{7}$

Combined probability that both names are those of girls = $\frac{5}{8} \times \frac{4}{7} = \frac{5}{14}$

Notice that the product law is used to solve problems which contain the words *and* or *both* and.

Exercise 19c

- 1 A card is chosen from a pack of playing cards then returned to the pack. A second card is chosen. What is the probability that both cards are black?
- 2 A coin is tossed and a die is thrown. What is the probability of getting a head and a perfect square?
- 3 If the arrow in Fig. 19.3 is spun twice, what is the probability of getting
 - (a) two 3s,
 - (b) two 4s,
 - (c) a 1 followed by a 2,
 - (d) a 4 followed by a 3?
- 4 Five cards are lettered A, B, C, D, E. Three cards are chosen at random, one after the other, without replacement, and are placed in the order shown in Fig. 19.5.

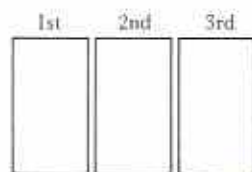


Fig. 19.5

What is the probability that the cards spell the word BED?

- 5 In a primary school 70% of the boys and 55% of the girls can ride a bicycle. If a boy and a girl are chosen at random what is the

probability that (a) both of them can ride a bicycle, (b) neither of them can ride a bicycle?

Outcome tables, tree diagrams

Example 7

Two dice are thrown at the same time. Find the probability of getting (a) at least one 5, (b) a total score divisible by 5.

Table 19.3 shows all the possible outcomes when two dice are thrown.

Table 19.3

		P ↓						
		6	7	8	9	10	11	12
second die	5	5	6	7	8	9	10	11
	4	4	5	6	7	8	9	10
	3	3	4	5	6	7	8	9
	2	2	3	4	5	6	7	8
	1	1	2	3	4	5	6	7
		1	2	3	4	5	6	
		first die						

The number of possible outcomes in Table 19.3 is $n(\mathcal{E})$ where $n(\mathcal{E}) = 36$

(a) Referring to the shaded column and row,

$P = \{\text{outcomes with 5 on the first die}\}$

$Q = \{\text{outcomes with 5 on the second die}\}$

$n(P \cup Q) = 11$

Probability of getting at least one five

$$= \frac{n(P \cup Q)}{n(\mathcal{E})} = \frac{11}{36}$$

(b) In Table 19.3, all of the total scores which are exactly divisible by 5 have been ringed.

Number of outcomes divisible by five

$$= 7$$

Probability of getting a total score divisible by five = $\frac{7}{36}$

In Example 7, notice how the table helps to overcome the problem of finding the various numbers of outcomes. A similar method was used in Example 5 on page 161.

Example 8

A bag contains 3 black balls and 2 white balls.

- (a) A ball is taken from the bag and then replaced. A second ball is chosen. What is the probability that (i) they are both black, (ii) one is black and one is white?
(b) Find out how those probabilities are affected if two balls are chosen without any replacement.

(a) The various possible ways of selecting the balls are shown in Fig. 19.6 on a **tree diagram**. In the diagram the branches of the tree show the different ways of choosing, with the related probabilities given as fractions.

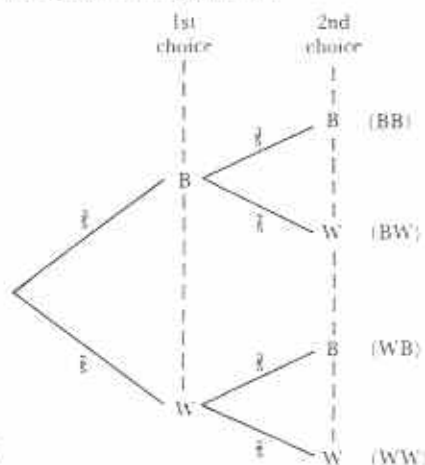


Fig. 19.6

By following the branches of the tree diagram,

- (i) Probability that the first two balls are black:

$$p(BB) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$$

- (ii) Probability that the first is black and the second is white:

$$p(BW) = \frac{3}{5} \times \frac{2}{5} = \frac{6}{25}$$

Probability that the first is white and the second is black:

$$p(WB) = \frac{2}{5} \times \frac{3}{5} = \frac{6}{25}$$

$p(BW)$ and $p(WB)$ are probabilities of mutually exclusive events. Hence the probability of getting a black ball and a white ball when the order does not matter,

$$= p(BW) + p(WB)$$

$$= \frac{6}{25} + \frac{6}{25} = \frac{12}{25}$$

- (b) If there is no replacement, then there are only 4 balls left after the first is taken. Compare the probability fractions in Fig. 19.6 with those of Fig. 19.7.

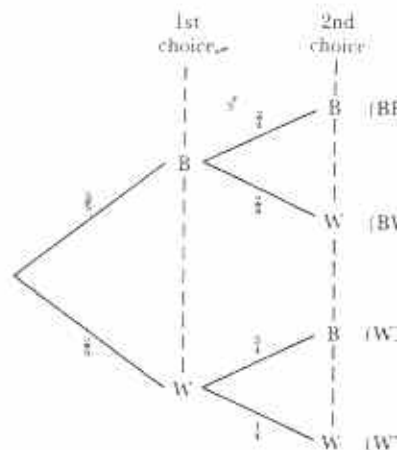


Fig. 19.7

From the tree diagram,

$$(i) \quad p(BB) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$

$$(ii) \quad p(BW) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$$

$$p(WB) = \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$$

Probability of getting a black and white regardless of order

$$= \frac{3}{10} + \frac{3}{10} = \frac{3}{5}$$

Notice in Example 8 that there are four possible outcomes: BB, BW, WB, WW and that in each case the sum of the probabilities of the outcomes is 1:

	(a)	(b)
BB	$\frac{9}{25}$	$\frac{3}{10}$
BW	$\frac{6}{25}$	$\frac{3}{10}$
WB	$\frac{6}{25}$	$\frac{3}{10}$
WW	$\frac{4}{25}$	$\frac{1}{10}$
Sum	1	1

This provides a useful check on calculation.

Example 9

If three cards are chosen from a pack without replacement, what is the probability of getting at least two spades?

Fig. 19.8, overleaf, shows the various ways of choosing the three cards.

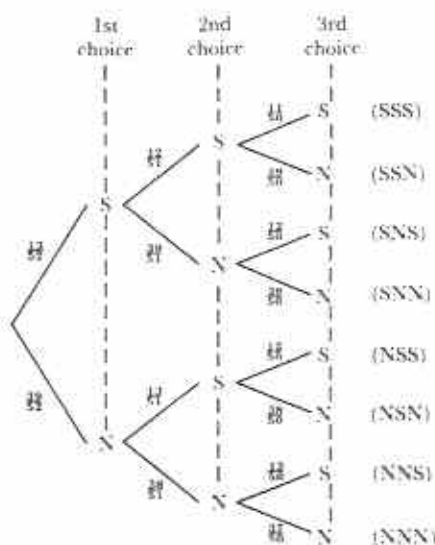


Fig. 19.8

Probability of choosing 3 spades

$$= \frac{13}{52} \times \frac{12}{51} \times \frac{11}{50} = \frac{11}{850}$$

Probability of choosing 2 spades

$$= \frac{13}{52} \times \frac{12}{51} \times \frac{39}{50} + \frac{13}{52} \times \frac{39}{51} \times \frac{12}{50} + \frac{39}{52} \times \frac{13}{51} \times \frac{12}{50}$$

$$= \frac{3 \times 12 \times 13 \times 39}{52 \times 51 \times 50} = \frac{117}{850}$$

Probability of getting at least 2 spades (i.e. 2 spades or 3 spades)

$$= \frac{11}{850} + \frac{117}{850}$$

$$= \frac{128}{850} = \frac{64}{425}$$

Exercise 19d

- 1 A pair of dice are thrown. What is the probability of getting (a) at least one six, (b) a total score of seven?
- 2 A game is played with a pentagonal spinner with sides marked 1 to 5. The score is on the side which comes to rest on the table. For example, in Fig. 19.9 the score is 3.

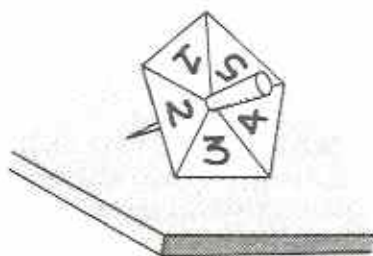


Fig. 19.9

In two spins, what is the probability of getting
(a) two 5s, (b) at least one 5,
(c) a total score of 5,
(d) a total score greater than 5?

- 3 When two dice are thrown what is the probability of the total score being a prime number?
- 4 In a large crowd, there are three times as many men as women. Three people are chosen at random. Assuming that there are so many people that choosing three has a negligible effect on the proportion of men to women, find the probability that they are
(a) all men, (b) 2 women and 1 man.
- 5 In a school, 4 out of 5 students have pens. If 2 students are picked at random, what is the probability that (a) both will have a pen, (b) one has a pen and the other has not?
- 6 M and N are the mid-points of opposite sides of square ABCD (Fig. 19.10).

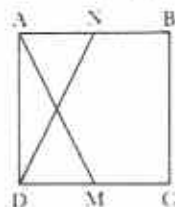


Fig. 19.10

A point is selected at random in the square. Find the probability that it lies

- (a) in $\triangle ADM$,
 - (b) in $\triangle ADM$ but not in $\triangle ADN$,
 - (c) neither in $\triangle ADM$ nor in $\triangle ADN$.
- 7 A ball is dropped at random into one of eight holes, numbered as shown in Fig. 19.11.

○	○	○	○	○	○	○	○
1	2	1	2	1	3	1	2

Fig. 19.11

The number under each hole gives the score obtained when the ball drops into that hole.

- (a) State the probability of scoring 1.
- (b) If the ball is dropped twice, find the probability of scoring (i) a total of 6, (ii) a total of 4. [Camb]

- 8 In order to choose an athletics team, 100 students were each timed over 1 500 metres. The data obtained is given in Table 19.4.

Table 19.4

time (min:sec)	4:00	4:30	5:00	5:30	6:00
number inside this time	7	28	65	88	100

- (a) A student is chosen at random. What is the probability that the student's time was inside 5 min?
- (b) A student is chosen at random from those whose time was inside 5 min 30 s. Find the probability that this student took less than 4 min 30 s.
- (c) Two students are chosen at random. Find the probability that both took 5 min 30 s or more.
- 9 The probability of a seed germinating is $\frac{1}{3}$. If 3 of the seeds are planted, what is the probability that
- none germinate,
 - at least one will germinate,
 - only one will germinate?
- 10 When three dice are thrown together what is the probability of getting a total score of 10?
- 11 A coin is tossed 3 times. What is the probability of getting
- 2 heads and 1 tail,
 - at least 1 head?
- 12 If two cards are drawn from a pack without replacement what is the probability of getting
- an Ace and a King,
 - two Aces?
- 13 It is assumed that when children are born they are equally likely to be boys or girls. What is the probability that a family of 4 children contains
- 3 boys and 1 girl,
 - 2 boys and 2 girls?
- 14 A bag contains 3 black balls, 4 white balls and 5 red balls. Three balls are removed without replacement. What is the probability of obtaining
- one of each colour,
 - at least two red balls?
- 15 A committee consists of 6 men and 4 women. A randomly chosen subcommittee is made up of 3 of the committee members. What is the probability that
- they are all women,
 - 2 of them are men?

Revision course

The remainder of this book is a revision course for students taking School Certificate/'O' level in mathematics.

It is assumed that readers have carefully worked through the previous chapters and books of the 'New General Mathematics' course. For this reason, explanations have been kept to a minimum. The revision course relies mainly on worked examples to show ways of solving problems and on exercises for further practice.

In order to make the revision course easy to use, each chapter covers both the elementary and more advanced parts of a topic.

At this stage the order in which the topics are studied is relatively unimportant. Therefore the revision chapters may be arranged to suit the needs of the reader. The certificate-level practice papers on pages 267 to 281 may be attempted after revision has been completed.

Contents of revision course

Chapter 21 General arithmetic	page 174	Chapter 23 Equations and inequalities	page 199
Chapter 22 Algebraic processes	page 187	Chapter 24 Properties of plane shapes, constructions, locus	page 209
		Chapter 25 Mensuration	page 224
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		Chapter 27 Matrices, transformations, vectors	page 241
		Chapter 28 Travel graphs, statistics, probability	page 248
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		Certificate-level practice examinations	page 267

Chapter 21

General arithmetic

Fractions, decimals, percentages

Example 1

Which is greater, $\frac{5}{6}$ or $\frac{7}{8}$?

24 is the LCM of the denominators, 6 and 8. Express each fraction with a common denominator of 24.

$$\frac{5}{6} = \frac{5 \times 4}{6 \times 4} = \frac{20}{24}$$

$$\frac{7}{8} = \frac{7 \times 3}{8 \times 3} = \frac{21}{24}$$

Since $\frac{21}{24} > \frac{20}{24}$, $\frac{7}{8}$ is greater than $\frac{5}{6}$.

Example 2

Evaluate $2\frac{1}{6} + (3\frac{3}{5} \div 1\frac{1}{8})$.

$$\begin{aligned} 2\frac{1}{6} + (3\frac{3}{5} \div 1\frac{1}{8}) &= \frac{13}{6} + \left(\frac{18}{5} \div \frac{9}{8}\right) \\ &= \frac{13}{6} + \left(\frac{18}{5} \times \frac{8}{9}\right) \\ &= \frac{13}{6} + \frac{16}{5} \\ &= \frac{65 + 96}{30} \\ &= \frac{161}{30} \\ &= 5\frac{11}{30} \end{aligned}$$

Example 3

Evaluate $6,2972 \times 0,251$ correct to 3 d.p.

$$\begin{aligned} 6,2972 \times 0,251 \\ &= \frac{62972}{10\,000} \times \frac{251}{1\,000} \end{aligned}$$

$$= \frac{62\,972 \times 251}{10\,000\,000}$$

$$= \frac{15\,805\,972}{10\,000\,000}$$

$$= 1,5805972 = 1,581 \text{ to 3 d.p.}$$

working:

62 972
251

62 972
3 148 60
12 591 4

15 805 972

Example 4

Find the value of $\frac{2,25 \times 7,5}{4,5}$

$$\frac{2,25 \times 7,5}{4,5}$$

$$= \frac{0,25 \times 1,5}{0,1} \text{ (after equal divisions by 9 and 5)}$$

$$= 0,25 \times 15 \text{ (multiplying num. and den. by 10)}$$

$$= 3,75$$

Example 5

Convert 550 Deutschmarks into Francs if £1 = 4,40 DM and £1 = 10,80 F.

$$4,40 \text{ DM} = 10,80 \text{ F (= £1)}$$

$$\Leftrightarrow 1 \text{ DM} = \frac{10,80}{4,40} \text{ F}$$

$$\Leftrightarrow 550 \text{ DM} = \frac{10,80}{4,40} \times 550 \text{ F} = 1\,350 \text{ F}$$

Example 6

When the length of a spring is increased by 8%, it becomes 351 mm long. What is its original length?

108% of the original length = 351 mm

$$108\% \text{ of the original length} = \frac{351}{108} \text{ mm}$$

$$\begin{aligned} \text{Original length (100\%)} &= \frac{351}{108} \times 100 \text{ mm} \\ &= 325 \text{ mm} \end{aligned}$$

Example 7

When a refrigerator is sold for \$558 the profit is 24%. What should be the selling price to make a profit of 28%?

124% of cost price = \$558

$$\begin{aligned} 128\% \text{ of cost price} &= \$558 \times \frac{128}{124} \\ &= \$558 \times \frac{32}{31} \\ &= \$18 \times 32 \\ &= \$576 \end{aligned}$$

Notice that it was not necessary to find the cost price.

Example 8

Chido sells a second-hand car to Thabo and makes a profit of 10%. Thabo then sells the car to Anna for \$4 180 making a loss of 5%. How much did Chido pay for the car?

Chido paid $\frac{100}{110}$ of what Thabo paid.

Thabo paid $\frac{100}{95}$ of what Anna paid.

Hence Chido paid

$$\frac{100}{110} \text{ of } \frac{100}{95} \text{ of what Anna paid}$$

$$= \frac{100}{110} \times \frac{100}{95} \times 4\,180$$

$$= \frac{10}{11} \times \frac{20}{19} \times 4\,180$$

$$= 10 \times 20 \times 20$$

$$= \$4\,000$$

Example 9

Find the rate of simple interest in per cent per annum at which \$142 will amount to \$295,36 in 12 years.

Amount = principal + interest

$$\$295,36 = \$142 + \text{interest}$$

$$\begin{aligned} \Leftrightarrow \text{interest} &= \$295,36 - \$142 \\ &= \$153,36 \end{aligned}$$

But $I = \frac{PRT}{100}$, where I is the interest, P is the principal, R is the rate per cent per annum and T is the time in years.

$$\text{Hence } \$153,36 = \frac{\$142 \times R \times 12}{100}$$

$$\begin{aligned} \Leftrightarrow R &= \frac{100 \times 153,36}{142 \times 12} \% \\ &= \frac{1\,278}{142} \% = \frac{639}{71} \% \\ &= 9\% \end{aligned}$$

Exercise 21a

- Arrange the following fractions in order of size from smallest to largest. $\frac{2}{3}, \frac{11}{15}, \frac{7}{10}, \frac{5}{6}, \frac{4}{5}$
- Evaluate the following.

(a) $2\frac{2}{3} + 1\frac{1}{2}$	(b) $5\frac{1}{3} - 2\frac{7}{8}$
(c) $1\frac{5}{8} + \frac{1}{2}$	(d) $1\frac{5}{6} + 2\frac{1}{3} - 3\frac{1}{2}$
(e) $\frac{3}{4} \times (1\frac{3}{5} \div \frac{1}{6})$	(f) $1\frac{1}{2} \times 6\frac{2}{5} \div \frac{2}{3}$
(g) $\frac{1\frac{2}{3} \times 7}{3\frac{1}{2}}$	(h) $2\frac{1}{2} \div (1\frac{1}{4} \div 3\frac{1}{3})$
- (a) $\frac{2}{3}$ of the students in a class do history. If 14 students do history, how many students are there in the class?

(b) In an election there were three candidates; $\frac{2}{3}$ of the electors voted for the first candidate, $\frac{1}{4}$ for the second candidate and the rest for the third. If the third got 3 290 votes, how many votes did the winner get?
- Calculate 129×54 and use the result to write down the value of

(a) $1,29 \times 5,4$	(b) $12,9 \times 0,54$
(c) $12,9 \times 0,0054$	(d) $0,129 \times 0,054$
- Calculate the following without using tables:

(a) $22,7 \times 0,38$	(b) $8,848 \div 0,28$
(c) $86,13 \div 2,7$	(d) $0,9916 \div 5,36$

- 6 Express the following quantities in terms of the units shown in brackets.
- 405 cm (km)
 - 158,7 m (km)
 - 905 g (kg)
 - 2,4 kg (g)
 - 7,03 km (m)
 - 1 305 ml (l)
- 7 Round off the following numbers to the degrees of accuracy shown in brackets.
- 3,7846 (3 d.p.)
 - 75,0794 (2 d.p.)
 - 144 (2 s.f.)
 - 9,84 (1 s.f.)
 - 34,625 (2 s.f.)
 - 34,625 (2 d.p.)
- 8 If £1 = Sh21,30 and £1 = 2,10 Roubles, convert Sh100 to Roubles. Give your answer correct to 2 d.p.
- 9 A woman changes \$62 for Lire. If £1 = \$1,55 and £1 = 2 200 lr, find the amount of Lire that she obtained.
- 10
- What percentage of 2 is 5?
 - The original area of a farm is 250 ha. The farmer sells 15% of his land. What area is left?
- 11 The length round a running track should be 400 m. The actual length is found to be 401,2 m. Calculate the percentage error in the length of the track.
- 12 A bicycle manufacturer requires wheel spokes to measure 260 mm, with a tolerance (acceptable error) of $\pm 0,5\%$. Calculate the acceptable range of length for the spokes.
- 13 A trader loses 12% by selling a watch for \$34,10. Find the cost price of the watch.
- 14 A trader makes a profit of $12\frac{1}{2}\%$ by selling some goods for \$94,50. Find her cash profit.
- 15 A boy spends 57% of his pocket money. If the amount that he spends is \$1,40 more than he has left, how much money had he originally?
- 16 When a car is sold for \$1 870 the profit is 10%. What should be the selling price to make a profit of 18%?
- 17 A woman's salary was \$12 600 in 1992 and was 15% more in 1993. If she paid a tax of $12\frac{1}{2}\%$ of her salary, how much tax did she pay in 1993?
- 18 A factory produced 3 456 radios in 1991 and 2 880 in 1992. (a) Calculate the percentage decrease in production from 1991 to 1992. (b) How many radios were produced in 1993 if there was a 15% increase over 1992?
- 19 A man's body-mass increases by 15%. He then goes on a diet and reduces his new body-mass by 15%. Is his final mass greater or less than the original, and by how much?
- 20 The total cost of a car service consists of a basic price plus a tax of 15%. Given that the total cost is \$690, calculate the basic price of the service. [Cambridge]
- 21 Find the simple interest on \$126 for 6 years at 7%.
- 22 Find the time in which \$168,40 will earn \$29,47 of interest at 5% per annum.
- 23 If \$206,40 amounts to \$237,36 in 2 years, find the rate of simple interest per annum.
- 24 Find the time in which \$108,33 will amount to \$123,50 at 8% per annum.
- 25 A salesman receives a basic monthly salary of \$580. He also gets $\frac{1}{2}\%$ commission on sales. Out of his monthly income he pays 1% as union fees and 3% as income tax. What was his net income for a month in which his sales were \$25 950?

Ratio, rate

A **ratio** is a numerical way of comparing quantities of the same kind. The quantities should be expressed in the same units. For example, the ratio of \$2 to 40c is 5:1, whether working in cents: $200:40 = 5:1$, or in Dollars: $2:\frac{2}{5} = 5:1$.

Example 10

Which ratio is greater, 3:4 or 6:11?

Express each ratio in the form $n:1$.

$$3:4 = \frac{3}{4}:1 = 0,75:1$$

$$6:11 = \frac{6}{11}:1 = 0,545 \dots:1$$

The ratio 3:4 is greater.

Example 11

Decrease \$73,35 in the ratio 5:9.

Express the ratio 5:9 as the fraction $\frac{5}{9}$.

Since the money is decreased, multiply \$73,35 by $\frac{5}{9}$.

$$\begin{aligned}\text{Required amount} &= \$73,35 \times \frac{5}{9} \\ &= \$8,15 \times 5 \\ &= \$40,75\end{aligned}$$

Example 12

If 9 men paint a building in 21 days, how long would 7 men take?

7 men take more time than 9 men.

The number of men is *decreased* in the ratio 7:9.

Hence the time taken is *increased* in the ratio 9:7.

Time is to be found, so time comes last in each line of working.

9 men take 21 days

$$\Rightarrow 7 \text{ men take } 21 \times \frac{9}{7} \text{ days} \\ = 27 \text{ days}$$

This example illustrates **inverse** ratio.

Example 13

9 notebooks cost \$12,15. How many can be bought for \$17,55?

Money is *increased* in the ratio 17,55:12,15.

The number of books will be *increased* in the same ratio.

The number of books is to be found, so this comes last in each line of working.

\$12,15 is the cost of 9 notebooks

$$\begin{aligned}\Rightarrow \$17,55 \text{ is the cost of } 9 \times \frac{17,55}{12,15} \text{ notebooks} \\ = \frac{1755}{35} \text{ notebooks} \\ = 13 \text{ notebooks}\end{aligned}$$

This is an example of **direct** ratio.

Quantities of different kinds may be connected in the form of a **rate**. For example, km/h, g/cm³ and number of people/km² are all examples of rates.

Example 14

A car travels 132 km in 1 h 15 min. Calculate the speed of the car.

1 h 15 min = $1\frac{1}{4}$ h.

In $1\frac{1}{4}$ h the car travels 132 km.

$$\begin{aligned}\text{In 1 h the car travels } \frac{132}{1\frac{1}{4}} \text{ km} &= \frac{132 \times 4}{5} \text{ km} \\ &= \frac{528}{5} \text{ km} \\ &= 105,6 \text{ km}\end{aligned}$$

The speed of the car is 105,6 km/h.

Example 15

A wooden cube has a mass of 50,48 g. If one edge of the cube measures 4 cm, calculate the density of the wood correct to 2 s.f.

$$\text{Volume of cube} = 4 \times 4 \times 4 \text{ cm}^3 = 64 \text{ cm}^3$$

$$\begin{aligned}\text{Density of wood} &= \frac{50,48}{64} \text{ g/cm}^3 = \frac{6,31}{8} \text{ g/cm}^3 \\ &= 0,78875 \text{ g/cm}^3 \\ &= 0,79 \text{ g/cm}^3 \text{ to 2 s.f.}\end{aligned}$$

Example 16

The population density of a village is 520 people/km². If the village has an area of about 3,3 km², find its population to the nearest 100 people.

The population density is 520 people/km², i.e. 1 km² contains 520 people

Hence 3,3 km² contain $520 \times 3,3$ people

$$= 1716 \text{ people}$$

The village contains 1 700 people (to the nearest 100 people).

Exercise 21b

1 Write the following as ratios in their simplest form.

(a) 16:20

(b) 150c to \$1

(c) $3\frac{1}{3}:8$

(d) 40 cm:1 m 20 cm

2 Express each of the following ratios in the form 1:n.

(a) 6:9

(b) 3:7

(c) 16:12

(d) 3,5:0,7

- 3 In each case, find out which one of the two ratios is greater.
 - (a) 17:8 or 15:6
 - (b) \$1,70:\$2 or \$3:\$4,80
 - (c) 1,5 g:2 kg or 0,5 kg:600 kg
- 4 The cost price, \$350, of a bicycle is reduced by \$105.
 - (a) Find the ratio by which the price is reduced.
 - (b) Express the price reduction as a rate of cents in the Dollar.
- 5 A car goes 60 km in 48 min. Find the speed of the car in km/h.
- 6 A shop sells oranges at 6 for \$1. A trader sells the same kind of oranges at 8 for \$1,20. Which price is cheaper, and by how much per orange?
- 7 A map is drawn on a scale of 1 cm to 5 km.
 - (a) Write the scale as a ratio in the form 1:n.
 - (b) What distance does 2,8 cm on the map represent?
- 8 A town has an area of 80 hectares and a population of 2 500 people. Calculate the population density of the town in people/ha correct to the nearest whole person.
- 9 The density of aluminium is $2,7 \text{ g/cm}^3$. Calculate the volume of a piece of aluminium of mass 13,5 g.
- 10 The telegraph poles along a road are 20 m apart and a car travels from the first to the fifteenth in $10\frac{1}{2}$ seconds. Calculate the speed of the car in km/h.
- 11 A train 180 m long travels through a station at 60 km/h. Find how many seconds it takes for the train to pass a man who is standing on the station platform.
- 12 The ages of a mother and daughter are in the ratio 8:3. If the daughter's age now is 12, what will be the ratio of their ages in 4 year's time?
- 13 A person's income is increased in the ratio 47:40. Find the increase per cent.
- 14 A tankful of water lasts 15 weeks if 3 litres a day are used. How long will the tankful last if 10 litres a day are used?
- 15 A train normally travels between two stations at v km/h. If its average speed is increased in the ratio $m:n$, will it take more or less time? In what ratio is the time changed?

Proportion, mixtures

Example 17

Three people share 30 eggs in the ratio 1:2:3. How many eggs does each get?

$$1 + 2 + 3 = 6$$

1st person gets $\frac{1}{6}$ of 30 eggs = 5 eggs

2nd person gets $\frac{2}{6}$ of 30 eggs = 10 eggs

3rd person gets $\frac{3}{6}$ of 30 eggs = 15 eggs

Check: $5 + 10 + 15 = 30$

Example 18

Mafa, Betty and Giya share \$180 so that for every \$1 that Mafa gets, Betty gets 50c, and for every \$1 that Betty gets, Giya gets \$3. Find Betty's share.

Assuming that Mafa has 1 share, then Betty gets $\frac{1}{2}$ share. Giya gets $1\frac{1}{2}$ times as much as Betty.

Hence Giya's share = $\frac{1}{2} \times 1\frac{1}{2} = \frac{3}{4}$.

They share the money in the ratio $1:\frac{1}{2}:\frac{3}{4}$

$$= 4:2:3$$

$$4 + 2 + 3 = 9$$

$$\text{Betty's share} = \frac{2}{9} \text{ of } \$180$$

$$= \$40$$

Example 19

Three numbers d, m, n are in the ratio 3:6:4. Find

the value of $\frac{4d - m}{m + 2n}$.

Since $d:m:n = 3:6:4$,

let $d = 3k, m = 6k, n = 4k$.

$$\text{Hence } \frac{4d - m}{m + 2n} = \frac{12k - 6k}{6k + 8k} = \frac{6k}{14k} = \frac{3}{7}$$

Example 20

A and B are partners in a business. A invests \$10 000 for one year and B invests \$25 000 for nine months. How should they share the first year's profit of \$4 600?

A's investment = \$10 000 for 12 months
= 120 000 Dollar-months

B's investment = \$25 000 for 9 months
= 225 000 Dollar-months

Ratio of investments, A:B = 120 000:225 000
= 8:15

$$+ 15 = 23$$

should get $\frac{8}{23}$ of \$4 600 = \$1 600

should get $\frac{15}{23}$ of \$4 600 = \$3 000

Example 21

A shopkeeper sells rice in bags. 30 bags of rice costing \$50 each are mixed with 40 bags of another rice costing \$58.75 per bag. If the mixture is sold at \$66 per bag, find the percentage gain.

Cost of first 30 bags = $\$50 \times 30 = \$1\,500$

Cost of other 40 bags = $\$58.75 \times 40 = \$2\,350$

Total cost of 70 bags

$$= \$1\,500 + \$2\,350 = \$3\,850$$

$$\text{Average cost of 1 bag} = \frac{\$3\,850}{70} = \$55$$

Profit in cash on 1 bag = $\$66 - \$55 = \$11$

$$\text{Gain \%} = \frac{11}{55} \times 100 = 20\%$$

Exercise 21c

Three children divide \$9.45 between them in the ratio 6:7:8. What is the size of the largest share?

A State lottery \$28 845 is divided between the 1st, 2nd and 3rd prizewinners in the ratio 4:3:2. How much does the 3rd prizewinner get?

\$90 is divided between Manasa, Bob and Sophie so that Sophie's share is $\frac{3}{8}$ of Bob's and Bob's share is twice that of Manasa. How much does Sophie receive?

If $a:b:c = 5:2:3$, evaluate

$$(a) \frac{a-2b}{3b-c} \quad (b) \frac{a+b+c}{5a}$$

$$(c) a-b:b+c \quad (d) 4a-b:a+2b-c$$

Find the value of the ratio $m:n$ if

$$(a) 4m = 7n \quad (b) 5m - n = 2m + 5n$$

10 kg of coffee costing \$24 per kg is mixed with 5 kg costing \$32 per kg. What should be the cost of the mixture per kg?

A shopkeeper has 5 shirts priced at \$23.20 each, 3 at \$22.00 each and 4 at \$16.00 each. He mixes the shirts up and sells them all at \$20 each. How much does he gain or lose altogether by doing this? What is his average gain or loss per shirt?

8 Tea costing \$12.90 and \$11.55 per kg are mixed together in the ratio 2:1. What is the value of the mixture per kg?

9 Copper sulphate is made from

32 parts of copper

16 parts of sulphur

32 parts of oxygen

45 parts of water.

Find the mass of copper, sulphur, oxygen and water in 4.5 kg of copper sulphate.

10 Concentrated acid costing 90c per litre is diluted with water in the ratio 1:5 by volume. The diluted acid is sold at 24c per litre. What is the percentage profit?

11 Farai invests \$15 000 in a business. Four months later Denga invests \$36 000 in the business. At the end of the first year the profit is \$7 865. How much should Farai and Denga get if the profit is shared according to the amount and duration of investment?

12 Wines at \$4.50 and \$5 per litre are mixed with a third wine in the ratio 2:3:4. If the mixture costs \$5.20 a litre, what is the cost of the third wine?

13 Sacks of rice costing \$62 and \$55 per sack are mixed together in the ratio 5:2. Find the selling price of the mixture per sack in order to make a profit of 35%.

14 A motorist averages 80 km/h for the first 60 km of a journey and 96 km/h for the next 120 km. What is her average speed for the whole journey?

15 Sugar costing \$1 215/tonne is mixed with sugar costing \$1 380/tonne in the ratio 3:8. If the mixture is sold at \$1 735.50/tonne calculate the percentage profit.

Indices

The following **laws of indices** are true for all non-zero values of a , b and x .

$$1 \quad x^a \times x^b = x^{a+b}$$

$$2 \quad x^a \div x^b = x^{a-b}$$

$$3 \quad x^0 = 1$$

$$4 \quad x^{-a} = \frac{1}{x^a}$$

$$5 \quad (x^a)^b = x^{ab}$$

$$6 \quad x^{\frac{1}{a}} = \sqrt[a]{x}$$

$$7 \quad x^{\frac{a}{b}} = \sqrt[b]{x^a} \text{ or } (\sqrt[b]{x})^a$$

Example 22 (Table 21.1)

Table 21.1

simplify	working	result
(a) $25^{\frac{1}{2}}$	$= \sqrt{25}$	$= \pm 5$
(b) $9^{\frac{1}{6}} \times 9^{\frac{1}{3}}$	$= 9^{\frac{1}{6} + \frac{1}{3}} = 9^{\frac{1}{2}} = \sqrt{9}$	$= \pm 3$
(c) $4^3 \div 4^5$	$= 4^{3-5} = 4^{-2} = \frac{1}{4^2}$	$= \frac{1}{16}$
(d) $2^6 \times 2^{-6}$	$= 2^{6+(-6)} = 2^{6-6} = 2^0$	$= 1$
(e) $(2^3)^{-2}$	$= 2^{-6} = \frac{1}{2^6}$	$= \frac{1}{64}$
(f) $27^{-\frac{2}{3}}$	$= \frac{1}{27^{\frac{2}{3}}} = \frac{1}{(\sqrt[3]{27})^2} = \frac{1}{3^2}$	$= \frac{1}{9}$
(g) $\left(\frac{81}{16}\right)^{-\frac{3}{4}}$	$= \sqrt[4]{\left(\frac{16}{81}\right)^3} = \left(\frac{\pm 2}{+3}\right)^3$	$= \pm \frac{8}{27}$

Example 23

Simplify (a) $9x^{-3} \times 2x^5$, (b) $(2d^3)^2$, (c) $2a^3 \div a^4$,
(d) $(32n)^{\frac{1}{5}}$.

$$\begin{aligned} \text{(a)} \quad 9x^{-3} \times 2x^5 &= 9 \times 2 \times x^{-3} \times x^5 \\ &= 18 \times x^{-3+5} \\ &= 18x^2 \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad (2d^3)^2 &= 2^2 \times (d^3)^2 \\ &= 4d^6 \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad (2a^3 \div a^4) &= 2(a^3 \div a^4) = 2a^{3-4} \\ &= 2a^{-1} = \frac{2}{a} \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad (32n)^{\frac{1}{5}} &= 32^{\frac{1}{5}} \times n^{\frac{1}{5}} \\ &= \sqrt[5]{32} \times \sqrt[5]{n} \\ &= 2 \times \sqrt[5]{n} \end{aligned}$$

Exercise 21d

Simplify the expressions in 1–28.

$$1 \quad 3^8 \times 3^3 \quad 2 \quad 2^{-3} \div 2^{57} \quad 3 \quad 5^3 \times 5^2$$

$$4 \quad 3^6 \div 3^2 \quad 5 \quad (2^3)^4 \quad 6 \quad (5^2)^{-1}$$

$$7 \quad 7^3 \times 7^{-3} \quad 8 \quad (32)^{\frac{2}{5}} \quad 9 \quad 5^{-2}$$

$$10 \quad 3^{-5} \div 3^{-3} \quad 11 \quad 2a^2 \times 5a \quad 12 \quad (2a)^2$$

$$13 \quad 2a^2 \times (5a)^3 \quad 14 \quad 2a^{-2} \times 5a$$

$$15 \quad (2a)^{-2} \times 5a \quad 16 \quad 2a^{-2} \times (5a)^3$$

$$17 \quad 2^{-4} \quad 18 \quad 8^{\frac{2}{3}} \quad 19 \quad 4^{-\frac{3}{2}}$$

$$20 \quad \sqrt{1\frac{9}{16}} \quad 21 \quad \left(\frac{1}{9}\right)^{-\frac{1}{2}} \quad 22 \quad 2^{\frac{2}{3}} \times 2^{\frac{1}{3}}$$

$$23 \quad 0.09^{\frac{1}{2}} \quad 24 \quad 5^x \times 5^{-x} \quad 25 \quad \sqrt[3]{4^{\frac{1}{2}}}$$

$$26 \quad \left(\frac{27}{48}\right)^{-\frac{3}{2}} \quad 27 \quad 0.216^{-\frac{2}{3}} \quad 28 \quad \sqrt[3]{8a^3}$$

29 Rewrite the following using positive indices only (e.g. $ab^{-2} = \frac{a}{b^2}$).

$$\text{(a)} \quad x^{-3} \quad \text{(b)} \quad xy^{-1} \quad \text{(c)} \quad (xy)^{-1}$$

$$\text{(d)} \quad a^{-2}b^3 \quad \text{(e)} \quad (ab^{-3})^2 \quad \text{(f)} \quad 3x^{-\frac{1}{2}}$$

30 Solve the following equations.

$$\text{(a)} \quad x^{\frac{1}{2}} = 3 \quad \text{(b)} \quad x^{\frac{1}{3}} = 2 \quad \text{(c)} \quad x^{-2} = 5$$

$$\text{(d)} \quad 3x^3 = 24 \quad \text{(e)} \quad x^{-\frac{2}{3}} = 9 \quad \text{(f)} \quad 5x = 80$$

Standard form

The number $A \times 10^n$ is said to be in **standard form** or in **scientific notation** if n is a positive or negative integer and $1 \leq A < 10$. For example, 5.3×10^6 and 9×10^{-2} are in standard form.

Example 24

Find the value of $\frac{1.26 \times 10^3}{7 \times 10^{-1}}$, expressing your answer in standard form.

$$\begin{aligned} \frac{1.26 \times 10^3}{7 \times 10^{-1}} &= \frac{1.26}{7} \times \frac{10^3}{10^{-1}} \\ &= 0.18 \times 10^{3-(-1)} = 0.18 \times 10^4 \\ &= (1.8 \times 10^{-1}) \times 10^4 \\ &= 1.8 \times 10^{-1+4} = 1.8 \times 10^3 \end{aligned}$$

Express the following numbers in standard form.

- (b) 9 500 (c) 0,95
(e) 23 (f) 0,000 23
(h) 23 000

Express the following numbers in ordinary form.

- (b) $7,01 \times 10^2$
(d) 8×10^{-5}
(f) $6,02 \times 10^3$
(h) $8,7 \times 10^{-1}$

Express $8 \times 10^{-3} + 5 \times 10^{-2} + 2 \times 10^{-1}$

as a decimal fraction, (b) in standard form.

Find the value of each of the following, expressing your answers in standard form.

- (a) $(6,2 \times 10^{-3}) + (5,08 \times 10^{-2})$
(b) $(7,3 \times 10^5) - (7,9 \times 10^4)$
(c) $(6,2 \times 10^5) \times (5 \times 10^2)$
(d) $(3,87 \times 10^{-2}) \div (9 \times 10^{-6})$

$$\frac{2,97 \times 10^4}{1,1 \times 10^{-4}}$$

Express in standard form

(i) 480, (ii) 0,016.

Evaluate $480 \div 0,016$, giving your answer in standard form. [Camb]

Given $m = 9,7 \times 10^4$ and $n = 8,3 \times 10^3$, evaluate (a) $m + n$, (b) $m - n$, giving your answers in standard form.

Given $p = 2,4 \times 10^3$ and $q = 6 \times 10^{-2}$ calculate

(a) pq , (b) $\frac{p}{q}$, expressing each of your answers in standard form.

If $r = 3,6 \times 10^3$ express (a) r^2 , (b) $\sqrt{r}$ as numbers in standard form.

Given that $a = 5 \times 10^4$ and $b = 3 \times 10^2$

find the value of (a) ab , (b) $a + b$, (c) $\frac{b}{a}$, expressing each of your answers in standard form.

[Camb]

The pages of a dictionary are numbered from 1 to 1 632. The dictionary is 6,4 cm thick (neglecting the covers).

How many thicknesses of paper make 1 632 numbered pages?

(b) Estimate the thickness of 1 page. Give your answer in metres in standard form correct to 1 significant figure.

Surds

Example 25

Simplify the following, making the number under the square root sign as small as possible.

$$(a) 2\sqrt{50} \quad (b) \frac{\sqrt{56}}{\sqrt{2}} \quad (c) \sqrt{72} \times \sqrt{75}$$

$$(a) 2\sqrt{50} = 2 \times \sqrt{25 \times 2} = 2 \times \sqrt{25} \times \sqrt{2} \\ = 2 \times 5 \times \sqrt{2} = 10\sqrt{2}$$

$$(b) \frac{\sqrt{56}}{\sqrt{2}} = \sqrt{\frac{56}{2}} = \sqrt{28} = \sqrt{4 \times 7} \\ = 2 \times \sqrt{7} = 2\sqrt{7}$$

$$(c) \sqrt{72} \times \sqrt{75} = \sqrt{36 \times 2} \times \sqrt{25 \times 3} \\ = 6\sqrt{2} \times 5\sqrt{3} \\ = 6 \times 5 \times \sqrt{2} \times \sqrt{3} \\ = 30\sqrt{6}$$

Example 26

Express each fraction with a rational denominator.

$$(a) \frac{2}{\sqrt{5}} \quad (b) \frac{\sqrt{7}}{\sqrt{3}} \quad (c) \frac{\sqrt{8}}{2\sqrt{3}}$$

$$(a) \frac{2}{\sqrt{5}} = \frac{2 \times \sqrt{5}}{\sqrt{5} \times \sqrt{5}} = \frac{2\sqrt{5}}{5}$$

$$(b) \frac{\sqrt{7}}{\sqrt{3}} = \frac{\sqrt{7} \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{7} \times \sqrt{3}}{3} = \frac{\sqrt{21}}{3}$$

$$(c) \frac{\sqrt{8}}{2\sqrt{3}} = \frac{\sqrt{8} \times \sqrt{3}}{2\sqrt{3} \times \sqrt{3}} = \frac{\sqrt{24}}{6} = \frac{2\sqrt{6}}{6} \\ = \frac{\sqrt{6}}{3}$$

Example 27Simplify $5\sqrt{18} - 3\sqrt{72} + 4\sqrt{50}$.

$$\begin{aligned}
 & 5\sqrt{18} - 3\sqrt{72} + 4\sqrt{50} \\
 &= 5\sqrt{9 \times 2} - 3\sqrt{36 \times 2} + 4\sqrt{25 \times 2} \\
 &= 5 \times 3\sqrt{2} - 3 \times 6\sqrt{2} + 4 \times 5\sqrt{2} \\
 &= 15\sqrt{2} - 18\sqrt{2} + 20\sqrt{2} \\
 &= 17\sqrt{2}
 \end{aligned}$$

Exercise 21f

- 1 Simplify the following by making the number under the square root sign as small as possible.

(a) $\sqrt{18}$ (b) $\sqrt{28}$ (c) $\sqrt{108}$ (d) $\sqrt{44}$

- 2 Express each of the following as the square root of a single number.

(a) $2\sqrt{6}$ (b) $3\sqrt{7}$ (c) $5\sqrt{6}$ (d) $12\sqrt{2}$

- 3 Simplify the following by rationalising the denominators.

(a) $\frac{2}{\sqrt{3}}$ (b) $\frac{8}{\sqrt{2}}$ (c) $\frac{21}{\sqrt{7}}$ (d) $\frac{10\sqrt{2}}{\sqrt{12}}$

(e) $\frac{2\sqrt{5}}{\sqrt{10}}$ (f) $\frac{20}{\sqrt{45}}$ (g) $\frac{\sqrt{72}}{\sqrt{75}}$

(h) $\frac{4\sqrt{5}}{3\sqrt{10}}$ (i) $\frac{12}{\sqrt{162}}$ (j) $\frac{2\sqrt{18}}{3\sqrt{12}}$

(k) $2\sqrt{3} - \frac{6}{\sqrt{3}} + \frac{3}{\sqrt{27}}$

(l) $\frac{3\sqrt{15} \times 2\sqrt{22}}{7\sqrt{2} \times \sqrt{165}}$

(m) $\frac{3\sqrt{8} \times 5\sqrt{3} \times \sqrt{7}}{\sqrt{42} \times 2\sqrt{3} \times \sqrt{15}}$

- 4 Simplify the following as far as possible.

(a) $\sqrt{20} + \sqrt{5}$ (b) $4\sqrt{3} - \sqrt{12}$

(c) $5\sqrt{7} - \sqrt{28}$

(d) $\sqrt{11} + \sqrt{44} - \sqrt{99}$

(e) $\sqrt{18} - \sqrt{32} + \sqrt{50}$

(f) $3\sqrt{27} - \sqrt{48} - 2\sqrt{75} + \sqrt{108}$

(g) $\sqrt{3} \times \sqrt{27}$ (h) $\sqrt{18} \times \sqrt{2}$

(i) $\sqrt{27} \times \sqrt{15}$

(j) $\sqrt{5} \times \sqrt{6} \times \sqrt{10} \times \sqrt{12}$

(k) $(\sqrt{3})^3$ (l) $(\sqrt{2})^6$ (m) $(3\sqrt{2})^2$

Number bases

Numbers are nearly always expressed in base ten. However, theoretically, they can be expressed in any base. In each case, the position of the digits indicates their value:

Base ten (denary)

$$693_{\text{ten}} = 6 \times 10^2 + 9 \times 10^1 + 3$$

Base five

$$4302_{\text{five}} = 4 \times 5^3 + 3 \times 5^2 + 0 \times 5^1 + 2$$

Base two (binary)

$$10110_{\text{two}} = 1 \times 2^4 + 0 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 0$$

Notice that the greatest digit in any number is 1 less than the base number.

Example 28

Express 271_{ten} as a base five number.

Use the method of repeated division.

$$\begin{array}{r}
 5 \overline{) 271} \\
 \underline{5 \ 54} + 1 \quad \text{i.e. } 54 \times 5^1 + 1 \times 1 \\
 5 \overline{) 10} + 4 \quad \text{i.e. } 10 \times 5^2 + 4 \times 5^1 \\
 5 \overline{) 2} + 0 \quad \text{i.e. } 2 \times 5^3 + 0 \times 5^2 \\
 \underline{0} + 2 \quad \text{i.e. } 0 \times 5^4 + 2 \times 5^3 \\
 \uparrow \qquad \qquad \qquad \uparrow
 \end{array}$$

Reading the remainders upwards gives

$$271_{\text{ten}} = 2041_{\text{five}}$$

Check:

$$\begin{aligned}
 2041_{\text{five}} &= 2 \times 5^3 + 0 \times 5^2 + 4 \times 5^1 + 1 \\
 &= 250 + 0 + 20 + 1 \\
 &= 271_{\text{ten}}
 \end{aligned}$$

Example 29

Convert 1101110_{two} to base five.

Convert 1010110_{two} to base ten.

$$\begin{aligned} 1010110_{\text{two}} &= 2^6 + 0 + 2^4 + 0 + 2^2 + 2^1 + 0 \\ &= 64 + 0 + 16 + 0 + 4 + 2 + 0 \\ &= 86_{\text{ten}} \end{aligned}$$

Convert 86_{ten} to base five.

$$\begin{aligned} 86 &= 75 + 10 + 1 \\ &= 3 \times 25 + 2 \times 5 + 1 \\ &= 3 \times 5^2 + 2 \times 5^1 + 1 \\ &= 321_{\text{five}} \end{aligned}$$

Example 30

Let $P = 242_{\text{five}}$ and $Q = 14_{\text{five}}$ calculate

(a) $P + Q$, (b) $P - Q$, (c) $P \times Q$, (d) $P \div Q$

Working:

$$\begin{array}{r} 242 \\ + 14 \\ \hline 311 \end{array}$$

method:

From the right,

$$2 + 4 = 6 = 1 \times 5 + 1$$

Write 1, carry 1,

$$4 + 1 + 1 = 6 = 1 \times 5 + 1$$

Write 1, carry 1

$$2 + 1 = 3$$

Working:

$$\begin{array}{r} 242 \\ - 14 \\ \hline 223 \end{array}$$

method:

$$1 \times 5 + 2 - 4 = 3$$

$$3 - 2 = 2$$

Working:

$$\begin{array}{r} 242 \\ \times 14 \\ \hline 2123 \\ 242 \\ \hline 10043 \end{array}$$

method:

$$4 \times 2 = 8 = 1 \times 5 + 3$$

Write 3, carry 1,

$$4 \times 4 + 1 = 17 = 3 \times 5 + 2$$

Write 2, carry 3.

... and so on

Working:

$$\begin{array}{r} 13 \\ 14 \overline{)242} \\ \underline{14} \\ 102 \\ \underline{102} \\ 0 \end{array}$$

method:

As in normal long

division with

subcalculations in base five.

Exercise 21g

What is the value of the 3 in 4312 if the number is in (a) base ten, (b) base five?

2 Arrange the following numbers in ascending order (i.e. from smallest to largest).

$$10111_{\text{two}}, 41_{\text{five}}, 22_{\text{ten}}$$

3 If $ab_{\text{ten}} = 32_{\text{five}}$, what is $a + b$?

4 What is the value, in base ten, of the digit 1 in each of the following?

$$(a) 10000_{\text{ten}} \quad (b) 214_{\text{five}} \quad (c) 1203_{\text{five}}$$

5 Convert the following numbers to base ten.

$$(a) 11011_{\text{two}} \quad (b) 230_{\text{five}}$$

$$(c) 413_{\text{five}} \quad (d) 1011_{\text{two}}$$

6 Express the following numbers in base two.

$$(a) 33_{\text{ten}} \quad (b) 31_{\text{ten}}$$

$$(c) 97_{\text{ten}} \quad (d) 111_{\text{ten}}$$

7 Express the following numbers in base five.

$$(a) 62_{\text{ten}} \quad (b) 312_{\text{ten}}$$

$$(c) 626_{\text{ten}} \quad (d) 555_{\text{ten}}$$

8 Express the following numbers in base two.

$$(a) 31_{\text{five}} \quad (b) 104_{\text{five}}$$

$$(c) 202_{\text{five}} \quad (d) 111_{\text{five}}$$

9 Express each of the following numbers in base five.

$$(a) 101010_{\text{two}} \quad (b) 1110110_{\text{two}}$$

$$(c) 1011111_{\text{two}} \quad (d) 1111001_{\text{two}}$$

10 Do the following additions in the given bases.

$$(a) 124 + 012 + 42 + 1124 \quad (\text{base five})$$

$$(b) 11011 + 1011 + 11110 \quad (\text{base two})$$

11 Subtract

$$(a) 2144_{\text{five}} \text{ from } 10023_{\text{five}},$$

$$(b) 110111_{\text{two}} \text{ from } 1110100_{\text{two}}.$$

12 Find the missing number if this addition is in base five:

$$\begin{array}{r} 1230 \\ 232 \\ * \\ + 123 \\ \hline 4011 \end{array}$$

13 Simplify the following. Express each answer in the given base.

$$(a) 302_{\text{five}} \times 13_{\text{five}}$$

$$(b) 11011_{\text{two}} \times 111_{\text{two}}$$

$$(c) 143_{\text{five}} \div 22_{\text{five}}$$

$$(d) 110110_{\text{two}} \div 1001_{\text{two}}$$

14 Evaluate the following binary multiplications.

$$(a) 1101 \times 11 \quad (b) 10111 \times 1011$$

$$(c) 1011101 \times 1110 \quad (d) 101010 \times 110$$

15 Evaluate the following binary divisions:

- (a) $110111 \div 1011$
- (b) $110010 \div 1010$
- (c) $110110 \div 1001$
- (d) $10000101 \div 10011$

16 Fig. 21.9 shows how a strip of graph paper may be used as a punch tape. The first two rows show how to space the 'holes'.

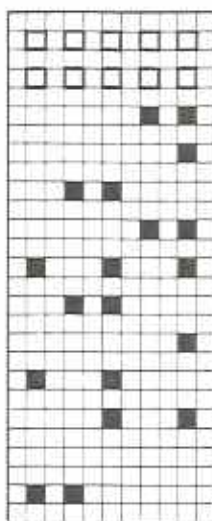


Fig. 21.1

The next row represents the first letter of the message. It shows the binary number 00011, equivalent to 3. This represents C, the 3rd letter of the alphabet.

- (a) Find out what the message is in Fig. 21.1.
- (b) Use a strip of graph paper to write the following messages in binary code.
 - (i) APPROXIMATE
 - (ii) COLLECT DATA.

Ready reckoners and tabulated data

A **ready reckoner** is a book which contains tables of numerical data which assist calculation. Sometimes ready reckoners are more convenient to use than calculators. Numerical information can be presented in many other ways, e.g. in simple lists and diagrams.

Exercise 21h contains extracts from a ready reckoner, conversion tables and other numerical information. Read the titles and column headings very carefully to find out what each is about.

Exercise 21h

Table 21.2 is a page from a ready reckoner. It gives the cost of from 1 to 140 articles at 69c each.

Table 21.2

69 cents

1	0.69	36	24.84	71	48.99	106	73.14
2	1.38	37	25.53	72	49.68	107	73.83
3	2.07	38	26.22	73	50.37	108	74.52
4	2.76	39	26.91	74	51.06	109	75.21
5	3.45	40	27.60	75	51.75	110	75.90
6	4.14	41	28.29	76	52.44	111	76.59
7	4.83	42	28.98	77	53.13	112	77.28
8	5.52	43	29.67	78	53.82	113	77.97
9	6.21	44	30.36	79	54.51	114	78.66
10	6.90	45	31.05	80	55.20	115	79.35
11	7.59	46	31.74	81	55.89	116	80.04
12	8.28	47	32.43	82	56.58	117	80.73
13	8.97	48	33.12	83	57.27	118	81.42
14	9.66	49	33.81	84	57.96	119	82.11
15	10.35	50	34.50	85	58.65	120	82.80
16	11.04	51	35.19	86	59.34	121	83.49
17	11.73	52	35.88	87	60.03	122	84.18
18	12.42	53	36.57	88	60.72	123	84.87
19	13.11	54	37.26	89	61.41	124	85.56
20	13.80	55	37.95	90	62.10	125	86.25
21	14.49	56	38.64	91	62.79	126	86.94
22	15.18	57	39.33	92	63.48	127	87.63
23	15.87	58	40.02	93	64.17	128	88.32
24	16.56	59	40.71	94	64.86	129	89.01
25	17.25	60	41.40	95	65.55	130	89.70
26	17.94	61	42.09	96	66.24	131	90.39
27	18.63	62	42.78	97	66.93	132	91.08
28	19.32	63	43.47	98	67.62	133	91.77
29	20.01	64	44.16	99	68.31	134	92.46
30	20.70	65	44.85	100	69.00	135	93.15
31	21.39	66	45.54	101	69.69	136	93.84
32	22.08	67	46.23	102	70.38	137	94.53
33	22.77	68	46.92	103	71.07	138	95.22
34	23.46	69	47.61	104	71.76	139	95.91
35	24.15	70	48.30	105	72.45	140	96.60

Use Table 21.2 to answer questions 1-5.

1 Find the cost of the following.

- (a) 8 articles at 69c each
- (b) 22 articles at 69c each
- (c) 49 articles at 69c each
- (d) 87 articles at 69c each
- (e) 133 articles at 69c each
- (f) 10 articles at 69c each

- 2 Find the cost of the following:
- 14 watches at \$69 each
 - 29 cups at \$6,90 each
 - 5 motorcycles at \$6 900 each
 - 69 eggs at 25c each
 - 690 pencils at 32c each
 - 6,9 metres of cloth at \$7,90 per metre
- 3 A trader bought 69 notebooks for 95c each and sold them all for \$1,25 each. Find the profit.
- 4 A shop bought two dozen books for \$122,50 altogether. The books were sold for \$6,90 each. Calculate the profit.
- 5 Calculate the cost of sending a telegram if it contains 69 words, each word costing 36c.

Table 21.3 gives salaries from 320 units per annum to 9 000 units per annum. Each salary is broken down into monthly and weekly rates.

Table 21.3
Annual salary 320 – 9 000 units

	Month	Week		Month	Week
320	26,667	6,154	1 600	133,333	30,769
325	27,083	6,250	1 700	141,667	32,692
330	27,500	6,346	1 750	145,833	33,654
340	28,333	6,538	1 800	150,000	34,615
350	29,167	6,731	1 900	158,333	36,538
360	30,000	6,923	2 000	166,667	38,462
370	30,833	7,115	2 100	175,000	40,385
375	31,250	7,212	2 200	183,333	42,308
380	31,667	7,308	2 250	187,500	43,269
390	32,500	7,500	2 300	191,667	44,231
400	33,333	7,692	2 400	200,000	46,154
425	35,417	8,173	2 500	208,333	48,077
450	37,500	8,654	2 600	216,667	50,000
475	39,583	9,135	2 700	225,000	51,923
500	41,667	9,615	2 750	229,167	52,885
525	43,750	10,096	2 800	233,333	53,846
550	45,833	10,577	2 900	241,667	55,769
575	47,917	11,058	3 000	250,000	57,692
600	50,000	11,538	3 100	258,333	59,615
650	54,167	12,500	3 200	266,667	61,538
700	58,333	13,462	3 300	275,000	63,462
750	62,500	14,423	3 400	283,333	65,385
800	66,667	15,385	3 500	291,667	67,308
850	70,833	16,346	4 000	333,333	76,923
900	75,000	17,308	4 500	375,000	86,538
950	79,167	18,269	5 000	416,667	96,154
1 000	83,333	19,231	5 500	458,333	105,769
1 100	91,667	21,154	6 000	500,000	115,385
1 200	100,000	23,077	6 500	541,667	125,000
1 250	104,167	24,038	7 000	583,333	134,615
1 300	108,333	25,000	7 500	625,000	144,231
1 400	116,667	26,923	8 000	666,667	153,846
1 500	125,000	28,846	9 000	750,000	173,077

Use Table 21.3 to answer questions 6–10. Give all answers to the nearest cent.

- 6 Find the monthly pay equivalent to an annual salary of
- \$950
 - \$3 100
 - \$8 000
 - \$8 400
 - \$18 000
 - \$17 360
- 7 Find the weekly wage equivalent to an annual pay of
- \$4 500
 - \$7 500
 - \$9 000
 - \$12 000
 - \$17 500
 - \$8 575
- 8 In 1992 the starting salary for a cleric officer was \$11 340. How much was this per month?
- 9 A company employs 1 caretaker at \$9 500 per annum, 1 receptionist at \$14 000 per annum and 3 clerks at \$16 500 per annum. Find the total wages paid to the workers each week. (Discuss possible reasons why the workers receive different wages.)
- 10 A Co-operative employs 69 workers who are each paid \$13 300 per annum. Use Table 21.2 and 21.3 to find the total wage bill for the Co-operative each week.

Table 21.4 is a *double conversion table* for converting between litres and gallons. The central bold numbers refer to either of the columns on each side. For example, 1 litre = 0,22 gallons and 1 gallon = 4,546 litres.

Table 21.4

litres		gallons
4,546	1	0,220
9,092	2	0,440
13,638	3	0,660
18,184	4	0,880
22,730	5	1,100
27,276	6	1,320
31,822	7	1,540
36,368	8	1,760
40,914	9	1,980

Use Table 21.4 to answer questions 11–13.

- 11 Express the following in litres, correct to one decimal place.
- 4 gallons
 - 7 gallons
 - 9 gallons
 - 10 gallons
 - 60 gallons
 - 0,8 gallons

- 12 Express the following in gallons, correct to one decimal place.
 (a) 3 litres (b) 5 litres (c) 8 litres
 (d) 10 litres (e) 90 litres (f) 6,5 litres
- 13 The petrol tank of a car contains 42 litres. How many gallons does it hold?

Table 21.5 converts speeds of km/h into either m/s (metres per second) or mph (miles per hour).

Table 21.5

km/h	m/s	mph
1	0,28	0,62
2	0,56	1,24
3	0,83	1,86
4	1,11	2,49
5	1,39	3,11
6	1,67	3,73
7	1,94	4,35
8	2,22	4,97
9	2,5	5,59

Use Table 21.5 to answer questions 14–16.

- 14 Express the following speeds in metres per second.
 (a) 5 km/h (b) 8 km/h
 (c) 60 km/h (d) 70 km/h
 (e) 89 km/h (f) 134 km/h
- 15 Express the following speeds in miles per hour.
 (a) 4 km/h (b) 7 km/h
 (c) 10 km/h (d) 80 km/h
 (e) 8,5 km/h (f) 44 km/h
- 16 Many years ago, when imperial units were used, the speed limit in built up areas was 30 mph. This was changed to 50 km/h. Which speed is slower (and therefore safer)?

Figure 21.2 may be used to convert between temperatures in degrees Celsius and Fahrenheit.

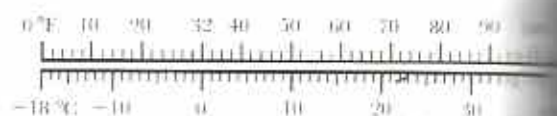


Fig. 21.2 Thermometer conversion

Use Fig. 21.2 to answer questions 17–19.

- 17 Convert the following temperatures to Celsius. Estimate your answers to the nearest °C.
 (a) 32 °F (b) 50 °F (c) 86 °F
 (d) 68 °F (e) 23 °F (f) 95 °F
- 18 Convert the following temperatures to Fahrenheit. Estimate your answers to the nearest °F.
 (a) -10 °C (b) 20 °C (c) 30 °C
 (d) 5 °C (e) 34 °C (f) 14 °C
- 19 (a) If the temperature rises by 10° on the Celsius scale, how much does it rise on the Fahrenheit scale?
 (b) Hence estimate what 50 °C is equivalent to on the Fahrenheit scale.
- 20 Table 21.6 is a table for converting Z\$ to US\$.

Table 21.6

Z\$	US\$
1	0,47
2	0,93
5	2,34
10	4,67
20	9,35
50	23,37
100	46,73

Use Table 21.6 to change the following amounts to US\$.

- (a) Z\$15 (b) Z\$40 (c) Z\$9 (d) Z\$20
 (e) Z\$135 (f) Z\$26 (g) Z\$38 (h) Z\$100

Algebraic processes

Sets

Refer to the list on page 285 for the symbols of set language and an explanation of their meanings.

Example 1

If $\mathcal{E} = \{1; 2; 3; 6; 9; 18\}$, $X = \{2; 6; 18\}$ and $Y = \{1; 3; 6\}$, list the elements of (a) $X' \cap Y$, (b) $X \cup Y'$, (c) $(X \cup Y)'$.

X' is the **complement** of X , i.e. those elements in $\mathcal{E}$ which are *not* in X .

$$X' = \{1; 3; 9\}$$

$$Y' = \{2; 9; 18\}$$

$$(a) \quad X' \cap Y = \{1; 3; 9\} \cap \{1; 3; 6\} \\ = \{1; 3\}$$

$$(b) \quad X \cup Y' = \{2; 6; 18\} \cup \{2; 9; 18\} \\ = \{2; 6; 9; 18\}$$

$$(c) \quad (X \cup Y)' \text{ is the complement of } X \cup Y. \\ X \cup Y = \{1; 2; 3; 6; 18\} \\ (X \cup Y)' = \{9\}$$

Example 2

Two sets A and B are such that $n(A) = 11$ and $n(B) = 6$. Given that $n(\mathcal{E}) = 15$ find (a) the smallest possible value of $n(A \cap B)$, (b) the largest possible value of $n(A \cup B)'$. [Camb]

Let $n(A \cap B) = x$
and $n(A \cup B)' = y$.

Fig. 22.1 is a **Venn diagram** showing the numbers of elements in each subset.

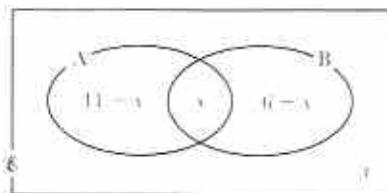


Fig. 22.1

Since $n(\mathcal{E}) = 15$,

$$(11 - x) + x + (6 - x) + y = 15 \\ 17 - x + y = 15 \quad (1)$$

(a) From (1), $x = 2 + y$

The smallest value of x occurs when y is least, i.e. when y is 0.

The smallest possible value of $n(A \cap B)$ is 2.

(b) From (1), $y = x - 2$.

The largest value of y occurs when x is greatest, i.e. when x is 6.

Note: 6 is the greatest value of x which will give non-negative values for the numbers of elements in the subsets.

The largest value of $n(A \cup B)' = 6 - 2 = 4$.

Example 3

A number of travellers were questioned about transport they used on a particular day. Each of them used one or more of the methods shown in the Venn diagram in Fig. 22.2.

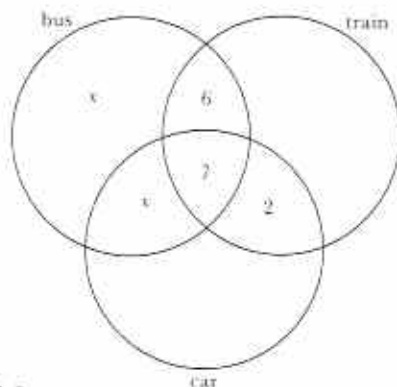


Fig. 22.2

Of those questioned, 6 said that they travelled by bus and train only, 2 by train and car only and 7 by bus and car only. The number x who travelled by bus, train and car was equal to the number who travelled by bus and car only. Given that 35 people used buses and 25 people used trains find (a) the value of x , (b) the number who

travelled by train only, (c) the number who travelled by at least two methods of transport.

Given also that 85 people were questioned altogether, calculate (d) the number who travelled by car only.

[Camb]

(a) If 35 people used buses, then

$$x + x + 6 + 7 = 35$$

$$\Leftrightarrow 2x + 13 = 35$$

$$\Leftrightarrow 2x = 22$$

$$\Leftrightarrow x = 11$$

(b) Let y people travel by train only. If 25 people used trains, then

$$6 + 7 + 2 + y = 25$$

$$\Leftrightarrow 15 + y = 25$$

$$\Leftrightarrow y = 10$$

10 people travelled by train only.

(c) The number who travelled by at least two methods (i.e. those using two or three methods)

$$= x + 6 + 2 + 7$$

$$= 11 + 15 = 26$$

(d) Let z people travel by car only, then

$$x + x + 6 + 7 + 2 + y + z = 85$$

$$11 + 11 + 15 + 10 + z = 85$$

$$\Leftrightarrow 47 + z = 85$$

$$\Leftrightarrow z = 38$$

38 people travel by car only.

Example 4

Given $\mathcal{E} = \{\text{all vehicles}\}$, $F = \{\text{four-wheeled vehicles}\}$, $E = \{\text{vehicles with engines}\}$, $C = \{\text{cars}\}$ and (i) all cars have four wheels, (ii) all cars have engines. Rewrite statements (i) and (ii) in set notation and draw as many Venn diagrams as possible which correctly illustrate them. Which, if any, of the following deductions follow from the given statements? (a) All four-wheeled vehicles have engines. (b) All vehicles with engines are either cars or have four wheels.

Fig. 22.3 shows all the possible Venn diagrams.

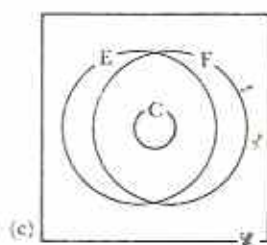
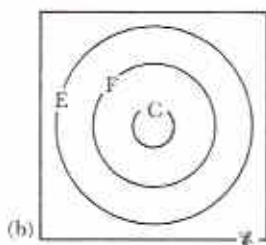
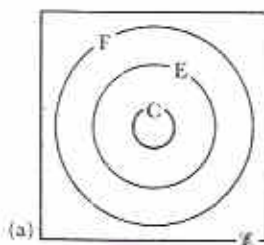


Fig. 22.3

(i): $C \subseteq F$

(ii): $C \subseteq E$

(a) FALSE The deduction is equivalent $F \subseteq E$ which is not true in either Fig. 22.3(b) or Fig. 22.3(c).

(b) FALSE The deduction is equivalent $E \subseteq C \cup F$ which is not true in either Fig. 22.3(b) or Fig. 22.3(c).

Exercise 22a

1 Given $\mathcal{E} = \{r; e; v; o; l; t; i; n; g\}$, $A = \{l; i; o; n\}$ and $B = \{t; i; g; e; r\}$, list the members of the following sets.

(a) $A \cup B$ (b) $A \cap B$ (c) A'

(d) B' (e) $A' \cup B'$ (f) $A' \cap B'$

(g) $(A \cup B)'$ (h) $(A \cap B)'$ (i) $A' \cap B$

2 For each of the following, make a copy of Fig. 22.4 then shade the given set.

(a) $(P \cup Q)'$ (b) $P' \cup Q'$ (c) $(P \cap Q)'$

(d) $P' \cap Q'$ (e) $P' \cup Q$ (f) $P \cap Q'$

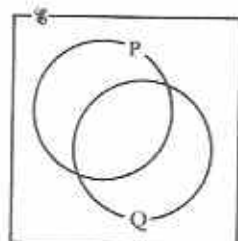


Fig. 22.4

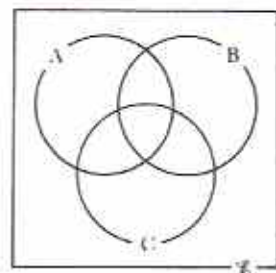


Fig. 22.5

For each of the following, make a copy of Fig. 22.5 then shade the given set.

- (a) $(A \cup B) \cap C$ (b) $(A \cap B) \cup C$
 (c) $A \cup (B \cap C')$ (d) $A \cap (B \cup C')$
 (e) $A \cap (B \cup C)'$ (f) $(A \cap C') \cup (B \cap C)$
 If $\mathcal{E} = \{1; 2; 3; \dots; 10\}$, list the members of the following subsets of $\mathcal{E}$.

- (a) $\{x: x = 2\}$
 (b) $\{x: x \text{ is a multiple of } 5\}$
 (c) $\{x: 3 < x < 8\}$
 (d) $\{x: 3x - 3 = 3\}$
 (e) $\{x: x \text{ is a factor of } 360\}$
 (f) $\{x: (x - 2)(x - 4) = 0\}$
 Given that $\mathcal{E} = \{x: x \text{ is an integer, } 2 \leq x \leq 10\}$, $A = \{x: x \text{ is a prime number}\}$, $B = \{x: x \text{ is a multiple of } 3\}$, (a) find $n(A \cup B)$, (b) list the elements of the set $A' \cap B'$.

[Camb]

- Given that $P = \{a; b; c; d\}$, $Q = \{b; c; d; e; f\}$ and $R = \{a; c; f; g\}$, mark the members of these sets clearly on the Venn diagram. (Make a copy of Fig. 22.6 in your exercise book.)

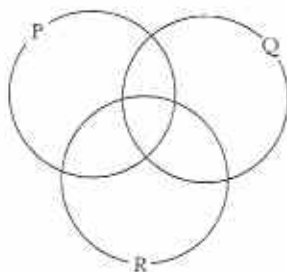


Fig. 22.6

Using your diagram, or otherwise, list the members of $(P \cup Q) \cap R$. [Camb]

- 7 $\mathcal{E} = \{\text{books}\}$, $A = \{\text{algebra books}\}$, $B = \{\text{books with brown covers}\}$. Show, by shading on a Venn diagram, the set of books with brown covers which are not algebra books.
 8 A company employs 79 people, 52 of whom are men, 38 people, including all the women, are clerical staff. Draw a suitable Venn diagram to show this information. Hence or otherwise find the number of men that are clerical staff.
 9 The numbers of elements of the subsets of the Venn diagram in Fig. 22.7 are as shown.

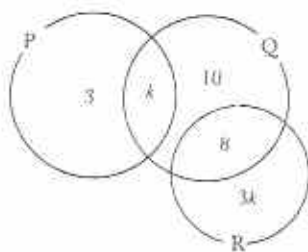


Fig. 22.7

If $\mathcal{E} = P \cup Q \cup R$ and $n(\mathcal{E}) = 33$, find

- (a) k (b) $n(P \cup R)$
 (c) $n(P \cap R)$ (d) $n(R' \cap Q)$
 10 Sets $\mathcal{E}$, P and Q are such that $n(\mathcal{E}) = 30$, $n(P) = 12$ and $n(Q) = 25$. Find (a) the smallest possible value of $n(P \cap Q)$, (b) the range of possible values of $n(P' \cap Q')$.
 11 In the Venn diagram of Fig. 22.8, $\mathcal{E}$ is the set of all children in a certain chosen group, $A = \{\text{children in Youth Club A}\}$ and $B = \{\text{children in Youth Club B}\}$. The letters p , q , x and y in the diagram represent the number of children in each subset.

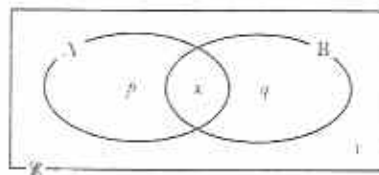


Fig. 22.8

Given that $n(\mathcal{E}) = 200$, $n(A) = 75$ and $n(B) = 35$,

- (a) express p in terms of x ,
 (b) find the smallest possible value of y ,
 (c) find the largest possible value of x ,
 (d) find the value of q if $p = 45$. [Camb]
 12 In a team of 30 athletes, some have a body-mass less than 70 kg, twice as many are over 60 kg and 12 have a body-mass between 60 kg and 70 kg. Show this information in a labelled Venn diagram. Find the number of athletes whose body-mass is (a) 60 kg or less, (b) 70 kg or more.
 13 There are 150 people at an International Medical Conference. Forty are Africans, 70 are women and 110 are doctors. Twelve of the women are Africans, 46 of the doctors are women and 31 of the Africans are doctors. If 5 of the African

men are not doctors, (a) how many of the African women are doctors, and (b) how many of the men are neither African nor doctors?

- 14 The 90 members of a sports club play at least one of the games tennis, football, volleyball. 10 play tennis and football, 19 play football and volleyball and 29 play tennis and volleyball. n people play all three games. $2n$ people each play only one game. How many play football altogether?
- 15 Use a Venn diagram to show that if $G \subseteq H$, then $H' \subseteq G'$ and $G \cap H' = \emptyset$. Hence state two of the conclusions that can be deduced from the following statements.
 $\mathcal{S} = \{\text{students}\}$, $G = \{\text{girls}\}$, $H = \{\text{happy students}\}$, $H' \neq \emptyset$ and all girls are happy students.

Simplification of algebraic expressions

Collecting like terms

Expressions such as $2x$, $3x$, $-7x$ are called **terms in x** . Since they are all terms in x they are called **like terms**. $2x$ and $3y$ are **unlike terms**. When simplifying algebraic expressions like terms are grouped together.

Example 5

Simplify $2x^2 - 3ax + 4x^2 - 6xa - x^2$.

$$\begin{aligned} 2x^2 - 3ax + 4x^2 - 6xa - x^2 \\ = 2x^2 + 4x^2 - x^2 - 3ax - 6ax \\ = 5x^2 - 9ax \end{aligned}$$

Notice that $6xa = 6ax$; the order of the letters is not important.

Exercise 22b

Simplify the following expressions.

- 1 $6x - 9y + 10x$
- 2 $2p^2 + 3q - 8q$
- 3 $7a - 2b - 3a - 8b - a$
- 4 $8 - 3x^2 + 21 + 18x^2 - 6$
- 5 $x^2 - xy - 9xy + 9y^2$
- 6 $5u^2v - 3vu^2 + uv^2 - 5v^2u$
- 7 $d^2 + 2ad - 2d^2 + 8ad + 3d^2 - 20ad$

- 8 $2a^2 - a + 6a^2 - 8a^2 - 6a + 5a^2 - + 10a$
- 9 $-3x^2 + 5x^2 + 6x^2 + 4x^3 + 2x^4 - x^2$
- 10 $m^2 - 2mn + 3n^2 + 5n^2 + 8mn - 4n^2 - 7mn$

Removing brackets

Example 6

Remove brackets from the expression

$$-3(2x - 5y + 6z).$$

$$\begin{aligned} & -3(2x - 5y + 6z) \\ & = (-3) \times (2x) + (-3) \times (-5y) + (-3) \times (6z) \\ & = -6x + 15y - 18z \end{aligned}$$

Notice that every term inside the bracket is multiplied by the quantity outside the bracket.

Example 7

Remove brackets and simplify

$$x(x - 2y) - 5y(3x - 6y).$$

$$\begin{aligned} x(x - 2y) - 5y(3x - 6y) \\ = x^2 - 2xy - 15xy + 30y^2 \\ = x^2 - 17xy + 30y^2 \end{aligned}$$

The expression $(a + b)(c + d)$ can be expanded in two ways:

$$\begin{aligned} \text{either: } (a + b)(c + d) &= (a + b)c + (a + b)d \\ &= ac + bc + ad + bd \end{aligned}$$

$$\begin{aligned} \text{or: } (a + b)(c + d) &= a(c + d) + b(c + d) \\ &= ac + ad + bc + bd \\ &= ac + bc + ad + bd \end{aligned}$$

The result is the same in both cases. Examples 8 and 9 show the two methods.

Example 8

Expand $(2n + 3)(3n - 7)$.

$$\begin{aligned} (2n + 3)(3n - 7) \\ = (2n + 3)3n + (2n + 3)(-7) \\ = 6n^2 + 9n - 14n - 21 \\ = 6n^2 - 5n - 21 \end{aligned}$$

Example 9

Expand $(2x - 3y)(5x - 2y)$.

$$\begin{aligned} (2x - 3y)(5x - 2y) \\ = 2x(5x - 2y) - 3y(5x - 2y) \\ = 10x^2 - 4xy - 15xy + 6y^2 \\ = 10x^2 - 19xy + 6y^2 \end{aligned}$$

By expanding brackets and collecting terms it can be shown that:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Example 10

Expand (a) $(4p + q)^2$, (b) $(5x - 3y)^2$.

$$\begin{aligned} \text{(a)} \quad (4p + q)^2 &= (4p)^2 + 2(4p)(q) + (q)^2 \\ &= 16p^2 + 8pq + q^2 \\ \text{(b)} \quad (5x - 3y)^2 &= (5x)^2 - 2(5x)(3y) + (3y)^2 \\ &= 25x^2 - 30xy + 9y^2 \end{aligned}$$

Exercise 22c

Expand brackets and collect like terms where possible.

- | | |
|-----------------------------|------------------------|
| 1 $4(9a + 1)$ | 2 $-5(6x - 2y + 1)$ |
| 3 $3x(14 - y)$ | 4 $-2a(5a - b + 3c)$ |
| 5 $8 - (x + 3)$ | 6 $11x^2 - x(3x + 4)$ |
| 7 $6x(x - 2) - 5(x - 2)$ | |
| 8 $5(6 - b^2) + 2(1 - b^2)$ | |
| 9 $(a + 2b)(3c + 4d)$ | 10 $(x + 1)(x + 8)$ |
| 11 $(2a + b)(5a + 8b)$ | 12 $(2a + b)(5a - 8b)$ |
| 13 $(2a - b)(5a + 8b)$ | 14 $(2a - b)(5a - 8b)$ |
| 15 $(2x - 5)(x - 3)$ | 16 $(3 - x)(4 + 5x)$ |
| 17 $(3a + 1)^2$ | 18 $(b - 4)^2$ |
| 19 $(2c - 5d)^2$ | 20 $(4m - 3n)^2$ |

Factorisation

Example 11

Factorise $(x - a)(3x + 2a) - (x - a)^2$.

$(x - a)$ is a **common factor** of the two parts of the expression.

$$\begin{aligned} &(x - a)(3x + 2a) - (x - a)^2 \\ &= (x - a)[(3x + 2a) - (x - a)] \\ &= (x - a)(3x + 2a - x + a) \\ &= (x - a)(2x + 3a). \end{aligned}$$

Example 12

Factorise $am + 3bn - an - 3bm$.

The terms am and an have a in common.

The terms $3bm$ and $3bn$ have $3b$ in common.

Grouping pairs in this way,

$$\begin{aligned} am + 3bn - an - 3bm &= am - an - 3bm + 3bn \\ &= a(m - n) - 3b(m - n) \\ &= (m - n)(a - 3b) \end{aligned}$$

Example 13

Factorise $10x^2 + 5x - 2x - 1$.

$$\begin{aligned} &10x^2 + 5x - 2x - 1 \\ &= 5x(2x + 1) - 1(2x + 1) \\ &= (2x + 1)(5x - 1) \end{aligned}$$

Example 14

Factorise $3x^2 - 13x - 10$.

1st step: Find the product of the first and last term

$$3x^2 \times (-10) = -30x^2$$

2nd step: Find two terms such that their product is $-30x^2$ and their sum is $-13x$ (the middle term). Since the middle term is negative, only consider those factors in which the negative term is numerically greater than the positive term.

factors of $-30x^2$	sum of factors
(a) $-30x$ and $+x$	$-29x$
(b) $-15x$ and $+2x$	$-13x$
(c) $-10x$ and $+3x$	$-7x$
(d) $-6x$ and $+5x$	$-x$

Of these, only (b) gives the required result.

3rd step: Replace $-13x$ in the given expression by $-15x + 2x$. Factorise by grouping.

$$\begin{aligned} 3x^2 - 13x - 10 &= 3x^2 - 15x + 2x - 10 \\ &= 3x(x - 5) + 2(x - 5) \\ &= (x - 5)(3x + 2) \end{aligned}$$

Notice in Example 14, that when the required result was found in line (b), it was not really necessary to do lines (c) and (d).

Example 15

Factorise $6 - 15x + 9x^2$.

3 is a common factor of all the terms. Take this out first.

$$\begin{aligned} 6 - 15x + 9x^2 &= 3(2 - 5x + 3x^2) \\ &= 3(2 - 2x - 3x + 3x^2)^* \\ &= 3[2(1 - x) - 3x(1 - x)] \\ &= 3(1 - x)(2 - 3x) \end{aligned}$$

* The reason why $-2x - 3x$ was substituted for $-5x$ is left as an exercise. See Example 14.

$$a^2 + 2ab + b^2 = (a + b)^2$$

$$a^2 - 2ab + b^2 = (a - b)^2$$

$$a^2 - b^2 = (a + b)(a - b)$$

$$a^2 + b^2 \text{ has no factors}$$

Example 16Factorise $d^2 - 10dm + 25m^2$. d^2 is the square of d . $25m^2$ is the square of $5m$. $10dm$ is twice the product of d and $5m$.

$$d^2 - 10dm + 25m^2 = (d - 5m)^2.$$

Example 17Factorise $16a^2 - 25b^2$.

$$\begin{aligned} 16a^2 - 25b^2 &= (4a)^2 - (5b)^2 \\ &= (4a + 5b)(4a - 5b) \end{aligned}$$

Example 18Factorise $5m^2 - 80$.

$$\begin{aligned} 5m^2 - 80 &= 5(m^2 - 16) \\ &= 5(m + 4)(m - 4) \end{aligned}$$

Exercise 22d

Factorise:

- 1 $(3c - d)(m - n) + (3c - d)(2m - 3n)$
- 2 $ac + bc + 2ad + 2bd$
- 3 $x^2 - 16a^2$
- 4 $4c^2 - 25d^2$
- 5 $a^2 - 3a - 10$
- 6 $a^2 - 3ab - 10b^2$
- 7 $a^2b^2 - 3ab - 10$
- 8 $5m^2 - 45n^2$
- 9 $hm - 2km - 2hn + 4kn$
- 10 $(2a + b)^2 - (2a + b)(a - 3b)$
- 11 $3m^2 - 10m + 3$
- 12 $c^2d^2 - 81$
- 13 $16x^2 - 9a^2m^2$
- 14 $6n^2 + 13n + 6$
- 15 $4a^2 + 20a + 25$
- 16 $9h^2 - 36k^2$
- 17 $(5a - 3b)(a - 2b) - (4a + b)(a - 2b)$
- 18 $25a^2b^2c^2 - 9d^2$
- 19 $\frac{m^2}{9} - \frac{n^2}{4}$
- 20 $3su + tu - 6sv - 2tv$
- 21 $9x^2 - 12xy + 4y^2$
- 22 $12d^2 + 5d - 2$
- 23 $x^4 - y^2$
- 24 $10m^2n^2 - 7mn - 12$
- 25 $16 - n^4$
- 26 $(a + b)^2 - c^2$
- 27 $x^2 - (m - n)^2$
- 28 $(m + 2n)(3a - b) - (a - 3b)(m + 2n)$
- 29 $2 - h - 15h^2$
- 30 $2am - bm + 3bn - 6an$
- 31 $a^2 - 15ab + 54b^2$
- 32 $m^2 - 15mn - 54n^2$
- 33 $(c - 2d)^2 - 9e^2$
- 34 $12x^2 + 35xy + 18y^2$

35 $(h - k)(2h - 3k) + (h - k)^2$

36 $6acx - 8ady + 4acy - 12adx$

37 $6a^2 - 19ax - 36x^2$

38 $25a^2 - 4(m + 2n)^2$

39 $25a^2 - 4(a - 2b)^2$

40 $9a^2 - (3a - 2b)^2$

Factorisation can often be used to simplify calculations.

Example 19Simplify $63 \times 29 + 37 \times 29$.

$$\begin{aligned} 63 \times 29 + 37 \times 29 &= 29(63 + 37) \\ &= 29 \times 100 = 2900 \end{aligned}$$

Example 20Evaluate $17.9^2 - 12.1^2$ by using factorisation.

$$\begin{aligned} 17.9^2 - 12.1^2 &= (17.9 + 12.1)(17.9 - 12.1) \\ &= 30 \times 5.8 = 174 \end{aligned}$$

Exercise 22e

Use factorisation to simplify the following.

- 1 $18 \times 57 + 18 \times 43$
- 2 $23 \times 119 - 23 \times 19$
- 3 $243 \times 4 + 243 \times 6$
- 4 $28 \times 752 + 28 \times 248$
- 5 $63 \times 47 - 43 \times 47$
- 6 $61 \times 127 - 77 \times 61$
- 7 $106^2 - 94^2$
- 8 $8.78^2 - 1.22^2$
- 9 $5 \times 9.2^2 - 5 \times 4.8^2$
- 10 $\pi R^2h - \pi r^2h$, where $\pi = 3\frac{1}{7}$, $R = 18$ cm,
 $r = 10$ cm and $h = 15$ cm.

Algebraic fractions**Example 21**Simplify $\frac{3x - 2}{5} + \frac{x + 1}{3}$.

$$\begin{aligned} \frac{3x - 2}{5} + \frac{x + 1}{3} &= \frac{3(3x - 2) + 5(x + 1)}{5 \times 3} \\ &= \frac{9x - 6 + 5x + 5}{15} \\ &= \frac{9x + 5x - 6 + 5}{15} \\ &= \frac{14x - 1}{15} \end{aligned}$$

Example 22

Simplify $\frac{a-4}{2a} - \frac{9b-2}{6b} + 1$.

$$\begin{aligned} \frac{a-4}{2a} - \frac{9b-2}{6b} + 1 &= \frac{3b(a-4) - a(9b-2) + 6ab(1)}{6ab} \\ &= \frac{3ab - 12b - 9ab + 2a + 6ab}{6ab} \\ &= \frac{2a - 12b}{6ab} = \frac{2(a-6b)}{6ab} = \frac{a-6b}{3ab} \end{aligned}$$

Example 23

Simplify $\frac{6-x-x^2}{x^2-4}$.

$$\begin{aligned} \frac{6-x-x^2}{x^2-4} &= \frac{(3+x)(2-x)}{(x+2)(x-2)} \\ &= -\frac{(3+x)(x-2)}{(x+2)(x-2)} = -\frac{3+x}{x+2} \end{aligned}$$

Example 24

Simplify $\frac{3a^3}{3a^2-6ab} + \frac{4b^3}{2b^2-ab}$.

$$\begin{aligned} \frac{3a^3}{3a^2-6ab} + \frac{4b^3}{2b^2-ab} &= \frac{3a^3}{3a(a-2b)} + \frac{4b^3}{b(2b-a)} \\ &= \frac{a^2}{a-2b} + \frac{4b^2}{2b-a} \\ &= \frac{a^2}{a-2b} - \frac{4b^2}{a-2b} \\ &= \frac{a^2-4b^2}{a-2b} = \frac{(a+2b)(a-2b)}{(a-2b)} = a+2b \end{aligned}$$

Example 25

Simplify $\frac{x}{x-3} - \frac{x-1}{x+2}$.

$$\begin{aligned} \frac{x}{x-3} - \frac{x-1}{x+2} &= \frac{x(x+2) - (x-1)(x-3)}{(x-3)(x+2)} \\ &= \frac{x^2+2x - (x^2-4x+3)}{(x-3)(x+2)} \\ &= \frac{x^2+2x-x^2+4x-3}{(x-3)(x+2)} \\ &= \frac{6x-3}{(x-3)(x+2)} \\ &= \frac{3(2x-1)}{(x-3)(x+2)} \end{aligned}$$

Exercise 22f

Simplify the following.

- 1 $\frac{x-4}{3} + \frac{x+3}{2}$
- 2 $\frac{b+5}{7} - \frac{2-b}{5}$
- 3 $\frac{3d+6}{10} + \frac{d+3}{3}$
- 4 $\frac{2x-5}{7} - \frac{8x-8}{6}$
- 5 $\frac{2}{3ab} - \frac{3}{4bc}$
- 6 $5 - \frac{p-q}{q}$
- 7 $\frac{ab+ac}{ab-ac}$
- 8 $\frac{a^2-b^2}{a^2-ab}$
- 9 $\frac{a^2+ab}{ab+b^2}$
- 10 $\frac{m^2-2mn+n^2}{m^2-n^2}$
- 11 $\frac{2x^2-x-1}{x-1}$
- 12 $\frac{x^2-5x+6}{x^2-9}$
- 13 $\frac{15-2x-x^2}{x^2-9}$
- 14 $\frac{9-a^2}{a^2+6a+9}$
- 15 $\frac{5xy}{5x^2-10xy} + \frac{8xy}{2y^2-xy}$
- 16 $\frac{2a+b}{a^2-ab} - \frac{2b+a}{ab-b^2}$
- 17 $\frac{2}{x+2} - \frac{x-6}{x^2-4}$
- 18 $\frac{b}{a^2-ab} + \frac{a+b}{ab}$
- 19 $\frac{x-5}{6} - \frac{6}{x-5}$
- 20 $\frac{1}{m-1} + \frac{9}{2m+3} - \frac{8}{m+4}$

Substitution

Example 26

Find the value of $3(x+y)$ if $x = -2$ and $y = 7$.

When $x = -2$ and $y = 7$,

$$3(x+y) = 3(-2+7) \\ = 3 \times 5 = 15$$

Example 27

Evaluate $ut - \frac{1}{2}ft^2$ when $t = 5$, $u = -20$ and $f = 10$.

$$ut - \frac{1}{2}ft^2 = (-20) \times 5 - \frac{1}{2}(10)(5)^2 \\ = -100 - 125 = -225$$

Example 28

Evaluate $\frac{ab^2 - c^2}{2bc} + \frac{a^2}{2b+c}$ when $a = 2$, $b = -3$ and $c = -2$.

$$\frac{ab^2 - c^2}{2bc} + \frac{a^2}{2b+c} \\ = \frac{2(-3)^2 - (-2)^2}{2(-3)(-2)} + \frac{2^2}{2(-3) + (-2)} \\ = \frac{2 \times 9 - 4}{12} + \frac{4}{-6-2} \\ = \frac{14}{12} + \frac{4}{-8} = \frac{7}{6} - \frac{1}{2} = \frac{7}{6} - \frac{3}{6} = \frac{2}{3}$$

Exercise 22g

- 1 Evaluate $u + at$ if $a = 10$, $u = 4$ and $t = 2$.
- 2 If $p = 42$ and $r = 3$, find the value of $\frac{p-r^2}{r}$.
- 3 Find $\sqrt{u^2 + 2av}$ when $u = 11$, $a = 4$ and $v = 13$.
- 4 If $x = 3$, $y = -5$ and $z = 2$, what is the value of $\frac{x^2(y-z)}{6z+y}$?
- 5 Evaluate $x^2 - 3x + 8$ when $x =$ (a) 3, (b) 0, (c) -1, (d) -4.
- 6 If $y = 3x^2 - 5x - 2$, find y when $x =$ (a) 0, (b) 1, (c) 2, (d) 3.

7 If $V = k\frac{R}{T}$, find k when $V = 14$, $R =$ and $T = 45$.

8 Evaluate $x^2y + y^2x$ when $x = 5$ and $y = -$

9 If $a = -4$, $b = 6$ and $c = -3$, find the values of (a) $5a - 3c + 2b$, (b) $9a^2c^2 -$

(c) $\frac{3ac}{b^2 - c^2}$.

10 If $s = u + 980t$, (a) find s when $u = 0$ and $t = 5$, (b) find t when $s = 10\,000$ and $u = 3\,000$.

Variation

Direct variation

y varies directly as x is written as $y \propto x$. $y \propto x$ means that $y = kx$, where k is a constant.

Example 29

If $P \propto R$ and $P = 10$ when $R = 6$, find the relationship between P and R . Hence find R when $P = 12.5$.

Let $P = kR$

$$10 = k6 \quad (\text{from 1st sentence})$$

$$\Leftrightarrow k = \frac{10}{6} = \frac{5}{3}$$

$\therefore P = \frac{5}{3}R$ is the required relationship between P and R .

When $P = 12.5$, $12.5 = \frac{5}{3}R$

$$\Leftrightarrow R = \frac{3}{5} \times 12.5 = 7.5$$

Fig. 22.9 is a graph of the relationship between R and P .

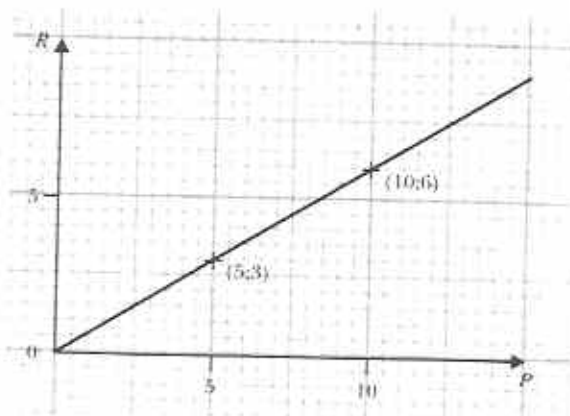


Fig. 22.9

Inverse variation

If y varies inversely as x then $y \propto \frac{1}{x}$ or $y = \frac{k}{x}$,

where k is a constant. Notice that if $y \propto \frac{1}{x}$, then

$x \propto \frac{1}{y}$. Similarly, if $x \propto y^3$, then $y \propto \sqrt[3]{x}$, and so on.

Example 30

Given that N varies as D^3 and that $N = 240$ when $D = 3$, find N when $D = 2$.

$$\text{Let } N = \frac{k}{D^3}$$

$$\text{then } 240 = \frac{k}{3^3}$$

$$\Leftrightarrow k = 240 \times 27$$

When $D = 2$,

$$N = \frac{240 \times 27}{2^3}$$

$$= \frac{240 \times 27}{8}$$

$$= 810$$

Example 31

$R \propto \sqrt{M}$ and $R = 6$ when $M = 16$. Find the law connecting R and M . Find R when $M = 6\frac{1}{4}$ and find M when $R = 15$.

$$\text{Let } R = k\sqrt{M}$$

$$\text{then } 6 = k\sqrt{16}$$

$$= 4k$$

$$\Leftrightarrow k = \frac{3}{2}$$

Hence $R = \frac{3}{2}\sqrt{M}$, which is the required law.

When $M = 6\frac{1}{4}$,

$$R = \frac{3}{2}\sqrt{6\frac{1}{4}} = \frac{3}{2}\sqrt{\frac{25}{4}} = \frac{3}{2} \times \frac{5}{2} = \frac{15}{4} = 3\frac{3}{4}$$

When $R = 15$,

$$15 = \frac{3}{2}\sqrt{M}$$

$$\Leftrightarrow \sqrt{M} = 15 \times \frac{2}{3} = 10$$

$$\Rightarrow M = 10^2 = 100$$

Notice that if $R = \frac{3}{2}\sqrt{M}$, then $R^2 = \frac{9}{4}M$ or $M = \frac{4}{9}R^2$. It is often convenient to use an alternative form of the original.

Joint variation

Joint variation involves three or more variables. The relationships between them can be in many forms:

$$M \propto R^2 T, \quad F \propto \frac{Mm}{d^2}, \quad W = kDS,$$

$$t = k \frac{mV^2}{r}$$

Example 32

The mass of a solid metal ball varies jointly as its specific gravity and the cube of its diameter. When the diameter is 6 cm and the specific gravity 7.5, the mass is 850 g. Find the mass of a ball of specific gravity 10.5 and diameter 8 cm.

Let M = mass in grams

D = diameter in cm

S = specific gravity

Then $M = kSD^3$

$$\therefore 850 = k \times 7.5 \times 6^3 \quad (1)$$

$$\text{and } M = k \times 10.5 \times 8^3 \quad (2)$$

Dividing (2) by (1),

$$\frac{M}{850} = \frac{10.5}{7.5} \times \left(\frac{8}{6}\right)^3 = \frac{7 \times 64}{5 \times 27}$$

$$\Leftrightarrow M = 850 \times \frac{7}{5} \times \frac{64}{27} \approx 2821$$

The mass is approximately 2821 g.

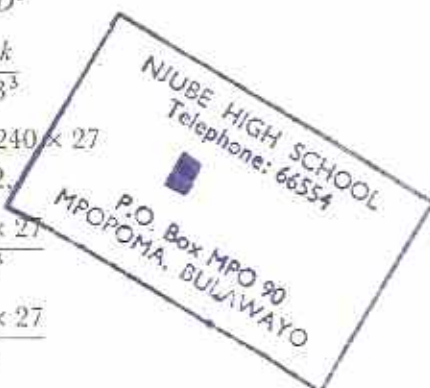
Example 32 shows a method that may be used when the value of k is not required.

Partial variation

Partial variation consists of two or more parts added together. For example:

$$S = a + bT, \quad R = a + bV^2,$$

$$E = aMH + bMV^2$$



Notice that in partial variation there are at least two constants, such as a and b above. These constants have to be found separately. See Examples 9 and 10 on pages 190 and 191.

Exercise 22h

- Express the following as relationships using the given letters and either the symbol $\propto$ or any constants which may be necessary.
 - The length, l , of a rectangle of constant area varies inversely as its breadth, b .
 - The resonance frequency, r , of a series circuit varies inversely as the square root of its capacitance, c .
 - The gravitational attraction, G , between two particles of mass m_1 and m_2 varies jointly as the product of their masses and the inverse of the square of their distance apart, d .
 - The energy, E , of a moving body varies partly as the height of the body above sea level, h , and partly as the square of its velocity, v .
- If $y \propto x$ and $y = 10$ when $x = \frac{1}{2}$ find the law of the variation. Find x if $y = 35$.
- $A \propto M$ and $A = 8$ when $M = 20$. Find A when $M = 15$ and M when $A = 7$.
- $P \propto Q$ and $P = 14$ when $Q = 8$. Find P when $Q = 6$ and Q when $P = 28$.
- $D \propto V$ and $D = 108$ when $V = 3$. Find D when $V = 3.75$ and V when $D = 189$.
- $P \propto Q^2$. $P = 27$ when $Q = 6$. Find the law, P when $Q = 10$, Q when $P = 18^{\frac{3}{2}}$.
- $x \propto \frac{1}{y}$. $x = 7\frac{1}{2}$ when $y = 4$. Find the law, x when $y = 12$, y when $x = 20$.
- $M \propto R^3$. $M = 40$ when $R = 4$. Find the law, M when $R = 10$, R when $M = 2.56$.
- $\sqrt{Y} \propto Z$. $Y = 4$ when $Z = 3$. Find the law, Y when $Z = 15$, Z when $Y = 16$.
- $A \propto BC$. When $B = 6$ and $C = 3$, $A = 7\frac{1}{2}$. Find A when $B = 8$ and $C = 9$; also B if $A = 25$ and $C = 8$.
- $P \propto \frac{Q}{R^2}$. When $Q = 5$ and $R = 3$, $P = 20$. Find P when $Q = 6$ and $R = 4$; also R when $P = 21.6$ and $Q = 15$.

- x is partly constant and partly varies as y . When $y = 5$, $x = 7$; and when $y = 7$, $x = 11$. Find the law of the variation and also when $y = 11$.
- x varies partly as y and partly as y^2 . When $y = 4$, $x = 52.8$; and when $y = 5$, $x = 65$. Find x when $y = 6$.
- $x \propto y$ and $y \propto z^2$. How does x vary with z ?
- $x \propto y^2$ and $y \propto \frac{1}{z}$. How does x vary with z ?
- A car takes 6 hours to travel from X to Y at a constant speed. How long does the same journey take for
 - a lorry travelling at half the speed of the car?
 - a helicopter travelling at 3 times the speed of the car?
 - an aeroplane travelling at 6 times the speed of the car?
- If y is inversely proportional to x , complete Table 22.1.

Table 22.1

x	10		20	25	30	
y		$\frac{1}{3}$	$\frac{1}{4}$			$\frac{1}{7}$

- The illumination of a small object by a lamp varies directly as the candlepower of the lamp and inversely as the square of the distance between the lamp and the object. If a light bulb of 8 candlepower, fixed 150 cm above a table, is replaced by a 16 candlepower bulb, how far must the new light be lowered to give the object the same illumination as before?
- The resistance to the motion of a vehicle is partly constant and partly varies as the square of its speed. At 30 km h^{-1} the resistance is 496 N, and at 50 km h^{-1} it is 656 N. Find the resistance at 60 km h^{-1} .
- The time taken for a committee meeting is partly constant and partly varies as the square of the number of members present. If there are twelve members present the meeting lasts only 56 minutes, but with twenty it takes exactly two hours. How long will it last if there are sixteen members?

Algebraic graphs

Example 33

Draw the graph of $y = x^2 + 3x - 4$ for values of x from -5 to $+2$. Find (a) y when $x = 1,5$, (b) x when $y = -5,2$, (c) x when $y = 4$, (d) the least value of $x^2 + 3x - 4$, (e) the value of x for which $y = 0$.

Obtain the values of y by adding the values of x^2 , $3x$ and -4 for each value of x . Table 22.2 shows the method.

Table 22.2

x	-5	-4	-3	-2	-1	0	1	2
x^2	25	16	9	4	1	0	1	4
$3x$	-15	-12	-9	-6	-3	0	3	6
-4	-4	-4	-4	-4	-4	-4	-4	-4
y	6	0	-4	-6	-6	-4	0	6

Fig. 22.10 is the graph of $y = x^2 + 3x - 4$. Note that scales of 2 cm to 1 unit on the x -axis and 1 cm to 1 unit on the y -axis would give a larger, clearer graph.

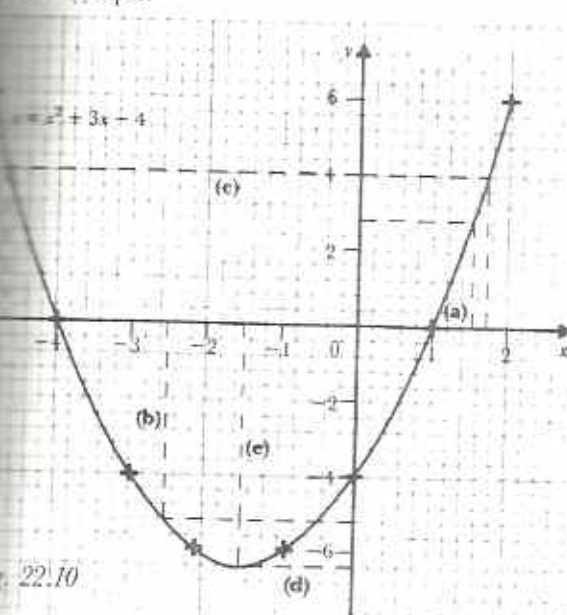


Fig. 22.10

The dotted construction lines (a), (b), (c), (d) and (e) on Fig. 22.10 are used to obtain the required results: (a) $y = 2,8$; (b) $y = -5,2$; (c) $x = 1,7$ or $-4,7$; (d) $-6,2$; (e) $x = -1,5$.

Example 34

On the same axes, draw the graphs of $y = \frac{3}{x-2}$ and $y = 2x - 3$ for values of x from -2 to $+4$. Find their points of intersection, if any.

Tables 22.3 and 22.4 give corresponding values of x and y for each graph.

Table 22.3 $y = \frac{3}{x-2}$

x	-2	-1	0	1	2	3	4	1,5	2,5
y	0,75	1	1,5	3	udf	3	1,5	6	6

Notice that no value of y is given for $x = 2$; y is undefined (udf) when $x = 2$. Values of y when $x = 1,5$ and $x = 2,5$ are found instead.

Table 22.4 $y = 2x - 3$

x	0	2	4
y	-3	1	5

Since $y = 2x - 3$ is a linear relation, three points are sufficient. Fig. 22.11 shows the graphs drawn on the same axes.

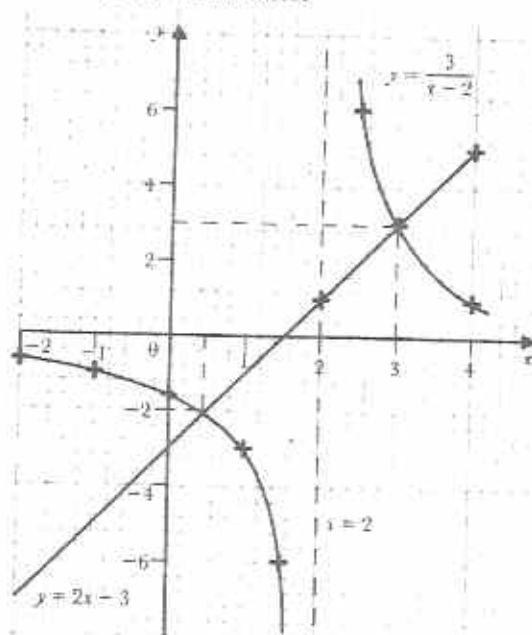


Fig. 22.11

The graphs intersect at the points $(3; 3)$ and $(\frac{1}{2}; -2)$.

Exercise 22i

- 1 Copy and complete Table 22.5, giving values for the relation $y = 3x^2 - 5x + 3$.

Table 22.5

x	-2	-1	0	1	2	3	4
y	25					15	31

- (a) Draw a graph of the relation using scales of 2 cm to 1 unit on the x -axis and 2 cm to 5 units on the y -axis.
 (b) What is the minimum value of $3x^2 - 5x + 3$?
 (c) Write down the equation of the line of symmetry of the curve.
 (d) Find the values of x when $y = 5$.
- 2 Draw the graph of $y = x^2 - 3x - 4$ for values of x from -2 to $+5$. Use scales of 2 cm to 1 unit on the x -axis and 1 cm to 1 unit on the y -axis. (a) Read off the value of y when $x =$ (i) 4.5; (ii) 2.5. (b) Find the values of x when $y =$ (i) 4; (ii) 0. (c) For what value of x is y a minimum?
- 3 Draw the graph of $y = 4x - x^2$ for values of x from -1 to $+5$. (a) Read off the value of y at the point where the line $x = -0.5$ cuts the curve. (b) Find the points of intersection of the curve with the line $y = -3$. (c) For what value of x does $4x - x^2$ have its greatest value, and what is this greatest value?
- 4 Choose a suitable scale and draw the graph of $y = x^2 + 4x + 1$ for values of x from -5 to $+1$. Find (a) the range of values of x for which y increases as x increases, and (b) the coordinates of the point at which y has its least value.
- 5 Draw the graph of $y = (x + 1)(x - 2)$ for values of x from -3 to $+4$. Find the range of values of x for which
 (a) y decreases as x increases?
 (b) y is negative.
- 6 Choose suitable scales and then draw the graph of
 (a) $y = \frac{3}{x+3}$ from -6 to 0.
 (b) $y = \frac{12}{x-2}$ from -1 to $+5$
 (c) $y = x(x-1)(x-3)$ from -2 to $+5$
 (d) $y = x^3 - 4x - 1$ from -3 to $+3$.
- 7 Without drawing the graph of the relation $y = (4-x)(2+x)$, write down the coordinates of the point where the curve cuts (a) the x -axis, (b) the y -axis.
 (c) State whether the curve has a maximum or minimum value of y and (d) write down the value of x at this point.
 (e) Hence calculate the corresponding value of y .
- 8 On the same axes, draw the graphs of $y = x^2 - 5x$ and $y = 5x - x^2$ for values of x from -1 to $+6$. (a) At what points do the curves intersect? (b) Write down the equations of two lines of symmetry of the completed curves.
- 9 On the same axes, draw the graphs of $y = \frac{2}{x}$ and $y = \frac{2}{3}x - 1$, for values of x from -3 to $+3$. Find the values of x and y at the points of intersection of the graphs.
- 10 On the same axes, draw the graphs of $y = \frac{-1}{x+2}$ and $x + y + 2 = 0$ for values of x from -5 to $+2$. Read off the values of x and y at the points of intersection of the graphs.

Chapter 23

Equations and inequalities

Linear equations

Example 1

Solve the equation $2(x + 3) = -11$.

$$2(x + 3) = -11$$

Remove brackets:

$$2x + 6 = -11$$

Subtract 6 from both sides:

$$2x = -17$$

Divide both sides by 2:

$$x = -8\frac{1}{2}$$

Example 2

Solve $\frac{8}{t} - 1 = 3$ for t .

$$\frac{8}{t} - 1 = 3$$

Multiply both sides by t :

$$8 - t = 3t$$

Add t to both sides:

$$8 = 4t$$

Divide both sides by 4:

$$2 = t$$

$$\text{or } t = 2$$

Exercise 23a

Solve the following equations for x .

$$1 \quad 3 - 2x = 7 \qquad 2 \quad 10 - 5x + 6 = 0$$

$$3 \quad x + 13 = 5x - 7 \qquad 4 \quad 2x - 13 = 5 + 6x$$

$$5 \quad -3(x - 1) = 9$$

$$6 \quad 2(3x - 1) - 10 = 0$$

$$7 \quad 6(3 - x) = 5(4 - x)$$

$$8 \quad \frac{11}{v} - 1\frac{1}{2} = \frac{5}{x} \qquad 9 \quad \frac{6}{v} + \frac{2}{3v} = 2$$

$$10 \quad 5(3x + \frac{1}{2}) = 2x - 17$$

Change of subject of a formula

Example 3

Make P the subject of the formula $R = \frac{Q^2 - PR}{Q + P}$.

$$R = \frac{Q^2 - PR}{Q + P}$$

Multiply both sides by $(Q + P)$ to clear fractions:

$$R(Q + P) = Q^2 - PR$$

Clear brackets:

$$RQ + PR = Q^2 - PR$$

Collect terms in P on the LHS of the equation:

$$2PR = Q^2 - RQ$$

Divide both sides by $2R$:

$$P = \frac{Q^2 - RQ}{2R} = \frac{Q(Q - R)}{2R}$$

Example 4

Make r the subject of the formula $b = \sqrt{\frac{4\pi r^3}{3h}}$.

$$b = \sqrt{\frac{4\pi r^3}{3h}}$$

Square both sides:

$$b^2 = \frac{4\pi r^3}{3h}$$

Multiply both sides by $\frac{3h}{4\pi}$:

$$\frac{3hb^2}{4\pi} = r^3 \quad \text{or } r^3 = \frac{3hb^2}{4\pi}$$

Take the cube root of both sides:

$$r = \sqrt[3]{\frac{3hb^2}{4\pi}}$$

Chapter 23

Equations and inequalities

Syllabus references 6.5.4, 6.6.4 and 6.5.4

Linear equations

Example 1

Solve the equation $2(x + 3) = -11$.

$$2(x + 3) = -11$$

Remove brackets:

$$2x + 6 = -11$$

Subtract 6 from both sides:

$$2x = -17$$

Divide both sides by 2:

$$x = -8\frac{1}{2}$$

Example 2

Solve $\frac{8}{t} - 1 = 3$ for t .

$$\frac{8}{t} - 1 = 3$$

Multiply both sides by t :

$$8 - t = 3t$$

Add t to both sides:

$$8 = 4t$$

Divide both sides by 4:

$$2 = t$$

$$\text{or } t = 2$$

Exercise 23a

Solve the following equations for x .

$$1 \quad 3 - 2x = 7 \qquad 2 \quad 10 - 5x + 6 = 0$$

$$3 \quad x + 13 = 5x - 7 \qquad 4 \quad 2x - 13 = 5 + 6x$$

$$5 \quad -3(x - 1) = 9$$

$$6 \quad 2(3x - 1) - 10 = 0$$

$$7 \quad 6(3 - x) = 5(4 - x)$$

$$8 \quad \frac{11}{x} - 1\frac{1}{2} = \frac{5}{x} \qquad 9 \quad \frac{6}{x} + \frac{2}{3x} = 2$$

$$10 \quad 5(3x + \frac{1}{2}) = 2x - 17$$

Change of subject of a formula

Example 3

Make P the subject of the formula $R = \frac{Q^2 - PR}{Q + P}$.

$$R = \frac{Q^2 - PR}{Q + P}$$

Multiply both sides by $(Q + P)$ to clear fractions:

$$R(Q + P) = Q^2 - PR$$

Clear brackets:

$$RQ + PR = Q^2 - PR$$

Collect terms in P on the LHS of the equation:

$$2PR = Q^2 - RQ$$

Divide both sides by $2R$:

$$P = \frac{Q^2 - RQ}{2R} = \frac{Q(Q - R)}{2R}$$

Example 4

Make r the subject of the formula $b = \sqrt{\frac{4\pi r^3}{3h}}$.

$$b = \sqrt{\frac{4\pi r^3}{3h}}$$

Square both sides:

$$b^2 = \frac{4\pi r^3}{3h}$$

Multiply both sides by $\frac{3h}{4\pi}$:

$$\frac{3hb^2}{4\pi} = r^3 \qquad \text{or } r^3 = \frac{3hb^2}{4\pi}$$

Take the cube root of both sides:

$$r = \sqrt[3]{\frac{3hb^2}{4\pi}}$$

Substitute 3 for x in equation (3):

$$y = 11 - 9 = 2$$

The solution is $x = 3, y = 2$.

Or by elimination:

$$2x - y = 4 \quad (1)$$

$$3x + y = 11 \quad (2)$$

Add (1) and (2) to eliminate y :

$$5x = 15$$

$$\Leftrightarrow x = 3$$

Substitute 3 for x in (1):

$$6 - y = 4$$

$$\Leftrightarrow y = 2$$

The solution is $x = 3, y = 2$.

Or by graph:

Fig. 22.2 shows the graphs of $2x - y = 4$ and $3x + y = 11$ drawn on the same axes.

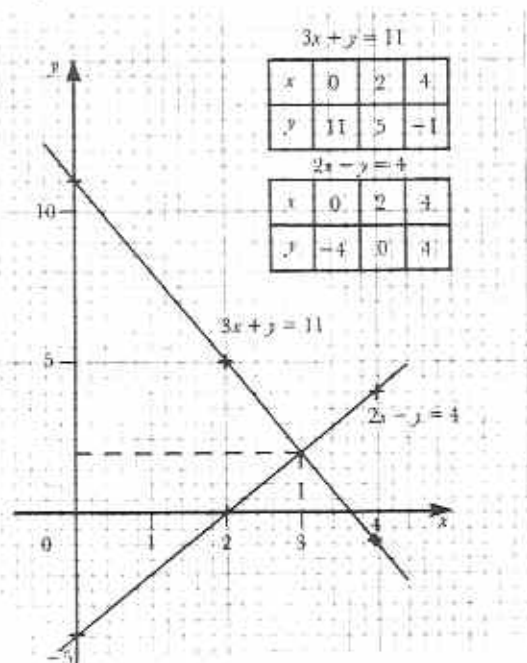


Fig. 22.2

The point of intersection of the two lines is $(3; 2)$. The solution of the equations is $x = 3, y = 2$.

Example 9

Solve the equations $\frac{2}{3}x - \frac{1}{2}y = 2, \frac{3}{4}x - \frac{1}{3}y = 3\frac{1}{6}$.

$$\frac{2}{3}x - \frac{1}{2}y = 2 \quad (1)$$

$$\frac{3}{4}x - \frac{1}{3}y = 3\frac{1}{6} \quad (2)$$

Simplify the equations by clearing the fractions

$$(1) \times 6: \quad 4x - 3y = 12 \quad (3)$$

$$(2) \times 12: \quad 9x - 4y = 38 \quad (4)$$

Solve in the usual way:

$$(3) \times 4: \quad 16x - 12y = 48 \quad (5)$$

$$(4) \times 3: \quad 27x - 12y = 114 \quad (6)$$

$$(6) - (5): \quad 11x = 66$$

$$\Leftrightarrow x = 6$$

Substituting 6 for x in (3):

$$24 - 3y = 12$$

$$\Leftrightarrow -3y = -12$$

$$\Leftrightarrow y = 4$$

The solution is $x = 6, y = 4$.

Example 10

Solve the equations

$$3x - 2y + 1 = 2x - 5y - 10 = 4x - 3y.$$

Pair the three expressions in any two different ways:

$$3x - 2y + 1 = 4x - 3y$$

$$\Leftrightarrow x - y = 1 \quad (1)$$

$$2x - 5y - 10 = 4x - 3y$$

$$\Leftrightarrow 2x - 8y = -10$$

$$\Leftrightarrow x - 4y = -5 \quad (2)$$

Solve equations (1) and (2) in the usual way.

This gives $x = 3$ and $y = 2$ as the solution.

Exercise 23d

Solve the following pairs of equations. Use either the substitution, elimination or graphical method, whichever is most suitable.

1 $x + y = 6$

$2x + 3y = 14$

3 $3x - 4y = 7$

$x - 2y = 5$

5 $5x - 2y = -23$

$3x + 4y = 7$

7 $x - 2y = 1$

$x + 2y = 9$

9 $4c - 3d = 1$

$2c + 4d = 17$

11 $5m - 2n = 15$

$3m + 5n = 9$

13 $3a + 4m = 0$

$a = 2m - 5$

2 $2x - y = 11$

$x + 2y = -7$

4 $3x + 2y = 7$

$7x - 3y = 1$

6 $3x - 2y = 4$

$2x - 7y = 31$

8 $a + 3b = -13$

$2a - 9b = 4$

10 $3x + 4y = -1$

$3x + 8y = 4$

12 $3x + 7y = -8$

$5y = 2x + 15$

14 $5d = 4n + 8$

$5n + 10 = 4d$

$$15 \quad 5x + 3y - 2 = 0 \quad 16 \quad \frac{3}{4}x + \frac{1}{5}y = 4$$

$$3x - 32 = 7y \quad \frac{1}{2}x = \frac{3}{5}y - 1$$

$$17 \quad 1,2x - 1,1y = 7,9 \quad 18 \quad \frac{d}{4} - \frac{n}{3} = 6$$

$$1,8x + 0,7y = 0,1$$

$$\frac{3d}{2} + \frac{5n}{6} = 2$$

$$19 \quad 3x - y + 12 = 5x + 2y + 4 = x + y$$

$$20 \quad 5(a + b) = 2(a + 3b) + 1$$

$$3(a + 2b) - 7 = a + 3b + 1$$

$$21 \quad 2,5d - 1,5g = -2$$

$$1,3d + g + 4,6 = 0$$

$$22 \quad \frac{2}{3}(a - 6) = m - 3$$

$$\frac{3}{5}(a + 2) = 2m + 1$$

$$23 \quad c + d + 6 = 2c - 5d - 3 = 3c - 4d + 2$$

$$24 \quad 3(2u - w) = 5u - w + 1$$

$$6u + 7w + 4 = 4(u + 2w)$$

$$25 \quad \frac{3}{4}(4x + y) = 4$$

$$2x = \frac{3}{5}(2x - 3y)$$

Simultaneous inequalities

Linear programming

This topic is fully covered in Chapter 17, pages 143 to 148.

Exercise 23e

1

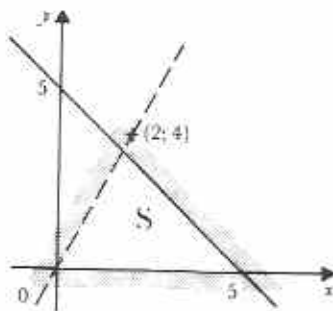


Fig. 23.3

Given Fig. 23.3, find

- the equation of the line through the origin and the point (2;4),
- the equation of the line through (0;5) and (5;0),
- the inequalities which define the triangular shape S.

- Write down the three inequalities which define the unshaded triangular area A in Fig. 23.4.

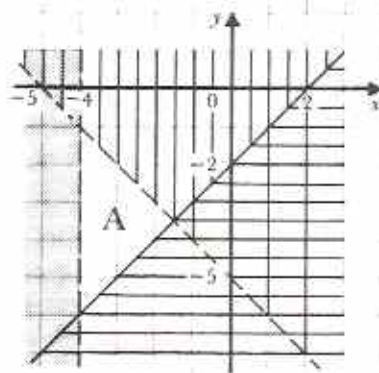


Fig. 23.4

- List the three inequalities which define the unshaded area labelled A in Fig. 23.5.

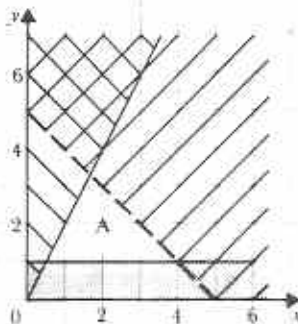


Fig. 23.5

- Write down the four inequalities which define the unshaded region R in Fig. 23.6.

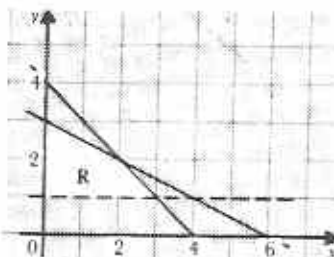


Fig. 23.6

- K is the set of points $(x; y)$ which satisfies the four inequalities $y - x \geq 1$, $2x \geq 5$, $5y > -4x$, $y \geq 2$. Show on a graph the region which represents K. Use your graph to find the greatest value of $(x + y)$.

- 6 (a) The set of points with coordinates $(x; y)$ satisfies the five inequalities

$$x \leq 0, y \leq 0, y + 2x \leq 8, \\ 4y + 3x \geq 24 \text{ and } 3y \geq 2x.$$

(i) Using 2 cm to represent 1 unit on each axis, construct accurately on graph paper, and clearly indicate by shading the unwanted regions, the region in which the set of points $(x; y)$ must lie.

(ii) Using your graph, find the least value of $(x - y)$ for points in the region.

(b) The owner of a large piece of land plans to divide it into not more than 36 plots and to build either a house or a bungalow on each plot. He decides that he will build at least 20 houses and that there will be at least twice as many houses as bungalows.

Taking h to represent the number of houses and b the number of bungalows, write down three inequalities, other than $h \leq 0$ and $b \leq 0$, which satisfy the above conditions. [Camb]

- 7 It takes 2 m of cloth to make a shirt and 3 m to make a dress. A tailor has 36 m of cloth and he needs to make at least 6 of each. If the profit on a shirt is the same as that on a dress, what arrangement of shirts and dresses gives the greatest profit?
- 8 A shopkeeper orders two sizes of notebooks, large at \$1.35 each and small at 60c each. She needs twice as many small ones as large ones, with minimum quantities of 10 large and 20 small. If she spends up to \$30, what is the maximum number she can buy? If the profit is 20c on a large notebook and 10c on a small one, what arrangement gives the greatest profit, and how much is that profit?
- 9 A company plans to spend \$500 000 on new machines. Table 23.1 shows the cost and necessary floor space for the two types of machine to be bought.

Table 23.1

machine	cost	floor space
A	\$20 000	6 m ²
B	\$25 000	4 m ²

More of machine A than of machine B are needed and there are only 120 m² of factory floor space available.

Which purchase arrangement (a) uses all the space available, (b) is more expensive?

- 10 A new book is to be published in both a hardback and a paperback edition. A bookseller agrees to purchase
- 15 or more hardback copies,
 - more than 25 paperback copies,
 - at least 45, but fewer than 60, copies altogether.

Using h to represent the number of hardback copies and p to represent the number of paperback copies, write down the inequalities which represent these conditions. The point $(h; p)$ represents h hardback copies and p paperback copies. Using a scale of 2 cm to represent 10 books, on each axis, construct, and indicate clearly by shading the *unwanted* regions, the region in which $(h; p)$ must lie.

Given that each hardback copy costs \$5 and that each paperback copy costs \$2, calculate the number of each sort that the bookseller must buy if he is prepared to spend between \$180 and \$200 altogether and he has to buy each sort in packets of five. [Camb]

Quadratic equations

Example 11

Solve the equation $y^2 - 4y = 0$.

$$\begin{aligned} y^2 - 4y &= 0 \\ \Leftrightarrow y(y - 4) &= 0 \\ \Leftrightarrow \text{either } y &= 0 \text{ or } y - 4 = 0 \\ \Leftrightarrow y &= 0 \text{ or } y = 4 \end{aligned}$$

Example 12

Solve the equation $3x^2 + 5x - 28 = 0$.

$$\begin{aligned} 3x^2 + 5x - 28 &= 0 \\ \text{Factorise the quadratic expression (see Chapter 22).} \end{aligned}$$

$$\begin{aligned} (x + 4)(3x - 7) &= 0 \\ \Leftrightarrow \text{either } x + 4 &= 0 \text{ or } 3x - 7 = 0 \\ \Leftrightarrow x &= -4 \text{ or } 3x = 7 \\ \text{i.e. } x &= -4 \text{ or } x = 2\frac{1}{3} \end{aligned}$$

Exercise 23f

Use factorisation to solve the following equations.

- 1 $x^2 - 10x + 21 = 0$ 2 $m^2 + 3m + 2 = 0$
 3 $a^2 + a - 6 = 0$ 4 $n^2 - 3n - 10 = 0$
 5 $x^2 + x - 2 = 0$ 6 $y^2 + 3y = 0$
 7 $2d^2 - 7d + 6 = 0$ 8 $4e^2 + 11e + 6 = 0$
 9 $a^2 - 4 = 0$ 10 $a^2 - 4a = 0$
 11 $2t^2 + 7t + 5 = 0$ 12 $3u^2 - 10u - 8 = 0$
 13 $8w^2 - 18w + 9 = 0$
 14 $12d^2 - 19d - 18 = 0$
 15 $8n^2 + 2n - 21 = 0$

Graphical solution of quadratic equations

Example 13

Solve $x^2 - 2x - 4 = 0$ graphically.

Let $y = x^2 - 2x - 4$. Make a table of corresponding values of x and y (Table 23.2). Fig. 23.7 is the graph of $y = x^2 - 2x - 4$.

Table 23.2 $y = x^2 - 2x - 4$.

x	-2	-1	0	1	2	3	4
x^2	4	1	0	1	4	9	16
$-2x$	4	2	0	-2	-4	-6	-8
-4	-4	-4	-4	-4	-4	-4	-4
y	4	-1	-4	-5	-4	-1	4

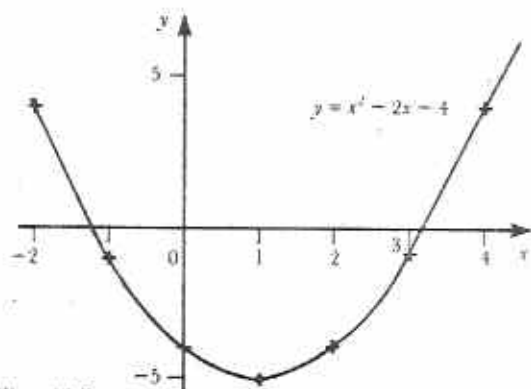


Fig. 23.7

$x^2 - 2x - 4 = 0$ when $y = 0$, i.e. where the curve intersects the x -axis at $x = 3.2$ and $x = -1.2$. These are the approximate solutions to the equation $x^2 - 2x - 4 = 0$.

Example 14

(a) Draw the graph of $y = 2x^2 + 3x - 6$ for values of x from -4 to 2 . (b) Use the graph to solve the equations (i) $2x^2 + 3x - 6 = 0$, (ii) $2x^2 + 3x - 3 = 0$. (c) By drawing the line $y = 2x + 1$ on the same axes, solve the equation $2x^2 + x - 7 = 0$.

(a) The values in Table 23.3 are used to draw the curve in Fig. 23.8.

Table 23.3 $y = 2x^2 + 3x - 6$

x	-4	-3	-2	-1	0	1	2
$2x^2$	32	18	8	2	0	2	8
$+3x$	-12	-9	-6	-3	0	3	6
-6	-6	-6	-6	-6	-6	-6	-6
y	14	3	-4	-7	-6	-1	8

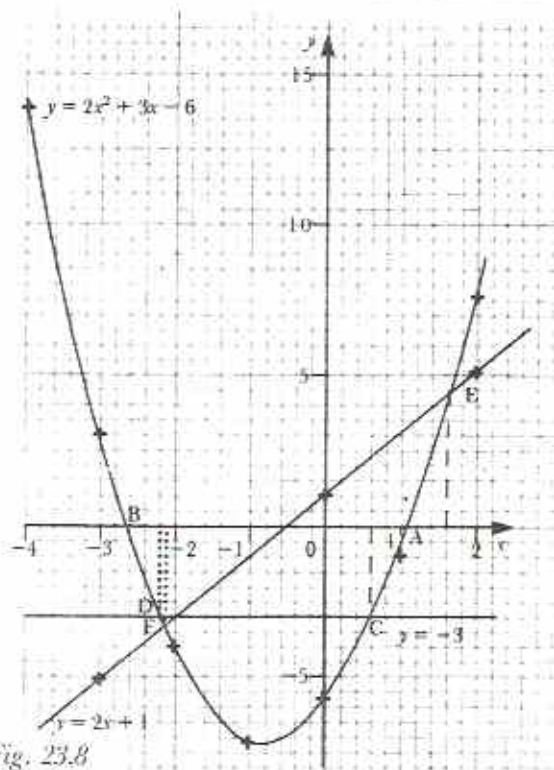


Fig. 23.8

(b) (i) $2x^2 + 3x - 6 = 0$ when $y = 0$, i.e. along the x -axis, at the points A and B in Fig. 23.8. Hence $x = 1.1$ and $x = -2.6$ are the approximate solutions.

$$\begin{aligned} \text{(ii)} \quad & 2x^2 + 3x - 3 = 0 \\ & 2x^2 + 3x = 3 \\ & 2x^2 + 3x - 6 = -6 \\ \text{i.e.} \quad & 2x^2 + 3x - 6 = -3 \end{aligned}$$

This is true at the points where the line $y = -3$ cuts the curve; i.e. at the points C and D in Fig. 23.8.

Hence $x = 0.7$ and $x = -2.2$ are the approximate solutions.

(c) Table 23.4 is used to draw the straight line $y = 2x + 1$ in Fig. 23.8.

Table 23.4

x	-3	0	2
y	-5	1	5

The curve and the straight line intersect at the points E and F in Fig. 23.8. At these points $y = 2x^2 + 3x - 6$ and $y = 2x + 1$
 $\Rightarrow 2x^2 + 3x - 6 = 2x + 1$
 $\Leftrightarrow 2x^2 + x - 7 = 0$

Hence $x = 1.6$ and $x = -2.1$ are the approximate solutions of $2x^2 + x - 7 = 0$.

Exercise 23g

- 1 Copy and complete Table 23.5 for the relation $y = x^2 - 4x + 2$.

Table 23.5

x	-2	-1	0	1	2	3	4	5
y	14			-1	-2		2	

Use a scale of 1 cm to 1 unit on both axes and draw a graph of the relationship. Use your graph to find (a) the solutions of $x^2 - 4x + 2 = 0$, (b) the least value of $x^2 - 4x + 2$.

- 2 Copy and complete Table 23.6 for the relation $y = 3x^2 - 6x + 1$.

Table 23.6

x	-2	-1	0	1	2	3	4
y	25			-2		10	

Using scales of 2 cm to 1 unit on the x -axis and 2 cm to 5 units on the y -axis, draw a graph of the relationship. Use the graph to solve $3x^2 - 6x + 1 = 0$.

- 3 Draw the graph of $y = x^2 - 3x - 2$, taking values of x from -1 to 4. Use the graph to read off the roots of the equation

$$\begin{aligned} \text{(a)} \quad & x^2 - 3x - 2 = 0 \\ \text{(b)} \quad & x^2 - 3x + 1 = 0 \\ \text{(c)} \quad & x^2 - 3x - 4 = 0. \end{aligned}$$

- 4 Draw the graph of $y = 3x^2 + 2x - 1$, taking values of x from -3 to 2. Use the same graph to read off the roots of the equations;

$$\begin{aligned} \text{(a)} \quad & 3x^2 + 2x = 0 \\ \text{(b)} \quad & 3x^2 + 2x - 7 = 0 \\ \text{(c)} \quad & 3x^2 + 2x = 3 \\ \text{(d)} \quad & 3x^2 + 2x - 12 = 0. \end{aligned}$$

- 5 Draw the graph of $y = 3x^2 - 5x + 3$ and use it to find the roots of (a) $3x^2 - 5x + 3 = 0$, (b) $3x^2 - 5x = 0$, (c) $3x^2 - 5x - 1 = 0$.

- 6 Draw the graph of $y = 2x$ to cut the curve drawn in question 5. Hence find the solution of the equation $3x^2 - 7x + 3 = 0$.

- 7 Draw the graph of $y = x(x - 1)$ for values of x from -2 to 3. Read off the values of x at the points where the curve cuts the line $y = 3 - x$. Of what equation in x are these values the roots?

- 8 Draw the graphs of $y = 2x^2 - 3x - 6$ and $y = 1 - 3x$ on the same axes for values of x from -2 to 3. Use the graph to find the roots of the equation $2x^2 - 7 = 0$.

- 9 Copy and complete Table 23.7 for the relation $y = 2 + x - x^2$.

Table 23.7

x	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2	2.5	3
y			0	1.25	2		2	1.25			-4

(a) Draw the graph of the relation using a scale of 2 cm to 1 unit on each axis.

(b) From your graph, find the greatest value of y and the value of x at which this occurs.

(c) Using the same axes, draw the graph of $y = 1 - x$.

(d) From your graphs, determine the roots of the equation $1 + 2x - x^2 = 0$.

- 10** Solve the equation $x^2 = \frac{1}{2}x + 4$ by drawing the graphs of $y = x^2$ and $y = \frac{1}{2}x + 4$ on the same axes for values of x from -3 to 3 . Check your result by drawing a separate graph of $y = x^2 - \frac{1}{2}x - 4$ for the same range of values of x .

Simultaneous linear and quadratic equations

Example 15

Solve the simultaneous equations

$$2x - 5y = 1; 4x^2 + 25y^2 = 41.$$

$$2x - 5y = 1 \quad (1)$$

$$4x^2 + 25y^2 = 41 \quad (2)$$

From (1): $x = \frac{1+5y}{2} \quad (3)$

Substitution $\frac{1}{2}(1+5y)$ for x in (2):

$$4 \times \left[\frac{1}{2}(1+5y)\right]^2 + 25y^2 = 41$$

$$4 \times \frac{1}{4}(1+10y+25y^2) + 25y^2 = 41$$

$$\Leftrightarrow 50y^2 + 10y - 40 = 0$$

$$\Leftrightarrow 5y^2 + y - 4 = 0$$

$$\Leftrightarrow (5y-4)(y+1) = 0$$

$$\Leftrightarrow y = \frac{4}{5} \text{ or } -1$$

Substitute for y in (3):

When $y = \frac{4}{5}$: $x = \frac{1+4}{2} = 2\frac{1}{2}$

When $y = -1$: $x = \frac{1-5}{2} = -2$

The solutions may be given in the form of ordered pairs: $(2\frac{1}{2}; \frac{4}{5})$ and $(-2; -1)$.

Notice that part (c) of Example 14 gives a graphical method of solving simultaneous linear and quadratic equations. Questions 6-10 of Exercise 23g provide practice in the use of graphical methods.

Exercise 23h

Solve the following pairs of equations.

$$\begin{array}{ll} 1 \quad 2x + y = 5 & 2 \quad 4x - y = 7 \\ x^2 + y^2 = 25 & xy = 15 \end{array}$$

$$3 \quad x + y = 3$$

$$x^2 - y^2 = -3$$

$$5 \quad 2x - y = 5$$

$$4x^2 - y^2 = 15$$

$$7 \quad 2x + 3y = 1$$

$$4x^2 - 9y^2 = -17$$

$$9 \quad x^2 + 2y^2 = 3$$

$$x - 3y = 2$$

$$11 \quad 25x^2 - 4y^2 = 36$$

$$5x - 2y = 2$$

$$13 \quad 3xy - y^2 = 2$$

$$2x - 3y = -4$$

$$15 \quad 9x^2 + 16y^2 = 52$$

$$3x - 4y = 2$$

$$4 \quad 4x^2 + y^2 = 61$$

$$2x + y = 1$$

$$6 \quad 2x^2 - y^2 = -2$$

$$3x + y = 1$$

$$8 \quad x + 2y = 2$$

$$x^2 + 2xy = 8$$

$$10 \quad -xy = 30$$

$$3x + y = 21$$

$$12 \quad x - 3y = 1$$

$$x^2 - 2xy - y^2 = 7$$

$$14 \quad 25x^2 - 7y^2 = 29$$

$$5x + 7y + 1 = 0$$

Word problems

Example 16

The ages of a man and his son add up to 39 years. In 3 years the man will be 4 times as old as his son. Find the ages of the man and his son.

Let the ages of the man and son be x and y years respectively. From the first sentence:

$$x + y = 39 \quad (1)$$

From the second sentence:

$$x + 3 = 4(y + 3)$$

$$x + 3 = 4y + 12$$

$$x - 4y = 9 \quad (2)$$

Solve equations (1) and (2) simultaneously.

Subtract (2) from (1):

$$5y = 30$$

$$y = 6$$

Substitute 6 for y in (1):

$$x = 33$$

The man is 33 years and the son is 6 years.

Check: $33 + 6 = 39$ (1st sentence)

$$36 = 4 \times 9 \quad (2nd \text{ sentence})$$

Example 16 led to simultaneous linear equations. Examples 17 and 18 which follow, lead to quadratic equations.

Example 17

Find two numbers whose difference is 4 and whose product is 192.

Let the smaller number be x .

Then the larger number is $x + 4$.

Their product is $x(x + 4)$.

$$\text{Hence } x(x + 4) = 192$$

$$\Leftrightarrow x^2 + 4x - 192 = 0$$

$$\Leftrightarrow (x - 12)(x + 16) = 0$$

$$\Leftrightarrow x = 12 \text{ or } -16$$

The other number is 4 more, i.e. $12 + 4$ or $-16 + 4$, i.e. 16 or -12 .

The two numbers are 12 and 16, or -16 and -12 .

Check: $12 \times 16 = 192$ and $-16 \times -12 = 192$.

Compare Example 17 with Example 18 which follows. Notice the use of units and the elimination of one root because it is not a sensible result.

Example 18

The length of a rectangular compound is 5 m more than the width. Its area is 500 m^2 . Find the width and length of the compound.

Let the width be $x \text{ m}$. Then, from the 1st sentence, the length is $(x + 5) \text{ m}$. The area is $x(x + 5) \text{ m}^2$. From the second sentence:

$$x(x + 5) = 500$$

$$\Leftrightarrow x^2 + 5x - 500 = 0$$

$$\Leftrightarrow (x - 20)(x + 25) = 0$$

$$\Rightarrow x = 20 \text{ or } -25$$

An answer of -25 m is clearly not sensible for the width of a compound. Hence the width is 20 m and the length, 5 m more, is 25 m.

Check: $20 \text{ m} \times 25 \text{ m} = 500 \text{ m}^2$.

Exercise 23i

- 1 A notebook and a pencil cost \$1.32. If the notebook costs 74¢ more than the pencil, find the cost of each.
- 2 Divide 27 into two parts such that their product is 180.
- 3 The area of a rectangle is 360 cm^2 and its length is 2 cm more than its width. Find the width.
- 4 A father is 28 years older than his daughter. In 2 years' time he will be 3 times as old as his daughter. Find their present ages.

- 5 The perimeter of a rectangle is 42 cm and its area is 68 cm^2 . Find its length and breadth.
- 6 The ages of two sisters are 11 and 8 years. In how many years' time will the product of their ages be 378?
- 7 When a man cycles for 1 hour at $x \text{ km/h}$ and 2 hours at $y \text{ km/h}$ he travels 32 km. When he cycles for 2 hours at $x \text{ km/h}$ and 1 hour at $y \text{ km/h}$ he travels 34 km. Find x and y .
- 8 A rectangular garden measures 12 m by 5 m. A path of uniform width runs along one side and one end. If the total area of the garden and path is 98 m^2 , find the width of the path.
- 9 A number is subtracted from 20 and from 17. The product of the numbers so obtained is 180. Find the original number.
- 10 If I subtract 1 from the numerator of a fraction, the fraction becomes $\frac{1}{2}$. If I add 1 to both the numerator and denominator of the fraction, it becomes $\frac{2}{3}$. What is the fraction?

(Hint: Let the fraction be $\frac{x}{y}$.)

- 11 A rectangular piece of cardboard measures 21 cm by 16 cm. When strips of equal width are cut off one side and one end, the area of the remaining piece is 234 cm^2 . Find the width of the strips.
- 12 A woman is 35 years old and her son is 12 years old. How many years ago was the product of their ages 174?
- 13 A table costs \$ x and a chair costs \$ y . If the price of each is raised by \$20, the ratio of their prices becomes 5:2 respectively. If the price of each is reduced by \$5, the ratio becomes 5:1. Express the ratio $x:y$ in its lowest terms.
- 14 Two rectangles have the same area of 24 cm^2 . The second rectangle is 4 cm shorter and 1 cm wider than the first. What is the length and breadth of the first rectangle?
- 15 A girl bought some pencils for \$3.60. If she had paid 4¢ less for each pencil she could have bought 3 more pencils. How many pencils did she buy?

Properties of plane shapes, constructions, locus

Syllabus reference 6.7

Angles

Angle is a measure of rotation or turning.

1 revolution = 360 degrees (1 rev = 360°)

1 degree = 60 minutes ($1^\circ = 60'$)

The names of angles change with their size (Fig. 24.1).

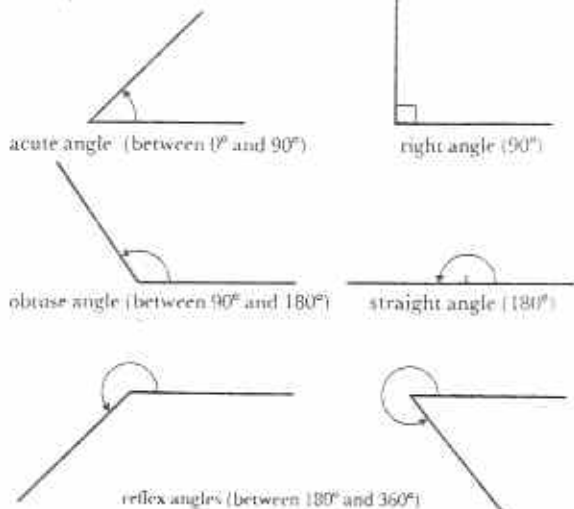


Fig. 24.1

Angles are formed when lines meet or intersect. Remember the following facts:

Angles at a point add up to 360° .

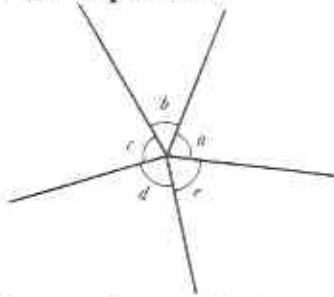


Fig. 24.2

In Fig. 24.2, $a + b + c + d + e = 360^\circ$.

Adjacent angles on a straight line add up to 180° .



Fig. 24.3

In Fig. 24.3, $a + b = 180^\circ$.

Vertically opposite angles are equal.

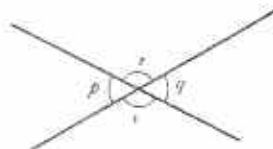


Fig. 24.4

In Fig. 24.4, $p = q$ and $r = s$.

Alternate angles are equal.

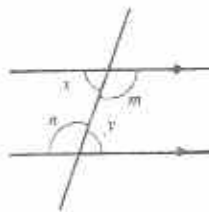


Fig. 24.5

In Fig. 24.5, $x = y$ and $m = n$.

Corresponding angles are equal.

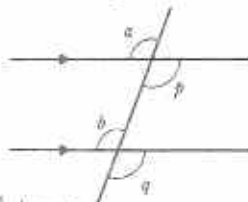


Fig. 24.6

In Fig. 24.6, $a = b$ and $p = q$.

Interior opposite angles add up to 180° .

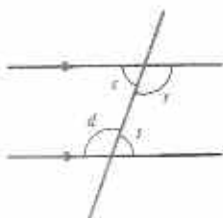


Fig. 24.7

In Fig. 24.7, $c + d = 180^\circ = r + s$.

Example 1

Find the angles marked a , b and c in Fig. 24.8.

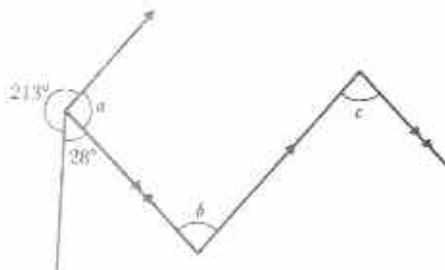


Fig. 24.8

$$\begin{aligned} a &= 360^\circ - (213^\circ + 28^\circ) && (\text{angles at a point}) \\ &= 360^\circ - 241^\circ = 119^\circ \\ b &= 180^\circ - a && (\text{int. opp.}) \\ &= 180^\circ - 119^\circ = 61^\circ \\ c &= b && (\text{alt. angles}) \\ &= 61^\circ \end{aligned}$$

Exercise 24a (oral or written)

Find the lettered angles in Fig. 24.9.

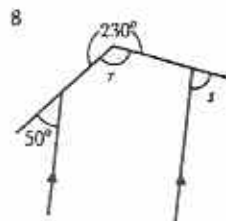
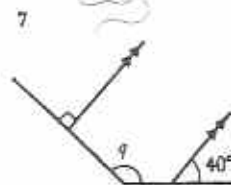
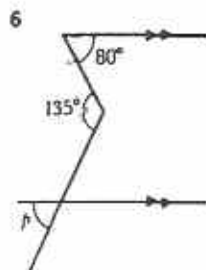
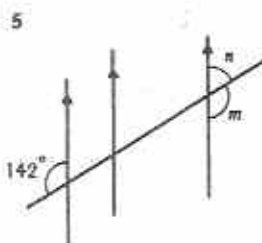
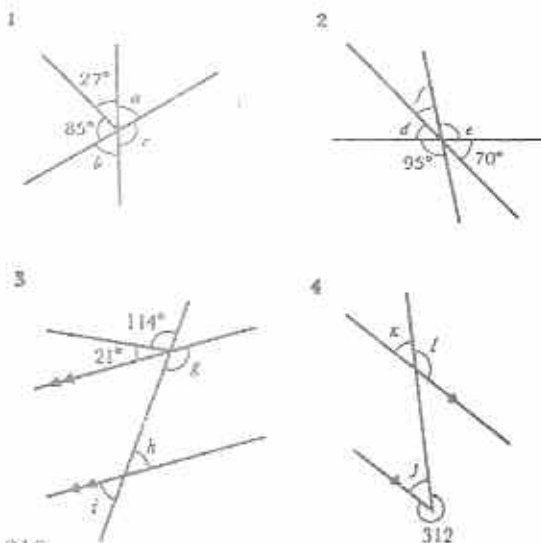


Fig. 24.9

Triangles

Angles in a triangle

The sum of the angles of a triangle is two right angles, or 180° .

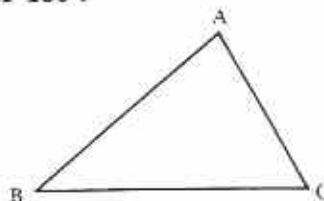


Fig. 24.10

In Fig. 24.10, $\hat{A} + \hat{B} + \hat{C} = 180^\circ$.

The exterior angle of a triangle equals the sum of the two interior opposite angles.

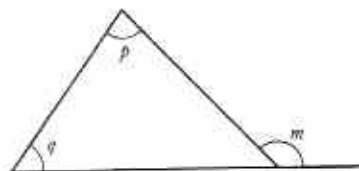


Fig. 24.11

In Fig. 24.11, $m = p + q$.

Types of triangle

Fig. 24.12 gives the names and properties of common types of triangle.

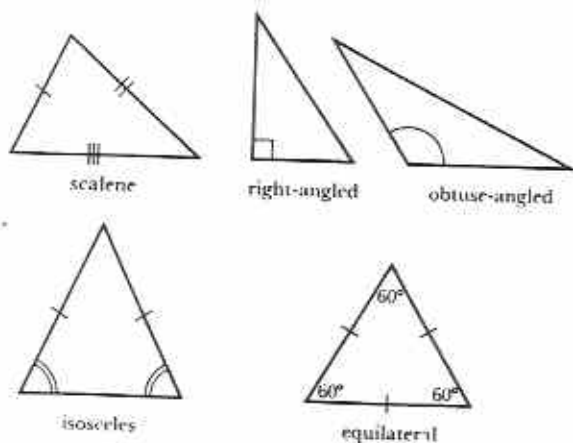


Fig. 24.12

Example 2

Find $\hat{P}$ in Fig. 24.13.

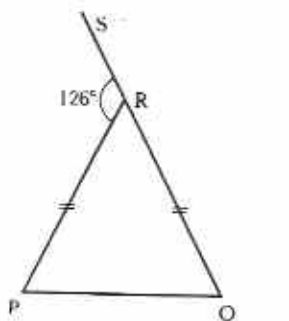


Fig. 24.13

$$\begin{aligned}\hat{P} &= \hat{Q} \\ \hat{P} + \hat{Q} &= 126^\circ \\ \therefore 2\hat{P} &= 126^\circ \\ \therefore \hat{P} &= 63^\circ\end{aligned}$$

(base $\angle$ s of isos. $\triangle$)
(ext. $\angle$ of $\triangle$)
($\hat{P} = \hat{Q}$)

Congruent triangles

Congruent means the same in all respects. Two triangles are congruent if:

- 1 **two sides and the included angle** of one are respectively equal to two sides and the included angle of the other (SAS), or
- 2 **two angles and a side** of one are respectively equal to two angles and a side of the other (ASA or AAS), or
- 3 **three sides** of one are respectively equal to three sides of the other (SSS), or
- 4 they are **right-angled** and have the **hypotenuse and another side** of one equal to the hypotenuse and side of the other (RHS).

Example 3

Given Fig. 24.14, name the triangle which is congruent to $\triangle XYZ$, keeping the letters in the correct order.

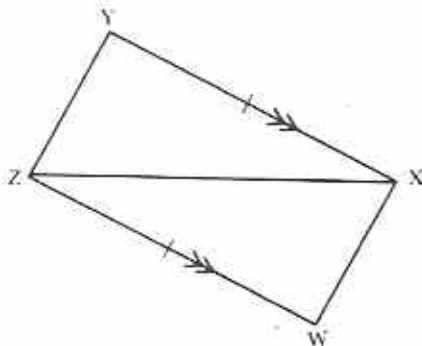


Fig. 24.14

$$\begin{aligned}\triangle XYZ &\equiv \triangle ZWX && (SAS) \\ \text{Reason: } XY &= ZW && (\text{given}) \\ \hat{YXZ} &= \hat{WZY} && (\text{alt. } \angle\text{s, } YX \parallel ZW) \\ XZ &= ZX && (\text{common side})\end{aligned}$$

Exercise 24b

- 1 Name and calculate the sizes of the exterior angles shown in Fig. 24.15.

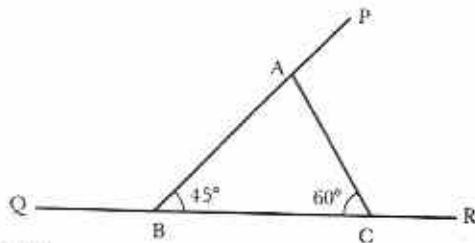
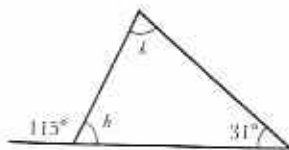


Fig. 24.15

- 2 In each of the following, two angles of a triangle are given. Find the third angle and name the type of triangle.

(a) $69^\circ, 46^\circ$	(b) $38^\circ, 71^\circ$
(c) $43^\circ, 94^\circ$	(d) $60^\circ, 60^\circ$
(e) $35^\circ, 55^\circ$	(f) $58^\circ, 25^\circ$
- 3 Calculate the sizes of the lettered angles in Fig. 24.16(a)–(f).

(a)



(b)



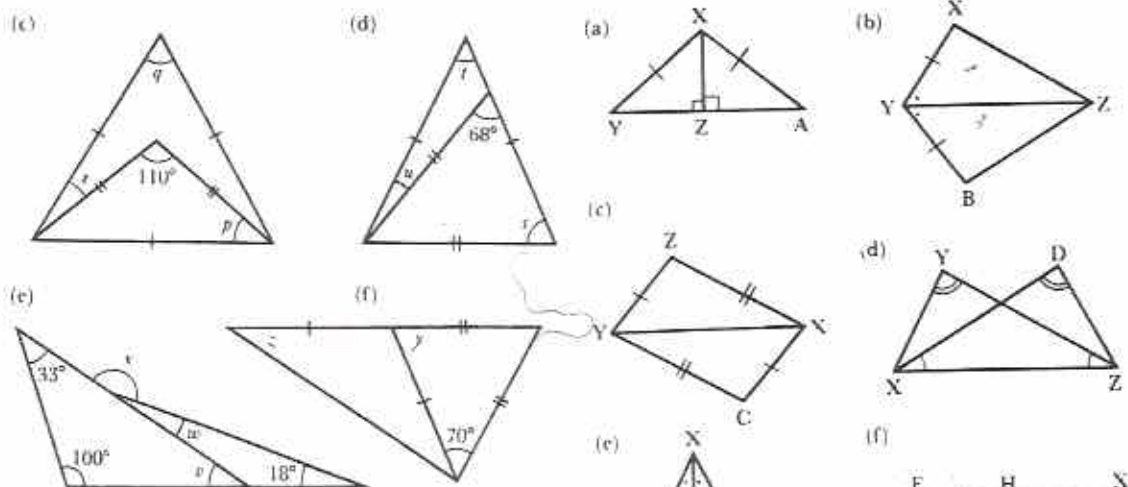


Fig. 24.16

- 4 Find the value of x in Fig. 24.17.

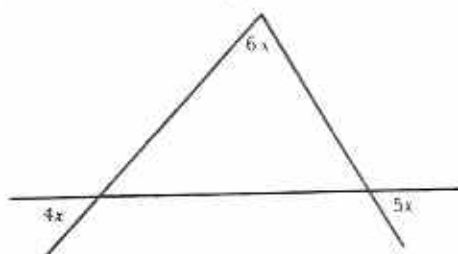


Fig. 24.17

- 5 Find the value of x in each part of Fig. 24.18. Hence state which type of triangle each is.

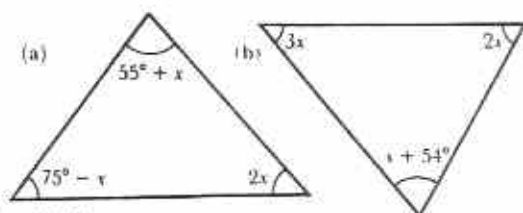


Fig. 24.18

- 6 In each part of Fig. 24.19, name the triangle which is congruent to $\triangle XYZ$. Keep the letters in the correct order and state the case of congruency using the abbreviations, *RHS*, *SSS*, *SAS*, *ASA* or *AAS* as appropriate.

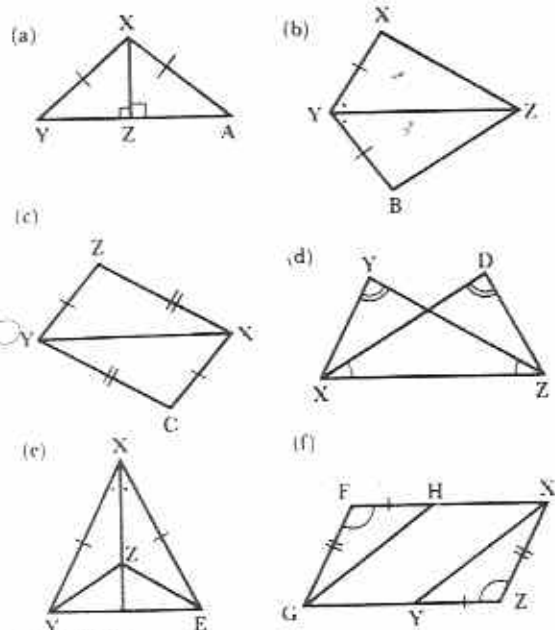


Fig. 24.19

- 7 In Fig. 24.20, $CD \parallel AB$ and AD bisects $\angle BAC$. If $\angle CDE = 27^\circ$ and $\angle ACB = 69^\circ$, find $\angle BED$.

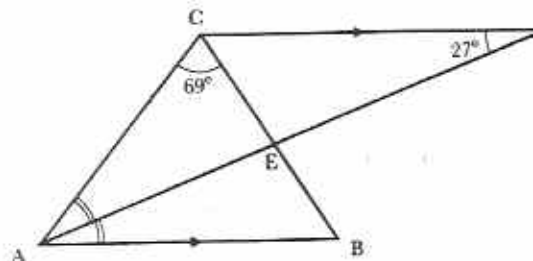
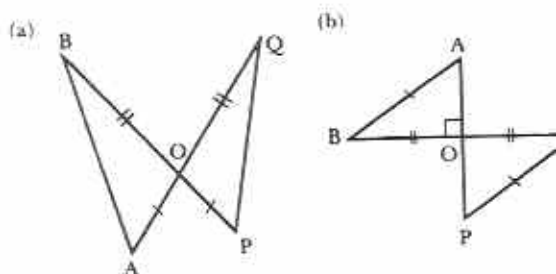
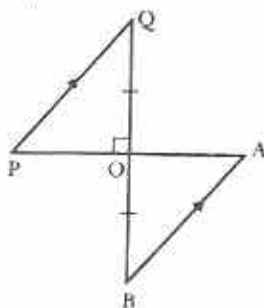


Fig. 24.20

- 8 In each part of Fig. 24.21, $\triangle AOB \cong \triangle POQ$. State the case of congruency for each.



(c)



(d)

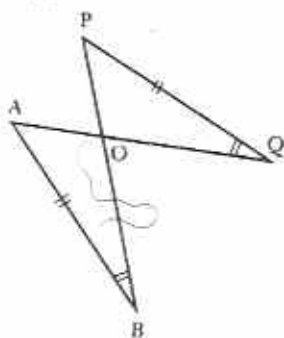


Fig. 24.21

- 9 In Fig. 24.22, $BC \parallel XY$, $\angle BXY = 50^\circ$, $\angle BYX = 28^\circ$ and $AB = BY$. Calculate $\angle ACB$.

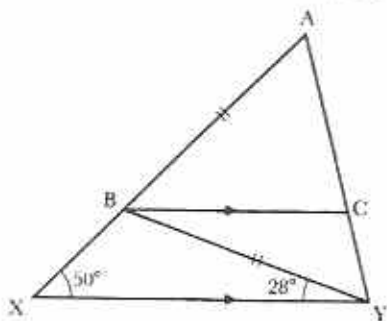


Fig. 24.22

- 10 Which of the triangles in Fig. 24.23 are congruent? State the case(s) of congruency.

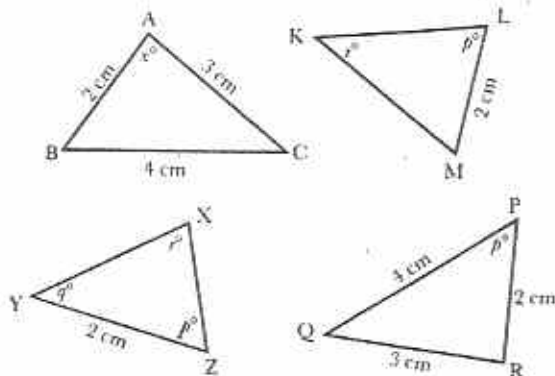
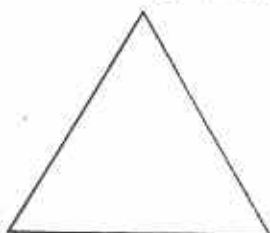


Fig. 24.23

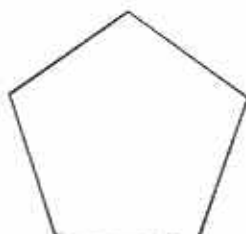
polygon. A **regular polygon** has all its sides of equal length and all its angles of equal size. Polygons are named after the number of sides they have. Fig. 24.24 gives the names of some common regular polygons.



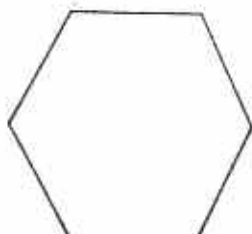
equilateral triangle



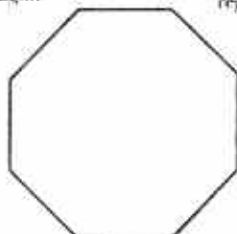
square



regular pentagon



regular hexagon



regular octagon

Fig. 24.24

Angles of a polygon

The sum of the angles of an n -sided polygon is $(n - 2) \times 180^\circ$ or $(2n - 4)$ right angles. The sum of the exterior angles of a polygon is 4 right angles, or 360° .

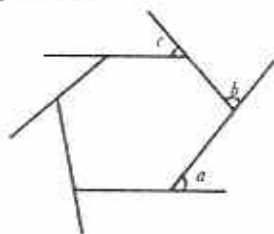


Fig. 24.25

In Fig. 24.25, $a + b + c + \dots = 360^\circ$.

Polygons

Any plane figure with straight sides is called a

Example 4

Calculate the interior angles of a regular pentagon.

A pentagon has 5 sides.
The five exterior angles add up to 360° .
Since the pentagon is regular, the exterior angles are equal.

$$\text{Each exterior angle} = \frac{360^\circ}{5} = 72^\circ$$

Hence each interior angle

$$\begin{aligned} &= 180^\circ - 72^\circ \quad (\angle\text{s on a str. line}) \\ &= 108^\circ \end{aligned}$$

Example 5

Each of the angles of a polygon is 140° . Find the number of sides that the polygon has.

Either:

Let the polygon have n sides.

$$\text{Sum of angles of polygon} = n \times 140^\circ = 140n^\circ$$

$$\text{Also, sum of angles} = (n - 2)180^\circ$$

$$\text{So } (n - 2)180^\circ = 140n^\circ$$

$$\Leftrightarrow 180n - 360 = 140n$$

$$\Leftrightarrow 40n = 360$$

$$\Leftrightarrow n = \frac{360}{40} = 9$$

or:

$$\text{Each exterior angle} = 180^\circ - 140^\circ = 40^\circ$$

$$\text{But the sum of the exterior angles} = 360^\circ$$

$$\text{Number of exterior angles} = \frac{360^\circ}{40^\circ} = 9$$

The polygon has 9 sides.

Properties of quadrilaterals

A **trapezium** is a quadrilateral which has one pair of opposite sides parallel (Fig. 24.26).



Fig. 24.26

A **parallelogram** is a quadrilateral which has both pairs of opposite sides parallel (Fig. 24.27).

In a parallelogram,

- 1 the opposite sides are parallel,
- 2 the opposite sides are equal,

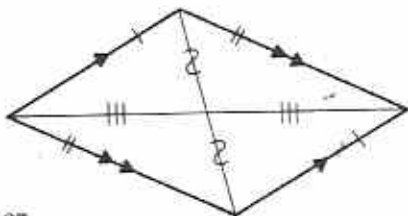


Fig. 24.27

- 3 the opposite angles are equal,
- 4 the diagonals bisect one another.

A **rhombus** is a quadrilateral which has all four sides equal (Fig. 24.28).

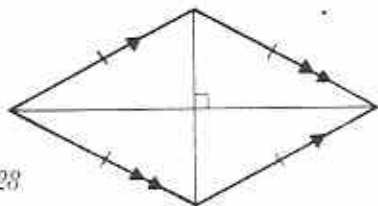


Fig. 24.28

In a rhombus,

- 1 all four sides are equal,
- 2 the opposite sides are parallel,
- 3 the opposite angles are equal,
- 4 the diagonals bisect one another at right angles,
- 5 the diagonals bisect the angles.

A **rectangle** is a quadrilateral in which every angle is a right angle (Fig. 24.29).

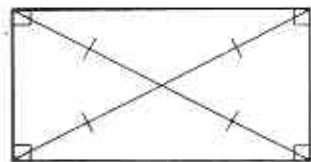


Fig. 24.29

In a rectangle, all of the facts given for a parallelogram are true. In addition:

- 1 all four angles are right angles,
- 2 the diagonals are of equal length.

A **square** is a rectangle which has all four sides equal (Fig. 24.30).

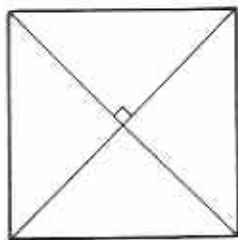


Fig. 24.30

In a square, all of the facts given for a rhombus are true. In addition:
 1 all four angles are right angles,
 2 the diagonals are of equal length,
 3 the diagonals meet the sides at 45° .

Exercise 24c

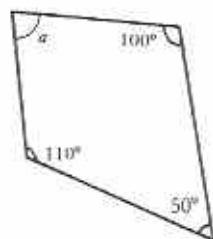
- 1 Use the $(2n - 4) \times 90^\circ$ formula to complete Table 24.1.

Table 24.1

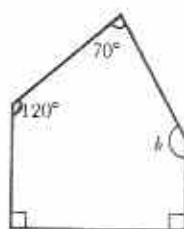
polygon	sum of interior angles
3 sides triangle	
4 sides quadrilateral	
5 sides pentagon	
6 sides hexagon	
7 sides heptagon	
8 sides octagon	
10 sides decagon	
12 sides dodecagon	

- 2 Calculate the sizes of the lettered angles in Fig. 24.31.

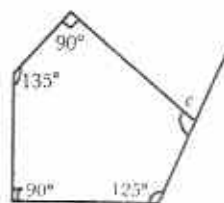
(a)



(b)



(c)



(d)

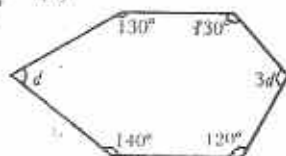


Fig. 24.31

- 3 Calculate the interior angles of a regular 15-sided polygon.
 4 A regular polygon has interior angles of 160° . How many sides has it?
 5 In Fig. 24.32, pentagon ABCDE is such that $AB \parallel ED$ and $BC \parallel AE$. If $\hat{ABC} = 118^\circ$ and $\hat{CDE} = 130^\circ$, calculate $\hat{BCD}$, $\hat{EAB}$ and $\hat{AED}$.

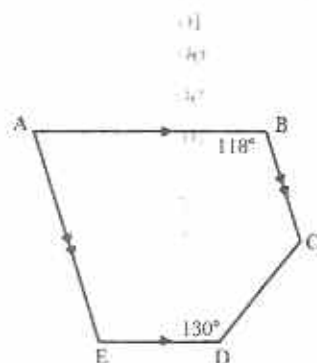


Fig. 24.32

- 6 In Fig. 24.33, QABR is a diagonal of rhombus PQRS and square PARB.

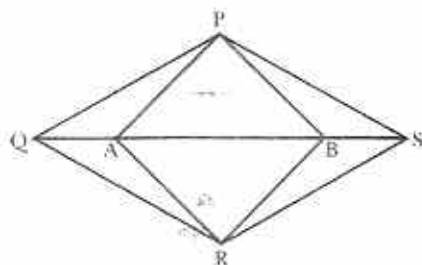


Fig. 24.33

If $\hat{PSR} = 68^\circ$, what is the size of $\hat{QPA}$?

- 7 In Fig. 24.34, ABDE and BCDE are parallelograms. Use the measurements given on the figure to calculate the perimeter of trapezium ABCDE.

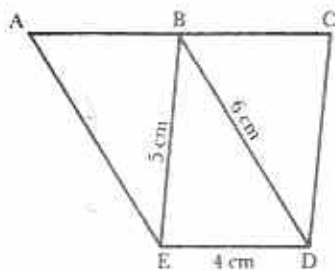


Fig. 24.34

- 8 ABCD is a trapezium such that $AB \parallel DC$. X is a point on CD such that $CX = BA$. If $\angle ABC = 102^\circ$ and $\angle DAX = 47^\circ$, calculate $\angle ADX$.
- 9 Two of the exterior angles of a polygon are 63° each. The remaining exterior angles are each 26° . How many sides has the polygon?
- 10 The angles of a pentagon are $4x$, $5x$, $6x$, $7x$ and $8x$. Find x and hence state the sizes of the angles of the pentagon.

Circles

Fig. 24.35 gives the names of the lines and regions of a circle.

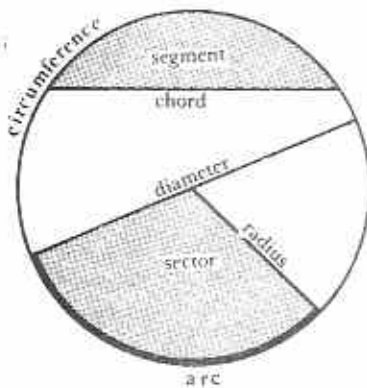


Fig. 24.35

Chords of a circle

- 1 The line joining the centre of a circle to the mid-point of a chord is perpendicular to the chord (Fig 24.36).

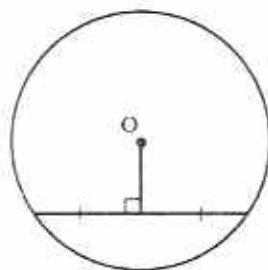


Fig. 24.36

- 2 Equal chords are equidistant from the centre of a circle, and conversely.
- 3 In equal circles equal chords are equidistant from the centres, and conversely.

Angle properties of circles

- 1 The angle which an arc of a circle subtends at the centre of a circle is twice that which it subtends at any point on the remaining part of the circumference.

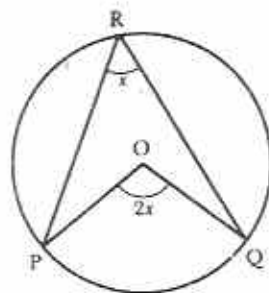


Fig. 24.37

In Fig. 24.37, $\angle POQ = 2 \times \angle PRQ$.

- 2 The angle in a semicircle is a right angle.

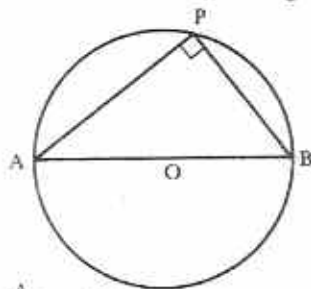


Fig. 24.38

In Fig. 24.38, $\angle APB = 90^\circ$.

3 Angles in the same segment are equal.

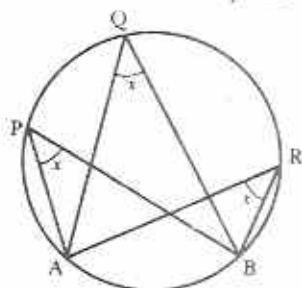


Fig. 24.39

In Fig. 24.39, $\hat{APB} = \hat{AQB} = \hat{ARB} = \dots$
4 Angles in opposite segments are supplementary.

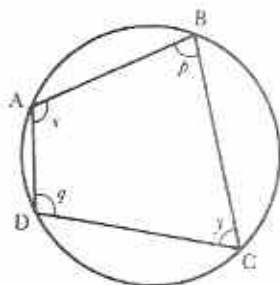


Fig. 24.40

In Fig. 24.40, $x + y = p + q = 180^\circ$. ABCD is known as a **cyclic quadrilateral**. Hence, opposite angles of a cyclic quadrilateral are supplementary. It follows that the exterior angle of a cyclic quadrilateral is equal to the interior opposite angle (Fig. 24.41).

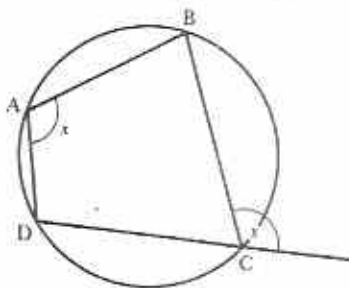


Fig. 24.41

Tangents to circles

A **tangent** to a circle is a line which meets the circle in one point only.

1 A tangent is perpendicular to the radius at the point of contact.

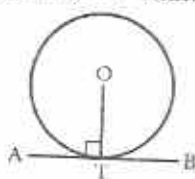


Fig. 24.42

In Fig. 24.42, $OT \perp ATB$.

2 The tangents to a circle from an external point are equal in length.

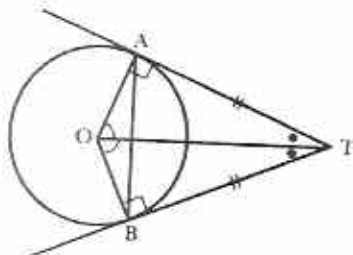


Fig. 24.43

In Fig. 24.43, $TA = TB$. Notice also that $\hat{ATO} = \hat{BTO}$, $\hat{AOT} = \hat{BOT}$ and OT bisects AB at right angles.

3 The angle between a tangent to a circle and a chord through the point of contact is equal to the angle in the alternate segment.

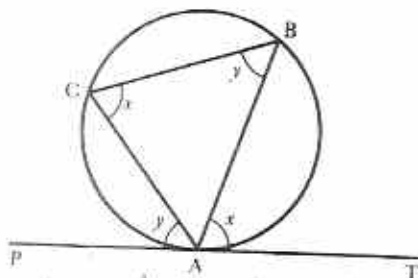
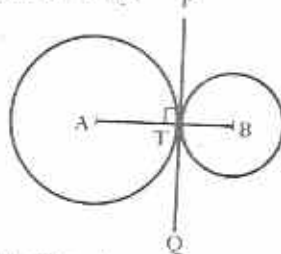


Fig. 24.44

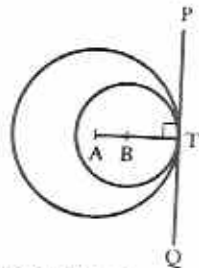
In Fig. 24.44, $\hat{TAB} = \hat{ACB}$ and $\hat{PAC} = \hat{ABC}$.

Contact of circles

Fig. 24.45 shows circles touching (a) externally, (b) internally.



(a) external contact



(b) internal contact

Fig. 24.45

In both cases the line joining the centres of the circles, AB, passes through their point of contact, T. The line of centres, AB, is at right angles to the common tangent, PTQ.

Exercise 24d

- 1 Redraw the figures in Fig. 24.46 and write in the sizes of the marked angles. (Some construction lines may be necessary.)

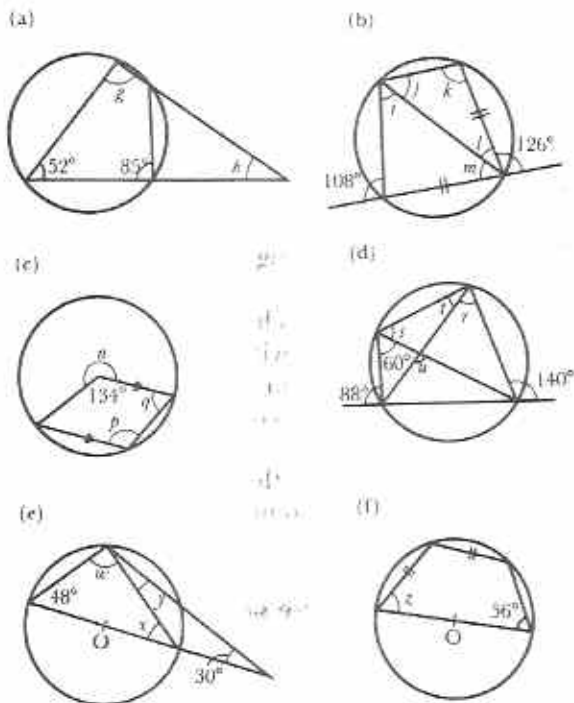


Fig. 24.46

- 2 Three circles, centres A, B and C with radii 4 cm, 3 cm and 2 cm respectively touch one another externally. Calculate the lengths of the sides of $\triangle ABC$.
- 3 Three circles touch one another externally. Their centres form a triangle with sides 10 cm, 9 cm and 7 cm. Find the radii of the circles.
- 4 In Fig. 24.47, the circle touches the sides of $\triangle ABC$ at X, Y, Z. If $BC = 11$ cm, $CA = 10$ cm and $AB = 9$ cm, find AY and BX .

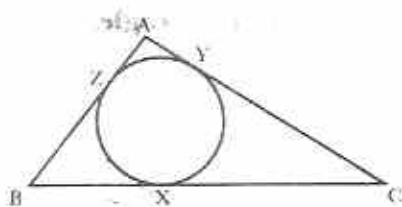


Fig. 24.47

- 5 In Fig. 24.48, lines drawn through T are tangents and O is the centre of any given circle. Find the lettered angles.

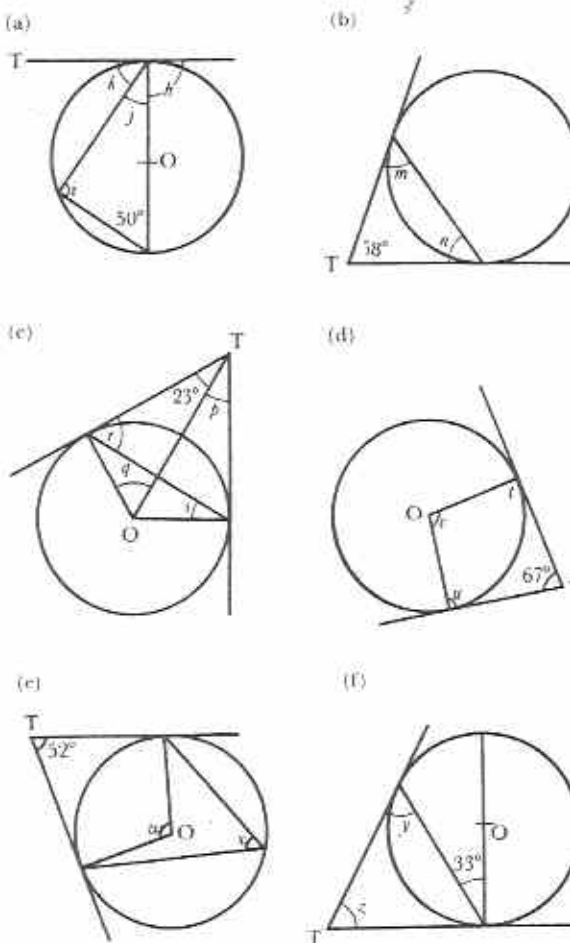
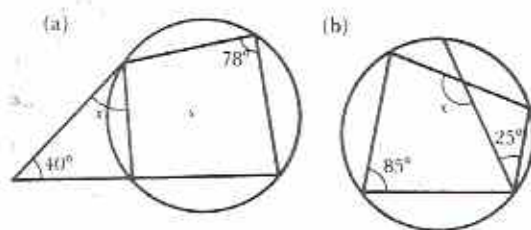
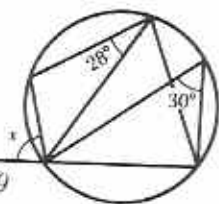


Fig. 24.48

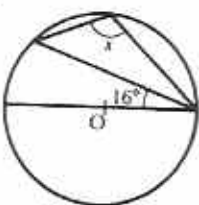
- 6 A circle is drawn inside a triangle ABC to touch the sides BC, CA and AB at P, Q and R respectively. If $\hat{A} = 56^\circ$ and $\hat{B} = 68^\circ$, find the angles of $\triangle PQR$.
- 7 In each part of Fig. 24.49, find the value of x . In part (d), O is the centre of the circle.



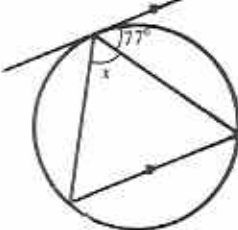
(c)



(d)



(e)



(d)

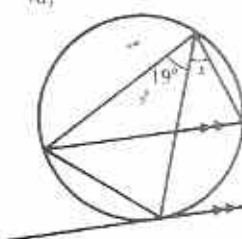
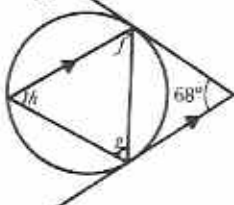


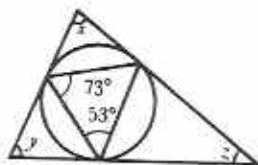
Fig. 24.49

8 In Fig. 24.50, find the sizes of the lettered angles.

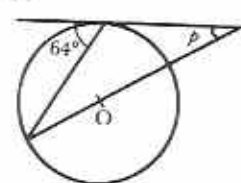
(a)



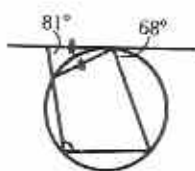
(b)



(c)



(d)



(e)

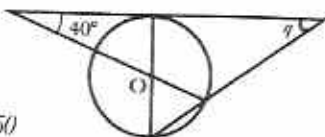


Fig. 24.50

9 In Fig. 24.51, express y in terms of x .

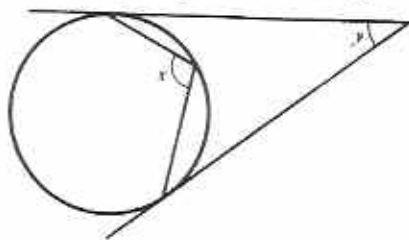
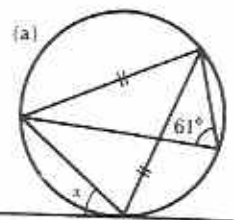


Fig. 24.51

10 In each part of Fig. 24.52, find the value of x .

(a)



(b)

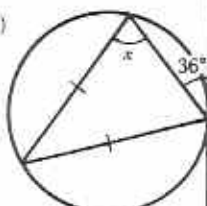


Fig. 24.52

Constructions

Remember the following when making constructions:

- 1 Make a rough sketch. This helps in anticipating problems associated with the construction.
- 2 Leave all construction lines visible. Do not rub off anything that contributes to the final result.
- 3 Use a hard pencil with a sharp point. This enables lines and points to be as fine and accurate as possible.

To bisect a given line segment, AB

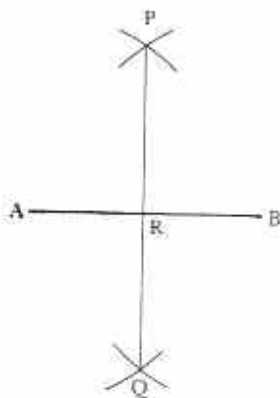


Fig. 24.53

With centres A and B and equal radii, draw arcs to cut each other at P and Q. Join PQ to cut AB at R (Fig. 24.53). R is the mid-point of AB.

To bisect a given angle, $\angle ABC$

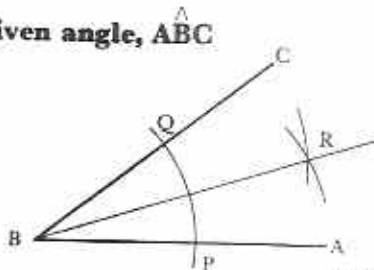


Fig. 24.54

With centre B and any radius, draw an arc to cut BA and BC at P and Q. With centres P and Q and equal radii, draw arcs to cut each other at R. Draw BR (Fig. 24.54). BR is the required bisector.

To construct a line perpendicular to a given straight line, AB, from a point, M, outside the line

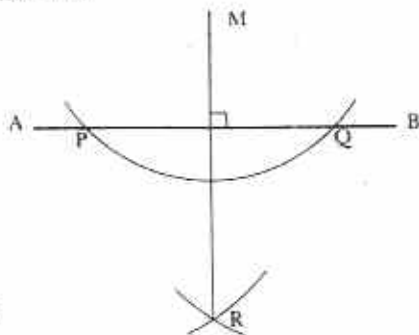


Fig. 24.55

With centre M and any radius, draw an arc to cut AB at P and Q. With centres P and Q and equal radii, draw arcs to cut each other at R (Fig. 24.55). MR is perpendicular to AB.

To construct parallel lines, using ruler and set square

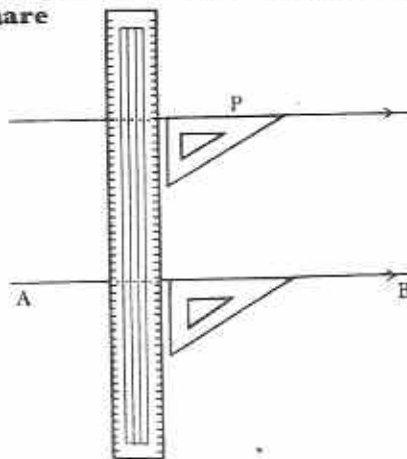


Fig. 24.56

Place a set square with one edge accurately along a given line AB. Place a ruler against one of the other edges of the set square. Holding the ruler firmly, slide the set square along its edge until the edge that was originally along AB passes through the required point P. Use that edge of the set square to draw a line parallel to AB (Fig. 24.56).

To construct an angle of 60°

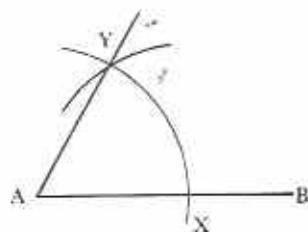


Fig. 24.57

With centre A and any convenient radius draw an arc cutting AB at X. With centre X and the same radius, draw an arc to cut the first arc at Y. Join AY to give $\angle BAY = 60^\circ$ (Fig. 24.57).

To construct an angle of 30° , first construct an angle of 60° as above and then bisect it. Further bisections will give angles of 15° , $7\frac{1}{2}^\circ$, etc.

To construct an angle of 45° , first construct a right angle and then bisect it. Further bisections will give angles of $22\frac{1}{2}^\circ$, etc.

To copy a given angle, $\angle ABC$

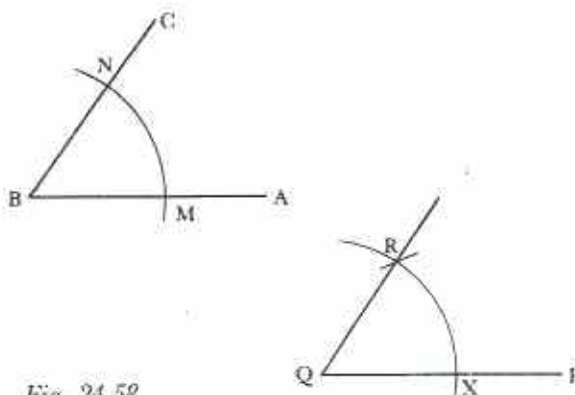


Fig. 24.58

Draw any line PQ. With centre B and any radius draw an arc to cut BA and BC at M and N. With centre Q and the same radius, draw an arc to cut QP at X. With centre X and radius MN, draw an arc to cut the arc through X at R (Fig. 24.58). Then $\angle PQR = \angle ABC$.

Exercise 24c

- Construct angles of 60° , 30° , 75° , 45° , 120° , $37\frac{1}{2}^\circ$, 135° .

- 2 Construct an equilateral triangle with sides of length 7.2 cm. Construct the perpendicular from a vertex to the opposite side and measure its length.
- 3 Construct $\triangle PQR$ such that $\hat{Q} = 90^\circ$, $\hat{P} = 60^\circ$ and $PQ = 8$ cm. Draw the bisectors of $\hat{P}$ and $\hat{R}$ to intersect at O. Measure OP.
- 4 Construct an isosceles triangle with the equal sides 9 cm long and the angle between them 45° . Measure the third side.
- 5 Construct $\triangle ABC$ such that $AB = 10.8$ cm, $BC = 6.4$ cm and $AC = 7$ cm. Draw the perpendicular bisectors of the three sides to meet at O. Draw the circumcircle of the triangle, i.e. the circle, centre O, which passes through A, B and C.
- 6 Construct the parallelogram ABCD in which $BD = 104$ mm, $DC = 48$ mm and $\hat{BCD} = 30^\circ$. Measure AC.
- 7 Construct a trapezium PQRS in which $PQ \parallel SR$, $PQ = 6$ cm, $PS = 5$ cm, $SR = 11$ cm and $QS = 9$ cm. Measure QR.
- 8 Construct a rhombus ABCD so that $AC = 6$ cm and $BD = 8$ cm. Measure a side of the rhombus.
- 9 Construct a triangle with sides of 5 cm, 6 cm, 7 cm. A rhombus with sides of length 4 cm has acute angles each equal to the smallest angle of the triangle. Copy the smallest angle of the triangle and hence construct the rhombus. Measure the longest diagonal of the rhombus.
- 10 Draw a circle of radius 3 cm. Draw a diameter RT and construct a tangent to the circle at T. Find a point P on the tangent such that $\triangle RTP$ is isosceles. Measure RP.

Locus

The simple idea of a locus is that it is the path traced out by a point which moves in accordance with a certain law. However, it is more correct to define a **locus as the set of all possible positions** occupied by an object which varies its position according to some given law.

Table 24.2 on page 222 gives some common **loci** (plural of locus) in 2 and 3 dimensions.

Example 6

$AB = 4$ cm and a point C moves so that the area of $\triangle ABC$ is 6 cm^2 and $\hat{ACB} = 60^\circ$. Construct two possible positions of C. Find AC and BC in each case.

Let C be a distance h cm from AB, then

$$\frac{1}{2} AB \times h = 6$$

$$\frac{1}{2} \times 4 \times h = 6$$

$$\Leftrightarrow h = 3$$

C is a distance 3 cm from AB.

Fig. 24.59 shows (a) sketch and (b) the accurate drawing giving two positions of C.

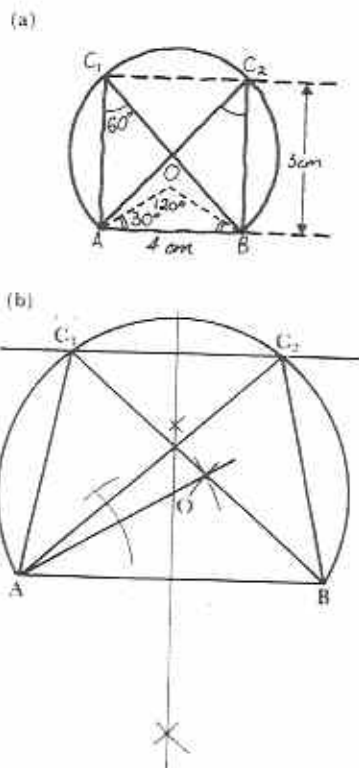
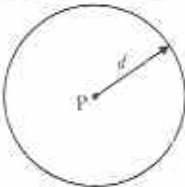
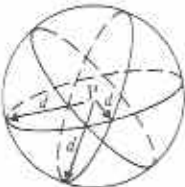
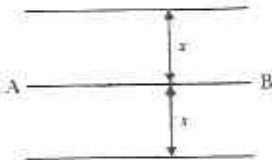
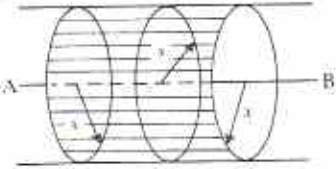
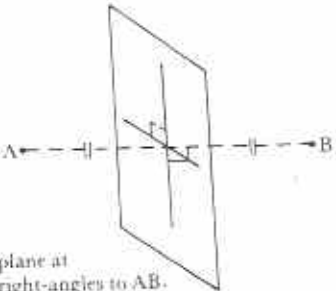
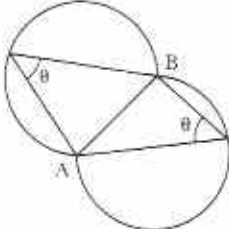


Fig. 24.59

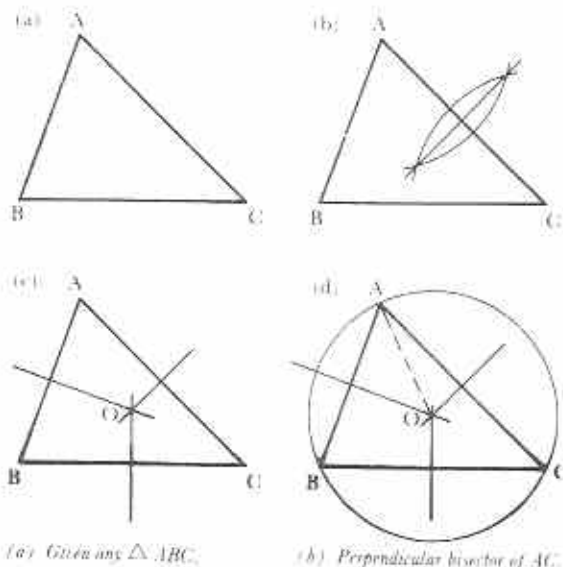
From Fig. 24.59,
either $AC = 3.1$ cm and $BC = 4.5$ cm,
or $AC = 4.5$ cm and $BC = 3.1$ cm.

Table 24.2

a set of points	locus	
	in 2 dimensions	in 3 dimensions
... which are a given distance, d , from a given point P	 circumference of a circle, centre P, radius d	 outer shell of a sphere, centre P, radius d
... which are a given distance, x , from a given straight line AB	 a pair of lines each parallel to AB	 a cylindrical surface with AB as its central axis
... which are equidistant from two given points AB	 the perpendicular bisector of AB	 plane at right-angles to AB, bisecting AB
... at which a given segment of a straight line AB subtends a given angle θ	 two segments of a circle on chord AB	non-applicable in 3 dimensions

Circumcircle

Fig. 24.60 shows how to construct the circumcircle of any triangle ABC .



(a) Given any $\triangle ABC$.

(b) Perpendicular bisector of AC .

(c) Perpendicular bisectors of all three sides meeting at O (lines of construction not shown here).

(d) Circumcircle, centre O , radius OA (or OB or OC).

Fig. 24.60

Exercise 24f

- 1 A pencil slides around inside a hemispherical bowl. Describe the locus of the mid-point of the pencil.
- 2 A pole stands vertically on horizontal ground. A wire is stretched tightly from the top of the pole to a point on the ground some distance from the foot of the pole. Describe the locus of the lower end of the wire.
- 3 Draw two intersecting lines. Construct the locus of points which are equidistant from the two lines.
- 4 A and B are fixed points. P can move so that $\angle APB = 45^\circ$. Use a 45° set square to plot several positions of P , and hence draw its locus.
- 5 A circle is drawn so that it passes through

two fixed points. Describe the locus of the centre of the circle as it varies in radius.

- 6 Draw two straight lines AXB and PXQ intersecting at X at an angle of 70° . Draw the locus of points which move so that they are (i) 2 cm from AB , (ii) 3 cm from PQ . How many points are common to both loci?
- 7 On the same diagram, draw the following loci: (a) a point P which moves so that $\angle APB = 90^\circ$ and $AB = 8$ cm; (b) points which are 3 cm from AB . How many points are common to both loci?
- 8 If $AB = 5$ cm, construct the locus of the set of points C such that the area of $\triangle ABC$ is 10 cm^2 . Use a suitable construction to find all the positions of C where $\angle ACB = 54^\circ$. Hence find two possible values for AC .

- 9 (a) Construct in a single diagram,
 - (i) triangle ABC such that $AB = 9$ cm, $AC = 7$ cm and $BC = 5$ cm,
 - (ii) the locus of points which are 4 cm from A ,
 - (iii) the circumcircle of triangle ABC .
 (b) If $\mathcal{E} = \{P: P \text{ lies inside the circumcircle of } \triangle ABC\}$
 $\mathcal{X} = \{P: P \text{ lies inside } \triangle ABC\}$
 $\mathcal{Y} = \{P: AP < 4 \text{ cm}\}$
 show the region $\mathcal{X}' \cap \mathcal{Y}'$ by shading in your diagram.

- 10 Construct the triangle XYZ in which $XY = 5$ cm, $\hat{X} = 60^\circ$ and $\hat{Y} = 90^\circ$. Measure and write down the length of YZ . On the same diagram,
 - (a) construct the circumcircle of $\triangle XYZ$.
 - (b) construct on the same side of XY as Z , the locus of the point P , such that the area of $\triangle XYP$ equals half the area of $\triangle XYZ$.
 - (c) mark, and label clearly, a point Q such that $\angle XQY = 30^\circ$ and the area of $\triangle XYQ$ is half the area of $\triangle XYZ$.

Given that M is a point such that $\angle XMY = 30^\circ$, find the largest possible area of $\triangle XMY$.
[Camb]

Mensuration

Syllabus references 6.4, 6.7.5 and 6.8.2

Perimeter

The **perimeter** of a plane shape is the distance round the edge of the shape. The perimeters of shapes are found by measurement. In some cases there are formulae which enable perimeters to be calculated:

Perimeter of rectangle in Fig. 25.1 = $2(l + b)$

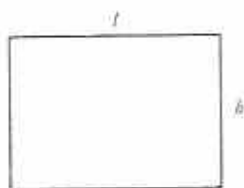


Fig. 25.1

Perimeter of circle in Fig. 25.2 = $2\pi r$

The perimeter of a circle is often called the **circumference**.



Fig. 25.2

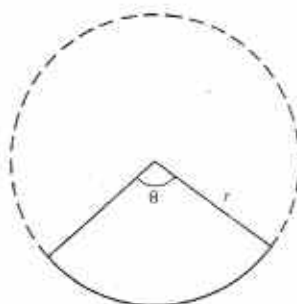


Fig. 25.3

Length of arc in Fig. 25.3 = $\frac{\theta}{360}$ of $2\pi r$

Perimeter of sector = $\frac{\theta}{360} \times 2\pi r + 2r$

Example 1

Find the perimeter of a sector of a circle of radius 3,5 cm, the angle of the sector being 144° .

Perimeter = length of arc + $2r$

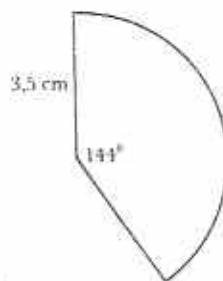


Fig. 25.4

$$\begin{aligned} \text{Length of arc} &= \frac{144}{360} \times 2\pi \times 3,5 \text{ cm} \\ &= \frac{144}{360} \times 2 \times \frac{22}{7} \times \frac{7}{2} \text{ cm} \\ &= \frac{44}{5} \text{ cm} = 8,8 \text{ cm} \end{aligned}$$

$$\text{Perimeter} = (8,8 + 2 \times 3,5) \text{ cm} = 15,8 \text{ cm}$$

Exercise 25a

Take the value of π to be $3\frac{1}{7}$, unless told otherwise.

- 1 Use the value 3,14 for π to calculate the circumference of a circle of diameter 20 cm.
- 2 The minute hand of a wall-clock is 10,5 cm long. How far does its tip travel in 24 hours?
- 3 Through what angle does the minute hand of a clock move in 25 min? If the minute hand is 6,3 cm long, how far does its tip move in 25 min?
- 4 How many revolutions does a bicycle wheel of diameter 70 cm make in travelling 110 m?
- 5 An arc subtends an angle of 72° at the centre of a circle of radius 17,5 cm. Find the length of the arc.
- 6 Two circles have circumferences of 10π cm and 12π cm. What is the difference in their radii?

- 7 A chord of a circle subtends an angle of 60° at the centre of a circle of radius 7 cm. Find the perimeter of the minor segment of the circle.
- 8 A piece of thread was wound tightly round a cylinder for 20 complete turns. The thread was found to be 3,96 m long. Calculate the diameter of the cylinder in cm.
- 9 Calculate the perimeter of a sector of a circle of radius 27 cm, the angle of the sector being 140° .
- 10 Fig. 25.5 is a sketch of a doorway. The arc at the top subtends an angle of 60° at the centre of a circle of radius 75 cm.

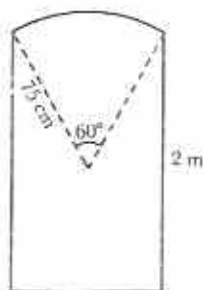


Fig. 25.5

Use the value 3,14 for π to calculate the perimeter of the doorway to the nearest $\frac{1}{2}$ -centimetre.

Area of plane shapes

The common units of area are cm^2 , m^2 and km^2 . The **hectare** (ha) is often used for land measure. $1 \text{ ha} = 10\,000 \text{ m}^2$. Formulae for the areas of the common plane shapes are given in Fig. 25.6.

Rectangle



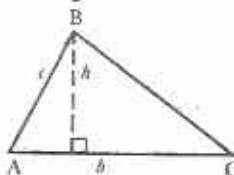
$$\begin{aligned}\text{Area} &= \text{base} \times \text{height} \\ &= b \times h\end{aligned}$$

Parallelogram



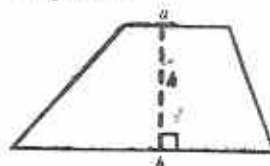
$$\begin{aligned}\text{Area} &= \text{base} \times \text{height} \\ &= b \times h\end{aligned}$$

Triangle



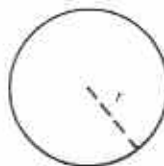
$$\begin{aligned}\text{Area} &= \frac{1}{2} \times \text{base} \times \text{height} \\ &= \frac{1}{2}bh \text{ or } \frac{1}{2}bc \sin A\end{aligned}$$

Trapezium



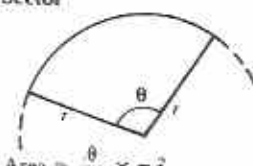
$$\text{Area} = \frac{1}{2}(a + b) \times h$$

Circle



$$\text{Area} = \pi r^2$$

Sector



$$\text{Area} = \frac{\theta}{360} \times \pi r^2$$

Fig. 25.6

Example 2

Calculate the area of $\triangle ABC$ in which $AB = 8 \text{ cm}$, $AC = 4 \text{ cm}$ and $BAC = 58^\circ$.

$$\begin{aligned}\text{Area of } \triangle ABC &= \frac{1}{2} \times 8 \times 4 \times \sin 58^\circ \text{ cm}^2 \\ &= 16 \times 0,848 \text{ cm}^2 \\ &= 13,6 \text{ cm}^2 \text{ to 3 s.f.}\end{aligned}$$

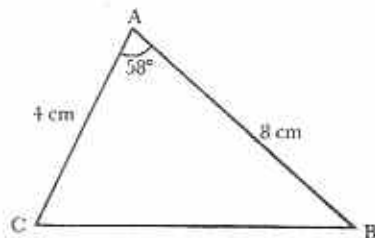


Fig. 25.7

Example 3

Calculate the area of the parallelogram in Fig. 25.8.

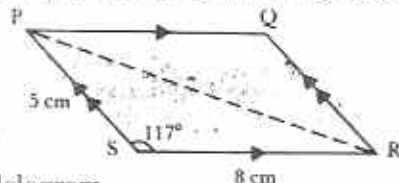


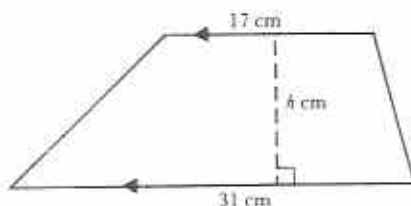
Fig. 25.8

$$\begin{aligned}\text{Area of parallelogram} &= 2 \times \text{area of } \triangle PRS \\ &= 2 \times \left(\frac{1}{2} \times 5 \times 8 \times \sin 117^\circ \right) \text{ cm}^2 \\ &= 5 \times 8 \times \sin 63^\circ \text{ cm}^2 \\ &= 40 \times 0,8910 \text{ cm}^2 = 35,6 \text{ cm}^2\end{aligned}$$

Example 4

The trapezium in Fig. 25.9, overleaf, has an area of 456 cm^2 . Find the distance between its parallel sides.

Fig. 25.9



$$\begin{aligned}\text{Area of trapezium} &= \frac{1}{2}(17 + 31)h \text{ cm}^2 \\ \frac{1}{2}(17 + 31) \times h &= 456 \\ \Leftrightarrow \frac{1}{2} \times 48 \times h &= 456 \\ \Leftrightarrow 24h &= 456 \\ \Leftrightarrow h &= \frac{456}{24} = 19\end{aligned}$$

The parallel sides are 19 cm apart.

Example 5

In Fig. 25.10, the chord AB subtends an angle of 120° at the centre of the circle, radius 7 cm. Use the value $\frac{22}{7}$ for π to calculate the area of the minor segment of the circle.

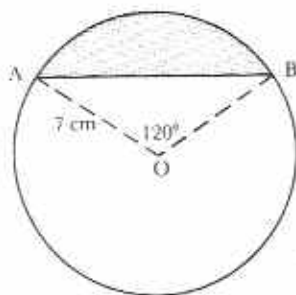


Fig. 25.10

$$\begin{aligned}\text{Area of minor segment} &= \text{area of sector AOB} - \text{area of } \triangle AOB \\ \text{Area of sector AOB} &= \frac{120}{360} \times \pi \times 7^2 \text{ cm}^2 \\ &= \frac{1}{3} \times 22 \times 7 \text{ cm}^2 \\ \text{Area of } \triangle AOB &= \frac{1}{2} \times 7 \times 7 \times \sin 120^\circ \text{ cm}^2 \\ &= \frac{1}{2} \times 7 \times 7 \times 0,8660 \text{ cm}^2 \\ &= 7 \times 7 \times 0,433 \text{ cm}^2 \\ \text{Area of segment} &= \frac{1}{3} \times 22 \times 7 - 7 \times 7 \times 0,433 \text{ cm}^2 * \\ &= 7(\frac{1}{3} \times 22 - 7 \times 0,433) \text{ cm}^2 \\ &= 7(7,333 - 3,031) \text{ cm}^2 = 7 \times 4,302 \text{ cm}^2 \\ &= 30,1 \text{ cm}^2 \text{ to 3 s.f.}\end{aligned}$$

* Alternatively the calculation may be done on a scientific calculator as follows.

$$\begin{aligned}\text{Area of segment} &= \frac{1}{3} \times 22 \times 7 - \frac{1}{2} \times 7 \times 7 \times \sin 120^\circ \text{ cm}^2 \\ &= \frac{154}{3} - 24,5 \times \sin 60^\circ \\ &= 51,333\ 333 - 24,5 \times \sin 60^\circ **\end{aligned}$$

On the calculator:

<i>Key</i>	<i>Display</i>
60 sin × 24.5 = +/-	-21.217622
+ 51.333333 =	30.115711

$$\text{Area of segment} = 30,1 \text{ cm}^2$$

** Where possible, always simplify calculations before using a calculator. This reduces the possibility of making keystroke errors.

Exercise 25b

Use the value $\frac{22}{7}$ for π unless told otherwise.

- 1 Calculate the area shaded in each of the shapes in Fig. 25.11.

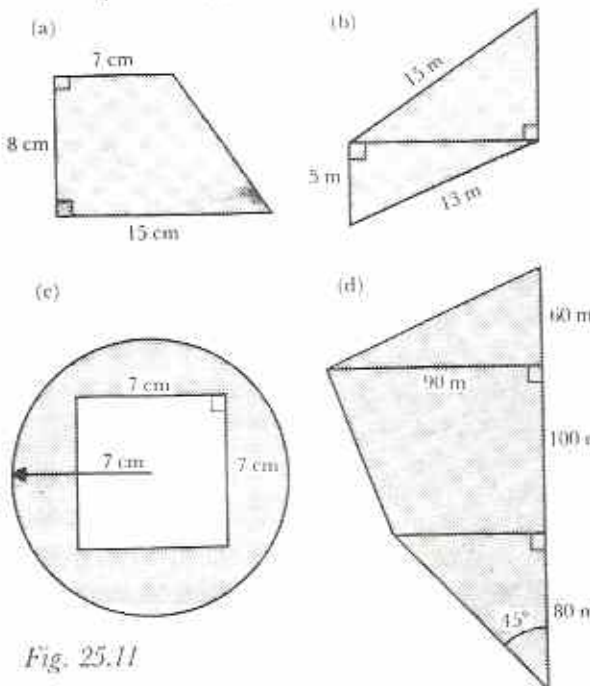


Fig. 25.11

- 2 The diagonals of a parallelogram are 6 cm and 8 cm long and they intersect at an angle of 55° . Find the area of the parallelogram.

- 3 Two sides of a triangular field are 120 m and 200 m. If the angle between the sides is 68° , find the area of the field in hectares.
- 4 Find the area of a circle of radius 35 cm. If a sector of angle 80° is removed from the circle, what area is left?
- 5 Circular discs of diameter 4 cm are punched out of a sheet of brass of mass 0.84 g/cm^2 . What is the mass of 500 discs?
- 6 If 350 of the discs in question 5 are punched from a square sheet of brass, 80 cm by 80 cm, what percentage of the sheet is not used?
- 7 What is the diameter, to the nearest metre, of a circular sports ground of area exactly one hectare?
- 8 A washer is 4.5 cm in diameter with a central hole of diameter 1.5 cm. Use the value 3.142 for π to calculate the surface area of the washer correct to 3 s.f.
- 9 Fig. 25.12 shows a cross-section of a tunnel in the form of a major segment of a 6 m diameter circle. The path at the base of the tunnel is 3 m wide. Use the value 3.142 for π to find the cross-sectional area of the tunnel correct to 3 s.f.

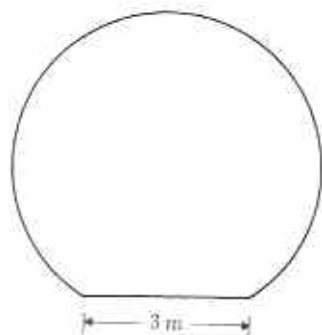


Fig. 25.12

- 10 Calculate the area of the doorway in Fig. 25.5 on page 225.
- 11 A chord subtends an angle of 140° at the centre of a circle of radius 10 cm. Calculate the area of the minor segment of the circle.
- 12 The floor of a classroom measures 6 m by 8 m. The walls are 3 m high and they contain four rectangular windows measuring 1.2 m by 75 cm. There is a door which measures 2 m by 80 cm. What percentage of the total wall area is taken up with windows and doors? (2 s.f.)

Surface area and volume of solids

Fig. 25.13 contains a summary of the formulae for surface area and volume of common solid shapes.

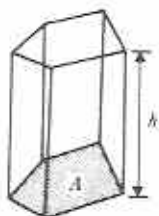
Cuboid

$$\begin{aligned}\text{volume} &= lbh \\ \text{surface area} &= 2(lb + lh + bh)\end{aligned}$$



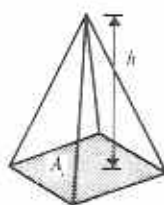
Prism

$$\begin{aligned}\text{volume} &= \text{base area} \times \text{perpendicular height} \\ &= A \times h\end{aligned}$$



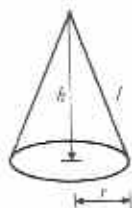
Pyramid

$$\begin{aligned}\text{volume} &= \frac{1}{3} \times \text{base area} \times \text{height} \\ &= \frac{1}{3}Ah\end{aligned}$$



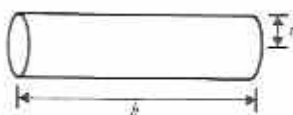
Cone

$$\begin{aligned}\text{volume} &= \frac{1}{3}\pi r^2 h \\ \text{curved surface area} &= \pi rl \\ \text{total surface area} &= \pi rl + \pi r^2\end{aligned}$$



Cylinder

$$\begin{aligned}\text{volume} &= \pi r^2 h \\ \text{curved surface area} &= 2\pi rh \\ \text{total surface area} &= 2\pi rh + 2\pi r^2 \\ &= 2\pi r(h + r)\end{aligned}$$



Sphere

$$\begin{aligned}\text{volume} &= \frac{4}{3}\pi r^3 \\ \text{surface area} &= 4\pi r^2\end{aligned}$$



Fig. 25.13

Example 6

An open rectangular box (Fig. 25.14, overleaf) measures externally 32 cm long, 27 cm wide and 15 cm deep. If the box is made of wood 1 cm thick, what volume of wood is used?

The internal dimensions are 30 cm, 25 cm and 14 cm respectively.

$$\begin{aligned}\text{External volume} &= 32 \times 27 \times 15 \text{ cm}^3 \\ &= 12\,960 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Internal volume} &= 30 \times 25 \times 14 \text{ cm}^3 \\ &= 10\,500 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\text{Volume of wood} &= 12\,960 \text{ cm}^3 - 10\,500 \text{ cm}^3 \\ &= 2\,460 \text{ cm}^3\end{aligned}$$

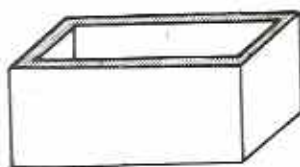


Fig. 25.14

Example 7

How many litres of oil does a cylindrical drum 28 cm in diameter and 50 cm deep hold?

$$\text{Volume of drum} = \pi r^2 h = \frac{22}{7} \times 14^2 \times 50 \text{ cm}^3$$

$$\begin{aligned}\text{Capacity of drum} &= \frac{22 \times 14^2 \times 50}{7 \times 1000} \text{ litres} \\ &= 30.8 \text{ litres}\end{aligned}$$

Example 8

A cylindrical metal bar 50 cm long and 6 cm in diameter is pulled out to form a wire of diameter 3 mm.

(a) What length is the wire? (b) How does the curved surface area of the wire compare with that of the bar?

(a) Let the length of the wire be x cm.

$$\text{Volume of wire} = \pi \left(\frac{3}{20} \right)^2 \times x \text{ cm}^3$$

$$\text{Volume of bar} = \pi \times 3^2 \times 50 \text{ cm}^3$$

$$\pi \frac{9x}{400} = \pi \times 9 \times 50$$

$$\Leftrightarrow x = \frac{9\pi \times 50 \times 400}{9\pi} = 20\,000$$

$$\text{length of wire} = 20\,000 \text{ cm} = 200 \text{ m}$$

(b) Considering curved surfaces only:

$$\text{area of bar} = 2\pi r \times 50 \text{ cm}^2 = 300\pi \text{ cm}^2$$

$$\text{area of wire} = 2\pi \frac{3}{20} \times 20\,000 \text{ cm}^2$$

$$\begin{aligned}&= 6\,000\pi \text{ cm}^2 = 20 \times 300\pi \text{ cm}^2 \\ &= 20 \times \text{area of bar}\end{aligned}$$

Notice in Example 8 that it was not necessary to use a numerical value for π .

Example 9

Find the capacity in litres of a bucket 24 cm in diameter at the top, 16 cm in diameter at the bottom and 18 cm deep.

The bucket is in the shape of a frustum of a cone. It is necessary to consider the whole cone. Complete the cone as in Fig. 25.15 and let the depth of the extension be x cm.

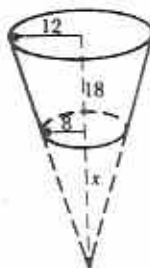


Fig. 25.15

In Fig. 25.15,

$$\frac{8}{x} = \frac{12}{x+18}$$

(similar Δ s)

$$8x + 144 = 12x$$

$$\Leftrightarrow 4x = 144$$

$$\Leftrightarrow x = 36$$

Volume of frustum

$$\begin{aligned}&= \frac{1}{3}\pi 12^2 \times 54 \text{ cm}^3 - \frac{1}{3}\pi 8^2 \times 36 \text{ cm}^3 \\ &= \frac{1}{3}\pi 4^2 \times 18(3^2 \times 3 - 2^2 \times 2) \text{ cm}^3 \\ &= \pi \times 16 \times 6 \times 19 \text{ cm}^3 \approx 5\,730 \text{ cm}^3\end{aligned}$$

$$\text{Capacity of bucket} \approx 5.73 \text{ litres}$$

Example 10

A solid is made up of a cylinder with a hemisphere on top as in Fig. 25.16. Calculate (a) the surface area (b) the volume of the solid.

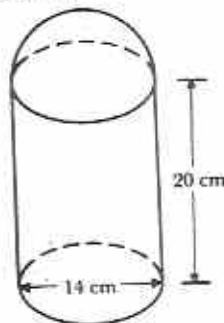


Fig. 25.16

$$\begin{aligned}
 \text{(a) Total surface area} &= \pi r^2 + 2\pi rh + \frac{1}{2}(4\pi r^2) \\
 &= \pi 7^2 + 2\pi \times 7 \times 20 + 2\pi \times 7^2 \text{ cm}^2 \\
 &= 7\pi(7 + 40 + 14) \text{ cm}^2 \\
 &= 7 \times \frac{22}{7} \times 61 \text{ cm}^2 \\
 &= 1342 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) Volume} &= \pi r^2 h + \frac{1}{2}(\frac{4}{3}\pi r^3) \\
 &= \pi \times 7^2 \times 20 + \frac{2}{3}\pi \times 7^3 \text{ cm}^3 \\
 &= 49\pi(20 + \frac{2}{3} \times 7) \text{ cm}^3 \\
 &= 49 \times \frac{22}{7} \times \frac{74}{3} \text{ cm}^3 \\
 &\approx 3800 \text{ cm}^3
 \end{aligned}$$

Exercise 25c

Use the value $\frac{22}{7}$ or 3,142 for π as appropriate.

- 1 Water in a 14 mm diameter pipe flows at 2 m/s. How many litres flow along the pipe in 1 min?
- 2 What is the mass in kg of a cylindrical metal bar 35 cm long and 3 cm in diameter if 1 cm^3 of the metal has a mass of 8 g?
- 3 154 litres of oil are poured into a cylindrical barrel of diameter 35 cm. To what depth is the drum filled?
- 4 Fig. 25.17 shows a casting with dimensions given in mm. What is the mass of the casting in kg if it is made of iron of density 7.2 g/cm^3 ?

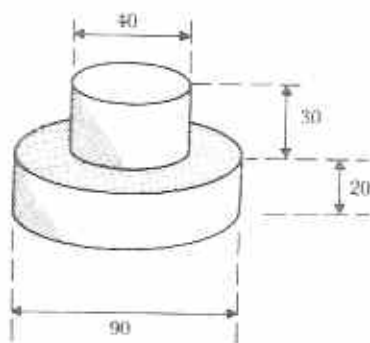


Fig. 25.17

- 5 Calculate the surface area of the casting in Fig. 25.17 in cm^2 .
- 6 6.6 mm of rain fall onto a rectangular roof 8 m long and 6 m wide. The rainwater drains into a cylindrical barrel of diameter 60 cm. How far does the water level rise in the barrel?

- 7 An open rectangular box made of wood 1.5 cm thick measures externally 63 cm long, 48 cm wide and 50 cm deep. Calculate (a) the volume of wood in the box, (b) the mass of the box if the density of the wood is 0.8 g/cm^3 .
- 8 The most economical shape for a cylindrical container is one in which the height and diameter are equal. Find the capacity in litres to 2 s.f. of such a tin which is 10 cm high.
- 9 A cylindrical tin 8 cm in diameter contains water to a depth of 4 cm. If a cylindrical wooden rod 4 cm in diameter and 6 cm long is placed in the tin, it floats exactly half submerged. What is the new depth of water?
- 10 Calculate (a) the slant height, (b) the curved surface area, (c) the volume, of a cone of height 8 cm and base diameter 12 cm. Leave the answers in terms of π .
- 11 If the cone in question 10 is made of paper which is cut and opened out into a sector of a circle, what is the angle of the sector?
- 12 A lampshade is in the shape of an open frustum of a cone. Its top and bottom diameters are 10 and 20 cm and its height is 12 cm. Find, in terms of π , the area of material required for the curved surface of the shade.
- 13 A pile of sand is in the form of a cone 20 m in diameter at the bottom and 6 m high. What is the mass of the sand in tonnes if 1 m^3 has a mass of 2.5 tonnes?
- 14 A lead ball of diameter 6 cm is melted down and cast in balls 5 mm in diameter. How many of the smaller balls are there?
- 15 A heavy 9 cm ball is placed in an empty cylindrical tin of diameter 12 cm. Enough water is poured into the tin to cover the ball. If the ball is then removed, how far does the water-level fall?
- 16 A pyramid 7 cm high stands on a base 12 cm square. Calculate the volume of the pyramid.
- 17 In Fig. 25.18, overleaf, the diameters and heights of the solids are all equal. (a) What fraction of the volume of the cylinder is (i) the sphere, (ii) the cone? (b) What is the ratio of the curved surface area of the sphere to that of the cylinder?

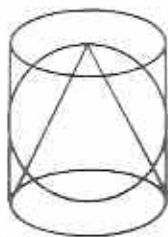


Fig. 25.18

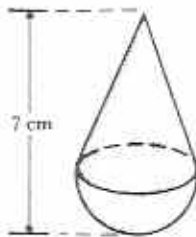


Fig. 25.19

- 18** A solid consists of a cone attached to a hemisphere as in Fig. 25.19. Calculate the volume of the solid if the diameter of the hemisphere is 3 cm and the overall height of the object is 7 cm.
- 19** A spherical retort 15 cm in diameter is half full of acid. The acid is poured into a tall cylindrical beaker of diameter 6 cm. How deep is the acid in the beaker?
- 20** A hollow metal sphere is made of metal 4 mm thick and has an external diameter of 12 cm. If the density of the metal is 8.8 g/cm^3 , what is the mass of the sphere?

Areas and volumes of similar shapes

If two similar shapes have corresponding lengths in the ratio $a:b$,

- the ratio of their areas is $a^2:b^2$,
- the ratio of their volumes is $a^3:b^3$.

Example 11

A board 1 m long and 80 cm wide costs \$4,80. What would be the cost of a similarly shaped board 75 cm long?

$$\text{Ratio of corresponding lengths} = \frac{75 \text{ cm}}{100 \text{ cm}} = \frac{3}{4}$$

Ratio of costs

$$= \text{ratio of corresponding areas} = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

$$\text{Cost of smaller piece} = \frac{9}{16} \text{ of } \$4,80 = \$2,70$$

Example 12

Two similarly shaped cans hold 2 litres and 6,75 litres respectively. If the smaller can is 16 cm in diameter, what is the diameter of the larger?

$$\text{Ratio of volumes} = \frac{6,75 \text{ litres}}{2 \text{ litres}} = \frac{27}{8} = \left(\frac{3}{2}\right)^3$$

$$\text{Ratio of corresponding lengths} = \frac{3}{2}$$

$$\text{Diameter of larger can} = \frac{3}{2} \text{ of } 16 \text{ cm} = 24 \text{ cm}$$

Exercise 25d

- A photograph is 20 cm long and 15 cm wide. The length of a small print of the photograph is 4 cm. Find (a) the width of the smaller print, (b) ratio of the areas of the two photographs.
- 8 kg of fertiliser are needed for a garden. How much fertiliser would be needed for a garden of double the linear dimensions?
- 1 kg of grass seed is needed for a rectangular plot 25 m long. How much seed would be needed for a similar plot which is 100 m long?
- The diameter of a sphere is 3 times that of another sphere. How many times greater is its surface area?
- A box of height 8 cm has a volume of 320 cm^3 . What is the volume of a similar box of height 6 cm?
- Two balls have diameters of 10 cm and 6 cm. Find (a) the ratio of their diameters in its simplest form, (b) the ratio of their surface areas, (c) the ratio of their volumes.
- A cylinder is 8 cm high and its base diameter is 4 cm. The height of a similar cylinder is 12 cm. (a) Find the diameter of the base of the larger cylinder. (b) What is the ratio of the volume of the larger cylinder to that of the smaller one?
- The area of a lake is 18 km^2 . It is represented by an area of 2 cm^2 on the map. (a) What area in km^2 is represented by 1 cm^2 on the map? (b) What length does 1 cm on the map represent? (c) What is the ratio of lengths on the map to actual lengths?
- A school has an area of $3\,025 \text{ m}^2$ and it is represented on a plan by an area of 144 cm^2 . Find the actual length of a wall which is shown on the plan by a line 8,4 cm long.
- Two similar boxes have volumes of 250 cm^3 and 54 cm^3 . What is the ratio of (a) their heights, (b) their surface areas?

Solution of triangles

Syllabus reference 6.8.1

Solving right-angled triangles

Pythagoras' theorem

In a right-angled triangle the square on the hypotenuse is equal to the sum of the squares on the other two sides.

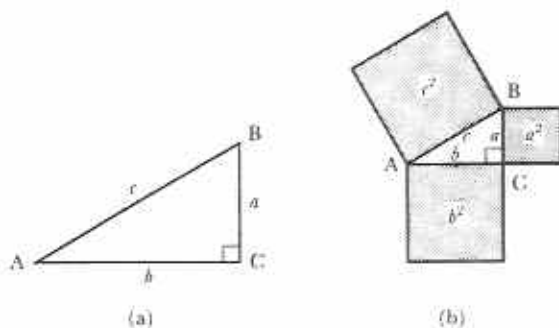


Fig. 26.1

In Fig. 26.1,

$$AB^2 = BC^2 + AC^2$$

$$\text{or } c^2 = a^2 + b^2$$

Notice also that

$$a^2 = c^2 - b^2$$

$$\text{and } b^2 = c^2 - a^2$$

Fig. 26.1(b) gives a geometrical interpretation of Pythagoras' theorem.

Example 1

In Fig. 26.2 if $AC = 12$ cm, $BC = 5$ cm, $CD = 11$ cm, find AD .

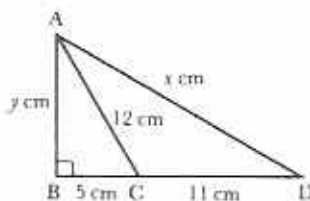


Fig. 26.2

Let $AD = x$ cm and $AB = y$ cm.

In $\triangle ABC$,

$$y^2 = 12^2 - 5^2 = 144 - 25 = 119$$

In $\triangle ABD$,

$$x^2 = y^2 + 16^2 = 119 + 256 = 375$$

$$\therefore x = \sqrt{375} \approx 19.4$$

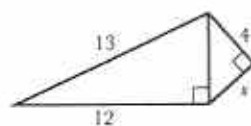
$$AD = 19.4 \text{ cm to 3 s.f.}$$

Notice that y represented an *intermediate* length. When y^2 was found to be 119 there was no need to find the value of y , since it was the value of y^2 that was needed in the subsequent working.

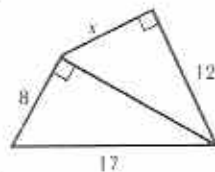
Exercise 26a

1 The dimensions in Fig. 26.3 are all cm. In each case x is a whole number of cm. Find the value of x in each part.

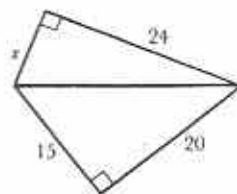
(a)



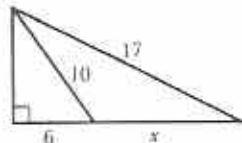
(b)



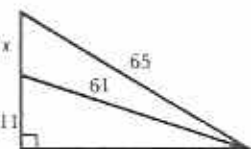
(c)



(d)



(e)



(f)

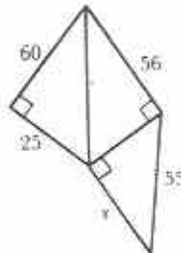


Fig. 26.3

- 2 A radio-mast 36 m high is supported by straight wires attached to its top and to points on the level ground 12 m from its base. Calculate the length of each wire.
- 3 A ladder 8.5 m long leans against a vertical wall so that its upper end is 7.5 m from the ground. How far is the foot of the ladder from the wall?
- 4 One side of a right-angled triangle is 24 cm long and its hypotenuse is 25 cm long. Calculate (a) the length of the third side of the triangle, (b) the area of the triangle.
- 5 A cone has a slant height of 29 cm and a circular base of diameter 42 cm. Calculate the vertical height of the cone.
- 6 In each part of Fig. 26.4, calculate the value of x correct to 3 s.f.

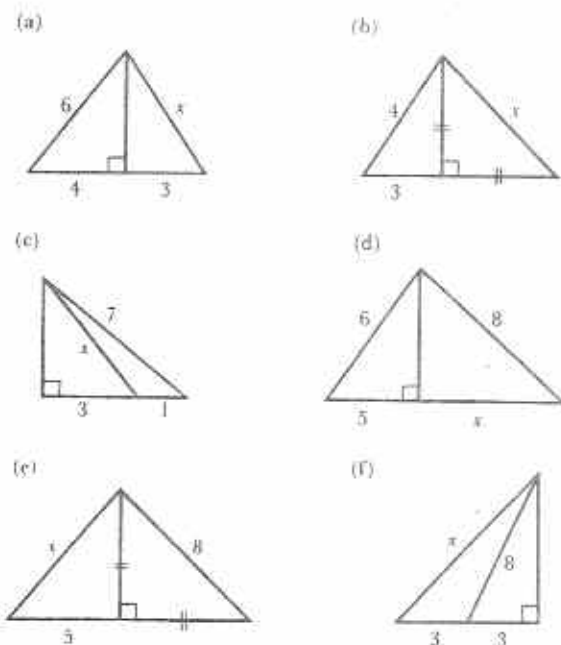


Fig. 26.4

- 7 A chord of a circle is 10 cm from the centre of the circle. Calculate the length of the chord given that the radius of the circle is 31 cm.
- 8 A rectangle is 4.3 cm long and the length of each diagonal is 5.1 cm. Calculate (a) the width, (b) the area of the rectangle, giving both answers correct to 3 s.f.

- 9 In $\triangle ABC$, $AC = 2$ m, $BC = 3$ m and C is obtuse. The perpendicular from A to BC produced is AD . If $CD = 1$ m, calculate AB .
- 10 $ABCD$ is a rectangle in which $AB = 2.8$ cm and $AD = 3.3$ cm. E is a point on DC produced such that $\triangle AED$ and rectangle $ABCD$ are equal in area. Calculate (a) DE , (b) AE .

Sine, cosine, tangent

The trigonometrical ratios sine, cosine and tangent are defined in terms of the hypotenuse, opposite and adjacent sides of a right-angled triangle as follows:

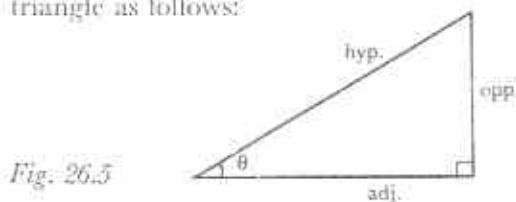


Fig. 26.5

$$\sin \theta = \frac{\text{opp.}}{\text{hyp.}}$$

$$\cos \theta = \frac{\text{adj.}}{\text{hyp.}}$$

$$\tan \theta = \frac{\text{opp.}}{\text{adj.}}$$

Example 2

Calculate angle x in Fig. 26.6.

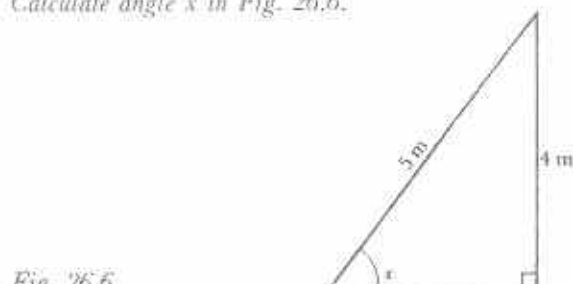


Fig. 26.6

$$\sin x = \frac{4}{5} = 0.8000$$

$$x = 53^\circ 8' \text{ or } 53.13^\circ \quad (\text{from sine tables})$$

Example 3

Calculate the lengths of the sides marked a and b in Fig. 26.7.

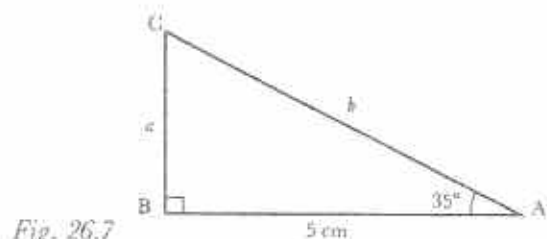


Fig. 26.7

$$\tan 35^\circ = \frac{a}{5}$$

$$a = 5 \times \tan 35^\circ \\ = 5 \times 0,7002 = 3,501$$

$$\cos 35^\circ = \frac{5}{b}$$

$$b = \frac{5}{\cos 35^\circ} \\ = 6,103$$

working:

No	Log
5	0,6990
$\cos 35^\circ$	1,9134
6,103	0,7856

On a scientific calculator:

Key

Display

35 cos 1/x × 5 =

6,1038729

$$b = 6,1$$



Note the use of the reciprocal key $1/x$ on the calculator. This saved the need to use the memory key.

The sides are 3,5 cm and 6,1 cm long (2 s.f.).

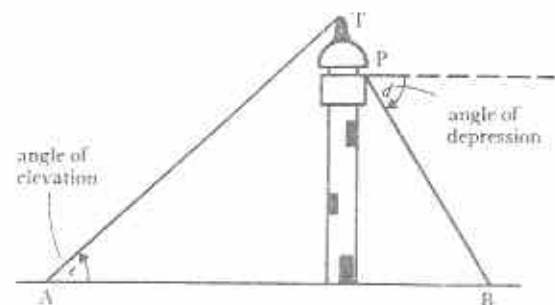


Fig. 26.8

Fig. 26.8 shows the **angle of elevation**, e , of the top of a tower, T, from a point A below.

Fig. 26.8 also shows the **angle of depression**, d , of a point B on the ground from a point, P, on the tower.

Example 4

From a window 10 m above level ground, the angle of depression of an object on the ground is $25,4^\circ$. Calculate the distance of the object from the foot of the building.

Note that the angle of depression is the angle between the horizontal and the line joining the window and the object (Fig. 26.9).

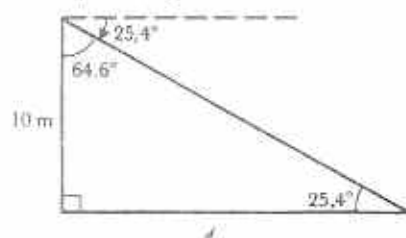


Fig. 26.9

$$\text{Either, } \tan 25,4^\circ = \frac{10}{d}$$

$$d = \frac{10}{\tan 25,4^\circ} = \frac{10}{0,4748}$$

$$= 21,06 \quad (\text{from recip. tables})$$

or, using the **complement** of $25,4^\circ$,

$$\tan 64,6^\circ = \frac{d}{10}$$

$$d = 10 \times \tan 64,6^\circ \\ = 10 \times 2,106 \\ = 21,06$$

The object is 21,1 m from the foot of the building (3 s.f.).

Angles of 45° , 60° , 30° , 0° , 90°

From Fig. 26.10,

$$\sin 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\cos 45^\circ = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$\tan 45^\circ = \frac{1}{1} = 1$$

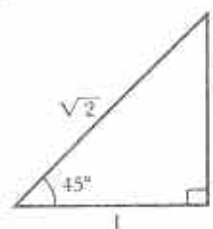


Fig. 26.10

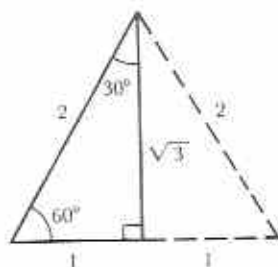


Fig. 26.11

From Fig. 26.11,

$$\sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 60^\circ = \frac{\sqrt{3}}{1} = \sqrt{3}$$

$$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

$$\sin 0^\circ = 0$$

$$\sin 90^\circ = 1$$

$$\cos 0^\circ = 1$$

$$\cos 90^\circ = 0$$

$$\tan 0^\circ = 0$$

$$\tan 90^\circ \text{ is undefined}$$

Example 5

Given Fig. 26.12, calculate RQ and RS.

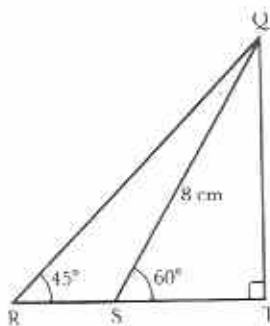


Fig. 26.12

In $\triangle QST$, the sides are in the ratio $1:2:\sqrt{3}$ (a $30^\circ, 60^\circ, 90^\circ$ $\triangle$).

$$ST = \frac{1}{2} \times SQ = \frac{1}{2} \times 8 \text{ cm} = 4 \text{ cm}$$

$$\text{and } TQ = \sqrt{3} \times ST = 4\sqrt{3} \text{ cm.}$$

In $\triangle QRT$, the sides are in the ratio $1:1:\sqrt{2}$ (a $45^\circ, 45^\circ, 90^\circ$ $\triangle$).

$$\begin{aligned} RQ &= \sqrt{2} \times TQ = \sqrt{2} \times 4\sqrt{3} \text{ cm} \\ &= 4\sqrt{6} \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{and } RS &= RT - ST \\ &= TQ - ST \\ &= 4\sqrt{3} \text{ cm} - 4 \text{ cm} \\ &= 4(\sqrt{3} - 1) \text{ cm} \end{aligned}$$

Exercise 26b

- 1 A plane takes off at an angle of $5^\circ 10'$ to the ground. How high is it when it has moved 2000 m horizontally from its take-off point?
- 2 The angle of elevation of the top of a vertical mast from a point on level ground 240 m from its foot is 31.5° . How high is the mast?
- 3 A town B is due north of A. A third town C is 10 km on a bearing 020° from A. If B is on a bearing of 290° from C, calculate (a) BC, (b) AB.
- 4 Calculate the values of a , b and c in Fig. 26.13.

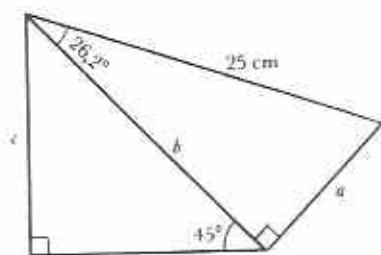
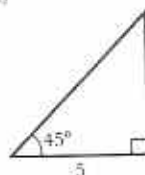


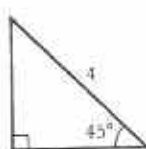
Fig. 26.13

- 5 A chord of a circle is 5 cm long and subtends an angle of 24.3° in the major segment. Calculate (a) the perpendicular distance of the chord from the centre, (b) the radius of the circle.
- 6 In each part of Fig. 26.14 the length of one side of a triangle is given in cm. Find the lengths of the other two sides, giving the answers in surd form with rational denominators where necessary.

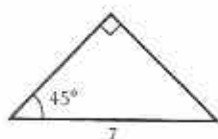
(a)



(b)



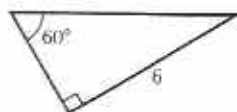
(c)



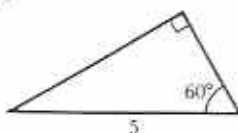
(d)



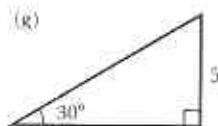
(e)



(f)



(g)



(h)

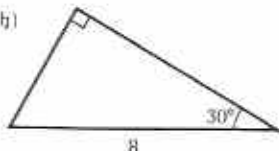
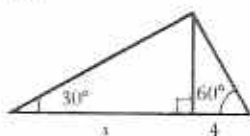


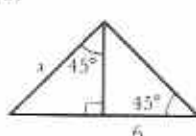
Fig. 26.14

- 7 In Fig. 26.15 the given lengths are in cm. In each part, find the length marked x , giving the answers in surd form with rational denominators where necessary.

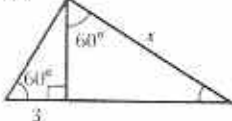
(a)



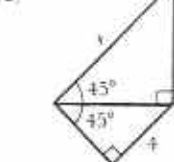
(b)



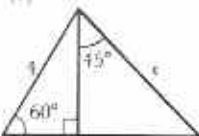
(c)



(d)



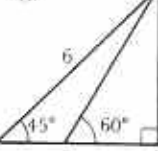
(e)



(f)



(g)



(h)

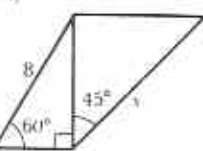


Fig. 26.15

- 8 The angle of elevation of the top of a flagpole from a point on level ground is 30° . From another point on the ground, 20 m nearer the flagpole, the angle of elevation is 60° . Calculate the height of the flagpole.
- 9 A tripod consists of three legs each 1.05 m long. The height of the top of the tripod above the ground is 90 cm. What is the inclination of each leg to the horizontal?
- 10 A man walks 11 km due north from A to B. He then walks 6.5 km due east from B to C. Calculate (a) the bearing of C from A, (b) AC.
- 11 A plank rests with one end on the ground and the other end on the back of a lorry 1.2 m above the ground. How long is the plank if it is inclined at 21° to the horizontal?
- 12 Farai walks 5 km from A to B on a bearing of 035° . She then walks 6 km from B to C on a bearing of 125° . Calculate (a) the distance, (b) the bearing of C from A.
- 13 A girl 160 cm tall stands 150 m from the foot of a building. She finds that the angle of elevation of the top of the building is 17° . Calculate the height of the building to the nearest $\frac{1}{2}$ -metre.
- 14 The angle of depression of a point on the 225 m contour line is 10.2° from the top of a hill 915 m high. Calculate the horizontal distance between the two points. Find the difference between this distance and the actual distance between the two points.
- 15 A point Y is 400 m north-west of X. A tree is north-east of Y and on a bearing 015° from X. Find the distance of the tree from (a) X, (b) Y.

Lengths and angles in solids

Revise Chapter 6, pages 41 to 52.

Example 6

A pyramid with a vertex O and edges OA, OB, OC, OD each 10 cm long stands on a square base ABCD of side 8 cm. Calculate (a) the height OP of the

pyramid, (b) the angle between an edge and the base,
(c) the angle between a sloping face and the base.

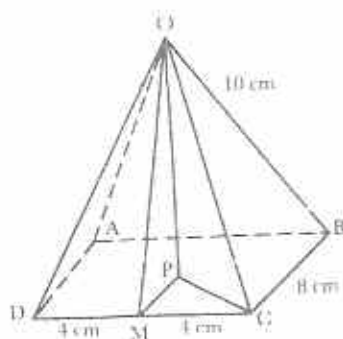


Fig. 26.16

In Fig. 26.16, M is the mid-point of the edge DC.

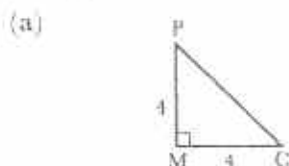


Fig. 26.17

In $\triangle PMC$ (Fig. 26.17),
 $PC^2 = 4^2 + 4^2$
 $= 16 + 16 = 32$

(Pythagoras)

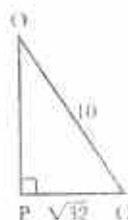


Fig. 26.18

In $\triangle OPC$ (Fig. 26.18),
 $OP^2 = OC^2 - PC^2$
 $= 100 - 32 = 68$

(Pythagoras)

$$OP = \sqrt{68} \text{ cm} = 8,246 \text{ cm}$$

(b) $\angle OCP$ is the angle between an edge OC and the base ABCD.

$$\text{In } \triangle OPC, \cos \angle OCP = \frac{\sqrt{32}}{10} = 0,5657$$

$$\angle OCP = 55,55^\circ$$

(c) $\angle OMP$ is the angle between the sloping face ODC and the base. Notice that OM and MP are both perpendicular to the common edge DC.



Fig. 26.19

In $\triangle OMP$ (Fig. 26.19),

$$\tan \angle OMP = \frac{8,246}{4} = 2,0615$$

$$\angle OMP = 64,1^\circ$$

Example 7

An area of sloping ground is 16 m wide, 12 m long and slopes at 25° to the horizontal as in Fig. 26.20. Find the angle of slope of the diagonal BD.

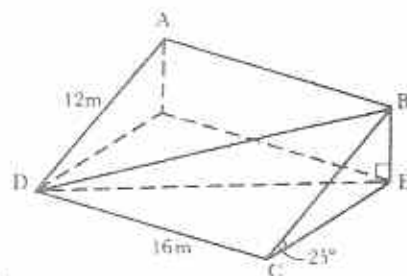


Fig. 26.20

In $\triangle BDE$,

$$BE = 12 \sin 25^\circ$$

(from $\triangle BCE$)

$$BD^2 = 12^2 + 16^2$$

(Pythagoras in $\triangle BCD$)

$$= 400$$

$$BD = 20 \text{ m}$$

$$\sin \angle BDE = \frac{BE}{BD} = \frac{12 \sin 25^\circ}{20}$$

$$= 0,6 \times 0,4226 = 0,2536$$

$$\angle BDE = 14^\circ 41'$$

Notice that Examples 6 and 7 were answered by solving the appropriate right-angled triangle. Example 6 shows the value of sketching the various triangles used.

Exercise 26c

- 1 A cube has edges of 4 cm length. Calculate
(a) the length of a diagonal of the cube,
(b) the angle between the diagonal of the cube and its base.
- 2 The vertex of a pyramid on a square base of side 12 cm is 7 cm above the base. Calculate
(a) the length of each sloping edge, (b) the angle between each sloping edge and the base, (c) the angle between each sloping face and the base.
- 3 A triangular prism like that of Fig. 26.20 has $BE = 4$ cm, $AB = 20$ cm and $AD = 16$ cm. Make a suitable sketch and calculate the slope of (a) BC , (b) BD .
- 4 A room in the shape of a cuboid has a floor which measures 5 m by 4 m. The longest diagonal of the room makes an angle of 35° with the floor. Find the height of the room.
- 5 A pyramid $OABCD$ stands on a square base $ABCD$ and $OA = OB = OC = OD = 16$ cm. $\angle BOD = 90^\circ$. Calculate the angle between a sloping face and the base.
- 6 A prism like that of Fig. 26.20 has $AB = 24$ cm, $AD = 7$ cm and $BE = 3$ cm. X is a point on AB such that $\angle ADX = 45^\circ$. Calculate the slope of BD and of DX .
- 7 A pole is resting in the corner of a room. The top of the pole is 2.5 m above the floor and the bottom is 1 m from each wall. Find the length of the pole and the angle which it makes with the floor.
- 8 A mast 45 m high is supported by 4 equal straight wires attached to the top of the mast and to the corners of a square of side 60 m on the level ground. Calculate the inclination of each wire to the horizontal.
- 9 Fig. 26.21 represents an open door. Assume that $ABCD$ and $ABEF$ are rectangles each measuring 2 m by 1.5 m.

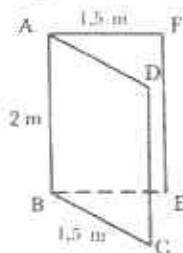


Fig. 26.21

If $\angle DAF = 60^\circ$ calculate the slopes of (a) DE , (b) $\triangle ACE$.

- 10 Fig. 26.22 shows an open rectangular box 7 cm long, 6 cm wide and 6 cm high. A rod 12 cm long rests with its lower end in one bottom corner and is supported by the opposite corner.

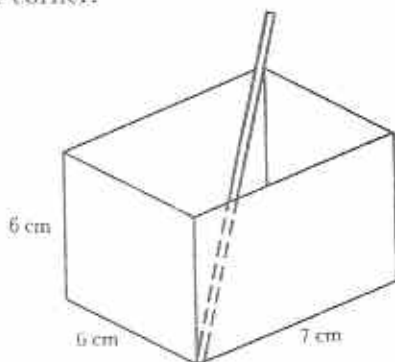


Fig. 26.22

Calculate (a) the inclination of the rod to the horizontal and (b) the height of its top end above the level of the base of the box.

Solving non-right-angled triangles

The sine rule

Revise Chapter 4, pages 25 to 31.

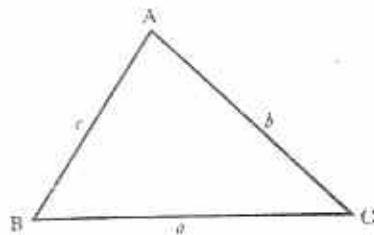


Fig. 26.23

The sine rule states that

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

or

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

for any triangle ABC in which A, B, C are the

angles of the triangle and a, b, c are the lengths of the sides opposite these angles.

In obtuse-angled triangles,
 $\sin \theta = \sin (180^\circ - \theta)$.

Example 8

Solve completely the $\triangle ABC$ in which $a = 12,4$ cm, $c = 14,7$ cm and $C = 72^\circ 4'$.

To 'solve completely' means to find all the unknown lengths and angles.

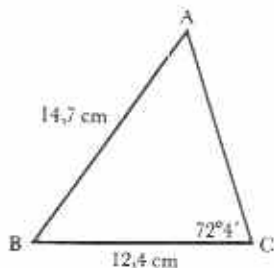


Fig. 26.24

In Fig. 26.24,

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

$$\frac{\sin A}{12,4} = \frac{\sin 72^\circ 4'}{14,7}$$

$$\Rightarrow \sin A = \frac{12,4 \sin 72^\circ 4'}{14,7}$$

$$A = 53^\circ 23' \text{ or } 126^\circ 37'$$

But $a < c$, therefore $A < C$.

$$A = 53^\circ 23'$$

and $B = 54^\circ 33'$ ($\angle$ s of $\triangle$)

$$\frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{b}{\sin 54^\circ 33'} = \frac{14,7}{\sin 72^\circ 4'}$$

$$b = \frac{14,7 \sin 54^\circ 33'}{\sin 72^\circ 4'}$$

$$\approx 12,6 \text{ cm}$$

working:

No	Log
12,4	1,0934
$\sin 72^\circ 4'$	1,9784
14,7	1,1673
$\sin 53^\circ 23'$	1,9045

working:

No	Log
14,7	1,1673
$\sin 54^\circ 33'$	1,9110
$\sin 72^\circ 4'$	1,9783
12,59	1,0999

This expression may be evaluated on a scientific calculator as follows:

Key

Display

$$4 \div 60 + 72 = \sin M+$$

$$0.951415$$

$$33 \div 60 + 54 = \sin$$

$$0.81462$$

$$\times 14.7 =$$

$$11.9749$$

$$= MR =$$

$$12.5864$$

$$b = 12,6 \text{ cm to 3 s.f.}$$



$$\text{Hence } A = 53^\circ 23', B = 54^\circ 33' \text{ and } b = 12,6 \text{ cm}$$

The cosine rule

Revise Chapter 11, pages 88 to 94.

The cosine rule is used for solving triangles in which two sides and the included angle are given.

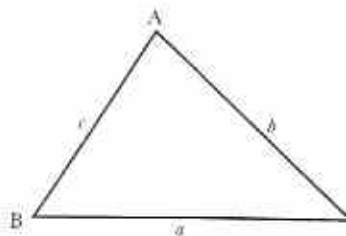


Fig. 26.25

In $\triangle ABC$,

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

These formulae are true for both acute and obtuse angles. In obtuse-angled triangles $\cos \theta = -\cos (180^\circ - \theta)$.

Example 9

Find x in Fig. 26.26.

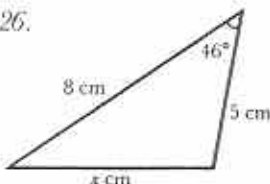


Fig. 26.26

$$\begin{aligned} x^2 &= 8^2 + 5^2 - 2 \times 8 \times 5 \times \cos 46^\circ \\ &= 64 + 25 - 80 \times 0,6947 \\ &= 89 - 55,576 = 33,424 \end{aligned}$$

$$x = \sqrt{33,42} = 5,781 \approx 5,78 \text{ cm}$$

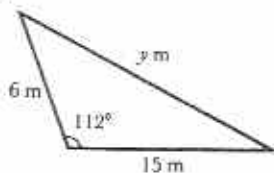
Example 10Find y in Fig. 26.27.

Fig. 26.27

$$\begin{aligned}
 y^2 &= 6^2 + 15^2 - 2 \times 6 \times 15 \times \cos 112^\circ \\
 &= 36 + 225 - 180 \times (-\cos 68^\circ) \\
 &= 261 + 180 \times 0,3746 \\
 &= 261 + 67,428 = 328,428
 \end{aligned}$$

$$y = \sqrt{328,4} = 18,12 \approx 18,1 \text{ m}$$

When all three sides of a triangle are known, the angles can be calculated by rearranging the basic formulae as follows:

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Example 11

Calculate the angles of a triangle which has sides 5 cm, 8 cm and 11 cm.

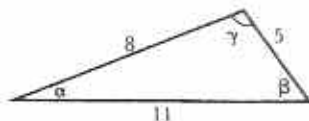


Fig. 26.28

Lettering the angles of the $\triangle$ as in Fig. 26.28,

$$\begin{aligned}
 \cos \alpha &= \frac{8^2 + 11^2 - 5^2}{2 \times 8 \times 11} \\
 &= \frac{160}{16 \times 11} = \frac{10}{11} = 0,909 \ 1
 \end{aligned}$$

$$\alpha = 24,6^\circ$$

$$\begin{aligned}
 \cos \beta &= \frac{5^2 + 11^2 - 8^2}{2 \times 5 \times 11} \\
 &= \frac{82}{110} = 0,745 \ 5
 \end{aligned}$$

$$\beta = 41,8^\circ$$

$$\cos \gamma = \frac{8^2 + 5^2 - 11^2}{2 \times 8 \times 5} = \frac{-32}{80} = -0,4$$

$$\therefore \gamma = 180^\circ - 66,4^\circ = 113,6^\circ$$

$$\text{Check: } \alpha + \beta + \gamma = 180^\circ$$

Examples which show how to use a scientific calculator with the sine and cosine rules are given in Chapter 20, pages 166 to 172.

Example 12

A village P is 10 km from a point X on a bearing 025° from X. Another village, Q, is 6 km from X on a bearing of 162° . Calculate the distance and bearing of P from Q.

Enter the details on a sketch such as Fig. 26.29.

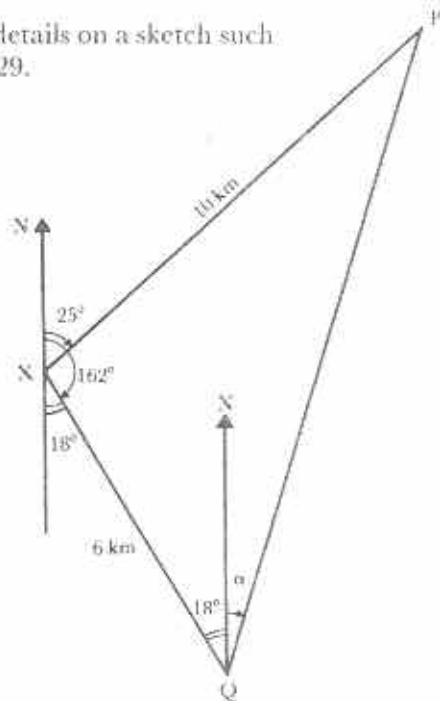


Fig. 26.29

$$\angle PXQ = 162^\circ - 25^\circ = 137^\circ$$

$$\begin{aligned}
 PQ^2 &= 10^2 + 6^2 - 2 \times 10 \times 6 \times \cos 137^\circ \\
 &= 100 + 36 - 120 \times (-\cos 43^\circ) \\
 &= 136 + 120 \times 0,731 \ 4 \\
 &= 136 + 87,768 = 223,768
 \end{aligned}$$

$$\begin{aligned}
 PQ &= \sqrt{223,8} \text{ km} = 14,96 \text{ km} \\
 &= 15,0 \text{ km to 3 s.f.}
 \end{aligned}$$

$$\frac{\sin Q}{10} = \frac{\sin 137^\circ}{14,96}$$

$$\sin Q = \frac{10 \sin 43^\circ}{14.96}^*$$

$$\hat{Q} = 27^\circ 8'$$

From Fig. 26.29,

α = bearing of P from Q

$$= 27^\circ 8' - 18^\circ$$

$$= 009^\circ 8'$$

working:

No	Log
10	1.0000
$\sin 43^\circ$	1.8338
	0.8338
14.96	1.1749
$\sin 27^\circ 8'$	1.6529

* This calculation may be done on a scientific calculator as follows.

Key

Display

43 $\sin \div 1.496 = \text{SHIFT} \sin^{-1}$ 27.12165

Exercise 26d

- In $\triangle ABC$, $\hat{B} = 38^\circ$, $\hat{C} = 48^\circ$, $c = 18.8$ cm. Find b .
- In $\triangle ABC$, $\hat{A} = 98^\circ$, $\hat{C} = 36^\circ$, $a = 34.4$ cm. Find c .
- In $\triangle ABC$, $\hat{B} = 29^\circ$, $b = 8.6$ cm, $c = 3.1$ cm. Find $\hat{C}$.
- In $\triangle ABC$, $\hat{A} = 96^\circ 13'$, $a = 39.4$ cm, $b = 11.2$ cm. Find $\hat{B}$.
- In $\triangle ABC$, $\hat{A} = 60^\circ$, $b = 5$ cm, $c = 3$ cm. Find a .
- In $\triangle ABC$, $\hat{B} = 22^\circ$, $a = 4$ cm, $c = 5$ cm. Find b .
- In $\triangle ABC$, $\hat{C} = 123^\circ$, $a = 3$ m, $b = 2$ m. Find c .
- In $\triangle ABC$, $\hat{B} = 143^\circ$, $a = 25$ cm, $c = 40$ cm. Find b .
- In $\triangle ABC$, $a = 5$ m, $b = 6$ m, $c = 7$ m. Find $\hat{B}$.
- In $\triangle ABC$, $a:b:c = 15:21:24$. Find $\hat{B}$.
- In $\triangle ABC$, $a = 8$ km, $b = 14$ km, $c = 18$ km. Find $\hat{C}$.
- In $\triangle ABC$, $a = 9$ m, $b = 6$ m, $c = 4.8$ m. Find $\hat{A}$.
- A boy starts from a point X and walks 220 m on a bearing 063° . He then walks to a point Y on a bearing 156° . If Y is due east of X, calculate XY.
- Three villages X, Y and Z are connected by straight level roads. XY = 5 km,

YZ = 4 km and $\hat{XYZ} = 160^\circ$. What distance is saved by walking from X to Z direct instead of through Y?

- 15 In Fig. 26.30 the lengths are in cm. Calculate d and α .

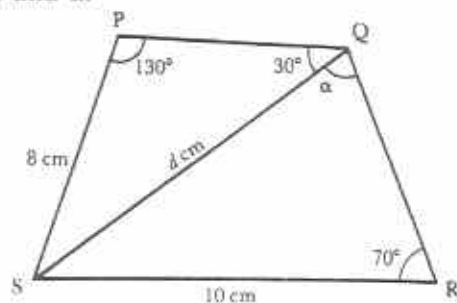


Fig. 26.30

- 16 In Fig. 26.31 the lengths are in m. Find x and θ .

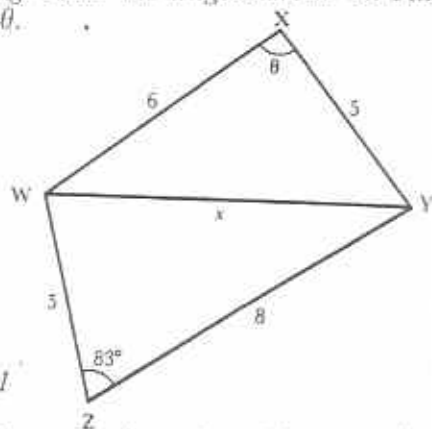


Fig. 26.31

- In the quadrilateral ABCD, $AD \parallel BC$, $AB = 15$ cm, $DC = 14$ cm, $\hat{DAB} = 60^\circ$ and $\hat{ADB} = 40^\circ$. Calculate (a) BD, to 2 s.f. (b) $\hat{BCD}$, correct to the nearest degree.
- ABCD is a cyclic quadrilateral in which $AB = 5$ cm, $BC = 4$ cm, $CD = 7$ cm, $DA = 6$ cm. Calculate $\hat{ABC}$.
- An aeroplane leaves an airport and flies due north for $1\frac{1}{2}$ hours at 500 km/h. It then flies 400 km on a bearing 053° . Calculate its final distance and bearing from the airport.
- Two explorers leave camp at the same time. One walks at 5 km/h on a bearing 039° . The other walks at 7.5 km/h on a bearing 265° . After two hours how far apart are they and what is the bearing of the second from the first?

Matrices, transformations, vectors

Matrices

A **matrix** is a rectangular pattern of elements, usually numbers. A matrix can be used as a store or as an operator.

Addition and subtraction

Matrices can be added and subtracted only if they have the same number of rows and columns. Only corresponding elements in each matrix may be combined.

Example 1

Simplify

$$\begin{pmatrix} 2 & 6 \\ 1 & -4 \end{pmatrix} - \begin{pmatrix} -3 & 2 \\ 0 & -5 \end{pmatrix} + \begin{pmatrix} -1 & 8 \\ -5 & 7 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 6 \\ 1 & -4 \end{pmatrix} - \begin{pmatrix} -3 & 2 \\ 0 & -5 \end{pmatrix} + \begin{pmatrix} -1 & 8 \\ -5 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 2 - (-3) + (-1) & 6 - 2 + 8 \\ 1 - 0 + (-5) & -4 - (-5) + 7 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 12 \\ -4 & 8 \end{pmatrix} = 4 \begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix}$$

In Example 1, notice that 4 is a factor of each element in the resultant matrix. The 4 can be taken out as a **scalar** factor.

Multiplication

Matrices can be multiplied only if the number of columns in the first or **pre-multiplying** matrix is the same as the number of rows in the second or **post-multiplying** matrix. A $p \times q$ matrix will multiply a $q \times r$ matrix to give a $p \times r$ product. In general $\mathbf{AB} \neq \mathbf{BA}$, where $\mathbf{A}$ and $\mathbf{B}$ are matrices.

To multiply matrices, find the sum of the products of corresponding elements in each row

of the pre-multiplying matrix and each column of the post-multiplying matrix. The following examples demonstrate the method of matrix multiplication.

Example 2

product	general case	example
$(1 \times 2) \times (2 \times 1)$	$a \ b \begin{pmatrix} p \\ q \end{pmatrix}$ $= (ap + bq)$	$-2 \ 3 \begin{pmatrix} 5 \\ 1 \end{pmatrix}$ $= (-2 \times 5 + 3 \times 1)$ $= -7$
$(2 \times 1) \times (1 \times 2)$	$\begin{pmatrix} a \\ b \end{pmatrix} p \ q$ $= \begin{pmatrix} ap & aq \\ bp & bq \end{pmatrix}$	$\begin{pmatrix} -4 \\ 7 \end{pmatrix} -1 \ 2$ $= \begin{pmatrix} 4 & -8 \\ -7 & 14 \end{pmatrix}$
$(2 \times 2) \times (2 \times 2)$	$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ $= \begin{pmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{pmatrix}$	$\begin{pmatrix} 5 & 4 \\ 2 & 7 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 3 & 1 \end{pmatrix}$ $= \begin{pmatrix} -7 & 4 \\ 17 & 7 \end{pmatrix}$

Identity and inverse

$\mathbf{I}$ is the 2×2 **identity matrix** where

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Any 2×2 matrix is left unchanged when pre- or post-multiplied by $\mathbf{I}$.

In most cases a matrix $\mathbf{M}$ has an **inverse**, $\mathbf{M}^{-1}$, such that $\mathbf{M} \times \mathbf{M}^{-1} = \mathbf{M}^{-1} \times \mathbf{M} = \mathbf{I}$.

$$\text{If } \mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\text{then } \mathbf{M}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

The cross-product-difference, $ad - bc$, is the **determinant** of the given matrix.

Example 3

Find the inverse of the following where possible.

(a) $\begin{pmatrix} 5 & -2 \\ 9 & -3 \end{pmatrix}$ (b) $\begin{pmatrix} 10 & 3 \\ 7 & 2 \end{pmatrix}$ (c) $\begin{pmatrix} 6 & 9 \\ 4 & 6 \end{pmatrix}$

(a) The determinant of the given matrix
 $= 5 \times (-3) - 9 \times (-2)$
 $= -15 + 18 = 3$

Its inverse is $\frac{1}{3} \begin{pmatrix} -3 & 2 \\ -9 & 5 \end{pmatrix}$.

(b) The determinant of the given matrix
 $= 10 \times 2 - 7 \times 3$
 $= 20 - 21 = -1$

Its inverse is $-1 \begin{pmatrix} 2 & -3 \\ -7 & 10 \end{pmatrix}$
or $\begin{pmatrix} -2 & 3 \\ 7 & -10 \end{pmatrix}$.

(c) The determinant of the given matrix
 $= 6 \times 6 - 9 \times 4$
 $= 36 - 36 = 0$

The inverse of the given matrix would contain the undefined fraction $\frac{1}{0}$. This is an example of a **singular** matrix; such a matrix has no inverse.

Further examples and information on matrices can be found in Chapter 13.

Exercise 27a

1 Find the determinant of the matrix $\begin{pmatrix} -5 & 2 \\ 4 & -3 \end{pmatrix}$. Hence write down the inverse of the matrix.

2 Find the value of x for which the matrix $\begin{pmatrix} 5x-3 & 7 \\ 2x+4 & 2 \end{pmatrix}$ has no inverse.

3 $\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 3 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$.

(a) Find $\mathbf{A} + 2\mathbf{B}$.

(b) Given that $\mathbf{A} \begin{pmatrix} x \\ 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 2y \end{pmatrix}$, find the value of x and y . [Camb]

4 Given that $\mathbf{A} = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & \frac{1}{2} \\ -1 & m \end{pmatrix}$,

$\mathbf{C} = \begin{pmatrix} -9 & 4 \\ 4 & n \end{pmatrix}$, find (a) $\mathbf{A}^2$, (b) m if $\mathbf{B} = \mathbf{A}^{-1}$, (c) n if $\mathbf{A}$ and $\mathbf{C}$ have equal determinants.

5 Given that the value of the determinant of the matrix $\begin{pmatrix} x & -3 \\ -1 & 2 \end{pmatrix}$ is 5, find the value of x . Hence write down the inverse of the matrix. [Camb]

6 Find a, b, c such that $\begin{pmatrix} a & b \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & 3 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 9 \\ 5 & 0 \end{pmatrix} - \begin{pmatrix} 4 & -6 \\ 3 & 2c \end{pmatrix}$.

7 (a) Given that $\mathbf{P} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -4 \\ 3 & 0 \end{pmatrix}$, find the matrix $\mathbf{P}$.

(b) Find the inverse of the matrix $\begin{pmatrix} 3 & 5 \\ 2 & 4 \end{pmatrix}$.

(c) Given that $\mathbf{R}$ is a 2×2 matrix such that $\mathbf{R} + \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{R} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, find $\mathbf{R}$.

8 Find two values of k such that $\begin{pmatrix} 2k+2 & k \\ 4k-3 & k+3 \end{pmatrix}$ is a singular matrix.

9 (a) Write down the inverse of the matrix $\begin{pmatrix} -1 & 4 \\ 1 & 3 \end{pmatrix}$.

(b) Hence or otherwise find x and y if $\begin{pmatrix} -1 & 4 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 13 \\ 1 \end{pmatrix}$.

10 Express the simultaneous equations $2x + 5y = 7$
 $x - y = 7$

as a single matrix equation

$$\mathbf{M} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \end{pmatrix},$$

where $\mathbf{M}$ is a 2×2 matrix. Pre-multiply both sides of the matrix equation by $\mathbf{M}^{-1}$ to find the values of x and y .

Geometrical transformations

A figure is **transformed** when its position and/or shape changes. The **image** of a shape is the figure obtained after a transformation.

If the image has the same dimensions as the original shape, the transformation is called a **congruency** or **isometry**. Fig. 27.1 shows a shaded triangle ABC and its images after a typical **translation**, T, **rotation**, R, **reflection**, M.

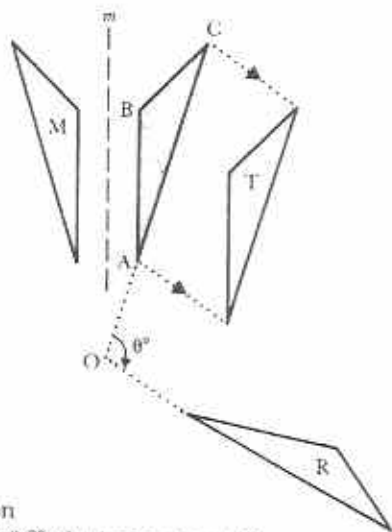


Fig. 27.1

T: translation

R: rotation of θ° clockwise about O

M: reflection in line m

Isometric (congruent) shapes have corresponding lengths and angles equal.

Fig. 27.2 shows triangle ABC and its images after **enlargements** E and E'.

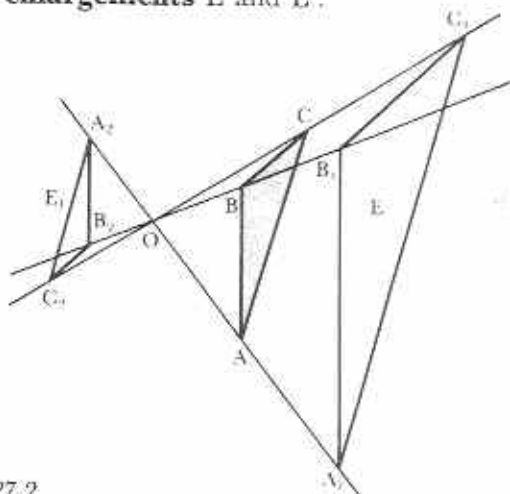


Fig. 27.2

E: enlargement of factor k where $k = \frac{OA_1}{OA} = \frac{OB_1}{OB}$

E': enlargement of factor K where $K = \frac{OA_2}{OA} = \frac{OB_2}{OB}$

Notice the following:

- Enlarged shapes are geometrically similar and have corresponding angles equal.
- If the enlargement factor is k then the original area will be enlarged by factor k^2 .
- In Fig. 27.2, K is negative and fractional.

Shear and **stretch** are shown in Fig. 27.3.

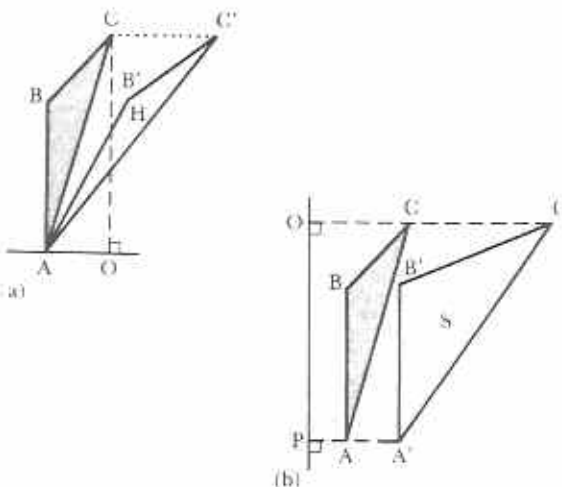


Fig. 27.3

H: shear of factor k where $k = \frac{CC'}{OC}$ and the line through AO is invariant.

S: stretch of factor K where $K = \frac{OC'}{OC} = \frac{OA'}{OA}$ and the line through OP is invariant.

Notice the following:

- Any shape has the same area as its image after shearing.
- If a shape is stretched with factor K , then its image has an area K times that of the shape.

Matrices can be used as operators which transform shapes drawn on the cartesian plane. Table 27.1, on page 244, gives a summary of the most common transformations of the cartesian plane and their related matrices.

Example 4

$\triangle PQR$ has coordinates $P(-3; 1)$, $Q(-2; 4)$, $R(0; 4)$.

(a) If $\triangle PQR$ is given a shear of factor 1 with the x -axis invariant, find by drawing or calculation the coordinates of its image, $\triangle P'Q'R'$.

(b) If $\triangle P'Q'R'$ is given a transformation represented by the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ find the coordinates of the new image $\triangle P''Q''R''$.

(a) Either by drawing as in Fig. 27.4:

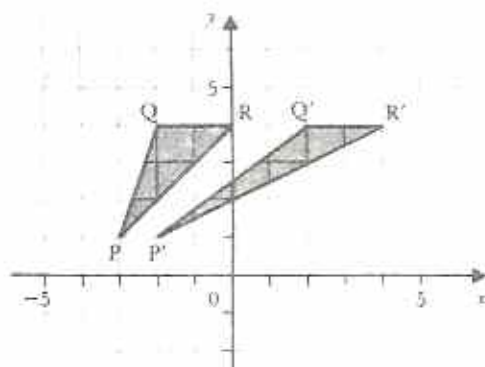


Fig. 27.4

$\triangle P'Q'R'$ has coordinates $P'(-2; 1)$, $Q'(2; 4)$, $R'(4; 4)$.

Or by calculation:

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is the matrix which represents the shear.

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} P & Q & R \\ -3 & -2 & 0 \\ 1 & 4 & 4 \end{pmatrix} = \begin{pmatrix} P' & Q' & R' \\ -2 & 2 & 4 \\ 1 & 4 & 4 \end{pmatrix}$$

$\triangle P'Q'R'$ has coordinates $P'(-2; 1)$, $Q'(2; 4)$, $R'(4; 4)$, as above

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} P' & Q' & R' \\ -2 & 2 & 4 \\ 1 & 4 & 4 \end{pmatrix} = \begin{pmatrix} P'' & Q'' & R'' \\ -2 & 2 & 4 \\ -1 & -4 & -4 \end{pmatrix}$$

$\triangle P''Q''R''$ has coordinates $P''(-2; -1)$, $Q''(2; -4)$, $R''(4; -4)$.

See Chapter 14, pages 113 to 123 for further coverage of matrices and transformations.

Table 27.1

Transformation	Sketch	Matrix
Identity		$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Translation		$\begin{pmatrix} a \\ b \end{pmatrix}$
Reflection in x -axis		$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Reflection in y -axis		$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
Rotation of 180° about $(0, 0)$		$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$
Enlargement centre $(0, 0)$		$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$
Shear x -axis invariant		$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$
Shear y -axis invariant		$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$
Stretch y -axis invariant		$\begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$
Stretch x -axis invariant		$\begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$

Exercise 27b

- 1 Describe completely the single transformation which maps $\triangle PQR$ onto $\triangle KLM$ in Fig. 27.5.

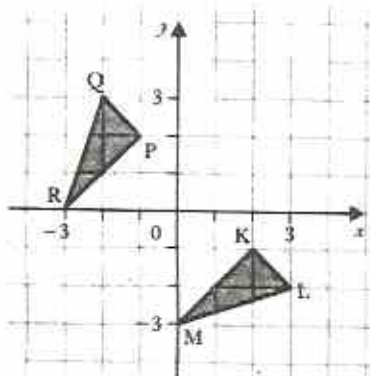


Fig. 27.5

- 2 In Fig. 27.6 $\triangle OPQ$ is enlarged to $\triangle OP'Q'$ by scale factor -1.4 with O as centre.

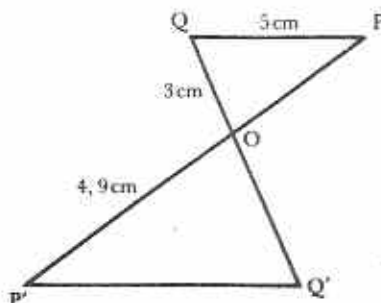


Fig. 27.6

If $PQ = 5$ cm, $OQ = 3$ cm and $OP' = 4.9$ cm calculate (a) $P'Q'$, (b) OQ' and (c) OP .

- 3 $\triangle ABC$ is as given in Fig. 27.7.

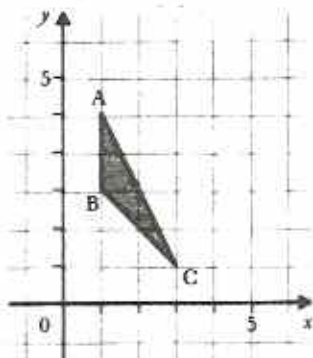


Fig. 27.7

S is a stretch of factor 3 in the x -direction with the y -axis invariant. If $\triangle PQR$ is the image of $\triangle ABC$ under S , find

- (a) the coordinates of $\triangle PQR$,
 (b) the numerical value of $\frac{\text{area of } \triangle ABC}{\text{area of } \triangle PQR}$.
- 4 The matrix $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$ represents a transformation Q .

- (a) Find the image of $(5; -2)$ under Q .
 (b) Find the image of $(5; 2)$ under Q .
 (c) Describe completely the transformation Q .
- 5 T is the translation $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and R is an anticlockwise rotation of 90° about the origin. A is the point $(2; -5)$, B is $(-1; 4)$ and C is $(-4; 4)$. Find the coordinates of (a) $T(A)$, (b) $R(B)$, (c) the point D if $RT(D) = C$. [Camb]

- 6 A shear H is represented by the matrix $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$. Line $A'B'$ is the image of line AB under H .
- (a) If the coordinates of B and A' are $(1; 4)$ and $(4; -3)$ respectively, find the coordinates of B' and A .
 (b) Write down the coordinates of any two points which remain invariant under H .

- 7 F is a transformation of matrix $\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}$.

The images of $A(2; 2)$, $B(-2; 4)$, $C(0; 8)$ under F are $A'(5; 5)$, $B'(-5; 10)$, $C'(0; 12)$.

- (a) Using a suitable scale, draw triangles ABC and $A'B'C'$ on the same graph.
 (b) Describe fully the transformation F and write down the value of k .
 (c) Find the coordinates of the vertices of the image of ABC after rotation of 270° clockwise about the point $(3; 2)$.
- 8 $P'(0; 0)$, $Q'(3; 13)$, $R'(-2; -11)$ are the images of $P(0; 0)$, $Q(3; 1)$, $R(-2; -3)$ under a transformation represented by a matrix

of the form $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

- (a) Find the transformation matrix.
 (b) Find the matrix which will transform $\triangle P'Q'R'$ back to $\triangle PQR$.

- 9 P is the point $(-1; 7)$ on the shape given in Fig. 27.8.

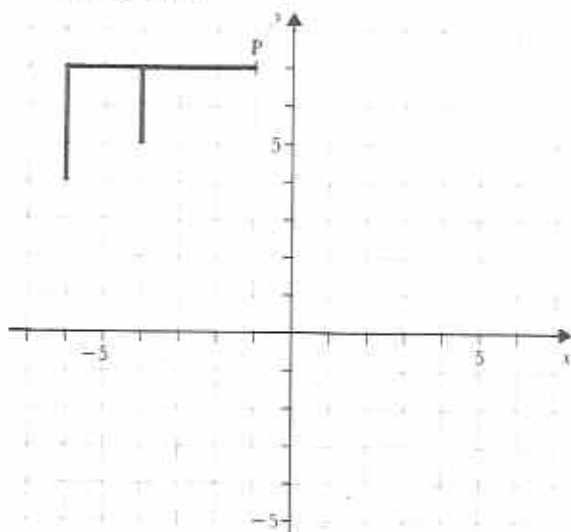


Fig. 27.8

M is a reflection in the line $2 - x = 0$.

N is a reflection in the line $x + y = 0$.

- (a) Find the image of P under (i) $MN(P)$,
 (ii) $NM(P)$.
 (b) Describe fully the single transformation K such that $K[MN(P)] = P$.

- 10 Answer the whole of this question on a sheet of graph paper.

The triangle ABC has vertices $A(2; 0)$, $B(4; 4)$, $C(0; 1)$. The triangle PQR has vertices $P(8; -2)$, $Q(4; 0)$, $R(7; -4)$. The triangle LMN has vertices $L(-2; -7)$, $M(-6; -9)$, $N(-3; -5)$. Draw these triangles on graph paper, using a scale of 1 cm to 1 unit on each axis, and label the vertices. $\triangle ABC$ can be mapped onto $\triangle PQR$ by an anticlockwise rotation about the origin followed by a translation.

- (a) State the angle of rotation.
 (b) Find the matrix which represents this rotation.
 (c) Find the column vector of the translation.
 (d) Given that $\triangle ABC$ can be mapped onto $\triangle PQR$ by a single rotation, find the coordinates of the centre of this rotation.

- (e) Given that $\triangle ABC$ can be mapped onto

$\triangle LMN$ by a translation of $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$ followed

by a reflection in the mirror line m , draw the line m on your graph and label it clearly.

- (f) Find the equation of m . [Camb]

Vectors

Chapter 18, pages 149 to 157, contains complete revision notes on vectors. Read the content and worked examples of Chapter 18 before attempting the following revision exercise.

Exercise 27c

- 1 Fig. 27.9 shows two vectors **a** and **b**.



Fig. 27.9

Copy Fig. 27.9 onto squared paper. Draw and clearly label the lines OP, OQ, OR, OS such that (a) $\mathbf{OP} = 4\mathbf{a}$, (b) $\mathbf{OQ} = -2\mathbf{b}$, (c) $\mathbf{OR} = 3\mathbf{a} + 2\mathbf{b}$, (d) $\mathbf{OS} = 5\mathbf{a} - 3\mathbf{b}$.

- 2 PQRS is a rhombus. $\mathbf{PQ}$ is shown in

Fig. 27.10 and $\mathbf{QR}$ is the column vector $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$.



Fig. 27.10

(a) On a copy of Fig. 27.10, mark and label the points R and S.

(b) Express the following as column vectors.

- (i) $\mathbf{SR}$ (ii) $\mathbf{PR}$ (iii) $\mathbf{QS}$

3 OABC is a trapezium such that O is the origin, $\mathbf{OA} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, $\mathbf{OC} = 2\mathbf{AB} = \begin{pmatrix} 6 \\ 2 \end{pmatrix}$.

(a) On squared paper, mark and label the points O, A, B and C.

(b) Express the following as column vectors.

- (i) $\mathbf{OB}$ (ii) $\mathbf{BC}$ (iii) $\mathbf{CA}$

4 $\mathbf{OA} = \begin{pmatrix} -4 \\ 9 \end{pmatrix}$, $\mathbf{OB} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$ and M is the mid-point of AB. Express the following as column vectors.

- (a) $\mathbf{AB}$ (b) $\mathbf{AM}$ (c) $\mathbf{OM}$

5 In Fig. 27.11, C, D and F are the mid-points of BE, CE and AE respectively. $\mathbf{AB} = \mathbf{a}$ and $\mathbf{AC} = \mathbf{a} + \mathbf{b}$.

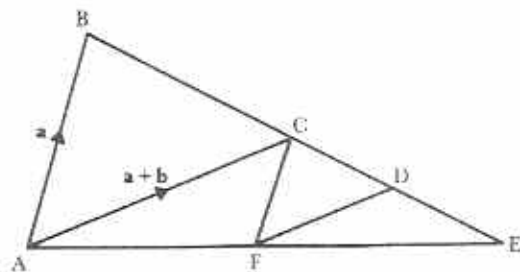


Fig. 27.11

Write down as simply as possible in terms of $\mathbf{a}$ and $\mathbf{b}$, expressions for the following.

- (a) $\mathbf{BC}$ (b) $\mathbf{CF}$ (c) $\mathbf{FD}$
(d) $\mathbf{DE}$ (e) $\mathbf{FE}$ (f) $\mathbf{BF}$

6 $\mathbf{OP} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$ and $\mathbf{OQ} = \begin{pmatrix} 1 \\ q \end{pmatrix}$. Find (a) $|\mathbf{OP}|$, (b) q if O, P and Q are vertices of a square OPQR.

7 It is given that $\mathbf{u} = 4\mathbf{a} + 3\mathbf{b}$, $\mathbf{v} = 5\mathbf{a} - \mathbf{b}$ and $\mathbf{w} = h\mathbf{a} + (h+k)\mathbf{b}$, where h and k are constants. If $\mathbf{w} = 3\mathbf{u} - 2\mathbf{v}$, calculate the value of h and k . [Camb]

8 If $\mathbf{OA} = 3\mathbf{p} - 2\mathbf{q}$, $\mathbf{OB} = \mathbf{p} + 7\mathbf{q}$ and $\mathbf{AB} = 2m\mathbf{p} + (m-n)\mathbf{q}$, find the values of m and n .

9 In Fig. 27.12, OABC is a square and X, Y, Z are the mid-points of OC, CB, BA respectively. $\mathbf{OA} = \mathbf{a}$ and $\mathbf{OC} = \mathbf{c}$.

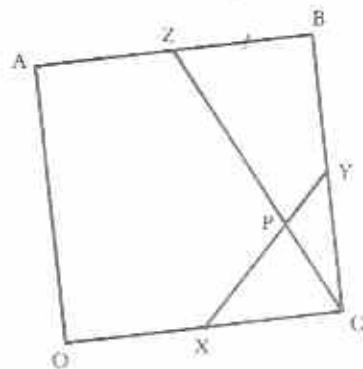


Fig. 27.12

- (a) Express $\mathbf{XY}$ and $\mathbf{CZ}$ in terms of $\mathbf{a}$ and $\mathbf{c}$.
(b) If $\mathbf{XP} = h\mathbf{XY}$, express $\mathbf{XP}$ in terms of $\mathbf{a}$, $\mathbf{c}$ and h .
(c) If $\mathbf{CP} = k\mathbf{CZ}$, express $\mathbf{CP}$ in terms of $\mathbf{a}$, $\mathbf{c}$ and k .
(d) Use the fact that $\mathbf{XP} = \mathbf{XC} + \mathbf{CP}$ to evaluate h and k .

10 (a) Given that $\mathbf{OK} = \begin{pmatrix} 16 \\ 2 \end{pmatrix}$, $\mathbf{OL} = \begin{pmatrix} 4 \\ -3 \end{pmatrix}$ and that M and N are the mid-points of OK and OL respectively, (i) express $\mathbf{MN}$ as a column vector, (ii) find the value of $|\mathbf{KL}|$.
(b)

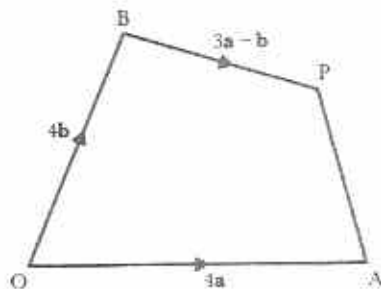


Fig. 27.13

Given that $\mathbf{OA} = 4\mathbf{a}$, $\mathbf{OB} = 4\mathbf{b}$ and $\mathbf{BP} = 3\mathbf{a} - \mathbf{b}$, express as simply as possible, in terms of $\mathbf{a}$ and $\mathbf{b}$, (i) $\mathbf{OP}$, (ii) $\mathbf{AP}$.

The lines OA produced and BP produced meet at Q. Given that $\mathbf{BQ} = m\mathbf{BP}$ and $\mathbf{OQ} = n\mathbf{OA}$, form an equation connecting m , n , $\mathbf{a}$ and $\mathbf{b}$. Hence deduce the values of m and n . [Camb]

Travel graphs, statistics, probability

Syllabus references 6.5.2 and 6.11

Travel graphs

Distance–time graphs

Fig. 28.1 is a typical distance–time graph of a journey which is in three stages.

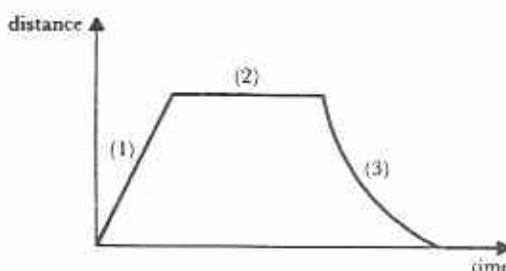


Fig. 28.1

In Fig. 28.1,

Stage 1: Distance is changing uniformly with time. The **gradient** of the line is a measure of the **speed**, or rate of change of distance with time.

Stage 2: Distance does not change with time. The object is at rest.

Stage 3: The gradient of the tangent at any point on the curve gives the speed at that point. During this stage the speeds are negative. This shows that the object is travelling in a direction opposite the original direction.

Example 1

At 0800 a cyclist left home and cycled at 13 km/h to a village 20 km away. She stayed for 2 hours at the village before cycling home at 9 km/h. Use a graphical method to find the time when she arrived home.

Fig. 28.2 is a graph of the cyclist's journey.

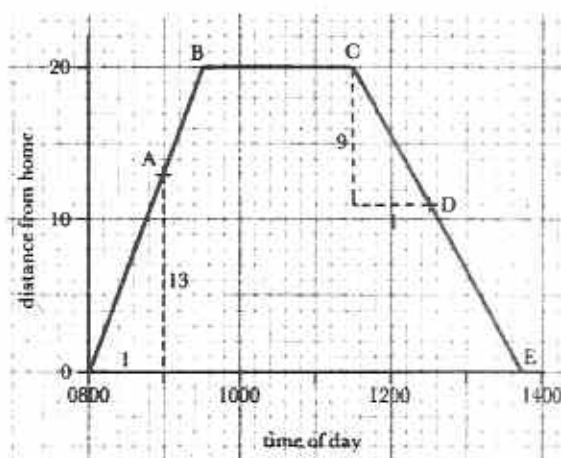


Fig. 28.2

From the graph, the time of arrival is found at E, approximately 1345.

Method

Choose suitable scales with time on the horizontal axis. Then:

- In 1 hour the cyclist covers 13 km. Plot the point A (0900; 13).
- Produce the line through OA to B, 20 km from the start.
- On the horizontal through B, mark C, 2 hours after B. C represents the starting point of the journey home.
- Going home the cyclist covers 9 km in 1 hour. Plot D 1 hour and 9 km from C.
- Produce the line through CD to cut the time axis at E.

Speed–time graphs

Fig. 28.3 is a typical speed–time graph.

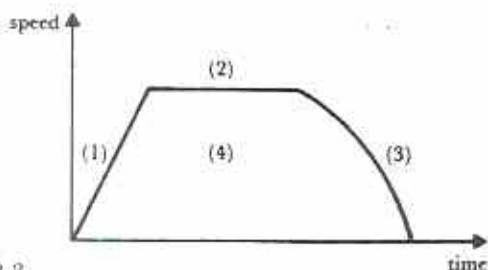


Fig. 28.3

In Fig. 28.3,

Stage 1: Speed is changing uniformly with time. The **gradient** of this line is a measure of the **acceleration**, or rate of change of speed with time.

Stage 2: Speed does not change with time. The horizontal line represents a period of constant speed.

Stage 3: Speed is decreasing with time. This gives a **negative acceleration** or **deceleration**. The gradient of the tangent at any point on the curve gives the deceleration at that point.

Region 4: The **area** under the graph is a measure of the **distance** travelled. The area can be estimated by counting squares or by considering the areas of trapeziums.

Chapter 8, pages 63 to 67, gives a full explanation of how to estimate the gradient at a point on a curve and how to use trapeziums to estimate the area under a curve. Also see Example 3 below.

Example 2

Fig. 28.4 is the speed–time graph of a train journey.

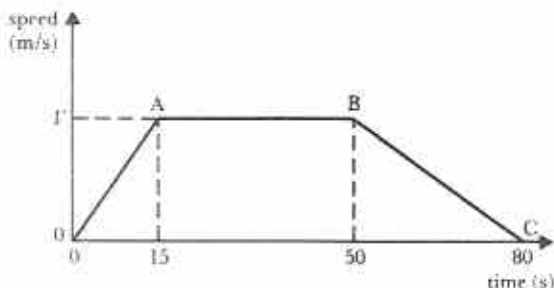


Fig. 28.4

If the total distance travelled in the 80 seconds is 920 m, calculate (a), V , (b) the acceleration of the train during the first 15 seconds, (c) the distance travelled in the final 40 seconds.

(a) Distance travelled

= area of trapezium OABC

$$= \frac{1}{2}(AB + OC)V = \frac{1}{2}(35 + 80)V$$

$$\text{Hence, } 920 = \frac{1}{2}(35 + 80)V$$

$$\Leftrightarrow 1840 = 115V$$

$$\Leftrightarrow V = \frac{1840}{115} = 16$$

(b) Acceleration during first 15 seconds

= gradient of OA

$$= \frac{V}{15} = \frac{16}{15} \text{ m/s}^2$$

$$= 1\frac{1}{15} \text{ m/s}^2$$

(c)

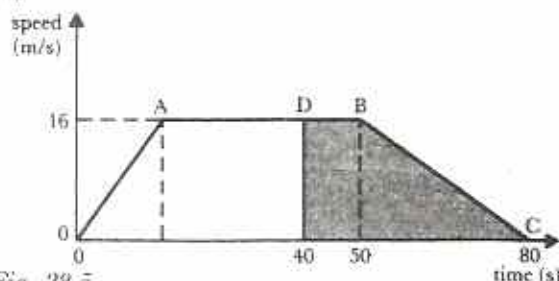


Fig. 28.5

Distance travelled during the last 40 seconds

= Area under DBC in Fig. 28.5

$$= \frac{1}{2}(DB + 40)16 \text{ m}$$

$$= \frac{1}{2}(10 + 40)16 \text{ m}$$

$$= 400 \text{ m}$$

Example 3

A particle moves in a straight line so that after t seconds its velocity v m/s is given by $v = 5t^2 - 12t + 8$.

Some corresponding values of v and t are given in Table 28.1.

Table 28.1

t	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3	$3\frac{1}{2}$	4
v	8	$3\frac{1}{4}$	p	$1\frac{1}{4}$	4	q	17	$27\frac{1}{4}$	40

- (a) Calculate p and q .
 (b) Draw the graph of $v = 5t^2 - 12t + 8$ for the range $0 \leq t \leq 4$.
 (c) Use the graph to estimate (i) the time at which the acceleration is zero, (ii) the acceleration when $t = 3$. (d) By using trapeziums with $\frac{1}{2}$ -second intervals, estimate the distance travelled during the 4 seconds.

(a) When $t = 1$,

$$p = 5(1)^2 - 12(1) + 8$$

$$= 5 - 12 + 8 = 1$$

When $t = 2\frac{1}{2}$,

$$q = 5(2\frac{1}{2})^2 - 12(2\frac{1}{2}) + 8$$

$$= 31\frac{1}{4} - 30 + 8 = 9\frac{1}{4}$$

- (b) Fig. 28.6 is the required graph.

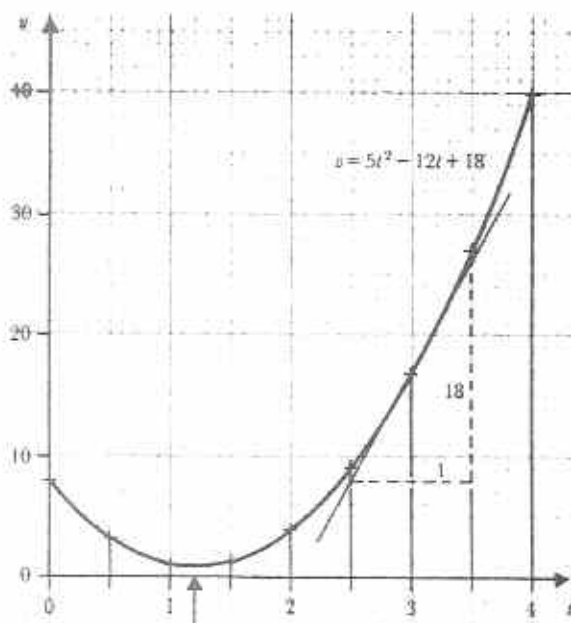


Fig. 28.6

- (c) (i) The acceleration is zero when the tangent to the curve is horizontal, i.e. at the lowest point of the curve in Fig. 28.6. At this point, $t = 1.2$ seconds.
 (ii) In Fig. 28.6 a tangent has been drawn at the point where $t = 3$.

Gradient of tangent = $\frac{18 \text{ m/s}}{1 \text{ s}} = 18 \text{ m/s}^2$

When $t = 3$, the acceleration is 18 m/s^2 .

- (d) Distance travelled
 = area under curve
 $\approx$ sum of areas of trapeziums in Fig. 28.6

$$= \frac{1}{2}(8 + 3\frac{1}{4})\frac{1}{2} + \frac{1}{2}(3\frac{1}{4} + 1)\frac{1}{2} + \frac{1}{2}(1 + 1\frac{1}{4})\frac{1}{2}$$

$$+ \dots + \frac{1}{2}(17 + 27\frac{1}{4})\frac{1}{2} + \frac{1}{2}(27\frac{1}{4} + 40)\frac{1}{2}$$

$$= \frac{1}{2}[1(8 + 40) + 3\frac{1}{4} + 1 + 1\frac{1}{4} + 4 + 9\frac{1}{4}$$

$$+ 17 + 27\frac{1}{4}] \text{ m}$$

$$= \frac{1}{2}[24 + 63] \text{ m}$$

$$= 43\frac{1}{2} \text{ m}$$

Exercise 28a

- 1 Fig. 28.7 shows the journey of a car from Harare to Kadoma and of a bus from Kadoma to Harare on the same road.

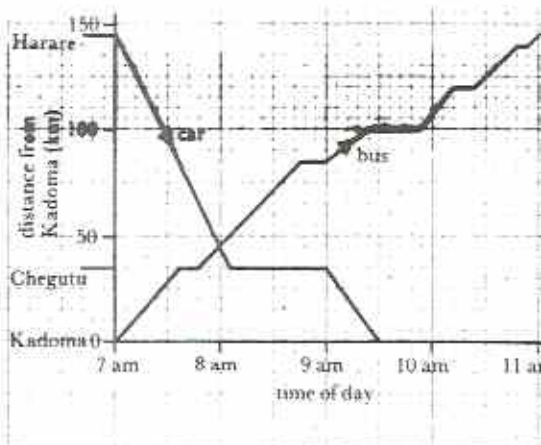


Fig. 28.7

- (a) How many times did the bus stop between Kadoma and Harare?
 (b) What is the distance between Kadoma and Harare?
 (c) What was the car's average speed for the whole journey?
 (d) What was the bus's average speed, to the nearest km/h for the whole journey?
 (e) What was the car's average speed between Harare and Chegutu?
 (f) What was the car's average speed between Chegutu and Kadoma?
 (g) How far were they both from Chegutu when they passed each other on the road?
 2 Fig. 28.8 shows how Sola walked from school to the Post Office and how Gogo walked from the Post Office to the school.

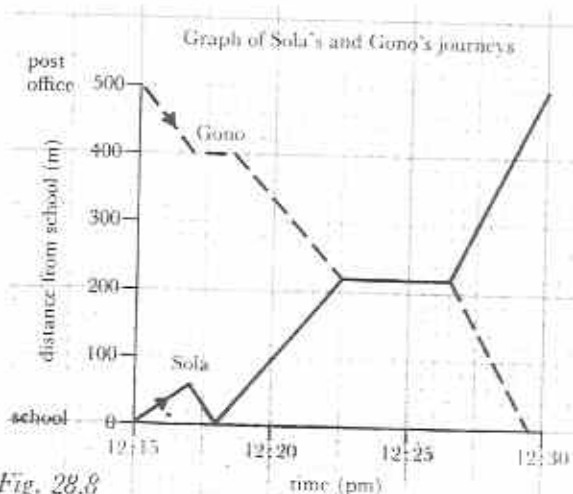


Fig. 28.8

- How far were they from the Post Office when they met?
 - How long did they stand talking?
 - After leaving school, Sola suddenly remembered a letter she had to post. She returned to school for the letter. At what time did she remember the letter?
 - Gono stopped for $1\frac{1}{2}$ min to buy some bread. How far is the bakery from the school?
 - After leaving Gono, what was Sola's walking speed in m/min?
 - How much further did Sola walk than Gono?
- 3 Three cars, A, B, C, start one after the other in alphabetical order at 5-minute intervals. They travel at 80, 100, 120 km/h respectively. Given that A leaves at midday, draw a distance-time graph which enables you to find when (a) B passes A, (b) C passes A, (c) C passes B.
- 4 An aircraft travels at an average speed of 800 km/h. It left airport A at 1230 and arrived at airport B at 1345. After stopping at airport B for 45 min it left for airport C, arriving there at 1900.

Draw a distance-time graph of its journey using a scale of 2 cm to 1 hour on the time axis and 1 cm to 400 km on the distance axis. Use your graph to find (a) the distance from A to C, (b) the distance of the plane from B at 1500.

- 5 A cyclist sets out at 0705 to reach his office 13 km away at 0800. After 5 min he finds he has forgotten his briefcase. He rides home again at 16 km/h and then takes 2 min to find the briefcase. Use a graphical method to answer the following. (a) How fast must he ride to get to work on time? (b) If his top speed is 17 km/h, how many minutes will he be late for work?
- 6 During a journey a train accelerates uniformly for 6 min. Its speed, v km/h, is given in 1-min intervals in Table 28.2.

Table 28.2

time (min)	0	1	2	3	4	5	6
v (km/h)	7	16	25	34	43	52	61

By drawing a speed-time graph find, (a) the train's acceleration in km/h per min, (b) the distance travelled in km during the whole 6 min.

- 7 Fig. 28.9 is the speed-time graph of a sprinter.

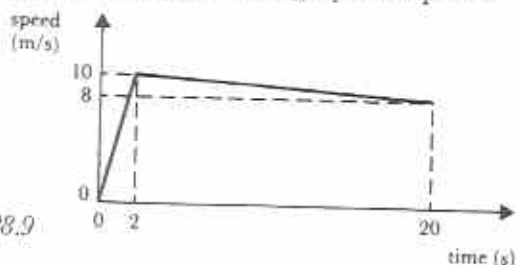


Fig. 28.9

Calculate (a) the acceleration of the sprinter over the first 2 seconds, (b) the deceleration of the sprinter over the next 18 seconds, (c) the distance covered in the 20 seconds.

- 8 Fig. 28.10 is the speed-time graph of a journey.

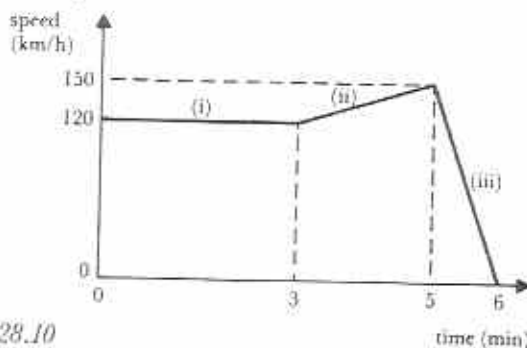


Fig. 28.10

- (a) Describe the three parts of the journey in your own words.
 (b) Calculate the accelerations represented by parts (ii) and (iii) of the graph (answers in km/h per minute).
 (c) Calculate the total distance travelled (answer in km).

9 Fig. 28.11 is the speed-time graph of a car journey.

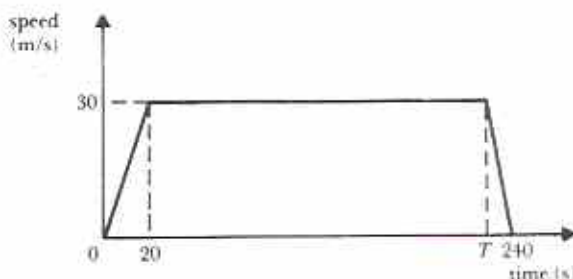


Fig. 28.11

- (a) Calculate the acceleration of the car during the first 20 seconds.
 (b) Given that the final deceleration is 0.5 m/s^2 , calculate (i) T , (ii) the total distance travelled by the car.
- 10** Fig. 28.12 is the speed-time graph of the last ten seconds of a car journey. It travels at a constant speed of 31 m/s for 4 seconds and slows down uniformly first to 16 m/s then to rest after a further 5 seconds and 1 second respectively.

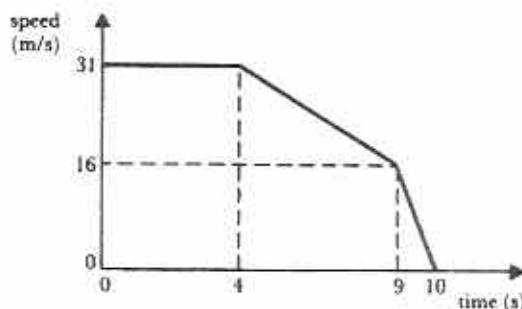


Fig. 28.12

Calculate (a) the speed of the car 2 seconds before it stopped, (b) the average speed of the car during the whole 10 seconds.

- 11** An aeroplane accelerates from rest at 40 km/h per minute until its speed is 840 km/h . It then travels at this speed for $2\frac{1}{2} \text{ h}$ before decelerating at an average of 15 km/h per minute until it comes to rest.

- (a) Sketch the journey on a speed (km/h)-time(min) graph.
 (b) Calculate the total time taken for the journey in minutes.
 (c) Calculate the distance travelled in km.
- 12** Fig. 28.13 is a speed-time curve of part of a taxi journey.

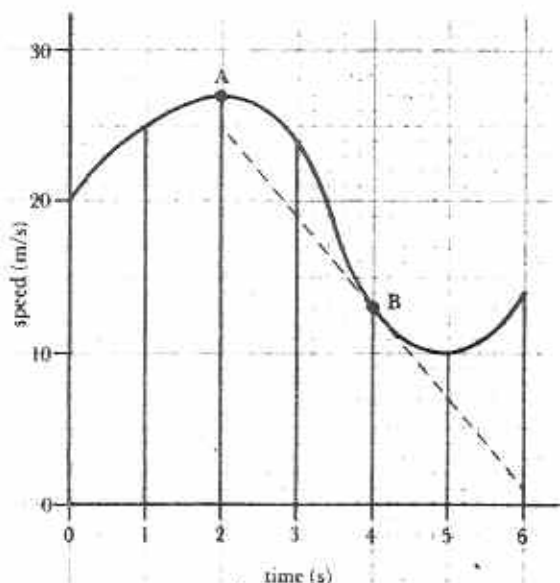


Fig. 28.13

- (a) What is the acceleration of the taxi at A?
 (b) Use the tangent drawn at B to estimate the acceleration of the taxi at that point.
 (c) By using suitable trapeziums or otherwise, estimate the distance travelled by the taxi in the 6 seconds.
- 13** Table 28.3 gives the speed, $v \text{ m/s}$, of an object taken at 1-second intervals.

Table 28.3

$t(\text{s})$	0	1	2	3	4	5	6	7	8
$v(\text{m/s})$	32	35	36	35	32	27	20	11	0

- (a) Draw a $v-t$ graph of the motion.
 (b) By constructing a suitable tangent, estimate the acceleration of the object after 5 seconds.
 (c) Estimate the total distance travelled during the 8 seconds.

- 14 The velocity, v m/s, of an object after t seconds is given by the equation
 $v = 4t^2 - 12t + 11$.

Table 28.4 contains some corresponding values of v and t .

Table 28.4

t (s)	0	$\frac{1}{2}$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3	$3\frac{1}{2}$	4
v (m/s)	11	6	3	m	32	6	11	n	27

- (a) Calculate m and n .
 (b) Taking 2 cm to represent 1 second on the horizontal axis and 2 cm to represent 5 m/s on the vertical axis, draw the graph of $v = 4t^2 - 12t + 11$ for the range $0 \leq t \leq 4$.
 (c) From your graph, find the times when the velocity is 8 m/s.
 (d) By drawing a tangent, find the acceleration of the object after 3 seconds.
 (e) Estimate the distance travelled during the 4 seconds.
- 15 The velocity, v m/s, of a car after t seconds is given by
 $v = 7 + 6t - t^2$.
 Table 28.5 contains some corresponding values of v and t .

Table 28.5

t	0	1	2	3	4	5	6	7
v	7	12	15	a	15	12	7	b

- (a) Calculate a and b .
 (b) Draw the graph of $v = 7 + 6t - t^2$ for the range $0 \leq t \leq 7$ using scales of 2 cm to 1 second horizontally and 1 cm to 1 m/s vertically.
 (c) Use your graph to find the speed of the car after 4.3 seconds.
 (d) By drawing suitable tangents, find the acceleration of the car after (i) 1 second,

- (ii) 6 seconds,
 (e) Estimate the distance travelled during the 7 seconds.

Statistics

Bar charts and pie charts

Example 4

Table 28.6 shows how a wage of \$150 is spent.

Table 28.6

item	amount
food	\$50
rent	\$25
clothing	\$20
savings	\$30
other expenses	\$25

Show this information on a bar chart.

Represent the items by bars of the same width. The height, or length, of each bar is proportional to the amount of money. Fig. 28.14 shows that the bars may be vertical or horizontal.

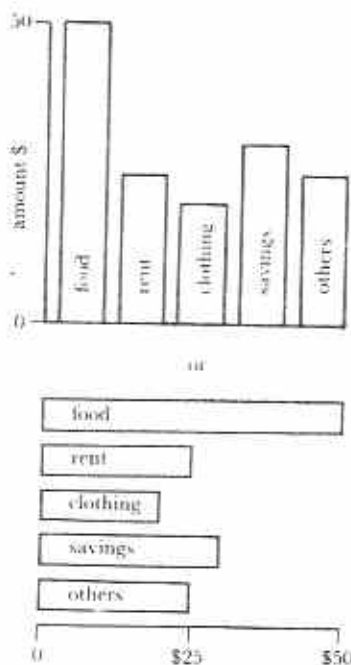


Fig. 28.14

Example 5

The number of students admitted to a university in a particular year is distributed among five faculties as follows:

Education, 350;

Medicine, 150;

Engineering, 200;

Law, 100;

Arts, 100.

Draw a pie chart to represent this information.

Table 28.7 shows how to calculate the angles of the sectors of the pie chart.

Table 28.7

faculty	number of students	angle of sector in pie chart
Education	350	$\frac{350}{900} \times 360^\circ = 140^\circ$
Medicine	150	$\frac{150}{900} \times 360^\circ = 60^\circ$
Engineering	200	$\frac{200}{900} \times 360^\circ = 80^\circ$
Law	100	$\frac{100}{900} \times 360^\circ = 40^\circ$
Arts	100	$\frac{100}{900} \times 360^\circ = 40^\circ$
totals	900	360°

Fig. 28.15 is the required pie chart.

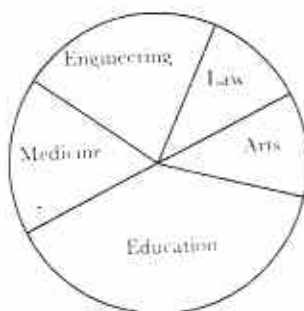


Fig. 28.15

Mean, median, mode

Given a set of values, the **mean** is the sum of all the values divided by the number of them.

If a set of numbers is arranged in order of size, the middle term is called the **median**. If there is an even number of terms the median is taken as the mean of the two middle terms.

The number of times any particular value occurs in a set is called its **frequency**. The number which occurs most often, i.e. the value that has the greatest frequency, is called the **mode**.

Example 6

Find (a) the mean, (b) the median, (c) the mode of the following set of numbers: 12; 16; 8; 11; 12; 8; 2; 8; 1; 14.

- (a) Sum of the numbers
 $= 12 + 16 + 8 + 11 + 12 + 8 + 2 + 8 + 1 + 14$
 $= 92$
Number of numbers = 10

$$\text{Mean} = \frac{92}{10} = 9.2$$

- (b) Arrange the numbers in ascending order:
1; 2; 8; 8; 8; 11; 12; 12; 14; 16

$$\text{Median} = \frac{8 + 11}{2} = 9.5$$

- (c) The mode is 8.

Example 7

The ages of 15 students in years and months are 14.5; 15.2; 14.3; 13.9; 14.10; 14.11; 13.8; 15.3; 14.6; 15.6; 15.8; 16.1; 15.4; 14.4; 14.7. Find the average age of the students to the nearest month.

The ages range from 13.8 to 16.1. Take 15.0 as a working mean. Make two columns of numbers. The one marked (+) shows all the deviations in months for ages over 15.0; the other, (-), gives the deviations in months for ages under 15.0.

(+)	(-)
2	7
3	9
6	15
8	2
13	1
-4	16
	6
	8
	5
36	69

33 (i.e. $69 - 36$)

The working shows that the total deviation for the 15 students is 33 months less than 15.0. Hence:

$$\begin{aligned}
 \text{mean age} &= 15 \text{ yr } 0 \text{ mo} + \left(\frac{-33}{15} \right) \text{ mo} \\
 &= 15 \text{ yr } 0 \text{ mo} - 2.2 \text{ mo} \\
 &= 14 \text{ yr } 10 \text{ mo to the nearest month}
 \end{aligned}$$

The method in Example 7 is useful when given a large set of numbers of roughly the same size.

Exercise 28b

- 1 A student spent a full day as shown in Table 28.8.

Table 28.8

activity	time
at lectures	5 h
reading	6 h
sleeping	7 h
sports	2 h
others	4 h

Show this data on a bar chart and a pie chart.

- 2 Table 28.9 shows how a wage of \$200 was spent.

Table 28.9

item	amount
food	\$60
rent	\$40
clothing	\$20
savings	\$40
others	\$30

Show this data on a bar chart and a pie chart.

- 3 A university admitted the following numbers of students from 1988 to 1992.

Table 28.10

year	number of students
1988	800
1989	1 200
1990	1 350
1991	1 570
1992	2 250

(a) Represent this information on a bar chart.

(b) Calculate the mean number of students admitted per year.

- 4 Table 28.11 shows the numbers of different types of books in a school library.

Table 28.11

subject	number of books
mathematics	48
science	110
engineering	54
novels	496
others	372

Draw a pie chart to show this information.

- 5 Find the mean, median and mode of the following sets of numbers:

(a) 2; 4; 4; 6; 7

(b) 3; 5; 5; 7; 7; 7; 9; 9

(c) 11; 9; 6; 4; 3; 12; 1; 6; 5

- 6 Use a working mean of 115 to find the mean of the following:
110; 120; 113; 116; 119;
127; 117; 118; 118; 113
- 7 The ages of 16 students in years and months are as follows:
17.2 17.10 18.2 19.5
18.0 17.11 18.7 19.7
19.3 19.8 16.11 17.9
17.10 17.5 18.5 18.1
Choose a suitable working mean and use it to find the average age of the students.
- 8 The heights, in cm, of 10 boys are
145 163 159 162 167
149 150 160 170 155
Calculate the mean and median heights.
- 9 The masses, to the nearest kg, of 15 girls are
45 38 51 44 43
60 55 47 45 42
52 46 41 50 53
Find the mean and median masses.
- 10 Table 28.12 gives the age distribution of members of a school choir.

Table 28.12

age	12	13	14	15	16	17
frequency	2	1	3	6	5	3

- (a) How many students are in the choir?
(b) What is the modal age?
(c) Draw a pie chart to show the age distribution.
(d) Calculate the mean age of the choir members.

Grouped data

Histogram, cumulative frequency curve

Example 8

Table 28.13 is the frequency distribution of the masses of 40 pupils in a class.

- (a) Draw a histogram of the distribution.
(b) State the modal class.

Table 28.13

mass (kg)	number of pupils
41–45	3
46–50	7
51–55	12
56–60	10
61–65	6
66–70	2

- (c) Calculate the mean mass of the pupils.
(d) Draw a cumulative frequency curve of the distribution.
(e) Hence estimate (i) the median, (ii) the semi-interquartile range.
- (a) Fig. 24.16 is the required histogram. Notice that the gaps between the bars have been closed by increasing the width of each rectangle by $\frac{1}{2}$ unit on both sides.

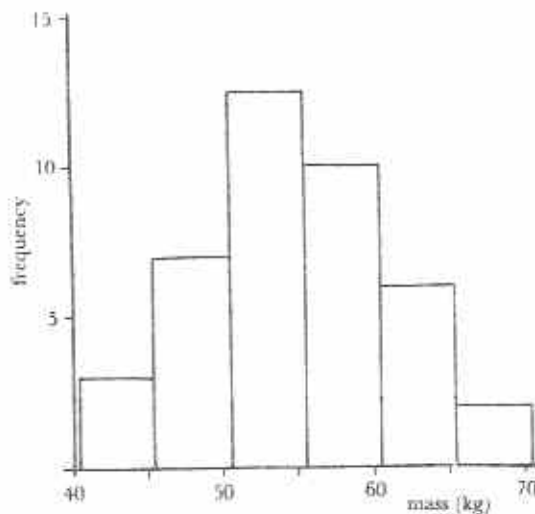


Fig. 28.16

- (b) The modal class is 51–55. This class has the highest frequency.
(c) The mid-value of each class can be taken to be representative of that class as a whole. Table 28.14 shows the mid-values and the corresponding deviations from a working mean of 53 kg.

Table 28.14

mass (kg)	frequency f	mid-value	deviation d	$f \times d$
41-45	3	43	-10	-30
46-50	7	48	-5	-35
51-55	12	53	0	0
56-60	10	58	+5	+50
61-65	6	63	+10	+60
66-70	2	68	+15	+30
total deviation = +75				

Either, using the mid-values only:
mean mass

$$= \frac{3 \times 43 + 7 \times 48 + 12 \times 53 + 10 \times 58 + 6 \times 63 + 2 \times 68}{3 + 7 + 12 + 10 + 6 + 2} \text{ kg}$$

$$= \frac{2195}{40} \text{ kg} = 54.875 \text{ kg}$$

or using a working mean of 53 kg, from Table 28.14:

$$\text{mean deviation} = \frac{+75}{40} \text{ kg} = 1.875 \text{ kg}$$

$$\text{mean mass} = 53 + (+1.875) \text{ kg} = 54.875 \text{ kg}$$

(d) Table 28.15 is used to draw the cumulative frequency curve in Fig. 28.17.

Table 28.15

mass (kg)	frequency	cumulative frequency
41-45	3	3
46-50	7	10
51-55	12	22
56-60	10	32
61-65	6	38
66-70	2	40

- (c) From Fig. 28.17, (i) median = $Q_2 = 54\frac{1}{2}$ kg
(ii) semi-interquartile range

$$= \frac{Q_3 - Q_1}{2} \text{ kg}$$

$$= \frac{59\frac{1}{2} - 50\frac{1}{2}}{2} \text{ kg}$$

$$= 4\frac{1}{2} \text{ kg}$$

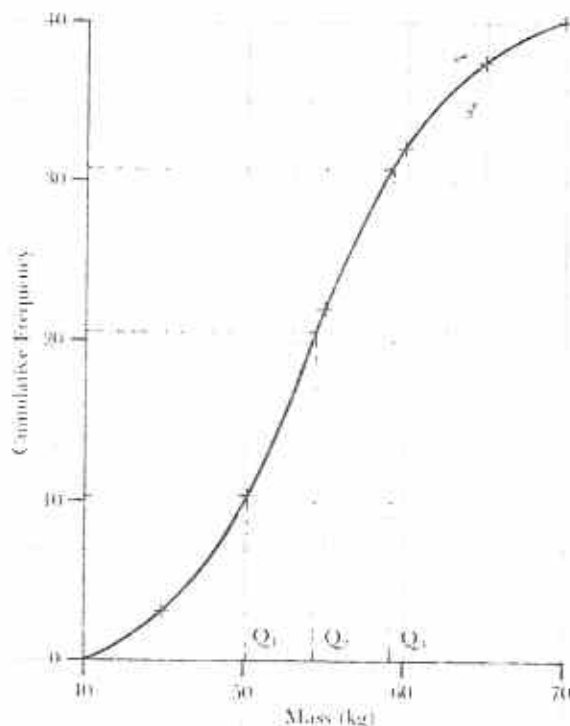


Fig. 28.17

Further information on frequency distributions, histograms and cumulative frequency curves can be found in Chapter 16, pages 133 to 142.

Exercise 28c

- 1 Table 28.16 shows the frequencies, f , of children of age x years in a hospital.

Table 28.16

x	1	2	3	4	5	6	7	8
f	3	4	5	6	7	6	5	4

- (a) What is the modal class?
(b) How many children are in the hospital?
(c) Calculate the mean age of the children.
- 2 Table 28.17, overleaf, shows the length of life of 200 electric light bulbs.

Table 28.17

length of life (hours)	number of bulbs
201-300	10
301-400	16
401-500	32
501-600	54
601-700	88

- (a) Draw a histogram of this distribution.
 (b) Use a working mean of 550.5 hours to calculate the mean life of the light bulbs.
3 Table 28.18 shows the number of work-days lost through illness among 500 factory employees during a 1-year period.

Table 28.18

number of days	number of employees
0-4	250
5-9	158
10-14	33
15-19	29
20-24	15
25-29	10
30-34	5

- (a) Draw a histogram of the distribution.
 (b) State the modal class.
 (c) Calculate the mean number of days lost.
4 The masses, in kg to the nearest kg, of 40 students are as follows:

59 54 51 56 59 61 60 61 59 58
 62 61 63 64 58 57 56 60 62 60
 61 65 58 57 54 52 62 67 69 49
 56 58 60 60 62 58 51 57 70 63

- (a) Take class intervals of 46-50, 51-55, ..., and make up a table of frequencies.
 (b) Draw the corresponding histogram.
 (c) Find the median mass.
 (d) Calculate the mean mass.

- 5** The examination marks of 50 students are as follows:

65 58 51 36 23 40 53 59 70 51
 46 59 50 67 46 39 61 62 73 60
 71 51 47 32 48 40 40 51 58 67
 60 69 43 52 37 26 38 50 59 40
 44 54 42 47 68 74 45 39 48 55

- (a) Make a frequency distribution using class intervals of 21-30, 31-40,
 (b) Draw a cumulative frequency curve.
 (c) Hence estimate (i) the median, (ii) the semi-interquartile range.
 (d) Find the percentage of students that got more than 45 marks.
6 Table 28.19 is the frequency distribution of the heights of 40 pupils.

Table 28.19

height (cm)	number of pupils
131-140	2
141-150	11
151-160	14
161-170	10
171-180	3

- (a) Draw a histogram of the distribution.
 (b) State the modal class.
 (c) Calculate the mean height of the pupils.
 (d) Draw a cumulative frequency curve of the distribution.
 (e) Hence estimate the median height of the pupils.

Probability

The **probability** of an event happening can be given a numerical value x where

$$x = \frac{\text{number of required outcomes}}{\text{number of possible outcomes}}$$

and $1 \geq x \geq 0$.

If $x = 1$, then the event is certain to happen.
 If $x = 0$, then the event cannot happen. $1 - x$ is the probability of the event *not* happening.

Example 9

Table 28.20 shows the numbers of students in each age group in a class.

Table 28.20

age (years)	16	17	18
number	7	22	13

What is the probability that a student chosen at random from the class is (a) 17 years old, (b) not 17 years old, (c) over 16 years old?

- (a) Probability that the student is 17 years old

$$\begin{aligned}
 &= \frac{\text{number of 17 year olds}}{\text{total number of students}} \\
 &= \frac{22}{7 + 22 + 13} = \frac{22}{42} = \frac{11}{21}
 \end{aligned}$$

- (b) Probability that the student is *not* 17 years old

$$= 1 - \frac{11}{21} = \frac{10}{21}$$

- (c) Probability that the student is over 16 years old

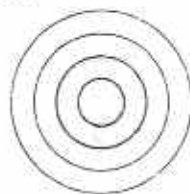
$$\begin{aligned}
 &= \frac{\text{number of students over 16}}{\text{total number of students}} \\
 &= \frac{22 + 13}{42} = \frac{35}{42} = \frac{5}{6}
 \end{aligned}$$

Probabilities of mutually exclusive events and independent events together with the use of outcome tables and tree diagrams are explained in Chapter 19, pages 158 to 165.

Exercise 28d

- A box contains 40 oranges, 12 of which are unripe. I pick one at random. What is the probability that it is (a) ripe, (b) unripe?
- A bag contains 10 white balls, 4 red balls and 6 black balls. If a ball is selected at random, what is the probability that it is (a) white, (b) red, (c) black?
- A card is picked at random from a pack of 52 playing cards. What is the probability that it is (a) the Queen of diamonds, (b) a black five, (c) a nine, (d) a black heart.

- A class contains 15 boys and 21 girls. A student is chosen at random. What is the probability that a boy is chosen?
- A fair 6-sided die is thrown. Find the probability of getting (a) a four, (b) an even number, (c) a number less than five.
- Three coins are tossed at the same time. (a) Write down all the possible ways that they can fall, using H for head and T for tail. (b) Find the probability of getting 1 head and 2 tails.
- A bag contains 24 tennis balls, some white and some green. If a ball is chosen at random, the probability of getting a green ball is $\frac{3}{8}$. How many white balls are in the bag?
- A worker is chosen at random from among the 500 employees in Table 28.18 on page 258. What is the probability that she had been absent for less than 10 days during the year in question?
- A number has three digits formed by arranging 4, 5 and 6 in a random order. Write down all the possible numbers. Hence find the probability that the number is divisible by (a) 3 (b) 4 (c) 5 (d) 6 (e) 7 (f) 8 (g) 9 (h) 13
- Fig. 28.18 shows a target made from four concentric rings of radii, r , $2r$, $3r$, $4r$. A bullet hits the target at random. Calculate the probability that it hits the shaded region.

*Fig. 28.18*

- Table 28.21 shows the numbers of students in each age group in a class.

Table 28.21

age (years)	16	17	18	19
frequency	9	11	11	5

(a) A student is chosen at random from the class. What is the probability that the age of the student is i. 16 years, ii. under 18 years, iii. not 19 years?

(b) Calculate the average age of the students in years and months.

(c) Two students are chosen at random. Find the probability that they are both 19.

- 12 $A = \{2; 3; 6; 9; 11\}$ and $B = \{13; 14; 15; 17; 19; 21\}$.

(a) If one element is selected at random from A , what is the probability that it is odd.

(b) If one element is selected from each set, calculate the probability that both elements are even, expressing your answer as a fraction in its lowest terms. [Camb]

- 13 A class of 53 students took a test which was marked out of 10. 27 of the students scored 7 marks or less.

(a) State the probability that a student chosen at random from the class scored more than 7.

A second class took the test and a quarter of them scored more than 7. If one student is chosen at random from each class, find the probability that

- (b) both scored more than 7,
(c) only one scored more than 7.

- 14 From a group of five children, consisting of three girls and two boys, one child is chosen at random. Write down the probability that the child chosen is a girl.

A second child is then chosen at random from the remaining four children. Given that the first child chosen is a girl, write down the probability that the second child chosen is also a girl.

Two children on another occasion are chosen at random from this same group of three girls and two boys. Calculate the probability that (a) both are girls, (b) both are boys, (c) they are of different sexes. [Camb]

- 15 Table 28.22 shows the distribution of grades obtained by 30 students in an examination.

Table 28.22

grade	A	B	C	D	E
frequency	5	6	9	7	3

A pupil is selected at random from the group. Find the probability that the student got (a) grade E, (b) either grade A or grade E. Two students are selected at random from the group. Find the probability that (c) both obtained C, (d) one obtained grade C.

Non-routine problems

In each section the questions have been categorised into problems, puzzles and investigations.

Problems can usually be solved in a fairly straightforward manner, using conventional methods.

Puzzles are less straightforward. You may have to consider unusual approaches. Do not be afraid to use the method of trial and error with puzzles. Allow time for thinking.

Investigations tend to be open-ended. Sometimes it is difficult to know when to stop. Investigations are best approached systematically. If a problem seems too complex, try a simpler example of the same problem. Find a helpful way of setting out your working and recording your results. This usually means making lists and/or tables. When enough results have been collected, it may then be possible to discover a rule.

Number, algebra and pattern

- 1 Arrange seven chairs in a row and ask three boys and three girls to sit on them as in Fig. 29.1.



Fig. 29.1

The aim is to change over the boys and girls. Here are the rules:

- a student can move into an adjacent empty chair,
 - a student can jump over one adjacent student of the opposite sex into an empty chair,
 - no backward moves are allowed.
- (a) What is the minimum number of moves needed? [Puzzle]

- (b) What is the least number of moves needed if there are 5 chairs, 2 boys and 2 girls? [Puzzle]

- (c) What if there were 9 chairs, 4 boys and 4 girls?

- (d) Generalise for n boys and n girls. [Investigation]

- (e) What if there were 2 empty chairs?

- (f) What if there were unequal numbers of boys and girls?

- (g) Investigate some ideas of your own. [Investigation]

- 2 (a) Choose any number less than 100 (e.g. 57).

- (b) Form a new number by squaring each of the digits and adding.

$$(57 \rightarrow 5^2 + 7^2 = 25 + 49 = 74)$$

- (c) Repeat:

$$(74 \rightarrow 7^2 + 4^2 = 49 + 16 = 65)$$

- (d) Carry on doing this. You will know when to stop!

- (e) Repeat the above with a different starting number.

- (f) Investigate for all numbers less than 100.

- 3 Choose any four-digit number, e.g. 1 952. [Investigation]

Write digits in descending order: 9 521

Write digits in ascending order: $\overline{-1\ 259}$

Subtract: 8 262

Repeat the process on the answer

$$(8\ 262 \rightarrow 8\ 622 - 2\ 268)$$

Keep on repeating. You will know when to stop! [Problem]

Investigate with numbers which have three digits, five digits, etc. [Investigation]

- 4 Find a four-digit number which is exactly four times greater when its digits are reversed. [Puzzle]

Make up a similar puzzle for 3- and 5-digit numbers. [Puzzle]

- 5 123×45 , 543×21 , 421×53 , $5 \times 41 \times 32$ are all products using the digits 1, 2, 3, 4,

5. Use a calculator to find the arrangement of these digits which gives the greatest product. [Problem]
 Extend the problem to the digits 1, 2, 3, 4, 5, 6. Try to find a rule. [Investigation]
 6 Make a copy of the grid in Fig. 29.2.

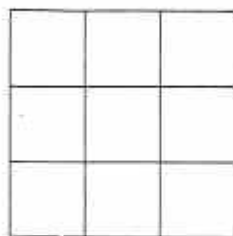


Fig. 29.2

- Place the numbers 2, 2, 2, 3, 3, 3, 4, 4, 4 in the cells of the grid so that when any line of three numbers is added up in any direction the total is always 9. [Puzzle]
 7 Make another copy of the grid in Fig. 29.2. Colour the squares either red, white or blue so that:
 each red touches a white
 each white touches a blue
 each blue touches a red [Puzzle]
 8 In a copy of Fig. 29.3 replace the asterisks by the numbers 1 to 12 in such a way that there are three numbers totalling 17 along each side. [Puzzle]

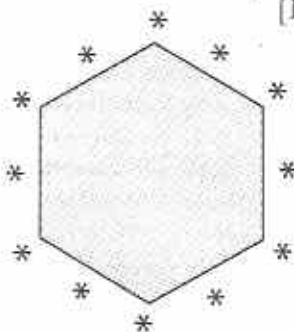


Fig. 29.3

- 9 x and y are numbers such that $xy = 1\,000\,000$. Find x and y if neither of them contain any 0's. [Problem]
 10

6	9	15	24	39
---	---	----	----	----

To make the above row of numbers, start with 6 and 9 and add them to get 15. Then add 9 and 15 to get 24; finally add 15 and 24 to get 39.

The same rule has been used in the next row, but the numbers in the middle are missing.

5				43
---	--	--	--	----

Find the missing numbers. [Puzzle]
 If the first number was m and the last number was n , what, in terms of m and n would be the middle number? [Problem]
 Investigate for rows of various length. [Investigation]

- 11 Refer to the following set of whole numbers: {1, 2, 5, 9, 16}
 (a) Which is the smallest prime number?
 (b) Which is a multiple of 4?
 (c) Write down all the square numbers.
 (d) Write down three factors of 18.
 (e) Find three numbers a, b, c such that $(a + b)^2 = c$.
 (f) Find three numbers x, y, z such that $x^2 = y + z$.
 (g) Find the numbers p, q, r such that $p^3 = q - r$. [Problem]
 12 (a) Extend Table 29.1 as far as 33.

Table 29.1

denary	binary	number of ones in binary
1	1	1
2	10	1
3	11	2
4	100	1
5	101	2
6	110	2
7	111	3
8	1000	1
9	1001	2
⋮	⋮	⋮
33	⋮	⋮

(b) Classify the denary numbers into sets, according to the number of 1's in their binary equivalents. For example:

$$S_1 = \{1, 2, 4, 8, \dots\}$$

$$S_2 = \{3, 5, 6, 9, \dots\}$$

$$S_3 = \{7, 11, \dots\}$$

$$S_4 = \{15, \dots\}$$

[Problem]

(c) Investigate the members of S_1, S_2, S_3, S_4, S_5 , etc.

(d) What patterns can you find?

[Investigation]

- 13 A tin of cooking oil costs \$10. If the oil is worth \$9 more than the tin, what is the value of the tin? [Puzzle]

- 14 A company proposes a choice of two pay plans to a Union negotiator:

(a) Initial salary of \$10 000, to be increased by \$500 after each 12 months;

(b) Initial salary of \$10 000, to be increased by \$125 after each 6 months.

Which plan should the Union negotiator recommend? [Puzzle]

- 15 What is the value of

$$(x-a)(x-b)(x-c)\dots(x-z)?$$

[Puzzle]

- 16 There are some goats and some hens on a farm. They have a total of 99 heads and legs. There are twice as many hens as goats. How many hens are there? [Problem]

- 17 A herd boy counts his cattle. When he counts them in threes there is one left over. When he counts them in fives there are two left over.

How many cattle might he have? [Puzzle]

Investigate the possible numbers of cattle he could have and suggest a rule.

[Investigation]

- 18 A student types some patterns using x's and m's. She makes the patterns according to a rule. Fig. 29.4 shows three of her patterns:

$$\begin{array}{ccc} \text{x x x} & \text{x x x x x} & \text{x x x x x x} \\ \text{m m m m} & \text{m m m m m m} & \text{m m m m m m m m} \\ \text{x x x} & \text{x x x x x} & \text{x x x x x x} \end{array}$$

Fig. 29.4

(a) Draw another way she could have typed the pattern.

(b) How many x's would she need to type if she made a pattern which had 15 m's?

(c) How many m's would she need to type if she had a pattern which had 90 x's?

(d) If X stands for the number of x's and M stands for the number of m's, find the rule, or formula, that connects X and M .
 $X = \dots$

(e) Another student types x's and m's according to the rule $X = \frac{1}{3}(2M + 2)$. Find the number of x's if this student types 29 m's.

(f) Find the value of X in the second student's rule when $M = 10$. What does your result tell you? [Problem]

- 19 Start with any two single-digit numbers e.g. 4 and 9. Here is a chain made by starting with these numbers:

$$4 \rightarrow 9 \rightarrow 3 \rightarrow 2 \rightarrow 5 \rightarrow 7 \rightarrow 2 \rightarrow$$

(a) How is the chain made? [Puzzle]

(b) Continue the chain. What happens? [Puzzle]

(c) Make similar chains with other starting numbers.

(d) Investigate what happens.

[Investigation]

(e) What happens if you use numbers in base five?

- 20 You have a large supply of 1c, 2c, 5c and 10c coins.

(a) How many ways can you make up a total of 10c? [Problem]

(b) How many ways can you make up a total of 15c? [Problem]

(c) Investigate for other sums of money.

[Investigation]

Spatial awareness and pattern

- 1 A farmer gives the field shown in Fig. 29.5 to his four children, provided they can divide it up according to his instructions.

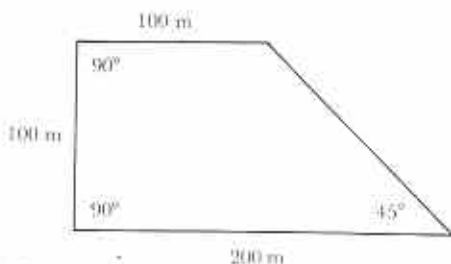


Fig. 29.5

All pieces should be equal in area. They must also be similar in shape to the original field. How can the children divide the field?

[Puzzle]

- 2 (a) In Fig. 29.6(a) find the ratio between the areas of the two circles.

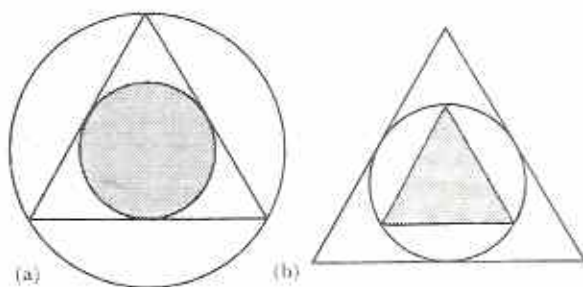


Fig. 29.6

- (b) In Fig. 29.6(b) find the ratio between the areas of the two triangles. [Problem]
 3 How many squares are there on an 8×8 chessboard? (64 is *not* the correct answer. Nor is it 65.) [Problem]

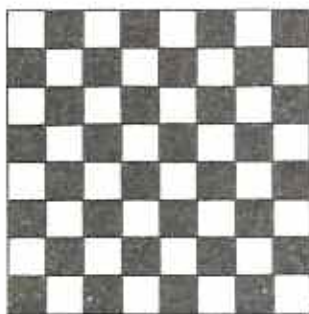


Fig. 29.7

It may help if you consider simpler cases, e.g. a 2×2 or a 3×3 chessboard (Fig. 29.8):



Fig. 29.8

- 4 Make a paper rectangle 8 cm by 3 cm like that in Fig. 29.9.

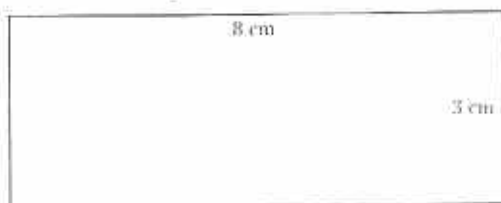


Fig. 29.9

Cut the paper into two pieces which can be rearranged to make a rectangle which measures 12 cm by 2 cm. [Puzzle]

- 5 Fig. 29.10 is a view of a crate which can hold 24 bottles of Cola.

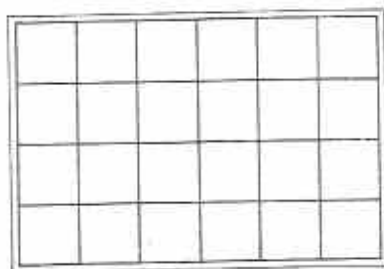


Fig. 29.10

- (a) Place 18 bottles of Cola in the crate so that each row and each column has an *even* number of bottles in it.
 (b) Find three different ways of doing this. [Puzzle]
 (c) Is it possible to do the problem with 17 bottles?
 (d) Investigate with different numbers of bottles.
 (e) Investigate with crates of different dimensions. [Investigation]
 6 How many acute angles can you see in Fig. 29.11? [Puzzle]

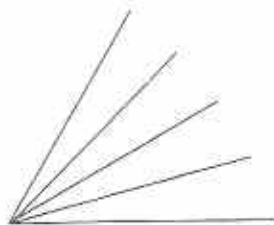


Fig. 29.11

- 7 Fig. 29.12 shows two dissections of a 4×4 square into rectangles (note that $\{\text{squares}\} \subset \{\text{rectangles}\}$).

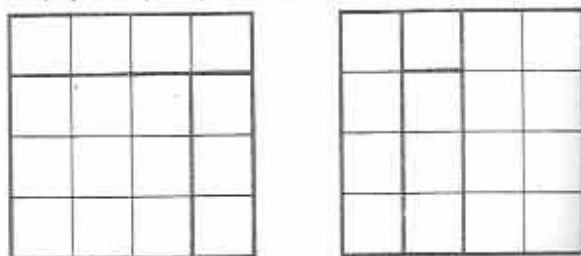


Fig. 29.12

The rules for dissection are:

(i) in each case all the rectangles must be different;

(ii) the edges of the rectangles are a whole number of units.

Find other dissections of a 4×4 rectangle. Investigate for other starting squares.

[Investigation]

- 8 Arrange some matches to make five squares as shown in Fig. 29.13.



Fig. 29.13

Change the positions of just two of the matches to reduce the number of squares to four. (No 'loose ends' are allowed.)

[Puzzle]

- 9 In Fig. 29.14, three squares and three diagonals are arranged as shown.

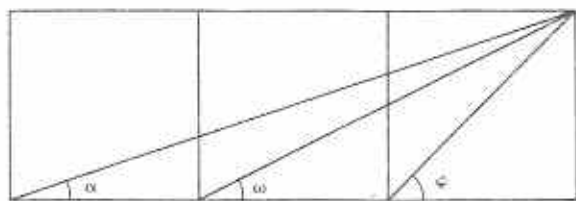


Fig. 29.14

There is a simple relationship between the three angles, α , ω and ϕ . Guess the relationship, then prove it.

[Puzzle]

- 10 Fig. 29.15 shows a 5×3 grid with one diagonal drawn.

The diagonal cuts 7 of the grid's squares. Investigate the numbers of squares cut by diagonals of grids of various dimensions. How many squares would be cut by the diagonal of an $m \times n$ grid? [Investigation]

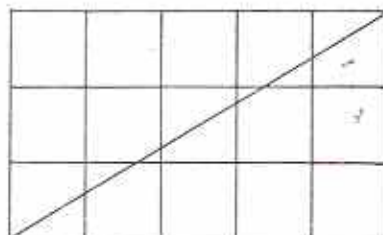


Fig. 29.15

Miscellaneous

- 1 A yellow ball, a blue ball and two red balls are placed in a bag. The bag is shaken. Someone takes two balls out of the bag, looks at them, and says, 'At least one of these balls is red.'

What is the probability that the other ball is also red?

[Puzzle]

- 2 It is recommended that the water in a swimming pool should be a 1% chlorine solution. How would you decide how much chlorine to add to a swimming pool near your school?

You may have to ask your chemistry teacher what a '1% chlorine solution' is.

[Real problem]

- 3 Nine people arrive at a meeting. Each person shakes hands once with every other person present. How many handshakes are there?

[Problem]

How many if only five people are at the meeting?

What if there were 500 people at the meeting?

[Investigation]

- 4 The maths teacher sets an exam paper for her class and left it in her desk overnight. Next morning it had disappeared. Four girls were in the classroom that morning: Aquilina, Bernadette, Cynthia and Dora. Of these students, only one of them tells the truth. Aquilina said, 'Cynthia took it'. Bernadette said, 'I didn't take it'. Cynthia said, 'Bernadette is lying'. Dora said, 'Cynthia is lying'.

Who took the exam paper?

[Puzzle]

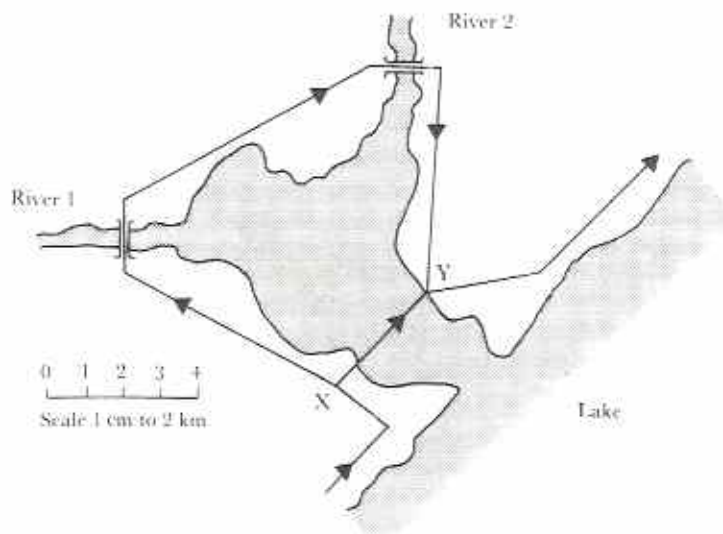


Fig. 29.6

5 Fig. 29.16 is a map showing two rivers flowing into a lake. A road runs northeast. To negotiate the estuary, a cyclist *either* has to make a detour, via two bridges, or can travel from X to Y with the help of a ferry.

(a) Measure the distance in cm from X to Y using the ferry.

(b) Measure the distance in cm from X to Y using the bridges.

(c) What are the actual distances in km for each route?

(d) What is the saving in km if the cyclist uses the ferry?

(e) The cyclist cycles at 20 km/h and the ferry travels at 8 km/h. How many minutes does the journey XY take by ferry? (Assume that there is no waiting for the ferry.)

[Problem]

6 A car has travelled 30 000 km altogether. It has one spare tyre. The tyres were changed at intervals so that each tyre had been used for the same number of km. For how many km had each tyre been used? [Puzzle]

7 One day four university students sat together at lunchtime. Their names were Tembo, Festus, John and Henry and they studied geology, history, biology and law though not necessarily in that order.

John and Henry ate fruit for lunch. The law student ate sandwiches. Tembo and Henry often play tennis with the biology student and the law student. Henry sat

between the biology student and the history student.

One of the students did not eat lunch. What was he studying? [Puzzle]

8 How many times will the long hand of a clock pass the short hand between midday today and midday tomorrow? (Since both hands are together at the starting time and finishing times, these do not count as passes.)

[Puzzle]

9 Two students, X and Y, run a race of 100 metres. X beats Y by 5 metres. They decide to race again, but this time to introduce a handicap system. X starts the race 5 metres behind the starting line, therefore giving Y a 5 metre start. If both students run at the same rate as before, what will be the result of the second race?

[Puzzle]

10 I am thinking of a whole number.

It is odd.

It is less than 1 000.

All the digits are different.

The sum of its digits is 12.

The difference between the first two digits is the same as the difference between the last two.

The hundreds digit is greater than the sum of the digits in the tens and units columns.

What is the number I am thinking of?

[Puzzle]

Certificate-level practice examinations

This chapter contains two full-scale practice examinations in Mathematics at School Certificate/O' level. They are included as a final revision of the senior secondary school course and to provide practice in examination technique. To be effective, each paper should be done under examination conditions, i.e. the times for each paper should be observed and neither the textbook nor notes should be referred to.

General advice

- 1 Be sure to read and understand the examination *rubric* (instructions). Typical rubric is given on the papers which follow. Note, however, that the Examinations Syndicate may change the rubric if it wishes. So, always check the rubric carefully.
- 2 Work out roughly how much time you can afford to spend on each question. Allow time for reading the questions and for checking your answers at the end. This advice is especially important in the short-answer paper where approximately 30 questions are to be answered in $2\frac{1}{2}$ hours (i.e. only about 5 minutes per question).
- 3 If any question involves drawing, make a rough sketch first. This helps you to position your final answer on the paper, whether it is a graph or a scale drawing.
- 4 Check your answers to see that they are sensible in terms of the given data. For example, a car costing \$27 or walking speed of 5600 km/h are sure signs that a slip has been made.
- 5 Show all your working in the body of the question you are answering. Do not be ashamed of your rough working. Examiners know what to look for. They may be able to give marks for such working, but only if they see it on the paper.

Examination 1

Paper 1 (2 h 30 min)

Instructions to candidates:

All questions may be attempted.

Answers are to be written in the spaces provided.*

If working is needed for any question, it must be shown in the space below that question.

Omission of essential working will result in loss of marks.

Questions 1 to 19 carry 3 marks each.

Questions 20 to 27 carry 4 marks each.

Question 28 carries 5 marks.

Question 29 carries 6 marks.

*The examination paper contains spaces for working and answers. To save space in this book, these spaces have been omitted.

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS MAY BE USED IN THIS PAPER.

1 Find the value of

(a) $1\frac{3}{4} - \frac{1}{3}$

(b) $2\frac{2}{5} \times 1\frac{2}{3}$

(c) $(\frac{1}{2} + \frac{1}{4}) \div \frac{1}{3}$

2 Find the value of

(a) $6,3 + 2,87$

(b) $0,8 \div 0,05$

(c) $0,056 \times 0,003$.

3 The universal set

$$U = \{u; n; i; v; e; r; s; a; t\},$$

$$R = \{r; i; v; e; r\} \text{ and } V = \{\text{vowels}\}.$$

(a) List the members of V .

(b) Find $n(R)$.

(c) List the members of $R \cup V$.

- 4 50 students were asked, 'How did you travel to school today?' An incomplete list of results is given in Table R1 overleaf.

Table R1

method of travel	bus	car	bicycle	walking
no. of students	13	4	8	

- (a) How many students walked to school?
 (b) What percentage travelled by bus?
- 5 (a) Express 924_{ten} in base five.
 (b) Find the sum of $10\ 100_{\text{two}}$ and $1\ 100_{\text{two}}$ as a binary number.
 (c) Find $2\ 033_{\text{five}} - 424_{\text{five}}$, giving the answer in base ten.
- 6 If $p = 9 \times 10^6$ and $q = 4 \times 10^7$, find the value of the following in standard form.
- (a) $p + q$ (b) $\sqrt{p}$ (c) $\frac{1}{q}$

7

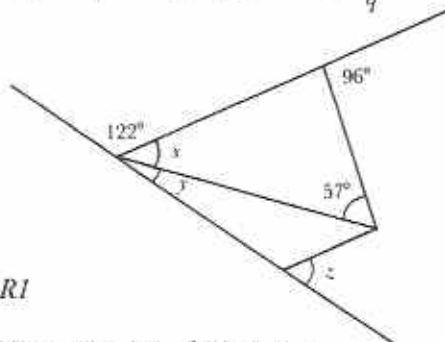


Fig. R1

- Given Fig. R1, find x , y , z .
- 8 What is
 (a) 20 mm as a fraction, in its lowest terms, of 9 m?
 (b) 0,004 09 correct to 3 decimal places?
 (c) 0,3125 as a common fraction in its lowest terms?
- 9 (a) Three students measure their body-mass to the nearest kg as 39 kg, 42 kg and 51 kg. What is the greatest possible value for their combined body-mass?
 (b) If x is an integer such that $3x < 40$ and $27 - 2x \leq 5$, list the possible values of x .
- 10 A straight line joins the points A(1; 3) and B(4; 7).
 (a) What is the length of AB?
 (b) What is the gradient of AB?
 (c) A line parallel to AB passes through the origin and the point (3; k). What is the value of k ?
- 11 Fig. R2 shows points O, P and Q drawn on the cartesian plane.

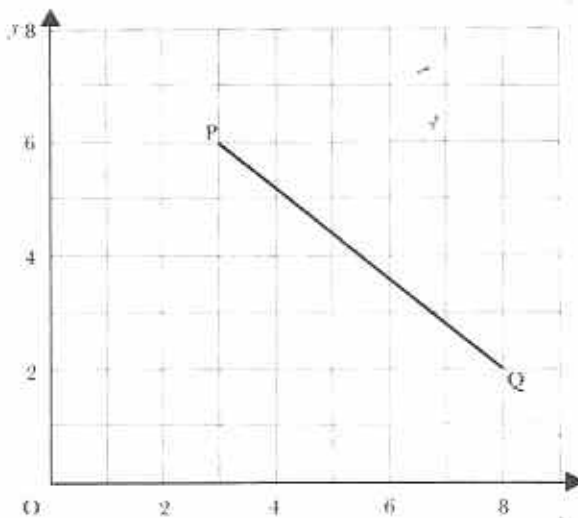


Fig. R2

- Express each of the following as a single column vector.
 (a) $\overrightarrow{PQ}$ (b) $\overrightarrow{OQ}$ (c) $\overrightarrow{OP} - \overrightarrow{OQ}$
- 12 Fig. R3 contains two regular pentagons A and B drawn on a common base-line as shown.

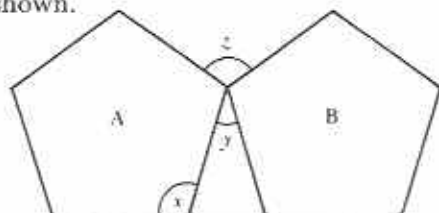


Fig. R3

- Calculate the angles x , y and z .
- 13 Fig. R4 shows the minimum and maximum temperatures recorded on a particular day.

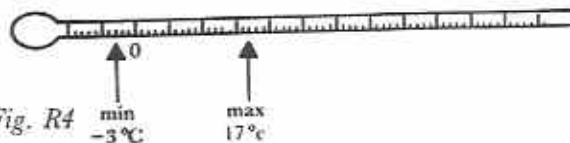


Fig. R4

- (a) How many degrees did the temperature change?
 (b) Calculate the mean of the two temperatures.
- 14 Factorise
 (a) $x^2 - 14x + 49$,
 (b) $6ab - 2b^2 - 4b + 12a$.

- 15 Solve the simultaneous equations:

$$2x + y - 2 = 0$$

$$4x - 3y - 19 = 0.$$

- 16 Fig. R5 is a graph showing the journey of a cyclist who left home at 11 am to visit a friend 5 km away, before returning home.

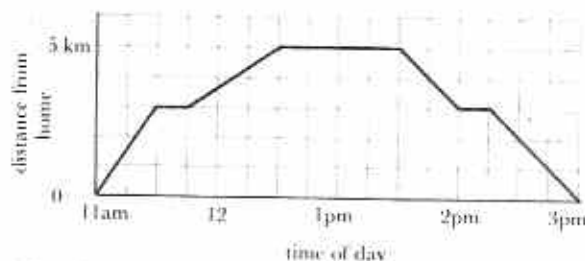


Fig. R5

- (a) When did the cyclist arrive at his friend's house?
 (b) On both the outward journey and the homeward journey, the cyclist called to greet his mother. How far does his mother live from his home?
 (c) What was the cyclist's average speed for the whole journey?
- 17 A mattress is priced at \$200 plus sales tax of 20%.
 (a) Find the total price of the mattress including sales tax.
 (b) A shop offers the mattress for sale with a discount of 20% of its total price. What is the discount price?
- 18 A drinking glass is in the shape of a cylinder of internal diameter 7 cm and height 12 cm. If the glass is two-thirds full, how many ml of liquid does it contain? [Take π to be $3\frac{1}{7}$ and assume that 1 ml of liquid occupies 1 cm³.]
- 19 A large cake is divided into n pieces, each of mass m grams.
 (a) Which of the following describes the relation between m and n ?
 (i) $m \propto n$ (ii) $m \propto \frac{1}{n}$ (iii) $\frac{1}{m} \propto \frac{1}{n}$
 (b) If $m = 360$ when $n = 15$, write down a formula connecting m and n .
 (c) If the reciprocal of m is 0,0025, calculate m .

- 20 A cinema can hold n people. If all the seats are taken, then 540 people are in the cinema.
 (a) Represent this information by an inequality in n .

(b) If a school party visits the cinema, the manager insists that there must be one teacher for every 30 students. If a school party of 120 students visits the cinema, how many seats will be left for members of the general public?

(c) Seats normally cost \$3,00 each, but the manager gives a 15% discount for parties of 20 or over. What would be the total bill for the school party in part (b)? (Include all the teachers.)

- 21 Fig. R6 is a sketch showing the relative positions of Mutare (M) and Harare (H).



Fig. R6

- (a) What is the bearing of Harare from Mutare? (b) What is the bearing of Mutare from Harare? (c) Use Fig. R6 and as much of the information below as is necessary to find how far Harare is north of Mutare ($\sin 37^\circ = 0,60$, $\cos 37^\circ = 0,80$, $\tan 37^\circ = 0,75$).
- 22 Fig. R7 shows a circle ABCT, centre O with PT a tangent at T.

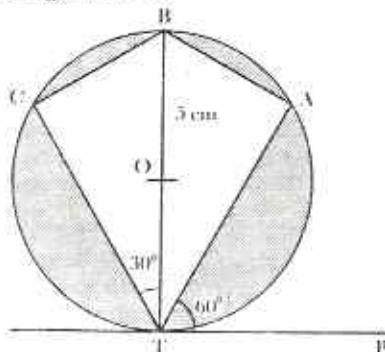


Fig. R7

(a) If $\hat{ATP} = 60^\circ$ and $\hat{CTB} = 30^\circ$, calculate $\hat{ABC}$.

(b) If the radius of the circle is 5 cm, show that the total area of the shaded regions is in the form $k(\pi - \sqrt{3})$. State the value of k .

23 In Fig. R8, $\triangle ABC$ is similar to $\triangle APQ$.

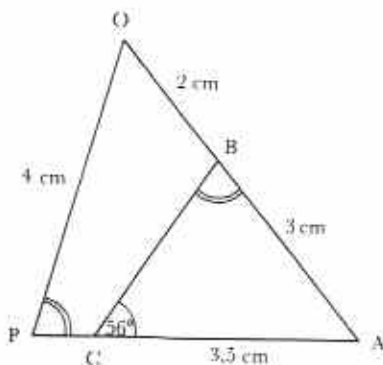


Fig. R8

Given $AB = 3$ cm, $BQ = 2$ cm, $AC = 3.5$ cm, $PQ = 4$ cm and $\hat{BCA} = 56^\circ$, and as much of the information given below as is necessary, calculate (a) BC , (b) the area of $\triangle APQ$. ($\sin 56^\circ = 0.829$, $\cos 56^\circ = 0.559$)

24 Solve the equations

(a) $\frac{x-1}{12} + \frac{x+1}{3} = \frac{2x-1}{4}$

(b) $(x+2)^2 = (2x-5)^2$

25 Table R2 shows the number of fish caught by a fisherman in one week.

Table R2

Sunday	0
Monday	0
Tuesday	8
Wednesday	19
Thursday	48
Friday	0
Saturday	2

For this data, calculate (a) the mode, (b) the median, (c) the mean.

26 A bag contains 5 red discs and 4 blue discs. Two discs are chosen at random, without replacement.

(a) Copy and complete the probability tree diagram in Fig. R9.

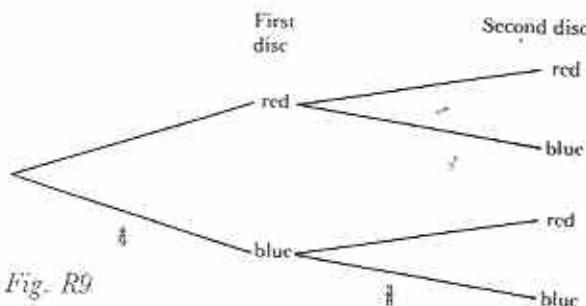


Fig. R9

(b) Find the probability that both discs are red.

27 Here is a number pattern:

$$1^3 = 1$$

$$2^3 = 3 + 5$$

$$3^3 = 7 + 9 + 11$$

$$4^3 = 13 + 15 + 17 + 19$$

$$5^3 =$$

(a) Write down the line of numbers for 5^3 .

(b) How many numbers will be in the line for 100^3 ?

(c) Use the pattern to find the value of $1^3 + 2^3 + 3^3 + 4^3 + 5^3 + 6^3$ without working out each cube separately.

28 The grid in Fig. R10 shows $\triangle ABC$. $A'(3; 2)$ is the image of $A(1; 2)$ after a one-way stretch with the y -axis invariant.

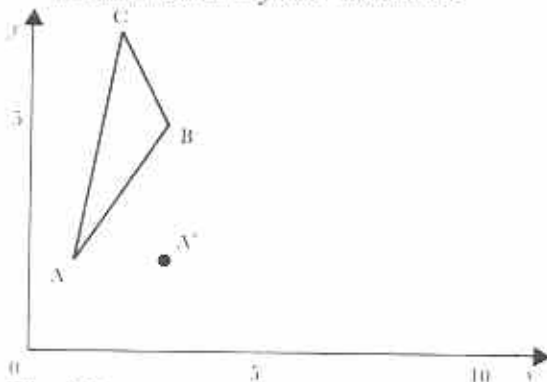


Fig. R10

(a) What is the stretch factor?

(b) What are the coordinates of B' and C' , the images of B and C under the same stretch?

(c) What is the matrix which represents the stretch?

29 Fig. R11 (overleaf) is a conversion graph which gives the rate of exchange from Z\$ to French Francs.

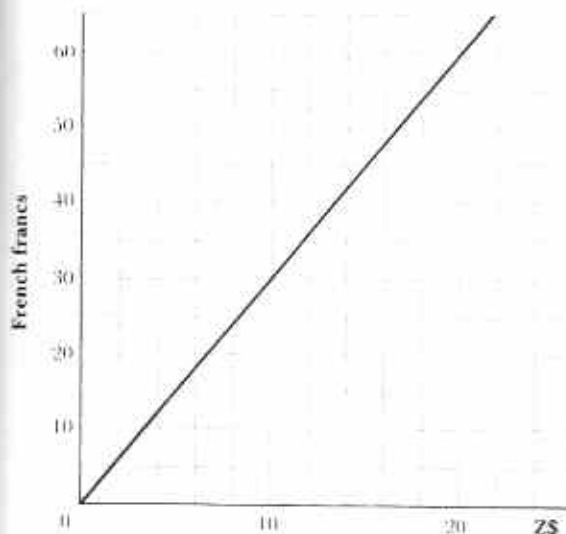


Fig. R11

Examination 1

Paper 2 (2 h 30 min)

Instructions to candidates:

Answer **all** the questions in Section A and any **three** questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

All working must be clearly shown. It should be done on the same sheet as the rest of the answer. Omission of essential working will result in loss of marks. If the degree of accuracy is not specified in the question and if the answer is not exact, three-figure accuracy is required.

Mathematical tables [or electronic calculators]* may be used to evaluate **explicit** numerical expressions.

Mathematical tables and graph paper are provided.

*The section within the square brackets is omitted in the non-calculator version of the paper.

Section A [64 marks]

Answer **all** the questions in this section.

- 1 (a) Ms Dube saved \$5 200 to buy a house. She estimates that she will have to use \$500 for legal costs and \$800 for moving expenses. She will use the remainder as a deposit for the house.

- (a) Use the graph to find
 (i) The number of French Francs equivalent to Z\$10.
 (ii) The cost in Z\$ of a watch bought in France for 55 Francs.
 (b) A new exchange rate gives Z\$1 = 3,25 Francs.
 (i) On a copy of the graph, draw a *new* line to represent this.
 (ii) Use your line to estimate the new cost in Z\$ of the watch above.

Table R3

Use this Ready Reckoner to decide which plan is best for you

	Plan A	Plan B	Plan C	Plan D
Minimum cover you get	\$10 000	\$20 000	\$35 000	\$50 000
Double cover if death is accidental	\$20 000	\$40 000	\$70 000	\$100 000
Age (nearest)	Amount you pay monthly			
20-30	\$5,00	\$6,00	\$7,35	\$10,50
31-35	\$5,00	\$6,20	\$9,10	\$13,00
36-40	\$5,40	\$8,40	\$12,95	\$18,50
41-45	\$6,00	\$12,00	\$19,25	\$27,50
46-50	\$9,30	\$18,60	\$30,80	—
51-55	\$14,70	\$29,40	—	—

(i) How much does she have for the deposit?

[1]

(ii) A building society offers her a maximum loan of 95% of the cost of the house. The other 5% represents the deposit, which she must pay. What would be the most expensive house she could afford?

[4]

(b) Table R3, on page 271, is a ready reckoner which can be used to work out the cost of life insurance premiums.

(i) Ms Dube, who is 29, chooses Plan B. How much is her monthly premium?

[1]

(ii) A 46-year-old man wants a minimum cover of \$35 000. What total premium will he pay in a year?

[2]

(iii) What is the maximum age of a person who can be insured for an accidental death benefit of \$100 000?

[2]

(iv) A teacher pays an *annual* premium of \$109,20. What is the age range of the teacher and what plan was chosen?

[2]

2 (a) In Fig. R12, the rectangle on the left is twice the area of that on the right.

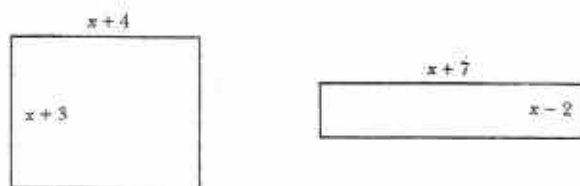


Fig. R12

(i) Form a quadratic equation in x and show that it reduces to $x^2 + 3x - 40 = 0$.

[3]

(ii) Solve the equation, stating which solution is realistic in terms of the given data.

[3]

(iii) Find the area of the larger rectangle.

[1]

(b) When the weather is cold, a strong wind makes it feel colder. The following formula gives the *effective* temperature, $e^\circ\text{C}$, when the *actual* temperature is $a^\circ\text{C}$ and the wind is blowing at w km/h:

$$e = \left(1 + \frac{w}{50}\right)a - \left(3 + \frac{3}{5}w\right)$$

For example, when the wind is 10 km/h the formula simplifies to

$$e = 1,2a - 9$$

(i) Find the simplified formula when the wind is 25 km/h.

[2]

(ii) If the actual temperature is 7°C when the wind speed is 30 km/h, find the effective temperature.

[3]

3 In Fig. R13, ABCD is a parallelogram in which $AB = 7$ cm, $BC = 10$ cm and diagonal $AC = 9$ cm.

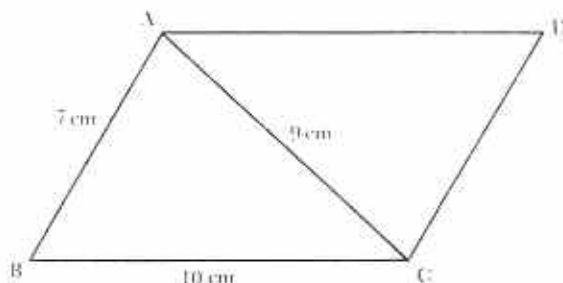


Fig. R13

Calculate:

(a) $\angle BAC$,

[4]

(b) the perpendicular distance between AB and DC.

[3]

4 (a) Ice lollipops are made of frozen fruit juice. They are in the shape of a triangular prism 11 cm long whose regular cross-section is an equilateral triangle of side 2,5 cm.

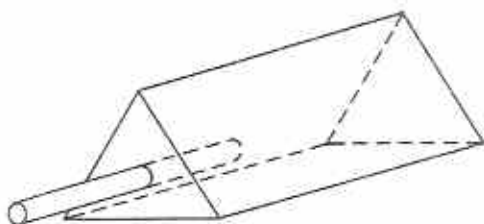


Fig. R14

Calculate

(i) the area of the triangular cross-section,

[3]

(ii) the volume of the prism,

[1]

- (iii) the volume of frozen fruit juice if the stick takes up 4 cm^3 of the prism, [1]
 (iv) the volume of liquid fruit juice required to make one lollipop if the fruit juice increases in volume by 6% when it is frozen. [3]
 (b) A model ship is made to a scale of 1:200.
 (i) The mast of the model is 9 cm high. What is the height of the mast on the ship in metres? [2]
 (ii) The volume of the part of the ship below the water line is $3.2 \times 10^4 \text{ m}^3$. Calculate the volume of the corresponding part on the model in cm^3 . [3]

- 5 (a) In Fig. R15, tangent PQ touches circle ABT at T. $\angle OAB = 12^\circ$ and $BO \parallel TA$.

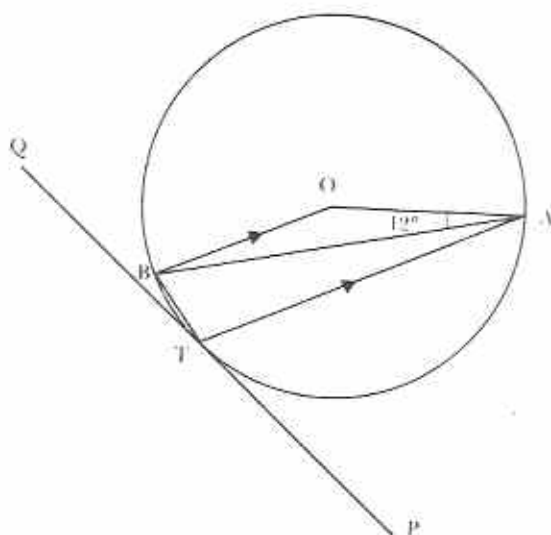


Fig. R15

Calculate, giving reasons,

- (i) $\angle OBA$, [1]
 (ii) reflex $\angle BOA$, [2]
 (iii) $\angle ATB$, [1]
 (iv) $\angle QTB$, [2]
 (v) $\angle ATP$. [1]
 (b) In Fig. R16, $OA = 3a$, $OB = 4b$, $OP:PA = 2:1$ and $OQ:QB = 3:1$. M is the mid-point of PQ and N is the mid-point of AB.
 (i) Express OP in terms of a . [1]
 (ii) Express QP in terms of a and b . [2]
 (iii) Find MN in terms of a and b . [3]

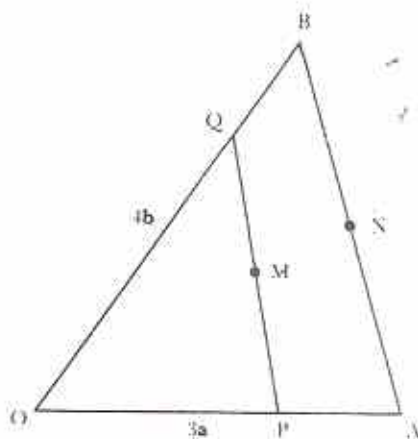


Fig. R16

- 6 The coordinates of the vertices of $\triangle ABC$ are $A(0;0)$, $B(2;0)$ and $C(1;2)$. $\triangle A_1B_1C_1$ is the image of $\triangle ABC$ under the transformation T_1 whose matrix is $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.
 (a) Find the coordinates of A_1 , B_1 and C_1 . [2]
 $\triangle A_2B_2C_2$ is the image of $\triangle A_1B_1C_1$ under transformation T_2 whose matrix is $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$.
 (b) Find the coordinates of A_2 , B_2 , C_2 . [2]
 (c) Find the matrix of the single transformation under which $\triangle A_2B_2C_2$ is the image of $\triangle ABC$. [3]

Section B [36 marks]

Answer **three** questions in this section.

Each question carries 12 marks.

- 7 A manager and a craftsman together form a small Co-operative making two models of chairs: Standards and Specials. The craftsman works for not more than 42 hours per week. Due to administration commitments, the manager cannot spend more than 34 hours a week in the workshop making chairs. The Standard chair requires 1 hour's work by the craftsman and 2 by the manager. The Special chair requires 3 hour's work by the craftsman and 1 by the manager. At least 6 of the Standard and 6 of the Special models are made each week.

(a) If, in a week, x Standard and y Special chairs are made, express the above information in the form of four inequalities in x and y . [4]

(b) Taking scales of 2 cm to 10 models on each axis, draw these inequalities. Show the region in which x and y are valid by leaving it clear. [4]

(c) If the profit on a Standard chair is \$24 and on a Special chair is \$40, find

(i) how many of each model should be made each week in order to obtain the greatest profit. [3]

(ii) the profit in this case. [1]

- 8 21 people work in an office. Fig. R17 gives some details about the people. In the Venn diagram,

$A = \{\text{audio-typists}\}$

$S = \{\text{shorthand-typists}\}$

$W = \{\text{word-processing users}\}$

Each person is a member of at least one of these sets.

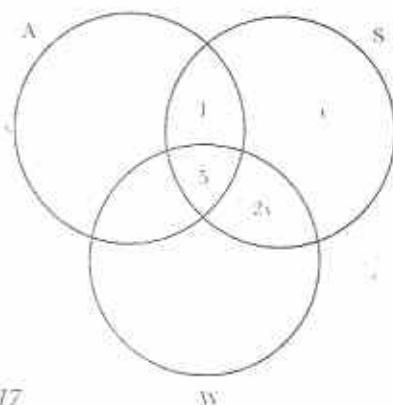


Fig. R17

(a) 15 people can do shorthand-typing. Find x . [2]

(b) 4 people cannot do either audio-typing or shorthand-typing. How many can do audio-typing? [4]

(c) No one does only audio-typing. Find the number of people who can

(i) use a word-processor, [2]

(ii) use a word-processor and do audio-typing. [2]

(d) Make a copy of the Venn diagram and shade the region $W \cap A' \cap S'$. [1]

Write a brief description of this set. [1]

- 9 VABCD is a right pyramid with a square base ABCD of side 5 cm. Each of the pyramid's four triangular faces is inclined at 75° to the base. Calculate

(a) the perpendicular height of the pyramid [2]

(b) the length of the slant edge VA, [4]

(c) the angle that VA makes with the horizontal, [2]

(d) the total surface area of the pyramid. [4]

- 10 The container of a petrol lorry is a cylinder with its axis horizontal. Its internal length is 7 m and its internal diameter is 3 m.

Fig. R18 shows a vertical section of the cylinder when the maximum depth of petrol it contains is 0.75 m. AB indicates the upper surface of the petrol.

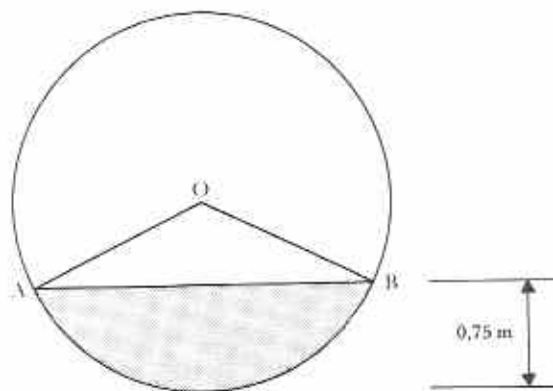


Fig. R18

Calculate

(a) the angle AOB, where O is the centre of the circular section, [1]

(b) the area of sector AOB, [1]

(c) the shaded area in Fig. R18, [1]

(d) the mass of petrol in the lorry, if 1 m³ of petrol has a mass of 700 kg. [1]

- 11 The chest measurements of 100 people were found to the nearest cm. Table R4 gives the result of the survey.

(a) Make a cumulative frequency table which shows the numbers of people with chest measurements of 80 cm or less, 85 cm or less, 90 cm or less, ..., 115 cm or less. [1]

Table R4

chest (cm)	76-80	81-85	86-90	91-95
frequency	3	10	14	20

chest (cm)	96-100	101-105	106-111	111-115
frequency	23	15	9	6

(b) Using a horizontal scale of 2 cm to represent 5 cm of chest measurement and a vertical scale of 2 cm to 10 people, draw a cumulative frequency graph of the survey measurements from 80 cm to 115 cm. [4]

(c) Use the graph to estimate the median chest measurement. [2]

(d) Use the graph to estimate the number of people with a chest measurement less than or equal to 103 cm. [1]

(e) Two people are chosen at random from those surveyed. What is the probability that both people have chest measurements greater than 100 cm? [3]

- 12 Given $y = \frac{9}{x}$, (a) copy and complete Table R5, giving values of y to 2 decimal places where necessary. [2]

Table R5

x	1	2	3	4	5	6	7	8	9
y	9	4,5		2,25				1,13	1

(b) Using scales of 1 cm to 1 unit on both axes, draw the graph of $y = \frac{9}{x}$ for $1 \leq x \leq 9$. [4]

(c) On the same axes draw the graph of $y = 8 - x$. [2]

(d) Find the values of x at the points where the straight line meets the curve. [2]

(e) Of which quadratic equation are the above values the solutions? Give your answer in the form $ax^2 + bx + c = 0$. [2]

Examination 2

Paper 1 (2 h 30 min)

Instructions to candidates:

All questions may be attempted.

Answers are to be written in the spaces provided.*

If working is needed for any question, it must be shown in the space below that question.

Omission of essential working will result in loss of marks.

Questions 1 to 19 carry 3 marks each.

Questions 20 to 27 carry 4 marks each.

Question 28 carries 5 marks.

Question 29 carries 6 marks.

* The examination paper contains spaces for working and answers. To save space in this book, these spaces have been omitted.

NEITHER MATHEMATICAL TABLES NOR SLIDE RULES NOR CALCULATORS MAY BE USED IN THIS PAPER.

1 Express 60 300

(a) to two significant figures,

(b) in standard form,

(c) as a product of prime factors.

2 Evaluate

(a) $(\frac{3}{4} - \frac{2}{3}) \times 1\frac{1}{5}$

(b) $\frac{0,6 + 0,75}{0,6 \times 0,75}$

3 If $x = 4$ and $y = -1$, evaluate

(a) $x + 4y$, (b) $x^2 - y^2$, (c) $(x - y)^2$.

4 Show the solution set of the inequality

$$3(2 + 3x) < 2x - 1$$

on a copy of the number line in Fig. R19.



Fig. R19

5 (a) factorise $a^2 + 5ax - 3ax - 15x^2$.

(b) Express m dollars and n cents in cents.

6 $\triangle ABC$ is right-angled at B. X is a point on AB such that $\angle AX = \angle XC$. If $\angle ACB = 71^\circ$ calculate (a) $\angle BAC$, (b) $\angle BXC$.

- 7 Each of the small circles in Fig. R20 is of radius r cm.

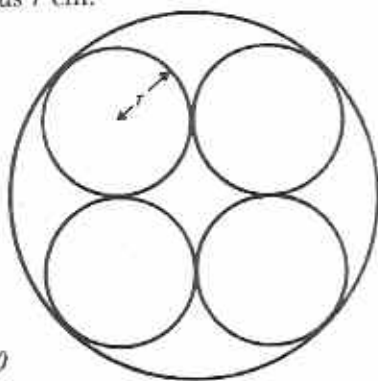


Fig. R20

- (a) Express the radius of the large circle in terms of r .
 (b) Express the ratio

$$\frac{\text{area of large circle}}{\text{area of one small circle}}$$

in its simplest form.

- 8 Write down the positive square roots of the following.
 (a) 0.0025 (b) $7\frac{1}{9}$ (c) $25d^{36}$
 9 Two geometrically similar tins have heights of 7 cm and 21 cm. If the smaller tin holds 250 g of milk powder, how many kg does the larger one hold?
 10 Find the value of
 (a) $32^{\frac{3}{2}}$; (b) $0.04^{-1\frac{1}{2}}$; (c) $25^{\frac{1}{2}} \times 8^{-\frac{2}{3}}$.
 11 In a family, the average mass of the five children is 34 kg, the mother is 59 kg and the father is 72 kg. Calculate the average mass of all 7 members of the family.
 12 A jug of water has a mass of 3.9 kg when full and 1.5 kg when quarter full. What is the mass of the jug when empty?

13

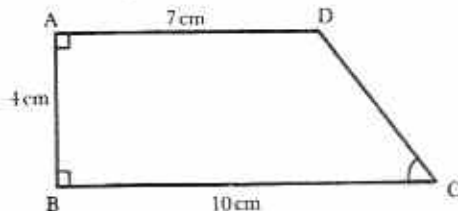


Fig. R21

Given Fig. R21, (a) calculate CD, (b) find the value of $\cos \angle BCD$, expressing your answer as a decimal.

- 14 Given that x is an even integer, find the values of x which satisfy both $x \geq 4$ and $3x + 7 > 10$.
 15 A hall contains 175 people. 12% of them are children and there are 56 men. How many women are in the hall?
 16 If $c = ab - \frac{b}{a}$,
 (a) find c when $a = \frac{2}{3}$ and $b = -6$,
 (b) express b in terms of a and c .
 17 Fig. R22 is a regular pentagram, i.e. a fully symmetrical five-pointed star as shown.

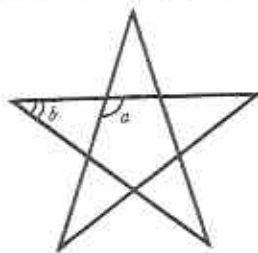


Fig. R22

Calculate the sizes of the angles marked a and b .

- 18 In Fig. R23, $BC = 10$ cm, $\angle A = 40^\circ$, $\angle C = 50^\circ$.

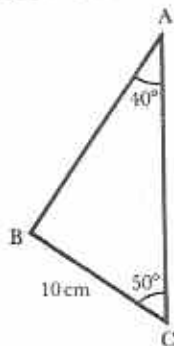


Fig. R23

Use as much of the given information as is necessary to calculate (a) AB, (b) AC.

$$(\sin 40^\circ = \cos 50^\circ = 0.64$$

$$\cos 40^\circ = \sin 50^\circ = 0.77$$

$$\tan 40^\circ = 0.84, \tan 50^\circ = 1.19)$$

- 19 A machine part is made of metal. The metal is a mixture of copper and zinc whose masses are in the ratio 13:7. If there are 147 g of zinc in the machine part, what is its total mass?
 20 Given that $S = \{1; 2; 3; \dots; 9; 10\}$
 $S = \{\text{perfect squares}\}$
 $T = \{1; 3; 6; 10\}$,
 (a) Mark the members of these sets on a copy of the Venn diagram in Fig. R24.

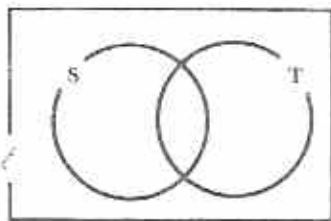


Fig. R24

(b) Using your diagram or otherwise (i) list the members of $S' \cap T'$, (ii) find $n(T' \cup S)$.

- 21 In Fig. R25, $\mathbf{AC}$ represents the column vector $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

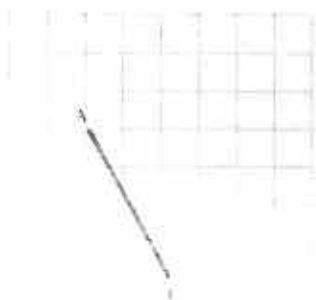


Fig. R25

$\mathbf{AC}$ is a diagonal of a square $ABCD$ such that $\mathbf{AD} = \begin{pmatrix} p \\ q \end{pmatrix}$ where p and q are both negative.

- (a) Sketch the square $ABCD$ on a copy of Fig. R25.
 (b) Write down the values of p and q .
 (c) Write down $\mathbf{AB}$ in column vector form.
 (d) Write down $\mathbf{BD}$ in column vector form.
- 22 $\begin{pmatrix} 4x & x-5 \\ 1-3x & x \end{pmatrix}$ is the inverse matrix of $\begin{pmatrix} x & 5-x \\ 3x-1 & 4x \end{pmatrix}$.

Find two values of x for which this is true.

- 23 A straight line passes through the points $(5; 0)$ and $(-4; 3)$. Find (a) the gradient of the line, (b) its equation, (c) the coordinates of the point where it cuts the line $x = 7$.
- 24 In Fig. R26, $ABCD$ is a trapezium such that $\angle ADB = \angle BCD$, $AB = 4 \text{ cm}$ and $BD = 6 \text{ cm}$.

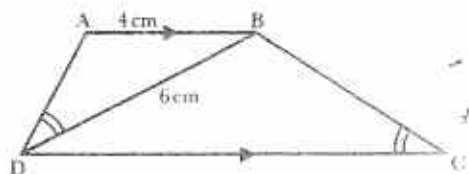


Fig. R26

- (a) Calculate the length of CD .
 (b) Find the ratio $\frac{\text{area of } \triangle ABD}{\text{area of } \triangle BCD}$.

- 25 In Fig. R27, O is the centre of the circle. $\angle APB$ and $\angle OAB = 34^\circ$.

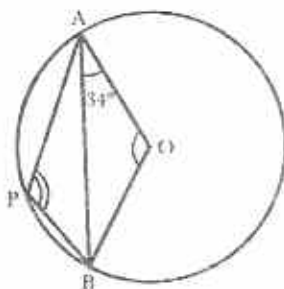


Fig. R27

Find the size of (a) $\angle AOB$, (b) $\angle APB$.

- 26 Fig. R28 shows a shape S drawn on a cartesian plane.

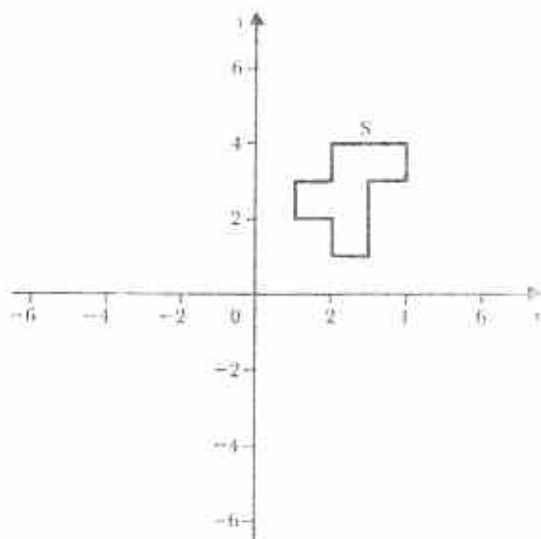


Fig. R28

On a copy of Fig. R28, draw the image of S under a

- translation by vector $\begin{pmatrix} -7 \\ 1 \end{pmatrix}$ (label the image T),
- reflection in the line $y = -x$ (label the image M),
- clockwise rotation of 90° about the point $(0; -2)$ (label the image R).

27 y varies inversely as the positive square root of x and $y = 12$ when $x = \frac{4}{9}$.

- Express y in terms of x .
- Find y when $x = 12\frac{1}{4}$.
- Find n such that $x = y = n$.

28 The curve in Fig. R29 is the graph of $y = 7 - x - x^2$ for values of x from -4 to 3 . The region above the x -axis between $x = -3$ and $x = 2$ has been shaded.

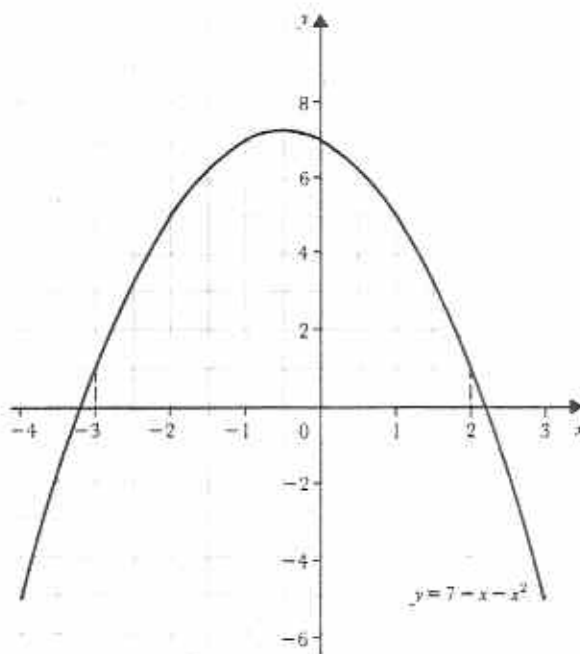


Fig. R29

Estimate

- the greatest value of $7 - x - x^2$,
- the solution of the equation $7 - x - x^2 = 0$,
- the area of the shaded region.

29 Fig. R30 is the speed-time graph of a car journey.

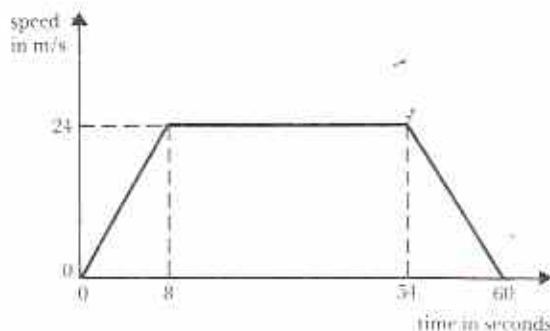


Fig. R30

Calculate

- the total distance travelled during the 60 seconds,
- the acceleration of the car during the first 8 s,
- the speed of the car at the end of the 56th second.

Examination 2

Paper 2 (2 h 30 min)

Instructions to candidates:

Answer **all** the questions in Section A and any **three** questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

All working must be clearly shown. It should be done on the same sheet as the rest of the answer. Omission of essential working will result in loss of marks. If the degree of accuracy is not specified in the question and if the answer is not exact, three figure accuracy is required.

Mathematical tables [or electronic calculators]* may be used to evaluate **explicit** numerical expressions. Mathematical tables and graph paper are provided.

*The section within the square brackets is omitted in the non-calculator version of the paper.

Section A [64 marks]

Answer **all** the questions in this section.

- (a) A farmer has 350 ha of land. After selling some land he is left with 329 ha. What percentage of his land did he sell? [2]

(b) $M = \begin{pmatrix} 4 & 0 \\ -2 & 2 \end{pmatrix}$ and $N = \begin{pmatrix} 1 & -5 \\ 1 & -2 \end{pmatrix}$.

Find

- (i) MN [2]
 (ii) NM [2]
 (iii) x and y if

$$M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \end{pmatrix}. \quad [4]$$

- 2 (a) In Fig. R31, O is the centre of circle PQRS, PQ is a diameter, $QR = RS$ and $\angle PSO = 42^\circ$.

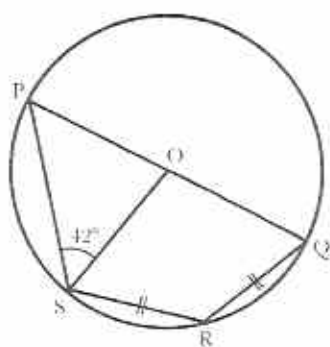


Fig. R31

Calculate the angles of quadrilateral PQRS. [4]

(b) The internal radius and height of a cylindrical oil drum are 30 cm and 90 cm respectively. The drum contains oil to a depth of 70 cm. Use the value 3.142 for π to calculate

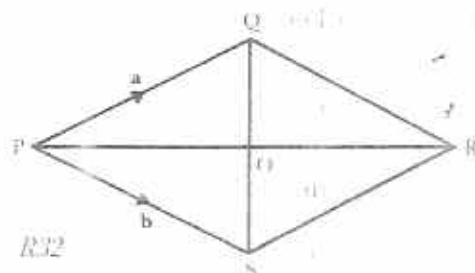
- (i) the area of the cylinder which is in contact with the oil, [4]
 (ii) the extra volume of oil required to fill the drum to the top. [4]

3 (a) Simplify $\frac{2}{x+2} - \frac{5}{3x-1}$. [3]

(b) Solve $\frac{2}{x+2} = \frac{5}{3x-1}$. [3]

(c) Factorise $16a^2 - 9$. Hence or otherwise express 1591 as the product of two prime numbers. [4]

- 4 (a) In Fig. R32, O is the centre of rhombus PQRS, $PQ = \mathbf{a}$ and $PS = \mathbf{b}$.



Write down expressions for the following in terms of $\mathbf{a}$ and $\mathbf{b}$.

- (i) $\mathbf{PR}$ [2]
 (ii) $\mathbf{OP}$ [2]
 (iii) $\mathbf{CQ}$ [3]

(b) X sold an article to Y at a profit of 20%. Y then sold it to Z at a loss of 20% of what it cost her. Calculate the ratio *final price:original price* in its simplest form. [3]

- 5 (a) A triangle has sides of length x cm, $(2x-1)$ cm and $(2x+1)$ cm. If its perimeter is 40 cm,

- (i) find the lengths of the sides of the triangle; [3]
 (ii) state the size of the largest angle of the triangle, giving a reason; [2]
 (iii) calculate the size of the smallest angle of the triangle. [3]

(b) Find the value of $\frac{2\sqrt{a}}{b}$ when

$a = 1.21 \times 10^{-4}$, $b = 4.4 \times 10^{-6}$, giving the answer in standard form. [4]

- 6 (a) In a lifting mechanism, the relation between the load W tonnes and the effort P kg is of the form $P = aW + b$, where a and b are constants. Table R6 gives the results of a lifting experiment.

Table R6

load, W tonnes	2	5
effort, P kg	2.32	3.19

- (i) Find a and b and the equation connecting P and W . [3]
 (ii) Sketch a graph of the relation showing where the line cuts the axes. [3]
 (b) Find the values of d for which $d - 2)^2 : (d + 3)^2 = 1:4$. [4]

Section B [36 marks]

Answer three questions in this section.

Each question carries 12 marks.

- 7 (a) The vertex V of a pyramid is vertically above the centre, O , of its rectangular base $PQRS$. If $PQ = 9$ cm, $QR = 12$ cm and $VO = 6$ cm, calculate
- $\angle VPO$. [3]
 - the angle between the faces PVS and QVR . [3]
- (b) The matrix $\begin{pmatrix} -3 & 0 \\ 0 & 1 \end{pmatrix}$ represents a transformation C .
- Find the image of $(2; 7)$ under C . [2]
 - If $(5; -1)$ is the image of point $(x; y)$ under C , find x and y . [2]
 - Describe C as completely as possible. [2]
- 8 The hourly earnings of 50 people are given in Table R7.

Table R7

earnings per hour (cents)	number of people (frequency)
151–175	1
176–200	4
201–225	17
226–250	15
251–275	11
276–300	2

- What is the modal class of hourly earnings? [1]
- Use mid-values to calculate the mean hourly earnings. [4]
- Construct a cumulative frequency table and draw the corresponding cumulative frequency curve. Use this to estimate the median hourly earnings. [5]
- One person is chosen at random. What is the probability that this person earns less than \$2.26 per hour? [1]
- Two people are chosen at random from the 50. Find the probability that
 - both are in the 226–250¢ group. [1]

- one is in that group and the other is not. [1]

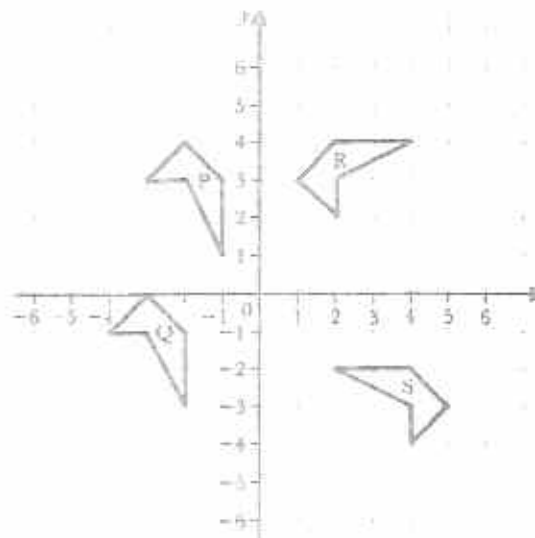
9 Shapes P , Q , R and S are as given in Fig. R33.

Fig. R33

- Write down the column vector representing the single translation which maps P onto Q .
 - R is the image of P under a clockwise rotation. Find
 - the angle of rotation,
 - the coordinates of the centre of rotation.
 - S is a reflection of P in a line m . Find the equation of m .
 - If P is transformed by a shear represented by the matrix $\begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix}$,
 - state the equation of the invariant line,
 - find the coordinates of the vertices of the image of P .
- 10 Fig. R34 represents a field $ABCD$.
- Use the given dimensions to calculate
 - BD ,
 - BC ,
 - the area of the field in m^2 .

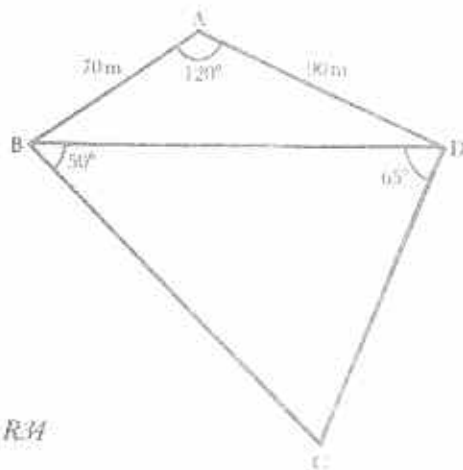


Fig. R34

- (b) If D is due East of B, find the bearing of
- B from C. [2]
 - C from B. [1]
 - A from B. [2]
- 11 An object moves in a straight line so that its velocity x m/s after t seconds is given by
- $$x = 2t^2 - 5t + 8.$$

Table R8 gives some corresponding values of x and t .

Table R8

t	0	1	2	3	4	5	6
x	8		6		20		50

- (a) Copy and complete Table R8 by finding the values of x which correspond to $t = 1; 3; 5$. [3]
- (b) Using 2 cm to 1 second horizontally and 5 m/s vertically, draw the graph of $v = 2t^2 - 5t + 8$ for the range $0 \leq t \leq 6$. [4]

- (c) Use your graph to estimate
- the period of time during which the object is decelerating, [2]
 - the velocity when the acceleration is zero, [1]
 - the acceleration after 4 seconds. [2]

- 12 Answer this question on a sheet of plain paper. Construct triangle ABC so that

- $BC = 10$ cm, $\angle ABC = 60^\circ$ and $\angle BCA = 15^\circ$.
- On the diagram write the length of AC. [3]
 - Draw the circumcircle of triangle ABC. [2]
 - Construct the locus of a set of points which are equidistant from A and B. [2]
 - Hence mark a point P such that $\angle APB = 45^\circ$ and $AP = PB$. [2]
 - Mark a point Q such that $\angle AQB = 45^\circ$ and $AB = AQ$. [2]
 - Measure PQ. [1]

Mensuration tables and formulae, four-figure tables

SI units

Mass

The **gram** is the basic unit of mass.

unit	abbreviation	basic units
1 kilogram	1 kg	1 000 g
1 hectogram	1 hg	100 g
1 decagram	1 dag	10 g
1 gram	1 g	g
1 decigram	1 dg	0.1 g
1 centigram	1 cg	0.01 g
1 milligram	1 mg	0.001 g

The **tonne (t)** is used for large masses. The most common measures of mass are the milligram, the gram, the kilogram and the tonne.

$$1 \text{ g} = 1\,000 \text{ mg}$$

$$1 \text{ kg} = 1\,000 \text{ g} = 1\,000\,000 \text{ mg}$$

$$1 \text{ t} = 1\,000 \text{ kg} = 1\,000\,000 \text{ g}$$

Time

The **second** is the basic unit of time.

unit	abbreviation	basic units
1 second	1 s	1 s
1 minute	1 min	60 s
1 hour	1 h	3 600 s

Length

The **metre** is the basic unit of length.

unit	abbreviation	basic unit
1 kilometre	1 km	1 000 m
1 hectometre	1 hm	100 m
1 decametre	1 dam	10 m
1 metre	1 m	1 m
1 decimetre	1 dm	0.1 m
1 centimetre	1 cm	0.01 m
1 millimetre	1 mm	0.001 m

The most common measures are the millimetre, the metre and the kilometre.

$$1 \text{ m} = 1\,000 \text{ mm}$$

$$1 \text{ km} = 1\,000 \text{ m} = 1\,000\,000 \text{ mm}$$

Area

The **square metre** is the basic unit of area. Units of area are derived from units of length.

unit	abbreviation	relation to other units of area
square millimetre	mm ²	
square centimetre	cm ²	1 cm ² = 100 mm ²
square metre	m ²	1 m ² = 10 000 cm ²
square kilometre	km ²	1 km ² = 1 000 000 m ²
hectare (for land measure)	ha	1 ha = 10 000 m ²

Volume

The **cubic metre** is the basic unit of volume. Units of volume are derived from units of length.

unit	abbrevi- ation	relation to other units of volume
cubic millimetre	mm ³	
cubic centimetre	cm ³	1 cm ³ = 1 000 mm ³
cubic metre	m ³	1 m ³ = 1 000 000 cm ³

Capacity

The **litre** is the basic unit of capacity. 1 litre takes up the same space as 1 000 cm³.

unit	abbrevi- ation	relation to other units of capacity	relation to units of volume
millilitre	ml		1 ml = 1 cm ³
litre	l	1 l = 1 000 ml	1 l = 1 000 cm ³
kilolitre	kl	1 kl = 1 000 l	1 kl = 1 m ³

Money

Some African currencies

Zimbabwe	100 cents (c) = 1 Dollar (\$)
Botswana	100 thebe (t) = 1 Pula (P)
Kenya	100 cents (c) = 1 Shilling (Sh)
Malawi	100 tambala (t) = 1 Kwacha (K)
Mozambique	100 centavos (c) = 1 Metical (M)
Nigeria	100 kobo (k) = 1 Naira (₦)
Zambia	100 ngwee (n) = 1 Kwacha (K)

Other currencies

Britain	100 pence (p) = 1 Pound (£)
USA	100 cents (c) = 1 Dollar (\$)

Exchange rates

At the time of going to press, \$1 (Zimbabwe) was equivalent to the following.

US dollar	\$0,50
UK sterling	£0,30
Botswana	P1,05
Kenya	Sh11,00
Mozambique	M318,00
Zambia	K5,20

Note: Exchange rates change from day to day. The above rates may be taken only as approximate.

The calendar

Remember this poem:

Thirty days has September,
April, June and November,
All the rest have thirty-one,
Excepting February alone;
This has twenty-eight days clear,
And twenty-nine in each Leap Year.

For a Leap Year, the date must be divisible by 4. Thus 1964 was a Leap Year.

Century year dates, such as 1900 and 2000, are Leap Years only if they are divisible by 400. Thus 1900 was not a Leap Year but 2000 will be a Leap Year.

Multiplication table

×	1	2	3	4	5	6	7	8	9	10
1	1	2	3	4	5	6	7	8	9	10
2	2	4	6	8	10	12	14	16	18	20
3	3	6	9	12	15	18	21	24	27	30
4	4	8	12	16	20	24	28	32	36	40
5	5	10	15	20	25	30	35	40	45	50
6	6	12	18	24	30	36	42	48	54	60
7	7	14	21	28	35	42	49	56	63	70
8	8	16	24	32	40	48	56	64	72	80
9	9	18	27	36	45	54	63	72	81	90
10	10	20	30	40	50	60	70	80	90	100

Divisibility tests

Any whole number is exactly divisible by

- 2 if its last digit is even
- 3 if the sum of its digits is divisible by 3
- 4 if its last two digits form a number divisible by 4
- 5 if its last digit is 5 or 0
- 6 if its last digit is even and the sum of its digits is divisible by 3
- 8 if its last three digits form a number divisible by 8
- 9 if the sum of its digits is divisible by 9
- 10 if its last digit is 0

Mensuration formulae

Plane shapes

	perimeter	area
square side s	$4s$	s^2
rectangle length l , breadth b	$2(l + b)$	lb
triangle base b , height h		$\frac{1}{2}bh$
parallelogram base b , height h		bh
trapezium height h , parallels a and b		$\frac{1}{2}(a + b)h$
circle radius r	$2\pi r$	πr^2
sector of circle radius r , angle θ	$2r + \frac{\theta}{360} 2\pi r$	$\frac{\theta}{360} \pi r^2$

Solid shapes

	surface area	volume
cube edge s	$6s^2$	s^3
cuboid length l , breadth b , height h	$2(lb + bh + lh)$	lbh
prism height h , base area A		Ah
cylinder radius r , height h	$2\pi rh + 2\pi r^2$	$\pi r^2 h$
cone radius r , slant height l , height h	$\pi rl + \pi r^2$	$\frac{1}{3}\pi r^2 h$
sphere radius r	$4\pi r^2$	$\frac{4}{3}\pi r^3$

Trigonometrical formulae

Right-angled triangles

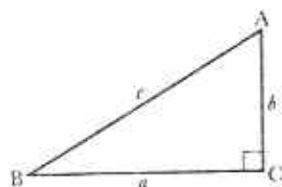


Fig. T1

In the right-angled triangle shown,
 $c^2 = a^2 + b^2$ (Pythagoras' theorem)

$$\tan B = \frac{b}{a} \quad \tan A = \frac{a}{b}$$

$$\sin B = \frac{b}{c} \quad \sin A = \frac{a}{c}$$

$$\cos B = \frac{a}{c} \quad \cos A = \frac{b}{c}$$

Obtuse angles

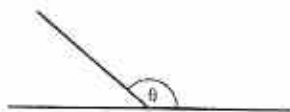


Fig. T2

$$\sin \theta = \sin (180^\circ - \theta)$$

$$\cos \theta = -\cos (180^\circ - \theta)$$

$$\tan \theta = -\tan (180^\circ - \theta)$$

Any triangle

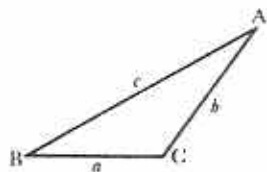
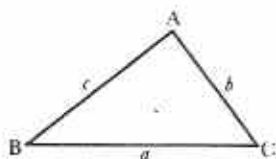


Fig. T3

In both triangles,

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

and

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{c^2 + a^2 - b^2}{2ca}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

(sine rule)

(cosine rule)

Symbols

symbol

=

≠

≈

≡

⇒

⇔

∝

>

<

≥

≤

°

°C

A, B, C,

AB

△ABC

||^{gm}ABCD

∠ABC

⊥

||

π

%

A = {p, q, r}

B = {1, 2, 3, ...}

C = {x: x is an integer}

n(A)

∈

∉

A'

{ } or ∅

∅

A ⊆ B

A ⊇ B

⊄, ⊈

A ∪ B

A ∩ B

a or $\vec{a}$ or $\overrightarrow{a}$

AB or $\overrightarrow{AB}$

|**AB**|

meaning

is equal to

is not equal to

is approximately equal to

is identical or

congruent to

leads to

is equivalent to

is proportional to

is greater than

is less than

is greater than or equal to

is less than or equal to

degree (angle)

degree Celsius

(temperature)

points (geometry)

the line through points A and B, or the distance between

points A and B

triangle ABC

parallelogram ABCD

angle ABC

is perpendicular to

is parallel to

pi

per cent

A is the set p, q, r

B is the infinite set 1, 2, 3 and so on

Set builder notation. C is the

set of numbers x such that x

is an integer

number of elements in set A

is an element of

is not an element of

complement of A

the empty set

the universal set

A is a subset of B

A contains B

negations of ⊆ and ⊇

union of A and B

intersection of A and B

vector **a**

vector **AB**

modulus of **AB**

Logarithms

x	Differences									
	0	1	2	3	4	5	6	7	8	9
16	0000	0043	0086	0128	0170	0212	0255	0294	0334	0374
17	0414	0453	0492	0531	0569	0607	0645	0682	0719	0755
18	0792	0829	0864	0899	0933	0967	1000	1032	1064	1095
19	1130	1171	1206	1239	1271	1303	1335	1367	1398	1429
20	1461	1492	1523	1553	1584	1614	1644	1673	1703	1732
21	1761	1790	1818	1847	1875	1903	1931	1959	1987	2014
22	2041	2068	2095	2122	2149	2175	2201	2227	2253	2279
23	2304	2330	2355	2380	2405	2430	2455	2480	2504	2529
24	2553	2577	2601	2625	2648	2672	2695	2718	2742	2765
25	2788	2810	2833	2856	2878	2900	2922	2944	2967	2989
26	3010	3032	3054	3075	3096	3118	3139	3160	3181	3201
27	3222	3243	3263	3284	3304	3324	3345	3365	3385	3404
28	3424	3444	3464	3483	3502	3521	3541	3560	3579	3598
29	3617	3636	3655	3674	3692	3711	3730	3749	3768	3786
30	3802	3820	3838	3856	3874	3892	3909	3927	3945	3962
31	3979	3997	4014	4031	4048	4065	4082	4099	4116	4133
32	4150	4166	4183	4200	4216	4232	4249	4265	4281	4298
33	4314	4330	4346	4362	4378	4393	4409	4425	4440	4456
34	4472	4487	4502	4518	4533	4548	4563	4579	4594	4609
35	4624	4639	4654	4669	4683	4698	4713	4728	4742	4757
36	4771	4786	4800	4814	4829	4843	4857	4871	4886	4900
37	4914	4928	4942	4955	4969	4983	4997	5011	5024	5038
38	5051	5065	5079	5092	5105	5119	5132	5145	5159	5172
39	5185	5198	5211	5224	5237	5250	5263	5276	5289	5302
40	5315	5328	5340	5353	5365	5378	5391	5403	5415	5428
41	5441	5453	5465	5478	5490	5502	5514	5527	5539	5551
42	5563	5575	5587	5599	5611	5623	5635	5647	5658	5670
43	5682	5694	5705	5717	5729	5740	5752	5763	5775	5786
44	5797	5809	5820	5832	5843	5855	5866	5877	5888	5899
45	5911	5922	5933	5944	5955	5966	5977	5988	5999	6010
46	6021	6031	6042	6052	6063	6073	6083	6093	6103	6113
47	6123	6133	6143	6153	6163	6173	6183	6193	6203	6212
48	6222	6232	6242	6252	6262	6272	6282	6292	6302	6311
49	6321	6331	6341	6351	6361	6371	6381	6391	6401	6411
50	6421	6431	6441	6451	6461	6471	6481	6491	6501	6510
51	6520	6530	6540	6550	6560	6570	6580	6590	6600	6609
52	6618	6628	6638	6648	6658	6668	6678	6688	6698	6707
53	6717	6727	6737	6747	6757	6767	6777	6787	6797	6806
54	6816	6826	6836	6846	6856	6866	6876	6886	6896	6905
55	6915	6925	6935	6945	6955	6965	6975	6985	6995	7004
56	7014	7024	7034	7044	7054	7064	7074	7084	7094	7103
57	7113	7123	7133	7143	7153	7163	7173	7183	7193	7202
58	7212	7222	7232	7242	7252	7262	7272	7282	7292	7301
59	7311	7321	7331	7341	7351	7361	7371	7381	7391	7400
60	7410	7420	7430	7440	7450	7460	7470	7480	7490	7500
61	7509	7519	7529	7539	7549	7559	7569	7579	7589	7598
62	7608	7618	7628	7638	7648	7658	7668	7678	7688	7697
63	7707	7717	7727	7737	7747	7757	7767	7777	7787	7796
64	7806	7816	7826	7836	7846	7856	7866	7876	7886	7895
65	7905	7915	7925	7935	7945	7955	7965	7975	7985	7994
66	8004	8014	8024	8034	8044	8054	8064	8074	8084	8093
67	8103	8113	8123	8133	8143	8153	8163	8173	8183	8192
68	8202	8212	8222	8232	8242	8252	8262	8272	8282	8291
69	8301	8311	8321	8331	8341	8351	8361	8371	8381	8391
70	8400	8410	8420	8430	8440	8450	8460	8470	8480	8490
71	8500	8510	8520	8530	8540	8550	8560	8570	8580	8590
72	8600	8610	8620	8630	8640	8650	8660	8670	8680	8690
73	8700	8710	8720	8730	8740	8750	8760	8770	8780	8790
74	8800	8810	8820	8830	8840	8850	8860	8870	8880	8890
75	8900	8910	8920	8930	8940	8950	8960	8970	8980	8990
76	9000	9010	9020	9030	9040	9050	9060	9070	9080	9090
77	9100	9110	9120	9130	9140	9150	9160	9170	9180	9190
78	9200	9210	9220	9230	9240	9250	9260	9270	9280	9290
79	9300	9310	9320	9330	9340	9350	9360	9370	9380	9390
80	9400	9410	9420	9430	9440	9450	9460	9470	9480	9490
81	9500	9510	9520	9530	9540	9550	9560	9570	9580	9590
82	9600	9610	9620	9630	9640	9650	9660	9670	9680	9690
83	9700	9710	9720	9730	9740	9750	9760	9770	9780	9790
84	9800	9810	9820	9830	9840	9850	9860	9870	9880	9890
85	9900	9910	9920	9930	9940	9950	9960	9970	9980	9990
86	10000	10010	10020	10030	10040	10050	10060	10070	10080	10090
87	10100	10110	10120	10130	10140	10150	10160	10170	10180	10190
88	10200	10210	10220	10230	10240	10250	10260	10270	10280	10290
89	10300	10310	10320	10330	10340	10350	10360	10370	10380	10390
90	10400	10410	10420	10430	10440	10450	10460	10470	10480	10490
91	10500	10510	10520	10530	10540	10550	10560	10570	10580	10590
92	10600	10610	10620	10630	10640	10650	10660	10670	10680	10690
93	10700	10710	10720	10730	10740	10750	10760	10770	10780	10790
94	10800	10810	10820	10830	10840	10850	10860	10870	10880	10890
95	10900	10910	10920	10930	10940	10950	10960	10970	10980	10990
96	11000	11010	11020	11030	11040	11050	11060	11070	11080	11090
97	11100	11110	11120	11130	11140	11150	11160	11170	11180	11190
98	11200	11210	11220	11230	11240	11250	11260	11270	11280	11290
99	11300	11310	11320	11330	11340	11350	11360	11370	11380	11390

x	Differences									
	0	1	2	3	4	5	6	7	8	9
55	7404	7417	7430	7443	7455	7467	7479	7490	7501	7512
56	7482	7497	7505	7517	7529	7538	7546	7554	7561	7567
57	7534	7546	7554	7562	7568	7575	7580	7586	7591	7597
58	7634	7647	7657	7664	7672	7679	7686	7694	7701	7707
59	7709	7719	7723	7731	7738	7745	7752	7760	7767	7774
60	7782	7795	7803	7810	7818	7825	7832	7839	7846	7853
61	7851	7860	7868	7875	7882	7889	7896	7903	7910	7917
62	7926	7931	7938	7945	7952	7959	7966	7973	7980	7987
63	7993	8000	8007	8014	8021	8028	8035	8041	8048	8055
64	8067	8069	8075	8082	8089	8096	8102	8109	8116	8122
65	8129	8136	8142	8148	8156	8162	8169	8176	8182	8189
66	8195	8202	8209	8215	8222	8228	8235	8241	8248	8254
67	8261	8267	8274	8280	8287	8293	8299	8306	8312	8319
68	8325	8331	8338	8344	8351	8357	8363	8370	8376	8382
69	8388	8395	8401	8407	8414	8420	8428	8432	8439	8445
70	8451	8457	8463	8470	8476	8482	8489	8494	8499	8506
71	8513	8519	8525	8531	8537	8543	8549	8555	8561	8567
72	8572	8578	8585	8591	8597	8603	8609	8615	8621	8627
73	8633	8639	8645	8651	8657	8663	8669	8675	8681	8686
74	8692	8698	8704	8710	8716	8722	8727	8733	8739	8745
75	8751	8756	8762	8768	8774	8779	8785	8791	8797	8802
76	8806	8811	8816	8822	8828	8833	8842	8848	8854	8859
77	8865	8871	8876	8882	8887	8893	8899	8904	8910	8915
78	8921	8927	8932	8938	8943	8949	8954	8960	8966	8971
79	8975	8982	8987	8993	8998	9004	9009	9015	9020	9025
80	9031	9036	9042	9047	9053	9058	9063	9069	9074	9079
81	9085	9090	9096	9101	9106	9112	9117	9122	9128	9133
82	9138	9143	9149	9154	9159	9165	9170	9175	9180	9185
83	9191	9196	9201	9206	9212	9217	9222	9227	9232	9238
84	9243	9248	9253	9258	9263	9269	9274	9279	9284	9289
85	9294	9299	9304	9309	9315	9320	9325	9330	9335	9340
86	9345	9351	9355	9360	9365	9370	9375	9380	9385	9390
87	9395	9400	9405	9410	9415	9420	9425	9430	9435	9440
88	9445	9450	9455	9460	9465	9470	9474	9478	9484	9489
89	9494	9499	9504	9509	9513	9518	9523	9528	9533	9538
90	9542	9547	9552	9557	9562	9566	9571	9576	9581	9586
91	9599	9603	9608	9613	9619	9624	9629	9634	9638	9643
92	9638	9643	9647	9653	9657	9663	9667	9671	9675	9680
93	9685	9689	9694	9699	9704	9708	9713	9717	9722	9726
94	9731	9736	9741	9745	9750	9754	9759	9763	9768	9773
95	9777	9782	9786	9791	9795	9800	9805	9809	9814	9819
96	9823	9827	9832	9836	9841	9845	9850	9854	9859	9863
97	9868	9872	9877	9881	9886	9890	9894	9899	9903	9908
98	9912	9917	9921	9925	9930	9934	9938	9943	9946	9952
99	9956	9961	9965	9969	9974	9978	9983	9987	9991	9996

x	Differences										Differences
	0	1	2	3	4	5	6	7	8	9	
00	1000	1002	1005	1007	1009	1012	1014	1016	1019	1021	
01	1023	1026	1028	1030	1033	1035	1038	1040	1042	1045	
02	1047	1050	1052	1054	1057	1059	1062	1064	1067	1069	
03	1072	1074	1076	1079	1081	1084	1086	1089	1091	1094	
04	1095	1097	1100	1102	1104	1107	1110	1112	1114	1117	
05	1122	1125	1127	1130	1132	1135	1138	1140	1143	1146	
06	1148	1151	1153	1156	1158	1161	1164	1167	1169	1172	
07	1175	1178	1180	1183	1186	1189	1191	1194	1197	1199	
08	1202	1205	1208	1211	1213	1216	1219	1222	1225	1227	
09	1230	1233	1236	1239	1242	1245	1247	1250	1253	1256	
10	1259	1262	1265	1268	1271	1274	1276	1279	1282	1285	
11	1288	1291	1294	1297	1300	1303	1306	1309	1312	1315	
12	1318	1321	1324	1327	1330	1334	1337	1340	1343	1346	
13	1349	1352	1355	1358	1361	1365	1368	1371	1374	1377	
14	1380	1384	1387	1390	1393	1396	1400	1403	1406	1409	
15	1413	1416	1419	1422	1426	1429	1432	1435	1437	1442	
16	1445	1449	1452	1455	1459	1462	1466	1469	1472	1476	
17	1479	1483	1486	1489	1492	1496	1499	1503	1507	1510	
18	1514	1517	1521	1524	1528	1531	1535	1538	1542	1545	
19	1549	1552	1556	1559	1563	1567	1570	1574	1576	1581	
20	1585	1589	1592	1596	1600	1603	1607	1611	1614	1618	
21	1622	1626	1629	1633	1637	1641	1644	1648	1652	1656	
22	1660	1664	1667	1671	1675	1679	1683	1687	1690	1694	
23	1698	1702	1706	1710	1714	1718	1722	1726	1730	1734	
24	1738	1742	1746	1750	1754	1758	1762	1766	1770	1774	
25	1778	1782	1786	1791	1795	1799	1803	1807	1811	1816	
26	1820	1824	1828	1832	1837	1841	1845	1849	1854	1858	
27	1862	1866	1871	1875	1879	1884	1888	1892	1897	1901	
28	1905	1910	1914	1919	1923	1928	1932	1936	1941	1945	
29	1951	1954	1959	1963	1968	1972	1977	1982	1986	1991	
30	1995	2000	2004	2009	2014	2018	2023	2028	2032	2037	
31	2042	2046	2051	2056	2061	2065	2070	2075	2080	2084	
32	2089	2094	2099	2104	2109	2113	2118	2123	2128	2133	
33	2138	2143	2148	2153	2158	2163	2168	2173	2178	2183	
34	2188	2193	2198	2203	2208	2213	2218	2223	2228	2234	
35	2239	2244	2249	2254	2259	2264	2269	2274	2279	2284	
36	2290	2296	2301	2307	2312	2317	2322	2328	2333	2339	
37	2344	2350	2355	2360	2366	2371	2377	2382	2388	2393	
38	2399	2404	2410	2415	2421	2427	2432	2438	2443	2449	
39	2455	2460	2466	2472	2477	2483	2489	2495	2500	2506	
40	2512	2518	2523	2529	2535	2541	2547	2553	2559	2564	
41	2570	2576	2582	2588	2594	2600	2606	2612	2618	2624	
42	2630	2636	2642	2648	2653	2659	2665	2671	2677	2683	
43	2690	2696	2702	2708	2714	2720	2726	2732	2738	2744	
44	2754	2760	2767	2773	2779	2786	2791	2798	2803	2810	
45	2816	2822	2828	2834	2841	2847	2854	2860	2867	2873	
46	2880	2887	2894	2901	2907	2914	2921	2928	2934	2941	
47	2948	2955	2962	2969	2975	2982	2989	2996	3003	3010	
48	3017	3024	3031	3038	3045	3052	3059	3066	3073	3080	
49	3087	3094	3101	3107	3114	3121	3128	3134	3141	3148	

Tangents of angles

$$\theta \rightarrow \tan \theta$$

θ	ADD Differences																54°	55°																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
0	0.0000	0.0175	0.0349	0.0521	0.0691	0.0860	0.1028	0.1195	0.1361	0.1526	0.1690	0.1853	0.2015	0.2176	0.2336	0.2495	0.2653	0.2810	0.2966	0.3121	0.3275	0.3428	0.3580	0.3731	0.3881	0.4031	0.4180	0.4328	0.4475	0.4621	0.4766	0.4910	0.5053	0.5195	0.5336	0.5475	0.5613	0.5750	0.5886	0.6021	0.6155	0.6288	0.6419	0.6549	0.6678	0.6806	0.6933	0.7059	0.7184	0.7308	0.7430	0.7551	0.7671	0.7790	0.7908	0.8025	0.8141	0.8256	0.8370	0.8483	0.8595	0.8706	0.8815	0.8923	0.9030	0.9136	0.9241	0.9345	0.9448	0.9550	0.9651	0.9751	0.9850	0.9948	1.0045	1.0141	1.0236	1.0330	1.0423	1.0515	1.0606	1.0696	1.0785	1.0873	1.0960	1.1046	1.1131	1.1215	1.1298	1.1380	1.1461	1.1541	1.1620	1.1698	1.1775	1.1851	1.1926	1.2000	1.2073	1.2145	1.2216	1.2286	1.2355	1.2423	1.2490	1.2556	1.2621	1.2685	1.2748	1.2810	1.2871	1.2931	1.2990	1.3048	1.3105	1.3161	1.3216	1.3271	1.3325	1.3378	1.3430	1.3481	1.3532	1.3582	1.3631	1.3679	1.3726	1.3773	1.3819	1.3864	1.3908	1.3952	1.3995	1.4037	1.4079	1.4120	1.4161	1.4201	1.4241	1.4280	1.4318	1.4356	1.4393	1.4429	1.4465	1.4500	1.4535	1.4569	1.4603	1.4636	1.4669	1.4701	1.4733	1.4764	1.4795	1.4825	1.4855	1.4884	1.4913	1.4941	1.4969	1.5000	1.5029	1.5057	1.5084	1.5111	1.5138	1.5164	1.5190	1.5215	1.5240	1.5265	1.5289	1.5313	1.5337	1.5360	1.5383	1.5406	1.5428	1.5450	1.5472	1.5493	1.5514	1.5535	1.5555	1.5575	1.5594	1.5613	1.5632	1.5650	1.5668	1.5686	1.5703	1.5720	1.5737	1.5753	1.5769	1.5784	1.5799	1.5814	1.5828	1.5842	1.5856	1.5870	1.5883	1.5896	1.5909	1.5922	1.5935	1.5947	1.5959	1.5971	1.5983	1.5994	1.6006	1.6017	1.6028	1.6038	1.6048	1.6058	1.6067	1.6076	1.6085	1.6094	1.6102	1.6110	1.6118	1.6126	1.6133	1.6140	1.6147	1.6154	1.6160	1.6167	1.6173	1.6179	1.6185	1.6190	1.6195	1.6200	1.6205	1.6210	1.6215	1.6219	1.6223	1.6227	1.6231	1.6235	1.6238	1.6241	1.6244	1.6247	1.6250	1.6253	1.6256	1.6258	1.6261	1.6263	1.6265	1.6267	1.6269	1.6271	1.6273	1.6275	1.6277	1.6278	1.6279	1.6280	1.6281	1.6282	1.6283	1.6284	1.6285	1.6286	1.6287	1.6288	1.6289	1.6290	1.6291	1.6292	1.6293	1.6294	1.6295	1.6296	1.6297	1.6298	1.6299	1.6300	1.6301	1.6302	1.6303	1.6304	1.6305	1.6306	1.6307	1.6308	1.6309	1.6310	1.6311	1.6312	1.6313	1.6314	1.6315	1.6316	1.6317	1.6318	1.6319	1.6320	1.6321	1.6322	1.6323	1.6324	1.6325	1.6326	1.6327	1.6328	1.6329	1.6330	1.6331	1.6332	1.6333	1.6334	1.6335	1.6336	1.6337	1.6338	1.6339	1.6340	1.6341	1.6342	1.6343	1.6344	1.6345	1.6346	1.6347	1.6348	1.6349	1.6350	1.6351	1.6352	1.6353	1.6354	1.6355	1.6356	1.6357	1.6358	1.6359	1.6360	1.6361	1.6362	1.6363	1.6364	1.6365	1.6366	1.6367	1.6368	1.6369	1.6370	1.6371	1.6372	1.6373	1.6374	1.6375	1.6376	1.6377	1.6378	1.6379	1.6380	1.6381	1.6382	1.6383	1.6384	1.6385	1.6386	1.6387	1.6388	1.6389	1.6390	1.6391	1.6392	1.6393	1.6394	1.6395	1.6396	1.6397	1.6398	1.6399	1.6400	1.6401	1.6402	1.6403	1.6404	1.6405	1.6406	1.6407	1.6408	1.6409	1.6410	1.6411	1.6412	1.6413	1.6414	1.6415	1.6416	1.6417	1.6418	1.6419	1.6420	1.6421	1.6422	1.6423	1.6424	1.6425	1.6426	1.6427	1.6428	1.6429	1.6430	1.6431	1.6432	1.6433	1.6434	1.6435	1.6436	1.6437	1.6438	1.6439	1.6440	1.6441	1.6442	1.6443	1.6444	1.6445	1.6446	1.6447	1.6448	1.6449	1.6450	1.6451	1.6452	1.6453	1.6454	1.6455	1.6456	1.6457	1.6458	1.6459	1.6460	1.6461	1.6462	1.6463	1.6464	1.6465	1.6466	1.6467	1.6468	1.6469	1.6470	1.6471	1.6472	1.6473	1.6474	1.6475	1.6476	1.6477	1.6478	1.6479	1.6480	1.6481	1.6482	1.6483	1.6484	1.6485	1.6486	1.6487	1.6488	1.6489	1.6490	1.6491	1.6492	1.6493	1.6494	1.6495	1.6496	1.6497	1.6498	1.6499	1.6500	1.6501	1.6502	1.6503	1.6504	1.6505	1.6506	1.6507	1.6508	1.6509	1.6510	1.6511	1.6512	1.6513	1.6514	1.6515	1.6516	1.6517	1.6518	1.6519	1.6520	1.6521	1.6522	1.6523	1.6524	1.6525	1.6526	1.6527	1.6528	1.6529	1.6530	1.6531	1.6532	1.6533	1.6534	1.6535	1.6536	1.6537	1.6538	1.6539	1.6540	1.6541	1.6542	1.6543	1.6544	1.6545	1.6546	1.6547	1.6548	1.6549	1.6550	1.6551	1.6552	1.6553	1.6554	1.6555	1.6556	1.6557	1.6558	1.6559	1.6560	1.6561	1.6562	1.6563	1.6564	1.6565	1.6566	1.6567	1.6568	1.6569	1.6570	1.6571	1.6572	1.6573	1.6574	1.6575	1.6576	1.6577	1.6578	1.6579	1.6580	1.6581	1.6582	1.6583	1.6584	1.6585	1.6586	1.6587	1.6588	1.6589	1.6590	1.6591	1.6592	1.6593	1.6594	1.6595	1.6596	1.6597	1.6598	1.6599	1.6600	1.6601	1.6602	1.6603	1.6604	1.6605	1.6606	1.6607	1.6608	1.6609	1.6610	1.6611	1.6612	1.6613	1.6614	1.6615	1.6616	1.6617	1.6618	1.6619	1.6620	1.6621	1.6622	1.6623	1.6624	1.6625	1.6626	1.6627	1.6628	1.6629	1.6630	1.6631	1.6632	1.6633	1.6634	1.6635	1.6636	1.6637	1.6638	1.6639	1.6640	1.6641	1.6642	1.6643	1.6644	1.6645	1.6646	1.6647	1.6648	1.6649	1.6650	1.6651	1.6652	1.6653	1.6654	1.6655	1.6656	1.6657	1.6658	1.6659	1.6660	1.6661	1.6662	1.6663	1.6664	1.6665	1.6666	1.6667	1.6668	1.6669	1.6670	1.6671	1.6672	1.6673	1.6674	1.6675	1.6676	1.6677	1.6678	1.6679	1.6680	1.6681	1.6682	1.6683	1.6684	1.6685	1.6686	1.6687	1.6688	1.6689	1.6690	1.6691	1.6692	1.6693	1.6694	1.6695	1.6696	1.6697	1.6698	1.6699	1.6700	1.6701	1.6702	1.6703	1.6704	1.6705	1.6706	1.6707	1.6708	1.6709	1.6710	1.6711	1.6712	1.6713	1.6714	1.6715	1.6716	1.6717	1.6718	1.6719	1.6720	1.6721	1.6722	1.6723	1.6724	1.6725	1.6726	1.6727	1.6728	1.6729	1.6730	1.6731	1.6732	1.6733	1.6734	1.6735	1.6736	1.6737	1.6738	1.6739	1.6740	1.6741	1.6742	1.6743	1.6744	1.6745	1.6746	1.6747	1.6748	1.6749	1.6750	1.6751	1.6752	1.6753	1.6754	1.6755	1.6756	1.6757	1.6758	1.6759	1.6760	1.6761	1.6762	1.6763	1.6764	1.6765	1.6766	1.6767	1.6768	1.6769	1.6770	1.6771	1.6772	1.6773	1.6774	1.6775	1.6776	1.6777	1.6778	1.6779	1.6780	1.6781	1.6782	1.6783	1.6784	1.6785	1.6786	1.6787	1.6788	1.6789	1.6790	1.6791	1.6792	1.6793	1.6794	1.6795	1.6796	1.6797	1.6798	1.6799	1.6800	1.6801	1.6802	1.6803	1.6804	1.6805	1.6806	1.6807	1.6808	1.6809	1.6810	1.6811	1.6812	1.6813	1.6814	1.6815	1.6816	1.6817	1.6818	1.6819	1.6820	1.6821	1.6822	1.6823	1.6824	1.6825	1.6826	1.6827	1.6828	1.6829	1.6830	1.6831	1.6832	1.6833	1.6834	1.6835	1.6836	1.6837	1.6838	1.6839	1.6840	1.6841	1.6842	1.6843	1.6844	1.6845	1.6846	1.6847	1.6848	1.6849	1.6850	1.6851	1.6852	1.6853	1.6854	1.6855	1.6856	1.6857	1.6858	1.6859	1.6860	1.6861	1.6862	1.6863	1.6864	1.6865	1.6866	1.6867	1.6868	1.6869	1.6870	1.6871	1.6872	1.6873	1.6874	1.6875	1.6876	1.6877	1.6878	1.6879	1.6880	1.6881	1.6882	1.6883	1.6884	1.6885	1.6886	1.6887	1.6888	1.6889	1.6890	1.6891	1.6892	1.6893	1.6894	1.6895	1.6896	1.6897	1.6898	1.6899	1.6900	1.6901	1.6902	1.6903	1.6904	1.6905	1.6906	1.6907	1.6908	1.6909	1.6910	1.6911	1.6912	1.6913	1.6914	1.6915	1.6916	1.6917	1.6918	1.6919	1.6920	1.6921	1.6922	1.6923	1.6924	1.6925	1.6926	1.6927	1.6928	1.6929	1.6930	1.6931	1.6932	1.6933	1.6934	1.6935	1.6936	1.6937	1.6938	1.6939	1.6940	1.6941	1.6942	1.6943	1.6944	1.6945	1.6946	1.6947	1.6948	1.6949	1.6950	1.6951	1.6952	1.6953	1.6954	1.6955	1.6956	1.6957	1.6958	1.6959	1.6960	1.6961	1.6962	1.6963	1.6964	1.6965	1.6966	1.6967	1.6968	1.6969	1.6970	1.

Sines of angles

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θ	α	Sines of angles										Sines of angles										θ	α	Sines of angles										ADD Differences																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
		0°	1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°			20°	0°	1°	2°	3°	4°	5°	6°	7°	8°		9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
0°	0.0000	0.0017	0.0034	0.0051	0.0068	0.0085	0.0102	0.0119	0.0136	0.0153	0.0170	0.0187	0.0204	0.0221	0.0238	0.0255	0.0272	0.0289	0.0306	0.0323	0.0340	0.0357	0.0374	0.0391	0.0408	0.0425	0.0442	0.0459	0.0476	0.0493	0.0510	0.0527	0.0544	0.0561	0.0578	0.0595	0.0612	0.0629	0.0646	0.0663	0.0680	0.0697	0.0714	0.0731	0.0748	0.0765	0.0782	0.0799	0.0816	0.0833	0.0850	0.0867	0.0884	0.0901	0.0918	0.0935	0.0952	0.0969	0.0986	0.1003	0.1020	0.1037	0.1054	0.1071	0.1088	0.1105	0.1122	0.1139	0.1156	0.1173	0.1190	0.1207	0.1224	0.1241	0.1258	0.1275	0.1292	0.1309	0.1326	0.1343	0.1360	0.1377	0.1394	0.1411	0.1428	0.1445	0.1462	0.1479	0.1496	0.1513	0.1530	0.1547	0.1564	0.1581	0.1598	0.1615	0.1632	0.1649	0.1666	0.1683	0.1700	0.1717	0.1734	0.1751	0.1768	0.1785	0.1802	0.1819	0.1836	0.1853	0.1870	0.1887	0.1904	0.1921	0.1938	0.1955	0.1972	0.1989	0.2006	0.2023	0.2040	0.2057	0.2074	0.2091	0.2108	0.2125	0.2142	0.2159	0.2176	0.2193	0.2210	0.2227	0.2244	0.2261	0.2278	0.2295	0.2312	0.2329	0.2346	0.2363	0.2380	0.2397	0.2414	0.2431	0.2448	0.2465	0.2482	0.2499	0.2516	0.2533	0.2550	0.2567	0.2584	0.2601	0.2618	0.2635	0.2652	0.2669	0.2686	0.2703	0.2720	0.2737	0.2754	0.2771	0.2788	0.2805	0.2822	0.2839	0.2856	0.2873	0.2890	0.2907	0.2924	0.2941	0.2958	0.2975	0.2992	0.3009	0.3026	0.3043	0.3060	0.3077	0.3094	0.3111	0.3128	0.3145	0.3162	0.3179	0.3196	0.3213	0.3230	0.3247	0.3264	0.3281	0.3298	0.3315	0.3332	0.3349	0.3366	0.3383	0.3400	0.3417	0.3434	0.3451	0.3468	0.3485	0.3502	0.3519	0.3536	0.3553	0.3570	0.3587	0.3604	0.3621	0.3638	0.3655	0.3672	0.3689	0.3706	0.3723	0.3740	0.3757	0.3774	0.3791	0.3808	0.3825	0.3842	0.3859	0.3876	0.3893	0.3910	0.3927	0.3944	0.3961	0.3978	0.3995	0.4012	0.4029	0.4046	0.4063	0.4080	0.4097	0.4114	0.4131	0.4148	0.4165	0.4182	0.4199	0.4216	0.4233	0.4250	0.4267	0.4284	0.4301	0.4318	0.4335	0.4352	0.4369	0.4386	0.4403	0.4420	0.4437	0.4454	0.4471	0.4488	0.4505	0.4522	0.4539	0.4556	0.4573	0.4590	0.4607	0.4624	0.4641	0.4658	0.4675	0.4692	0.4709	0.4726	0.4743	0.4760	0.4777	0.4794	0.4811	0.4828	0.4845	0.4862	0.4879	0.4896	0.4913	0.4930	0.4947	0.4964	0.4981	0.4998	0.5015	0.5032	0.5049	0.5066	0.5083	0.5100	0.5117	0.5134	0.5151	0.5168	0.5185	0.5202	0.5219	0.5236	0.5253	0.5270	0.5287	0.5304	0.5321	0.5338	0.5355	0.5372	0.5389	0.5406	0.5423	0.5440	0.5457	0.5474	0.5491	0.5508	0.5525	0.5542	0.5559	0.5576	0.5593	0.5610	0.5627	0.5644	0.5661	0.5678	0.5695	0.5712	0.5729	0.5746	0.5763	0.5780	0.5797	0.5814	0.5831	0.5848	0.5865	0.5882	0.5899	0.5916	0.5933	0.5950	0.5967	0.5984	0.5999	0.6016	0.6033	0.6050	0.6067	0.6084	0.6101	0.6118	0.6135	0.6152	0.6169	0.6186	0.6203	0.6220	0.6237	0.6254	0.6271	0.6288	0.6305	0.6322	0.6339	0.6356	0.6373	0.6390	0.6407	0.6424	0.6441	0.6458	0.6475	0.6492	0.6509	0.6526	0.6543	0.6560	0.6577	0.6594	0.6611	0.6628	0.6645	0.6662	0.6679	0.6696	0.6713	0.6730	0.6747	0.6764	0.6781	0.6798	0.6815	0.6832	0.6849	0.6866	0.6883	0.6900	0.6917	0.6934	0.6951	0.6968	0.6985	0.7002	0.7019	0.7036	0.7053	0.7070	0.7087	0.7104	0.7121	0.7138	0.7155	0.7172	0.7189	0.7206	0.7223	0.7240	0.7257	0.7274	0.7291	0.7308	0.7325	0.7342	0.7359	0.7376	0.7393	0.7410	0.7427	0.7444	0.7461	0.7478	0.7495	0.7512	0.7529	0.7546	0.7563	0.7580	0.7597	0.7614	0.7631	0.7648	0.7665	0.7682	0.7699	0.7716	0.7733	0.7750	0.7767	0.7784	0.7801	0.7818	0.7835	0.7852	0.7869	0.7886	0.7903	0.7920	0.7937	0.7954	0.7971	0.7988	0.8005	0.8022	0.8039	0.8056	0.8073	0.8090	0.8107	0.8124	0.8141	0.8158	0.8175	0.8192	0.8209	0.8226	0.8243	0.8260	0.8277	0.8294	0.8311	0.8328	0.8345	0.8362	0.8379	0.8396	0.8413	0.8430	0.8447	0.8464	0.8481	0.8498	0.8515	0.8532	0.8549	0.8566	0.8583	0.8600	0.8617	0.8634	0.8651	0.8668	0.8685	0.8702	0.8719	0.8736	0.8753	0.8770	0.8787	0.8804	0.8821	0.8838	0.8855	0.8872	0.8889	0.8906	0.8923	0.8940	0.8957	0.8974	0.8991	0.9008	0.9025	0.9042	0.9059	0.9076	0.9093	0.9110	0.9127	0.9144	0.9161	0.9178	0.9195	0.9212	0.9229	0.9246	0.9263	0.9280	0.9297	0.9314	0.9331	0.9348	0.9365	0.9382	0.9399	0.9416	0.9433	0.9450	0.9467	0.9484	0.9501	0.9518	0.9535	0.9552	0.9569	0.9586	0.9603	0.9620	0.9637	0.9654	0.9671	0.9688	0.9705	0.9722	0.9739	0.9756	0.9773	0.9790	0.9807	0.9824	0.9841	0.9858	0.9875	0.9892	0.9909	0.9926	0.9943	0.9960	0.9977	0.9994	1.0011	1.0028	1.0045	1.0062	1.0079	1.0096	1.0113	1.0130	1.0147	1.0164	1.0181	1.0198	1.0215	1.0232	1.0249	1.0266	1.0283	1.0300	1.0317	1.0334	1.0351	1.0368	1.0385	1.0402	1.0419	1.0436	1.0453	1.0470	1.0487	1.0504	1.0521	1.0538	1.0555	1.0572	1.0589	1.0606	1.0623	1.0640	1.0657	1.0674	1.0691	1.0708	1.0725	1.0742	1.0759	1.0776	1.0793	1.0810	1.0827	1.0844	1.0861	1.0878	1.0895	1.0912	1.0929	1.0946	1.0963	1.0980	1.0997	1.1014	1.1031	1.1048	1.1065	1.1082	1.1099	1.1116	1.1133	1.1150	1.1167	1.1184	1.1201	1.1218	1.1235	1.1252	1.1269	1.1286	1.1303	1.1320	1.1337	1.1354	1.1371	1.1388	1.1405	1.1422	1.1439	1.1456	1.1473	1.1490	1.1507	1.1524	1.1541	1.1558	1.1575	1.1592	1.1609	1.1626	1.1643	1.1660	1.1677	1.1694	1.1711	1.1728	1.1745	1.1762	1.1779	1.1796	1.1813	1.1830	1.1847	1.1864	1.1881	1.1898	1.1915	1.1932	1.1949	1.1966	1.1983	1.2000	1.2017	1.2034	1.2051	1.2068	1.2085	1.2102	1.2119	1.2136	1.2153	1.2170	1.2187	1.2204	1.2221	1.2238	1.2255	1.2272	1.2289	1.2306	1.2323	1.2340	1.2357	1.2374	1.2391	1.2408	1.2425	1.2442	1.2459	1.2476	1.2493	1.2510	1.2527	1.2544	1.2561	1.2578	1.2595	1.2612	1.2629	1.2646	1.2663	1.2680	1.2697	1.2714	1.2731	1.2748	1.2765	1.2782	1.2799	1.2816	1.2833	1.2850	1.2867	1.2884	1.2901	1.2918	1.2935	1.2952	1.2969	1.2986	1.3003	1.3020	1.3037	1.3054	1.3071	1.3088	1.3105	1.3122	1.3139	1.3156	1.3173	1.3190	1.3207	1.3224	1.3241	1.3258	1.3275	1.3292	1.3309	1.3326	1.3343	1.3360	1.3377	1.3394	1.3411	1.3428	1.3445	1.3462	1.3479	1.3496	1.3513	1.3530	1.3547	1.3564	1.3581	1.3598	1.3615	1.3632	1.3649	1.3666	1.3683	1.3700	1.3717	1.3734	1.3751	1.3768	1.3785	1.3802	1.3819	1.3836	1.3853	1.3870	1.3887	1.3904	1.3921	1.3938	1.3955	1.3972	1.3989	1.4006	1.4023	1.4040	1.4057	1.4074	1.4091	1.4108	1.4125	1.4142	1.4159	1.4176	1.4193	1.4210	1.4227	1.4244	1.4261	1.4278	1.4295	1.4312	1.4329	1.4346	1.4363	1.4380	1.4397	1.4414	1.4431	1.4448	1.4465	1.4482	1.4499	1.4516	1.4533	1.4550	1.4567	1.4584	1.4601	1.4618	1.4635	1.4652	1.4669	1.4686	1.4703	1.4720	1.4737	1.4754	1.4771	1.4788	1.4805	1.4822	1.4839	1.4856	1.4873	1.4890	1.4907	1.4924	1.4941	1.4958	1.4975	1.4992	1.5009	1.5026	1.5043	1.5060	1.5077	1.5094	1.5111	1.5128	1.5145	1.5162	1.5179	1.5196	1.5213	1.5230	1.5247	1.5264	1.5281	1.5298	1.5315	1.5332	1.5349	1.5366	1.5383	1.5400	1.5417	1.5434	1.5451	1.5468	1.5485	1.5502	1.5519	1.5536	1.5553	1.5570	1.5587	1.5604	1.5621	1.5638	1.5655	1.5672	1.5689	1.5706	1.5723	1.5740	1.5757	1.5774	1.5791	1.5808	1.5825	1.5842	1.5859	1.5876	1.5893	1.5910	1.5927	1.5

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Cosines of angles

θ	θ	SUBTRACT Differences										SUBTRACT Differences									
		1	2	3	4	5	6	7	8	9	10	1	2	3	4	5	6	7	8	9	10
0°	0.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999	0.999
1	0.9998	9998	9997	9996	9995	9994	9993	9992	9991	9990	9989	9988	9987	9986	9985	9984	9983	9982	9981	9980	9979
2	0.9994	9993	9991	9989	9987	9985	9983	9981	9979	9977	9975	9973	9971	9969	9967	9965	9963	9961	9959	9957	9955
3	0.9986	9984	9981	9978	9975	9972	9969	9966	9963	9960	9957	9954	9951	9948	9945	9942	9939	9936	9933	9930	9927
4	0.9975	9973	9969	9965	9961	9957	9953	9949	9945	9941	9937	9933	9929	9925	9921	9917	9913	9909	9905	9901	9897
5	0.9962	9960	9955	9950	9945	9940	9935	9930	9925	9920	9915	9910	9905	9900	9895	9890	9885	9880	9875	9870	9865
6	0.9945	9943	9937	9931	9925	9919	9913	9907	9901	9895	9889	9883	9877	9871	9865	9859	9853	9847	9841	9835	9829
7	0.9925	9923	9916	9909	9902	9895	9888	9881	9874	9867	9860	9853	9846	9839	9832	9825	9818	9811	9804	9797	9790
8	0.9903	9900	9892	9884	9876	9868	9860	9852	9844	9836	9828	9820	9812	9804	9796	9788	9780	9772	9764	9756	9748
9	0.9879	9876	9867	9858	9849	9840	9831	9822	9813	9804	9795	9786	9777	9768	9759	9750	9741	9732	9723	9714	9705
10	0.9854	9851	9841	9831	9821	9811	9801	9791	9781	9771	9761	9751	9741	9731	9721	9711	9701	9691	9681	9671	9661
11	0.9828	9824	9813	9802	9791	9780	9769	9758	9747	9736	9725	9714	9703	9692	9681	9670	9659	9648	9637	9626	9615
12	0.9800	9796	9784	9772	9760	9748	9736	9724	9712	9700	9688	9676	9664	9652	9640	9628	9616	9604	9592	9580	9568
13	0.9771	9767	9754	9741	9728	9715	9702	9689	9676	9663	9650	9637	9624	9611	9598	9585	9572	9559	9546	9533	9520
14	0.9741	9736	9723	9709	9695	9681	9667	9653	9639	9625	9611	9597	9583	9569	9555	9541	9527	9513	9499	9485	9471
15	0.9709	9704	9690	9675	9660	9645	9630	9615	9600	9585	9570	9555	9540	9525	9510	9495	9480	9465	9450	9435	9420
16	0.9675	9669	9654	9638	9622	9606	9590	9574	9558	9542	9526	9510	9494	9478	9462	9446	9430	9414	9398	9382	9366
17	0.9640	9633	9617	9600	9583	9566	9549	9532	9515	9498	9481	9464	9447	9430	9413	9396	9379	9362	9345	9328	9311
18	0.9603	9596	9578	9560	9542	9524	9506	9488	9470	9452	9434	9416	9398	9380	9362	9344	9326	9308	9290	9272	9254
19	0.9565	9557	9538	9519	9500	9481	9462	9443	9424	9405	9386	9367	9348	9329	9310	9291	9272	9253	9234	9215	9196
20	0.9526	9517	9497	9477	9457	9437	9417	9397	9377	9357	9337	9317	9297	9277	9257	9237	9217	9197	9177	9157	9137
21	0.9486	9476	9455	9435	9414	9393	9372	9351	9330	9309	9288	9267	9246	9225	9204	9183	9162	9141	9120	9100	9079
22	0.9445	9434	9413	9392	9371	9350	9329	9308	9287	9266	9245	9224	9203	9182	9161	9140	9119	9098	9077	9056	9035
23	0.9403	9392	9370	9348	9326	9304	9282	9260	9238	9216	9194	9172	9150	9128	9106	9084	9062	9040	9018	8996	8974
24	0.9360	9348	9325	9302	9279	9256	9233	9210	9187	9164	9141	9118	9095	9072	9049	9026	9003	8980	8957	8934	8911
25	0.9316	9303	9280	9256	9232	9208	9184	9160	9136	9112	9088	9064	9040	9016	8992	8968	8944	8920	8896	8872	8848
26	0.9271	9258	9234	9209	9184	9159	9134	9109	9084	9059	9034	9009	8984	8959	8934	8909	8884	8859	8834	8809	8784
27	0.9225	9212	9187	9162	9137	9111	9086	9060	9035	9009	8983	8958	8932	8906	8881	8855	8830	8804	8778	8753	8727
28	0.9179	9165	9140	9114	9088	9062	9036	9010	8984	8958	8932	8906	8880	8854	8828	8802	8776	8750	8724	8698	8672
29	0.9132	9117	9091	9065	9039	9013	8987	8961	8935	8909	8883	8857	8831	8805	8779	8753	8727	8701	8675	8649	8623
30	0.9085	9069	9042	9016	8990	8964	8938	8912	8886	8860	8834	8808	8782	8756	8730	8704	8678	8652	8626	8600	8574
31	0.9037	9020	8993	8967	8940	8914	8887	8861	8835	8809	8783	8757	8731	8705	8679	8653	8627	8601	8575	8549	8523
32	0.8989	8971	8944	8917	8890	8863	8836	8810	8783	8757	8731	8705	8679	8653	8627	8601	8575	8549	8523	8497	8471
33	0.8940	8922	8894	8867	8840	8813	8786	8759	8732	8705	8678	8651	8624	8597	8570	8543	8516	8490	8463	8436	8410
34	0.8891	8872	8844	8816	8788	8761	8734	8707	8680	8653	8626	8599	8572	8545	8518	8491	8464	8437	8410	8383	8356
35	0.8842	8822	8794	8766	8738	8710	8682	8654	8626	8598	8570	8542	8514	8486	8458	8430	8402	8375	8347	8320	8292
36	0.8793	8773	8744	8716	8687	8659	8630	8602	8574	8546	8518	8489	8461	8433	8405	8377	8349	8321	8293	8265	8237
37	0.8743	8723	8694	8665	8636	8607	8578	8549	8520	8491	8462	8433	8404	8375	8346	8317	8288	8260	8231	8202	8173
38	0.8693	8673	8643	8614	8585	8556	8526	8497	8468	8439	8410	8381	8352	8323	8294	8265	8236	8207	8178	8149	8120
39	0.8643	8622	8592	8563	8533	8504	8474	8445	8415	8386	8357	8328	8299	8270	8241	8212	8183	8154	8125	8096	8067
40	0.8593	8572	8542	8512	8482	8452	8422	8392	8362	8332	8303	8273	8244	8214	8185	8155	8126	8096	8067	8037	8008
41	0.8543	8521	8491	8461	8431	8401	8371	8341	8311	8281	8251	8221	8191	8161	8131	8101	8072	8042	8012	7982	7952
42	0.8493	8471	8440	8410	8380	8350	8320	8290	8260	8230	8200	8170	8140	8110	8080	8050	8020	7990	7960	7930	7900
43	0.8443	8420	8389	8358	8328	8297	8267	8236	8206	8175	8145	8115	8085	8055	8025	7995	7965	7935	7905	7875	7845
44	0.8393	8370	8339	8308	8277	8246	8215	8185	8154	8124	8093	8063	8033	8003	7973	7943	7913	7883	7853	7823	7793

Logarithms of tangents

$$\theta \rightarrow \log \tan \theta$$

θ	M.D. Differences									
	0	1	2	3	4	5	6	7	8	9
0°	—	1.2419	3211	3556	3681	3781	3860	3920	3960	3982
1	2.2419	3833	3211	3556	3681	3781	3860	3920	3960	3982
2	2.2431	3845	3845	3845	3845	3845	3845	3845	3845	3845
3	2.2434	3845	3845	3845	3845	3845	3845	3845	3845	3845
4	2.2445	3854	3854	3854	3854	3854	3854	3854	3854	3854
5	2.2449	3856	3856	3856	3856	3856	3856	3856	3856	3856
6	2.2458	3860	3860	3860	3860	3860	3860	3860	3860	3860
7	2.2461	3861	3861	3861	3861	3861	3861	3861	3861	3861
8	2.2463	3862	3862	3862	3862	3862	3862	3862	3862	3862
9	2.2465	3863	3863	3863	3863	3863	3863	3863	3863	3863
10	2.2467	3864	3864	3864	3864	3864	3864	3864	3864	3864
11	2.2469	3865	3865	3865	3865	3865	3865	3865	3865	3865
12	2.2471	3866	3866	3866	3866	3866	3866	3866	3866	3866
13	2.2473	3867	3867	3867	3867	3867	3867	3867	3867	3867
14	2.2475	3868	3868	3868	3868	3868	3868	3868	3868	3868
15	2.2477	3869	3869	3869	3869	3869	3869	3869	3869	3869
16	2.2479	3870	3870	3870	3870	3870	3870	3870	3870	3870
17	2.2481	3871	3871	3871	3871	3871	3871	3871	3871	3871
18	2.2483	3872	3872	3872	3872	3872	3872	3872	3872	3872
19	2.2485	3873	3873	3873	3873	3873	3873	3873	3873	3873
20	2.2487	3874	3874	3874	3874	3874	3874	3874	3874	3874
21	2.2489	3875	3875	3875	3875	3875	3875	3875	3875	3875
22	2.2491	3876	3876	3876	3876	3876	3876	3876	3876	3876
23	2.2493	3877	3877	3877	3877	3877	3877	3877	3877	3877
24	2.2495	3878	3878	3878	3878	3878	3878	3878	3878	3878
25	2.2497	3879	3879	3879	3879	3879	3879	3879	3879	3879
26	2.2499	3880	3880	3880	3880	3880	3880	3880	3880	3880
27	2.2501	3881	3881	3881	3881	3881	3881	3881	3881	3881
28	2.2503	3882	3882	3882	3882	3882	3882	3882	3882	3882
29	2.2505	3883	3883	3883	3883	3883	3883	3883	3883	3883
30	2.2507	3884	3884	3884	3884	3884	3884	3884	3884	3884
31	2.2509	3885	3885	3885	3885	3885	3885	3885	3885	3885
32	2.2511	3886	3886	3886	3886	3886	3886	3886	3886	3886
33	2.2513	3887	3887	3887	3887	3887	3887	3887	3887	3887
34	2.2515	3888	3888	3888	3888	3888	3888	3888	3888	3888
35	2.2517	3889	3889	3889	3889	3889	3889	3889	3889	3889
36	2.2519	3890	3890	3890	3890	3890	3890	3890	3890	3890
37	2.2521	3891	3891	3891	3891	3891	3891	3891	3891	3891
38	2.2523	3892	3892	3892	3892	3892	3892	3892	3892	3892
39	2.2525	3893	3893	3893	3893	3893	3893	3893	3893	3893
40	2.2527	3894	3894	3894	3894	3894	3894	3894	3894	3894
41	2.2529	3895	3895	3895	3895	3895	3895	3895	3895	3895
42	2.2531	3896	3896	3896	3896	3896	3896	3896	3896	3896
43	2.2533	3897	3897	3897	3897	3897	3897	3897	3897	3897
44	2.2535	3898	3898	3898	3898	3898	3898	3898	3898	3898

θ	M.D. Differences									
	0	1	2	3	4	5	6	7	8	9
45°	0.0000	4815	6033	6843	7461	8000	8481	8916	9316	9688
46	0.0152	0167	0182	0197	0212	0228	0243	0259	0275	0290
47	0.0303	0319	0334	0349	0364	0379	0395	0410	0425	0440
48	0.0456	0471	0486	0501	0517	0532	0547	0562	0578	0593
49	0.0608	0624	0639	0654	0670	0685	0700	0716	0731	0746
50	0.0762	0777	0793	0808	0824	0839	0854	0870	0885	0900
51	0.0916	0932	0947	0963	0978	0994	1010	1025	1041	1056
52	0.1072	1088	1103	1119	1135	1150	1166	1182	1197	1213
53	0.1229	1245	1260	1276	1292	1308	1324	1340	1356	1371
54	0.1387	1403	1419	1435	1451	1467	1483	1499	1515	1531
55	0.1546	1562	1578	1594	1610	1626	1642	1658	1674	1690
56	0.1710	1726	1742	1758	1774	1790	1806	1822	1838	1854
57	0.1875	1891	1907	1923	1939	1955	1971	1987	2003	2019
58	0.2042	2058	2074	2090	2106	2122	2138	2154	2170	2186
59	0.2212	2228	2244	2260	2276	2292	2308	2324	2340	2356
60	0.2386	2402	2418	2434	2450	2466	2482	2498	2514	2530
61	0.2562	2578	2594	2610	2626	2642	2658	2674	2690	2706
62	0.2742	2758	2774	2790	2806	2822	2838	2854	2870	2886
63	0.2950	2966	2982	2998	3014	3030	3046	3062	3078	3094
64	0.3110	3126	3142	3158	3174	3190	3206	3222	3238	3254
65	0.3313	3329	3345	3361	3377	3393	3409	3425	3441	3457
66	0.3514	3530	3546	3562	3578	3594	3610	3626	3642	3658
67	0.3721	3737	3753	3769	3785	3801	3817	3833	3849	3865
68	0.3936	3952	3968	3984	4000	4016	4032	4048	4064	4080
69	0.4158	4174	4190	4206	4222	4238	4254	4270	4286	4302
70	0.4369	4385	4401	4417	4433	4449	4465	4481	4497	4513
71	0.4630	4646	4662	4678	4694	4710	4726	4742	4758	4774
72	0.4832	4848	4864	4880	4896	4912	4928	4944	4960	4976
73	0.5147	5163	5179	5195	5211	5227	5243	5259	5275	5291
74	0.5455	5471	5487	5503	5519	5535	5551	5567	5583	5599
75	0.5719	5735	5751	5767	5783	5799	5815	5831	5847	5863
76	0.6032	6048	6064	6080	6096	6112	6128	6144	6160	6176
77	0.6366	6382	6398	6414	6430	6446	6462	6478	6494	6510
78	0.6572	6588	6604	6620	6636	6652	6668	6684	6700	6716
79	0.7115	7131	7147	7163	7179	7195	7211	7227	7243	7259
80	0.7537	7553	7569	7585	7601	7617	7633	7649	7665	7681
81	0.8003	8019	8035	8051	8067	8083	8099	8115	8131	8147
82	0.8322	8338	8354	8370	8386	8402	8418	8434	8450	8466
83	0.8710	8726	8742	8758	8774	8790	8806	8822	8838	8854
84	0.9104	9120	9136	9152	9168	9184	9200	9216	9232	9248
85	0.9500	9516	9532	9548	9564	9580	9596	9612	9628	9644
86	1.0000	1016	1032	1048	1064	1080	1096	1112	1128	1144
87	1.0500	1066	1082	1098	1114	1130	1146	1162	1178	1194
88	1.1000	1116	1132	1148	1164	1180	1196	1212	1228	1244
89	1.1500	1216	1232	1248	1264	1280	1296	1312	1328	1344

Logarithms of sines

θ	θ'	θ''	θ'''	θ''''	θ'''''	ADD Differences				
						1	2	3	4	5
0	∞	3.2419	34.28	71.90	94.99	94.00	0.00	14.50	1.00	
1	3.2419	26.32	33.16	33.58	33.80	41.79	44.59	47.23	49.71	52.06
2	2.5428	26.40	34.42	34.85	35.07	43.97	46.77	49.31	51.69	53.94
3	2.1868	27.30	34.66	35.09	35.31	45.77	48.57	51.01	53.29	55.44
4	2.0435	28.43	34.67	35.10	35.31	47.57	50.37	52.81	55.09	57.24
5	2.0403	29.89	34.73	35.15	35.36	49.37	52.17	54.61	56.89	59.04
6	2.0392	30.64	34.73	35.15	35.36	51.17	53.97	56.41	58.69	60.84
7	2.0392	31.69	34.73	35.15	35.36	52.97	55.77	58.21	60.49	62.64
8	2.0392	32.64	34.73	35.15	35.36	54.77	57.57	60.01	62.29	64.44
9	2.0392	33.59	34.73	35.15	35.36	56.57	59.37	61.81	64.09	66.24
10	2.0392	34.54	34.73	35.15	35.36	58.37	61.17	63.61	65.89	68.04
11	2.0392	35.49	34.73	35.15	35.36	60.17	62.97	65.41	67.69	69.84
12	2.0392	36.44	34.73	35.15	35.36	61.97	64.77	67.21	69.49	71.64
13	2.0392	37.39	34.73	35.15	35.36	63.77	66.57	69.01	71.29	73.44
14	2.0392	38.34	34.73	35.15	35.36	65.57	68.37	70.81	73.09	75.24
15	2.0392	39.29	34.73	35.15	35.36	67.37	70.17	72.61	74.89	77.04
16	2.0392	40.24	34.73	35.15	35.36	69.17	71.97	74.41	76.69	78.84
17	2.0392	41.19	34.73	35.15	35.36	70.97	73.77	76.21	78.49	80.64
18	2.0392	42.14	34.73	35.15	35.36	72.77	75.57	78.01	80.29	82.44
19	2.0392	43.09	34.73	35.15	35.36	74.57	77.37	79.81	82.09	84.24
20	2.0392	44.04	34.73	35.15	35.36	76.37	79.17	81.61	83.89	86.04
21	2.0392	44.99	34.73	35.15	35.36	78.17	80.97	83.41	85.69	87.84
22	2.0392	45.94	34.73	35.15	35.36	79.97	82.77	85.21	87.49	89.64
23	2.0392	46.89	34.73	35.15	35.36	81.77	84.57	86.91	89.29	91.44
24	2.0392	47.84	34.73	35.15	35.36	83.57	86.37	88.81	91.09	93.24
25	2.0392	48.79	34.73	35.15	35.36	85.37	88.17	90.61	92.89	95.04
26	2.0392	49.74	34.73	35.15	35.36	87.17	89.97	92.41	94.69	96.84
27	2.0392	50.69	34.73	35.15	35.36	88.97	91.77	94.01	96.49	98.64
28	2.0392	51.64	34.73	35.15	35.36	90.77	93.57	95.81	98.29	100.44
29	2.0392	52.59	34.73	35.15	35.36	92.57	95.37	97.61	100.09	102.24
30	2.0392	53.54	34.73	35.15	35.36	94.37	97.17	99.61	101.89	104.04
31	2.0392	54.49	34.73	35.15	35.36	96.17	98.97	101.41	103.69	105.84
32	2.0392	55.44	34.73	35.15	35.36	97.97	100.77	103.01	105.49	107.64
33	2.0392	56.39	34.73	35.15	35.36	99.77	102.57	104.61	107.29	109.44
34	2.0392	57.34	34.73	35.15	35.36	101.57	104.37	106.21	109.09	111.24
35	2.0392	58.29	34.73	35.15	35.36	103.37	106.17	107.81	110.89	113.04
36	2.0392	59.24	34.73	35.15	35.36	105.17	107.97	109.41	112.69	114.84
37	2.0392	60.19	34.73	35.15	35.36	106.97	109.77	111.01	114.49	116.64
38	2.0392	61.14	34.73	35.15	35.36	108.77	111.57	112.61	116.29	118.44
39	2.0392	62.09	34.73	35.15	35.36	110.57	113.37	114.21	118.09	120.24
40	2.0392	63.04	34.73	35.15	35.36	112.37	115.17	115.81	119.89	122.04
41	2.0392	63.99	34.73	35.15	35.36	114.17	116.97	117.41	121.69	123.84
42	2.0392	64.94	34.73	35.15	35.36	115.97	118.77	119.01	123.49	125.64
43	2.0392	65.89	34.73	35.15	35.36	117.77	120.57	120.61	125.29	127.44
44	2.0392	66.84	34.73	35.15	35.36	119.57	122.37	122.21	127.09	129.24

$$\theta \rightarrow \log \sin \theta$$

θ	θ'	θ''	θ'''	θ''''	θ'''''	ADD Differences				
						1	2	3	4	5
45	1.8495	45.02	45.10	45.17	45.24	85.32	85.40	85.47	85.55	85.62
46	1.8509	46.07	46.15	46.22	46.29	86.37	86.45	86.52	86.60	86.67
47	1.8523	47.12	47.20	47.27	47.34	87.42	87.50	87.57	87.65	87.72
48	1.8537	48.17	48.25	48.32	48.39	88.47	88.55	88.62	88.70	88.77
49	1.8551	49.22	49.30	49.37	49.44	89.52	89.60	89.67	89.75	89.82
50	1.8565	50.27	50.35	50.42	50.49	90.57	90.65	90.72	90.80	90.87
51	1.8579	51.32	51.40	51.47	51.54	91.62	91.70	91.77	91.85	91.92
52	1.8593	52.37	52.45	52.52	52.59	92.67	92.75	92.82	92.90	92.97
53	1.8607	53.42	53.50	53.57	53.64	93.72	93.80	93.87	93.95	94.02
54	1.8621	54.47	54.55	54.62	54.69	94.77	94.85	94.92	95.00	95.07
55	1.8635	55.52	55.60	55.67	55.74	95.82	95.90	95.97	96.05	96.12
56	1.8649	56.57	56.65	56.72	56.79	96.87	96.95	97.02	97.10	97.17
57	1.8663	57.62	57.70	57.77	57.84	97.92	98.00	98.07	98.15	98.22
58	1.8677	58.67	58.75	58.82	58.89	98.97	99.05	99.12	99.20	99.27
59	1.8691	59.72	59.80	59.87	59.94	99.92	100.00	100.07	100.15	100.22
60	1.8705	60.77	60.85	60.92	60.99	100.97	101.05	101.12	101.20	101.27
61	1.8719	61.82	61.90	61.97	62.04	101.92	102.00	102.07	102.15	102.22
62	1.8733	62.87	62.95	63.02	63.09	102.97	103.05	103.12	103.20	103.27
63	1.8747	63.92	64.00	64.07	64.14	103.92	104.00	104.07	104.15	104.22
64	1.8761	64.97	65.05	65.12	65.19	104.87	104.95	105.02	105.10	105.17
65	1.8775	66.02	66.10	66.17	66.24	105.82	105.90	105.97	106.05	106.12
66	1.8789	67.07	67.15	67.22	67.29	106.77	106.85	106.92	107.00	107.07
67	1.8803	68.12	68.20	68.27	68.34	107.72	107.80	107.87	107.95	108.02
68	1.8817	69.17	69.25	69.32	69.39	108.67	108.75	108.82	108.90	108.97
69	1.8831	70.22	70.30	70.37	70.44	109.62	109.70	109.77	109.85	109.92
70	1.8845	71.27	71.35	71.42	71.49	110.57	110.65	110.72	110.80	110.87
71	1.8859	72.32	72.40	72.47	72.54	111.52	111.60	111.67	111.75	111.82
72	1.8873	73.37	73.45	73.52	73.59	112.47	112.55	112.62	112.70	112.77
73	1.8887	74.42	74.50	74.57	74.64	113.42	113.50	113.57	113.65	113.72
74	1.8901	75.47	75.55	75.62	75.69	114.37	114.45	114.52	114.60	114.67
75	1.8915	76.52	76.60	76.67	76.74	115.32	115.40	115.47	115.55	115.62
76	1.8929	77.57	77.65	77.72	77.79	116.27	116.35	116.42	116.50	116.57
77	1.8943	78.62	78.70	78.77	78.84	117.22	117.30	117.37	117.45	117.52
78	1.8957	79.67	79.75	79.82	79.89	118.17	118.25	118.32	118.40	118.47
79	1.8971	80.72	80.80	80.87	80.94	119.12	119.20	119.27	119.35	119.42
80	1.8985	81.77	81.85	81.92	82.00	120.07	120.15	120.22	120.30	120.37
81	1.8999	82.82	82.90	82.97	83.04	121.02	121.10	121.17	121.25	121.32
82	1.9013	83.87	83.95	84.02	84.10	121.97	122.05	122.12	122.20	122.27
83	1.9027	84.92	85.00	85.07	85.14	122.92	123.00	123.07	123.15	123.22
84	1.9041	85.97	86.05	86.12	86.20	123.87	123.95	124.02	124.10	124.17
85	1.9055	87.02	87.10	87.17	87.24	124.82	124.90	124.97	125.05	125.12
86	1.9069	88.07	88.15	88.22	88.30	125.77	125.85	125.92	126.00	126.07
87	1.9083	89.12	89.20	89.27	89.34	126.72	126.80	126.87	126.95	127.02
88	1.9097	90.17	90.25	90.32	90.40	127.67	127.75	127.82	127.90	127.97
89	1.9111	91.22	91.30	91.37	91.44	128.62	128.70	128.77	128.85	128.92

Logarithms of cosines

$$\theta \rightarrow \log \cos \theta$$

θ	SUBTRACT Differences				
	1	2	3	4	5
0°	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.9999	0.9999	0.9999	0.9999	0.9999
2	0.9997	0.9997	0.9997	0.9997	0.9997
3	0.9994	0.9994	0.9994	0.9994	0.9994
4	0.9990	0.9990	0.9990	0.9990	0.9990
5	0.9985	0.9985	0.9985	0.9985	0.9985
6	0.9979	0.9979	0.9979	0.9979	0.9979
7	0.9972	0.9972	0.9972	0.9972	0.9972
8	0.9964	0.9964	0.9964	0.9964	0.9964
9	0.9955	0.9955	0.9955	0.9955	0.9955
10	0.9945	0.9945	0.9945	0.9945	0.9945
11	0.9934	0.9934	0.9934	0.9934	0.9934
12	0.9922	0.9922	0.9922	0.9922	0.9922
13	0.9909	0.9909	0.9909	0.9909	0.9909
14	0.9895	0.9895	0.9895	0.9895	0.9895
15	0.9880	0.9880	0.9880	0.9880	0.9880
16	0.9864	0.9864	0.9864	0.9864	0.9864
17	0.9847	0.9847	0.9847	0.9847	0.9847
18	0.9829	0.9829	0.9829	0.9829	0.9829
19	0.9810	0.9810	0.9810	0.9810	0.9810
20	0.9790	0.9790	0.9790	0.9790	0.9790
21	0.9769	0.9769	0.9769	0.9769	0.9769
22	0.9747	0.9747	0.9747	0.9747	0.9747
23	0.9724	0.9724	0.9724	0.9724	0.9724
24	0.9700	0.9700	0.9700	0.9700	0.9700
25	0.9676	0.9676	0.9676	0.9676	0.9676
26	0.9651	0.9651	0.9651	0.9651	0.9651
27	0.9625	0.9625	0.9625	0.9625	0.9625
28	0.9599	0.9599	0.9599	0.9599	0.9599
29	0.9572	0.9572	0.9572	0.9572	0.9572
30	0.9545	0.9545	0.9545	0.9545	0.9545
31	0.9517	0.9517	0.9517	0.9517	0.9517
32	0.9488	0.9488	0.9488	0.9488	0.9488
33	0.9459	0.9459	0.9459	0.9459	0.9459
34	0.9429	0.9429	0.9429	0.9429	0.9429
35	0.9398	0.9398	0.9398	0.9398	0.9398
36	0.9367	0.9367	0.9367	0.9367	0.9367
37	0.9335	0.9335	0.9335	0.9335	0.9335
38	0.9303	0.9303	0.9303	0.9303	0.9303
39	0.9270	0.9270	0.9270	0.9270	0.9270
40	0.9237	0.9237	0.9237	0.9237	0.9237
41	0.9203	0.9203	0.9203	0.9203	0.9203
42	0.9169	0.9169	0.9169	0.9169	0.9169
43	0.9134	0.9134	0.9134	0.9134	0.9134
44	0.9099	0.9099	0.9099	0.9099	0.9099

θ	SUBTRACT Differences				
	1	2	3	4	5
45°	0.8451	0.8451	0.8451	0.8451	0.8451
46	0.8418	0.8418	0.8418	0.8418	0.8418
47	0.8383	0.8383	0.8383	0.8383	0.8383
48	0.8347	0.8347	0.8347	0.8347	0.8347
49	0.8310	0.8310	0.8310	0.8310	0.8310
50	0.8272	0.8272	0.8272	0.8272	0.8272
51	0.8234	0.8234	0.8234	0.8234	0.8234
52	0.8195	0.8195	0.8195	0.8195	0.8195
53	0.8155	0.8155	0.8155	0.8155	0.8155
54	0.8115	0.8115	0.8115	0.8115	0.8115
55	0.8074	0.8074	0.8074	0.8074	0.8074
56	0.8033	0.8033	0.8033	0.8033	0.8033
57	0.7991	0.7991	0.7991	0.7991	0.7991
58	0.7949	0.7949	0.7949	0.7949	0.7949
59	0.7906	0.7906	0.7906	0.7906	0.7906
60	0.7863	0.7863	0.7863	0.7863	0.7863
61	0.7819	0.7819	0.7819	0.7819	0.7819
62	0.7775	0.7775	0.7775	0.7775	0.7775
63	0.7730	0.7730	0.7730	0.7730	0.7730
64	0.7685	0.7685	0.7685	0.7685	0.7685
65	0.7639	0.7639	0.7639	0.7639	0.7639
66	0.7593	0.7593	0.7593	0.7593	0.7593
67	0.7546	0.7546	0.7546	0.7546	0.7546
68	0.7499	0.7499	0.7499	0.7499	0.7499
69	0.7451	0.7451	0.7451	0.7451	0.7451
70	0.7403	0.7403	0.7403	0.7403	0.7403
71	0.7354	0.7354	0.7354	0.7354	0.7354
72	0.7305	0.7305	0.7305	0.7305	0.7305
73	0.7255	0.7255	0.7255	0.7255	0.7255
74	0.7205	0.7205	0.7205	0.7205	0.7205
75	0.7154	0.7154	0.7154	0.7154	0.7154
76	0.7103	0.7103	0.7103	0.7103	0.7103
77	0.7051	0.7051	0.7051	0.7051	0.7051
78	0.7000	0.7000	0.7000	0.7000	0.7000
79	0.6947	0.6947	0.6947	0.6947	0.6947
80	0.6895	0.6895	0.6895	0.6895	0.6895
81	0.6842	0.6842	0.6842	0.6842	0.6842
82	0.6788	0.6788	0.6788	0.6788	0.6788
83	0.6734	0.6734	0.6734	0.6734	0.6734
84	0.6679	0.6679	0.6679	0.6679	0.6679
85	0.6624	0.6624	0.6624	0.6624	0.6624
86	0.6568	0.6568	0.6568	0.6568	0.6568
87	0.6512	0.6512	0.6512	0.6512	0.6512
88	0.6456	0.6456	0.6456	0.6456	0.6456
89	0.6400	0.6400	0.6400	0.6400	0.6400

Reciprocals

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$$x \rightarrow \frac{1}{x}$$

x	SUBTRACT DIFFERENCES										x	SUBTRACT DIFFERENCES										SUBTRACT DIFFERENCES																		
	0	1	2	3	4	5	6	7	8	9		1	2	3	4	5	6	7	8	9																				
1.0	1.0000	9901	9804	9709	9615	9524	9434	9346	9259	9174	9	10	28	37	46	55	64	74	83	5.5	0.1810	1815	1812	1808	1805	1802	1799	1795	1792	1789	0	1	1	1	1	1	1	1	1	
1.1	0.9091	9090	8929	8850	8772	8696	8621	8547	8475	8405	8	15	23	31	38	46	53	61	69	5.6	0.1786	1783	1779	1776	1773	1770	1767	1764	1761	1757	0	1	1	1	1	1	1	1	1	
1.2	0.8333	8364	8192	8131	8065	8000	7937	7874	7813	7752	7	11	20	26	31	36	41	46	52	59	5.7	0.1754	1751	1748	1745	1742	1739	1736	1733	1730	1727	0	1	1	1	1	1	1	1	1
1.3	0.7692	7684	7536	7510	7481	7447	7417	7383	7349	7316	6	11	17	22	26	31	35	39	44	50	5.8	0.1724	1721	1718	1715	1712	1709	1706	1704	1701	1698	0	1	1	1	1	1	1	1	1
1.4	0.7143	7082	7042	6990	6944	6895	6849	6800	6757	6711	5	10	14	19	24	29	33	38	43	5.9	0.1695	1692	1689	1686	1684	1681	1678	1675	1672	1669	0	1	1	1	1	1	1	1	1	
1.5	0.6667	6623	6579	6536	6494	6452	6410	6368	6327	6286	4	9	13	17	21	25	29	33	38	6.0	0.1667	1664	1661	1658	1656	1653	1650	1647	1644	1642	0	1	1	1	1	1	1	1	1	
1.6	0.6250	6211	6173	6135	6098	6061	6024	5988	5952	5917	3	7	11	15	19	23	27	31	35	6.1	0.1639	1637	1634	1631	1629	1626	1623	1620	1617	1615	0	1	1	1	1	1	1	1	1	
1.7	0.5882	5848	5814	5780	5747	5714	5682	5650	5618	5587	2	6	10	13	16	20	23	26	29	6.2	0.1615	1611	1608	1605	1603	1600	1597	1595	1592	1590	0	1	1	1	1	1	1	1	1	
1.8	0.5556	5525	5495	5464	5433	5403	5373	5343	5313	5283	1	5	8	11	14	17	20	23	26	6.3	0.1592	1589	1586	1584	1582	1580	1577	1575	1572	1570	0	1	1	1	1	1	1	1	1	
1.9	0.5263	5236	5200	5164	5128	5093	5057	5022	4987	4952	0	4	7	9	11	13	16	19	22	6.4	0.1562	1560	1558	1555	1553	1550	1548	1546	1544	1541	0	1	1	1	1	1	1	1	1	
2.0	0.5000	4975	4950	4926	4902	4878	4854	4831	4808	4785	0	3	5	7	10	12	14	17	19	6.5	0.1538	1536	1534	1531	1529	1527	1524	1522	1520	1517	0	1	1	1	1	1	1	1	1	
2.1	0.4762	4739	4717	4695	4673	4651	4629	4608	4587	4566	0	2	4	6	9	11	13	15	17	6.6	0.1515	1513	1511	1508	1506	1504	1502	1499	1497	1495	0	1	1	1	1	1	1	1	1	
2.2	0.4545	4525	4505	4484	4464	4444	4424	4405	4386	4367	0	1	3	5	8	10	12	14	16	6.7	0.1493	1491	1489	1486	1484	1481	1478	1477	1475	1473	0	1	1	1	1	1	1	1	1	
2.3	0.4348	4329	4310	4292	4274	4256	4237	4219	4202	4184	0	1	2	4	6	8	10	12	14	6.8	0.1471	1468	1466	1464	1462	1460	1458	1456	1454	1451	0	1	1	1	1	1	1	1	1	
2.4	0.4167	4149	4132	4115	4098	4082	4065	4049	4032	4016	0	0	2	3	5	7	9	11	13	6.9	0.1449	1447	1445	1443	1441	1439	1437	1435	1433	1431	0	1	1	1	1	1	1	1	1	
2.5	0.4000	3984	3968	3953	3937	3922	3906	3891	3876	3861	0	0	1	2	4	6	8	10	12	7.0	0.1425	1423	1421	1419	1417	1415	1413	1411	1410	1408	0	1	1	1	1	1	1	1	1	
2.6	0.3846	3831	3817	3802	3788	3774	3759	3745	3731	3717	0	0	1	2	4	6	7	9	11	7.1	0.1400	1400	1400	1400	1400	1399	1397	1395	1393	1391	0	1	1	1	1	1	1	1	1	
2.7	0.3704	3690	3676	3663	3650	3636	3623	3610	3597	3584	0	0	1	2	4	5	7	8	10	7.2	0.1389	1387	1385	1383	1381	1379	1377	1375	1374	1372	0	1	1	1	1	1	1	1	1	
2.8	0.3571	3559	3546	3534	3521	3509	3497	3484	3472	3460	0	0	1	2	4	5	6	7	9	7.3	0.1370	1368	1366	1364	1362	1361	1359	1357	1355	1353	0	1	1	1	1	1	1	1	1	
2.9	0.3448	3436	3425	3413	3401	3390	3378	3367	3356	3344	0	0	1	2	3	5	6	7	8	7.4	0.1351	1350	1348	1346	1344	1342	1340	1339	1337	1335	0	1	1	1	1	1	1	1	1	
3.0	0.3333	3322	3312	3302	3292	3282	3272	3262	3252	3242	0	0	1	2	3	4	5	6	7	7.5	0.1333	1332	1330	1328	1326	1325	1323	1321	1319	1318	0	1	1	1	1	1	1	1	1	
3.1	0.3226	3215	3205	3195	3185	3175	3165	3155	3145	3135	0	0	1	2	3	4	5	6	7	7.6	0.1316	1314	1312	1311	1309	1307	1305	1304	1302	1300	0	1	1	1	1	1	1	1	1	
3.2	0.3120	3115	3106	3096	3086	3077	3067	3057	3048	3039	0	0	1	2	3	4	5	6	7	7.7	0.1299	1297	1295	1294	1292	1290	1289	1287	1285	1284	0	1	1	1	1	1	1	1	1	
3.3	0.3020	3021	3012	3003	2994	2985	2976	2967	2958	2950	0	0	1	2	3	4	5	6	7	7.8	0.1282	1280	1279	1277	1276	1274	1272	1271	1269	1267	0	1	1	1	1	1	1	1	1	
3.4	0.2924	2923	2914	2905	2897	2888	2879	2870	2861	2852	0	0	1	2	3	4	5	6	7	7.9	0.1266	1264	1263	1261	1259	1258	1256	1255	1253	1252	0	1	1	1	1	1	1	1	1	
3.5	0.2832	2849	2841	2833	2825	2817	2809	2801	2793	2786	0	0	1	2	3	4	5	6	7	8.0	0.1250	1248	1247	1245	1244	1242	1241	1239	1238	1236	0	1	1	1	1	1	1	1	1	
3.6	0.2740	2770	2762	2755	2747	2740	2732	2725	2717	2710	0	0	1	2	3	4	5	6	7	8.1	0.1235	1233	1232	1230	1229	1227	1225	1224	1222	1221	0	1	1	1	1	1	1	1	1	
3.7	0.2650	2685	2688	2681	2674	2667	2660	2653	2646	2639	0	0	1	2	3	4	5	6	7	8.2	0.1220	1218	1217	1215	1214	1212	1211	1209	1208	1206	0	1	1	1	1	1	1	1	1	
3.8	0.2562	2575	2578	2571	2564	2557	2550	2543	2536	2529	0	0	1	2	3	4	5	6	7	8.3	0.1205	1203	1202	1200	1199	1197	1196	1195	1193	1192	0	1	1	1	1	1	1	1	1	
3.9	0.2476	2459	2451	2443	2435	2428	2421	2414	2407	2400	0	0	1	2	3	4	5	6	7	8.4	0.1190	1189	1188	1186	1185	1183	1182	1181	1179	1178	0	1	1	1	1	1	1	1	1	
4.0	0.2500	2494	2488	2481	2475	2469	2463	2457	2451	2445	0	0	1	2	3	4	5	6	7	8.5	0.1176	1175	1174	1172	1171	1170	1168	1167	1166	1164	0	1	1	1	1	1	1	1	1	
4.1	0.2430	2433	2427	2421	2415	2410	2404	2398	2392	2387	0	0	1	2	3	4	5	6	7	8.6	0.1163	1161	1160	1159	1157	1156	1155	1153	1152	1151	0	1	1	1	1	1	1	1	1	
4.2	0.2380	2375	2370	2365	2360	2355	2350	2344	2338	2331	0	0	1	2	3	4	5	6	7	8.7	0.1142	1140	1140	1140	1140	1140	1140	1140	1140	1139	1138	0	1	1	1	1	1	1	1	
4.3	0.2326	2320	2315	2310	2305	2300	2294	2289	2283	2277	0	0	1	2	3	4	5	6	7	8.8	0.1126	11231																		

Differences											Differences										
x	0	1	2	3	4	5	6	7	8	9	1	2	3	4	5	6	7	8	9	10	
10	1000	1020	1040	1060	1080	1100	1124	1145	1166	1188	2	4	6	8	10	13	15	17	19	21	
11	1210	1232	1254	1277	1299	1321	1346	1368	1392	1416	3	5	7	9	11	14	16	18	21	23	
12	1440	1464	1488	1513	1538	1563	1588	1613	1638	1664	4	6	8	10	12	15	17	20	22	24	
13	1690	1716	1742	1769	1796	1823	1850	1877	1904	1932	5	7	9	11	13	16	19	21	24	26	
14	1960	1986	2016	2043	2074	2103	2132	2161	2190	2220	6	8	10	12	14	17	20	23	26	28	
15	2250	2280	2310	2341	2372	2403	2434	2465	2496	2528	7	9	11	13	15	19	22	25	28	31	
16	2560	2592	2624	2657	2690	2723	2756	2789	2822	2856	8	10	12	14	16	20	23	26	30	33	
17	2890	2924	2958	2993	3027	3063	3098	3133	3168	3204	9	11	13	15	17	21	24	28	31	34	
18	3240	3276	3312	3349	3386	3423	3460	3497	3534	3572	10	12	14	16	18	22	26	30	33	36	
19	3610	3648	3686	3725	3764	3803	3842	3881	3920	3960	11	13	15	17	19	23	27	31	34	37	
20	4000	4040	4080	4121	4162	4203	4244	4285	4326	4368	12	14	16	18	20	24	28	32	35	38	
21	4410	4452	4494	4537	4580	4623	4666	4709	4752	4796	13	15	17	19	21	25	29	33	36	39	
22	4840	4884	4928	4973	5018	5063	5108	5153	5198	5244	14	16	18	20	22	26	30	34	37	40	
23	5290	5336	5382	5429	5476	5523	5570	5617	5664	5712	15	17	19	21	23	27	31	35	38	41	
24	5760	5808	5856	5905	5954	6003	6052	6101	6150	6200	16	18	20	22	24	28	32	36	39	42	
25	6250	6300	6350	6401	6452	6503	6554	6605	6656	6708	17	19	21	23	25	29	33	37	40	43	
26	6760	6812	6864	6917	6970	7023	7076	7128	7182	7236	18	20	22	24	26	30	34	38	41	44	
27	7290	7344	7398	7453	7508	7563	7618	7673	7728	7784	19	21	23	25	27	31	35	39	42	45	
28	7840	7896	7952	8009	8066	8123	8180	8237	8294	8352	20	22	24	26	28	32	36	40	43	46	
29	8410	8468	8526	8585	8644	8703	8762	8821	8880	8940	21	23	25	27	29	33	37	41	44	47	
30	9000	9060	9120	9181	9242	9303	9364	9425	9486	9548	22	24	26	28	30	34	38	42	45	48	
31	9610	9672	9734	9797	9860	9923	9986	—	—	—	23	25	27	29	31	35	39	43	46	49	
32	—	—	—	—	—	—	—	—	—	—	24	26	28	30	32	36	40	44	47	50	
33	1024	1037	1053	1070	1086	1103	1120	1136	1152	1169	25	27	29	31	33	37	41	45	48	51	
34	1136	1165	1194	1224	1254	1284	1314	1344	1374	1404	26	28	30	32	34	38	42	46	49	52	
35	1225	1257	1290	1324	1358	1392	1426	1460	1494	1528	27	29	31	33	35	39	43	47	50	53	
36	1360	1403	1446	1490	1534	1578	1622	1666	1710	1754	28	30	32	34	36	40	44	48	51	54	
37	1464	1518	1572	1626	1680	1734	1788	1842	1896	1950	29	31	33	35	37	41	45	49	52	55	
38	1584	1640	1696	1752	1808	1864	1920	1976	2032	2088	30	32	34	36	38	42	46	50	53	56	
39	1721	1780	1839	1898	1957	2016	2075	2134	2193	2252	31	33	35	37	39	43	47	51	54	57	
40	1876	1938	2000	2062	2124	2186	2248	2310	2372	2434	32	34	36	38	40	44	48	52	55	58	
41	2049	2114	2179	2244	2309	2374	2439	2504	2569	2634	33	35	37	39	41	45	49	53	56	59	
42	2240	2308	2376	2444	2512	2580	2648	2716	2784	2852	34	36	38	40	42	46	50	54	57	60	
43	2448	2518	2588	2658	2728	2798	2868	2938	3008	3078	35	37	39	41	43	47	51	55	58	61	
44	2676	2748	2820	2892	2964	3036	3108	3180	3252	3324	36	38	40	42	44	48	52	56	59	62	
45	2924	2998	3072	3146	3220	3294	3368	3442	3516	3590	37	39	41	43	45	49	53	57	60	63	
46	3192	3268	3344	3420	3496	3572	3648	3724	3800	3876	38	40	42	44	46	50	54	58	61	64	
47	3480	3558	3636	3714	3792	3870	3948	4026	4104	4182	39	41	43	45	47	51	55	59	62	65	
48	3788	3868	3948	4028	4108	4188	4268	4348	4428	4508	40	42	44	46	48	52	56	60	63	66	
49	4116	4198	4280	4362	4444	4526	4608	4690	4772	4854	41	43	45	47	49	53	57	61	64	67	
50	4464	4548	4632	4716	4800	4884	4968	5052	5136	5220	42	44	46	48	50	54	58	62	65	68	
51	4832	4918	5004	5090	5176	5262	5348	5434	5520	5606	43	45	47	49	51	55	59	63	66	69	
52	5220	5308	5396	5484	5572	5660	5748	5836	5924	6012	44	46	48	50	52	56	60	64	67	70	
53	5628	5718	5808	5898	5988	6078	6168	6258	6348	6438	45	47	49	51	53	57	61	65	68	71	
54	6048	6140	6232	6324	6416	6508	6600	6692	6784	6876	46	48	50	52	54	58	62	66	69	72	

Square roots from 1 to 10

$$x \rightarrow \sqrt{x}$$

x	Differences										Differences
	0	1	2	3	4	5	6	7	8	9	
1.0	1.000	1.000	1.010	1.015	1.020	1.025	1.030	1.035	1.040	1.045	0.010
1.1	1.049	1.054	1.058	1.061	1.064	1.067	1.070	1.073	1.076	1.079	0.010
1.2	1.095	1.100	1.104	1.108	1.111	1.114	1.117	1.120	1.123	1.126	0.010
1.3	1.116	1.121	1.125	1.128	1.131	1.134	1.137	1.140	1.143	1.146	0.010
1.4	1.183	1.187	1.191	1.194	1.197	1.200	1.203	1.206	1.209	1.212	0.010
1.5	1.225	1.229	1.233	1.236	1.239	1.242	1.245	1.248	1.251	1.254	0.010
1.6	1.265	1.269	1.272	1.275	1.278	1.281	1.284	1.287	1.290	1.293	0.010
1.7	1.304	1.308	1.311	1.314	1.317	1.320	1.323	1.326	1.329	1.332	0.010
1.8	1.342	1.345	1.348	1.351	1.354	1.357	1.360	1.363	1.366	1.369	0.010
1.9	1.376	1.379	1.382	1.385	1.388	1.391	1.394	1.397	1.400	1.403	0.010
2.0	1.414	1.418	1.421	1.424	1.427	1.430	1.433	1.436	1.439	1.442	0.010
2.1	1.453	1.457	1.460	1.463	1.466	1.469	1.472	1.475	1.478	1.481	0.010
2.2	1.483	1.487	1.490	1.493	1.496	1.499	1.502	1.505	1.508	1.511	0.010
2.3	1.517	1.520	1.523	1.526	1.529	1.532	1.535	1.538	1.541	1.544	0.010
2.4	1.548	1.552	1.555	1.558	1.561	1.564	1.567	1.570	1.573	1.576	0.010
2.5	1.581	1.585	1.588	1.591	1.594	1.597	1.600	1.603	1.606	1.609	0.010
2.6	1.612	1.615	1.618	1.621	1.624	1.627	1.630	1.633	1.636	1.639	0.010
2.7	1.641	1.644	1.647	1.650	1.653	1.656	1.659	1.662	1.665	1.668	0.010
2.8	1.671	1.674	1.677	1.680	1.683	1.686	1.689	1.692	1.695	1.698	0.010
2.9	1.701	1.704	1.707	1.710	1.713	1.716	1.719	1.722	1.725	1.728	0.010
3.0	1.732	1.735	1.738	1.741	1.744	1.747	1.750	1.753	1.756	1.759	0.010
3.1	1.761	1.764	1.767	1.770	1.773	1.776	1.779	1.782	1.785	1.788	0.010
3.2	1.788	1.791	1.794	1.797	1.800	1.803	1.806	1.809	1.812	1.815	0.010
3.3	1.817	1.820	1.823	1.826	1.829	1.832	1.835	1.838	1.841	1.844	0.010
3.4	1.844	1.847	1.850	1.853	1.856	1.859	1.862	1.865	1.868	1.871	0.010
3.5	1.871	1.874	1.877	1.880	1.883	1.886	1.889	1.892	1.895	1.898	0.010
3.6	1.897	1.900	1.903	1.906	1.909	1.912	1.915	1.918	1.921	1.924	0.010
3.7	1.924	1.927	1.930	1.933	1.936	1.939	1.942	1.945	1.948	1.951	0.010
3.8	1.949	1.952	1.955	1.958	1.961	1.964	1.967	1.970	1.973	1.976	0.010
3.9	1.975	1.978	1.981	1.984	1.987	1.990	1.993	1.996	1.999	2.002	0.010
4.0	2.000	2.003	2.006	2.009	2.012	2.015	2.018	2.021	2.024	2.027	0.010
4.1	2.025	2.028	2.031	2.034	2.037	2.040	2.043	2.046	2.049	2.052	0.010
4.2	2.050	2.053	2.056	2.059	2.062	2.065	2.068	2.071	2.074	2.077	0.010
4.3	2.074	2.077	2.080	2.083	2.086	2.089	2.092	2.095	2.098	2.101	0.010
4.4	2.100	2.103	2.106	2.109	2.112	2.115	2.118	2.121	2.124	2.127	0.010
4.5	2.121	2.124	2.127	2.130	2.133	2.136	2.139	2.142	2.145	2.148	0.010
4.6	2.145	2.148	2.151	2.154	2.157	2.160	2.163	2.166	2.169	2.172	0.010
4.7	2.168	2.171	2.174	2.177	2.180	2.183	2.186	2.189	2.192	2.195	0.010
4.8	2.191	2.194	2.197	2.200	2.203	2.206	2.209	2.212	2.215	2.218	0.010
4.9	2.214	2.217	2.220	2.223	2.226	2.229	2.232	2.235	2.238	2.241	0.010
5.0	2.236	2.239	2.242	2.245	2.248	2.251	2.254	2.257	2.260	2.263	0.010
5.1	2.258	2.261	2.264	2.267	2.270	2.273	2.276	2.279	2.282	2.285	0.010
5.2	2.280	2.283	2.286	2.289	2.292	2.295	2.298	2.301	2.304	2.307	0.010
5.3	2.302	2.305	2.308	2.311	2.314	2.317	2.320	2.323	2.326	2.329	0.010
5.4	2.324	2.327	2.330	2.333	2.336	2.339	2.342	2.345	2.348	2.351	0.010

Square roots from 10 to 100

$$x \rightarrow \sqrt{x}$$

x	Difference									
	0	1	2	3	4	5	6	7	8	9
10	3.162	3.178	3.194	3.209	3.225	3.240	3.256	3.271	3.286	3.302
11	3.317	3.332	3.347	3.362	3.376	3.391	3.406	3.421	3.435	3.450
12	3.464	3.479	3.493	3.507	3.521	3.536	3.550	3.564	3.578	3.592
13	3.606	3.619	3.633	3.647	3.661	3.674	3.688	3.701	3.715	3.728
14	3.762	3.775	3.788	3.802	3.815	3.828	3.841	3.854	3.867	3.880
15	3.892	3.905	3.918	3.931	3.944	3.957	3.969	3.982	3.994	4.007
16	4.000	4.012	4.025	4.037	4.050	4.062	4.074	4.087	4.099	4.111
17	4.124	4.135	4.147	4.159	4.171	4.183	4.195	4.207	4.219	4.231
18	4.243	4.254	4.266	4.278	4.289	4.301	4.312	4.324	4.335	4.347
19	4.359	4.370	4.382	4.393	4.405	4.416	4.427	4.438	4.450	4.461
20	4.472	4.483	4.494	4.506	4.517	4.528	4.539	4.550	4.561	4.572
21	4.583	4.594	4.604	4.615	4.626	4.637	4.648	4.658	4.669	4.680
22	4.690	4.701	4.712	4.722	4.733	4.743	4.754	4.765	4.775	4.785
23	4.796	4.806	4.817	4.827	4.837	4.848	4.858	4.868	4.879	4.889
24	4.899	4.909	4.919	4.929	4.940	4.950	4.960	4.970	4.980	4.990
25	5.000	5.010	5.020	5.030	5.040	5.050	5.060	5.070	5.079	5.089
26	5.099	5.109	5.119	5.128	5.138	5.148	5.158	5.167	5.177	5.187
27	5.196	5.206	5.215	5.225	5.234	5.244	5.254	5.263	5.273	5.282
28	5.292	5.301	5.311	5.320	5.329	5.339	5.348	5.357	5.367	5.376
29	5.385	5.394	5.404	5.413	5.422	5.431	5.440	5.450	5.459	5.468
30	5.477	5.486	5.495	5.505	5.514	5.523	5.532	5.541	5.550	5.559
31	5.568	5.577	5.586	5.595	5.604	5.613	5.622	5.630	5.639	5.648
32	5.657	5.666	5.675	5.684	5.692	5.701	5.710	5.718	5.727	5.736
33	5.745	5.753	5.762	5.771	5.779	5.788	5.797	5.805	5.814	5.822
34	5.831	5.840	5.848	5.857	5.865	5.874	5.882	5.891	5.899	5.908
35	5.916	5.925	5.933	5.941	5.950	5.958	5.967	5.975	5.983	5.992
36	6.000	6.009	6.017	6.025	6.033	6.042	6.050	6.058	6.066	6.075
37	6.083	6.091	6.099	6.107	6.116	6.124	6.132	6.140	6.148	6.156
38	6.164	6.171	6.179	6.187	6.195	6.203	6.211	6.219	6.227	6.235
39	6.243	6.251	6.258	6.266	6.273	6.281	6.289	6.297	6.305	6.313
40	6.321	6.329	6.336	6.344	6.352	6.360	6.367	6.375	6.383	6.391
41	6.400	6.407	6.415	6.423	6.431	6.438	6.446	6.453	6.461	6.469
42	6.477	6.484	6.491	6.499	6.506	6.513	6.521	6.527	6.535	6.542
43	6.550	6.557	6.564	6.571	6.578	6.585	6.592	6.599	6.606	6.613
44	6.621	6.628	6.635	6.642	6.648	6.655	6.662	6.669	6.676	6.683
45	6.690	6.696	6.703	6.709	6.716	6.723	6.729	6.736	6.742	6.749
46	6.756	6.762	6.768	6.774	6.780	6.786	6.792	6.798	6.804	6.810
47	6.816	6.822	6.828	6.834	6.840	6.846	6.852	6.858	6.864	6.870
48	6.876	6.882	6.888	6.894	6.899	6.905	6.911	6.917	6.923	6.929
49	6.935	6.940	6.946	6.952	6.957	6.963	6.969	6.975	6.981	6.987
50	6.993	6.998	7.004	7.009	7.015	7.021	7.026	7.032	7.038	7.044
51	7.050	7.055	7.061	7.066	7.072	7.077	7.083	7.088	7.094	7.100
52	7.105	7.111	7.116	7.122	7.127	7.132	7.138	7.143	7.149	7.154
53	7.160	7.165	7.171	7.176	7.182	7.187	7.192	7.198	7.203	7.209
54	7.214	7.219	7.225	7.230	7.235	7.241	7.246	7.251	7.257	7.262
55	7.267	7.272	7.278	7.283	7.288	7.294	7.299	7.304	7.309	7.315
56	7.320	7.325	7.330	7.336	7.341	7.346	7.351	7.356	7.361	7.367
57	7.372	7.377	7.382	7.387	7.392	7.397	7.402	7.407	7.412	7.417
58	7.423	7.428	7.433	7.438	7.443	7.448	7.453	7.458	7.463	7.468
59	7.473	7.478	7.483	7.488	7.493	7.498	7.503	7.508	7.513	7.518
60	7.523	7.528	7.533	7.538	7.543	7.548	7.553	7.558	7.563	7.568
61	7.573	7.578	7.583	7.588	7.593	7.598	7.603	7.608	7.613	7.618
62	7.623	7.628	7.633	7.638	7.643	7.648	7.653	7.658	7.663	7.668
63	7.673	7.678	7.683	7.688	7.693	7.698	7.703	7.708	7.713	7.718
64	7.723	7.728	7.733	7.738	7.743	7.748	7.753	7.758	7.763	7.768
65	7.773	7.778	7.783	7.788	7.793	7.798	7.803	7.808	7.813	7.818
66	7.823	7.828	7.833	7.838	7.843	7.848	7.853	7.858	7.863	7.868
67	7.873	7.878	7.883	7.888	7.893	7.898	7.903	7.908	7.913	7.918
68	7.923	7.928	7.933	7.938	7.943	7.948	7.953	7.958	7.963	7.968
69	7.973	7.978	7.983	7.988	7.993	7.998	8.003	8.008	8.013	8.018
70	8.023	8.028	8.033	8.038	8.043	8.048	8.053	8.058	8.063	8.068
71	8.073	8.078	8.083	8.088	8.093	8.098	8.103	8.108	8.113	8.118
72	8.123	8.128	8.133	8.138	8.143	8.148	8.153	8.158	8.163	8.168
73	8.173	8.178	8.183	8.188	8.193	8.198	8.203	8.208	8.213	8.218
74	8.223	8.228	8.233	8.238	8.243	8.248	8.253	8.258	8.263	8.268
75	8.273	8.278	8.283	8.288	8.293	8.298	8.303	8.308	8.313	8.318
76	8.323	8.328	8.333	8.338	8.343	8.348	8.353	8.358	8.363	8.368
77	8.373	8.378	8.383	8.388	8.393	8.398	8.403	8.408	8.413	8.418
78	8.423	8.428	8.433	8.438	8.443	8.448	8.453	8.458	8.463	8.468
79	8.473	8.478	8.483	8.488	8.493	8.498	8.503	8.508	8.513	8.518
80	8.523	8.528	8.533	8.538	8.543	8.548	8.553	8.558	8.563	8.568
81	8.573	8.578	8.583	8.588	8.593	8.598	8.603	8.608	8.613	8.618
82	8.623	8.628	8.633	8.638	8.643	8.648	8.653	8.658	8.663	8.668
83	8.673	8.678	8.683	8.688	8.693	8.698	8.703	8.708	8.713	8.718
84	8.723	8.728	8.733	8.738	8.743	8.748	8.753	8.758	8.763	8.768
85	8.773	8.778	8.783	8.788	8.793	8.798	8.803	8.808	8.813	8.818
86	8.823	8.828	8.833	8.838	8.843	8.848	8.853	8.858	8.863	8.868
87	8.873	8.878	8.883	8.888	8.893	8.898	8.903	8.908	8.913	8.918
88	8.923	8.928	8.933	8.938	8.943	8.948	8.953	8.958	8.963	8.968
89	8.973	8.978	8.983	8.988	8.993	8.998	9.003	9.008	9.013	9.018
90	9.023	9.028	9.033	9.038	9.043	9.048	9.053	9.058	9.063	9.068
91	9.073	9.078	9.083	9.088	9.093	9.098	9.103	9.108	9.113	9.118
92	9.123	9.128	9.133	9.138	9.143	9.148	9.153	9.158	9.163	9.168
93	9.173	9.178	9.183	9.188	9.193	9.198	9.203	9.208	9.213	9.218
94	9.223	9.228	9.233	9.238	9.243	9.248	9.253	9.258	9.263	9.268
95	9.273	9.278	9.283	9.288	9.293	9.298	9.303	9.308	9.313	9.318
96	9.323	9.328	9.333	9.338	9.343	9.348	9.353	9.358	9.363	9.368
97	9.373	9.378	9.383	9.388	9.393	9.398	9.403	9.408	9.413	9.418
98	9.423	9.428	9.433	9.438	9.443	9.448	9.453	9.458	9.463	9.468
99	9.473	9.478	9.483	9.488	9.493	9.498	9.503	9.508	9.513	9.518

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Answers

Exercise 1a (p. 2)

- (a) 3 m (b) 1 m (c) 86 ha
 (d) 342 km (e) 496 km (f) 165 mm
 (g) 53 (h) \$18 (i) 130 litres
 (j) \$100
 (a) 0,1 (b) 18,0 (c) 19,0
 (d) 0,8 (e) 7,9 (f) 20,1
 (a) 37 (b) 110 (c) 0,009 3
 (d) 5,0 (e) 0,024 (f) 86 000
 (g) 140 (h) 9,0 (i) 3,0
 (j) 0,030
 (a) 7 580 (b) 52 100 (c) 352 000
 (d) 1 790 (e) 0,089 2 (f) 170
 (g) 83,4 (h) 0,906 (i) 828 000
 (j) 8,01

\$389 million, \$390 million,
 Almost \$400 million, \$0,39 billion

(a)

town	population
Bulawayo	500 000
Chitungwiza	200 000
Harare	700 000

(b)

town	population
Bulawayo	500 000
Chitungwiza	170 000
Harare	660 000

\$1,45 billion and \$1,55 billion

(a)

1984	1985	1986	1987	1988
00 000	400 000	400 000	500 000	500 000

- (b) 2 100 000 (true value is 2 141 872)
 (a) 28 (b) 300 (c) 900 (d) 0,002
 (a) 34,62 (b) 327,9 (c) 742,8 (d) 0,002 135

Exercise 1b (p. 4)

- 1 (a) 8,5 cm to 9,5 cm (b) 1,745 m to 1,755 m
 (c) 6 350 km to 6 450 km (d) 9,75 m to 9,85 m
 (e) 8,5 million to 9,5 million
 (f) 7,45 litres to 7,55 litres
 (g) \$19 500 to \$20 500 (h) 75,55 kg to 75,65 kg
 (i) 30,15 °C to 30,25 °C
 (j) 859,5 cm³ to 860,5 cm³
 2 (a) 0,667% (b) 0,83% (c) 0,4%
 (d) 2% (e) 2,5% (f) 0,806%
 (g) 6,25% (h) 2,38% (i) 1,25%
 (j) 5,6%
 3 (a) 4 000 g (4 kg) (b) 4 140 g (4,14 kg)
 (c) -3,4%
 4 (a) \$350 (b) \$340,06
 (c) +2,9%
 5 -8,0% 6 102 g, 100 g
 7 244,8 kg, 235,2 kg
 8 6 175 m², 7 875 m²
 9 (a) 11π cm, 13π cm (b) 30 $\frac{1}{2}$ π cm², 42 $\frac{1}{4}$ π cm²
 10 (a) 337,5 cm², 433,5 cm²
 (b) 421,875 cm³, 614,125 cm³

Exercise 1c (p. 6)

- 1 (a) 1 $\frac{3}{4}$ h (b) 60 km/h
 2 39 000 cm³
 3 132 cm
 4 (a) 5,442 (b) 33,236 (c) 14 (d) 0,11
 5 (a) 0,095 or 0,10 (b) 0,83 or 0,8 (c) 23
 6 250 mm to 280 mm, 590 to 670
 7 (a) 5,6 m (b) 5,9 m
 8 800 km/h
 9 1 000 cm³
 10 2 800 cm²

Exercise 2a (p. 7)

- 1 (c) XY = 4,1 cm, CY = 6,25 cm
 2 (d) a rhombus (e) 11,9 cm, 3,3 cm
 3 (c) 3,5 cm 4 (b) 8,7 cm
 5 (b) 8,25 cm 6 (b) 9,9 cm

Exercise 2b (p. 8)

- 1 (c) The altitudes meet at one point inside the triangle.
 2 (c) Yes (outside the triangle)

- 3 (c) 43 mm 4 (c) 8,9 cm
 5 (d) The circle touches all three sides of $\triangle ABC$
 (c) 2,05 cm
 6 (c) 1,7 cm
 7 (d) (i) 3,4 cm, (ii) the circle touches all four sides of ABCD
 8 (c) X, Y, Z lie in a straight line (d) yes

Exercise 2c (p. 10)

- 1 a spiral
 2 an oval shape or **ellipse**
 3 an arc of a circle
 4 (a), (b): major arc of a circle; (c) a semicircle
 5 a circle of diameter 2 cm
 6 (a) a semicircle (b) a circle
 (c) an arc of a circle subtending 30° at its centre
 (d) a straight line (e) a circular arc
 (f)



Fig. A1

- 7 a straight line at an angle to the direction in which the car is moving
 8 a hemispherical surface (smaller than that of the bowl)
 9 a straight line 1,5 cm from AB
 10 a circle with the foot of the pole at its centre
 11 a semicircle
 12 the perpendicular bisector of the line joining the two points
 13 two straight lines, parallel to and on either side of AB
 14 42 m
 15 88 cm

Exercise 2d (p. 13)

- 1 two; 6,7 cm 2 four
 3 2,7 cm, 8,7 cm 4 1,8 cm, 6,2 cm
 5 23° , $56\frac{1}{2}^\circ$ 6 5,7 cm
 7 6 cm 8 42 m
 10 8 cm 11 (b) 11,3 cm

Exercise 2e (p. 16)

- 1 P is the point of intersection of side YZ and the bisector of X
 2 two
 4 3,2 cm or 9,2 cm
 8 AB = 4,1 cm; AC = 6,4 cm
 10 (c) 0,8 cm or 1,1 cm

- 12 the circumcentres lie (a) inside (b) outside
 (c) at the mid-point of the hypotenuse of the triangles respectively
 13 PZ = 4,6 cm
 14 (b) AC = 9,6 cm

Exercise 3a (p. 18)

- 1 (a) 18° (b) 45° (c) 63° (d) 70°
 2 (a) 36° (b) 42° (c) 124° (d) 70°
 3 (a) 15 cm (b) 5 cm
 4 (a) 43° (b) 63° (c) 48° (d) 32°
 5 4,5 cm 6 3,3 cm
 7 141° 8 8 cm
 9 (a) (i) $(90 - x)^\circ$, (ii) $(90 - x)^\circ$
 (b) $\angle OBA = \angle ABC$
 10 (a) 90° (b) $(90 - x)^\circ$
 (c) $\angle ADB = x^\circ$ (sum of angles of $\triangle ADB$)
 $\therefore \angle BAT = \angle ADB = x^\circ$

Exercise 3a (p. 18)

- 1 (a) 36° (b) 57° (c) 46° (d) 12 cm
 2 $\angle AOB = \angle BOC = 90^\circ$ (radius $\perp$ tangent)
 $\therefore \angle AOB$ is cyclic (opposite angles are supplementary)
 3 43° 4 106°
 5 59° or 121° 10 134°

Exercise 3c (p. 21)

- 1 5 cm; 4 cm; 2 cm
 2 2 cm; 3 cm; 4 cm
 3 5 cm; 6 cm; 14 cm
 4 (a) 14 cm (b) 4 cm
 8 86 cm

Exercise 3d (p. 23)

- 1 (a) $\angle ABY = \angle ACY$ (b) $\angle CBY$, $\angle CAZ$
 (c) $\angle BCY$ (d) $\angle BAY$
 (e) 58° (f) 112°
 (g) 55° (h) 80°
 2 (a) 56° (b) 52° (c) 43°
 (d) 102° (e) 78°
 3 ZYQ 4 84°
 5 72° 6 82° ; 49° ; 49°
 7 54° 8 40°
 9 101° ; 74° 10 34° ; 77° ; 69°
 11 50° ; 60° ; 70° 12 44° ; 108°

Exercise 4a (p. 26)

- 1 0,939 7 2 -0,342 0 3 -2,747
 4 0,454 0 5 0,990 3 6 -0,275 6
 7 -0,788 0 8 -0,230 9 9 -19,08

- | | | |
|-------------|-------------|-------------|
| 10 -0,615 7 | 11 0,398 7 | 12 -0,113 9 |
| 13 -0,947 8 | 14 -1,122 | 15 0,998 4 |
| 16 -0,714 5 | 17 -0,226 7 | 18 0,877 1 |
| 19 0,028 8 | 20 -21,45 | 21 0,271 9 |
| 22 -0,482 2 | 23 -0,522 8 | 24 0,996 0 |

Exercise 4b (p. 27)

- | | | |
|--------------------------------|--------------------------------|-------------------|
| 1 36° | 2 144° | 3 75° |
| 4 105° | 5 $67^\circ, 113^\circ$ | 6 117° |
| 7 160° | 8 $25^\circ, 155^\circ$ | 9 98° |
| 10 148° | 11 $136,7^\circ$ | |
| 12 $37,5^\circ, 142,5^\circ$ | 13 $74,7^\circ, 105,3^\circ$ | |
| 14 $115,4^\circ$ | 15 $162,43^\circ$ | 16 $115,17^\circ$ |
| 17 $56,4^\circ, 123,6^\circ$ | 18 $144,13^\circ$ | |
| 19 $73,62^\circ, 106,38^\circ$ | 20 $141,68^\circ$ | |
| 21 $120,8^\circ$ | 22 $15,23^\circ, 164,77^\circ$ | |
| 23 $48,15^\circ, 131,85^\circ$ | 24 $94,6^\circ$ | |

Exercise 4c (p. 29)

$$1 \quad \frac{x}{\sin 80^\circ} = \frac{4}{\sin 30^\circ} \quad \text{or} \quad x = \frac{4 \sin 80^\circ}{\sin 30^\circ}$$

The answer to questions 2-12 have been rounded to 3 s.f.

- | | | |
|--|----------------|-----------|
| 2 13,0 cm | 3 14,5 cm | 4 21,5 cm |
| 5 $48,8^\circ$ | 6 $41,6^\circ$ | |
| 7 $B = 26,5^\circ, C = 38,5^\circ, c = 44,6$ m | | |
| 8 $A = 30^\circ, a = 23,8$ m, $b = 20,5$ m | | |
| 9 $X = 98,8^\circ, y = 9,9$ cm, $z = 15,5$ cm | | |
| 10 $B = 29^\circ 9', C = 112^\circ 33', c = 376$ m | | |
| 11 $A = 67,4^\circ, B = 16,4^\circ, a = 36,6$ cm | | |
| 12 $A = 29^\circ 26', B = 34^\circ 58'$ | | |

Exercise 4d (p. 30)

- | | |
|---|----------------|
| 1 38,1 m | 2 8,15 km |
| 3 $294,8^\circ, 188$ m | 4 190 m |
| 5 4,06 m | 6 240 km |
| 7 23,6 m, 21,7 m | 8 165 m, 116 m |
| 9 2,54 km | |
| 10 (a) 386 km | (b) 141 km |
| 11 (a) $63^\circ 3'$ | (b) 7,51 cm |
| 12 (a) 060° (b) 340° (c) 9,54 n.mi. (d) 5 n.mi. | |

Exercise 5a (p. 33)

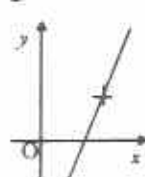
- | | | | | |
|-----------------|-----------------|------------------|------------------|-------------------|
| 1 $\frac{2}{7}$ | 2 $\frac{3}{2}$ | 3 $\frac{3}{5}$ | 4 $-\frac{4}{5}$ | 5 $-\frac{4}{3}$ |
| 6 $\frac{2}{7}$ | 7 -2 | 8 $-\frac{3}{4}$ | 9 3 | 10 $-\frac{1}{2}$ |

Exercise 5b (p. 34)

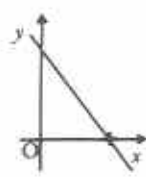
- | | | | | |
|------------------|-----------------|-----------------|-----------------|-------------------|
| 1 3 | 2 3 | 3 -2 | 4 2 | 5 $-\frac{2}{3}$ |
| 6 $-\frac{2}{3}$ | 7 $\frac{4}{3}$ | 8 $\frac{2}{5}$ | 9 $\frac{5}{2}$ | 10 $-\frac{1}{4}$ |

Exercise 5c (p. 35)

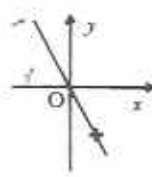
1



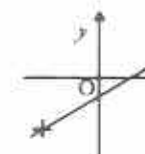
(a)



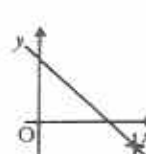
(b)



(c)



(d)



(e)

Fig. A2

- 2 (a) 2 (b) $\frac{1}{3}$ (c) $\frac{5}{4}$ (d) $-\frac{3}{7}$ (e) $\frac{4}{7}$
 3 (a) (0; 2), (1; 0) (b) (0; 1), (-3; 0)
 (c) (0; -6), (10; 0) (d) (0; $\frac{2}{3}$), ($\frac{1}{2}$; 0)
 (e) (0; $\frac{4}{3}$), ($\frac{1}{2}$; 0)
 4 (a) 1 (b) -2 (c) $1\frac{1}{3}$ (d) $-\frac{5}{4}$
 5 (a) -1 (b) undefined
 (c) 1 (d) 0 (e) $\frac{1}{2}$

Exercise 5d (p. 37)

- 1 (a) $y = 3x - 3$ (b) $y = 3x$
 (c) $x + y = 6$ (d) $3x + 4y = 18$
 (e) $4x + y + 5 = 0$ (f) $2y = 5x + 9$
 2 (a) $7x - 3y = 0$ (b) $7x + 3y = 0$
 (c) $x + y = 3$ (d) $3x - 10y = 42$
 (e) $5x + 16y = 67$ (f) $5x + 2y + 12 = 0$
 3 (a) $\frac{1}{3}$ (b) $y = \frac{1}{3}x - \frac{7}{3}$
 4 (a) $2y = 9x - 42$ (b) $2y = 9x + 1$
 5 $k: 3y = x - 13$ $m: 5y = 13x - 3$
 6 (a) 2 (b) $4x + y = 17$
 7 (a) -1
 8 (a) $\frac{k-1}{2}; k = 2\frac{1}{2}$ (b) $2x + y + 3 = 0$
 9 AB: $y = 0$, BC: $3x + y = 21$, AC: $y = \frac{1}{5}x$
 10 (a) $k = -4$ (b) $2y = x + 14$

Exercise 5e (p. 39)

- 1 (a) $x = \frac{1}{2}$ (b) $x = 0$
 2 (a) (i) 6 (ii) -2 (b) -5 (c) $x = 3$
 3 (a) (i) 2 (ii) -4 (b) 4 (c) $x = -1$
 4 (c) (i) 3 (ii) -1 (d) $x = 1\frac{1}{2}$ (e) $2\frac{1}{4}$
 5 (a) 1,2 (b) 0,6 (c) -0,8
 6 (a) $x = 1$ (b) 8 (c) 1
 7 (a) 4 (b) 0 (c) -4

- 8 (a) (i) 5 (ii) -1 (iii) -7 (b) $x = 1\frac{1}{2}$
 9 (a) 2 (b) 0 (c) -3 (d) -4
 10 (a) -4 (b) -2 (c) 0 (d) 3

Exercise 6a (p. 42)

- 1 $\hat{AOK}$ ($\hat{NOK}$), $\hat{AOL}$ ($\hat{NOL}$), $\hat{AOM}$ ($\hat{NOM}$)
 2 (a) $\hat{VOA}$, $\hat{VOB}$, $\hat{VOC}$ (b) $\hat{VBO}$
 3 (a) $\hat{OA}$, $\hat{OB}$, $\hat{OC}$, $\hat{OD}$, $\hat{OK}$, $\hat{OL}$, $\hat{OM}$, $\hat{ON}$
 (b) $\hat{VBD}$ (c) $\hat{VKM}$
 (d) $\hat{VBO}$ (or $\hat{VBD}$) (e) $\hat{VAO}$, $\hat{VCO}$, $\hat{VDO}$
 (f) $\hat{VKO}$ (or $\hat{VKM}$) (g) $\hat{VLO}$, $\hat{VMO}$, $\hat{VNO}$
 (h) $\hat{MVO}$
 4 (a) $PQRS$, $TUVW$ (b) SP , RQ , WV , TU
 (c) all except QVS
 (d) (i) VT (ii) QW (iii) RT
 (e) (i) $\hat{PRL}$ (ii) $\hat{RUS}$ (iii) $\hat{RUQ}$
 5 (a) $BCDF$ (b) $\hat{AFD}$
 (c) (i) $\hat{AD}$ (ii) $\hat{CF}$
 (d) (i) $\hat{BEC}$ (ii) $\hat{EBD}$
 6 (a) true (b) true (c) false
 (d) true (e) true

Exercise 6b (p. 44)

- 1 $\hat{DEH}$ (or $\hat{AFG}$), 90°
 2 $\hat{ECH}$ 3 $\hat{AOB}$
 4 (a) $\hat{VOD}$ (or $\hat{VOB}$) (b) $\hat{VOM}$ (or $\hat{VOK}$)
 (c) $\hat{VMO}$ (d) $\hat{VNO}$
 (e) $\hat{AON}$ (or $\hat{COL}$) (f) $\hat{NVL}$
 5 (a) $\hat{DCB}$ (b) $\hat{BAC}$ (or $\hat{FED}$)
 (c) $\hat{ACB}$ (or $\hat{EDF}$) (d) $\hat{ABC}$ (or $\hat{EFD}$)
 6 (c) $\hat{QWX}$ (d) $\hat{QWP}$ (e) 90°
 7 (a) $\hat{BPQ}$ (or $\hat{CSR}$) (b) $\frac{2}{3}$
 8 (a) $\hat{PNO}$ (b) 0.3 ($0.3333\ldots$)
 9 (a) $\hat{QMO}$ (or $\hat{QMN}$) (b) $\frac{1}{2}$ (i.e. $\frac{3}{6}$)
 (c) $\hat{PNO}$ (or $\hat{PNY}$) (d) $\frac{3}{4}$
 10 (b) $\hat{VMO}$ (c) $\hat{QNS}$

Exercise 6c (p. 49)

- 1 (a) 19 cm (b) 6 cm
 (c) 15 cm (d) 21 cm
 2 (a) 10 cm (b) 15.6 cm ($\sqrt{244}$)
 (c) 33.7°
 3 18.4° 4 41.8°
 5 (a) 38.9° (b) 28.3 cm
 6 25°
 7 (a) 50 cm (b) 67.4° (c) 130 cm

- 8 (a) 44.7 cm (b) 63.4°
 9 (a) 7 cm (b) (i) 51.1° (ii) 60.3°
 10 (a) 5 cm (b) 53.1° (c) 57°
 11 (a) 49.2° (b) 60.3°
 12 (a) 15 cm (b) 17 cm
 (c) (i) 28.1° (ii) 44.9° (iii) 32°
 (d) 41.6° (e) 56.3°
 13 (a) 1.5 m (b) 2.5 m (c) 57°
 14 (a) 10.3 cm (b) 61.1° (c) 343 cm^2
 15 (a) 144 cm^2 (b) 8.5 cm
 (c) 407 cm^3 (d) 45°
 16 (b) 12.7 cm (c) 48.3° (d) 57.8°
 17 (a) 26.6° (b) 36.9°
 18 (a) 3 m (b) 36.9°
 19 (a) 6 cm (b) 3.07 cm (c) $10 - 4\sqrt{3}$
 (d) 48.6°
 20 (a) 100 cm (b) 88° (c) 56°

Exercise 6d (p. 52)

- 1 (a) AB or EF (b) PB or SC
 (c) (i) NP (ii) MQ
 (d) (i) AV (ii) NV
 (e) ST (f) CB or DA
 2 (a) AV or DV (b) AP or BP
 (c) PA (d) SA or RB
 3 45° ; 31° 4 (a) 23.6° (b) 22.2°
 5 8° 6 16.3° ; 11.2°

Exercise 7a (p. 54)

- 1 $\frac{m}{x}$ 2 $\frac{4x}{5r}$ 3 No simpler form
 4 $\frac{b+c}{d+c}$ 5 $\frac{a+b}{a+c}$ 6 No simpler form
 7 $\frac{u}{v}$ 8 $\frac{h-k}{k}$ 9 $\frac{1}{3dnr}$
 10 $-\frac{c}{d}$ 11 $-\frac{a+b}{b}$ 12 $\frac{1}{x-y}$
 13 $\frac{4cd^2}{3e^2}$ 14 No simpler form 15 $\frac{c-d}{c}$
 16 $\frac{m+n}{m-n}$ 17 $\frac{c+3}{c+2}$ 18 $\frac{d+3}{d-4}$
 19 $-\frac{m}{n}$ 20 $\frac{y}{x-y}$ 21 No simpler form
 22 No simpler form 23 $\frac{h+k}{h-k}$ 24 $\frac{a-2r}{a+4r}$

$$\begin{array}{lll}
 25 \frac{x+3y}{x-y} & 26 -\frac{x+3}{x+5} & 27 -\frac{3a+m}{a+m} \\
 28 -\frac{a+4}{2a+1} & 29 \frac{a-n}{a+n} & 30 \frac{a-n}{a+n} \\
 31 \frac{w+u-v}{w-u+v} & 32 \frac{a+b+c}{b-a-c} & 33 \frac{m(m-a)}{a(m+a)} \\
 34 \frac{3}{5} & 35 \frac{b+c}{b-c} & 36 \frac{a-b}{c-a}
 \end{array}$$

Exercise 7b (p. 55)

$$\begin{array}{llll}
 1 \frac{a}{c} & 2 \frac{8\pi^2}{9c^2} & 3 \frac{n}{3} & 4 \frac{4}{3u} \\
 5 \frac{b}{2} & 6 \frac{a+2}{a} & 7 \frac{m+3}{m} & 8 \frac{3u^3v}{16m^2} \\
 9 -2a & 10 \frac{d(d-2)}{2} & 11 \frac{a}{2} & 12 \frac{n-3}{n} \\
 13 -\frac{n}{m} & 14 \frac{a+2b}{a+3b} & 15 2c & 16 -\frac{c^2}{d^2} \\
 17 \frac{2e(e-2)}{3(e-1)} & 18 \frac{u-2}{3u} & & \\
 19 \frac{(x+1)(x+2)}{x(x-2)} & 20 \frac{a}{b} & &
 \end{array}$$

Exercise 7c (p. 56)

$$\begin{array}{lll}
 1 \frac{8a+9c}{6abc} & 2 \frac{5c-a+b}{c} & 3 \frac{3c-14b}{30bc} \\
 4 \frac{7}{6(x+y)} & 5 \frac{2}{a-2b} & 6 \frac{3a-b}{a-b} \\
 7 \frac{x+4y}{x+2y} & 8 \frac{9}{20(u-v)} & 9 \frac{u-9v}{2u+3v} \\
 10 \frac{6a-b}{2(2a+b)} & 11 \frac{19mn}{6(m^2+n^2)} & 12 \frac{3}{2(2x-y)} \\
 13 \frac{c-a}{ac} & 14 \frac{u(u-2v)}{v^2} & 15 \frac{5d+7}{6(d-4)} \\
 16 \frac{5a+7}{(a+1)(a+2)} & 17 \frac{3x^2+2x+4}{(x-1)(x+2)} & \\
 18 -\frac{4}{(e+2)(e+3)} & 19 \frac{4m}{(m-n)^2} & \\
 20 \frac{c-d}{(2c+d)^2} & 21 \frac{2a-b}{(a-2b)^2} & 22 \frac{3}{c-d}
 \end{array}$$

$$\begin{array}{lll}
 23 \frac{2}{4m+3n} & 24 -\frac{m+n}{n} & 25 \frac{2}{(c+5)(c-2)} \\
 26 \frac{7d-6}{(d+2)(d-2)(d-4)} & 27 -\frac{a+11b}{6(a+b)(a-b)} & \\
 28 \frac{7d+12}{(d-4)(d+4)^2} & 29 \frac{2}{x-3} & \\
 30 \frac{a^2+7a+5}{(a-2)(a-3)(a+4)} & &
 \end{array}$$

Exercise 7d (p. 58)

$$\begin{array}{llll}
 1 \frac{1}{5} & 2 1 & 3 8\frac{1}{2} & 4 21 \\
 5 d & 6 \frac{4a+5}{5a+8} & 7 -\frac{5}{3m} & \\
 8 \frac{3w-8}{8w-5} & 9 \frac{5-a}{7a+4} & 10 \frac{m+3}{5m+1} &
 \end{array}$$

Exercise 7e (p. 59)

$$\begin{array}{llll}
 1 3; -1 & 2 \frac{1}{2}; 2 & 3 3; -\frac{2}{3} & 4 1; -3 \\
 5 -2; 1\frac{1}{3} & 6 -1; -6 & 7 -\frac{1}{3}; 2 & 8 5; 1 \\
 9 -1; -2 & 10 0; 2 & 11 5; -\frac{2}{3} & 12 1 \\
 13 2\frac{1}{5} & 14 1; 3 & 15 6; -1\frac{1}{3} & 16 1 \\
 17 -\frac{1}{4} & 18 5; 8\frac{1}{2} & 19 3; 1\frac{3}{4} & 20 -2; 1\frac{3}{4} \\
 21 3; -7\frac{3}{8} & 22 3; 1\frac{1}{10} & 23 2\frac{1}{2} & 24 2; -3 \\
 25 -\frac{3}{4} & 26 -1\frac{1}{6} & 27 0 & 28 -9 \\
 29 -\frac{1}{3} & 30 -2 & &
 \end{array}$$

Exercise 7f (p. 61)

$$\begin{array}{llll}
 1 3 & 2 4 & 3 -7 & 4 0 \\
 5 2\frac{1}{2} & 6 -5 & 7 6\frac{2}{3} & 8 0; -2 \\
 9 0; \frac{1}{2} & 10 -4; -3 & 11 \pm 1 & 12 6 \\
 13 0; -4; 9 & 14 1; 2 & 15 10; -2 & \\
 16 5; -2 & 17 -6 & 18 0; \pm 3 & \\
 19 0; \pm 2 & 20 1; -3; -5 & &
 \end{array}$$

Exercise 7g (p. 62)

$$\begin{array}{lll}
 1 \frac{y+z}{y-z} & 2 \frac{2m-5}{m-3}; 3 & 3 \frac{3(2a-1)}{2(a-1)}; 1 \\
 4 \frac{6-x^2+10x}{2x} & & 5 \frac{5}{6(x-2)} \\
 6 (a) -\frac{a+2b}{b(a-b)} & (b) \frac{1}{a+b} & 7 \frac{3x+2}{4} \\
 8 \frac{2}{x} & 9 \frac{2y-5x}{4} & 10 \frac{6m+1}{2m+3}
 \end{array}$$

11 $m = 1$ (Note: $m = 2$ is not a solution since the terms of the equation are not defined when $m = 2$)

12 $x = \frac{1}{2}$

13 $a = 4$ or $4\frac{1}{2}$

14 10

15 (a) -3

(b) $x < 4$ (but not -3)

16 (a) $-5; 4$

(b) $-5\frac{1}{2}$

17 (a) $a = 6$

(b) $\frac{3(a-6)}{(a+4)(a-2)}$

18 (a) $\frac{3(b-5)}{2(b+1)(b-3)}$

(b) $b = 5$

19 $\frac{1}{a+2}; -2$

20 (a) -2 (b) (i) $0; 3$ (ii) $k = -2$

Exercise 8a (p. 64)

1 4.5 unit^2 2 2.25 unit^2 3 10.7 unit^2

4 14.9 unit^2 5 36 unit^2 6 9 unit^2

Exercise 8b (p. 66)

- 1 (b) 0.9 m/s^2 ; 0.15 m/s^2 (c) 1600 m
 2 (b) 1100 m^2 (c) $1.2 \times 10^6 \text{ litres/second}$
 3 (b) $3; -1$
 (c) velocity in m/s ; the object is not moving
 4 (b) 0.18 cm/s ; 0.10 cm/s
 (c) round-bottomed flask with thin neck
 5 when $t = 1$, $v = -2$; when $t = 7$, $v = 40$
 (i) 0.65 ; 3.85 (ii) 2.25 s (iii) 15 m/s^2
 6

t	0	1	2	3	4	5	6
v	6	10	12	12	10	6	0

- (c) $v = 12.25 \text{ m/s}$ when $t = 2.5 \text{ s}$
 (d) -5 m/s^2 (a deceleration) (e) 31.5 m
 7 (b) 1.2 m/s^2 ; 3.68 m/s^2 (c) 6.4 m
 8 (a) 29.8 m/s when $t = 27$
 (b) 0.38 m/s^2 (c) 1360 m
 9 0.067 m/s^2 ; -0.047 m/s^2 ; 5600 m
 10 600 m ; 0.60 m/s^2 ; -0.15 m/s^2 ; 45.3 km/h after 43 s

Exercise 9a (p. 69)

- 1 $25x \text{ g}$ 2 $54y$ 3 $15t \text{ km}$ 4 $16\pi \text{ cents}$
 5 $8d \text{ ml}$ 6 $C = 7\pi$ 7 $D = 16t$ 8 $x = \frac{1}{2}y$
 9 $d = 4s$ 10 $a = 0.8b$
 11 (a) $D = 4s$ (b) 44
 12 (a) $x = 2\frac{1}{2}y$ (b) 25 (c) 5.6
 13 (a) $P = \frac{3}{8}Q$ (b) 6 (c) 6.4

14 (a) 0.9

(b) $3\frac{1}{3}$

15 (a) 0.36

(b) 55

16 (a) $D = 4S$

(b) $S = 48$

17 (a) $y = \frac{1}{5}x$

(b) $y = 8$ (c) $x = 50$

18 $p = 25q$

(b) $x = 20$

19 (a) $x = \frac{3}{3.2}y$

(b) $x = 20$

20 225 cm^2

21 $y \propto x$ (or $x \propto y$)

22

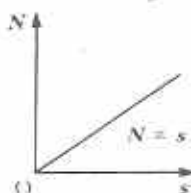


Fig. A3

23 (b) $B = 56\frac{2}{3}$

24 (b) $\text{£}1 = \$1.60$ (c) $\$11.20$ (d) $\text{£}5$

25 (b) $H = \frac{13}{15}V$ (or, by graph, $H = 0.9V$)

(c) $V = 7.5$ (d) $H = 3.9$

Exercise 9b (p. 71)

- 1 $a = 2$ 2 $y = 50$
 3 (a) $x = 5y^2$ (b) $x = 80$ (c) $y = 5$
 4 (a) $A = \frac{1}{2}B^3$ (b) $A = 108$ (c) $B = 3$
 5 (a) $P = \frac{5}{2}\sqrt{Q}$ (b) $P = 7\frac{1}{2}$ (c) $Q = \frac{9}{16}$
 6 (a) $z^2 = 3Y$ (b) $z = 12$ (c) $Y = 12$
 7 (a) $V = \frac{1}{2}D^3$ (b) $V = 13\frac{1}{2}$ (c) $D = 1.6$
 8 (a) $D = \sqrt{\frac{3H}{2}}$ (b) $D = 15$ (c) $H = 73\frac{1}{2}$

9 $y \propto x^2$

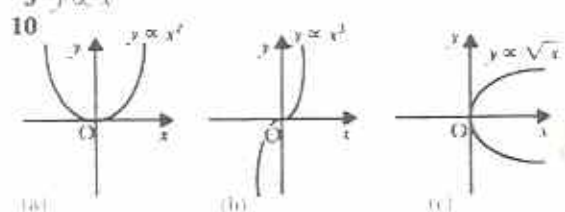


Fig. A4

11

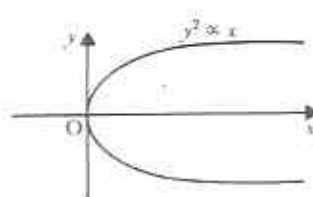


Fig. A5

12 $y = 4x^2$, $y = 1$ when $x = \frac{1}{2}$

13 8.8 amps 14 35 km

15 (a) x increases by 21%

(b) x decreases by 19%

16 x increases by 20%

17 72,8% 18 $27\frac{3}{4}\%$

Exercise 9c (page 73)

1 $d \propto \frac{1}{t}$

2 (a) inversely (b) $n \propto \frac{1}{l}$

3 (a) $l = \frac{A}{b}$ (b) $b = \frac{A}{l}$ (c) inversely

4 $x = \frac{66}{y}$ 5 $R = \frac{32}{T}$

6 $y = 1$ 7 $Q = \frac{4}{s}$

8 $y = \frac{1}{\sqrt{s}}$ 9 $x \propto \frac{1}{r^2}$

10 $y = k + \frac{h}{x^2}$ 11 $R = \frac{k}{r^2}$

12 $v = 2,25$

13

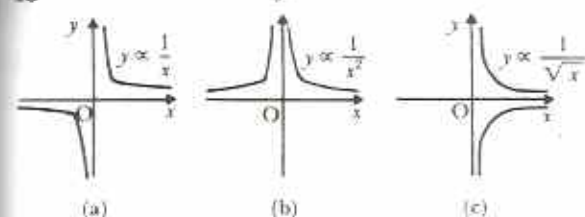


Fig. A6

14 $\gamma = 4$

15 11,25 km

Exercise 9d (p. 75)

1 (a) $x = 5yz$ (b) 120

2 (a) $x = \frac{6y}{z}$ (b) 7

3 (a) $p = \frac{6q}{r^2}$ (b) $p = 37\frac{1}{2}$

4 $h = \frac{kV}{r^2}$

5 (a) $A = 5$ (b) $C = 8$ (c) A decreases by 1%

6 (a) $x = 37\frac{1}{2}$ (b) $y = \pm 4$ (c) x is doubled

7 $x = 900$ 8 $P \propto \frac{1}{R^2}$

9 $x \propto z^3$

10 $x \propto z^4$

11 $A \propto \frac{1}{C}$

13 (a) $y = \frac{1}{2x}$ (b) $z = \frac{x^3}{24}$ (c) 33,1%

14 26,1 kg

15 From the given values, $E = \frac{4R}{d^2}$

(a) $R = 36$ (b) $d = \sqrt{24}$ (c) 9,2%

Exercise 9e (p. 76)

1 (a) $x = 20 + 5y$ (b) 35

2 (a) $x = 2 + 3y$ (b) 32

3 (a) $x = 5 + \frac{2}{3}y$ (b) $7\frac{2}{3}$

4 (a) $D = 30 + 3V$ (b) 249

5 \$35,50 6 $A = 100$

7 (a) $k = 2$ (b) $y = 10$

8 (a) $P = \frac{1}{2}Q - 20$

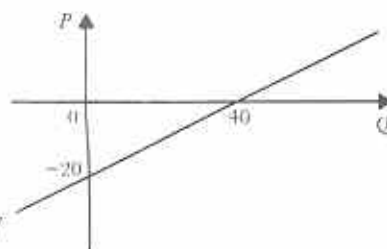


Fig. A7

9 costs: \$75 000, profit: \$15 000

10 \$1 600 11 \$63 500

12 45 km/h 13 90 m

14 \$795 15 $15\frac{4}{5}$

Exercise 10a (p. 79)

1 (a) 120 cm^3 (b) 616 cm^3 (c) 480 cm^3

(d) 48 cm^3 (e) 77 cm^3 (f) 420 cm^3

2 (a) 158 cm^2 (b) 484 cm^2 (c) 528 cm^2

3 152 litres 4 108 times 5 16 cm

6 (a) $5 000 \text{ cm}^3$ (b) $5 500 \text{ cm}^3$ (c) $7 500 \text{ cm}^3$

7 1,44 tonnes

8 (a) 27 cm^3 (b) $1\frac{2}{3} \text{ g/cm}^3$ (c) 19 g

9 $42\pi \text{ cm}^2$ 10 396 cm^2 11 9 cm

12 (a) 275 cm^3 (b) $1,09 \text{ g/cm}^3$

13 $6,2 \text{ g/cm}^3$ 14 9 cm 15 28,8 cm

16 64 17 (a) $15 400 \text{ cm}^3/\text{s}$ (b) 924 t/min

18 5 19 1,4 cm 20 64 cm^3

21 (a) $16\pi \text{ cm}^3$ (b) 5 cm

(c) $20\pi \text{ cm}^2$ (d) 288°

22 $1 232 \text{ cm}^3, 550 \text{ cm}^2$

23 $100,8^\circ$ 24 $6 750 \text{ m}^3$ 25 $38,9^\circ$

Exercise 10b (p. 82)

- 1 (a) $4\,190\text{ cm}^3$, $1\,260\text{ cm}^3$ (b) $2\,140\text{ cm}^3$, 804 cm^3
 (c) $16,8\text{ cm}^3$; $37,7\text{ cm}^3$ (d) 191 cm^3 , 191 cm^3
 2 $7,06\text{ kg}$ 3 27 cm 4 $1\,728$
 5 (a) $6,20\text{ cm}$ (b) 484 cm^2

Exercise 10c (p. 84)

- 1 $38\,125\text{ cm}^3$
 2 (a) $3\,000\text{ litres}$ (b) $1,224\text{ cm}^3$
 3 $121\frac{1}{2}\text{ tonnes}$ 4 $57,6\text{ kg}$
 5 (a) $7,08\text{ litres}$ (b) $3,24\text{ kg}$
 6 $67,0\text{ cm}^3$ 7 352 cm^3 8 (b) $V = 880$
 9 $52,4\% \left(\frac{\pi}{6} \times 100\% \right)$
 10 $9\,610\text{ cm}^3$ 11 $16,5\text{ kg}$ 12 $11,4\text{ cm}$
 13 9 cm 14 $31\frac{1}{4}\text{ cm}$ 15 $2\frac{1}{4}\text{ cm}$
 16 $3\frac{25}{7}\text{ cm}$ 17 1 cm 18 2 cm
 19 420 g 20 $0,343\text{ kg}$

Exercise 10d (p. 87)

- 1 $218\pi\text{ cm}^3$
 2 (a) 500 m^3 (b) 468 m^3
 3 $1\,552\text{ cm}^3$
 4 (a) $195\pi\text{ cm}^2$ (b) $700\pi\text{ cm}^3$
 5 cone: 185 ml , frustum: $4\,815\text{ ml}$
 6 16 m^3 7 $12,7\text{ cm}$ 8 $5\frac{1}{3}\text{ cm}$

Exercise 11a (p. 89)

- 1 $a = 3,79\text{ cm}$ 2 $b = 1,49\text{ cm}$
 3 $c = 1,58\text{ m}$ 4 $a = 2,31\text{ m}$
 5 $a = 16,6\text{ cm}$ 6 $b = 4,18\text{ m}$
 7 $c = 5,72\text{ m}$ 8 $b = 12,1\text{ cm}$
 9 $a = 6,68\text{ cm}$ 10 $c = 18,2\text{ m}$

Exercise 11b (p. 90)

- 1 $a = 4,97\text{ cm}$ $B = 47,3^\circ$ $C = 66,7^\circ$
 2 $c = 9,87\text{ cm}$ $B = 29,2^\circ$ $A = 76,8^\circ$
 3 $c = 12,2\text{ m}$ $A = 25,4^\circ$ $B = 15,6^\circ$
 4 $b = 375\text{ m}$ $C = 32,1^\circ$ $A = 52,9^\circ$
 5 $a = 9,08\text{ m}$ $B = 69,3^\circ$ $C = 52,6^\circ$
 6 $b = 9,45\text{ cm}$ $C = 28,7^\circ$ $A = 25,3^\circ$
 7 $c = 3,23\text{ cm}$ $B = 28,1^\circ$ $A = 126,2^\circ$
 8 $a = 65,3\text{ m}$ $B = 26,2^\circ$ $C = 13,6^\circ$
 9 $c = 5,81\text{ cm}$ $A = 13,7^\circ$ $B = 23^\circ$
 10 $b = 3,21\text{ cm}$ $C = 29,6^\circ$ $A = 115,9^\circ$

Exercise 11c (p. 91)

- 1 $29,9^\circ$; $63,9^\circ$; $86,2^\circ$ 2 $81,8^\circ$; $38,2^\circ$; 60°
 3 $62,7^\circ$; 81° ; $36,3^\circ$ 4 $108,2^\circ$; $22,3^\circ$; $49,5^\circ$
 5 $33,6^\circ$; $50,7^\circ$; $95,7^\circ$ 6 $110,9^\circ$; $43,2^\circ$; $25,9^\circ$
 7 $41,4^\circ$; $55,8^\circ$; $82,8^\circ$ 8 $46,5^\circ$; $39,4^\circ$; $94,1^\circ$
 9 $98,2^\circ$; $50,3^\circ$; $31,5^\circ$ 10 $45,7^\circ$; 79° ; $55,3^\circ$

Exercise 11d (p. 92)

- 1 (a) $\frac{2}{3}$ (b) 7 cm
 2 (a) 117° (b) $9,40\text{ cm}$
 3 $x = 6,36$; $\theta = 61,4^\circ$
 4 (a) $120,4^\circ$ (b) $29,8^\circ$
 5 $-\frac{23}{40}$; $4,24\text{ cm}$ 6 $\frac{1}{16}$; $7,42\text{ m}$
 7 $7,63\text{ cm}$, $3,12\text{ cm}$ 8 $y = 5$
 9 $4:1:1$
 10 (a) $82,8^\circ$ (b) $8,89\text{ cm}$

Exercise 11e (p. 93)

- 1 $5,99\text{ km}$ 2 226 m
 3 11° 4 171 km
 5 $82,0\text{ m}$ 6 $067,2^\circ$
 7 $36,1\text{ km}$, $073,2^\circ$ 8 $19,2\text{ km}$
 9 (a) $8,72\text{ km}$ (b) $054,1^\circ$
 10 154 m 11 $0,706\text{ m}$, $56,4^\circ$
 12 (a) 700 km (b) $021,8^\circ$
 13 (a) $44,5\text{ km}$ (b) $134,2^\circ$
 14 193 km , $339,9^\circ$
 15 $5,65\text{ km}$, $N\,9,5^\circ W$ (or $350,5^\circ$)

Exercise 12a (p. 96)

- 1 three piece suite: \$960, table and chairs: \$420, black pots: \$16,80, bicycle: \$418,80
 2 \$2; \$1; \$4; \$1,50; \$3; \$4,40; \$5,11; \$0,37 respectively
 3 \$15,22 4 \$45
 5 (a) (i) \$900 (ii) \$1 140
 (b) (i) \$1 620 (ii) \$1 980
 (c) (i) \$3 220 (ii) \$3 780
 (d) (i) \$1 240 (ii) \$1 540
 (e) (i) \$1 516 (ii) \$1 860
 (f) (i) \$2 134 (ii) \$2 565,20
 6 (a) \$2 200 (b) \$4 800 (c) \$4 000
 (d) \$6 800
 7 \$1 265
 8 (a) \$1 800 (b) \$3 508 (c) \$268
 (d) \$3 240 (e) \$486 (f) \$3 726
 (g) \$9 594
 9 (a) \$5 800 (b) \$9 688,75 (c) \$1 509,27
 10 \$46 912,82

Exercise 12b (p. 98)

- 1 (a) \$53,10 (b) \$367,50 (c) \$23,80
 (d) \$36
 2 \$165
 3 (a) 35% (b) \$25,35
 4 \$1,52
 5 \$1,96 and \$1,30, a saving of 66 cents per kg
 6 \$28,25 7 \$38 700
 8 (a) \$825 (b) \$126

- 9 \$858 and \$988 (the higher price reflects the longer hire period)
 10 \$833.65

Exercise 12c (p. 101)

- 2 \$29.24 3 \$87.11
 5 \$14.93 6 \$28.15
 7 (a) \$35.95 (b) \$33.95 (c) \$40.25
 8 \$142.03 9 \$156.80
 10 \$356.90 (i.e. \$250.32 rates, \$32.35 water and \$74.23 electricity)

Exercise 12d (p. 103)

- 1 \$37.80 each
 2 benefit C, \$3 422 (= \$2 215 + \$1 207)
 3 \$858
 4 (a) \$910 (b) \$364
 5 (a) \$67 (b) \$99 (c) \$164
 6 (a) \$173.10 (b) \$2 077.20 (c) \$51 930
 7 (a) \$132.50 (b) \$11.04
 8 (a) \$45 400 (b) \$90 975.60
 9 \$66 375
 10 (a) \$45 000 (b) \$622.35 (c) \$149 364

Exercise 12e (p. 105)

- 1 (a) mortgage, insurance, school costs
 (b) \$320.78 (c) \$150.84
 2 (a) \$19 593.33 (b) \$980
 (c) \$15 401.20 (d) \$3 608.28
 (e) \$124.90
 3 (b) \$6 380.70
 (c) operating loss of \$3 283.37
 4 (a) \$1 386.7 million (b) 1988/89
 (c) income tax (d) 1987/88
 5 (a) yes (b) yes (c) no (d) no
 6 it is probably a rounding off error, for example,

<i>possible real value</i>	<i>rounded value</i>
210.77	210.8
+ 786.06	786.1
996.83 (sum)	996.9

 7 \$4 831.3 million 8 \$13 078 million
 9 (a) 21.1% (b) 11.6%
 (c) the proportion of deficit financed by foreign loan was virtually halved during the two-year period
 10 (a) 142.5% (b) 134.9%
 (c) both are rising; however, revenue (income) is rising at a greater rate than expenditure; this means that Government had a firm control of Zimbabwe's financial affairs during the given two-year period

Exercise 13a (p. 108)

- 1 (a) $\begin{pmatrix} 7 & 9 \\ -4 & 3 \end{pmatrix}$ (b) $\begin{pmatrix} 12 \\ 9 \end{pmatrix}$
 (c) $\begin{pmatrix} 3 & -7 \\ 0 & 6 \\ -5 & 7 \end{pmatrix}$ (d) $\begin{pmatrix} -10 & -7 \\ 3 & 1 \end{pmatrix}$
 (e) $\begin{pmatrix} 5 \\ 10 \end{pmatrix}$ (f) $\begin{pmatrix} 0 & -3 \\ -8 & 6 \end{pmatrix}$
 2 (a) $\begin{pmatrix} -3 & 15 \\ 6 & 9 \end{pmatrix}$ (b) $\begin{pmatrix} 12 & 0 \\ 8 & -16 \end{pmatrix}$
 (c) $\begin{pmatrix} 2 & -10 \\ -4 & -6 \end{pmatrix}$ (d) $\begin{pmatrix} 3 & 0 \\ 2 & -4 \end{pmatrix}$
 (e) $\begin{pmatrix} 4 & 10 \\ 8 & -12 \end{pmatrix}$ (f) $\begin{pmatrix} -19 & 5 \\ -10 & 27 \end{pmatrix}$
 3 (a) (29) (b) (-8)
 (c) (3x - y) (d) (4x + 5y + 6z)
 4 x = 8, y = -5
 5 (a) $\begin{pmatrix} -7 \\ -10 \end{pmatrix}$ (b) (7; 4; 11)
 (c) $\begin{pmatrix} 4 & 3 \\ -2 & -11 \end{pmatrix}$ (d) $\begin{pmatrix} -2 & 24 \\ 2 & -5 \end{pmatrix}$
 (e) $\begin{pmatrix} 33 & 24 \\ -8 & 1 \end{pmatrix}$ (f) $\begin{pmatrix} 2 & 3 & 0 \\ 5 & -1 & 6 \end{pmatrix}$

Exercise 13b (p. 110)

- 1 $\frac{1}{9} \begin{pmatrix} 2 & -3 \\ -1 & 6 \end{pmatrix}$ 2 $\frac{1}{9} \begin{pmatrix} 3 & -3 \\ -2 & 5 \end{pmatrix}$
 3 $-\frac{1}{2} \begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}$ 4 $-\frac{1}{6} \begin{pmatrix} 3 & -3 \\ -5 & 4\frac{1}{2} \end{pmatrix}$
 5 $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$ 6 $-\begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix}$
 7 singular 8 $-\frac{1}{17} \begin{pmatrix} 4 & -3 \\ 1 & -5 \end{pmatrix}$
 9 $\frac{1}{60} \begin{pmatrix} -6 & 9 \\ -8 & 2 \end{pmatrix}$ 10 $-\frac{1}{8} \begin{pmatrix} 0 & 4 \\ 2 & -5 \end{pmatrix}$
 11 $-\frac{1}{2} \begin{pmatrix} 3 & -1 \\ -16 & 4 \end{pmatrix}$ 12 $\begin{pmatrix} 0 & 3 \\ -2 & 0 \end{pmatrix}$

Exercise 13c (p. 111)

$$1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$$

$$4 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

$$5 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2\frac{1}{2} \\ \frac{1}{2} \end{pmatrix}$$

$$6 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}$$

$$7 \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

$$8 \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 12 \\ 6 \end{pmatrix}$$

$$9 \begin{pmatrix} s \\ t \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ \frac{1}{6} \end{pmatrix}$$

$$10 \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ 5 \end{pmatrix}$$

Exercise 13d (p. 111)

$$1 \begin{pmatrix} -7 & -6 \\ 0 & 12 \end{pmatrix}$$

$$2 \text{ (a) } 2 \quad \text{(b) } \frac{1}{2} \begin{pmatrix} 4 & 6 \\ 1 & 2 \end{pmatrix}$$

$$3 \text{ } 14, \frac{1}{14} \begin{pmatrix} 3 & 4 \\ -5 & -2 \end{pmatrix}$$

$$4 \text{ (a) } x = 3 \quad \text{(b) } \begin{pmatrix} 3 & 2 \\ 4 & 5 \end{pmatrix} \quad 5 \text{ } k = 5$$

$$6 \text{ (a) } \begin{pmatrix} -6 & 15 & 0 \\ 2 & -5 & 0 \\ -4 & 10 & 0 \end{pmatrix} \quad \text{(b) } (11)$$

$$7 \text{ (a) } \begin{pmatrix} 14 \\ -4 \\ 7 \end{pmatrix} \quad \text{(b) } \begin{pmatrix} 0 & 12 & 15 & 3 \\ 0 & 3 & 3 & 0 \end{pmatrix}$$

$$8 \text{ } m = 5, n = 3 \quad 9 \text{ } P = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix}$$

$$10 \text{ (a) } \begin{pmatrix} 5 & -2 \\ 0 & -4 \end{pmatrix} \quad \text{(b) } p = 9, q = -6$$

$$11 \text{ } p = 1, q = -3, r = 5 \quad 12 \text{ } a = 5, b = 2$$

$$13 \text{ (a) } \begin{pmatrix} 16 & 14 \\ 0 & 9 \end{pmatrix} \quad \text{(b) } k = -\frac{1}{6} \\ \text{(c) } m = -2$$

$$14 \text{ (a) } \frac{1}{41} \begin{pmatrix} 7 & 4 \\ -5 & 3 \end{pmatrix} \quad \text{(b) } x = 2, y = -1$$

$$15 \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, x = 9 \text{ and } y = -14$$

Exercise 14a (p. 116)

1	transformation	matrix
	identity	$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
	reflection in x -axis	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
	reflection in y -axis	$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$
	rotation of 180° about origin	$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

$$2 \text{ (a) } \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{(b) } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

$$3 \text{ } (1; -1), (4; -2), (7; -3)$$

$$4 \text{ } P(-1; 5)$$

$$5 \text{ } (0; 0), (-1; 1), (-1; -1)$$

$$6 \text{ } (0; 0), (-1; -1), (-1; 1)$$

$$7 \text{ (a) } (2; -2), (1; 1), (-2; 2), (-1; -1) \\ \text{(b) } (-2; 2), (-1; -1), (2; -2), (-1; 1)$$

$$8 \text{ (a) } \begin{pmatrix} 7 \\ -5 \end{pmatrix} \quad \text{(b) (i) } 270^\circ,$$

$$\text{(ii) } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{(c) } y = -1$$

$$10 \text{ (a) } (-2; 1) \quad \text{(b) } (-3; -4) \quad \text{(c) } C(-1; 7)$$

Exercise 14b (p. 119)

$$1 \text{ } A'(-3; -3), B'(-6; -9), C'(-12; -6)$$

$$2 \text{ } A'(4; 1), B'(11; 3), C'(10; 2), \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix}$$

$$3 \text{ } A'(2; 5), B'(4; 15), C'(8; 10), \text{ a two-way stretch of factor 2 in the } x\text{-direction and factor 5 in the } y\text{-direction.}$$

$$4 \text{ } \begin{pmatrix} 1\frac{1}{2} & 0 \\ 0 & 1\frac{1}{2} \end{pmatrix}$$

$$5 \text{ } (0; 0), (4\frac{1}{2}; 0), (0; 3), (4\frac{1}{2}; 3)$$

$$6 \text{ (a) } (2; 15) \quad \text{(b) } (-2; -5)$$

$$\text{(c) a shear of factor 5 in the } y\text{-direction with the } y\text{-axis invariant}$$

$$7 \text{ (a) } -3 \quad \text{(b) } 2 \quad \text{(c) } \begin{pmatrix} -3 & 0 \\ 0 & 2 \end{pmatrix}$$

- (a) an enlargement of scale factor 4 with the origin as centre; $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$

$P_2(16; 4), Q_2(23; 5)$

(a) $\begin{pmatrix} -4; -2 \end{pmatrix}$ (b) $\begin{pmatrix} -5; 4 \end{pmatrix}$ (c) $y = 0$

$P_1(0; 0), Q_1(4; 2), R_1(-2; 6)$

$P_2(0; 0), Q_2(2; -4), R_2(6; 2)$

Exercise 14c (p. 122)

$X_1(-1; 2), Y_1(4; 5), Z_1(7; 4)$

$X_2(-2; 15), Y_2(-5; 31), Z_2(-4; 21)$

- (a) enlargement, scale factor 4 with origin as centre followed by a translation of vector $\begin{pmatrix} -3 \\ 1 \end{pmatrix}$; shear of factor 3 with $y = 0$

invariant followed by a two-way stretch

(b) $\begin{pmatrix} -7; 17 \end{pmatrix}, (22; 12)$

(a) $\begin{pmatrix} -10; 30 \end{pmatrix}$ (b) $\begin{pmatrix} -4; 14 \end{pmatrix}$ (c) $\begin{pmatrix} -1\frac{1}{2}; \frac{2}{3} \end{pmatrix}$

(a) $\begin{pmatrix} -5; 13 \end{pmatrix}$ (b) $\begin{pmatrix} -11; -3 \end{pmatrix}$ (c) $\begin{pmatrix} -22; 5 \end{pmatrix}$

(a) 90° (b) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (c) $\begin{pmatrix} -1 \\ -7 \end{pmatrix}$

(d) $x + y = -1$

(a) 2 (b) $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ (c) $\begin{pmatrix} -3 \\ -7 \end{pmatrix}$

(d) $\begin{pmatrix} -1\frac{1}{2} \\ -3\frac{1}{2} \end{pmatrix}$ (e) $h = 7, k = 3$

$(-1; 0)$

(a) $(-1; 5)$ (b) $(10; 7)$ (c) $m = -4, n = 3$

(a)

- (b) U is an enlargement of scale factor 3 with the origin as centre.
(c) V is a reflection in the line $y = -x$.

Exercise 15a (p. 126)

1 (c) $x \approx 3,7$

2 $x = -2; -0,4$ or $2,4$

3 (c) $x \approx 0,5$

4 $x \approx -2,5; 0,7; 1,8$

5 $x \approx 1,4$

6 (b) $x \approx -3,1; -0,4; 3,5$

7 $x \approx 3,8$

8 (a) $x = 0; 1$ or 3 (b) $1 < x < 3$ (c) (i) 3 (ii) 13

9 (a) 32 cm (b) 3,9 s

10 (c) (i) 27 cm^3 (ii) $3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm}$

Exercise 15b (p. 128)

2 (a) $x \approx 2,1$

(b) $x^3 - x^2 - 5 = 0$

3 (c) $y \approx 3,9$

4 (b) $x \approx -1,45; 3,45$

5 $x \approx 0,2; -2,7$

6 $x \approx -0,6; -6,4$

7 (b) $x \approx -3,1; 1,4$ or $1,7$

8 $x \approx 2, 6$

9 (b) $x \approx 0,2; 4,3$

10 (a) $x = 5; 10$

(b) $-1,8$

11 (a) $y \approx 3,2$

(b) $x \approx 4,5$

12 (a) $x \approx 2,2; 0,1; 1,55$ (b) 1,5 (c) 1,4

Exercise 15c (p. 131)

2 (a) $5x - 6y = 30$

(b) $2x + 3y = 18$

(c) $x + 2y = -8$

(d) $4x - y = -12$

3 $-11 \quad 5$ (a) $x = 3$ (b) $\tan \angle OMN = 2$

6 $y = x^2 - x - 2$

7 (a) $y = 4 - 3x - x^2$ (b) $y = 2x^2 + 2x - 12$

9 (a) $y = \frac{1}{2-x}$ (b) $y = \frac{1}{x+7}$

10 $l: y = x + 1; p: y = x^2 - 2x - 3; h: y = \frac{1}{x+1}$

Exercise 16a (p. 133)

1 (a) music

(b) 3 hours

(c) 16 hours

(d) $\frac{1}{4}$

2 (a) 6 km

(b) 28 students

(c) 305 km

(d) 4 km

3 (b) mode = median = 2

(c) 2,1

4 (b) 31 people

(c) 42 marks

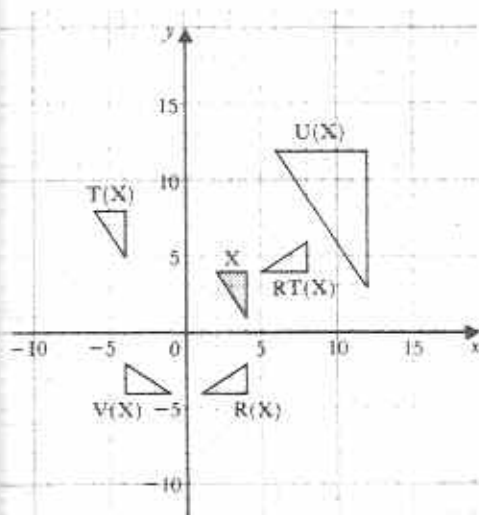
5 (b) 52,85%

(c) mode = 60%, median = 50%

Exercise 16b (p. 135)

1	1-100	101-200	201-300	301-400	401-500
	5	17	19	5	4

There is very little change in the pattern of the distribution.



2

0-49	50-99	100-149	150-199	200-249
0	5	7	9	11

250-299	300-349	350-399	400-449	450-499
8	4	2	2	2

3

55-59	60-64	65-69	70-74	75-79	80-84
3	7	16	14	6	4

Exercise 16c (p. 138)

- 1 11-15
 2 12,7
 3 25,5
 4 mode: 32, mean: 31,4
 5 6 plants
 6 \$224,50
 7 69,5"
 8 22,2 years
 9 $27\frac{3}{4}$ absentees (28, to nearest whole person)
 10 (a) 25 students (b) 50,1%
 11 yes (mean = 199,75 = 200 to the nearest whole nail)

12 (a)

40-44	45-49	50-54	55-59	60-65
5	9	10	5	1

(c) 50 kg

13 (a)

135-144	144-154	155-164	165-174
4	12	8	6

(b) 154,8 cm

14 (a)

51-55	56-60	61-65	66-70
4	6	10	13

71-75	76-80	81-85
9	4	4

(c) 67,9 kg

15 (a) 70 mm to 79 mm

(b)

70-79	80-89	90-94	95-99
4	9	12	16

100-104	105-109	110-119	120-129
22	15	8	4

(d) modal group 100 mm-104 mm,
mode $\approx$ 102 mm

Exercise 16d (p. 141)

- 1 (a) 28% (b) 20 students (c) 5%
 2

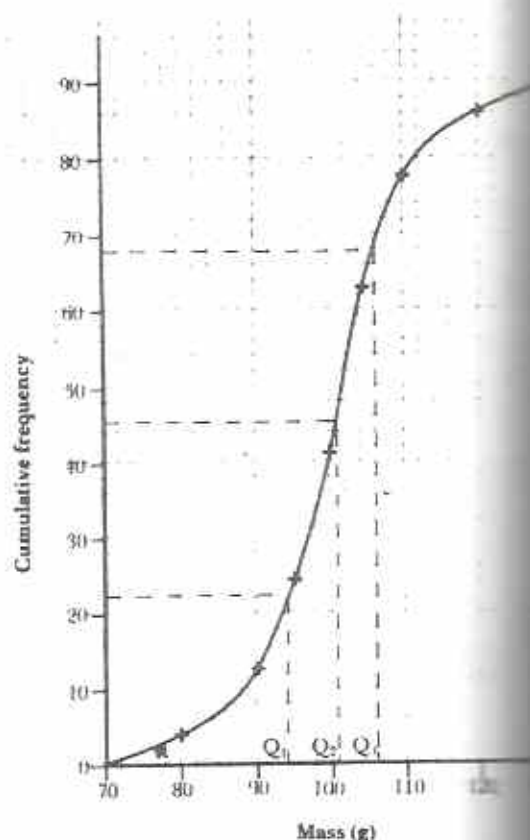
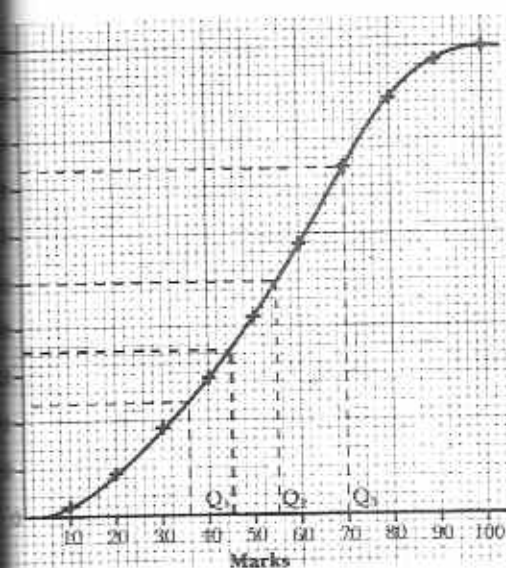


Fig. A9

(a) 101 mm (b) about $6\frac{1}{4}$ mm

class interval	frequency	cumulative frequency
1-10	2	2
11-20	7	9
21-30	9	18
31-40	11	29
41-50	13	42
51-60	16	58
61-70	16	74
71-80	15	89
81-90	8	97
91-100	3	100



110

Q_2 (median) ≈ 55 , $Q_1 \approx 36\frac{1}{2}$, $Q_3 \approx 70\frac{1}{2}$
 17 marks
 65%

4 (a)

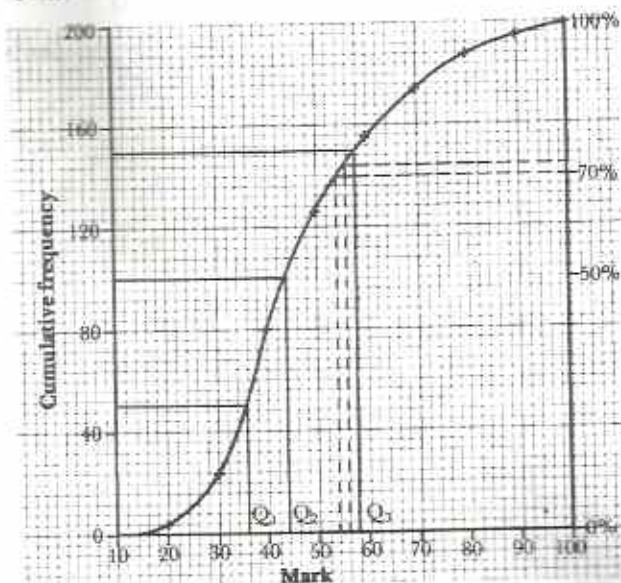


Fig. A.11

- (b) 44; 11 (c) 28% (d) 54
 5 (a)

class interval	tally	frequency	cumulative frequency
551-560	I	2	2
561-570	II	2	4
571-580	III	4	8
581-590	IIII II	7	15
591-600	IIII IIII IIII	15	30
601-610	IIII II	8	38
611-620	IIII	5	43
621-630	IIII	5	48
631-640	II	2	50

modal group: 591-600, mode: 596 h, to nearest hour

(b)

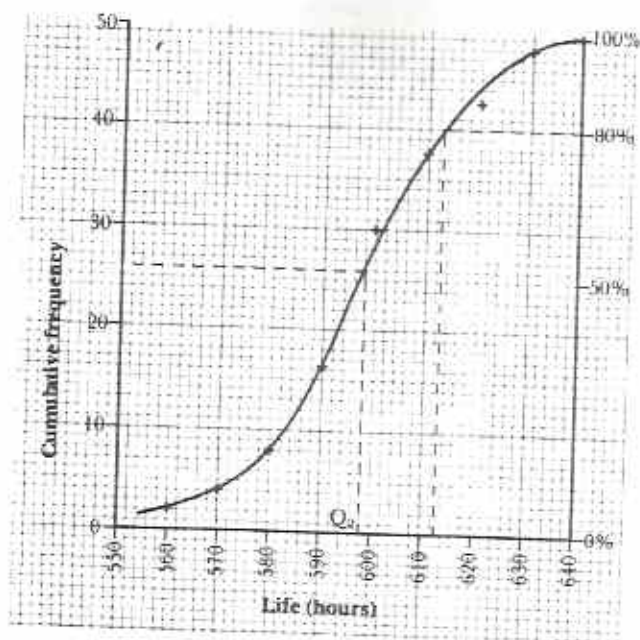


Fig. A12

median: 597 h, to nearest hour

80th percentile: 613 h, to nearest hour

6 (a) 3

(b)

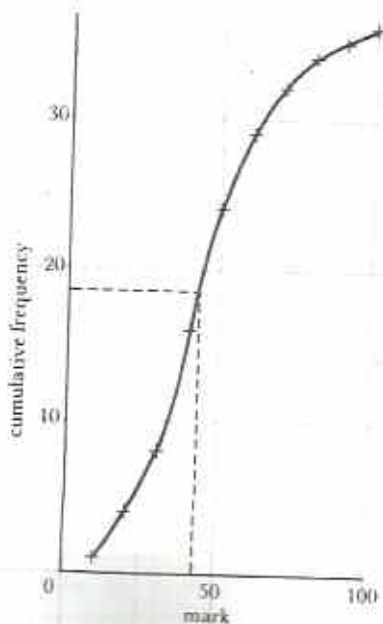


Fig. A13

(c) median mark = 43 (approx)

Exercise 17a (p. 144)

- 2 (a) $\{(1; 2), (1; 3), (2; 3)\}$ (b) $\{(0; 2)$
 3 $x > 1, x + y < 5, 3y > x$
 4 $y > 2, y > 3x, x + y > -2$
 5 (a) $P(0; 4)$ (b) $y = -2x$ (or $2x + y = 0$)
 (c) $x \leq 0; 2x + y \geq 0; 2y < x + 8$

Exercise 17b (p. 146)

- 1 14 buses, 11 minibuses
 2 (a) two ways: 3 notebooks, 5 pencils or 4 notebooks, 3 pencils
 (b) yes, the 2nd way: 12 c change
 3 (a) 93; either (72; 21) or (73; 20)
 (b) 60 cheap; 30 dear
 4 (a) (i) (96; 34) (ii) (140; 20) (b) \$300
 5 (a) $4x + y > 20, 4x + 3y > 30$
 (c) 4 Feelgood pills and 5 Getbetter pills
 6 15 lorries (5 Landmasters; 10 Sandromas)
 7 (a) 37 (10; 27) (b) either (10; 27) or (20; 27)
 (c) \$7.05
 8 (a) 10 (b) 4 of A and 6 of B
 9 (a) 20
 (b) (i) (16; 38) (ii) either (20; 30) or (30; 30)
 10 $x + y \leq 1000; y \geq 2x; x \geq 100; y < 800$
 333 cans of Kula, 667 cans of Sundown

Exercise 18a (p. 150)

- 1 $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$
 2 (a) $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ (b) $\begin{pmatrix} -4 \\ 5 \end{pmatrix}$
 (c) $\begin{pmatrix} 8 \\ 6 \end{pmatrix}$ (d) $\begin{pmatrix} -2 \\ -7 \end{pmatrix}$
 3 (a) $\sqrt{89}$ (b) $\begin{pmatrix} 8 \\ -5 \end{pmatrix}$
 4 (a) $\begin{pmatrix} 7 \\ 5 \end{pmatrix}$ (b) $\sqrt{74}$
 5 $\mathbf{PR} = \begin{pmatrix} -3 \\ 7 \end{pmatrix}, \mathbf{RP} = \begin{pmatrix} 3 \\ -7 \end{pmatrix}$
 6 (a) $\begin{pmatrix} 15 \\ 20 \end{pmatrix}$ (b) $\begin{pmatrix} -9 \\ 3 \end{pmatrix}$
 (c) $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ (d) $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$
 (e) $\begin{pmatrix} -7 \\ -4 \end{pmatrix}$ (f) $\begin{pmatrix} 7 \\ 4 \end{pmatrix}$

$$(g) \begin{pmatrix} 5 \\ 12 \end{pmatrix} \quad (h) \begin{pmatrix} -3 \\ 6 \end{pmatrix}$$

$$(i) \begin{pmatrix} -1 \\ 24 \end{pmatrix} \quad (j) \begin{pmatrix} 36 \\ 11 \end{pmatrix}$$

$$7 \text{ (a) } 5 \quad (b) \sqrt{10} \quad (c) 4$$

$$(d) \sqrt{17} \quad (e) \sqrt{2} \quad (f) 5$$

$$8 \text{ (a) } A'(4; 4), B'(3; 6), C'(4; 7)$$

$$(b) 5$$

$$9 \text{ (a) } \begin{pmatrix} -8 \\ 6 \end{pmatrix} \quad (b) 10$$

$$10 \text{ P}(6; 6), Q(1; 3), |OA| = 5$$

$$7 \text{ } OA = CB \left[= \begin{pmatrix} 4 \\ 0 \end{pmatrix} \right]$$

$$\Rightarrow OA \parallel CB \text{ and } OA = CB$$

$$\Rightarrow OABC \text{ is a parallelogram}$$

$$8 \text{ } PQ = SR \left[= \begin{pmatrix} 6 \\ 3 \end{pmatrix} \right]$$

$$\Rightarrow PQ \parallel SR \text{ and } PQ = SR$$

$$\Rightarrow PQRS \text{ is a parallelogram}$$

$$\text{diagonals intersect at } (7; 5)$$

$$9 \text{ Q}(12; 8), \text{ diagonals intersect at } (6; 4)$$

$$10 \text{ (a) } \begin{pmatrix} 8 \\ 6 \end{pmatrix} \quad (b) 10 \quad (c) P(-7; -10)$$

Exercise 18b (p. 153)

$$1 \text{ (a) } \begin{pmatrix} -5 \\ -9 \end{pmatrix} \quad (b) \begin{pmatrix} 5 \\ 9 \end{pmatrix}$$

$$2 \text{ (a) } \begin{pmatrix} 3 \\ 8 \end{pmatrix} \quad (b) \begin{pmatrix} 10 \\ 3 \end{pmatrix} \quad (c) \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$(d) \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad (e) \begin{pmatrix} 7 \\ -5 \end{pmatrix} \quad (f) \begin{pmatrix} -6 \\ -5 \end{pmatrix}$$

$$3 \text{ (a) } \begin{pmatrix} -1 \\ 3 \end{pmatrix} \quad (b) \begin{pmatrix} 8 \\ -1 \end{pmatrix}$$

$$(c) \begin{pmatrix} 6 \\ 1 \end{pmatrix} \quad (d) \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

$$(e) \begin{pmatrix} 5 \\ 4 \end{pmatrix} \quad (f) \begin{pmatrix} 9 \\ -4 \end{pmatrix}$$

$$(g) \begin{pmatrix} 5 \\ 4 \end{pmatrix} \quad (h) \begin{pmatrix} 8 \\ -1 \end{pmatrix}$$

$$(i) \begin{pmatrix} -3 \\ 5 \end{pmatrix} \quad (j) \begin{pmatrix} -8 \\ 1 \end{pmatrix}$$

$$4 \text{ (a) } A(1; 4), B(6; 6), C(5; 2)$$

$$(b) \text{ (i) } \begin{pmatrix} 6 \\ 6 \end{pmatrix} \quad \text{(ii) } \begin{pmatrix} -4 \\ 2 \end{pmatrix}$$

$$5 \text{ } QR = \begin{pmatrix} 4 \\ -1 \end{pmatrix} \text{ and } PS = \begin{pmatrix} 8 \\ -2 \end{pmatrix} = 2 \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

$$\Rightarrow QR \parallel PS$$

$$6 \text{ } AB = BC = CD = DA (= \sqrt{125}) \Rightarrow ABCD \text{ is a rhombus}$$

Exercise 18c (p. 155)

$$1 \text{ (a) } PR \quad (b) PS \quad (c) PT \quad (d) PT$$

$$(e) PS \quad (f) PS \quad (g) PT \quad (h) PS$$

$$2 \text{ } \frac{1}{2}(a + b)$$

$$3 \text{ } XY = b - a, YZ = c - b, ZX = a - c$$

$$4 \text{ (a) } 2b - a \quad (b) a + b$$

$$5 \text{ (a) } t \quad (b) -r \quad (c) -\frac{1}{4}r$$

$$(d) t - \frac{1}{4}r \quad (e) \frac{3}{4}r + t$$

$$6 \text{ (a) } b - a \quad (b) \frac{1}{2}b \quad (c) \frac{1}{2}b - a$$

$$7 \text{ (a) } 2a \quad (b) b - a$$

$$(c) 1\frac{1}{2}a - \frac{1}{2}b \quad (d) 1\frac{1}{2}a + \frac{1}{2}b$$

$$8 \text{ (a) } 2x \quad (b) x + y \quad (c) x + y$$

$$(d) x + 2y \quad (e) 2x + 2y \quad (f) 2x + y$$

$$9 \text{ (a) } \frac{4}{5}x \quad (b) \frac{1}{2}(x + y) \quad (c) 1\frac{1}{2}y$$

$$(d) \frac{1}{2}y - \frac{3}{10}x \quad (e) y - \frac{1}{2}x$$

$$10 \text{ (a) } 2a + b \quad (b) -b \quad (c) a$$

$$13 \text{ (a) } b - a, \frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}(b - a)$$

$$(b) MN \parallel AB \text{ and } MN = \frac{1}{2}AB$$

$$16 \text{ (a) } RP = a + b, QS = 3a - b$$

$$17 \text{ (a) (i) } b - a, \text{ (ii) } \frac{1}{4}a + \frac{3}{4}b, \text{ (iii) } \frac{1}{2}a - b$$

$$(b) \frac{1}{2}ha + (1 - h)b$$

$$(c) h = \frac{2}{5}, k = \frac{4}{5} \quad (d) \frac{1}{5}a + \frac{3}{5}b$$

$$18 \text{ (a) (i) } \frac{1}{2}a, \text{ (ii) } c, \text{ (iii) } c - a, \text{ (iv) } \frac{2}{3}(c - a)$$

$$(b) \frac{3}{14}a + \frac{2}{7}c \quad (c) \frac{1}{2}a + pc$$

$$(d) \frac{3}{14}qa + \frac{2}{7}qc$$

$$(e) p = \frac{2}{3}, q = 2\frac{1}{3}, AY:YB = 2:1$$

$$19 \text{ (a) } a - b \quad (c) 3a + 2b$$

$$(d) h = \frac{3}{5}, k = \frac{1}{5} \quad (e) \frac{2}{3}$$

$$20 \text{ (a) } \frac{1}{2}(x + y) \quad (b) \frac{1}{2}hx + \frac{1}{2}hy$$

$$(c) \frac{1}{2}y - x \quad (d) (1 - k)x + \frac{1}{2}ky$$

$$(e) h = k = \frac{2}{3}$$

$$(f) \text{ the lines meet each other at a single point and divide each other in the ratio } 2:1$$

$$(i) \frac{2\sqrt{2}}{3} \quad (j) \frac{\sqrt{6}}{3} \quad (k) \frac{\sqrt{3}}{3} \quad (l) \frac{6}{7}$$

$$(m) \sqrt{5}$$

- 4 (a) $3\sqrt{5}$ (b) $2\sqrt{3}$ (c) $3\sqrt{7}$ (d) 0
 (e) $4\sqrt{2}$ (f) $\sqrt{3}$ (g) 9 (h) 12
 (i) $9\sqrt{5}$ (j) 60 (k) $3\sqrt{3}$ (l) 8
 (m) $54\sqrt{2}$

Exercise 21g (p. 183)

- 1 (a) $300 (3 \times 10^2)$ (b) $75 (3 \times 5^2)$
 2 $41_{\text{five}}, 22_{\text{ten}}, 1011_{\text{two}}$
 3 8
 4 (a) 16 (b) 5 (c) 625
 5 (a) 27 (b) 65 (c) 108 (d) 11
 6 (a) 100 001 (b) 11 111 (c) 1 100 001
 (d) 1 100 111
 7 (a) 222 (b) 2 222 (c) 10 001
 (d) 4 210
 8 (a) 10 000 (b) 11 101 (c) 110 11
 (d) 11 111
 9 (a) 131 (b) 433 (c) 340
 (d) 441
 10 (a) 14 125 (b) 1 000 100
 11 (a) 2 324 (b) 111 101
 12 1 321
 13 (a) 1 021 (b) 10 111 101 (c) 4
 (d) 110
 14 (a) 100 111 (b) 11 111 101
 (c) 10 100 010 110
 (d) 11 111 100
 15 (a) 101 (b) 101 (c) 110 (d) 111
 16 (a) CALCULATE X
 (b) (i) (ii)
 00001 00011
 10000 01111
 10000 01100
 10010 01100
 01111 00101
 11000 00011
 01001 10100
 01101 00000
 00001 00100
 10100 00001
 00101 10100
 00001

Exercise 21h (p. 184)

(Answers given to ready-reckoner accuracy only.)

- 1 (a) \$5.52 (b) \$15.18 (c) \$33.81
 (d) \$60.03 (e) \$91.77 (f) \$73.83
 2 (a) \$966 (b) \$200.10 (c) \$34,500
 (d) \$17.25 (e) \$220.80 (f) \$54.51

- 3 \$20.70 4 \$43.10 5 \$24.84
 6 (a) \$79.17 (b) \$258.33 (c) \$666.67
 (d) \$700 (e) \$1 500 (f) \$1 446.67
 7 (a) \$86.54 (b) \$144.23 (c) \$173.08
 (d) \$230.77 (e) \$336.54 (f) \$164.90
 8 \$945
 9 \$1 403.89
 10 \$17 648.08
 11 (a) 18.2 (b) 31.8 (c) 40.9
 (d) 45.5 (e) 272.8 (f) 3.6
 12 (a) 0.7 (b) 1.1 (c) 1.8
 (d) 2.2 (e) 19.8 (f) 1.4
 13 9.24 gallons
 14 (a) 1.39 (b) 2.22 (c) 16.7
 (d) 19.4 (e) 24.7 (f) 37.4
 15 (a) 2.49 (b) 4.35 (c) 6.2
 (d) 49.7 (e) 5.28 (f) 27.4
 16 30 mph (since $50 \text{ km/h} = 31.1 \text{ mph}$)
 17 (a) 0 (b) 10 (c) 30
 (d) 20 (e) -5 (f) 37
 18 (a) 14 (b) 68 (c) 86
 (d) 41 (e) 93 (f) 57
 19 (a) 18°F (b) 122°F
 20 (a) US\$7.01 (b) US\$18.70
 (c) US\$4.20 (d) US\$10.28
 (e) US\$63.09 (f) US\$12.16
 (g) US\$17.76 (h) US\$35.99

Exercise 22a (p. 188)

- 1 (a) {l; i; o; n; t; g; e; r} (b) {i}
 (c) {r; e; v; t; g} (d) {v; o; l; n}
 (e) {r; e; v; o; l; t; n; g} (f) {v}
 (g) {v} (h) {r; e; v; o; l; t; n; g}
 (i) {t; g; e; r}

2

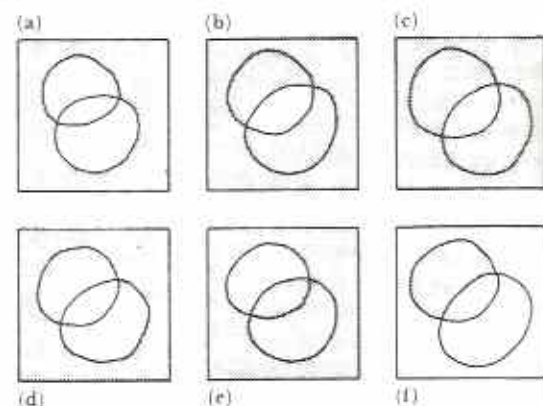


Fig. A14

3

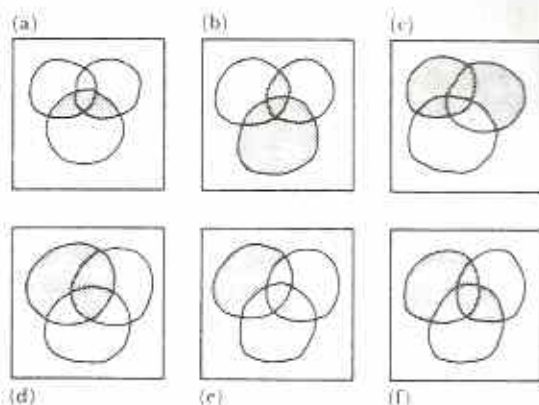


Fig. A15

- 4 (a) {1; 2} (b) {5; 10}
 (c) {4; 5; 6; 7} (d) {2}
 (e) {1; 2; 3; 4; 5; 6; 8; 9; 10} (f) {2; 4}
- 5 (a) 6 (b) {4; 8; 10}
- 6

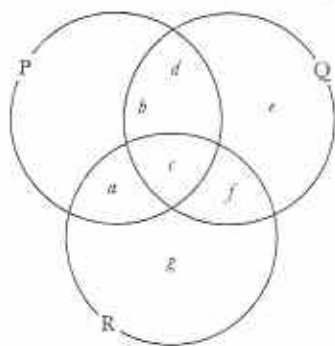


Fig. A16

- {a; e; f}
- 7

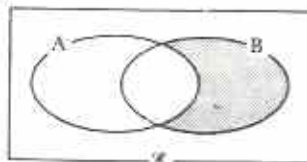


Fig. A.17

- 8 11 men
- 9 (a) $k = 3$ (b) 23 (c) 0 (d) 13
- 10 (a) 7 (b) $\{x: 0 \leq x \leq 5, x \text{ is an integer}\}$
- 11 (a) $p = 75 - x$ (b) 90 (c) 35 (d) 5
- 12 (a) 2 (b) 16
- 13 (a) 8 (b) 11
- 14 37
- 15 some boys are unhappy; no girls are unhappy

Exercise 22b (p. 190)

- 1 $16x - 9y$ 2 $2b^2 - 5q$
 3 $3a - 10b$ 4 $23 + 15x^2$
 5 $x^2 - 10xy + 9y^2$ 6 $2u^2v - 4uv^2$

- 7 $2d^2 - 10ad$
 9 $7x^2 + 4x^3 + 2x^4$

- 8 $5a^2 + 3a$
 10 $m^2 - mn + 4n^2$

Exercise 22c (p. 191)

- 1 $36a + 4$ 2 $-30x + 10y - 5$
 3 $42x - 3xy$ 4 $-10a^2 + 2ab - 6ac$
 5 $5 - x$ 6 $8x^2 - 4x$
 7 $6x^2 - 17x + 10$ 8 $32 - 7b^2$
 9 $3ac + 4ad + 6bc + 3bd$
 10 $x^2 + 9x + 8$ 11 $10a^2 + 21ab + 8b^2$
 12 $10a^2 - 11ab - 8b^2$ 13 $10a^2 - 11ab - 8b^2$
 14 $10a^2 - 21ab + 8b^2$ 15 $2x^2 - 11x + 15$
 16 $12 + 11x - 5x^2$ 17 $9a^2 + 6a + 1$
 18 $b^2 - 8b + 16$ 19 $4x^2 - 20cd + 25d^2$
 20 $16m^2 - 24mn + 9n^2$

Exercise 22d (p. 192)

- 1 $(3c - d)(3m - 4n)$ 2 $(a + b)(c + 2d)$
 3 $(x + 4a)(x - 4a)$ 4 $(2c + 5d)(2c - 5d)$
 5 $(a + 2)(a - 5)$ 6 $(a + 2b)(a - 5b)$
 7 $(ab + 2)(ab - 5)$ 8 $5(m - 3n)(m + 3n)$
 9 $(h - 2k)(m - 2n)$ 10 $(2a + b)(a + 4b)$
 11 $(3m - 1)(m - 3)$ 12 $(cd - 9)(cd + 9)$
 13 $(4x + 3am)(4x - 3am)$ 14 $(2n + 3)(3n + 2)$
 15 $(2a + 5)^2$ 16 $9(h - 2k)(h + 2k)$
 17 $(a - 2b)(a - 4b)$
 18 $(5abc + 3d)(5abc - 3d)$
 19 $\left(\frac{m}{3} - \frac{n}{2}\right)\left(\frac{m}{3} + \frac{n}{2}\right)$
 20 $(3x + t)(u - 2v)$
 21 $(3x - 2y)^2$ 22 $(4d - 1)(3d + 2)$
 23 $(x^2 + y)(x^2 - y)$ 24 $(2mn - 3)(5mn + 4)$
 25 $(4 + n^2)(2 + n)(2 - n)$
 26 $(a + b + c)(a + b - c)$
 27 $(x + m - n)(x - m + n)$
 28 $2(m + 2n)(a + b)$
 29 $(2 + 5h)(1 - 3h)$
 30 $(2a - b)(m - 3n)$
 31 $(a - 6b)(a - 9b)$
 32 $(m - 18n)(m + 3n)$
 33 $(c - 2d - 3a)(c - 2d + 3a)$
 34 $(3x + 2y)(4x + 9y)$
 35 $(h - k)(3h - 4k)$
 36 $2a(3x + 2y)(c - 2d)$
 37 $(2a - 9x)(3a + 4x)$
 38 $(5a + 2m + 4n)(5a - 2m - 4n)$
 39 $(7a - 4b)(3a + 4b)$
 40 $4b(3a - b)$

Exercise 22e (p. 192)

- 1 1800 2 2300 3 2430 4 28000 5 940
 6 3050 7 2400 8 75.6 9 308 10 10560

Exercise 22f (p. 193)

- 1 $\frac{5x+1}{6}$ 2 $\frac{12b+11}{35}$ 3 $\frac{19d+48}{30}$
 4 $\frac{13-22x}{21}$ 5 $\frac{8c-9a}{12abc}$ 6 $\frac{6q-p}{q}$
 7 $\frac{b+c}{b-c}$ 8 $\frac{a+b}{a}$ 9 $\frac{a}{b}$
 10 $\frac{m+n}{m-n}$ 11 $2x+1$ 12 $\frac{x+2}{x+3}$
 13 $-\frac{x+5}{x+3}$ 14 $\frac{3-a}{3+a}$ 15 $\frac{y-8x}{x-2y}$
 16 $-\frac{a+b}{ab}$ 17 $\frac{1}{x-2}$ 18 $\frac{a}{b(a-b)}$
 19 $\frac{(x-11)(x+1)}{6(x-5)}$ 20 $\frac{5m(6-m)}{(m-1)(2m+3)(m+4)}$

Exercise 22g (p. 194)

- 1 24 2 11 3 ± 15 4 -9
 5 (a) 8 (b) 8 (c) 12 (d) 36
 6 (a) -2 (b) -4 (c) 0 (d) 10
 7 18 8 -30
 9 (a) 1 (b) 1260 (c) $1\frac{1}{3}$
 10 (a) 4900 (b) $7\frac{1}{2}$

Exercise 22h (p. 196)

- 1 (a) $l \propto \frac{1}{b}$ or $l = \frac{k}{b}$
 (b) $r \propto \frac{1}{\sqrt{c}}$ or $r = \frac{k}{\sqrt{c}}$
 (c) $G \propto \frac{m_1 m_2}{d^2}$ or $G = \frac{km_1 m_2}{d^2}$
 (d) $E = ah + bv^2$
 2 $y = 20x$, $x = 1\frac{3}{4}$ 3 $A = 6$, $M = 17\frac{1}{2}$
 4 $P = 10\frac{1}{2}$, $Q = 16$ 5 $D = 135$, $V = 5\frac{1}{4}$
 6 $P = \frac{3}{4}Q^2$; $P = 75$, $Q = 5$
 7 $xy = 30$; $x = 2\frac{1}{2}$, $y = 1\frac{1}{2}$
 8 $M = \frac{5}{8}R^3$; $M = 625$, $R = 1,6$
 9 $\sqrt{Y} = \frac{2}{3}Z$; $Y = 100$, $Z = 6$
 10 $A = 30$, $B = 7\frac{1}{2}$ 11 $P = 13\frac{1}{2}$, $R = 5$
 12 $x = 4\frac{1}{2} + \frac{1}{2}y$; $x = 10$ 13 $x = 115,2$

- 14 $x \propto z^2$ 15 $x \propto \frac{1}{z^2}$

- 16 (a) 12 h (b) 2 h (c) 1 h

17	x	10	15	20	25	30	35
	y	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{5}$	$\frac{1}{6}$	$\frac{1}{7}$

- 18 31,4 cm 19 766 N 20 84 min

Exercise 22i (p. 198)

1	x	-2	-1	0	1	2	3	4
	y	25	11	3	1	5	15	31

- (b) 0,92 (c) $x = 0,83$
 (d) $x = 2$, $x = -0,3$
 2 (a) (i) 2,75 (ii) -5,25
 (b) (i) 4,7 or -1,7 (ii) -1 or 4
 (c) $x = 1,5$
 3 (a) -2,25 (b) 4,65 or -0,65 (c) $x = 2; 4$
 4 (a) $x > -2$ (b) $(-2, -3)$
 5 (a) $x < \frac{1}{2}$ (b) $-1 < x < 2$
 7 (a) (4; 0) and (-2; 0) (b) (0; 8)
 (c) maximum (d) $x = 1$
 (e) $y = 9$
 8 (a) (0; 0) and (5; 0)
 (b) $y = 0$ (the x -axis), $x = 2\frac{1}{2}$
 9 (2,64; 0,76), (-1,14; -1,75)
 10 (-1; -1), (-3; 1)

Exercise 23a (p. 199)

- 1 -2 2 $3\frac{1}{5}$ 3 5 4 $-4\frac{1}{2}$ 5 -2
 6 2 7 -2 8 4 9 $3\frac{1}{3}$ 10 $-1\frac{1}{2}$

Exercise 23b (p. 200)

- 1 $T = \frac{100I}{PR}$ 2 $h = \sqrt{\frac{4V}{\pi d^2}}$, $d = 2\sqrt{\frac{V}{\pi h}}$
 3 $t = \frac{mv - mu}{F}$ 4 $M = N - RD$
 5 $N = \sqrt{\frac{T-a}{b}}$ 6 $r = \sqrt[3]{\frac{3V}{4\pi}}$
 7 $h = \frac{S - 2\pi r^2}{2\pi r}$
 8 $a = \frac{x}{\sqrt{1-w^2}}$, $w = \sqrt{1 - \frac{x^2}{a^2}} = \frac{\sqrt{a^2 - x^2}}{a}$
 9 $f = \frac{uv}{u+v}$, $u = \frac{vf}{v-f}$

$$10 \ a = \frac{2s}{n} - (n-1)d$$

$$11 \ Q = \frac{P(y+3x)}{x-3y}$$

$$12 \ s = \frac{u^2 - v^2}{2a}, \ u = \pm \sqrt{v^2 + 2as}$$

$$13 \ u = \pm \sqrt{v^2 - \frac{2E_g}{m}}$$

$$14 \ \rho = \frac{P(1 - \epsilon T)}{1 - \epsilon t}$$

$$15 \ W = \frac{Tgx}{gx + v^2}$$

$$16 \ g = \frac{4\pi^2 l}{t^2}$$

$$17 \ h = \frac{V}{\pi r^2} - \frac{2}{3}r$$

$$18 \ h = \frac{v^2 - gd}{3g}, \ d = \frac{v^2 - 3gh}{g}$$

$$19 \ h = \pm \sqrt{\frac{A^2}{\pi^2 r^2} + r^2}$$

$$20 \ Q = \frac{P}{d^3} + P, \ P = \frac{d^3 Q}{1 + d^3}$$

$$21 \ (a) \ T = \frac{100}{R} \left(\frac{A}{P} - 1 \right) \quad (b) \ 8 \text{ years}$$

$$22 \ (b) \ 39\frac{1}{3} \text{ cm} \quad 23 \ a = \frac{x}{\sqrt{1-w^2}}, \pm \frac{20\sqrt{3}}{3}$$

$$24 \ (a) \ a = \frac{1}{l} \left(\frac{l}{L} - 1 \right)$$

$$(b) \ \frac{1}{24000}$$

$$25 \ (a) \ r = \frac{C}{2\pi} \quad (b) \ V = \frac{C^2 h}{4\pi}$$

Exercise 23c (p. 201)

- 1 (a) $x \leq 1$ (b) $x < 4$ (c) $x \geq 2$
 (d) $x \geq 4$ (e) $x > -6$
 (f) $x > -5$
 2 (a) $x \leq 2$ (b) $x \geq 6$ (c) $p < 47$
 (d) $x \leq 3$ (e) $x > 1$ (f) $x < 11$
 (g) $x \leq -10$ (h) $y < -47$ (i) $x \leq 4$
 (j) $z < 1\frac{1}{2}$

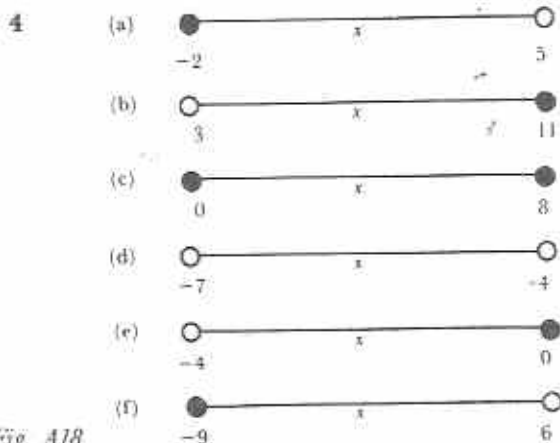


Fig. A18

- 5 C
 6 (a) $-3 \leq x < 1$
 7 (a) $\{-6; -5; -4\}$ (b) $\{-2; -1; 0; 1; 2\}$
 8 $\{-1; 0; 1; 2; 3; 4; 5\}$
 9 (a) (i) 6 (ii) -14 (iii) -9 (iv) -45
 (b) (i) 14 (ii) -6 (iii) 16 (iv) 27
 10 17, 19, 23

Exercise 23d (p. 202)

- 1 4; 2 2 3; -5 3 -3; -4 4 1; 2
 5 -3; 4 6 -2; -5 7 5; 2 8 -7; -2
 9 $2\frac{1}{2}; 3$ 10 -2; $1\frac{1}{4}$ 11 3; 0 12 5; 1
 13 -2; $1\frac{1}{2}$ 14 0; -2 15 $2\frac{1}{2}; -3\frac{1}{2}$ 16 4; 5
 17 2; -5 18 8; -12 19 -2; 4 20 1; 2
 21 -2; -2 22 3; 1 23 -3; -2 24 -3; -2
 25 $1\frac{1}{2}; -\frac{2}{3}$

Exercise 23e (p. 203)

- 1 (a) $y = 2x$ (b) $x + y = 5$
 (c) $y < 2x, x + y \geq 5, y \leq 0$
 2 $x > -4, x + y < -5, y \leq x - 2$
 3 $y \leq 1, x + y < 5, y \geq 2x$
 4 $x \leq 0, y > 1, x + y \geq 4, x + 2y \geq 6$
 5 greatest value is $4\frac{1}{2}$ at $(2\frac{1}{2}, 2)$
 6 (a) (i)

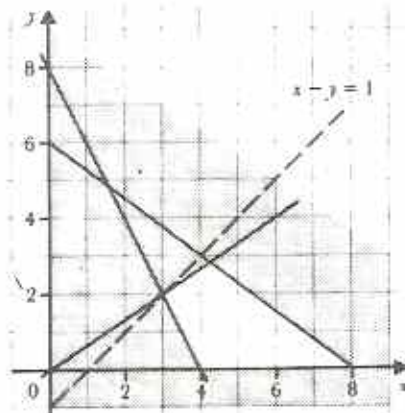


Fig. A19

- (ii) The least value of $x - y$ is 1
 (b) $h + b \geq 36$, $h \leq 20$, $h \leq 2b$
 7 9 shirts, 6 dresses
 8 Maximum 37 (10 large; 27 small)
 Greatest profit from either (10 large; 27 small)
 or (11 large; 25 small) \$4,70
 9 (a) (14 of A; 9 of B) (b) (12 of A; 11 of B)
 10 (i) $h \leq 15$ (ii) $p > 25$ (iii) $45 \geq h + p < 60$

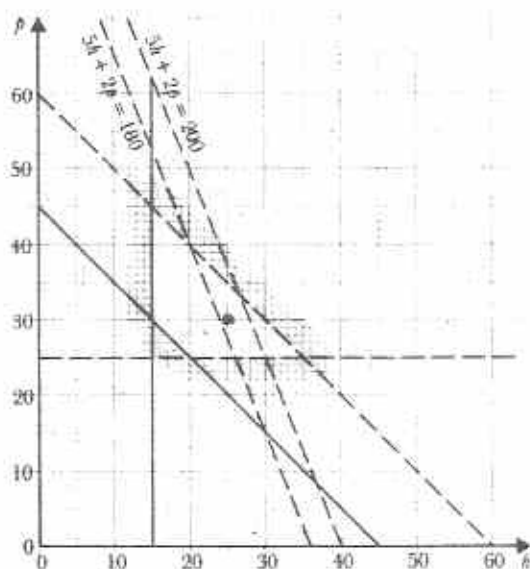


Fig. A20

25 hardback, 30 paperback

Exercise 23f (p. 205)

- 1 3; 7 2 -1; -2 3 2; -3
 4 5; -2 5 1; -2 6 0; -3
 7 2; $1\frac{1}{2}$ 8 -2; $-\frac{3}{4}$ 9 2; -2
 10 0; 4 11 -1; $-2\frac{1}{2}$ 12 4; $-\frac{2}{3}$
 13 $1\frac{1}{2}$; $\frac{3}{4}$ 14 $2\frac{1}{4}$; $-\frac{2}{3}$ 15 $1\frac{1}{2}$; $-1\frac{3}{4}$

Exercise 23g (p. 206)

1	x	-2	-1	0	1	2	3	4	5
	y	14	7	2	-1	-2	-1	2	7

- (a) $x = 3.41$ or 0.59 (b) -2

2	x	-2	-1	0	1	2	3	4
	y	25	10	1	-2	1	10	25

- $x = 0.18; 1.82$

- 3 (a) 3.56; 0.56 (b) 2.62; 0.38
 (c) 4; -1
 4 (a) 0; -0.67 (b) 1.23; -1.9
 (c) 0.72; -1.39 (d) 1.69; -2.36
 5 (a) imaginary roots (b) 0; 1.67
 (c) 1.85; -0.18
 6 $x = 1.77$ or 0.57 7 ± 1.73 , $x^2 - 3 = 0$
 8 ± 1.87

9	x	-2	-1.5	-1	-0.5	0	0.5
	y	-4	-1.75	0	1.25	2	2.25

x	1	1.5	2	2.5	3
y	2	1.25	0	-1.75	-4

- (b) $y_{\max} = 2.25$ when $x = 0.5$ (d) 2.41; -0.41
 10 2.27; -1.77

Exercise 23h (p. 207)

- 1 (0; 5), (4; -3) 2 (3; 5), $(-1\frac{1}{4}; -12)$
 3 (1; 2) 4 (3; -5), $(-2\frac{1}{2}; 6)$
 5 (2; -4) 6 (1; -2), $(-\frac{1}{2}; 1\frac{1}{2})$
 7 (-4; 3) 8 (4; -1)
 9 (-1; 1), $(1\frac{8}{11}; -\frac{1}{11})$ 10 (5; 6), (2; 15)
 11 (2; 4) 12 (4; 1), (-8; -3)
 13 (1; 2), $(-2\frac{3}{5}; -\frac{2}{5})$ 14 $(1\frac{1}{3}; -1)$, $(-1\frac{2}{15}; \frac{2}{3})$
 15 (2; 1), $(-1\frac{1}{3}; -1\frac{1}{2})$

Exercise 23i (p. 208)

- 1 \$1.03; 29c 2 12; 15 3 18 cm
 4 40 yr, 12 yr 5 17 cm, 4 cm 6 10 yr
 7 $x = 12$, $y = 10$ 8 2 m
 9 5 10 $\frac{5}{6}$ 11 3 cm
 12 6 years ago 13 4:1
 14 12 cm, 2 cm 15 15

Exercise 24a (p. 210)

- 1 $a = 68^\circ$, $b = 68^\circ$, $c = 112^\circ$
 2 $d = 70^\circ$, $e = 95^\circ$, $f = 15^\circ$
 3 $g = 135^\circ$, $h = 45^\circ$, $i = 45^\circ$
 4 $j = 48^\circ$, $k = 48^\circ$, $l = 132^\circ$
 5 $n = 38^\circ$, $m = 142^\circ$
 6 $p = 55^\circ$ 7 $q = 130^\circ$
 8 $r = 130^\circ$, $s = 80^\circ$

Exercise 24b (p. 211)

- 1 $\hat{PAC} = 105^\circ$, $\hat{QBA} = 135^\circ$, $\hat{RCA} = 120^\circ$
 2 (a) 65° , scalene (b) 71° , isosceles
 (c) 43° , isosceles (d) 60° , equilateral
 (e) 90° , right-angled (f) 97° , obtuse-angled
 3 (a) $h = 65^\circ$, $k = 84^\circ$ (b) $m = n = 71^\circ$
 (c) $p = 35^\circ$, $q = 60^\circ$, $r = 25^\circ$
 (d) $s = 68^\circ$, $t = 44^\circ$, $u = 24^\circ$
 (e) $v = 47^\circ$, $w = 29^\circ$, $x = 151^\circ$
 (f) $y = 70^\circ$, $z = 35^\circ$
 4 $x = 12^\circ$
 5 (a) $x = 25^\circ$, isosceles (b) $x = 21^\circ$, scalene
 6 (a) $\triangle XAZ$ (RHS) (b) $\triangle BYZ$ (SAS)
 (c) $\triangle YXC$ (SSS) (d) $\triangle ZDX$ (AAS)
 (e) $\triangle XEZ$ (SAS) (f) $\triangle GHF$ (SAS)
 7 84°
 8 (a) SAS (b) RHS (c) ASA or AAS (d) ASA
 9 79°
 10 $\triangle ABC = \triangle RPQ$ (SSS),
 $\triangle KLM = \triangle XZY$ (AAS)

Exercise 24c (p. 215)

1	polygon	sum of interior angles
	triangle (3)	180°
	quadrilateral (4)	360°
	pentagon (5)	540°
	hexagon (6)	720°
	heptagon (7)	900°
	octagon (8)	1080°
	decagon (10)	1440°
	dodecagon (12)	1800°

- 2 (a) 100° (b) 170° (c) 80° (d) 50° , 150°
 3 156° each 4 18 sides
 5 $\hat{BCD} = 112^\circ$, $\hat{EAB} = 62^\circ$, $\hat{AED} = 118^\circ$
 6 11 7 23 cm 8 55° 9 11 sides
 10 $x = 18^\circ$; 72° , 90° , 108° , 126° , 144°

Exercise 24d (p. 218)

- 1 (a) $g = 95^\circ$, $h = 33^\circ$
 (b) $i = j = 63^\circ$, $k = 108^\circ$, $l = 9^\circ$, $m = 45^\circ$
 (c) $n = 226^\circ$, $p = 113^\circ$, $q = 67^\circ$
 (d) $r = 60^\circ$, $s = 80^\circ$, $t = 28^\circ$, $u = 72^\circ$
 (e) $w = 90^\circ$, $x = 42^\circ$, $y = 12^\circ$
 (f) $z = 62^\circ$
 2 7 cm, 6 cm, 5 cm 3 3 cm, 4 cm, 6 cm
 4 $AY = 4$ cm, $BX = 5$ cm
 5 (a) $h = 90^\circ$, $i = 90^\circ$, $j = 40^\circ$, $k = 50^\circ$
 (b) $m = n = 61^\circ$

- (c) $p = 23^\circ$, $q = 67^\circ$, $r = 67^\circ$, $s = 23^\circ$
 (d) $t = u = 90^\circ$, $v = 113^\circ$
 (e) $w = 128^\circ$, $x = 64^\circ$
 (f) $y = 57^\circ$, $z = 66^\circ$
 6 62° , 56° , 62°
 7 (a) 62° (b) 120° (c) 58° (d) 106°
 8 (a) $f = 56^\circ$, $g = 68^\circ$, $h = 56^\circ$
 (b) $x = 76^\circ$, $y = 70^\circ$, $z = 34^\circ$
 (c) $p = 38^\circ$ (d) $r = 86^\circ$ (e) $q = 25^\circ$
 9 $y = 2x - 180^\circ$
 10 (a) 58° (b) 72° (c) 26° (d) 19°

Exercise 24e (p. 220)

- 2 6.2 cm 3 5.9 cm 4 6.9 cm
 6 185 mm 7 5.8 cm 8 10 cm
 9 7.4 cm 10 8.5 cm

Exercise 24f (p. 223)

- 1 hemispherical surface 2 circle
 3 construct the bisectors of the angles between the lines
 5 perpendicular bisector of line joining the fixed points
 6 4 7 4 8 4 cm or 6.2 cm
 10 $YZ = 8.7$ cm; maximum area of $\triangle XMY$ is 23.3 cm²

Exercise 25a (p. 224)

- 1 62.8 cm 2 15.84 m (1584 cm)
 3 150° , 16.5 cm 4 50
 5 22 cm 6 1 cm
 7 $14\frac{1}{2}$ cm 8 6.3 cm
 9 120 cm 10 553.5 cm

Exercise 25b (p. 226)

- 1 (a) 88 cm² (b) 84 m²
 (c) 105 cm² (d) $14\,400$ m²
 2 19.7 cm² 3 1.113 ha
 4 $3\,850$ cm², 294 cm²
 5 5.28 kg 6 $31\frac{1}{4}\%$
 7 113 m 8 14.2 cm²
 9 27.5 m 10 $15\,510$ cm²
 11 90.1 cm² 12 6.19%

Exercise 25c (p. 229)

- 1 18.48 litres 2 1.98 kg
 3 1.6 m (160 cm) 4 1.188 kg
 5 221.57 (70.5π) cm² 6 1.12 m
 7 (a) $20\,250$ cm³ (b) 16.2 kg
 8 0.79 litres 9 $4\frac{3}{4}$ cm
 10 (a) 10 cm (b) 60π cm² (c) 96π cm³
 11 216° 12 195π cm²
 13 $1\,571$ (500π) l 14 1 728

- 15 $3\frac{3}{8}$ cm
 17 (a) (i) $\frac{2}{3}$ (ii) $\frac{1}{3}$
 18 20,0 cm²
 20 1,49 kg

- 16 336 cm²
 (b) 1; 1
 19 $31\frac{1}{4}$ cm

- 4 4,48 m
 6 $6^{\circ}54'$, $17^{\circ}38'$
 8 $46^{\circ}41'$
 9 (a) $53^{\circ}8'$ (b) 57°
 10 (a) $33^{\circ}4'$ (b) $6\frac{6}{11}$ cm

Exercise 25d (p. 230)

- 1 (a) 3 cm (b) 25:1
 3 16 kg 4 9 times 5 135 cm³
 6 (a) 5:3 (b) 25:9 (c) 125:27
 7 (a) 6 cm (b) 27:8
 8 (a) 9 km² (b) 3 km (c) 300 000:1
 9 38,5 cm 10 (a) 5:3 (b) 25:9

Exercise 26a (p. 231)

- 1 (a) 3 (b) 9 (c) 7 (d) 9 (e) 14 (f) 44
 2 37 m 3 4 m
 4 (a) 7 cm (b) 84 cm² 5 20 cm
 6 (a) 5,39 (b) 3,74 (c) 6,48
 (d) 7,28 (e) 7,55 (f) 9,54
 7 58,7 cm 8 (a) 2,74 cm (b) $11,8$ cm²
 9 4,36 cm 10 (a) 5,6 cm (b) 6,5 cm

Exercise 26b (p. 234)

- 1 181 m 2 147 m
 3 (a) 3,64 km (b) 10,6 km
 4 $a = 11,0$ cm, $b = 11,4$ cm, $c = 15,9$ cm
 5 (a) 5,54 (b) 6,08 cm
 6 (a) 5; $5\sqrt{2}$ (b) $2\sqrt{2}$; $2\sqrt{2}$
 (c) $\frac{7\sqrt{2}}{2}$; $\frac{7\sqrt{2}}{2}$ (d) 6; $3\sqrt{3}$
 (e) $2\sqrt{3}$; $4\sqrt{3}$ (f) $2\frac{1}{2}$; $2\frac{1}{2}\sqrt{3}$
 (g) 10; $5\sqrt{3}$ (h) 4; $4\sqrt{3}$
 7 (a) 12 (b) $6\sqrt{2}$
 (c) $6\sqrt{3}$ (d) 8
 (e) $2\sqrt{6}$ (f) $\sqrt{6}$
 (g) $3\sqrt{2} - \sqrt{6}$ (h) $4\sqrt{6}$
 8 $10\sqrt{3}$ m 9 59°
 10 (a) $030^{\circ}35'$ (b) 12,8 km 11 3,35 m
 12 (a) 7,81 km (b) $085,2^{\circ}$
 13 $47\frac{1}{2}$ m (45,9 m + 1,6 m)
 14 3 835 m, 61 m
 15 (a) 800 m (b) $400\sqrt{3}$ m

Exercise 26c (p. 237)

- 1 (a) 6,93 cm ($\sqrt{48}$ cm) (b) $35^{\circ}21'$
 2 (a) 11 cm (b) $39^{\circ}31'$ (c) $49^{\circ}24'$
 3 (a) $14,5^{\circ}$ (b) 9°

Exercise 26d (p. 240)

- 1 13,1 cm 2 20,4 cm 3 $10^{\circ}4'$
 4 $16^{\circ}25'$ 5 4,36 cm 6 1,98 cm
 7 4,42 m 8 61,8 cm 9 $57,1^{\circ}$
 10 60° 11 $106,6^{\circ}$ 12 $112,4^{\circ}$
 13 240,5 m 14 135 m
 15 $d = 12,3$ cm, $\alpha = 50^{\circ}3'$
 16 $x = 8,90$ m, $\theta = 107,7^{\circ}$
 17 (a) 20,2 cm (b) $68,1^{\circ}$ 18 $110,8^{\circ}$
 19 1 041 km, $017^{\circ}52'$ 20 23,1 km, $246^{\circ}51'$

Exercise 27a (p. 242)

- 1 $7, \frac{1}{7} \begin{pmatrix} -3 & -2 \\ -4 & -5 \end{pmatrix}$ 2 $x = -8\frac{1}{2}$
 3 (a) $\begin{pmatrix} 4 & 4 \\ 1 & 7 \end{pmatrix}$ (b) $x = 4, y = 7$
 4 (a) $\begin{pmatrix} 7 & -3 \\ 6 & -2 \end{pmatrix}$ (b) $m = 1\frac{1}{2}$ (c) $n = -2$
 5 $x = 4, \frac{1}{2} \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$
 6 $a = 4, b = -3, c = 1$
 7 (a) $\begin{pmatrix} 1 & -8 \\ 3 & 0 \end{pmatrix}$ (b) $\frac{1}{2} \begin{pmatrix} 4 & -5 \\ -2 & 3 \end{pmatrix}$
 (c) $\frac{1}{3} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$
 8 $k = -\frac{1}{2}$ or 6
 9 (a) $-\frac{1}{7} \begin{pmatrix} 3 & -4 \\ -1 & -1 \end{pmatrix}$ (b) $\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -5 \\ 2 \end{pmatrix}$
 10 $\begin{pmatrix} 2 & 5 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ 7 \end{pmatrix}, \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ -1 \end{pmatrix}$

Exercise 27b (p. 245)

- 1 Reflection in the line $x = y$
 2 (a) 7 cm (b) 4,2 cm (c) 3,5 cm
 3 (a) P (3; 5), Q (3; 3), R (9; 1) (b) $\frac{1}{3}$
 4 (a) (9; -2) (b) (1; 2)
 (c) A shear of factor -2 with the x -axis invariant
 5 (a) (5; -3) (b) (-4; -1) (c) (1; 2)

- 6 (a) $B'(-7; 4)$, $A(-2; -3)$
 (b) any point of the form $(k; 0)$, i.e. any point on the x -axis
 7 (b) enlargement, centre at origin, factor k
 $k = 2\frac{1}{2}$
 (c) $A''(3; 1)$, $B''(1; -3)$, $C''(-3; 2)$
 8 (a) $\begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}$
 9 (a) (i) $(7; 5)$, (ii) $(-3; -1)$
 (b) Rotation of 90° anticlockwise about $(2; 2)$

- 10 (a) 90° (b) $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (c) $\begin{pmatrix} -8 \\ -4 \end{pmatrix}$
 (d) $(6; 2)$ (f) $x + y = -5$

Exercise 27c (p. 246)

1

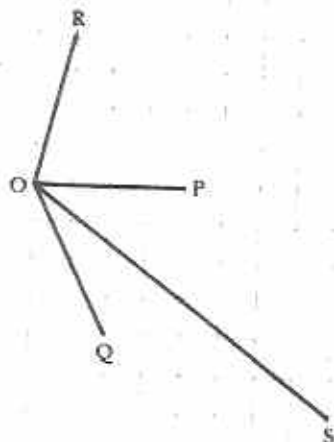


Fig. A21

- 2 (b) (i) $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$, (ii) $\begin{pmatrix} 3 \\ -3 \end{pmatrix}$, (iii) $\begin{pmatrix} 5 \\ 5 \end{pmatrix}$
 3 (b) (i) $\begin{pmatrix} 5 \\ 6 \end{pmatrix}$, (ii) $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$, (iii) $\begin{pmatrix} -4 \\ 3 \end{pmatrix}$
 4 (a) $\begin{pmatrix} 10 \\ -12 \end{pmatrix}$ (b) $\begin{pmatrix} 5 \\ -6 \end{pmatrix}$ (c) $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$
 5 (a) $\mathbf{b}$ (b) $-\frac{1}{2}\mathbf{a}$ (c) $\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$
 (d) $\frac{1}{2}\mathbf{b}$ (e) $\frac{1}{2}\mathbf{a} + \mathbf{b}$ (f) $\mathbf{b} - \frac{1}{2}\mathbf{a}$
 6 (a) 5 (b) $q = 7$
 7 $h = 2$, $k = 9$
 8 $m = -1$, $n = -10$
 9 (a) $\mathbf{XY} = \frac{1}{4}\mathbf{a} + \frac{1}{2}\mathbf{c}$, $\mathbf{CZ} = \mathbf{a} - \frac{1}{2}\mathbf{c}$

(b) $\mathbf{XP} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}$

(c) $\mathbf{CP} = k\mathbf{a} - \frac{1}{2}k\mathbf{c}$

(d) $h = \frac{2}{3}$, $k = \frac{1}{3}$

10 (a) (i) $\begin{pmatrix} -6 \\ -2\frac{1}{2} \end{pmatrix}$, (ii) 13

(b) (i) $3\mathbf{a} - 3\mathbf{b}$, (ii) $-\mathbf{a} + 3\mathbf{b}$

$4\mathbf{b} + m(3\mathbf{a} - \mathbf{b}) = n(4\mathbf{a})$; $m = 4$, $n = 3$

Exercise 28a (p. 250)

- 1 (a) 5 (b) 145 km (c) 58 km/h (d) 36 km/h
 (e) 100 km/h (f) 70 km/h (g) 10 km
 2 (a) 280 m (b) 4 min (c) 1217
 (d) 400 m (e) 80 m/min (f) 120 m
 3 (a) 1250 (b) 1300 (c) 1310
 4 (a) 4600 km (b) 400 km
 5 (a) 17.9 km/h (b) 2 min
 6 (a) 9 km/h (b) 3.4 km
 7 (a) 5 m/s^2 (b) $\frac{1}{9} \text{ m/s}^2$ (c) 172 m
 8 (a) (i) constant speed of 120 km/h for 3 min
 (ii) uniform acceleration from 120 km/h to 150 km/h in 2 min
 (iii) uniform deceleration from 150 km/h to rest in 1 min
 (b) (i) 15 km/h per min
 (ii) $-150 \text{ km/h per min}$
 (c) $11\frac{3}{4} \text{ km}$
 9 (a) 1.5 m/s^2 (b) (i) 225 (ii) 6675 km
 10 (a) 19 m/s (b) 24.95 m/s
 11 (b) 227 min (c) 2639 km
 12 (a) 0 m/s^2 (b) -6 m/s^2 (c) 116 m
 13 (a) -6 m/s^2 (b) 212 m
 14 (a) $m = 2$, $n = 18$ (c) 0.3 s and 2.7 s
 (d) 12 m/s^2 (e) 33 m
 15 (a) $a = 16$, $b = 0$ (c) 14.3 m/s
 (d) (i) 4 m/s^2 , (ii) -6 m/s^2 (e) 82 m

Exercise 28b (p. 255)

- 3 (b) 1434
 5 (a) 4, 6; 4; 4 (b) 6, 5; 7; 7 (c) $6\frac{1}{3}$; 6; 6
 6 117.1 7 18 yr 3 mo.
 8 158 cm, 159.5 cm 9 $47\frac{7}{15} \text{ kg}$, 46 kg
 10 (a) 20 students (b) 15 yr (d) 15 yr

Exercise 28c (p. 257)

- 1 (a) 5 yr (b) 40 children (c) 4.7 yr
 2 (a) 547.5 h
 3 (b) 0-4 days (c) 6.51 days
 4 (c) 60 kg (d) $59\frac{1}{4} \text{ kg}$
 5 (c) (i) 51 (ii) 9.5 (d) 68%
 6 (b) 151-160 cm (c) 155.75 cm (e) 155 cm

Exercise 28d (p. 259)

- 1 (a) $\frac{7}{10}$ (b) $\frac{3}{10}$
 2 (a) $\frac{1}{2}$ (b) $\frac{1}{5}$ (c) $\frac{3}{10}$ (d) 0
 3 (a) $\frac{1}{32}$ (b) $\frac{1}{26}$ (c) $\frac{1}{13}$ (d) 0
 4 $\frac{5}{12}$
 5 (a) $\frac{1}{6}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$
 6 (a) HHH, HHT, HTH, HTT, THH, THT, TTH, TTT (b) $\frac{3}{8}$
 7 15 8 $\frac{102}{125}$
 9 (a) 1 (b) $\frac{1}{3}$ (c) $\frac{1}{3}$ (d) $\frac{2}{3}$
 (e) $\frac{1}{3}$ (f) $\frac{1}{6}$ (g) 0 (h) $\frac{1}{6}$
 10 $\frac{3}{16}$
 11 (a) (i) $\frac{1}{4}$, (ii) $\frac{3}{9}$, (iii) $\frac{31}{36}$
 (b) 17 yr 4 mo (c) $\frac{1}{63}$
 12 (a) $\frac{3}{5}$ (b) $\frac{1}{15}$
 13 (a) $\frac{2}{11}$ (b) $\frac{1}{22}$ (c) $\frac{15}{44}$
 14 $\frac{3}{5}, \frac{1}{2}$ (a) $\frac{3}{10}$ (b) $\frac{1}{10}$ (c) $\frac{3}{5}$
 15 (a) $\frac{1}{10}$ (b) $\frac{4}{15}$ (c) $\frac{12}{145}$ (d) $\frac{63}{145}$

Chapter 29

Note: answers to investigations are generally not given.

Number, algebra and pattern (p. 261)

- 1 (a) 15 (b) 8 (c) 24
 (d) $n^2 + 2n$ or $(n+1)^2 - 1$
 (e) 21 for 3 boys and 3 girls
 4 2 178
 5 431×52
 6

4	2	3
2	3	4
3	4	2

7 here are two solutions; there are others

B	R	B
W	W	W
R	B	R

R	W	B
W	B	R
R	B	W

8

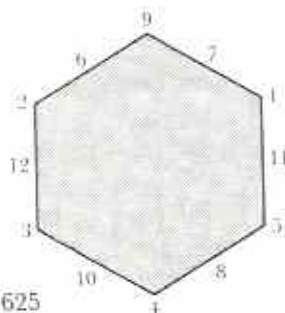


Fig. A22

- 9 $64 \times 15\ 625$
 10 11, 16, 27; $\frac{(m+n)}{3}$
 11 (a) 2 (b) 16 (c) 1, 9, 16
 (d) 1, 2, 9 (e) 1, 2, 9 (f) 5, 9, 16
 (g) 2, 9, 1
 12 (b) $S_1 = \{1; 2; 4; 8; 16; 32\}$
 $S_2 = \{3; 5; 6; 9; 10; 12; 17; 18; 20; 24; 33\}$
 $S_3 = \{7; 11; 13; 14; 19; 21; 22; 25; 26; 28\}$
 $S_4 = \{15; 23; 27; 29; 30\}$
 $S_5 = \{31\}$
 13 50 cents
 14 plan (b)
 15 0
 16 18 hens
 17 7 cattle or 22 cattle or 37 cattle or ... $7 + 15n$
 cattle where $n = 0, 1, 2, 3, \dots$
 18 (a) $\begin{matrix} \times \times \\ m \ m \ m \end{matrix}$ or $\begin{matrix} \times \times \times \times \\ m \ m \ m \ m \ m \end{matrix}$ etc.
 (b) 28 (c) 46 (d) $X = 2M - 2$ (e) 20
 (f) $7\frac{1}{3}$, it would be impossible to make this pattern
 19 (a) the first two numbers are added and only
 the units digit of their sum is written down to
 become the 3rd number; the 2nd and 3rd
 numbers are added, writing down the
 units digit of their sum as the 4th number;
 and so on
 (b) the chain repeats itself after a while
 20 (a) 11 (b) 44

Spatial awareness and pattern (p. 263)

1

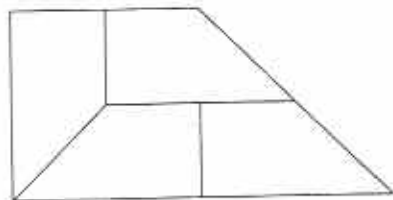


Fig. A23

- 2 (a) 1:4 (b) 1:4
 3 $1^2 + 2^2 + 3^2 + \dots + 8^2 = 204$
 4

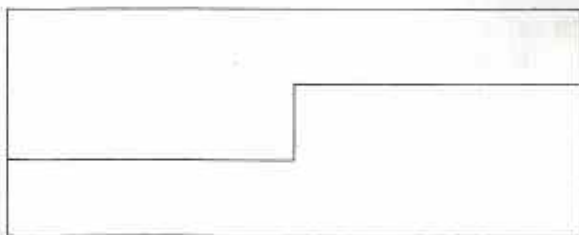


Fig. A24

- 5 (a) here is one of many solutions:

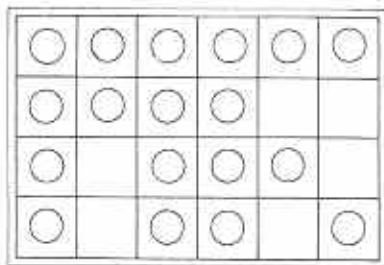


Fig. A25

- 6 10

8

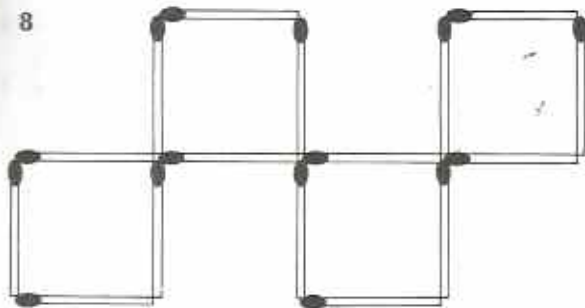


Fig. A26

- 9 $x + w = \phi$
 10 $m + n =$ (highest common factor of m and n)

Miscellaneous (p. 265)

- 1 1 in 5
 3 $36, 10, 124750 \left(= \frac{499 \times 500}{2} \right)$
 4 Bernadette took the exam paper (Cynthia is truthful)
 5 (a) 1,5 cm (b) 11,8 cm (c) 3,2 km, 23,6 km
 (d) 20,4 km (e) $10\frac{1}{2}$ min
 6 24000 km
 7 Tembo, the history student, did not eat lunch
 8 only 10 times
 9 X overtakes Y inside the last 5 m and wins
 10 741

Examination 1

Paper 1 (p. 267)

- 1 (a) $1\frac{5}{12}$ (b) 4 (c) $2\frac{1}{4}$
 2 (a) 9,17 (b) 16 (c) 0,000 168
 3 (a) {a; e; i; u} (b) 4 (c) {a; e; i; u; r; v}
 4 (a) 25 (b) 26%
 5 (a) 12 144_{five} (b) 1 000 00_{two} (c) 154_{ten}
 6 (a) $4,9 \times 10^7$ (b) 3×10^3 (c) $2,5 \times 10^{-8}$
 7 $x = 39, y = 19, z = 58$
 8 (a) $\frac{1}{450}$ (b) 0,004 (c) $\frac{5}{16}$
 9 (a) 133,5 kg (b) 11; 12 or 13
 10 (a) 5 units (b) $1\frac{1}{3}$ (c) $k = 4$
 11 (a) $\begin{pmatrix} 5 \\ -4 \end{pmatrix}$ (b) $\begin{pmatrix} 8 \\ 2 \end{pmatrix}$ (c) $\begin{pmatrix} -5 \\ 4 \end{pmatrix}$
 12 $x = 108^\circ, y = 36^\circ, z = 108^\circ$
 13 (a) 20°C (b) 7°C
 14 (a) $(x-7)(x-7)$ or $(x-7)^2$
 (b) $(3a-b)(2b+4) = 2(3c-b)(b+z)$
 15 $x = 2,5; y = -3$
 16 (a) 2,30 pm (b) 3 km (c) 2,5 km/h
 17 (a) \$240 (b) \$192
 18 308 ml
 19 (a) (ii) (b) $m = \frac{5400}{n}$ (c) 400
 20 (a) $n \geq 540$ (b) 416 (c) \$316,20
 21 (a) 323° or $\text{N}37^\circ\text{W}$ (b) 143° or $\text{S}37^\circ\text{E}$
 (c) 176 km
 22 (a) 120° (b) $k = 25$
 23 (a) 2,8 cm (b) $8,29 \text{ cm}^2$
 24 (a) $x = 6$ (b) $x = 1$ or 7
 25 (a) 0 (b) 2 (c) 11
 26 (a)

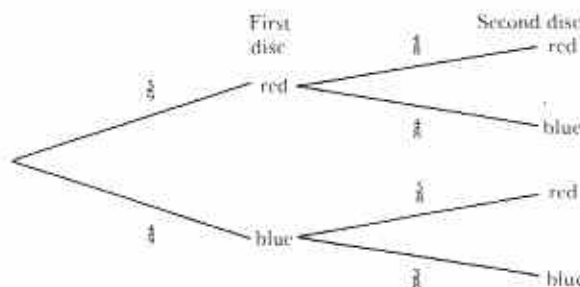


Fig. A27

- (b) $\frac{5}{18}$
 27 (a) $21 + 23 + 25 + 27 + 29$
 (b) 100 (c) 441

- 28 (a) 3 (b) $B'(9,5), C'(6,7)$

$$(c) \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}$$

- 29 (a) (i) 30 Francs (ii) Z\$18,30 approx.
 (b) (i) line should join origin to (20,65)
 (ii) Z\$16,90 approx.

Paper 2 (p. 271)

- 1 (a) (i) \$3900 (ii) \$78 000
 (b) (i) \$6,00 (ii) \$369,60 (iii) 45 years
 (iv) 31–35 years, Plan C
 2 (a) (i) $(x+3)(x+4) = 2(x+7)(x-2)$
 (ii) $x = 5$ or -8 , only $x = 5$ is realistic
 (iii) 72 units²
 (b) (i) $c = 1,5a - 18$ (ii) $-9,8^\circ\text{C}$
 3 (a) 85° (b) 8,97 cm
 4 (a) (i) $5,41 \text{ cm}^2$ (ii) $59,5 \text{ cm}^3$
 (iii) $55,5 \text{ cm}^3$ (iv) $52,4 \text{ cm}^3$
 (b) (i) 18 m (ii) 4000 cm^3
 5 (a) (i) 12° (base angles, isos $\triangle$)
 (ii) $204^\circ (= 360^\circ - \text{obtuse } \angle O\hat{A}A = 360^\circ - 156^\circ \text{ (angles of } \triangle O\hat{A}A \text{)})$
 (iii) 102° (angle at circumference = half angle at centre)
 (iv) $12^\circ (= \angle BAT \text{ (alt. segment)} = \angle O\hat{B}A \text{ (alt. angles)})$
 (v) 66° (angles on a straight line)
 (b) (i) $2a$ (ii) $2a - 3b$ (iii) $\frac{1}{2}a + \frac{1}{2}b$
 6 (a) $A_1(0;0), B_1(2;0), C_1(3;2)$
 (b) $A_2(0;0), B_2(6;0), C_2(9;6)$
 (c) $\begin{pmatrix} 3 & 3 \\ 0 & 3 \end{pmatrix}$
 7 (a) $x \leq 6, y \leq 6, 2x + y \geq 34, x + 3y < 42$
 (b)

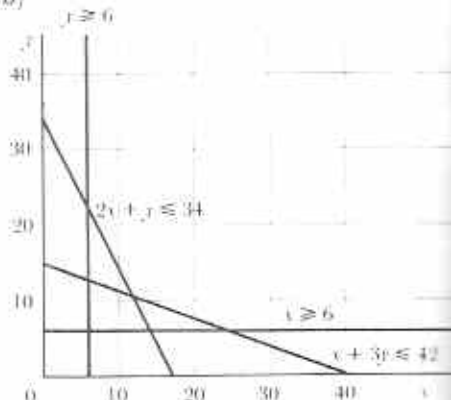


Fig. A28

- 8 (c) (i) 12 Standard and 10 Special (ii) \$688
 (a) $x = 3$ (b) 8 (c) (i) 17, (ii) 7
 (d)

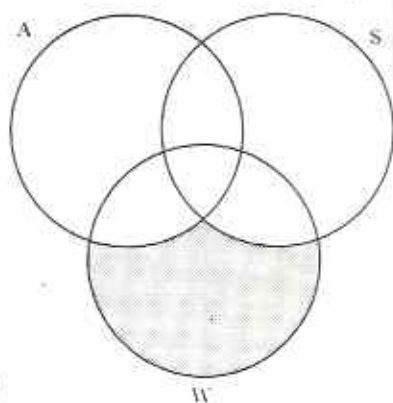


Fig. A29

those who can do only word-processing

- 9 (a) 9,33 cm (b) 9,98 cm
 (c) $69,25^\circ$ or $69^\circ 15'$ (d) $121,6 \text{ cm}^2$

- 10 (a) 120° (b) $2,36 \text{ m}^2$ (c) $1,38 \text{ m}^2$
 (d) 6 760 kg (or 6,76 tonnes)
 11 (a)

chest (cm)	≥ 80	≥ 85	≥ 90	≥ 95	≥ 100	≥ 105	≥ 110	≥ 115
cum. freq.	3	13	27	47	70	85	94	100

- (c) 96 cm (d) 77 people (e) $\frac{29}{330}$

- 12 (a)

x	1	2	3	4	5	6	7	8	9
y	9	4,5	3	2,25	1,8	1,5	1,29	1,13	1

- (d) $x = 1,35$ and $6,65$ (e) $x^2 - 8x + 9 = 0$

Examination 2

Paper 1 (p. 275)

- 1 (a) 60 000
(b) $6,03 \times 10^4$
(c) $2^2 \times 3^2 \times 5^2 \times 67$
- 2 (a) $\frac{1}{10}$
(b) 3
- 3 (a) 0
(b) 15
(c) 25
- 4 $x < -1$
- 5 (a) $(a + 5x)(a - 3x)$
(b) $100m + n$
- 6 (a) 19°
(b) 38°
- 7 (a) $r + r\sqrt{2}$
(b) $3 + 2\sqrt{2}$
- 8 (a) 0,05
(b) $2\frac{2}{3}$
(c) $5d^{18}$
- 9 6,75 kg
- 10 (a) 8
(b) 125
(c) $1\frac{1}{4}$
- 11 43 kg
- 12 0,7 kg (700 g)
- 13 (a) 5 cm
(b) 0,6
- 14 2 or 4
- 15 98 women
- 16 (a) 5
(b) $b = \frac{ac}{a^2 - 1}$
- 17 $a = 108^\circ$, $b = 36^\circ$
- 18 (a) 11,9 cm
(b) 15,6 cm
- 19 420 g
- 20 (b) (i) $\{2; 5; 7; 8\}$, (ii) 7
- 21 (b) $p = -1$, $q = -3$
(c) $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ (d) $\begin{pmatrix} 4 \\ 2 \end{pmatrix}$
- 22 $x = 2$ or $\frac{2}{3}$
- 23 (a) $-\frac{1}{3}$
(b) $y = -\frac{1}{3}x + 1\frac{2}{3}$ (i.e. $x + 3y = 5$)
(c) $(7; -\frac{2}{3})$
- 24 (a) 9 cm (b) $\frac{4}{9}$
- 25 (a) 112° (b) 124°

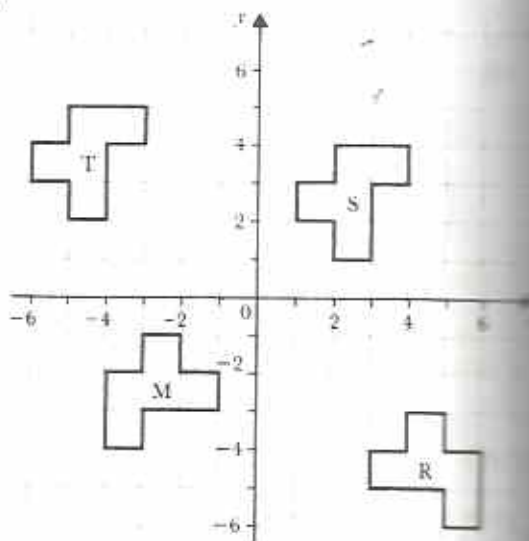


Fig. A30

- 27 (a) $y = \frac{8}{\sqrt{x}}$ (b) $y = 2\frac{2}{7}$ (c) $n = 4$
- 28 (a) 7,25 (b) $-3,2; 2,2$
(c) area in the range 25–26 unit²
- 29 (a) 1 272 m (b) 3 m/s^2 (c) 16 m/s

Paper 2 (p. 278)

- 1 (a) 6%
(b) (i) $\begin{pmatrix} 4 & -20 \\ 0 & 6 \end{pmatrix}$ (ii) $\begin{pmatrix} -6 & -10 \\ 8 & -4 \end{pmatrix}$
(iii) $x = -\frac{1}{2}$, $y = 3$
- 2 (a) $\hat{P} = 42^\circ$, $\hat{Q} = 69^\circ$, $\hat{R} = 138^\circ$, $\hat{PSR} = 117^\circ$
(b) (i) 16 000 (16 024,2) cm²
(ii) 56 600 (56 556) cm³
- 3 (a) $\frac{x-12}{(x+2)(3x-1)}$ (b) $x = 12$
(c) $(4a-3)(4a+3)$
 $1591 = 1600 - 9 = (40-3)(40+3)$
 $= 37 \times 43$
- 4 (a) (i) $\mathbf{a} + \mathbf{b}$ (ii) $-\frac{1}{2}\mathbf{a} - \frac{1}{2}\mathbf{b}$ (iii) $\frac{1}{2}\mathbf{a} - \frac{1}{2}\mathbf{b}$
(b) 24:25
- 5 (a) (i) 8 cm, 15 cm, 17 cm
(ii) 90° ($17^2 = 15^2 + 8^2 \Rightarrow$ rt-angled $\triangle$)
(iii) $28^\circ 4'$ or $28,07^\circ$
(b) 5×10^3

- 6 (a) (i) $a = 0,29$; $b = 1,74$; $P = 0,29W + 1,74$
 (ii) straight line cutting W -axis at $(-6; 0)$
 and P -axis at $(0; 1,74)$
- (b) $d = \frac{1}{3}$ or 7
- 7 (a) (i) $38^{\circ}40'$ or $38,66^{\circ}$ (ii) $73^{\circ}44'$ or $73,74^{\circ}$
 (b) (i) $(-6; 7)$ (ii) $r = -2, s = -1$
 (iii) one-way stretch of factor -3 parallel
 to the x -axis with the y -axis invariant
- 8 (a) $101c - 125c$ (b) $131,5c$ (c) $131c$
 (d) $\frac{11}{25}$ (e) (i) $\frac{3}{35}$, (ii) $\frac{3}{7}$
- 9 (a) $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$ (b) (i) 270° , (ii) $(0; 5)$
 (c) $y = x - 1$
- (d) (i) $x = 0$, (ii) $(-1; -2)$, $(-1; 0)$
 $(-2; -3)$, $(-2; -2)$, $(-3; -6)$
- 10 (a) (i) 139 m ,
 (ii) 139 m ,
 (iii) $10\,000 \text{ m}^2$
 (b) (i) 320° (or $N40^{\circ}W$),
 (ii) 140° (or $S40^{\circ}E$),
 (iii) $055,87^{\circ}$ (or $N55^{\circ}52'E$)
- 11 (a) $(1; 5)$, $(3; 11)$, $(5; 33)$
 (c) (i) during first $1,25$ seconds,
 (ii) $4,9 \text{ m/s}$,
 (iii) 11 m/s^2
- 12 (a) $9,0 \text{ cm}$
 (f) $2,7 \text{ cm}$ (answers to nearest $0,1 \text{ cm}$)